Springer Undergraduate Mathematics Series

S U M S

Rafael López

Point-Set Topology

A Working Textbook

Springer

Springer Undergraduate Mathematics Series

The Springer Undergraduate Mathematics Series (SUMS) is a series designed for undergraduates in mathematics and the sciences worldwide. From core foundational material to final year topics, SUMS books take a fresh and modern approach. Textual explanations are supported by a wealth of examples, problems and fully-worked solutions, with particular attention paid to universal areas of difficulty. These practical and concise texts are designed for a one- or two-semester course but the self-study approach makes them ideal for independent use.

Rafael López

Point-Set Topology

A Working Textbook

Springer

Rafael López
Departamento de Geometría y Topología
Universidad de Granada
Granada, Spain

ISSN 1615-2085 ISSN 2197-4144 (electronic)
Springer Undergraduate Mathematics Series
ISBN 978-3-031-58512-8 ISBN 978-3-031-58513-5 (eBook)
https://doi.org/10.1007/978-3-031-58513-5

Mathematics Subject Classification: 54-01

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

Preface

Learning mathematics requires doing mathematics. The set of definitions, theorems, corollaries and so on cannot be apprehended by simply looking at a book. One does not attain the real knowledge of a definition or a result if one does not bring it to bear on the resolution of problems. And this goes in both directions; practice using the concepts feeds back into a better understanding of them. Having a full solution allows one to identify an approach to the problem and to acquire tools for the following exercises, hence some exercises seem repeated. From repetition the reader acquires the ability to learn and subsequently face more difficult exercises. If students are never satisfied by the number of examples and solved exercises that lecturers can realistically present in a lecture course, this book provides more than enough exercises to acquire a reasonable knowledge of point-set topology.

This was the motivation for the writing of this book. In each chapter, there are a large number of worked exercises that use the concepts, results and techniques introduced in the chapter. These exercises can also introduce some more advanced concepts, generalizations and new results. Moreover, the theoretical contents of each chapter are accompanied by a large number of examples. Many of these examples are fully developed and can therefore also be seen as solved exercises. Since much of the material presented in this book can be found in many books, I have chosen to omit proofs of standard results in order to focus on the exercises.

The book covers an introductory course in point-set topology, also known as general topology, for undergraduate students. The material includes connectedness and compactness as important topological invariants, and methods of construction of topological spaces such as the relative topology, product topology and quotient topology. The last chapter delves into algebraic topology with an introduction to the notion of fundamental group. I include more than 160 examples in the main body of the text, illustrating concepts and results. As a second class of examples, there are numerous fully worked exercises at the end of each chapter, almost 300 in total.

The prerequisites for the reader are elements of set theory, including knowledge of standard operations between subsets, properties of maps and basic notions of cardinality. Nevertheless, reminders of relevant definitions and properties will be included in the text as they arise. Other prerequisites are calculus of one variable, including basics of convergence of sequences and continuity of functions, and familiarity with the notions of (real) vector spaces and groups. It generally assumed that the reader is familiar with mathematical proofs and logical reasonings, in particular, proof by induction, by contrapositive and by contradiction.

The book is designed to be used for a year-long course divided into two semesters. In the first semester, I recommend covering Chaps. 1–5, which introduce the core concepts of point-set topology. The remaining chapters can be covered in the second semester with the possible exception of Chap. 10, which can be omitted if time is short. It is not necessary to cover all of the exercises in the text, especially if the instructor feels that the relevant skill has already been acquired. Thus, the instructor may choose which exercises to include at their discretion, leaving others as reading for the students.

Each chapter is divided into sections. In general, a definition is followed by several examples that illustrate the concept. Each example ends with the symbol $\Diamond$. In every chapter, definitions, propositions, examples, theorems, etc. are marked x,y,z, where x is the chapter number, y is the section and z is the proposition number within the section. Only the Examples and the Worked Exercises follow their own numbering, so in case of a cross-reference the reader will know if the exercise is solved. The end of a proof or of a worked exercise is marked by the symbol $\Box$. We will use the usual notation of the sets of numbers, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{I}$, $\mathbb{R}$ and $\mathbb{C}$.

Granada, Spain
February 2024

Rafael López

Contents

Chapter 1
Introduction

Briefly,

Topology is the study of continuous maps.

In this book, we will explain the meaning of this statement, and see how the study of continuous functions between sets provides insights into their structure or *topology*. Of course, the reader is assumed to be already familiar with continuous functions from calculus, and these provide us with invaluable motivation. Our goal is not only to extend the idea of continuous functions, but also the sets where the functions are defined, the so-called topological spaces.

Let us begin by briefly explaining how topology generalizes the concept of continuity. In Fig. 1.1 (left), the function f is continuous at x_0. However, in Fig. 1.1 (right), the function f is not continuous at x_0 because the graph of f is "broken" around the point x_0. In calculus, one uses the standard Euclidean distance between two real numbers to measure how far and close points are to one another. The continuity of f at x_0 is written in $\varepsilon - \delta$ language, as we now recall. The function f is continuous at x_0 if (in symbolic notation)

$$\forall\varepsilon > 0, \exists\delta > 0 : |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon. \tag{1.1}$$

If we examine this expression, the numbers $|x-x_0|$ and $|f(x)-f(x_0)|$ represent the distance between the numbers x and x_0 and $f(x)$ and $f(x_0)$, respectively. However, for both functions of Fig. 1.1 it is clear that we do not need to know the explicit value of ε and δ to realize that the left function is continuous and the right one is not. The key notion is the *proximity* between the numbers x and x_0 and between their images under f. This is the point of view of a topologist; quantitative information about their distances is not required.

The proximity can be written in terms of sets and inclusions of sets. For f continuous at x_0, let the interval of length 2ε around $f(x_0)$ be viewed as a vicinity V; similarly, let the vicinity U correspond to the interval of length 2δ around x_0. Thus

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_1

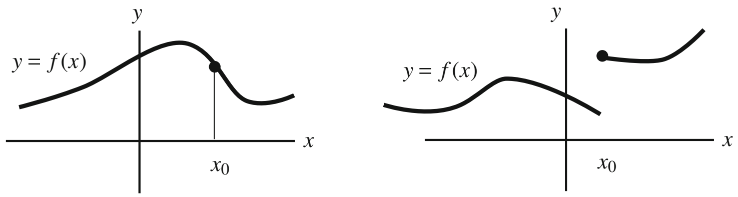


Fig. 1.1 The function f on the left is continuous at x_0, but on the right, the function f is not continuous at x_0

we say that f is continuous at x_0 if for any vicinity V of $f(x_0)$ we find a vicinity U of x_0 such that $f(U) \subset V$. We have replaced the distance and the inequalities in (1.1) by a statement in terms of set theory. This is the birth of topology.

A topological space is nothing other than a set and a function assigning to each element (point) of the set a collection of subsets called *neighborhoods* which will indicate the proximity of points in the set. The notions of continuity of mapping will be expressed in terms of neighborhoods, and *we will use only set theory to define and construct the framework of point-set topology.*

Returning to the initial statement, topology studies those properties preserved by continuous maps. One illustration of how topology helps in mathematics is the following. Mathematicians solve problems, or at least give strategies and methods to solve them. Think of a problem for which we want to find a solution. If we are able to write the problem in terms of continuous maps, then we can apply topological techniques which can give us the solution, or at least a method for how to find it. We offer here an elementary but instructive example whose consequences in mathematics are of high importance. The simplicity of the problem does not diminish its interest and serves as a model for much more complex problems. The context appears in a first course of calculus without any previous knowledge of topology. We ask if there is a solution of the equation

$$e^x + \sin(x) - 2 = 0.$$

One immediately realizes that it is not possible to find an explicit solution from simple algebraic operations. Even more, we do not know yet if there exists a solution of this equation. We define the function $f(x) = e^x + \sin(x) - 2$, just the left-hand side of the equation. The problem is to find a real number c such that the function f evaluated at $x = c$ is 0. If we regard f as a function $f : \mathbb{R} \to \mathbb{R}$, then f is *continuous*. How can we exploit the topological techniques? A famous result in calculus is Bolzano's theorem that asserts that if a continuous function $f : I \to \mathbb{R}$ defined on an interval $I \subset \mathbb{R}$ satisfies $f(a) < 0$ and $f(b) > 0$ for some $a, b \in I$, then there is $c \in I$ such that $f(c) = 0$ (Fig. 1.2). For this f, we have $f(-\pi) = e^{-\pi} - 2 < 0$ and $f(\pi) = e^{\pi} - 2 > 0$ and thus there is a solution in the interval $(-\pi, \pi)$. Although Bolzano's theorem tells if a given equation admits a

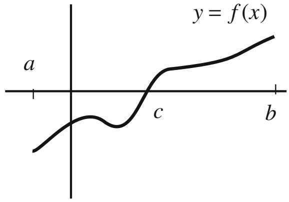


Fig. 1.2 Bolzano's theorem. A continuous function defined on the bounded closed interval $[a, b]$ that takes negative and positive values must vanish at some point

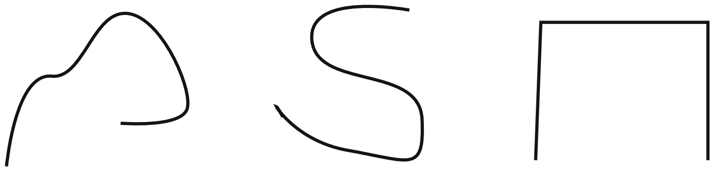


Fig. 1.3 All figures are equal for a topologist, because each one can be deformed into the others

solution, it does not give us the explicit solution. However on many occasions it is sufficient to show the mere existence of solutions.

Which topological properties must the interval I have for Bolzano's theorem to be true? The key is that an interval I is formed by "one piece", or in topological terminology, the set I is *connected*. Bolzano's theorem asserts that any real continuous function defined on a connected space and assuming both positive and negative values must vanish at some point. Summarizing, the strategy has been to formulate the problem in terms of continuous maps defined on connected spaces.

Topology studies topological spaces and the properties preserved by continuous maps. Two spaces are topologically equivalent if there is a one-to-one mapping between them such that the mapping and its inverse are continuous. These mappings are called *homeomorphisms*. For subsets of the Euclidean space, one may think of a homeomorphism as a deformation of one space to another. This is why it is often said that topology is the study of the mathematics of rubber objects. If a space X made of rubber can be deformed into another space Y, in the eyes of a topologist X is equal to Y. The allowed deformations consist of stretching, contracting or moving, but cutting or gluing is forbidden. For example, if we cut, close points in X may correspond with far points in Y, losing the continuity. In case that we glue, we may identify two points of X to the same point in Y, losing the bijectivity of the transformation. Hence all the curves of Fig. 1.3 are equal for a topologist. Therefore, for a topologist the size is not important and what is really significant is that proximity of points is preserved.

Chapter 2
Topological Spaces

The purpose of this chapter is to introduce the concept of a topological space and to study its properties. In principle, we will ignore the intuitive concepts of proximity that we discussed in the introduction. The only tool we need is set theory; the axioms utilized in the definition of a topological space are simple operations of sets. A topological space is a set together with a collection of subsets endowed with various properties of the union and intersection of its elements. Despite this simplicity, this set-theoretic approach to topology, the so-called *point-set-topology*, is the foundation of all other disciplines of topology.

2.1 Topological Spaces and Examples

Definition 2.1.1 A *topological space* is a pair (X, τ), where X is a nonempty set and τ is a collection of subsets of X, called *open sets*, satisfying the following properties:

(i) The trivial sets $\emptyset$ and X are open sets.
(ii) The union of arbitrarily many open sets is an open set.
(iii) The intersection of two open sets is an open set.

The collection τ of open sets is said to be the *topology* of X.

The same underlying set X may be equipped with distinct topologies, resulting in different topological spaces. Given a bare set X, *to topologize* X is to assign a collection τ of subsets such that (X, τ) is a topological space. To simplify the notation and language, we say a *space* X instead of *a topological space* (X, τ) if the topology τ is understood. The following observations are worth pointing out.

1. The intersection of a finite number of open sets is open. This is immediate by an induction argument in (iii).

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_2

2. We can replace the open sets in (ii) and (iii) with specifically open sets distinct from $\emptyset$ and X; if one of the subsets is $\emptyset$ or X, that subset does not affect the verification of (ii) and (iii).

Example 2.1

1. On a set X, consider $\tau_T = \{\emptyset, \ X\}$. The topology τ_T is called the *trivial topology*. This topology is clearly the smallest one that can exist on X.
2. On a set X, consider $\tau_D = \mathbf{P}(X)$, where $\mathbf{P}(X)$ is the collection of all subsets of X. The topology τ_D is called the *discrete topology* . This topology is the largest one that can support X. Only in case that X has one element do the trivial and discrete topologies coincide.

$\diamondsuit$

Example 2.2 The discrete topology is characterized by the property that all singletons (subsets with only one element) are open sets. Indeed, we need to prove that any subset of X is open. Let $A \subset X$. Then $A = \bigcup_{x \in A}\{x\}$. Since $\{x\}$ is open for each $x \in A$, A is open because it is union of open sets. $\diamondsuit$

Example 2.3 If $X = \{a, b\}$, define the *Sierpinski topology* by $\tau = \{\emptyset, X, \{a\}\}$. Another topology on X is $\tau' = \{\emptyset, X, \{b\}\}$. For a subset with only two elements the only topologies that can support X are τ_T, τ, τ' and τ_D. $\diamondsuit$

Example 2.4 On a set X, let $\tau_{CF} = \{\emptyset\} \cup \{X \setminus F : \ F \subset X \text{ is finite}\}$. Here $X \setminus F = \{x \in X : x \notin F\}$ stands for the complement of F. This topology is called the *cofinite topology*. If X is finite, then the cofinite topology coincides with the discrete topology. Throughout this book, we will consider the cofinite topology when X is infinite, so $\tau_{CF} \neq \tau_D$. To check that τ_{CF} obeys the properties to be a topology, we use De Morgan laws. If $\{X \setminus F_i : i \in I\} \subset \tau_{CF}$, then

$$\bigcup_{i \in I}(X \setminus F_i) = X \setminus \bigcap_{i \in I} F_i \in \tau_{CF}$$

because $\bigcap_{i \in I} F_i$ is finite. Also

$$(X \setminus F_1) \cap (X \setminus F_2) = X \setminus (F_1 \cup F_2) \in \tau_{CF}$$

because $F_1 \cup F_2$ is finite. In the cofinite topology the intersection of arbitrary open sets may be not open. For example, if $X = \mathbb{R}$ and $O_n = \mathbb{R} \setminus \{n\} \in \tau_{CF}$ for $n \in \mathbb{N}$, then $\bigcap_{n \in \mathbb{N}} O_n = \mathbb{R} \setminus \mathbb{N}$ is not open. $\diamondsuit$

Example 2.5 Let X be a set and let $p \in X$ be a distinguished point of X. Define

$$\tau_{in}^p = \{\emptyset\} \cup \{O \subset X : \ p \in O\}, \quad \tau_{ex}^p = \{X\} \cup \{O \subset X : p \notin O\}.$$

We shall call τ_{in}^p and τ_{ex}^p the *included point topology* and the *excluded point topology* for p, respectively. See Fig. 2.1. In general, we will omit the superscript if p is

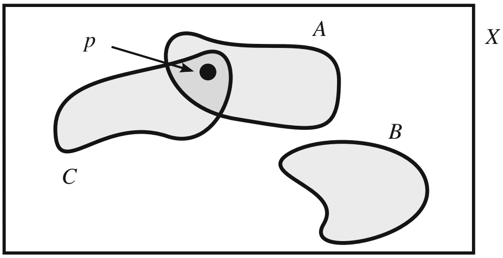


Fig. 2.1 The included point topology and the excluded point topology. With respect to p, $A, C \in \tau_{in}$ and $B \in \tau_{ex}$

understood. In both topologies, the intersection of an arbitrary number of open sets is open. $\diamondsuit$

Example 2.6 Let X be a set with two topologies τ_1 and τ_2. The elements of $\tau_1 \cap \tau_2$ are the sets that are open in both topologies (not the intersection of elements of τ_1 with those of τ_2). It is immediate that $\tau_1 \cap \tau_2$ is a topology on X. However, the union of two topologies is not a topology in general. For instance, let $X = \{a, b, c\}$, $\tau_1 = \{\emptyset, X, \{a, b\}\}$ and $\tau_2 = \{\emptyset, X, \{a, c\}\}$. Then $\tau_1 \cup \tau_2 = \{\emptyset, X, \{a, b\}, \{a, c\}\}$ is not a topology because $\{a, b\} \cap \{a, c\} = \{a\} \notin \tau_1 \cup \tau_2$. $\diamondsuit$

We now introduce the concept of a closed set in a space.

Definition 2.1.2 A subset F in a topological space X is said to be *closed* if its complement $X \setminus F$ is open. Denote by $\mathscr{F}$ the collection of all closed sets of the space.

The trivial subsets $\emptyset$ and X are both open and closed. Further, a subset of a space need not be either open or closed. This happens in $(\mathbb{R}, \tau_{CF})$, where $\mathbb{N}$ is not open because its complement is not finite and $\mathbb{N}$ is not closed because $\mathbb{N}$ is infinite.

Example 2.7

1. In the trivial topology, $\mathscr{F}_T = \tau_T$ and in the discrete topology, $\mathscr{F}_D = \tau_D$ again.
2. In the Sierpinski topology, the family of closed sets is $\mathscr{F} = \{\emptyset, X, \{b\}\}$. In this space, $\{b\}$ is closed but not open.
3. In the cofinite topology, $\mathscr{F}_{CF} = \{F \subset X : F \text{ is finite}\} \cup \{X\}$. The only open and closed sets are the trivial sets.
4. If $p \in X$ and τ_{in} and τ_{ex} are the included and excluded point topologies for p, respectively, then $\mathscr{F}_{in} = \tau_{ex}$ and $\mathscr{F}_{ex} = \tau_{in}$. $\diamondsuit$

Example 2.8 It is relatively easy to find spaces with nontrivial subsets that are both open and closed. For instance, if $X = \{a, b, c\}$ and $\tau = \{\emptyset, X, \{a\}, \{b, c\}\}$, then $\tau = \mathscr{F}$. The question of which spaces satisfy that the only open and closed subsets are the trivial ones will be studied in Chap. 8. $\diamondsuit$

Coming back to the concept of closed sets, and thanks to the properties of open sets and De Morgan laws of set theory, the statements of open sets that appeared in the definition of a topology can be dualized. Thus we could begin the chapter defining a topological space using the concept of closed sets.

Theorem 2.1.3 *Suppose that $\mathscr{C}$ is a collection of subsets of a set X satisfying:*

(i) The trivial sets $\emptyset$ and X belong to $\mathscr{C}$.
(ii) The intersection of arbitrary many elements of $\mathscr{C}$ belongs to $\mathscr{C}$.
(iii) The union of two elements of $\mathscr{C}$ belongs to $\mathscr{C}$.

Then there is a unique topology τ on X such that its family of closed sets is $\mathscr{C}$.

Let us complete this section by comparing two topologies on a given set in terms of inclusions.

Definition 2.1.4 Let τ_1 and τ_2 be two topologies on a given set X. The topology τ_1 is said to be *finer* than τ_2 if $\tau_2 \subset \tau_1$. We also say that the topology τ_2 is *coarser* than τ_1. Two topologies are *comparable* if any one is finer than the other.

It is clear that τ_1 is finer than τ_2 if and only if $\mathscr{F}_2 \subset \mathscr{F}_1$.

It is obvious that the discrete topology is the finest topology on a given set and that the trivial topology is the coarsest one.

2.2 Basis for a Topology

It would be desirable to have a small number of open sets that would allow us to find the rest of the open sets. This role is played by the notion of a basis for a topology.

Definition 2.2.1 A *basis* for the topology of a space is a collection β of open sets with the property that every open set is the union of some subcollection of β. The elements of a basis are called *basic elements* or *basic open sets*.

Proposition 2.2.2 *A collection β of open sets is a basis for τ if and only if for every open set O and $x \in O$, there is $B \in \beta$ such that $x \in B \subset O$. As a consequence, $A \subset X$ is open if and only if for each $x \in A$, there is $B \in \beta$ such that $x \in B \subset A$.*

Proof Assume that β is a basis for τ and let $O \in \tau$ and $x \in O$. Since $O = \bigcup_{i \in I} B_i$ for a certain set of indices I and $B_i \in \beta$, there is $j \in I$ such that $x \in B_j$. Then $x \in B_j \subset O$.

Reciprocally, let $O \in \tau$. By hypothesis, for each $x \in O$ there is an element $B \in \beta$ such that $x \in B \subset O$. This B is not necessarily unique. We fix one of them and denote it by B_x, so $x \in B_x \subset O$. Then

$$\bigcup_{x \in O} \{x\} \subset \bigcup_{x \in O} B_x \subset O.$$

Because $\bigcup_{x \in O} \{x\} = O$, we conclude $O = \bigcup_{x \in O} B_x$ and this finishes the proof. $\square$

For a given basis, if we add more open sets, the resulting family remains a basis. One may ask if a space has a smallest basis in terms of cardinality. Such a basis is called an *efficient basis*.

Example 2.9

1. A basis for the discrete topology (X, τ_D) is $\beta = \{\{x\} : x \in X\}$ because if $O \in \tau_D$, then $O = \bigcup_{x \in O}\{x\}$. Moreover, β is the smallest basis for τ_D. Certainly, let β' be another basis. For each $x \in X$, the unitary set $\{x\}$ is open. As $x \in \{x\}$ and β' is a basis, there is $B' \in \beta'$ such that $x \in B' \subset \{x\}$. This yields $B' = \{x\}$, proving the inclusion $\beta \subset \beta'$.
2. If a basis is finite, then the collection of open sets, the topology, is finite. Thus if the topology is infinite, any basis must be infinite. Examples of spaces with an infinite number of open sets are spaces where X is infinite and the singletons are closed (for instance, the cofinite topology). In such a case, the collection of open sets $\{X \setminus \{x\} : x \in X\}$ is infinite.
3. In the cofinite topology, the family of open sets $\gamma = \{X \setminus \{x\} : x \in X\}$ is not a basis. Indeed, $O = X \setminus \{x_1, x_2\}$ is open and let $y \in O$. If γ were a basis, there would be $x \in X$ such that

$$y \in X \setminus \{x\} \subset X \setminus \{x_1, x_2\},$$

but the last inclusion would imply that $\{x_1, x_2\} \subset \{x\}$, which is a contradiction.
4. For the included point topology τ_{in} with distinguished point p, the family of open sets

$$\beta_{in} = \{\{x, p\} : x \in X\}$$

is a basis, for if $O \in \tau_{in}$ and $x \in O$, then $x \in \{x, p\} \subset O$. We see that β_{in} is the smallest basis for τ_{in}. If β' is another basis, let $\{x, p\} \in \beta_{in}$. Then there is $B' \in \beta'$ such that $x \in B' \subset \{x, p\}$. Since B' is open, then $p \in B'$, hence $B' = \{x, p\}$. This proves that $\{x, p\} \in \beta'$.
5. For the excluded point topology τ_{ex}, the family of open sets

$$\beta_{ex} = \{\{x\} : x \neq p, x \in X\} \cup \{X\}$$

is a basis for τ_{ex}. Indeed, for each $O \in \tau_{ex}$ and each $x \in O$, we have: (i) if $O = X$ then $x \in X \subset O$ and $X \in \beta_{ex}$; (ii) if $O \neq X$ then $p \notin O$ so $x \neq p$, $\{x\} \in \beta_{ex}$ and $x \in \{x\} \subset O$. Also β_{ex} is the smallest basis for τ_{ex}. Let us observe that $\sigma = \{\{x\} : x \neq p\}$ is not a basis for τ_{ex} because X is open, $p \in X$ but there is no $B \in \sigma$ such that $p \in B \subset X$. $\diamondsuit$

Now we ask if it is possible to topologize an abstract set X by using the notion of basis for the topology.

Question: *Given a set X, under what conditions may we consider a collection γ of subsets of X to be a basis for some topology τ on X?*

Theorem 2.2.3 *Let X be a set and let γ be a collection of subsets of X with the following properties:*

(i) $X = \bigcup_{B \in \gamma} B$.
(ii) For every $B_1, B_2 \in \gamma$ *and every* $x \in B_1 \cap B_2$*, there is* $B_3 \in \gamma$ *such that* $x \in B_3 \subset B_1 \cap B_2$.

Then γ is a basis for a unique topology for X, denoted by $\tau(\gamma)$. This topology is called the topology generated by γ. A sufficient condition that implies (ii) is

(ii) the intersection of any two elements of γ is either empty or another member of γ.*

Throughout this book, and in order to verify (ii*), we will prove that the intersection of two elements $B_1, B_2 \in \gamma$ with $B_1 \cap B_2 \neq \emptyset$ is another element of γ; otherwise, if $B_1 \cap B_2 = \emptyset$, there is nothing to check.

Example 2.10 For each $x \in \mathbb{R}$, let $B_x = \{x\} \cup (x+1, \infty)$. We see that $\beta = \{B_x : x \in \mathbb{R}\}$ constitutes a basis for some topology on $\mathbb{R}$. First notice that β does not satisfy (ii*); for example, $B_1 \cap B_2 = (3, \infty) \notin \beta$. We must test the conditions of Theorem 2.2.3. Since every $x \in \mathbb{R}$ is contained in B_x, then $X = \bigcup_{x \in \mathbb{R}} B_x$. Now let $z \in B_x \cap B_y$. The intersection $B_x \cap B_y$ is easily computed. So, without loss of generality we assume that $x < y$. Then the intersection is $(y+1, \infty)$ if $y \leq x+1$ or B_y if $x+1 < y$. In the first case, $y+1 < z$, so $z \in B_z \subset (y+1, \infty)$. The second case is immediate because the intersection is an element of β. ♢

Example 2.11 Let $\{A_i : i \in I\}$ be a collection of subsets of a set X that are pairwise disjoint. Then $\beta = \{A_i : i \in I\} \cup \{X\}$ is a basis for some topology because (ii*) is trivial. Notice that $\beta \neq \tau(\beta)$; that is, there are open sets that are not basic open sets because, for instance, $A_i \cup A_j \in \tau(\beta)$ but it is not an element of β. Consequently, any partition of a set constitutes a basis for some topology. For example, the discrete topology results from the partition formed by all singletons, and the trivial topology results from the partition formed by the trivial subsets. ♢

We apply Theorem 2.2.3 to produce many topologies on the underlying set $\mathbb{R}$ of real numbers by taking collections of subsets where (ii*) holds. We are assuming that the reader is familiar with the concepts of supremum and infimum.

Example 2.12

1. Consider the family of all bounded closed intervals $\beta_c = \{[a, b] : a \leq b, a, b \in \mathbb{R}\}$ of the real line $\mathbb{R}$. The topology determined by β_c is discrete because any singleton $\{x\}$ is the interval $[x, x] \in \beta_c$, so an open set.
   If we replace β_c by $\gamma = \{[a, b] : a < b, a, b \in \mathbb{R}\}$, then γ is not a basis for any topology. For example, $1 \in [0, 1] \cap [1, 2]$ but there is no $[a, b] \in \gamma$ such that

$$1 \in [a, b] \subset [0, 1] \cap [1, 2] = \{1\}.$$

Fig. 2.2 Basic open sets in the Sorgenfrey topology (left) and in the right order topology (right)

2. Let $\beta_S = \{[a, b) : a, b \in \mathbb{R}\}$. This family of intervals is a basis for a topology τ_S called the *Sorgenfrey topology* and $(\mathbb{R}, \tau_S)$ is called the *Sorgenfrey line* (Fig. 2.2, left). Here $\beta_S \subsetneq \tau_S$ because, for example, $(0, 1) = \bigcup_{n\in\mathbb{N}}[\frac{1}{n}, 1) \in \tau_S$ but $(0, 1)$ is not an element of β_S.
3. Define the *strictly right order topology* on $\mathbb{R}$ as the topology generated by $\beta_{sr} = \{(a, \infty) : a \in \mathbb{R}\}$. Then

$$\bigcup_{i\in I}(a_i, \infty) = \begin{cases} (a, \infty), & \text{if } a = \inf\{a_i : i \in I\} \\ \mathbb{R}, & \text{if } \{a_i : i \in I\} \text{ is not bounded from below.} \end{cases}$$

Thus $\tau_{sr} = \{\emptyset, \mathbb{R}\} \cup \beta_{sr}$.
4. Consider $\beta_r = \{[a, \infty) : a \in \mathbb{R}\}$. Then β_r is a basis for a topology τ_r called the *right order topology* on $\mathbb{R}$ (Fig. 2.2, right). If $a \in \mathbb{R}$, then $(a, \infty) = \bigcup_{b>a}[b, \infty) \in \tau_r$ and this proves that $\tau_{sr} \subset \tau_r$. It is also possible to calculate all open sets in $(\mathbb{R}, \tau_r)$, namely, if $A = \{a_i : i \in I\}$, then

$$\bigcup_{i\in I}[a_i, \infty) = \begin{cases} [a, \infty), & \text{if } A \text{ is bounded from below and } a = \min(A) \\ (a, \infty), & \text{if } A \text{ is bounded from below and } a = \inf(A), a \notin A \\ \mathbb{R}, & \text{if } A \text{ is not bounded from below.} \end{cases}$$

As a conclusion, $\tau_r = \{\emptyset, \mathbb{R}\} \cup \beta_r \cup \{(a, \infty) : a \in \mathbb{R}\}$.
5. Following the preceding examples, define new topologies on $\mathbb{R}$ by the bases

$$\beta_S^l = \{(a, b] : a, b \in \mathbb{R}\},$$

$$\beta_{sl} = \{(-\infty, a) : a \in \mathbb{R}\},$$

$$\beta_l = \{(-\infty, a] : a \in \mathbb{R}\}.$$

The topologies τ_{sl} and τ_l determined by β_{sl} and β_l, are called the *strictly left order topology* and the *left order topology*, respectively. ♢

Example 2.13 We compare the above topologies. Observe first that in general if β_1 is a basis for τ_1 such that $\beta_1 \subset \tau_2$, then arbitrary unions of elements of β_1 are open in τ_2, so $\tau_1 \subset \tau_2$. Coming back to the above topologies, the elements (a, ∞) are open in the Sorgenfrey topology because $(a, \infty) = \bigcup_{n\in\mathbb{N},a<b}[b, b + n)$, hence $\tau_{sr} \subset \tau_S$. It is also obvious that $[a, \infty) \in \tau_S$ because $[a, \infty) = \bigcup_{n\in\mathbb{N}}[a, a + n)$. This demonstrates that $\tau_r \subset \tau_S$.

On the other hand, τ_S and τ_S^l are not comparable. So $[0, 1) \in \tau_S$ but $[0, 1) \notin \tau_S^l$ and, similarly, $(0, 1] \in \tau_S^l$ but it is not an element of τ_S. For instance, if $[0, 1) \in \tau_S^l$, there is $(b, c] \in \beta_S^l$ such that $0 \in (b, c] \subset [0, 1)$ because $0 \in [0, 1)$. This implies $b < 0$ which is not possible. ♢

In general, a space may have many different bases that generate the same topology.

Definition 2.2.4 Two bases on a given underlying set are called *equivalent* if both bases generate the same topology.

The equivalence of bases establishes an equivalence relation $\sim$ on the set of all bases defined on a given set by declaring $\beta_1 \sim \beta_2$ if $\tau(\beta_1) = \tau(\beta_2)$. Consequently, the set of all bases related to a given basis β coincides with all bases of the topology $\tau(\beta)$. The next result investigates the question of when two bases are equivalent.

Theorem 2.2.5 (Hausdorff) *If β_1 and β_2 are bases for τ_1 and τ_2 respectively, then $\tau_1 \subset \tau_2$ if and only if for every $B_1 \in \beta_1$ and $x \in B_1$, there is $B_2 \in \beta_2$ such that $x \in B_2 \subset B_1$.*

Example 2.14 The right order topology τ_r is included in the Sorgenfrey topology τ_S because if $[a, \infty) \in \beta_r$ and $x \in [a, \infty)$ then $[x, x+1) \in \beta_S$ and $x \in [x, x+1) \subset [a, \infty)$. ♢

2.3 The Euclidean Topology

In this section we present a remarkable topology on $\mathbb{R}^n$, the so-called Euclidean topology. We begin by defining the Euclidean topology on $\mathbb{R}$ and later on $\mathbb{R}^n$. The reader will have noticed that bounded open intervals were not employed in Example 2.12.

Definition 2.3.1 The *Euclidean topology* (or *usual topology*) on $\mathbb{R}$ is the topology generated by the collection of all bounded open intervals

$$\beta_u = \{(a, b) : a, b \in \mathbb{R}\}.$$

The topology is denoted τ_u and we say that $(\mathbb{R}, \tau_u)$ is the *Euclidean real line*.

The hypothesis (ii*) of Theorem 2.2.3 is clearly fulfilled. However, β is not a topology since the union of two disjoint open intervals is not an interval. Also intersection of an arbitrary family of open sets is not necessarily an open set. For example,

$$\{x\} = \bigcap_{n\in\mathbb{N}} (x - \tfrac{1}{n}, x + \tfrac{1}{n})$$

and $\{x\}$ is not open because there are no elements of β_u inside $\{x\}$.

As a way to become familiar with the Euclidean topology, we investigate which intervals of $\mathbb{R}$ are open and which ones are closed. Recall that an *interval* I of $\mathbb{R}$ is a subset satisfying the following property: for every $s, t \in I$ such that $s < t$, we have $\{x \in \mathbb{R} : s \leq x \leq t\} \subset I$. Or, equivalently, if for $s, t \in I$, $s < t$, we have $[s, t] \subset I$. The types of intervals are the following:

1. $(a, b) = \{x \in \mathbb{R} : a < x < b\}$.
2. $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$.
3. $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$.
4. $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$.
5. $(a, \infty) = \{x \in \mathbb{R} : a < x\}$.
6. $[a, \infty) = \{x \in \mathbb{R} : a \leq x\}$.
7. $(-\infty, a) = \{x \in \mathbb{R} : x < a\}$.
8. $(-\infty, a] = \{x \in \mathbb{R} : x \leq a\}$.
9. $\mathbb{R} = (-\infty, \infty)$.

The intervals (1), (5) and (7) are called *open intervals* and (2), (6) and (8) are called *closed intervals*. The intervals $[a, b)$ and $(a, b]$ of (3) and (4) are called *closed-open* and *open-closed* intervals, respectively.

Example 2.15 We show that an open interval is an open set but it is not closed. A bounded open interval (a, b) is open because it is an element of β_u. The rest of open intervals are unions of elements of β_u,

$$(a, \infty) = \bigcup_{n \in \mathbb{N}} (a, a + n), \quad (-\infty, a) = \bigcup_{n \in \mathbb{N}} (a - n, a).$$

We prove that (a, b) is not a closed set, or equivalently, that $A = (-\infty, a] \cup [b, \infty)$ is not an open set. Otherwise and in view of $a \in A$, there would be $(c, d) \in \beta_u$ such that $a \in (c, d) \subset A$. Then $a < d$. Since $a < b$, there is $z \in \mathbb{R}$ such that $a < z < \min\{b, d\}$. Thus $a < z < b$, which a contradiction because $z \in A$.

Similarly, (a, ∞) is not a closed set because $(-\infty, a]$ is not open: since $a \in (-\infty, a]$, there would be $(b, c) \in \beta_u$ such that $a \in (b, c) \subset (-\infty, a]$, which is not possible (Fig. 2.3, left).

$\diamondsuit$

Example 2.16 We now see that a closed interval is a closed set but it is not open. In particular, every singleton $\{x\}$ is closed. The complement of a bounded closed interval is union of open intervals and the complement of an unbounded closed interval is an open interval. This proves that a closed interval is a closed set. On

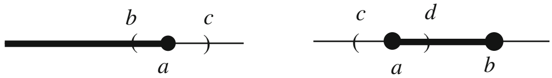


Fig. 2.3 The sets $(-\infty, a]$ (left) and $[a, b]$ (right) are not open in the Euclidean topology

the other hand, if $[a, b]$ was an open set, there would exist $(c, d) \in \beta_u$ such that $a \in (c, d) \subset [a, b]$ because $a \in [a, b]$. This implies $c < a$ and $a \leq c$ (Fig. 2.3, right.) In the same fashion, $[a, \infty)$ and $(-\infty, a]$ are not open sets.

$\diamondsuit$

Example 2.17 We prove that an interval $[a, b)$ or $(a, b]$ is neither open nor closed. Since $a \in [a, b)$, if $[a, b)$ is an open set, there is $(c, d) \in \beta_u$ such that $a \in (c, d) \subset [a, b)$. Again, this is not possible. If $[a, b)$ is a closed set, its complement $(-\infty, a) \cup [b, \infty)$ is an open set. Then there would be $(c, d) \in \beta_u$ such that $b \in (c, d) \subset (-\infty, a) \cup [b, \infty)$, which is a contradiction again. Similarly, $(a, b]$ is neither open nor closed.

$\diamondsuit$

These examples tells that the terminology of "open interval" and "closed interval" is in accordance with the topological terms "open" and "closed", respectively. The next remarks are interesting.

(i) Since bounded and non-bounded open intervals are open sets, the collection of all open intervals is a basis for the Euclidean topology because it contains β_u.
(ii) The real line $\mathbb{R} = (-\infty, \infty)$ is the only interval that is both open and closed. In Chap. 8, it will be proved that *the only open and closed subsets in the Euclidean real line are the trivial sets* $\emptyset$ *and* $\mathbb{R}$.
(iii) One may think that $[a, b]$ is not open with the incorrect argument that its complement $\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, \infty)$ is union of two non-closed sets. This reasoning fails as for example $[-1, 1) \cup (0, 2] = [-1, 2]$.

Example 2.18 We compare the Euclidean topology with the topologies of Example 2.12. The Sorgenfrey topology is finer than the Euclidean one because the members of β_u are open in τ_S. In fact, for $(a, b) \in \beta_u$, if $x \in (a, b)$, then $[x, b) \in \tau_S$ and $x \in [x, b) \subset (a, b)$. However, both topologies do not coincide because $[a, b) \in \tau_S$ but $[a, b) \notin \tau_u$.

The right order topology τ_r is, by contrast, not comparable with the Euclidean topology. So $[0, \infty)$ is open in τ_r but not in the Euclidean topology. Similarly, $(0, 1)$ is open in the Euclidean topology but $(0, 1) \notin \tau_r$. $\diamondsuit$

The Euclidean topology is a clear example of a space where the number of open sets is surprisingly large and complex. Let us show one of these open sets, which will be described by its complement. This set, defined by Georg Cantor in 1883, is famous in mathematics.

Example 2.19 The *Cantor set* is the set

$$\mathfrak{C} = \bigcap_{n=1}^{\infty} C_n, \tag{2.1}$$

where the sets C_n are defined by a recursive procedure as follows (Fig. 2.4). Let us start with the interval $[0, 1]$. To define C_1, remove the middle third interval $J_1 =$

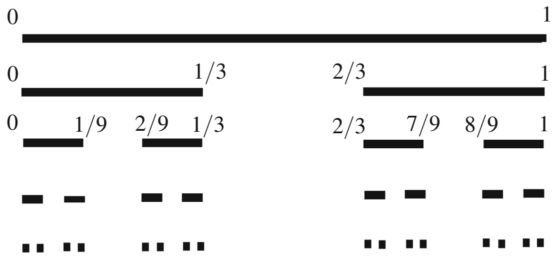


Fig. 2.4 The Cantor set $\mathfrak{C} = \bigcap_{n=1}^{\infty} C_n$

$J_{11} = (\frac{1}{3}, \frac{2}{3})$ of $[0, 1]$, obtaining two closed intervals $[0, 1] \setminus (\frac{1}{3}, \frac{2}{3}) = C_{11} \cup C_{12}$, with $C_{11} = [0, \frac{1}{3}]$ and $C_{12} = [\frac{2}{3}, 1]$. Let

$$C_1 = [0, 1] \setminus J_1 = C_{11} \cup C_{12}.$$

To define C_2, we iterate this process for C_{11} and C_{12} removing the middle third intervals $J_{21} = (\frac{1}{9}, \frac{2}{9})$ and $J_{22} = (\frac{7}{9}, \frac{8}{9})$. After removing $J_2 = J_{21} \cup J_{22}$, the resulting four intervals are

$$C_{21} = \left[0, \frac{1}{3^2}\right], \quad C_{22} = \left[\frac{2}{3^2}, \frac{3}{3^2}\right], \quad C_{23} = \left[\frac{6}{3^2}, \frac{7}{3^2}\right], \quad C_{24} = \left[\frac{8}{3^2}, 1\right].$$

Let

$$C_2 = [0, 1] \setminus J_1 \cup J_2 = \bigcup_{j=1}^{4} C_{2j}.$$

The construction is continued inductively by defining $\{J_n : n \in \mathbb{N}\}$ and $\{C_n : n \in \mathbb{N}\}$, where J_n is disjoint union of 2^{n-1} open intervals $J_{n1}, \ldots, J_{n2^{n-1}}$ and C_n is disjoint union of 2^n closed intervals $C_{n1}, \ldots, C_{n2^n}$,

$$C_n = \bigcup_{j=1}^{2^n} C_{nj}, \quad C_{nj} = \left[\frac{a_j}{3^n}, \frac{a_j + 1}{3^n}\right],$$

for suitable numbers a_j (that we do not need to be precise). Note that $\mathfrak{C} \neq \emptyset$ because the endpoints of the closed intervals C_{ij} belong to $\mathfrak{C}$.

Each one of the sets C_n is closed because it is a finite union of closed intervals, hence $\mathfrak{C}$ is closed. In each nth-stage, we remove 2^{n-1} intervals of length $1/3^n$ and the sum of the lengths of these intervals is $2^{n-1}/3^n$. Thus, the total length removed from the interval $[0, 1]$ is given by the series

$$\sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = 1.$$

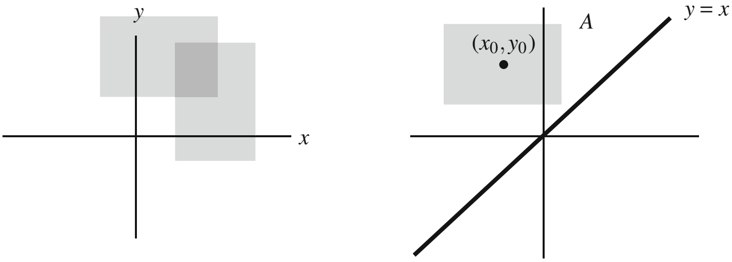


Fig. 2.5 Elements of the basis of the Euclidean topology on $\mathbb{R}^2$ (left). The set $A = \{(x, y) \in \mathbb{R}^2 : x < y\}$ is open in $\mathbb{R}^2$ (right)

This would indicate that $\mathfrak{C}$ has zero length. Therefore the Cantor set cannot contain open intervals. $\Diamond$

We generalize the Euclidean topology to the Cartesian product $\mathbb{R}^n = \mathbb{R} \times \overset{n}{\ldots} \times \mathbb{R}$ of n copies of $\mathbb{R}$. In order to visualize definitions and arguments, many of the pictures will be done in the Euclidean plane $\mathbb{R}^2$. The idea is very simple. If in $\mathbb{R}$ we have used bounded open intervals, then in $\mathbb{R}^n$ we employ the Cartesian product of bounded open intervals (Fig. 2.5).

Definition 2.3.2 The *Euclidean topology* on $\mathbb{R}^n$ (or *usual topology*) is the topology, denoted τ_u, generated by the basis

$$\beta = \{(a_1, b_1) \times \ldots \times (a_n, b_n) : a_i, b_i \in \mathbb{R}, 1 \leq i \leq n\}.$$

We say that $(\mathbb{R}^n, \tau_u)$ is the n-dimensional *Euclidean space*.

Throughout this book, we will understand that the topology on $\mathbb{R}^n$ is the Euclidean topology unless otherwise specified.

Example 2.20 We see that the Cartesian product of open sets in $\mathbb{R}$ is open in $\mathbb{R}^n$. Indeed, let I_i^j stands for an open interval of $\mathbb{R}$. Then

$$(\bigcup_{i_1 \in I_1} I_{i_1}^1) \times \ldots \times (\bigcup_{i_n \in I_n} I_{i_n}^n) = \bigcup_{i_1 \in I_1, \ldots, i_n \in I_n} (I_{i_1}^1 \times \ldots \times I_{i_n}^n).$$

$\Diamond$

Example 2.21 There are open sets of $\mathbb{R}^n$ that are not Cartesian products of open sets of $(\mathbb{R}, \tau_u)$. This is because the following inclusion holds

$$\bigcup_{i \in I} (I_i^1 \times \ldots I_i^n) \subset (\bigcup_{i \in I} I_i^1) \times \ldots \times (\bigcup_{i \in I} I_i^n)$$

and the equality is false in general. For example,

$$((0,1) \times (0,1)) \cup ((2,3) \times (2,3))$$

is open but it is not the Cartesian product of open sets of $(\mathbb{R}, \tau_u)$. $\Diamond$

Example 2.22 The set $A = \{(x, y) \in \mathbb{R}^2 : x < y\}$ is open in $\mathbb{R}^2$. To show this, we use Proposition 2.2.2. Let $(x_0, y_0) \in A$, which means that $x_0 < y_0$, and set $r = (y_0 - x_0)/2 > 0$. We see that $(x_0 - r, x_0 + r) \times (y_0 - r, y_0 + r) \subset A$ (Fig. 2.5, right). If (x, y) is a point of this rectangle, then $-r < x - x_0 < r$ and $-r < y - y_0 < r$. Then $x < x_0 + r$ and $y_0 - r < y$. By the value of r,

$$x < x_0 + \frac{y_0 - x_0}{2} = \frac{y_0 + x_0}{2} = y_0 - \frac{y_0 - x_0}{2} < y.$$

It can be demonstrated similarly that $\{(x, y) \in \mathbb{R}^2 : x > y\}$ is open. In conclusion, their complements $\{(x, y) \in \mathbb{R}^2 : x \geq y\}$ and $\{(x, y) \in \mathbb{R}^2 : x \leq y\}$ are closed. $\Diamond$

We present a special basis for the Euclidean topology. In $\mathbb{R}^n$ the notion of proximity between two points x and y is gathered by the Euclidean distance, which is merely the modulus $|x - y|$ of the difference vector $x - y$ (Fig. 2.6, left).

Definition 2.3.3 The *Euclidean distance* between $x, y \in \mathbb{R}^n$ is defined by

$$d(x, y) = |x - y| = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}, \tag{2.2}$$

where $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$. If $r > 0$ and $x \in \mathbb{R}^n$, the *ball of radius r and center x* is defined by (Fig. 2.6, right)

$$B_r(x) = \{y \in \mathbb{R}^n : |y - x| < r\}.$$

As in (2.2), we utilize the symbol $|.|$ to stand for the (Euclidean) modulus of a vector in Euclidean space $\mathbb{R}^n$ as well as the absolute value of a real number. When

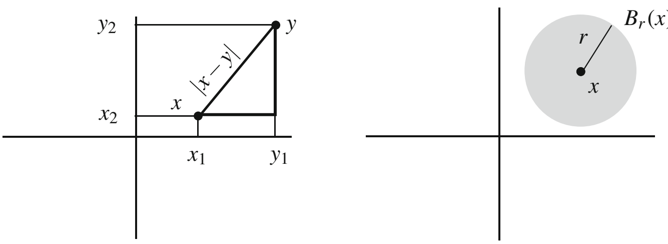


Fig. 2.6 The distance between $x = (x_1, x_2)$ and $y = (y_1, y_2)$ (left). A ball in $\mathbb{R}^2$ (right)

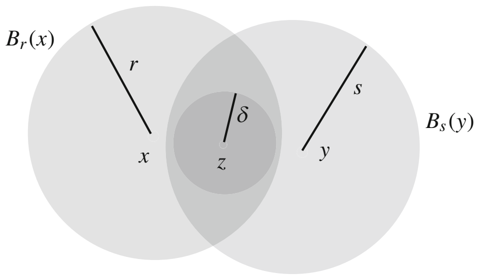


Fig. 2.7 Proof of (ii) of Theorem 2.2.3 for the collection of the balls of $\mathbb{R}^n$

$n = 1$, $B_r(x) = (x - r, x + r)$. In $\mathbb{R}^2$, $B_r(x)$ is the planar round disc centered at x and of radius r without its border.

Theorem 2.3.4 *A basis for the Euclidean topology on* $\mathbb{R}^n$ *is* $\beta = \{B_r(x) : x \in \mathbb{R}^n, r > 0\}$.

Proof First, we show that β is a basis for some topology on $\mathbb{R}^n$. Let us notice that in general, the intersection of two basic elements is not another basic element, or in plain language and for $\mathbb{R}^2$, the intersection of two round discs is not a round disc. Thus, we verify (ii) of Theorem 2.2.3 (Fig. 2.7). Let $B_r(x), B_s(y) \in \beta$ and $z \in B_r(x) \cap B_s(y)$. We claim that $B_\delta(z) \subset B_r(x) \cap B_s(y)$ where

$$\delta = \min\{r - |z - x|, s - |z - y|\}.$$

Note that $\delta > 0$. We require the use of the *triangle inequality*

$$|x - y| \leq |x - z| + |z - y| \tag{2.3}$$

of the Euclidean distance, for all $x, y, z \in \mathbb{R}^n$. If $w \in B_\delta(z)$, we have

$$|w - x| \leq |w - z| + |z - x| < \delta + |z - x| \leq (r - |z - x|) + |z - x| = r,$$

which shows that $w \in B_r(x)$. Likewise, $w \in B_s(y)$.

It remains to prove that the topology $\tau(\beta)$ generated by all balls and the Euclidean topology coincide. Let β_u be the standard basis for the Euclidean topology, and we will use the Hausdorff criterion to compare both bases (Fig. 2.8).

(i) Let $(a_1, b_1) \times \ldots \times (a_n, b_n) \in \beta_u$ and $x = (x_1, \ldots, x_n) \in (a_1, b_1) \times \ldots \times (a_n, b_n)$. Let $r = \min\{x_i - a_i, b_i - x_i : 1 \leq i \leq n\} > 0$. We see that $x \in B_r(x) \subset (a_1, b_1) \times \ldots \times (a_n, b_n)$. If $y = (y_1, \ldots, y_n) \in B_r(x)$, then

$$r > |y - x| = \sqrt{\sum_{i=1}^n (y_i - x_i)^2} \geq |y_1 - x_1|.$$

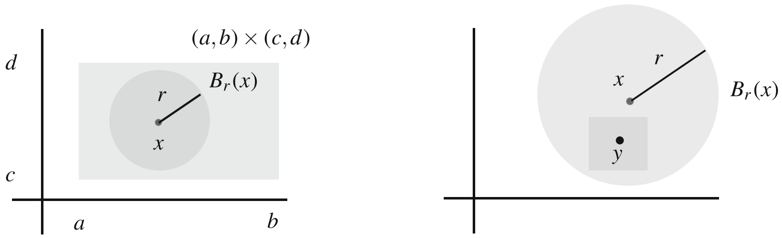


Fig. 2.8 The basis β_u consisting of all rectangles of $\mathbb{R}^2$ and the basis β of all balls are equivalent

From the definition of r, we then infer

$$|y_1 - x_1| < x_1 - a_1 \text{ and } |y_1 - x_1| < b_1 - x_1,$$

proving that $y_1 \in (a_1, b_1)$. Similarly, $y_i \in (a_i, b_i)$ so $y \in (a_1, b_1) \times \ldots \times (a_n, b_n)$.

(ii) Let $B_r(x) \in \beta$ and $y \in B_r(x)$. Let $\delta = (r - |x - y|)/\sqrt{n} > 0$ and we see that $y \in (y_1 - \delta, y_1 + \delta) \times \ldots \times (y_n - \delta, y_n + \delta) \subset B_r(x)$. Indeed, if $z = (z_1, \ldots, z_n) \in (y_1 - \delta, y_1 + \delta) \times \ldots \times (y_n - \delta, y_n + \delta)$, then

$$|z - y| = \sqrt{\sum_{i=1}^{n} (z_i - y_i)^2} < \sqrt{\sum_{i=1}^{n} \delta^2} = \sqrt{n}\delta = r - |x - y|.$$

Thus $|x - y| + |y - z| < r$. This inequality and the triangle inequality imply $|z - x| \leq |z - y| + |y - x| < r$, so $z \in B_r(x)$.

□

We close this section by going back to the triangle inequality. The triangle inequality tells us what everybody knows by daily experience: the distance between two points is less than the sum of distances through a third point. This inequality holds in the general context of Euclidean metric (real) vector spaces. An *Euclidean metric vector space* (V, g) is a vector space equipped with a *metric* $g\colon V \times V \to \mathbb{R}$, a positive definite symmetric bilinear map:

1. $g(v, v) \geq 0$ (nonnegative) and $g(v, v) = 0$ if and only if $v = 0$ (positive definite);
2. $g(u, v) = g(v, u)$ (symmetry); and
3. $g(\lambda u + \mu v, w) = \lambda g(u, w) + \mu g(v, w)$, for all $u, v, w \in V$ and $\lambda, \mu \in \mathbb{R}$.

If $v \in V$ is a vector, the *modulus* of v is defined by $|v| = \sqrt{g(v, v)}$. For example, the *Euclidean metric* in $\mathbb{R}^n$ is defined by $g(x, y) = \sum_{i=1}^{n} x_i y_i$, and consequently

$$|x - y| = \sqrt{g(x - y, x - y)} = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

For this metric, $|x| = \sqrt{\sum_{i=1}^{n} x_i^2}$ and the Euclidean distance $d(x, y)$ is just $|x - y|$.

The proof of the triangle inequality appeals to the *Cauchy-Schwarz inequality*.

Proposition 2.3.5 *In a Euclidean vector space (V, g), we have for all $u, v \in V$,*

$$|g(u, v)| \leq |u||v| \quad \textit{(Cauchy-Schwarz inequality)} \tag{2.4}$$

and equality holds if and only if u and v are proportional. As a consequence, we have for all $u, v, w \in V$,

$$|u - v| \leq |u - w| + |w - v| \quad \textit{(triangle inequality.)} \tag{2.5}$$

Proof If $u, v \in V$ then $|u + \lambda v|^2 \geq 0$ for all $\lambda \in \mathbb{R}$. Then

$$0 \leq |u + \lambda v|^2 = g(u + \lambda v, u + \lambda v) = |u|^2 + 2\lambda g(u, v) + \lambda^2 |v|^2$$

for all λ. If we regard this inequality on λ, then the discriminant of the corresponding second degree polynomial must be nonpositive, so $g(u, v)^2 - |u|^2|v|^2 \leq 0$ and this gives the desired inequality. Further, the case of equality in the discriminant is equivalent to $|u + \lambda v| = 0$ for a certain $\lambda \in \mathbb{R}$, concluding that $u + \lambda v = 0$, hence the vectors u and v are proportional. □

2.4 Subbasis for a Topology

In the above section we have shown how a topology is determined from a basis. However, the collection of open sets may in turn be specified by a smaller collection of open sets.

Definition 2.4.1 A *subbasis* for a space is a family $\mathscr{S}$ of open sets such that the collection of all finite intersections of elements of $\mathscr{S}$ is a basis for the topology.

If we denote this basis by $\beta(\mathscr{S})$, then

$$\beta(\mathscr{S}) = \{\bigcap_{j \in J} B_j : B_j \in \mathscr{S}, J \text{ is a finite set of indices}\}.$$

Thus the topology is

$$\tau = \left\{ \bigcup_{i \in I} \left(\bigcap_{j \in J} B_j \right) : B_j \in \mathscr{S}, J \text{ is a finite set, } I \text{ is a set of indices} \right\}.$$

It is worth noting the following observation from set theory. For any subbasis $\mathscr{S}$, the entire set X belongs to $\beta(\mathscr{S})$. This is because among all finite intersections of elements of β, we have the case $J = \emptyset$. Then $X = \bigcap_{i \in \emptyset} \{B_i : B_i \in \mathscr{S}\}$: see (3) after Definition 2.1.1.

Example 2.23 On a space, any basis for the topology is a subbasis. Certainly, if β is a basis and letting $\mathscr{S} = \beta$, all elements of $\beta(\mathscr{S})$ are open because every element is a finite (one) intersection of open sets. Next, compute all finite intersections of elements of β. If we make the intersections with only one element of $\mathscr{S}$, we obtain all elements of β showing that $\beta \subset \beta(\mathscr{S})$. Thus $\beta(\mathscr{S})$ is a family of open sets containing a basis, so $\beta(\mathscr{S})$ is a basis for τ. $\Diamond$

However, a subbasis may not be a basis, as is shown in the next examples.

Example 2.24

1. In the Sierpinski topology (X, τ), the family $\mathscr{S} = \{\{a\}\}$ is a subbasis since $\beta(\mathscr{S}) = \{\{a\}, X\}$ is a basis, but $\mathscr{S}$ itself is not a basis because X is not the union of elements of $\mathscr{S}$.
2. In the cofinite topology, $\mathscr{S} = \{X \setminus \{x\} : x \in X\}$ is a subbasis. Indeed, each member of $\mathscr{S}$ is an open set and every finite intersection of elements of $\mathscr{S}$ can be expressed as

$$\bigcap_{i=1}^{n}(X \setminus \{x_i\}) = X \setminus \{x_1, \ldots, x_n\}.$$

   Thus $\beta(\mathscr{S}) = \{X \setminus F : F \text{ is finite}\} = \tau_{CF} \setminus \{\emptyset, X\}$ and it is clear that it is a basis for τ_{CF}. Recall that $\mathscr{S}$ is not a basis (Example 2.9).
3. Let $X = \{1, 2, 3\}$ and $\tau = \{\emptyset, X, \{1, 2\}, \{2, 3\}, \{2\}\}$. Then $\mathscr{S} = \{\{1, 2\}, \{2, 3\}\}$ is a subbasis because $\beta(\mathscr{S}) = \{X, \{1, 2\}, \{2, 3\}, \{2\}\}$. However, $\mathscr{S}$ is not a basis because $\{2\}$ is open, but there is no $B \in \beta(\mathscr{S})$ such that $2 \in B \subset \{2\}$. $\Diamond$

As we did with bases, we can pose a similar question regarding sufficient conditions for subbases.

Question: *Given a set X, under what conditions may we consider a collection of subsets $\mathscr{S}$ of X to be a subbasis for some topology on X?*

The answer is that there are no requirements, as is proven in the next result.

Proposition 2.4.2 *If $\mathscr{S}$ is a collection of subsets in X, then there is a unique topology τ on X such that $\mathscr{S}$ is a subbasis for τ.*

Proof We prove that the finite intersections of elements of $\mathscr{S}$ form a basis for a topology. First, $X = \bigcap_{i\in\emptyset}\{B : B \in \mathscr{S}\}$. Second, condition (ii*) of Theorem 2.2.3 is true because the intersection of two finite intersections of elements of $\mathscr{S}$ is again a finite intersection of elements of $\mathscr{S}$. $\square$

Thus any collection of subsets, without further conditions, is always a subbasis for a topology. We demonstrate this procedure.

Example 2.25

1. Let $X = \{a, b, c, d\}$ and $\mathscr{S} = \{\{d\}, \{a, b\}, \{b, c, d\}\}$. Note that $\mathscr{S}$ is neither a topology nor a basis for some topology. For instance, $b \in \{a, b\} \cap \{b, c, d\} = \{b\}$

but there is no any element of $\mathscr{S}$ between b and this intersection. The finite intersection of elements of $\mathscr{S}$ is $\beta = \mathscr{S} \cup \{\{b\}, X\}$. The arbitrary unions of elements of β are

$$\tau = \{\emptyset, X, \{b\}, \{d\}, \{a, b\}, \{b, d\}, \{a, b, d\}, \{b, c, d\}\}.$$

Here we have proper inclusions $\mathscr{S} \subset \beta(\mathscr{S}) \subset \tau$. ♢

2. Let $\mathscr{S}$ be the collection of all bounded open intervals of $\mathbb{R}$. Let I_1 and I_2 be two elements of $\mathscr{S}$. If $I_1 \cap I_2 \neq \emptyset$, then $I_1 \cap I_2$ is another bounded open interval, so $I_1 \cap I_2 \in \mathscr{S}$. Thus $\beta(\mathscr{S}) = \mathscr{S} \cup \{\mathbb{R}\}$. This proves that the topology is the Euclidean topology.
3. In $\mathbb{R}$, let $\mathscr{S} = \{\{x\} : x \neq 0, x \in \mathbb{R}\}$. If $x \neq y$, then $\{x\} \cap \{y\} = \emptyset$. Thus $\beta(\mathscr{S}) = \{\mathbb{R}\} \cup \mathscr{S}$ and the arbitrary unions of elements of $\beta(\mathscr{S})$ are

$$\tau(\beta(\mathscr{S})) = \{\emptyset, \mathbb{R}\} \cup \{A \subset \mathbb{R} : 0 \notin A\}.$$

This topology is the excluded point topology for $p = 0$. ♢

2.5 Topological Subspaces

The purpose of this section is to define, in a natural manner, a topology on a subset of a topological space.

Question: *Given a topological space (X, τ), in what sense may we define a topology on a subset A of X that has some relation with the topology τ?*

A candidate definition may be to declare the open sets of A as those open sets of X included in A,

$$\gamma_A = \{O \in \tau : O \subset A\} \cup \{A\}. \tag{2.6}$$

We need to add A in order to satisfy (1) of Definition 2.1.1 because we do not know if A is open in X. It is easy to verify that γ_A obeys the axioms to be a topology on A. However, the closed sets in the topology γ_A are not the subsets of A that are closed in X.

What, then, should be the nature of the open sets in a subset of a topological space? Better than considering open sets of X included in A, which is a strong condition, one can capture the information of the topology of X with respect to A by taking all intersections of open sets of X with A. See Fig. 2.9.

Definition 2.5.1 Let (X, τ) be a topological space and $A \subset X$. The *relative topology* (or *induced topology*) on A, denoted $\tau_{|A}$, is defined by

$$\tau_{|A} = \{O \cap A : O \in \tau\}.$$

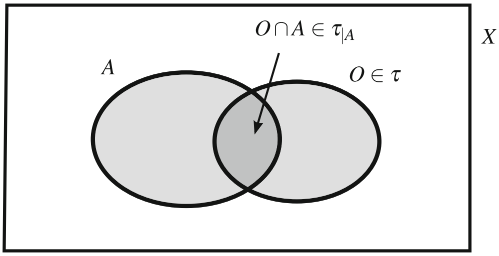


Fig. 2.9 The relative topology on $A \subset X$

We say that $(A, \tau_{|A})$ is a *topological subspace* of (X, τ).

Henceforth and if it is not specified, we will assume that a subset of a topological space has the relative topology.

Theorem 2.5.2 *Let (X, τ) be a space and $A \subset X$.*

1. *If $B \subset A$, then $\tau_{|B} = (\tau_{|A})_{|B}$.*
2. *The collection of closed sets of $\tau_{|A}$ is $\mathscr{F}_{|A} = \{F \cap A : F \in \mathscr{F}\}$.*
3. *If β is a basis for τ, then $\beta_A = \{B \cap A : B \in \beta\}$ is a basis for $\tau_{|A}$.*

Example 2.26

1. Let (X, τ_T) be the trivial space and $A \subset X$. If we intersect the two open sets X and $\emptyset$ with A, we obtain A and $\emptyset$. Thus $\tau_{T|A}$ is the trivial topology on A.
2. Let (X, τ_D) be the discrete space and $A \subset X$. The intersection of all subsets of X with A yields all subsets of A, hence $\tau_{D|A}$ is the discrete topology on A.
3. Let (X, τ_{CF}) be the cofinite space and $A \subset X$. The family of closed sets of A is formed by the intersection of all finite sets of X with A, obtaining all finite sets of A. Thus $\tau_{CF|A}$ is the cofinite topology of A.
4. Let $p \in X$ and let τ_{in} be the included point topology for the point p. We describe the relative topology of a subset A of X.

   (a) If $p \notin A$ then $\tau_{in|A}$ is the discrete topology on A because any singleton $\{a\}$ writes as $\{a\} = \{a, p\} \cap \{a\} \in \tau_{in|A}$.

   (b) If $p \in A$ then we see that $\tau_{in|A}$ coincides with the included point topology on A for the particular point p. We denote the latter topology by τ_{in}^A. Then the equality $\tau_{in|A} = \tau_{in}^A$ is by double inclusion. If $B \in \tau_{in|A}$, then there is $O \in \tau_{in}$ such that $B = O \cap A$. Since $p \in O$ and $p \in A$, we have $p \in B$, so $B \in \tau_{in}^A$. Reciprocally, if $B \in \tau_{in}^A$ then $p \in B$, so $B \in \tau_{in}$. Thus $B = B \cap A \in \tau_{in}^A$. $\Diamond$

We provide sufficient conditions to determine if a set is open or closed in the relative topology. Also, we give conditions for an open set in the relative topology to be open in the ambient space.

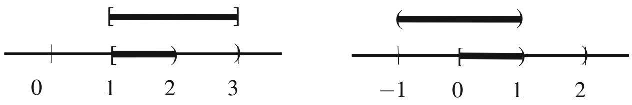


Fig. 2.10 In the interval $[0, 2)$, the subset $[1, 2)$ is closed and the subset $[0, 1)$ is open

Proposition 2.5.3 *Let (X, τ) be a space and $A \subset X$.*

1. *If O is open in X and $O \subset A$, then $O \in \tau_{|A}$.*
2. *If $O \in \tau_{|A}$ and A is open in X, then O is open in X. Thus $\tau_{|A} = \{O \in \tau : O \subset A\}$.*

In the same fashion, we have analogous statements for closed sets.

The open sets of item (1) coincide with the elements of the family γ_A defined in (2.6). Thus $\gamma_A \subset \tau_{|A}$. In the case that A is open in X, then $\gamma_A = \tau_{|A}$ by (2).

If A is a subset of the Euclidean space $(\mathbb{R}^n, \tau_u)$, the relative topology is called the *Euclidean topology* on A.

Example 2.27

1. Consider $A = [0, 2)$ (Fig. 2.10).
   (a) The subset $[1, 2)$ is closed in A because $[1, 2) = [1, 3] \cap A$ and $[1, 3]$ is closed. Notice that $[1, 2)$ is not closed in $\mathbb{R}$.
   (b) The interval $[0, 1)$ is open in A because $[0, 1) = (-1, 1) \cap A$ and $(-1, 1)$ is open. Notice that $[0, 1)$ is not open in $\mathbb{R}$.
2. The relative topology on $A = \{1/n : n \in \mathbb{N}\}$ is discrete because every singleton of A is an open set (in the relative topology). Indeed, if $n = 1$, $\{1\} = (\frac{1}{2}, 2) \cap A$ and if $n > 1$,
$$\left\{\frac{1}{n}\right\} = \left(\frac{1}{n+1}, \frac{1}{n-1}\right) \cap A.$$
3. The relative topology on $A = \{0\} \cup \{1/n : n \in \mathbb{N}\}$ is not discrete. If $\{0\}$ is open in $\tau_{u|A}$, there is $(a, b) \in \beta_u$ such that $0 \in (a, b) \cap A \subset \{0\}$. However, in the interval (a, b) there are infinitely many elements of A.
4. The Euclidean topology on $\mathbb{Z}$ is discrete (the same occurs for $\mathbb{N}$). Certainly, if $n \in \mathbb{Z}$ then $\{n\} = (n - 1, n + 1) \cap \mathbb{Z}$ and $(n - 1, n + 1) \in \tau_u$.
5. The Euclidean topology on $\mathbb{Q}$ is not discrete. If $\{x\}$ is open in $\tau_{u|\mathbb{Q}}$, there is $(a, b) \in \beta_u$ such that $x \in (a, b) \cap \mathbb{Q} \subset \{x\}$. This inclusion is not possible because (a, b) has infinitely many rationals.
6. In $\mathbb{Q}$, there are nontrivial subsets that are open and closed. For example, $\mathbb{Q} \cap (-\sqrt{2}, \sqrt{2})$ is both open and closed in $\mathbb{Q}$ because
$$\mathbb{Q} \cap (-\sqrt{2}, \sqrt{2}) = \mathbb{Q} \cap [-\sqrt{2}, \sqrt{2}].$$

$\Diamond$

2.6 Worked Exercises

Exercise 2.1 (Many Topologies on $\mathbb{R}^2$) Prove that

$$\beta = \beta_u \times \{\mathbb{R}\} = \{(a, b) \times \mathbb{R} : a, b \in \mathbb{R}\}$$

is a basis for a topology on $\mathbb{R}^2$. Compare with the Euclidean topology.

Solution We see that β satisfies the requirements of Theorem 2.2.3. It is clear that

$$\bigcup_{a,b\in\mathbb{R}} ((a, b) \times \mathbb{R}) = \left(\bigcup_{a,b\in\mathbb{R}} (a, b) \right) \times \mathbb{R} = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2.$$

Property (ii*) holds because $((a, b)\times\mathbb{R})\cap((c, d)\times\mathbb{R}) = ((a, b)\cap(c, d))\times\mathbb{R} \in \beta_u \times \{\mathbb{R}\}$. To compare $\tau(\beta)$ with the Euclidean topology, we prove the inclusion $\beta \subset \tau_u$ using the characterization of an open set in terms of a basis (Proposition 2.2.2). Use caution in this exercise, as the symbol (a, b) may be used to denote an interval in $\mathbb{R}$ or a point of $\mathbb{R}^2$. Let $(x, y) \in (a, b) \times \mathbb{R}$. Take intervals $B_1 = (a, b)$, $B_2 = (y - 1, y + 1)$. Then $B_1 \times B_2 \in \beta_u$ and $(x, y) \in B_1 \times B_2 \subset (a, b) \times \mathbb{R}$. From the inclusion $\beta \subset \tau_u$, we deduce $\tau(\beta) \subset \tau_u$, so the Euclidean topology is finer than $\tau(\beta)$.

However, the inclusion $\tau(\beta) \subset \tau_u$ is proper. For instance, $A = (0, 2) \times (0, 2)$ is open in the Euclidean topology but it is not open in $\tau(\beta)$. If A were open in $\tau(\beta)$, given $(1, 1) \in A$, there would be $(a, b) \times \mathbb{R} \in \beta$ such that $(1, 1) \subset (a, b) \times \mathbb{R} \subset A$. This, however, is absurd because $(a, b) \times \mathbb{R}$ is not bounded, in contrast with A.

This topology on $\mathbb{R}^2$ can be extended by considering the Cartesian product of other types of intervals. With the notation employed in this chapter, consider $\Gamma = \{\beta_u, \beta_c, \beta_S, \beta_S^l, \beta_{sr}, \beta_r, \beta_{sl}, \beta_l, \{\mathbb{R}\}\}$. Then we have that $\beta_1 \times \beta_2$ is a basis for a topology on $\mathbb{R}^2$ for all $\beta_1, \beta_2 \in \Gamma$. See Fig. 2.11. This is because (ii*) holds in all cases for two reasons. First, the intersection of two elements of β_i belongs to β_i and second, from the identity

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D).$$

If now $A, C \in \beta_1$ and $B, D \in \beta_2$, then $A \cap C \in \beta_1$ and $B \cap D \in \beta_2$. □

Exercise 2.2 On $\mathbb{N}$, define two topologies.

1. For each $n \in \mathbb{N}$, let $A_n = \{1, 2, \dots, n\}$. Prove that $\tau = \{\emptyset, \mathbb{N}\} \cup \{A_n : n \in \mathbb{N}\}$ is a topology on $\mathbb{N}$. Prove that $\gamma = \{A_n : n \in \mathbb{N}\}$ is the smallest basis for τ.
2. For each $n \in \mathbb{N}$, let $B_n = \{n, n + 1, \dots\}$. Prove that $\tau' = \{\emptyset\} \cup \{B_n : n \in \mathbb{N}\}$ is a topology and that its only basis is τ' itself.
3. Describe the family of closed sets in these topologies.

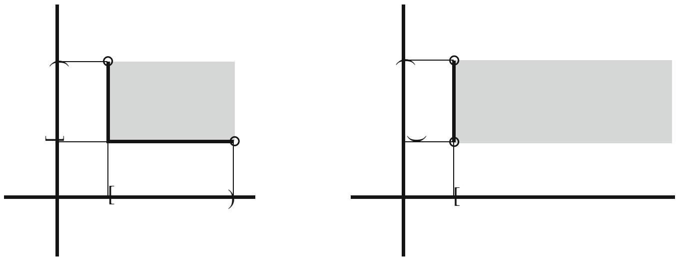


Fig. 2.11 Basic open sets of $\beta_S \times \beta_S$ (left) and of $\beta_r \times \beta_u$ (right)

Solution

1. (a) It is obvious that $A_n \cap A_m = A_{\min\{n,m\}}$. On the other hand, if $\{A_{n_i} : i \in I\}$ is a family of open sets, then

$$\bigcup_{i\in I} A_{n_i} = \begin{cases} \mathbb{N}, & \text{if } \{n_i : i \in I\} \text{ is not bounded} \\ A_m, & \text{if } \{n_i : i \in I\} \text{ is bounded and } m \text{ is its maximum.} \end{cases}$$

   (b) It is clear that the elements of γ are open and that arbitrary unions of elements of γ give γ again, together with $\emptyset$ and X. Let β be a basis for τ and we show the inclusion $\gamma \subset \beta$. Let $A_n \in \gamma$ and $n \in A_n$. In view of Proposition 2.2.2, there is $B \in \beta$ such that $n \in B \subset A_n$. Using that B is open, $B = A_m$ for some $m \in \mathbb{N}$; in particular, $n \in A_m \subset A_n$. As $n \in A_m$, then $n \leq m$ and from the inclusion $A_m \subset A_n$, we obtain $m \leq n$. Thus $m = n$, so $B = A_n$. This proves that $A_n \in \beta$.
2. (a) If I is a set of indices, then $\bigcup_{i\in I} B_{n_i} = B_m$, where m is the minimum of $\{n_i : i \in I\}$. (Recall that any subset of natural numbers has a minimum.) For the intersection, it is clear that $B_n \cap B_m = B_{\max\{n,m\}}$.
   (b) Let β be a basis for τ'. We show the inclusion $\tau' \subset \beta$. Let $B_n \in \tau'$. As $n \in B_n$, there is $B \in \beta$ such that $n \in B \subset B_n$. Hence $B = B_m$ for some $m \in \mathbb{N}$. If $n \in B_m$, then $m \leq n$. Since $B_m \subset B_n$, we must have $n \leq m$. This proves that $m = n$, so $B = B_n \in \beta$.
3. It is clear that $\mathbb{N} \setminus A_n = \{n+1, n+2, \ldots\} = B_{n+1}$. Thus $\mathscr{F} = \tau'$ and $\mathscr{F}' = \tau$.

Notice that these topologies are simply the relative topologies on $\mathbb{N}$ of topologies already defined on $\mathbb{R}$. So the topology τ is the relative topology on $\mathbb{N}$ from the left order topology τ_l on $\mathbb{R}$, $\tau = \tau_{l|\mathbb{N}}$. Recall that the topology τ_l consists of the intervals of type $(-\infty, a]$ and of type $(-\infty, a)$ and when intersected with $\mathbb{N}$ we obtain all elements A_n of τ. Likewise, the topology τ' is the relative topology on $\mathbb{N}$ from the right order topology on $\mathbb{R}$, $\tau' = \tau_{r|\mathbb{N}}$. □

In the next exercise we shall construct a topology on an ordered set.

Exercise 2.3 Let X be a set equipped with an order relation $\leq$. Assume that this order is total, which means that for any $x, y \in X$, either $x \leq y$ or $y \leq x$. For $a, b \in X$, define the set $]a, b[$ as

$$]a, b[= \{x \in X : a < x\} \cap \{x \in X : x < b\}.$$

In case that there is a minimum m (resp. a maximum M), let $[m, a[= \{x \in X : m \leq x < a\}$ (resp. $]a, M] = \{x \in X : a < x \leq M\}$). Consider

$$\beta_o = \{]a, b[: a, b \in X\} \cup \{[m, a[: a \in X\} \cup \{]a, M] : a \in X\}. \tag{2.7}$$

Prove that β_o is a basis for a topology on X. This topology, written τ_o, is called the *order topology*. With the usual order on $\mathbb{R}$ and on $[0, 1]$, compare τ_o with the Euclidean topology in each space.

Solution In case that X has only one or two elements, the proof is trivial. Indeed, if there is one element, then the topology is trivial. If $X = \{a, b\}$ with $a < b$, then $m = a$; $M = b$ and $]a, b[= \emptyset$, $[m, b[= \{a\}$, $]a, M] = \{b\}$. Thus the topology is discrete. We proceed with the case that X has more than two elements.

Let us apply Theorem 2.2.3. For the first property, let $x \in X$. If $x = m$ is the minimum (provided one exists), then for any $a \in X$, $x \in [x, a[\in \beta_o$. If $x = M$ is the maximum (provided one exists), then $x \in]a, M] \in \beta_o$. Suppose $x \neq m, M$. Pick another $a \in X$ and suppose, without loss of generality, that $a < x$. Because x is not the maximum, there is $b \in X$ such that $x < b$. Thus $x \in]a, b[\in \beta_o$.

We prove that the intersection of two elements of β_o belongs to β_o independently of the existence of m and M. For example, if $]a, b[\cap]c, d[\neq \emptyset$, then it is immediate by the properties of $\leq$ that

$$]a, b[\cap]c, d[=] \max\{a, c\}, \min\{b, d\}[.$$

Note that $\max\{a, c\}$ and $\min\{b, d\}$ exist because the order is total. The other possibilities are:

(i) $]a, b[\cap]c, M] =] \max\{a, c\}, b[$,
(ii) $[m, c[\cap]a, b[=]a, \min\{b, c\}[$,
(iii) $[m, a[\cap]b, M] =]b, a[$,
(iv) $[m, a[\cap[m, b[= [m, \min\{a, b\}[$,
(v) $]a, M]\cap]b, M] =] \max\{a, b\}, M]$.

Thus, the requirement (ii*) of Theorem 2.2.3 is valid. We turn to the topology τ_o on $\mathbb{R}$ and on $[0, 1]$. In $\mathbb{R}$ there is neither a minimum nor a maximum. Thus the only elements of β_o are of type $]a, b[$. It is clear that $]a, b[$ coincides with the open interval (a, b). This proves that $\beta_o = \beta_u$ and the order topology and the Euclidean topology coincide.

If $X = [0, 1]$ then $m = 0$ and $M = 1$. The basis (2.7) is

$$\beta_o = \{(a, b) : a, b \in [0, 1]\} \cup \{[0, a) : a \in (0, 1]\} \cup \{(a, 1] : a \in [0, 1)\}.$$

This basis coincides with the intersection of $[0, 1]$ with the usual basis β_u of the Euclidean topology. Thus β_o determines the Euclidean topology of $[0, 1]$. □

Exercise 2.4 (Continuation) Consider the set $A = (0, 1) \cup [2, 3)$ and the usual order of real numbers. Let τ_o^A be the order topology on A and τ_o the corresponding topology on $\mathbb{R}$. Compare τ_o^A with $\tau_{o|A}$.

Solution We have seen in the previous exercise that on $\mathbb{R}$, we have $\tau_o = \tau_u$. Thus $\tau_{o|A}$ is the Euclidean topology on A, $\tau_{u|A}$. In the set A there is neither a minimum nor a maximum. Therefore the basis for the order topology is $\beta_o^A = \{]a, b[: a, b \in A\}$. We see that for $a, b \in A$, $]a, b[$ is an open set of $\tau_{u|A}$. Indeed, by definition of $]a, b[$,

$$]a, b[= (a, b) \cap A \in \tau_{u|A}. \tag{2.8}$$

Thus $\tau_o^A \subset \tau_{u|A}$ and the Euclidean topology is finer than the order topology. However, this inclusion is proper and there are open sets in $\tau_{u|A}$ that do not belong to the order topology τ_o^A. An example is $[2, 3)$. The set $[2, 3)$ is open in $\tau_{u|A}$ because $[2, 3) = (1, 3) \cap A$. Notice that $[2, 3)$ is not an element of the basis β_o^A described in (2.7), but suppose that it is open in the order topology τ_o^A. If $[2, 3)$ were open in τ_o^A and since $2 \in [2, 3)$, there would be $B \in \beta_o^A$ such that $2 \in B \subset [2, 3)$. This element B must be $]a, b[$ with $a, b \in A$. Since $2 \in]a, b[$, then $a < 2 < b$ and because $a \in A$, then $a < 1$. Let $c \in (a, 1)$. Then $c \in]a, b[$. The inclusion $]a, b[\subset [2, 3)$ would imply $c \in [2, 3)$, so $2 \leq c$, a contradiction. □

This exercise proves that the order topology on a subset does not coincide, in general, with its relative topology on the ambient space. We define new topologies on an ordered set.

Exercise 2.5 Let $(X, \leq)$ be an ordered set. For each $x \in X$, define

$$B^x = [x, \rightarrow [= \{y \in X : x \leq y\}.$$

Let $\beta_r = \{B^x : x \in X\}$. Prove that β_r is a basis for a topology τ_r. This topology is called the *right order topology*. Prove that arbitrary intersection of open sets is an open set.

Solution

1. Since for all $x \in X$, $x \in B^x$, it follows that $X = \bigcup_{x \in X} B^x$. On the other hand, let $z \in B^x \cap B^y$. We prove that $z \in B^z \subset B^x \cap B^y$. Let $t \in B^z$, so $z \leq t$. Since $x \leq z$ and $y \leq z$, by transitivity, $x \leq t$ and $y \leq t$. Thus $t \in B^x$ and $t \in B^y$, and t belongs to the intersection $B^x \cap B^y$.

2. We prove in two steps that the arbitrary intersection of open sets is open.
   (a) We first prove the property for the elements of the family β_r. Consider $\{B^{x_i} : i \in I\}$, a collection of elements of β_r. Let $A = \{y \in X : x_i \leq y \text{ for all } i \in I\}$. Let us check that

$$\bigcap_{i\in I} B^{x_i} = \bigcup_{y\in A} B^y.$$

   In such a case, the intersection is open in (X, τ_r). We show this equality by the usual method of showing that each of the sets is contained in the other.

   i. Let $x \in \bigcap_{i\in I} B^{x_i}$. Then $x \in B^{x_i}$ for all $i \in I$, so $x_i \leq x$. Thus $x \in A$ and $x \in B^x \subset \bigcup_{y\in A} B^y$.
   ii. Let $x \in \bigcup_{y\in A} B^y$. Then there is $y \in A$ such that $x \in B^y$. Then $y \leq x$ and by the transitive property, $x_i \leq x$ for all $i \in I$. This means that $x \in \bigcap_{i\in I} B^{x_i}$.

   (b) We show the property holds in general, so let $\{O_i : i \in I\}$ be a family of open sets. To establish that $\bigcap_{i\in I} O_i$ is an open set, we make use of Proposition 2.2.2. If $x \in \bigcap_{i\in I} O_i$, there is $B^{y_i^x} \in \beta_r$ such that $x \in B^{y_i^x} \subset O_i$ for all $i \in I$. Thus

$$x \in \bigcap_{i\in I} B^{y_i^x} \subset \bigcap_{i\in I} O_i.$$

   Considering all elements of $\bigcap_{i\in I} O_i$, we deduce

$$\bigcap_{i\in I} O_i = \bigcup_{x\in\bigcap_{i\in I} O_i} \left(\bigcap_{i\in I} B^{y_i^x}\right),$$

   which is an open set because it is union of sets which are open by (a).

Some final observations.

1. If the order is total, then $B^x \cap B^y = B^z$, where $z = \max\{x, y\}$.
2. If $X = \mathbb{R}$ and $\leq$ is the usual order, the set $[x, \rightarrow[$ coincides with the interval $[x, \infty)$. The right order topology is just the topology defined in (4) of Example 2.12.
3. If $]x, \rightarrow[= \{y \in X : x < y\}$, then $\beta_{sr} = \{]x, \rightarrow[: x \in X\}$ is also a basis for a topology on X. This topology is not τ_r. For example, if we consider $\mathbb{R}$ with the usual topology, this topology coincides with β_{sr} of Example 2.12. □

We examine the relative topology of the right order topology.

Exercise 2.6 (Continuation) Let τ_r be the right order topology on an ordered set X. If $A \subset X$, let τ_r^A be the right order topology on A with the same order. Prove that $\tau_r^A = \tau_{r|A}$.

Solution First, we fix notation. For the topology τ_r we use $[x, \rightarrow)$ and for τ_r^A, $[a, \rightarrow^A)$. By Theorem 2.5.2, a basis for $\tau_{r|A}$ is $\beta_{r|A} = \{[x, \rightarrow) \cap A : x \in X\}$. On the other hand, a basis for τ_r^A is $\gamma = \{[a, \rightarrow^A) : a \in A\}$. Observe that $[a, \rightarrow^A) = \{x \in A : a \leq x\} = [a, \rightarrow) \cap A$. Because of this identity, we have the inclusion $\gamma \subset \beta_{r|A}$. Then the topology τ_r^A determined by γ is included in the topology generated by $\beta_{r|A}$. This proves the inclusion $\tau_r^A \subset \tau_{r|A}$.

For the other inclusion, apply the Hausdorff criterion. Let $[x, \rightarrow) \cap A \in \beta_{r|A}$, where $x \in X$, and let $a \in [x, \rightarrow) \cap A$. We prove $a \in [a, \rightarrow^A) \subset [x, \rightarrow) \cap A$. Let $y \in [a, \rightarrow^A)$. Then $y \in A$ and $a \leq y$. Since $x \leq a$, then $x \leq y$, so $y \in [x, \rightarrow) \cap A$. □

Exercise 2.7 Let $\{I_i : i \in I\}$ be a collection of (bounded or not) intervals of $\mathbb{R}$ such that $I_i \cap I_j \neq \emptyset$ for all $i, j \in I$. Prove that $\bigcup_{i \in I} I_i$ is an interval. Deduce that every open set of $\mathbb{R}$ is the countable union of pairwise disjoint open intervals.

Solution

1. Let $x, y \in \bigcup_{i \in I} I_i$, and we assume $x \leq y$. Our job is to show that if $z \in \mathbb{R}$ satisfies $x \leq z \leq y$, then z belongs to $\bigcup_{i \in I} I_i$. We know that $x \in I_j$ and $y \in I_k$ for certain $j, k \in I$. Because $I_j \cap I_k \neq \emptyset$, let $t \in I_j \cap I_k$. If $z = t$ then z belongs to I_j, hence to $\bigcup_{i \in I} I_i$. Suppose that $z \neq t$ and assume that $z < t$ (the case $t < z$ is analogous). Then $x \leq z < t$. As $x, t \in I_j$ and I_j is an interval, necessarily $z \in I_j$, hence $z \in \bigcup_{i \in I} I_i$.
2. Let O be an open set of $\mathbb{R}$. We use as basis β for τ_u all (bounded or not) open intervals. Let $x \in O$ be an arbitrary but fixed point. Since β is a basis, there is $I \in \beta$ such that $x \in I \subset O$. Let us pick all such intervals I, that is, let $O_x = \bigcup \{I_k \in \beta : x \in I_k \subset O\}$. We prove that $O = \bigcup_{x \in O} O_x$ and that this union is countable and formed by pairwise disjoint open intervals.

   (a) The set O_x is an interval by (1), which is union of open sets. This proves that $O_x \in \beta$.
   (b) The inclusion $O \subset \bigcup_{x \in O} O_x$ is clear because if $x \in O$, then $x \in O_x$. For the other inclusion, all I_k of O_x are included in O, so $O_x \subset O$ and this holds for the union as well.
   (c) If $O_x \cap O_y \neq \emptyset$, then $O_x \cup O_y$ is an open interval by (1), hence $O_x \cup O_y \in \beta$. By definition of O_x and O_y, we deduce $O_x = O_x \cup O_y = O_y$. From this, we derive that the collection of O_x are pairwise disjoint.
   (d) We conclude by proving that $\{O_x : x \in O\}$ is countable. For each O_x, pick a rational number $q_x \in \mathbb{Q}$, which we fix. Then the map

   $$\phi \colon \{O_x : x \in O\} \to \mathbb{Q}, \quad \phi(O_x) = q_x$$

   is injective. Indeed, if $\phi(O_x) = \phi(O_y)$, then $q_x = q_y \in O_x \cap O_y$, so $O_x \cap O_y \neq \emptyset$. This forces that $O_x = O_y$ by (c). Finally, since ϕ is injective, the set $\{O_x : x \in O\}$ is countable. □

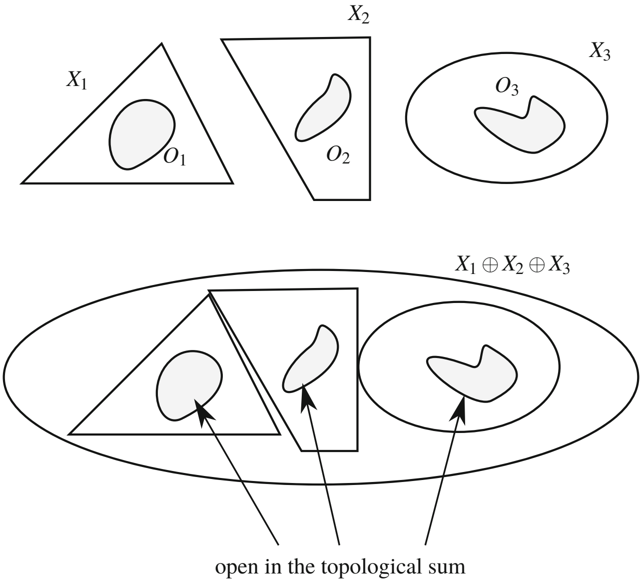


Fig. 2.12 The topological sum of topological spaces

In the following exercise, we construct a new topology using the concept of the disjoint union from set theory. If $\{X_i : i \in I\}$ is a collection of pairwise disjoint sets, the *disjoint union* of all X_i is the set $X = \bigcup_{i \in I} X_i$. The disjoint union is a set where we put all elements of all X_i in such a way that if we pick one of them, we are able to discern what X_i it belongs to.

Exercise 2.8 Let $\{(X_i, \tau_i) : i \in I\}$ be a collection in which the X_i are pairwise disjoint and let $X = \bigcup_{i \in I} X_i$ be its disjoint union.

1. Prove that $\beta = \bigcup_{i \in I} \tau_i$ is a basis for a topology on X. This topology is called the *sum topology* and denoted by $\sum_{i \in I} \tau_i$. The topological space X is denoted by $\bigoplus_{i \in I} X_i$ (Fig. 2.12).
2. Prove that O is open in $\bigoplus_{i \in I} X_i$ if and only if $O \cap X_i \in \tau_i$ for all $i \in I$.
3. Prove that $(\sum_{i \in I} \tau_i)_{|X_j} = \tau_j$ for all $j \in I$.

Solution

1. The first requirement of Theorem 2.2.3 is clear since $X_i \in \tau_i$ and $X = \bigcup_{i \in I} X_i \in \beta$. On the other hand, let $x \in O \cap G$, where $O, G \in \beta$. Since $O \cap G \neq \emptyset$ and because X is the disjoint union of all X_i, then necessarily O and G belong to the same topology, say τ_{i_0} for some $i_0 \in I$. As τ_{i_0} is a topology, then $O \cap G \in \tau_{i_0}$, so $O \cap G \in \beta$.

2. (a) Assume that O is open in $\bigoplus_{i\in I} X_i$. Then there is a set of indices J such that $O = \bigcup_{j\in J} B_j$, where $B_j \in \beta$. Let $i \in I$ be a fixed, but arbitrary, index of I. Intersecting O with X_i, we have

$$O \cap X_i = (\bigcup_{j\in J} B_j) \cap X_i = \bigcup_{j\in J} (B_j \cap X_i).$$

We use that all X_k are pairwise disjoint. In case that $B_j \in \tau_i$, we have $B_j \cap X_i = B_j$; otherwise, $B_j \cap X_i = \emptyset$. If $J_i \subset J$ is the set of indices such that $B_j \in \tau_i$ for $j \in J_i$, then

$$O \cap X_i = \bigcup_{j\in J_i} (B_j \cap X_i) = \bigcup_{j\in J_i} B_j$$

belongs to τ_i because it is union of elements of τ_i.

(b) Assume that $O \cap X_i \in \tau_i$ for all $i \in I$. Then

$$O = O \cap X = \bigcup_{i\in I} (O \cap X_i)$$

is open in the sum topology because it is union of elements of τ_i, so $O \cap X_i \in \tau_i$.

3. Let $O \in (\sum_{i\in I} \tau_i)_{|X_j}$. Then there is $G \in \sum_{i\in I} \tau_i$ such that $O = G \cap X_j$. By (2), $G \cap X_j \in \tau_j$, so $O \in \tau_j$. Reciprocally, let $O \in \tau_j$. Then $O \in \beta$ so O is an element of $\sum_{i\in I} \tau_i$. Since $O \subset X_j$, then $O \in (\sum_{i\in I} \tau_i)_{|X_j}$ by Proposition 2.5.3. □

Assume now that we have a partition of a topological space. It is natural to ask if the topology of the space is captured by the sum topology of the subspaces of the partition. The following exercise analyzes this situation.

Exercise 2.9 Let (X, τ) be a space and let $\{X_i : i \in I\}$ be a partition of X. Prove that $\tau = \sum_{i\in I} \tau_{|X_i}$ if and only if X_i is open in (X, τ) for all $i \in I$.

Solution

1. Assume that $\tau = \sum_{i\in I} \tau_{|X_i}$. Since $X_j \in \tau_{|X_j}$ for all $j \in I$, it follows $X_j \in \bigcup_{i\in I} \tau_{|X_i}$. Thus $X_j \in \sum_{i\in I} \tau_{|X_i} = \tau$.
2. Assume that all X_i are open sets. Let O be an element of the basis $\bigcup_{i\in I} \tau_{|X_i}$ of the sum topology, which means $O \in \tau_{|X_j}$ for some $j \in I$. Since X_j is open in X, then O is open in X. This proves that all elements of $\bigcup_{i\in I} \tau_{|X_i}$ belong to τ, so the sum topology is included in τ.

   For the other inclusion, let $O \in \tau$. As the sets $\{X_i : i \in I\}$ constitute a partition, we have $O = \bigcup_{i\in I} (O \cap X_i)$, where $O \cap X_i \in \tau_{|X_i}$. Since each set $O \cap X_j$ belongs to $\bigcup_{i\in I} \tau_{|X_i}$, the open set O is union of elements of $\bigcup_{i\in I} \tau_{|X_i}$, hence O is open in the sum topology. Observe that we do not use that the X_i are open, so this inclusion holds in general. □

In the following exercise, we prove that the set of all prime numbers is infinite. The proof employs an argument due to Furstenberg (see H. Furstenberg, On the infinitude of primes, *Amer. Math. Monthly*, **62** (1955), 353).

Exercise 2.10 Let $\mathbb{Z}$ be the set of integer numbers. If $a, b \in \mathbb{Z}$ with $a \neq 0$, denote

$$A(a, b) = a\mathbb{Z} + b = \{an + b : n \in \mathbb{Z}\}.$$

1. Prove that $\beta = \{A(a, b) : a, b \in \mathbb{Z}, a \neq 0\}$ is a basis for a topology τ on $\mathbb{Z}$.
2. Prove that the set of prime numbers is infinite.

Solution A first observation is that if $c \in A(a, b)$, then $A(a, b) = A(a, c)$. This set-theoretical identity is shown by double inclusion. Let us write $c = am + b$ with $m \in \mathbb{Z}$. If $x \in A(a, b)$ then $x = an + b = an + c - am = a(n - m) + c \in A(a, c)$. Conversely, if $x \in A(a, c)$ then $x = an+c = an+am+b = a(n+m)+b \in A(a, b)$. Thus if we take an element of $A(a, b)$, we can assume that this element is just b.

1. Using Theorem 2.2.3 we check (ii*) for β, observing that if $b \in A(a_1, b) \cap A(a_2, b)$, then

$$A(a_1, b) \cap A(a_2, b) = A(a, b),$$

where $a = \text{l.c.m}\{a_1, a_2\}$. We see this identity by double inclusion. Let $x \in A(a_1, b) \cap A(a_2, b)$. Then $x = a_1 n + b = a_2 m + b$ for some $n, m \in \mathbb{Z}$. Then $a_1 n = a_2 m$. Since this number is a multiple of a_1 and of a_2, it is also a multiple of a, so there is $k \in \mathbb{Z}$ such that $a_1 n = a_2 m = ak$. Thus $x = ak + b \in A(a, b)$. Reciprocally, if $x \in A(a, b)$ then $x = an + b$ for some $n \in \mathbb{Z}$. Since $a = a_1 p$ for some $p \in \mathbb{Z}$, then $x = a_1(pn) + b \in A(a_1, b)$. Analogously, $x \in A(a_2, b)$.
We collect a few properties of this topology.

(i) For $a \neq 0$, we have $A(a, b) = A(-a, b)$.
(ii) The set $A(a, 0)$ is the set of multiples of a.
(iii) The only finite open set is $\emptyset$ because every $A(a, b)$ is infinite.
(iv) The set $A(a, b)$ is closed. If $a = \pm 1$ then $A(a, b) = \mathbb{Z}$, which is closed because it is the entire space. If $a \neq \pm 1$, we show that

$$\mathbb{Z} \setminus A(a, b) = \bigcup_{n=1}^{|a|-1} A(a, b + n),$$

hence an open set. Suppose that $p \notin A(a, b)$. Then $p \neq am + b$ for all $m \in \mathbb{Z}$. Then there is $n \in \mathbb{Z}$ such that $an + b < p < a(n + 1) + b$. Thus $an + b < p < an + b + a$, so $p = an + b + q$ for some natural number q in the interval $[1, |a| - 1]$).
Let $p \in \bigcup_{n=1}^{|a|-1} A(a, b+n)$. Then $p = am+b+n$ with $n \leq |a|-1$. If $p \in A(a, b)$, then $p = ak + b$, hence $n = a(k - m)$. As $k \neq m$ because $n \geq 1$, we have $n \geq |a|$, which is not possible. This proves that $p \in \mathbb{Z} \setminus A(a, b)$.

2. We proceed to show that the set of prime numbers $\mathscr{P}$ is infinite. The proof is by contradiction. Because the only integers that are not multiples of prime numbers are 1 and -1,

$$\mathbb{Z} \setminus \{-1, 1\} = \bigcup_{p \in \mathscr{P}} A(p, 0). \tag{2.9}$$

The right-hand side is closed because $A(p, 0)$ is closed (by (1.b)) and $\mathscr{P}$ is finite. Then looking the left-hand side, we deduce that $\{-1, 1\}$ is open, which is not possible by (1 iii). This contradiction proves the result. □

If A is a subset of a set X, then $\tau = \{\emptyset, X, A\}$ is a topology on X. We extend this idea under certain conditions by considering two subsets of X.

Exercise 2.11 If A and B are two nontrivial subsets of a set X, find the properties of A and B to ensure that $\tau = \{\emptyset, X, A, B\}$ is a topology on X.

Solution The properties of the intersection and union of the topology require that $A \cup B \in \tau$ and $A \cap B \in \tau$. Starting from $A \cup B \in \tau$, we have three possible cases, and within each we examine the requirement that $A \cap B \in \tau$.

1. Case $A \cup B = A$. Then $B \subset A$. In such a case, $A \cap B = B$ and $B \in \tau$.
2. The case $A \cup B = B$ is analogous.
3. Case $A \cup B = X$. In order to ensure that $A \cap B \in \tau$ and because τ has only four elements, we have the following alternatives:
   (a) Case $A \cap B = \emptyset$. Then A and B constitute a partition of the space.
   (b) Case $A \cap B = A$. Then $A \subset B$. But in such a case, $A \cup B = B \neq X$, which is not possible.
   (c) The case $A \cap B = B$ is analogous and also not possible.
   (d) Case $A \cap B = X$. Then $A = B = X$, and that is not possible either.

Then the collection of subsets τ is a topology on X if and only if $A \subset B$, $B \subset A$ or $\{A, B\}$ is a partition of X. □

Motivated by the included point topology, we replace the particular point by a fixed subset and the open sets will be those containing this particular subset.

Exercise 2.12 Fix a subset A of a set X and define

$$\tau = \{O \subset X : A \subset O\} \cup \{\emptyset\}.$$

Prove that τ is a topology. This topology is called the *included set topology*. Characterize the closed sets. Find a basis for the topology. If $C \subset X$, prove that the relative topology on C is trivial if $C \subset A$ and discrete if $C \cap A = \emptyset$.

Solution

1. It is obvious $X \in \tau$. If $\{O_i : i \in I\} \subset \tau$ then each one of the sets O_i contains A, so this also holds for $\bigcup_{i \in I} O_i$ and $\bigcap_{i \in I} O_i$.

2. A set F is closed if $X \setminus F$ is open and this means that $A \subset X \setminus F$. Thus F is a closed set if and only if $A \cap F = \emptyset$.
3. For each $x \in X$, let $B_x = \{x\} \cup A$. Notice that $B_x = A$ if $x \in A$. We prove $\beta = \{B_x : x \in X\}$ is a basis for τ. It is clear that all B_x are open sets. On the other hand, let $O \in \tau$ and $x \in O$. Then $A \subset O$. Thus O contains the point x and the set A, so $\{x\} \cup A = B_x \subset O$.
4. Suppose $C \subset A$. If $O \in \tau$, $O \neq \emptyset$ then $A \subset O$, so $O \cap C = C$. This proves that $\tau_{|C} = \{\emptyset, C\}$.
5. Suppose $C \cap A = \emptyset$. For any $F \subset C$, $F \cap A = \emptyset$, so F is closed in X by (2), so also closed in C. Since all subsets of C are closed, the topology is discrete. Another way is to check that all singletons are open in $\tau_{|C}$. If $x \in C$, then $A \cup \{x\} \in \tau$, so $(A \cup \{x\}) \cap C \in \tau_{|C}$, but this intersection is just $\{x\}$ because $C \cap A = \emptyset$. □

Exercise 2.13 If $X = [-1, 1]$, define

$$\tau = \{O \subset X : 0 \notin O\} \cup \{O \subset X : (-1, 1) \subset O\}.$$

Prove that τ is a topology on X and find the family of closed sets. Find an efficient basis for τ. Study the relative topology on $(0, 1)$ and on $[0, 1)$.

Solution There are two types of open sets, $\tau_1 = \{O \subset X : 0 \notin O\}$ and $\tau_2 = \{O \subset X : (-1, 1) \subset O\}$. Thus τ is the union of two topologies, where τ_1 together with X is the topology of the excluded point topology and τ_2 together with $\emptyset$ is the included set topology. Although the union of two topologies is not, in general, a topology, in this particular case, it is. Indeed, it is obvious that $\emptyset \in \tau_1$ and $X \in \tau_2$. Moreover,

1. (a) It is clear that the properties of the union and intersection hold if all elements belong to τ_1 or if all belong to τ_2. Let $\{O_i : i \in I\}$ be a collection of elements of τ where there are elements of the two types. Let O_{i_0} be any of τ_2. Then $(-1, 1) \subset O_{i_0}$, so $(-1, 1) \subset O_{i_0} \subset \bigcup_{i \in I} O_i$, proving that $\bigcup_{i \in I} O_i \in \tau_2$, hence in τ.
   (b) If $O_1 \in \tau_1$, $O_2 \in \tau_2$, then $0 \notin O_1$, so $0 \notin O_1 \cap O_2$ and $O_1 \cap O_2 \in \tau_1$.
2. We show that

$$\beta = \{\{x\} : x \neq 0\} \cup \{(-1, 1)\}$$

is a basis for τ. It is obvious that each element of β is an open set. Let $O \in \tau$ and $x \in O$. If $O \in \tau_1$ then necessarily $x \neq 0$ and the element $B = \{x\} \in \beta$ satisfies $x \in B \subset O$. Let $O \in \tau_2$. Then $(-1, 1) \subset O$. If $x = 0$, let $B = (-1, 1) \in \beta$, so $0 \in B \subset O$. If $x \neq 0$, take $B = \{x\} \in \beta$ and $x \in B \subset O$.
3. Let $A = (0, 1)$. For all $x \in (0, 1)$, the set $\{x\}$ is open in X so also in $\tau_{|A}$. This proves that $\tau_{|A}$ is discrete.
4. Let $C = [0, 1)$. We use the above basis β. A basis for $\tau_{|C}$ is $\beta_C = \{B \cap C : B \in \beta\}$. Making all these intersections, we obtain $\beta_C = \{\{x\} : x \neq 0\} \cup \{C\}$ and the arbitrary unions of elements of β_C is $\tau_{|C} = \mathbf{P}((0, 1)) \cup \{C\}$. □

We can construct a new topological space from a given space by adding one point and declaring that the entire set is the only open set containing that point.

Exercise 2.14 Let (X, τ) be a topological space and $\omega \notin X$. On $X^* = X \cup \{\omega\}$, define $\tau^* = \tau \cup \{X^*\}$. Prove that τ^* is a topology on X^*, called the *Big Brother topology*. Prove that the relative topology on X is τ.

Solution

1. It is obvious that $\emptyset$ and X^* belong to τ^*. Let $O_1, O_2 \in \tau^*$. If they belong to τ, then the intersection is an element of τ because τ is a topology. If, say $O_1 = X^*$, then $O_1 \cap O_2 = O_2 \in \tau$, so belongs to τ^*.
   Let $\{O_i : i \in I\} \subset \tau^*$. If some $O_i = X^*$ then the union of all them is X^*, which belongs to τ^*. Otherwise, all O_i are elements of τ and thus the union belongs to τ because τ is a topology, hence also to τ^*.
2. The relative topology $\tau^*{}_{|X}$ is derived by intersecting all elements of τ^* with X. If $O \in \tau$, $O \cap X = O$, and if $O = X^*$ then $O \cap X = X$. Thus we obtain τ, the topology on X. $\square$

Exercise 2.15 On $X = \mathbb{N} \setminus \{1\}$, define the *topology of divisors* τ as follows. A subset $O \subset X$ is open if satisfies the following property:

$$\text{if } n \in O \text{ then all divisors of } n \text{ belong to } O.$$

Show that τ is a topology on X and find an efficient basis for the topology.

Solution When m is a divisor of n, we simply write $m|n$.

1. The trivial sets $\emptyset$ and X clearly belong to τ. Let $\{O_i : i \in I\} \subset \tau$ and $n \in \bigcup_{i \in I} O_i$. Suppose that $m|n$. Then there is $i_0 \in I$ such that $n \in O_{i_0}$. Because $m|n$ and $O_{i_0} \in \tau$, $m \in O_{i_0}$, so $m \in \bigcup_{i \in I} O_i$. Let O_1 and O_2 be two elements of τ and let $n \in O_1 \cap O_2$ be given. Let $m \in \mathbb{N}$ such that $m|n$. Since $n \in O_1$ and $O_1 \in \tau$, we must have $m \in O_1$. Similarly, $m \in O_2$, proving that $m \in O_1 \cap O_2$. Notice that the above argument remains valid for an arbitrary family of open sets, so the intersection of arbitrary open sets is open.
2. For each n, let U_n be the set of all divisors of n. We prove that $\beta = \{U_n : n \in \mathbb{N}\}$ is a basis for τ. First, we prove that U_n is an open set. Let $m \in U_n$. We must see that every divisor of m belongs to U_n. But if $p|m$ then $p|n$ because $m|n$. Then $p \in U_n$.
   Let $O \in \tau$ and $n \in O$. We prove that $n \in U_n \subset O$. Let $m \in U_n$. Then m is a divisor of n. As $n \in O$ and O contains all divisors of n, then $m \in O$. This proves the inclusion $U_n \subset O$ and the proof is completed.
   Let γ be another basis for τ. We see that $U_n \in \gamma$. There is $B \in \gamma$ such that $n \in B \subset U_n$. Let $m \in U_n$. Then $m|n$ and as $n \in B$ and B is open, then $m \in B$. This proves that $U_n = B \in \gamma$, so $\beta \subset \gamma$. $\square$

Exercise 2.16 Define a topology τ on the set of natural numbers $\mathbb{N}$ by declaring a set $O \subset \mathbb{N}$ to be open if and only if satisfies the following property:

$$\text{if } 2n+1 \notin O, \text{ then } 2n \notin O \text{ and } 2n+2 \notin O.$$

Prove that τ is a topology and find the relative topology on the set of even numbers, the set of odd numbers and on $\{3, 4, 5, 6\}$.

Solution

1. It is elementary that the trivial sets belong to τ. Now let $\{O_i : i \in I\} \subset \tau$. Let $2n + 1 \notin \bigcup_{i \in I} O_i$. Then $2n + 1 \notin O_i$ for all $i \in I$; in particular and because $O_i \in \tau$, we have $2n \notin O_i$ and $2n+2 \notin O_i$ for all i. Thus $2n, 2n+2 \notin \bigcup_{i \in I} O_i$. Let O_1, O_2 two elements of τ. If $2n + 1 \notin O_1 \cap O_2$ then $2n + 1 \notin O_1$ or $2n + 1 \notin O_2$. Suppose, for instance, that $2n + 1 \notin O_1$. Since $O_1 \in \tau$, we have $2n, 2n + 2 \notin O_1$ and therefore $2n, 2n + 2 \notin O_1 \cap O_2$.
2. Let A be the set of even numbers. If $2n$ is an arbitrary element of A, then $\{2n - 1, 2n, 2n+1\} \in \tau$. Thus $A \cap \{2n-1, 2n, 2n+1\}$ is open in A. But this intersection is just $\{2n\}$. Therefore, all singletons of A are open, so the relative topology is discrete.
3. Let B be the set of odd numbers. Let $2n + 1 \in B$ be given. Since the set $\{2n + 1\}$ is open in $\mathbb{N}$, the relative topology on B is discrete.
4. Let $C = \{3, 4, 5, 6\}$. We know that the sets $\{2n-1, 2n, 2n+1\}$ and the singletons of odd numbers are open sets. When we intersect these open sets with C, the resulting sets are $\{3\}$, $\{5\}$, $\{3, 4, 5\}$ and $\{5, 6\}$. Notice that there are no open sets in $\mathbb{N}$ formed by only two consecutive numbers. Thus the open sets in C with two elements are $\{3, 5\}$ and $\{5, 6\}$.
   In the next stage, we examine the open sets with three elements. By the union of the previous open sets, we deduce that $\{3, 4, 5\}$ and $\{3, 5, 6\}$ are open in C. The other sets of three elements are $\{3, 4, 6\}$ and $\{4, 5, 6\}$. The set $\{3, 4, 6\}$ is not open in C because the smallest open set in $\mathbb{N}$ containing 4 is $\{3, 4, 5\}$. So if $\{3, 4, 6\} \in \tau_{|C}$, then $4 \in \{3, 4, 5\} \cap C \subset \{3, 4, 6\}$ that is not possible. Likewise, $\{4, 5, 6\} \notin \tau_{|C}$ because we would have $4 \in \{3, 4, 5\} \cap C \subset \{4, 5, 6\}$ which is also not possible. In conclusion, the relative topology is

$$\tau_{|C} = \{\emptyset, C, \{3\}, \{5\}, \{3, 5\}, \{5, 6\}, \{3, 4, 5\}, \{3, 5, 6\}\}.$$

□

Exercise 2.17 Let X be an infinite set and $p \in X$ a particular point. Prove that $\tau = \tau_{ex} \cup \tau_{CF}$ is a topology on X. This topology is called *strong topology*. Study the relative topology on any subset of X.

Solution

1. The properties of the union and intersection of elements where all elements belong to τ_{ex} or all to τ_{CF} hold in this case.

   (a) Since $p \notin \emptyset$ and $X \setminus X$ is finite, we have $\emptyset, X \in \tau$.

(b) Let $\{O_i : i \in I\}$ be a family of elements of τ, where there are elements of τ_{ex} and of τ_{CF}. There is $i_0 \in I$ such that $O_{i_0} \in \tau_{CF}$, so by De Morgan's laws,

$$X \setminus \bigcup_{i\in I} O_i = \bigcap_{i\in I}(X \setminus O_i) \subset X \setminus O_{i_0},$$

and $X \setminus O_{i_0}$ is finite. Thus $X \setminus \bigcup_{i\in I} O_i$ is also finite, so $\bigcup_{i\in I} \in \tau_{CF} \subset \tau$.

(c) Let $O_1, O_2 \in \tau$. Suppose that $O_1 \in \tau_{ex}$ and $O_2 \in \tau_{CF}$. Then $p \notin O_1$. In particular, $p \notin O_1 \cap O_2$, proving that $O_1 \cap O_2 \in \tau_{ex} \subset \tau$.

2. Let $A \subset X$ be given. We distinguish two cases.

(a) Case $p \notin A$. If $x \in A$ then $\{x\} \in \tau_{ex}$, so $\{x\}$ is open in τ. This proves that the relative topology on A is discrete.

(b) Case $p \in A$. If we intersect all open sets of τ_{ex} with A, we get all subsets of A that do not contain the point p. If we intersect the elements of τ_{CF} with A, we obtain subsets of A whose complement, with the exception of A, is a finite set. Thus we have proved that $\tau_{|A} = \mathbf{P}(A \setminus \{p\}) \cup \tau^A_{CF}$, where τ^A_{CF} is the cofinite topology on A. □

In Exercise 2.1 we defined the topology generated by the basis $\{\mathbb{R}\} \times \beta_{sr}$ whose topology is formed by the trivial sets and the horizontal halfplanes $\mathbb{R} \times (x, \infty)$. We generalize this topology, changing horizontal halfplanes for sloped halfplanes.

Exercise 2.18 Consider the collection of subsets of $\mathbb{R}^2$ defined by

$$\tau = \{\emptyset, \mathbb{R}^2\} \cup \{G_a : a \in \mathbb{R}\},$$

where $G_a = \{(x, y) \in \mathbb{R}^2 : y > x + a\}$ as it is represented in Fig. 2.13. Show that τ is a topology. Compare with the Euclidean topology on $\mathbb{R}^2$. Characterize the closed sets and check if $\beta = \{G_a : a \in \mathbb{Q}\}$ is a basis for τ.

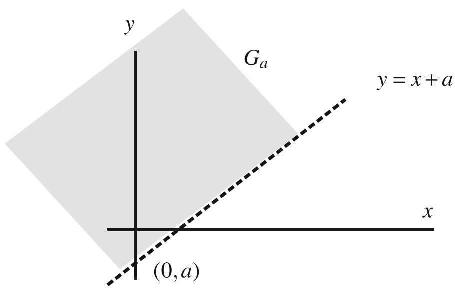


Fig. 2.13 The sets G_a of Exercise 2.18

Solution

1. Let $\{G_{a_i} : i \in I\}$. If the set $\{a_i : i \in I\}$ is not bounded from below, then we see that $\bigcup_{i\in I} G_{a_i} = \mathbb{R}^2$. Indeed, if $(x, y) \in \mathbb{R}^2$, let $i_0 \in I$ such that $y - x > a_{i_0}$. Then clearly $(x, y) \in G_{a_{i_0}}$. In case that $\{a_i : i \in I\}$ is bounded from below, we claim that $\bigcup_{i\in I} G_{a_i} = G_a$, where $a = \inf\{a_i : i \in I\}$. Let $(x, y) \in \bigcup_{i\in I} G_{a_i}$. Then there is $j \in I$ such that $(x, y) \in G_j$, hence $y > x + a_j$. Since $a_j \geq a$, then $y > x + a$, proving that $(x, y) \in G_a$. Reciprocally, let $(x, y) \in G_a$. Then $y - x > a$. By the definition of infimum, there is a_j such that $y - x > a_j \geq a$, so $(x, y) \in G_{a_j}$.
   For the intersection, it is obvious that $G_a \cap G_b = G_{\max\{a,b\}} \in \tau$.
2. An argument as in Example 2.22 demonstrates that all G_a are open in the Euclidean topology. However, the open set $(0, 1) \times (0, 1) \in \tau_u$ is not open in τ because it would then contain an element G_a, which is not possible because $(0, 1) \times (0, 1)$ is bounded and the G_a is not.
3. The closed sets are the complements of the open sets, so $\mathscr{F} = \{\emptyset, \mathbb{R}^2\} \cup \{F_a : a \in \mathbb{R}\}$, where $F_a = \{(x, y) \in \mathbb{R}^2 : y \leq x + a\}$.
4. It is clear that each element of β is open. Let G_a and $(x_0, y_0) \in G_a$. Then $y_0 - x_0 > a$. Let $q \in \mathbb{Q}$ be a rational number such that $y_0 - x_0 > q > a$. Then $(x_0, y_0) \in G_q$. We see that $(x_0, y_0) \in G_q \subset G_a$. But this is clear for if $(x, y) \in G_q$, then $y - x > q > a$. □

Exercise 2.19 For each $(x, y) \in \mathbb{R}^2$, define

$$B_{x,y} = (\{x\} \times \mathbb{R}) \cup (\mathbb{R} \times \{y\});$$

that is, the two lines parallel to the coordinate axes through the point (x, y). Prove that $\mathscr{S} = \{B_{x,y} : (x, y) \in \mathbb{R}^2\}$ is not a basis for a topology. Prove that the topology that has $\mathscr{S}$ as a subbasis is discrete.

Solution

1. If we intersect two elements of $\mathscr{S}$ we obtain two points of $\mathbb{R}^2$, a vertical line or a horizontal line,

$$B_{x,y} \cap B_{z,t} = \begin{cases} \{(x,t), (z,y)\}, & x \neq z, y \neq t \\ \{x\} \times \mathbb{R}, & x = z \\ \mathbb{R} \times \{y\}, & y = t. \end{cases}$$

   Therefore, we cannot find an element of $\mathscr{S}$ inside $B_{x,y} \cap B_{z,t}$.
2. It suffices to verify that all singletons of $\mathbb{R}^2$ are open in the topology determined by $\tau(\mathscr{S})$. Let $(x, y) \in \mathbb{R}^2$ be given. Then the set

$$O = B_{x-1,y} \cap B_{x,y+1} = \{(x, y), (x - 1, y + 1)\}$$

   is open because it is intersection of two elements of the subbasis. Now, $O \cap B_{x,y}$ is open again, but this intersection is $\{(x, y)\}$ because $(x - 1, y + 1) \notin B_{x,y}$. □

Exercise 2.20 On $\mathbb{R}$, consider the collection of subsets

$$\tau = \{\emptyset, \mathbb{R}\} \cup \{(-a, a) : a \in \mathbb{R}, a > 1\}.$$

Prove that τ is a topology and describe the collection of closed sets. Describe the relative topology on the intervals $[-1, 1]$ and $[1, 2]$.

Solution

1. Let $\{(-a_i, a_i) : i \in I\}$, with $a_i > 1$. If the set $\{a_i : i \in I\}$ is not bounded from above, then the union of all them is $\mathbb{R}$. If $\{a_i : i \in I\}$ is bounded from above, then $\bigcup_{i \in I}(-a_i, a_i) = (-\alpha, \alpha)$, with $\alpha = \sup\{a_i : i \in I\}$. Note that $\alpha > 1$. On the other hand, it is easily checked that

$$(-a_1, a_1) \cap (-a_2, a_2) = (-\min\{a_1, a_2\}, \min\{a_1, a_2\}) \in \tau$$

because $\min\{a_1, a_2\} > 1$.
2. A closed set is the complement of an open set. Thus

$$\mathscr{F} = \{\emptyset, \mathbb{R}\} \cup \{(-\infty, -a] \cup [a, \infty) : a \in \mathbb{R}, a > 1\}.$$

3. For the set $[-1, 1]$, if we intersect an interval $(-a, a)$, $a > 1$, with $[-1, 1]$, we get $[-1, 1]$. Thus the relative topology is trivial.
4. For the set $[1, 2]$, the intersection of all intervals $(-a, a)$, $a > 1$, with $[1, 2]$ yields $\tau_{|[1,2]} = \{\emptyset, [1, 2]\} \cup \{[1, a) : 1 < a \leq 2\}$. □

Exercise 2.21 The *scattered line* is the space $(\mathbb{R}, \tau)$ where

$$\tau = \{O \cup A : O \in \tau_u,\ A \subset \mathbb{R} - \mathbb{Q}\}.$$

Prove that τ is a topology and study the relative topologies on $\mathbb{Q}$ and on $\mathbb{R} - \mathbb{Q}$.

Solution

1. If $O = \mathbb{R} \in \tau_u$ and $A = \emptyset$, then $\mathbb{R} \in \tau$. If $O = A = \emptyset$, then $\emptyset \in \tau$.
For unions of elements of τ,

$$\bigcup_{i \in I}(O_i \cup A_i) = (\bigcup_{i \in I} O_i) \cup (\bigcup_{i \in I} A_i)$$

and $\bigcup_{i \in I} O_i \in \tau_u$ and $\bigcup_{i \in I} A_i \subset \mathbb{R} - \mathbb{Q}$. For the intersection of elements of τ, by the distributive property of unions and intersections, we have

$$(O_1 \cup A_1) \cap (O_2 \cup A_2) = (O_1 \cap O_2) \cup (O_1 \cap A_2) \cup (O_2 \cap A_1) \cup (A_1 \cap A_2).$$

The first set $O_1 \cap O_2$ is open in the Euclidean topology and each one of the others is a subset of $\mathbb{R} - \mathbb{Q}$, so its union is a subset of $\mathbb{R} - \mathbb{Q}$.

2. An open set in $\tau_{|\mathbb{Q}}$ is of type $(O \cup A) \cap \mathbb{Q}$, where $O \in \tau_u$ and $A \subset \mathbb{R} - \mathbb{Q}$. Then
$$(O \cup A) \cap \mathbb{Q} = (O \cap \mathbb{Q}) \cup (A \cap \mathbb{Q}) = (O \cap \mathbb{Q}) \cup \emptyset = O \cap \mathbb{Q}.$$
Thus $\tau_{|\mathbb{Q}}$ is the relative topology on $\mathbb{Q}$ from the Euclidean topology.
3. Let $x \in \mathbb{R} - \mathbb{Q}$ be given. If we choose $O = \emptyset$ and $A = \{x\}$, then $\{x\} = O \cup A \in \tau$ so $\{x\}$ is open in the relative topology on $\mathbb{R} - \mathbb{Q}$. Thus the topology on $\mathbb{R} - \mathbb{Q}$ is discrete. □

Exercise 2.22 Let $X = \mathbb{R}$ and $\mathscr{C} = \{A \subset \mathbb{R} : A \text{ is bounded}\} \cup \{\mathbb{R}\}$. Prove that $\mathscr{C}$ is the collection of closed sets for a topology τ. Determine if $\beta = \{\mathbb{R} \setminus [a, b] : a, b \in \mathbb{R}\}$ is a basis for τ. Compare with the Euclidean topology on $\mathbb{R}$.

Solution

1. It is obvious that $\mathscr{C}$ obeys the axioms to be the closed sets of a topology on $\mathbb{R}$ because the union of two bounded sets is bounded and arbitrary intersections of bounded sets are bounded.
2. The elements of β are open sets because the complement of $\mathbb{R} \setminus [a, b]$ is $[a, b]$, a bounded set of $\mathbb{R}$. However, β does not satisfy (ii) of Theorem 2.2.3. We can see this with the following example. The set $(0, 1)$ is bounded, so $A = \mathbb{R} \setminus (0, 1)$ is open in τ. Let $0 \in A$. If β is a basis, there would be $a \leq b$ such that $0 \in \mathbb{R} \setminus [a, b] \subset A$; that is,
$$0 \in (-\infty, a) \cup (b, \infty) \subset (-\infty, 0] \cup [1, \infty).$$
We discuss two alternatives:
   (a) Case $0 < a$. Let $x < 1$ such that $0 < x < a$; in particular, $x \notin A$. But $x \in (-\infty, a) \subset (-\infty, a) \cup (b, \infty) \subset A$, which is not possible.
   (b) Case $a \leq 0$. Since $0 \in (-\infty, a) \cup (b, \infty)$, we deduce $b < 0$. Now $x = 1/2$ satisfies $1/2 \in (-\infty, a) \cup (b, \infty)$, but $1/2 \notin A$.
3. The topologies are not comparable. The set $(0, 1)$ is closed in $(\mathbb{R}, \tau)$ but not in the Euclidean topology. On the other hand, $(-\infty, 0]$ is closed in Euclidean topology but not in $(\mathbb{R}, \tau)$ because it is not bounded. □

Exercise 2.23 The *cocompact topology* on $\mathbb{R}$ is defined by
$$\tau' = \{O \subset \mathbb{R} : \mathbb{R} \setminus O \text{ is closed in } \tau_u \text{ and bounded}\} \cup \{\emptyset\}.$$
Prove that τ' is a topology and compare with the Euclidean topology. What is the relative topology on $\mathbb{Z}$?

Solution

1. Instead of proving that τ' satisfies the axioms of being a topology, the family τ' is simply $\tau_u \cap \tau$, where τ is the topology defined in the foregoing exercise. Thus τ'

is a topology because it is intersection of two topologies (Example 2.6). Notice that the collection of closed sets is

$$\mathscr{F}' = \{F \subset \mathbb{R} : F \text{ is closed in } \tau_u \text{ and bounded}\} \cup \{\mathbb{R}\}.$$

2. By definition of τ', we have $\tau' \subset \tau_u$. However, this inclusion is proper because $(-\infty, 0]$ is closed in the Euclidean topology but $(-\infty, 0] \notin \mathscr{F}'$ because it is not bounded.
3. We find the closed sets of $\tau'_{|\mathbb{Z}}$. If F is a closed set in $\mathbb{R}$, then $F \cap \mathbb{Z}$ is bounded, so it is finite. Conversely, if $A \subset \mathbb{Z}$ is finite, then it is closed in τ_u, so closed in $\tau'_{|\mathbb{Z}}$. This proves that the relative topology on $\mathbb{Z}$ is the cofinite topology. □

Exercise 2.24 Consider $\beta = \{B_{x,n} : x \in \mathbb{R}, n \in \mathbb{N}\}$, where

$$B_{x,n} = \left(x - \frac{1}{n}, x + \frac{1}{n}\right) \cup (n, \infty).$$

1. Prove that β is a basis for some topology τ on $\mathbb{R}$.
2. Compare with the Euclidean topology.
3. If $A \subset \mathbb{R}$ is a bounded set, prove that the relative topology is Euclidean.
4. Find a subset of $\mathbb{R}$ where the relative topology does not coincide with the Euclidean one.

Solution

1. It is obvious that the union of all elements of β is $\mathbb{R}$. For the second property, let us observe that the intersection of two elements of β is not an element of β. For example,

$$B_{0,2} \cap B_{1,2} = \left(\left(\tfrac{-1}{2}, \tfrac{1}{2}\right) \cup (2, \infty)\right) \cap \left(\left(\tfrac{1}{2}, \tfrac{3}{2}\right) \cup (2, \infty)\right) = (2, \infty).$$

Let $z \in B_{x,n} \cap B_{y,m}$. Without loss of generality, suppose $m \geq n$. Note that

$$\begin{aligned} B_{x,n} \cap B_{y,m} = &\left(\left(x - \tfrac{1}{n}, x + \tfrac{1}{n}\right) \cap \left(y - \tfrac{1}{m}, x + \tfrac{1}{m}\right)\right) \\ &\cup \left(\left(x - \tfrac{1}{n}, x + \tfrac{1}{n}\right) \cap (m, \infty)\right) \\ &\cup \left((n, \infty) \cap \left(y - \tfrac{1}{m}, x + \tfrac{1}{m}\right)\right) \cup (m, \infty). \end{aligned}$$

Then z is contained in some of these four subsets and all them are open in the Euclidean topology. Thus there is $k \in \mathbb{N}$ such that the interval $(z - 1/k, z + 1/k)$ is contained in that subset. If we choose $k \geq m$, then $(k, \infty) \subset (m, \infty)$. This proves that $B_{z,k} \subset B_{x,n} \cap B_{y,m}$.
2. Because each element $B_{x,n}$ is union of two open intervals, the set $B_{n,x}$ is open in the Euclidean topology. Consequently, the Euclidean topology is finer than

the topology generated by β. Nevertheless, there are elements of the Euclidean topology that do not belong to $\tau(\beta)$. For example, $(a, b) \notin \tau(\beta)$ because all elements of β are not bounded and the same follows for $\tau(\beta)$.

3. Let $A \subset \mathbb{R}$ be a bounded set. We will prove that $\tau_{|A} = \tau_{u|A}$. Since $\tau \subset \tau_u$, then $\tau_{|A} \subset \tau_{u|A}$. Conversely, let $O \cap A \in \tau_{u|A}$. We see that $O \cap A$ is open in $\tau_{|A}$. Let $x \in O \cap A$. Since O is open in τ_u, there is $n \in \mathbb{N}$ such that

$$x \in \left(x - \tfrac{1}{n}, x + \tfrac{1}{n}\right) \subset O.$$

On the other hand, A is bounded, so we may select m sufficiently large so $A \subset (-m, m)$. If $k = \max\{n, m\}$, then $1/k \leq 1/n$ and $(k, \infty) \subset (m, \infty)$. Thus $(x - 1/k, x + 1/k) \subset O$. Then $B_{x,k} \cap A \in \tau_{|A}$ and we see that $x \in B_{x,n} \cap A \subset O \cap A$:

$$B_{x,k} \cap A = \left(\left(x - \tfrac{1}{k}, x + \tfrac{1}{k}\right) \cup (k, \infty)\right) \cap A = \left(x - \tfrac{1}{k}, x + \tfrac{1}{k}\right) \cap A \subset O \cap A.$$

4. We know that $\tau_{u|\mathbb{Z}}$ is discrete. We show that $\tau_{|\mathbb{Z}}$ is not discrete. Let $m \in \mathbb{Z}$. If $\{m\}$ is open in $\tau_{|\mathbb{Z}}$, there are $x \in \mathbb{R}$ and $n \in \mathbb{N}$ such that $m \in B_{x,n} \cap \mathbb{Z} \subset \{m\}$. However, the set $B_{x,n} \cap \mathbb{Z}$ is infinite because it contains $(n, \infty) \cap \mathbb{Z}$. □

Exercise 2.25 The *nested interval topology* on $(0, 1)$ is defined by

$$\tau = \left\{\left(0, 1 - \frac{1}{n}\right) : n \in \mathbb{N}\right\} \cup \{(0, 1)\}.$$

Prove that τ is a topology and compare with the Euclidean topology on $(0, 1)$. Study the relative topology on $(0, 1/2)$ and on $[1/2, 2/3]$.

Solution

1. If $n = 1$, then $\emptyset \in \tau$ and by definition $(0, 1) \in \tau$. Let $\{(0, 1 - \frac{1}{n_i}) : i \in I\}$. If the set $\{n_i : i \in I\}$ is not bounded from above, then $\bigcup_{i \in I}(0, 1 - \frac{1}{n_i}) = (0, 1)$. In case that $\{n_i : i \in I\}$ is bounded from above, there is a maximum m. Now $\bigcup_{i \in I}(0, 1 - \frac{1}{n_i}) = (0, 1 - \frac{1}{m})$. For the intersection,

$$\left(0, 1 - \frac{1}{n}\right) \cap \left(0, 1 - \frac{1}{m}\right) = \left(0, 1 - \frac{1}{\min\{n, m\}}\right) \in \tau.$$

2. The elements of τ are open intervals of $\mathbb{R}$, so they are open in the Euclidean topology. Taking into account that all of them are included in $(0, 1)$, they are open in the relative topology on $(0, 1)$. Thus the Euclidean topology on $(0, 1)$ is finer than τ. However, the inclusion $\tau \subset \tau_u$ is proper. For instance, $(1/2, 1)$ is open in the Euclidean topology but it is not an element of τ.

3. The elements of τ are $\{(0, \frac{1}{2}), (0, \frac{2}{3}), (0, \frac{3}{4}), \ldots\}$. When we intersect with $(0, \frac{1}{2})$, we obtain $(0, \frac{1}{2})$, so the relative topology on $(0, \frac{1}{2})$ is trivial. For the set $[\frac{1}{2}, \frac{2}{3}]$, the intersection is

$$\tau_{|[\frac{1}{2},\frac{2}{3}]} = \left\{\emptyset, \left[\frac{1}{2}, \frac{2}{3}\right], \left[\frac{1}{2}, \frac{2}{3}\right)\right\}.$$

□

Exercise 2.26 Let G be an abelian group. For each $A \subset G$, define

$$r(A) = \{x \in G : \text{there is } n \in \mathbb{N} \text{ such that } x^n \in A\}.$$

We are using the notation of multiplication for the group operation: $x^n = x \cdot \overset{n}{\smile\!\!\!\!\ldots} \cdot x$. Prove that there is a unique topology on G such that the family of closed sets is $\mathscr{F} = \{A \subset G : r(A) = A\}$. Prove that the arbitrary union of closed sets is closed and characterize the open sets. Particularize to the case $G = \mathbb{Z}$.

Solution Observe that $A \subset r(A)$ for all $A \subset G$.

1. We see that the family $\mathscr{F}$ satisfies the properties of Theorem 2.1.3, proving also that arbitrary union of closed sets is closed. It is elementary that $\emptyset, G \in \mathscr{F}$.
   (a) Let $\{A_i : i \in I\}$ be a family of closed sets and let $x \in r(\bigcup_{i\in I} A_i)$. Then there is $n \in \mathbb{N}$ such that $x^n \in \bigcup_{i\in I} A_i$ and thus there is $i_0 \in I$ such that $x^n \in A_{i_0}$. Then $x \in r(A_{i_0}) = A_{i_0} \subset \bigcup_{i\in I} A_i$.
   (b) Let $\{A_i : i \in I\} \subset \mathscr{F}$, and we prove that $\bigcap_{i\in I} A_i \in \mathscr{F}$. Let $x \in r(\bigcap_{i\in I} A_i)$. Then there is $n \in \mathbb{N}$ such that $x^n \in \bigcap_{i\in I} A_i$, so $x^n \in A_i$ for all $i \in I$. Since $r(A_i) = A_i$, we deduce that $x \in A_i$ for all $i \in I$, hence $x \in \bigcap_{i\in I} A_i$.
2. A set O is open if $G \setminus O = r(G \setminus O)$. Since the inclusion $G \setminus O \subset r(G \setminus O)$ is always true, then equality says that if $x^n \in G \setminus O$ for some $n \in \mathbb{N}$, then $x \in G \setminus O$. By contraposition, O is open if and only if for every $x \in O$, we have $x^n \in O$ for all $n \in \mathbb{N}$.
3. Consider $\mathbb{Z}$ as an abelian group with respect to the sum of integers. A subset $O \subset \mathbb{Z}$ is open if for all $x \in O$, $x + \overset{n}{\smile\!\!\!\!\ldots} + x = nx \in O$ for all $n \in \mathbb{Z}$. Thus O is open if and only if O contains all multiples of every number of O. In particular, the sets $M_x = \{nx : n \in \mathbb{Z}\}$ are open. □

Exercise 2.27 Define $\tau = \{\mathbb{R} \setminus A : A \text{ is a finite set of } \mathbb{Z}\} \cup \{\emptyset\}$. Prove that τ defines a topology on $\mathbb{R}$ and compare with the cofinite topology. If $A \subset \mathbb{R}$, describe the relative topology when $A \subset \mathbb{Z}$ and $A \cap \mathbb{Z} = \emptyset$.

Solution

1. Instead of proving the axioms of being a topology, we see that the collection of all complements

$$\mathscr{F} = \{F \subset \mathbb{Z} : F \text{ is finite}\} \cup \{\mathbb{R}\}$$

obeys the properties to be the collection of closed sets for a topology. But this is obvious because the arbitrary intersection of finite sets is finite and the union of two finite sets is finite (this argument is entirely similar to that of the cofinite topology up to the difference that the finite sets are included in $\mathbb{Z}$). Observe that the idea behind this exercise can be extended to any set X, fixing $A \subset X$ and declaring the family of closed sets $\mathscr{F} = \{C \subset A : C \text{ is finite}\} \cup \{X\}$.

2. Since $\mathscr{F} \subset \mathscr{F}_{CF}$, the cofinite topology is finer than τ. However, the inclusion is proper. For example, $\{\sqrt{2}\}$ is closed in the cofinite topology but not in τ.
3. (a) If $A \subset \mathbb{Z}$, any subset of A is included in $\mathbb{Z}$, so F is closed if and only if F is finite. Thus $\mathscr{F}_{|A} = \mathscr{F}_{CF|A}$.
   (b) If $A \cap \mathbb{Z} = \emptyset$ then for all $F \in \mathscr{F}$, $F \cap A = \emptyset$ if $F \neq \mathbb{R}$ or $F \cap A = A$ if $F = \mathbb{R}$. Thus the relative topology is trivial. □

2.7 Suggested Exercises

2.7.1 Find all topologies on a set with three elements.

2.7.2 Let $X = \{a, b, c, d\}$ and $A = \{a, b\}$. Find all topologies on X such that A is both open and closed.

2.7.3 Find, if possible, efficient bases for τ_r and τ_{sr} on $\mathbb{R}$.

2.7.4 Let β_1 and β_2 be two bases for the same topology. Determine if $\beta_1 \cap \beta_2$, $\beta_1 \cup \beta_2$ and $\{B_1 \cap B_2 : B_1 \in \beta_1, B_2 \in \beta_2\}$ are also bases.

2.7.5 If τ_u^n be the Euclidean topology on $\mathbb{R}^n$, prove that $\beta = \{O_1 \times \ldots \times O_n : O_i \in \tau_u^1\}$ is a basis for τ_u^n. With the identity $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$, prove that $\beta = \{O_1 \times O_2 : O_1 \in \tau_u^n, O_2 \in \tau_u^m\}$ is a basis for τ_u^{n+m}.

2.7.6 Let (X, τ) be a space and let $f : X \to Y$. Prove that $\tau(f) = \{O' \subset Y : f^{-1}(O') \in \tau\}$ is a topology on Y called the *final topology* on Y determined by f. Here the symbol $f^{-1}(O')$ denotes the preimage of $O \subset Y$, the set defined by $f^{-1}(O') = \{x \in X : f(x) \in O'\}$. If f is bijective, $\tau(f)$ is called the *direct image topology*.

2.7.7 Let (Y, τ') be a space and let $f : X \to Y$. Prove that $\tau'_f = \{f^{-1}(O') : O' \in \tau'\}$ is a topology on X called the *initial topology* on X determined by f. Using the notation of the above exercise, study if $\tau'_f(f) = \tau'$. Observe that the relative topology on a subspace is the initial topology for the inclusion.

2.7.8 Prove that $\beta = \{(a, b) : a, b \in \mathbb{Q}\}$ is a basis for the Euclidean topology on $\mathbb{R}$, but $\beta' = \{[a, b) : a, b \in \mathbb{Q}\}$ is not a basis for the Sorgenfrey topology. Consider the same exercise replacing $\mathbb{Q}$ by $\mathbb{R} - \mathbb{Q}$.

2.7.9 In Exercise 2.18, we replace G_a by $H_a = \{(x, y) \in \mathbb{R}^2 : y \geq x + a\}$. Prove that $\{\emptyset, \mathbb{R}\} \cup \{H_a : a \in \mathbb{R}\}$ is not a topology on $\mathbb{R}^2$, but show that the above collection of subsets is a basis for a topology. Determine if $\gamma = \{H_a : a \in \mathbb{Q}\}$ is a basis for the same topology. Compare with the topology of Exercise 2.18.

2.7.10 Determine if the following collections of bounded open intervals are bases for the Euclidean topology.

1. Let $r > 0$ be a fixed number and $\beta = \{(x, x + r) : x \in \mathbb{R}\}$.
2. $\gamma = \{(a, b) : b - a \in \mathbb{Q}^+\}$.
3. $\sigma = \{(a, b) : b - a \in (\mathbb{R} - \mathbb{Q})^+\}$.

2.7.11 Motivated by the included set topology of Exercise 2.12, analyze if, for a fixed a subset A in a set X, the collection $\{O \subset X : A \not\subset O\} \cup \{X\}$ is a topology on X.

2.7.12 Determine if $\{\emptyset, \mathbb{R}\} \cup \{[-a, a] : a \in \mathbb{R}, a > 1\}$ is a basis for some topology on $\mathbb{R}$ and calculate the topology.

2.7.13 Describe the open sets in $\mathbb{R}$ of the topologies where $\mathscr{S}$ is a subbasis:

(i) $\mathscr{S} = \{(a, \infty) : a \in \mathbb{R}\}$.
(ii) $\mathscr{S} = \{(a, \infty) : a > 0, a \in \mathbb{R}\}$.
(iii) $\mathscr{S} = \{(a, \infty) : a \in \mathbb{R}\} \cup \{(-\infty, b) : b \in \mathbb{R}\}$.

2.7.14 Let $\mathscr{S}$ be a collection of subsets in a given space. Show that the topology that has $\mathscr{S}$ as a subbasis is the smallest topology that contains $\mathscr{S}$ as open sets.

2.7.15 For each $a \in \mathbb{R}$, let $P_a = \{(x, y) \in \mathbb{R}^2 : x \geq a\}$. Find the topology that has $\mathscr{S} = \{P_a : a \in \mathbb{R}\}$ as a subbasis.

2.7.16 Prove that the relative topology on $\mathbb{Z}^n$ from $\mathbb{R}^n$ is discrete. Show that this does not occur for $\mathbb{Q}^n$.

2.7.17 (Converse of Theorem 2.5.2). Let (X, τ) be a space, $A \subset X$ and let β' be a basis for $\tau_{|A}$. For each $V \in \beta'$, since $V \in \tau_{|A}$ there is $O \in \tau$ such that $O \cap A = V$. Among all these open sets, we fix one and we denote it by O_V. Find an example where $\beta = \{O_V : V \in \beta'\}$ is not a basis for τ.

2.7.18 Continuing from the previous exercise, investigate if $\beta = \{O \in \tau : O \cap A \in \beta'\}$ is basis for τ.

Chapter 3
Proximity on a Topological Space

Up to now, the concept of a topological space has lacked any notion of proximity. In this chapter, we establish equivalent ways to define a topological space by using the concepts of neighborhoods, convergence of sequences, and interior points. Historically, mathematicians such as Hausdorff, Fréchet and Riesz wanted to break away from the constraint of the notion of distance while working with continuous maps. The concept of a "vicinity" of a point arose assigning to each point a collection of subsets with certain properties. As the field developed, people became convinced that it was better to approach the concept of topological space by assigning to a set a collection of subsets, the so-called open sets, with certain properties. This is the framework by which we typically define a topological space, as was done in the previous chapter.

3.1 Neighborhoods of a Point

The first notion of vicinity is that of a neighborhood of a point. In plain language, a neighborhood is the region around or near something. See Fig. 3.1.

Definition 3.1.1 Let X be a space and $x \in X$. A subset $U \subset X$ is called a *neighborhood* of x if there is an open set O such that $x \in O \subset U$. The collection of all neighborhoods of x is denoted by $\mathscr{U}_x$.

A first example of a neighborhood of $x \in X$ is any open set O containing x. Further, because O is open, then O is a neighborhood not only of x but of all its points. We can think of this, in plain language, as the point x having other points (the points of O) as "neighbors".

If U is a neighborhood of x, then any set V containing U is also a neighborhood of x. This indicates that the collection $\mathscr{U}_x$ of neighborhoods is large and turns out

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_3

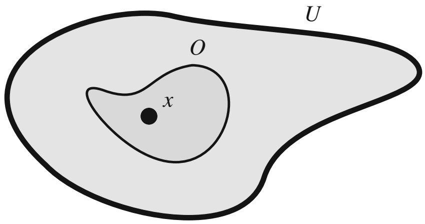


Fig. 3.1 The set U is a neighborhood of x. The open set O is also a neighborhood of x

to be rather difficult to handle. For that reason, it should be noted that some authors require that a neighborhood be an open set. We distinguish these open sets and let

$$\tau_x = \tau \cap \mathscr{U}_x = \{O \in \tau : x \in O\}.$$

Example 3.1

1. In the trivial topology (X, τ_T), the entire set X is the only neighborhood of each $x \in X$, because it is the only non-empty open set. Thus $\mathscr{U}_x = \{X\} = \tau_x$.
2. In the discrete topology (X, τ_D), every subset is open and thus any subset is a neighborhood of all its points. This proves that $\mathscr{U}_x = \{A \subset X : x \in A\} = \tau_x$.
3. Consider the included point topology (X, τ_{in}) for the particular point p. Since a neighborhood U of $x \in X$ must contain an open set, then U also contains p. Thus $\mathscr{U}_x = \{A \subset X : x, p \in A\}$, which coincides with τ_x.
4. Consider the Big Brother topology (Exercise 2.14). If $x \in X$, the neighborhoods in τ^* coincide with that of τ together with the space X^* itself. For the point ω, the only open set containing ω is X^*, so $\mathscr{U}_\omega = \{X^*\}$.
5. In the Euclidean real line, there are neighborhoods that are not open sets. For example, $[-1, 1)$ is a non-open neighborhood of $x = 0$ because $0 \in (-1, 1) \subset [-1, 1)$. ◊

We know that an open set is a neighborhood for all its points. We reverse this statement, obtaining a characterization of open sets in terms of neighborhoods.

Theorem 3.1.2 *A subset in a topological space is an open set if and only if it is a neighborhood for all its points.*

If β a basis for the topology, then U is a neighborhood of x if and only if there is $B \in \beta$ such that $x \in B \subset U$.

As in the previous chapter, we wish to topologize an abstract set by using the notion of neighborhoods.

Question: *Given a set X, under what conditions may we assign, for each $x \in X$, a collection of subsets $\mathscr{V}_x$ of X to be the neighborhoods for some topology on X?*

Theorem 3.1.3 *Suppose that X is a set such that each $x \in X$ has assigned a nonempty collection $\mathscr{V}_x$ of subsets of X with the following properties:*

(i) *$x \in V$ for all $V \in \mathscr{V}_x$;*
(ii) *if $V_1, V_2 \in \mathscr{V}_x$ then $V_1 \cap V_2 \in \mathscr{V}_x$;*
(iii) *if $U \in \mathscr{V}_x$ and $U \subset V$, then $V \in \mathscr{V}_x$;*
(iv) *for every $U \in \mathscr{V}_x$, there is $W \in \mathscr{V}_x$ such that $U \in \mathscr{V}_y$ for all $y \in W$.*

Then there exists a unique topology on X such that the neighborhoods of x coincide with $\mathscr{V}_x$ for all $x \in X$.

Example 3.2 The role of the property (iii) is crucial and it cannot be discarded. For example, for each $x \in \mathbb{R}$ consider the collection $\mathscr{V}_x$ of the open intervals containing x. Then $\mathscr{V}_x$ satisfies all properties of Theorem 3.1.3 except (iii), because a set that contains an open interval is not, in general, an open interval. Thus $\{\mathscr{V}_x : x \in \mathbb{R}\}$ does not define some topology on $\mathbb{R}$. Note that $\mathscr{V}_x$ does not coincide with the neighborhoods of x in the Euclidean topology, nor with $(\tau_u)_x$. $\diamondsuit$

3.2 Basis of Neighborhoods

As in the case of open sets, it is desirable to work with a smaller number of neighborhoods, so the concept of a basis of neighborhoods arises naturally.

Definition 3.2.1 A collection β_x of neighborhoods of x is said to be a *basis of neighborhoods* at x if every neighborhood of x contains an element of β_x.

Example 3.3

1. The collection of all neighborhoods is a basis of neighborhoods. The collection $\tau_x = \tau \cap \mathscr{U}_x$ is also a basis of neighborhoods. Likewise, if β is a basis for τ, then

$$\beta_x = \beta \cap \mathscr{U}_x = \{B \in \beta : x \in B\} \tag{3.1}$$

   is also a basis of neighborhoods of x.
2. If a point x has a neighborhood U which is included in every neighborhood, then $\{U\}$ is a basis of neighborhoods. For instance, in the discrete topology, a basis of neighborhoods is $\beta_x = \{\{x\}\}$. Another example is a space where arbitrary intersection of open sets is open, because $V = \bigcap\{O \in \tau : x \in O\}$ is the smallest neighborhood of x.
3. In Euclidean space $\mathbb{R}^n$, the set of all balls containing $x \in \mathbb{R}^n$ is a basis of neighborhoods thanks to (3.1). However, there is a smaller basis formed by all balls centered at x, $\beta_x = \{B_r(x) : r > 0\}$. This is not obvious and requires a proof. Let U be a neighborhood of x. From the characterization of neighborhoods in terms of elements of a basis for the topology, there is $y \in X$ and $r > 0$ such that $x \in B_r(y) \subset U$. Let $\varepsilon = r - |x - y| > 0$. Then $B_\varepsilon(x) \subset B_r(y)$ because if $z \in B_\varepsilon(x)$, by the triangle inequality we have

$$|z - y| \leq |x - z| + |x - y| < \varepsilon + |x - y| = r.$$

4. Other bases of neighborhoods on $\mathbb{R}^n$ are the collections of balls $\{B_r(x) : 0 < r < 1\}$, $\{B_{1/m}(x) : m \in \mathbb{N}\}$ and $\{B_r(x) : r \in \mathbb{Q}^+\}$.

$\Diamond$

Example 3.4 The standard basis β_u for the Euclidean topology on $\mathbb{R}^n$ is formed by Cartesian products of bounded open intervals. By (3.1), the elements of β_u that contain a given point $x \in \mathbb{R}^n$ form a basis of neighborhoods. A basis with fewer elements is

$$\{(x_1 - r_1, x_1 + r_1) \times \ldots \times (x_n - r_n, x_n + r_n) : r_1, \ldots, r_n > 0\},$$

where $x = (x_1, \ldots, x_n)$. However, an even smaller basis is

$$\beta_x = \{(x_1 - r, x_1 + r) \times \ldots \times (x_n - r, x_n + r) : r > 0\}.$$

If $n = 2$, the basic elements are squares of side $2r$, and cubes of side $2r$ if $n = 3$. Even more, we can consider only those positive numbers r that belong to $\mathbb{Q}^+$, or of the form $1/m$, with $m \in \mathbb{N}$,

$$\beta_x = \{(x_1 - \tfrac{1}{m}, x_1 + \tfrac{1}{m}) \times \ldots \times (x_n - \tfrac{1}{m}, x_n + \tfrac{1}{m}) : m \in \mathbb{N}\}.$$

$\Diamond$

The next result characterizes bases of neighborhoods in terms of a basis for the topology. Also open sets are characterized in terms of bases of neighborhoods.

Proposition 3.2.2

1. *Let X be a space, $x \in X$, β_x a collection of neighborhoods at x and β a basis for τ. Then β_x is a basis of neighborhoods of x if and only if for every $B \in \beta$ and $x \in B$, there is $V \in \beta_x$ such that $V \subset B$.*
2. *If β is a basis for the topology, then $\beta_x = \{B \in \beta : x \in B\}$ is a basis of neighborhoods of x.*
3. *If β_x is a basis of neighborhoods for each x and all elements of β_x are open sets, then $\beta = \bigcup_{x \in X} \beta_x$ is a basis for the topology.*
4. *If $A \subset X$, then A is open if and only if for every $x \in A$, there is $V \in \beta_x$ such that $V \subset A$.*

We give a description of all neighborhoods in the relative topology and a method of constructing bases of neighborhoods (compare with Theorem 2.5.2).

Proposition 3.2.3 *Let A be a subspace of a space X and $a \in A$.*

1. *The collection of neighborhoods of a in $(A, \tau_{|A})$ is $\mathscr{U}_a^A = \{U \cap A : U \in \mathscr{U}_a\}$.*
2. *If β_a is a basis of of neighborhoods of $a \in A$ in X, then $\beta_a^A = \{B \cap A : B \in \beta\}$ is a basis of neighborhoods of a in $(A, \tau_{|A})$.*

Example 3.5 In the topology of the divisors on $\mathbb{N} \setminus \{1\}$ (Exercise 2.15), we see that $\beta_n = \{U_n\}$ is a basis of neighborhoods of n, where $U_n = \{\text{divisors of } n\}$. It was proved that $\beta = \{U_m : m \in \mathbb{N}\}$ is a basis for the topology, so taking those ones containing n, we have a basis of neighborhoods of n. We see that U_n is the smallest one. If $n \in U_m$, then n is a divisor of m, so all divisors of n also are divisors of m, proving $U_n \subset U_m$. ♢

Example 3.6 In the included point topology where p is the particular point, $\beta_{in} = \{\{x, p\} : x \in X\}$ is a basis for the topology (Example 2.9). Thus $\beta_x = \{\{x, p\}\}$ is a basis of neighborhoods of x. Similarly, in the excluded point topology a basis of neighborhoods is

$$\beta_x = \begin{cases} \{\{x\}\}, & \text{if } x \neq p \\ \{X\}, & x = p. \end{cases}$$

♢

Example 3.7 In the Sorgenfrey topology, a basis of neighborhoods of $x \in \mathbb{R}$ is formed by the elements of β_S containing x, that is $\{[a, b) : a \leq x < b\}$. However, smaller bases are $\{[x, x + r) : r > 0\}$ and $\{[x, x + \frac{1}{n}) : n \in \mathbb{N}\}$. ♢

Example 3.8 A basis of neighborhoods on the right order topology τ_r (Exercise 2.5) is $\beta_x = \{[x, \rightarrow [\}$. Indeed, since $[x, \rightarrow [$ is open, it is a neighborhood of x. Now let $U \in \mathscr{U}_x$. Then there is $a \in \mathbb{R}$ such that $x \in [a, \rightarrow [\subset U$; in particular, $a \leq x$. From this inequality, it follows that $[x, \rightarrow [\subset [a, \rightarrow [$ and thus $[x, \rightarrow [\subset U$. ♢

As in the case of neighborhoods, we pose the following question:

Question: *Suppose that for each point x of a given set X, we have assigned a collection of subsets γ_x. Under what conditions γ_x is a basis of neighborhoods for some topology on X?*

Theorem 3.2.4 *Suppose that X is a set such that for each $x \in X$ there is a nonempty collection γ_x of subsets of X with the following properties:*

(i) $x \in V$ for all $V \in \gamma_x$;
(ii) if $V_1, V_2 \in \gamma_x$, then there is $V_3 \in \gamma_x$ such that $V_3 \subset V_1 \cap V_2$;
(iii) if $V \in \gamma_x$ then there is $V_0 \in \gamma_x$ such that for each $y \in V_0$, there is $V_y \in \gamma_y$ with $V_y \subset V$.

Then there is a unique topology on X such that γ_x are bases of neighborhoods.

Let us observe that (ii) is fulfilled if the intersection of any two elements of γ_x belongs to γ_x. In many situations, the element V_0 of (iii) will simply be V.

Example 3.9 For each $(x, y) \in \mathbb{R}^2$, let

$$\beta_{(x,y)} = \{(x - r, x + r) \times \{y\} : r > 0\}.$$

We see that all $\beta_{(x,y)}$ defines a basis of neighborhoods for a topology on $\mathbb{R}^2$. Property (i) is immediate. For (ii),

$$((x-r_1, x+r_1) \times \{y\}) \cap ((x-r_2, x+r_2) \times \{y\}) = (x-r, x+r) \times \{y\},$$

where $r = \min\{r_1, r_2\}$. We prove (iii). Let $V = (x-r, x+r) \times \{y\}$ and choose $V_0 = V$. For each $(z, y) \in V$, let $\delta = r - |x-z| > 0$. Then $(z-\delta, z+\delta) \times \{y\} \subset V$ for if $t \in (z-\delta, z+\delta)$,

$$|t-x| \leq |t-z| + |z-x| < \delta + |z-x| = r.$$

$\lozenge$

Having arrived at this point, it is worth to collect all the ways that we have adopted to topologize a given point set: with open sets, closed sets, bases for a topology, subbases, neighborhoods and bases of neighborhoods. Figure 3.2 simplifies all scenarios.

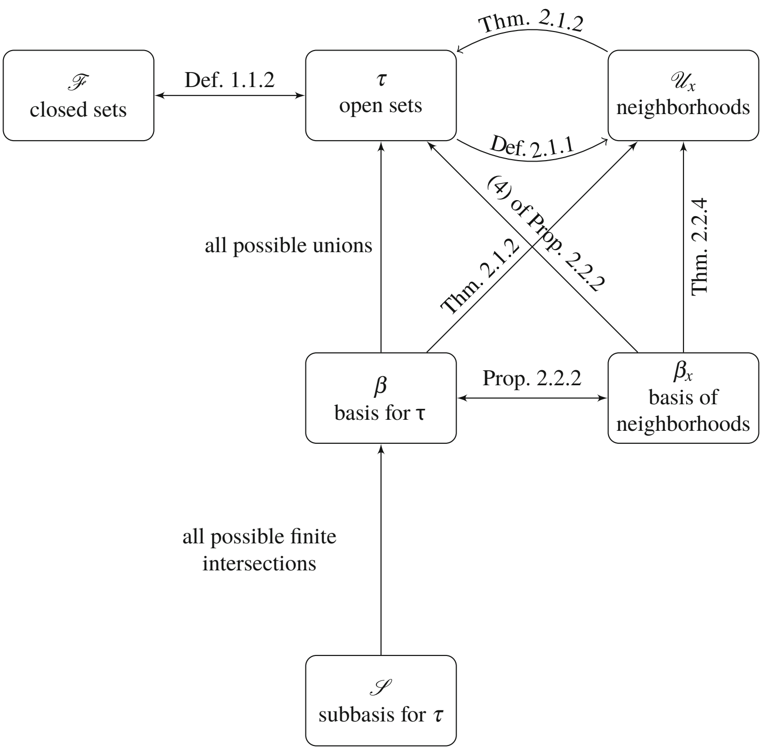


Fig. 3.2 Different equivalent ways to topologize a set

3.3 Interior and Closure

If in the foregoing sections we expressed the notion of vicinity, we proceed to formalize the concept of proximity.

Definition 3.3.1 Let A be a subset of a space X and $x \in X$ (Fig. 3.3).

1. The point x is an *interior point* of A if there is $U \in \mathscr{U}_x$ such that $U \subset A$. The set of all interior points is denoted by $\overset{\circ}{A}$ (or $\operatorname{int}(A)$) and it is called the *interior of* A.
2. The point x is an *exterior point* of A if there is $U \in \mathscr{U}_x$ such that $U \subset X \setminus A$. The set $\operatorname{ext}(A)$ of all exterior points is called the *exterior of* A.
3. The point x is a *boundary point* of A if $U \cap A \neq \emptyset$ and $U \cap (X \setminus A) \neq \emptyset$ for every neighborhood U of x. The *boundary of* A is the set $\operatorname{Bd}(A)$ of all boundary points.

Proposition 3.3.2 *Let A be a subset of a space X.*

1. $X = \overset{\circ}{A} \cup \operatorname{ext}(A) \cup \operatorname{Bd}(A)$ *is a disjoint union.*
2. $\overset{\circ}{A} \subset A$ *and* $\operatorname{ext}(A) \subset X \setminus A$.
3. $\overset{\circ}{A} = A \cap (X \setminus \operatorname{Bd}(A))$ *and* $\operatorname{ext}(A) = (X \setminus A) \cap (X \setminus \operatorname{Bd}(A)$.
4. $\overset{\circ}{A} = \operatorname{ext}(X \setminus A)$ *and* $\operatorname{ext}(A) = \operatorname{int}(X \setminus A)$.
5. $\operatorname{Bd}(A) = \operatorname{Bd}(X \setminus A)$.
6. $\overset{\circ}{A} = \bigcup\{O \in \tau : O \subset A\}$. *As a consequence,* $\overset{\circ}{A}$ *is the largest open set included in A.*

As a consequence, if we know the interior of *all* the subsets in a space, we can find the exterior and the boundary of any subset. This is because the exterior is the interior of the complement by (4), hence we also know the boundary. The same occurs if we know the exterior of *all* the subsets or the boundary of *all* the subsets.

Proposition 3.3.3 *Let β be a basis for τ and β_x a basis of neighborhoods for each $x \in X$. If $A \subset X$, then $x \in \overset{\circ}{A}$ if and only if it holds any of the following properties:*

(i) *There is $O \in \tau$ such that $x \in O \subset A$.*
(ii) *There is $B \in \beta$ such that $x \in B \subset A$.*
(iii) *There is $V \in \beta_x$ such that $V \subset A$.*

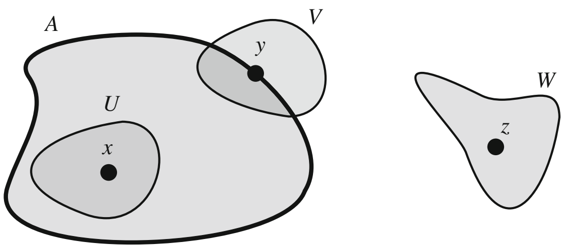


Fig. 3.3 Given the set A, the point x is interior, y is a boundary point and z is an exterior point

Table 3.1 The interior, exterior and boundary of all intervals of $\mathbb{R}$

	Interior	Exterior	Boundary
(a, b)	(a, b)	$(-\infty, a) \cup (b, \infty)$	$\{a, b\}$
$[a, b)$	(a, b)	$(-\infty, a) \cup (b, \infty)$	$\{a, b\}$
$(a, b]$	(a, b)	$(-\infty, a) \cup (b, \infty)$	$\{a, b\}$
$[a, a] = \{a\}$	$\emptyset$	$(-\infty, a) \cup (a, \infty)$	$\{a\}$
$[a, \infty)$	(a, ∞)	$(-\infty, a)$	$\{a\}$
$(-\infty, a]$	$(-\infty, a)$	(a, ∞)	$\{a\}$
(a, ∞)	(a, ∞)	$(-\infty, a)$	$\{a\}$
$(-\infty, a)$	$(-\infty, a)$	(a, ∞)	$\{a\}$
$\mathbb{R}$	$\mathbb{R}$	$\emptyset$	$\emptyset$

Example 3.10 Find the interior, exterior and boundary of a closed interval $A = [a, b]$ with $a < b$. As (a, b) is an open set inside of A, then (i) of Proposition 3.3.3 yields $(a, b) \subset \overset{\circ}{A}$. Because $\overset{\circ}{A} \subset A$, it only remains to decide if a and b are interior points. If $a \in \overset{\circ}{A}$ and taking the standard basis β_u, there are $c, d \in \mathbb{R}$ such that $a \in (c, d) \subset [a, b]$. From $a \in (c, d)$, we deduce $c < a$ and from $(c, d) \subset [a, b]$, that $a < c$. This is not possible. Likewise, $b \notin \overset{\circ}{A}$. We can also utilize the characterization by bases of neighborhoods. By Example 3.3, a basis of neighborhoods of b is $\{(b - r, b+r) : r > 0\}$. If b is an interior point of A, there is $r > 0$ such that $(b-r, b+r) \subset [a, b]$; in particular, $b + r \leq b$, which is absurd. Definitively $\overset{\circ}{A} = (a, b)$.

We calculate the exterior of A or equivalently, $\mathrm{ext}(A) = \mathrm{int}((-\infty, a) \cup (b, \infty))$. Since $(-\infty, a) \cup (b, \infty)$ is an open set, part (i) of Proposition 3.3.3 gives $(-\infty, a) \cup (b, \infty) \subset \mathrm{ext}(A)$. As the other inclusion is always valid, we conclude $\mathrm{ext}(A) = (-\infty, a) \cup (b, \infty)$, hence $\mathrm{Bd}(A) = \{a, b\}$. ♢

Similar arguments can be employed for the rest of the types of intervals. In Table 3.1, we show the interior, exterior and boundary of all them.

Proposition 3.3.4 *Let X be a space and $A, B \subset X$.*

1. $\overset{\circ}{X} = X$, $\overset{\circ}{\emptyset} = \emptyset$.
2. *If $A \subset B$, then $\overset{\circ}{A} \subset \overset{\circ}{B}$.*
3. $\mathrm{int}(A \cap B) = \overset{\circ}{A} \cap \overset{\circ}{B}$.
4. $\overset{\circ}{A} \cup \overset{\circ}{B} \subset \mathrm{int}(A \cup B)$.
5. $\mathrm{int}(\mathrm{int}(A)) = \mathrm{int}(A)$.

Example 3.11 The inclusion (4) is, in general, proper. For an example, in $\mathbb{R}$ let $A = \mathbb{Q}$ and $B = \mathbb{R} - \mathbb{Q}$. Since no open bounded interval is included in either $\mathbb{Q}$ or $\mathbb{R} - \mathbb{Q}$, we have $\overset{\circ}{A} = \overset{\circ}{B} = \emptyset$. However, $\mathrm{int}(A \cup B) = \overset{\circ}{\mathbb{R}} = \mathbb{R}$. Another example is $A = (0, 1)$ and $B = [1, 2)$. Then $\overset{\circ}{A} \cup \overset{\circ}{B} = (0, 1) \cup (1, 2)$ but $\mathrm{int}(A \cup B) = \mathrm{int}((0, 2)) = (0, 2)$. ♢

Theorem 3.3.5 *A subset is open if and only if it coincides with its interior.*

This theorem indicates that it is possible to introduce the axioms of topological spaces in terms of the concept of interior. If the open sets have been characterized in terms of interior, it seems plausible to characterize closed sets in terms of vicinity.

Definition 3.3.6 A point x is an *adherent* of A if $U \cap A \neq \emptyset$ for every neighborhood U of x. The set of all adherent points of A, denoted $\overline{A}$, is called the *closure* of A.

As in Proposition 3.3.3, there are similar characterizations and properties of an adherent point.

Proposition 3.3.7 *Let X be a space and $A,\ B \subset X$.*

1. $A \subset \overline{A}$, $\mathrm{Bd}(A) \subset \overline{A}$, $\mathrm{ext}(A) \cap \overline{A} = \emptyset$ *and* $\overline{A} = A \cup \mathrm{Bd}(A)$.
2. $\overline{A} = \overset{\circ}{A} \cup \mathrm{Bd}(A) = A \cup \mathrm{Bd}(A)$.
3. $\mathrm{Bd}(A) = \overline{A} \cap \overline{X \setminus A}$.
4. $X \setminus \overset{\circ}{A} = \overline{X \setminus A}$ *and* $X \setminus \overline{A} = \mathrm{int}(X \setminus A)$.
5. $\overline{\emptyset} = \emptyset$, $\overline{X} = X$.
6. *If* $A \subset B$, *then* $\overline{A} \subset \overline{B}$.
7. $\overline{A \cup B} = \overline{A} \cup \overline{B}$ *and* $\overline{A \cap B} \subset \overline{A} \cap \overline{B}$.
8. $\overline{\overline{A}} = \overline{A}$.
9. *If* $B \subset A$, *then* $\overline{B}^A = \overline{B} \cap A$, *where* $\overline{B}^A$ *stands for the closure of* B *in* $(A, \tau_{|A})$.

The statements analogous to (9) for the interior and boundary are not true.

1. We have the inclusion $\overset{\circ}{B} \cap A \subset \mathrm{int}(B)^A$. So, if $x \in \overset{\circ}{B} \cap A$ then there is $U \in \mathscr{U}_x$ such that $U \subset B$, hence $U \cap A \subset B \cap A = B$ and $U \cap A \in \mathscr{U}_x^A$. It may occur that the inclusion is proper. For example, let $A = (0, 1) \cup \{2\}$ and $B = \{2\}$. Then $\overset{\circ}{B} = \emptyset$ so $\overset{\circ}{B} \cap A = \emptyset$. However, $\mathrm{int}(B)^A = B = \{2\}$ because $B = (1, 3) \cap A \in \tau_{|A}$.
2. In general, $\mathrm{Bd}(B)^A \subset \mathrm{Bd}(B) \cap A$ because

$$\mathrm{Bd}(B)^A = \overline{B}^A \cap \overline{A \setminus B}^A = (\overline{B} \cap A) \cap (\overline{A \setminus B} \cap A) \subset \overline{B} \cap \overline{X \setminus B} \cap A = \mathrm{Bd}(B) \cap A.$$

This inclusion may be proper, as is shown by $A = \{0, 1\}$ and $B = \{0\}$, where $\mathrm{Bd}(B)^A = \emptyset$ and $\mathrm{Bd}(B) \cap A = \{0\}$.

From (4) of Proposition 3.3.7, if we know the closures of all subsets then we also know the interiors of all them, because $\overset{\circ}{A} = X \setminus (\overline{X \setminus A})$. In fact, the concepts of interior and closure are closely related, establishing a certain duality between them.

Proposition 3.3.8 *A subset $A \subset X$ is closed if and only if $\overline{A} = A$. Furthermore, $\overline{A} = \bigcap\{F \in \mathscr{F} : A \subset F\}$; in particular, $\overline{A}$ is the smallest closed subset of X that contains A.*

Example 3.12

1. In the trivial space (X, τ_T), let $A \subset X$ be a nontrivial subset. Then $\overset{\circ}{A} = \emptyset$ and $\overline{A} = X$.

2. Let $A \subset X$ in the discrete space (X, τ_D). Since A is open and closed, we have $\overset{\circ}{A} = A = \overline{A}$.
3. Consider the right order topology τ_r on $\mathbb{R}$. We see that $\overline{\{x\}} = (-\infty, x]$. First, notice that $(-\infty, x]$ is closed because its complement (x, ∞) is open. Thus $\overline{\{x\}} \subset (-\infty, x]$. If $y > x$ then $[y, \infty)$ is a neighborhood of y with $[y, \infty) \cap \{x\} = \emptyset$, so y is not an adherent point. This proves the desired identity.
4. We calculate the interior and closure of the set of rational numbers $\mathbb{Q}$ in $\mathbb{R}$. Since any open interval contains rationals and irrationals, there are no open intervals included in $\mathbb{Q}$. Then $\emptyset$ is the only open set included in $\mathbb{Q}$, so $\overset{\circ}{\mathbb{Q}} = \emptyset$. On the other hand, $\overline{\mathbb{Q}} = \mathbb{R}$. Certainly, if $x \in \mathbb{R}$ and considering the basis of neighborhoods $\{(x - r, x + r) : r > 0\}$, any such interval contains rationals.
5. The interior of the Cantor set $\mathfrak{C}$ is $\emptyset$ because $\mathfrak{C}$ cannot contain open intervals. On the other hand, it was proved in Example 2.19 that $\mathfrak{C}$ is closed and thus $\overline{\mathfrak{C}} = \mathfrak{C}$.

$\diamondsuit$

We complete this section with two new definitions concerning the relative position of a point with respect to a subset of the space.

Definition 3.3.9 Let A be a subset of a space X and $x \in X$.

1. The point x is a *point of accumulation* of A if $(U \setminus \{x\}) \cap A \neq \emptyset$ for all $U \in \mathscr{U}_x$. The set of all points of accumulation of A, denoted A', is called the *derived set* of A.
2. The point x is an *isolated point* of A if there is a neighborhood U of x such that $U \cap A = \{x\}$. The set of all isolated points is denoted by $\mathrm{Ais}(A)$.

Example 3.13

1. In the discrete topology, if $A \subset X$, every element of A is isolated because $U = \{a\}$ is a neighborhood of $a \in A$ and $U \cap A = \{a\}$.
2. In $\mathbb{R}$ we find the points of accumulation and isolated points of $A = (0, 1) \cup \{3\}$. First, $x = 3$ is an isolated point because $U = (2, 4)$ is a neighborhood of 3 and $U \cap A = \{3\}$. We see that the rest of the points of A are points of accumulation. If $x \in (0, 1)$ and $(x - r, x + r)$ is any basic neighborhood, then $(x - r, x + r) \cap A \supset (x - r, x + r) \cap (0, 1)$ contains infinite points, hence x is not isolated. Thus $\mathrm{Ais}(A) = \{3\}$ and $A' = \overline{A} \setminus \mathrm{Ais}(A) = [0, 1]$.
3. Consider the right order topology on $\mathbb{R}$ and $A = [0, 1]$. We see that $x = 1$ is the only isolated point of A. Since $\beta_x = \{[x, \infty)\}$ is a basis of neighborhoods, x is isolated if and only if $[x, \infty) \cap [0, 1] = \{x\}$. This occurs only when $x = 1$. For the accumulation points and because $\overline{A} = (-\infty, 1]$, we deduce that $A' = (-\infty, 1)$.
4. The Cantor set has no isolated points. In effect, if $x \in \mathfrak{C}$, given $\varepsilon > 0$, it was proved in Example 2.19 the existence of a set C_{ij} such that $x \in C_{ij}$ and the length of C_{ij} is less than ε. In particular, the endpoints of C_{ij}, which are points of $\mathfrak{C}$, are included in the neighborhood $(x - \varepsilon, x + \varepsilon) \cap \mathfrak{C}$. Consequently, the topology of the Cantor set is not discrete.

$\diamondsuit$

3.4 Convergence of Sequences

One of the notions of proximity in calculus is the convergence of sequences of reals. In calculus, $(x_n) \to x$ if

$$\text{for all } \varepsilon > 0, \text{ there is } \nu \in \mathbb{N} \text{ such that } |x_n - x| < \varepsilon \text{ for all } n \geq \nu.$$

This definition also holds in $\mathbb{R}^m$, where $|x_n - x|$ indicates the Euclidean distance between x_n and x. The inequality $|x_n - x| < \varepsilon$ writes $x_n \in B_\varepsilon(x)$ and taking into account that the set of all the balls centered at x is a basis of neighborhoods, it is now clear how one should define convergence in a topological space.

Definition 3.4.1 A sequence (x_n) is said to be *convergent* to x, written $(x_n) \to x$, if for every neighborhood U of x, there is $\nu \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq \nu$. We say that x is a *limit* of (x_n).

Let us observe that we call x "a" limit of the sequence rather than "the" limit because, in principle, nothing more could be said regarding the uniqueness of the limit.

Sequences that are constant from a term are convergent. These sequences are called *eventually constant*.

It is interesting to see how the convergent sequences change in the same underlying set X but endowed with different topologies.

Example 3.14

1. In the trivial space (X, τ_T), every sequence is convergent and converges to any element of X. This is immediate because $\mathscr{U}_x = \{X\}$.
2. In the cofinite topology $(\mathbb{R}, \tau_{CF})$, the sequence (n) converges to any real number $x \in \mathbb{R}$ because if $U = \mathbb{R} \setminus \{x_1, \ldots, x_m\}$ is a neighborhood of x, we have $n \in U$ for all $n > \nu = \max\{|x_1|, \ldots, |x_m|\}$.
3. Let $\mathbb{R}$ be endowed with the strictly right order topology $\tau_{sr} = \{\emptyset, \mathbb{R}\} \cup \{(x, \infty) : x \in \mathbb{R}\}$. The sequence $(1/n)$ converges to every x with $x \leq 0$. Certainly, if $U \in \mathscr{U}_x$, let $a \in \mathbb{R}$ such that $x \in (a, \infty) \subset U$. In consequence, $a < x$. Because $x \leq 0$ then $a < 0$, so the sequence $(1/n)$ is entirely contained in (a, ∞), hence in U.

   Also, $(n) \to x$ for every x because given $(a, \infty) \in \tau_{sr}$ with $x \in (a, \infty)$, there is $\nu \in \mathbb{N}$ such that $\nu > x$. Thus for $n \geq \nu$, $n > x > a$, so $n \in (a, \infty)$ (Fig. 3.4). ♢

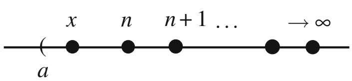


Fig. 3.4 The sequence (n) converges to any real number x in the strictly right order topology

Example 3.15 Consider the included point topology in a space X for a particular point p. Let $(x_n) \to x$. Since $\{p, x\}$ is the smallest neighborhood of x, then there is $\nu \in \mathbb{N}$ such that $x_n \in \{x, p\}$ for all $n \geq \nu$. Thus we deduce that convergent sequences are those ones such that after some term of the sequence, x_n is either x or p. ♢

Example 3.16 In the excluded point topology, let $(x_n) \to x$. If $x \neq p$, then $\{x\}$ is the smallest neighborhood of x. Since from some term of the sequence, all elements must belong to $\{x\}$, then (x_n) is eventually constant. If $x = p$, the only neighborhood of p is X itself, so there are no restrictions. This proves that all convergent sequences in X either have limit p or are eventually constant. ♢

We provide (necessary or sufficient depending on the case) conditions to be an interior, adherent, exterior or boundary point in terms of convergence of sequences.

Proposition 3.4.2 *Let A be a subset in a space X and $x \in X$.*

1. *If $x \in \overset{\circ}{A}$, then for every $(x_n) \to x$, there is $\nu \in \mathbb{N}$ such that $x_n \in A$ whenever $n \geq \nu$.*
2. *If there exists a sequence $(a_n) \subset A$ such that $(a_n) \to x$, then $x \in \overline{A}$.*
3. *If $x \in \mathrm{ext}(A)$, then for every sequence $(x_n) \to x$, there is $\nu \in \mathbb{N}$ such that $x_n \in X \setminus A$ whenever $n \geq \nu$.*
4. *If there is a sequence $(a_n) \subset A$ such that $(a_n) \to x$ and $\{y_n\} \subset X \setminus A$ with $(y_n) \to x$, then $x \in \mathrm{Bd}(A)$.*

Conditions (1) and (3) are necessary while (2) and (4) are sufficient. There are also spaces where the equivalences are not valid.

We have seen topological spaces where convergent sequences have many limits, which it is against our intuition. We introduce a class of spaces where the limit is unique. By its importance, these spaces are worth highlighting.

Definition 3.4.3 A space X is said to be *Hausdorff* (or T_2) if for each $x, y \in X$, there are $U \in \mathscr{U}_x$ and $V \in \mathscr{U}_y$ such that $U \cap V = \emptyset$. As a consequence, in a Hausdorff topological space, the limit of a convergent sequence is unique.

Example 3.17

1. If any two open sets are not disjoint, then the space is not Hausdorff. For example, the trivial topology, the cofinite topology, the included point topology and the right order topology are not Hausdorff.
2. The discrete topology is Hausdorff because singletons are open sets.
3. The Euclidean space $\mathbb{R}^n$ is Hausdorff. Indeed, let $x, y \in \mathbb{R}^n$. If $r = |x - y|/2$ then $B_r(x) \cap B_r(y) = \emptyset$, for if z is an element of the intersection, the triangle inequality yields

$$|x - y| \leq |x - z| + |z - y| < \frac{r}{2} + \frac{r}{2} = r = |x - y|,$$

which is a contradiction.

4. The Hausdorff property is hereditary in the sense that any topological subspace of a Hausdorff space is Hausdorff. In effect, if $A \subset X$ and $a, b \in A$, then the intersection of disjoint neighborhoods of each point with A provide disjoint neighborhoods in the relative topology.
5. In a Hausdorff space, singletons are closed sets. Indeed, if $y \neq x$, there is $V \in \mathcal{U}_y$ such that $x \notin V$, which means that $\{x\} \cap V = \emptyset$. Thus $y \notin \overline{\{x\}}$. $\Diamond$

3.5 Worked Exercises

Exercise 3.1 Find the interior, exterior, boundary and closure of the interval $A = (0, 1)$ in $(\mathbb{R}, \tau_{CF})$.

Solution If $x \in \overset{\circ}{A}$, there is a finite set $F \subset \mathbb{R}$ such that $x \in \mathbb{R} \setminus F \subset (0, 1)$. Thus $\mathbb{R} \setminus (0, 1)$ is included in the finite set F, which is not possible. This proves that $\text{int}((0, 1)) = \emptyset$. Analogously, $\text{ext}((0, 1)) = \emptyset$, hence $\text{Bd}((0, 1)) = \mathbb{R}$. Finally, $\overline{A} = A \cup \text{Bd}(A) = \mathbb{R}$.

Instead of solving point-by-point, another way of calculating the closure is to find the smallest closed set containing A. Here, we begin with the closure because the family of closed sets is more manageable. Since there are no finite sets containing $(0, 1)$, the only closed set containing $(0, 1)$ is $\mathbb{R}$. This proves that $\overline{A} = \mathbb{R}$. Then $\text{ext}(A) = \mathbb{R} \setminus \overline{A} = \emptyset$. The next set to compute is the boundary of A because it is intersection of two closures, $\text{Bd}(A) = \overline{A} \cap \overline{\mathbb{R} \setminus A}$. Since the complement of A is not finite, we deduce $\overline{\mathbb{R} \setminus A} = \mathbb{R}$ again. Thus $\text{Bd}(A) = \mathbb{R}$ and finally, $\overset{\circ}{A} = \mathbb{R} \setminus \overline{\mathbb{R} \setminus A} = \emptyset$. $\square$

Exercise 3.2 Find the interior, exterior, boundary and closure of the interval $A = (-\infty, 0]$ in $(\mathbb{R}, \tau_r)$.

Solution A basis of neighborhoods is formed by only one element, $\beta_x = \{[x, \infty)\}$. Thus all we have to do is check whether or not $[x, \infty)$ satisfies the property in consideration. We show three different manners of solving the exercise.

1. Since no interval $[x, \infty)$ is included in A, we have $\overset{\circ}{A} = \emptyset$. For the exterior of A, we know that $\text{ext}(A) = \text{int}((0, \infty))$. Then for each $x \in (0, \infty)$, the neighborhood $[x, \infty)$ is included in $(0, \infty)$. Thus $\text{ext}(A) = (0, \infty)$, hence $\text{Bd}(A) = (-\infty, 0]$ and $\overline{A} = A \cup \text{Bd}(A) = A$.
2. Here we use the characterization of the interior as the largest open set contained in the set. This is because we explicitly have the family of open sets,

$$\tau_r = \{(a, \infty) : a \in \mathbb{R}\} \cup \{[a, \infty) : a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}.$$

For the interior of $(-\infty, 0]$, the empty set $\emptyset$ is the only open set included in $(-\infty, 0]$, hence $\overset{\circ}{A} = \emptyset$. For the exterior of A, which is the interior of the complement, we see that $\mathbb{R} \setminus A = (0, \infty)$ is a member of the topology. Thus $\operatorname{ext}(A) = (0, \infty)$, so the boundary and the closure of A are as before.

3. By using the characterization of the closure as the smallest closed set containing the set. Again, the key point is that the family of closed sets is known explicitly,

$$\mathscr{F}_r = \{(-\infty, a] : a \in \mathbb{R}\} \cup \{(-\infty, a) : a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}.$$

Since $A \in \mathscr{F}_r$, then $\overline{A} = A$. For the boundary, we use $\operatorname{Bd}(A) = \overline{A} \cap \overline{\mathbb{R} \setminus A}$. Now $\mathbb{R} \setminus A = (0, \infty)$ and $\mathbb{R}$ is the only closed set containing A. Then $\overline{\mathbb{R} \setminus A} = \mathbb{R}$, so $\operatorname{Bd}(A) = A$. Finally, $\overset{\circ}{A} = \overline{A} \setminus \operatorname{Bd}(A) = \emptyset$ and $\operatorname{ext}(A) = \mathbb{R} \setminus \overline{A} = (0, \infty)$. □

Exercise 3.3 Consider the space $(\mathbb{R}, \tau_r)$.

1. Prove that 0 is not a limit of the sequence $(-\frac{1}{n})$.
2. Prove that $(1/n) \to -2$.
3. Let (x_n) be a sequence bounded from below by α. Prove that $(x_n) \to x$ for all $x \le \alpha$.
4. Show that $(n) \to x$ for all $x \in \mathbb{R}$.
5. As a generalization of (4), prove that every sequence that is not bounded from above converges to any real number.

Solution A basis of neighborhoods of x is $\beta_x = \{V_x = [x, \infty)\}$.

1. It suffices to see that there are no points of the sequence $(-\frac{1}{n})$ in $V_0 = [0, \infty)$.
2. For the neighborhood $V_{-2} = [-2, \infty)$, all elements of the sequence $(\frac{1}{n})$ are included in V_{-2}.
3. We know that $\alpha \le x_n$ for all $n \in \mathbb{N}$. Let $x \le \alpha$ and $V_x = [x, \infty)$. Then the sequence is included in V_x, so $(x_n) \to x$.
4. It is clear that the sequence (n), excepting fo finitely terms, is contained in V_x for all $x \in \mathbb{R}$. If we want to be precise, let $\nu = E(x) + 1$, $E(x)$ being the integer part of x. Then $n \in V_x$ for $n \ge \nu$.
5. Let (x_n) be a sequence that is not bounded from above. Given $x \in \mathbb{R}$, and because (x_n) is not bounded from above, there is $\nu \in \mathbb{N}$ such that $x_n > x$ whenever $n \ge \nu$; in particular, $x_n \in V_x$. □

The next exercise shows that the implications in Proposition 3.4.2 cannot be reversed in general.

Exercise 3.4 On $\mathbb{R}$, define the topology

$$\tau_{CC} = \{\mathbb{R} \setminus N : N \subset \mathbb{R} \text{ is a countable set}\} \cup \{\emptyset\}.$$

This topology is called the *co-countable topology.* Let $A = \mathbb{R} \setminus \{0\}$.

1. Prove that $\overline{A} = \mathbb{R}$.
2. Prove that there is not a sequence in A converging to 0.
3. Prove that the convergent sequences in $(\mathbb{R}, \tau_{CC})$ are eventually constant sequences.

Solution The proof that τ_{CC} is a topology is entirely similar to that of the cofinite topology.

1. Since the complement of A is only one point, the work reduces to testing if 0 is an adherent point of A. Let $O \in \tau$ with $0 \in O$, where $O = \mathbb{R} \setminus N$ and N is a countable set. Then

$$A \cap (\mathbb{R} \setminus N) = \mathbb{R} \setminus (\{0\} \cup N)$$

is not empty because $\{0\} \cup N$ is countable. This proves that $0 \in \overline{A}$. Another method is finding the smallest closed set containing A. The family of closed sets is

$$\mathscr{F}_{CC} = \{N \subset \mathbb{R} : N \text{ is countable}\} \cup \{\mathbb{R}\}.$$

Because A is not countable, among all the elements of $\mathscr{F}_{CC}$, the set $\mathbb{R}$ is the only closed set containing A, hence that $\mathbb{R}$ is the closure of A.
2. Suppose that there is a sequence $(a_n) \subset A$ such that $(a_n) \to 0$. Let $U = \mathbb{R} \setminus \{a_n : n \in \mathbb{N}\}$, the complement on $\mathbb{R}$ of the sequence (a_n). Then U is an open set containing 0 because $0 \neq a_n$. By the convergence of (a_n), there would be $\nu \in \mathbb{N}$ such that $a_m \in \mathbb{R} \setminus \{a_n : n \in \mathbb{N}\}$ for $m \geq \nu$, but this is not possible.
3. Let $(x_n) \to x$. Suppose that (x_n) is not eventually constant. Then the set $J = \{n \in \mathbb{N} : x_n \neq x\}$ is infinite. Let $U = \mathbb{R} \setminus \{x_j : j \in J\}$. This set is open and contains x, so U is a neighborhood of x. By convergence, there is $\nu \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq \nu$. Then $x_n = x$ for all $n \geq \nu$, which contradicts that J is infinite. □

Exercise 3.5 Consider the sequence $((-1)^n) = \{-1, 1, -1, \ldots\}$.

1. Determine the convergence in the Sorgenfrey topology.
2. Determine the convergence in the right order topology.
3. Let $A = \{-1, 1\}$ and define the topology $\tau = \{O \subset \mathbb{R} : A \subset O\} \cup \{\emptyset\}$ (see Exercise 2.12). Prove that every real number is a limit of the sequence.

Solution

1. The sequence $((-1)^n)$ is not convergent in $(\mathbb{R}, \tau_S)$ because it is not in $(\mathbb{R}, \tau_u)$ and τ_S is finer than τ_u. If one prefers to use directly the Sorgenfrey topology, let $x \in \mathbb{R}$ be a limit of the sequence. Let $y > x$ such that the basic open set $[x, y)$ contains at most one element of the set $\{-1, 1\}$. Indeed, if $x \neq 1, -1$, choose y so the interval contains neither -1 nor 1. If $x = 1$, let $y = 2$. If $x = -1$, let $y = 0$. In any case, there would be $\nu \in \mathbb{N}$ such that $x_n \in [x, y)$ whenever $n \geq \nu$. In particular, -1 and 1 would belong to $[x, y)$, which is not possible.
2. A basis of neighborhoods of x is $\beta_x = \{[x, \infty)\}$. Then for all $x \leq -1$ the sequence lies entirely contained in the basic neighborhood $[x, \infty)$, so x is a limit of the sequence (see also (3) of Exercise 3.3). If $x > -1$ then $[x, \infty)$ does not

contains -1, which appears infinitely in the sequence. Thus x is not a limit. Definitively, the set of all limits is $(-\infty, -1]$.

3. Let $x \in \mathbb{R}$ and $U \in \mathscr{U}_x$. Then there is an open set O such that $x \in O \subset U$; in particular, $\{-1, 1\} \subset O \subset U$. Thus $((-1)^n) \to x$. $\square$

Exercise 3.6 On $\mathbb{R}^2$, consider the topology generated by the basis $\beta = \{\mathbb{R} \times \{y\} : y \in \mathbb{R}\}$. Prove that $((\frac{1}{n}, \frac{1}{n}))$ does not converge to $(0, 0)$. Characterize convergent sequences and show that this space is not Hausdorff.

Solution The basic open sets consist of all horizontal straight lines of $\mathbb{R}^2$. The collection β is a basis for a topology τ because the intersection of any two members of β is empty. It is possible to describe the topology because

$$\bigcup_{i \in I} (\mathbb{R} \times \{y_i\}) = \mathbb{R} \times \bigcup_{i \in I} \{y_i\},$$

hence $\tau = \{\mathbb{R} \times A : A \subset \mathbb{R}\} = \tau_T \times \tau_D$. On the other hand, by (3.1), the only element of β containing (x, y) is $\mathbb{R} \times \{y\}$. Thus, letting $B_y = \mathbb{R} \times \{y\}$, we deduce that $\beta_{(x,y)} = \{B_y\}$ is a basis of neighborhoods of (x, y).

1. Let B_0 be the basic neighborhood of $(0, 0)$. However, no element of $((\frac{1}{n}, \frac{1}{n}))$ belongs to B_0, hence the sequence does not converge to $(0, 0)$.
2. Let $((x_n, y_n)) \to (x, y)$. Taking again B_y, there is $\nu \in \mathbb{N}$ such that $(x_n, y_n) \in B_y = \mathbb{R} \times \{y\}$ whenever $n \geq \nu$. Because $x_n \in \mathbb{R}$ is obvious, the convergence is equivalent to saying that $y_n \in \{y\}$ for all $n \geq \nu$. This means that (y_n) is eventually constant and constant to y. Definitively, we have proved that the convergent sequences in $\mathbb{R}^2$ are those sequences $((x_n, y_n))$ where (y_n) is eventually constant. As a consequence, there are infinite limits because the sequence $((x_n, y))$ converges to any (x, y).
3. The space is not Hausdorff because if $x \neq x'$ and $y \in \mathbb{R}$, then $(x, y) \neq (x', y)$, but the basic neighborhoods for both points coincide because they are B_y. $\square$

In the previous exercise, the basis for the topology is formed by Cartesian products of the basis $\{\mathbb{R}\}$ of the trivial topology on $\mathbb{R}$ by the basis $\{\{y\} : y \in \mathbb{R}\}$ of the discrete topology. In the next exercise, we replace the trivial topology by the Euclidean topology.

Exercise 3.7 Consider on $\mathbb{R}^2$ the topology generated by the basis $\beta = \{(a, b) \times \{y\} : a, b, y \in \mathbb{R}\}$. Characterize the convergent sequences and show that the space is Hausdorff.

Solution The family β is a basis for some topology because the intersection of any two members of β is also in β. Indeed, if $(a, b) \times \{y_1\}, (c, d) \times \{y_2\} \in \beta$ which intersect, then necessarily $y_1 = y_2 = y$ and, furthermore,

$$((a, b) \times \{y\}) \cap ((c, d) \times \{y\}) = (\max\{a, c\}, \min\{b, d\}, y) \in \beta.$$

By (3.1), a basis of neighborhoods of (x, y) is $\{(a, b) \times \{y\} : x \in (a, b), a, b \in \mathbb{R}\}$, but we can reduce the number of elements considering the intervals (a, b) centered at x,

$$\beta_{(x,y)} = \{(x - r, x + r) \times \{y\} : r > 0\}.$$

Assume $((x_n, y_n)) \to (x, y)$. For $r > 0$, there is $\nu \in \mathbb{N}$ such that $(x_n, y_n) \in (x - r, x + r) \times \{y\}$ for all $n \geq \nu$. This is equivalent to saying two things. First, that $x_n \in (x - r, x + r)$, or in other words, $(x_n) \to x$ in the Euclidean topology. Second, that $y_n \in \{y\}$ and in consequence (y_n) is eventually constant and constant to y. We have proved that the convergent sequences $((x_n, y_n))$ are those such that (x_n) is convergent in the Euclidean topology and (y_n) is eventually constant. Thus convergent sequences have a unique limit.

To prove that the space is Hausdorff, let $(x, y) \neq (x', y')$. If $y \neq y'$, it suffices to take the neighborhoods $\mathbb{R} \times \{y\}$ and $\mathbb{R} \times \{y'\}$ for each one of the points. If $y = y'$, then $x \neq x'$. Since the Euclidean real line is Hausdorff, let $r > 0$ such that $(x - r, x + r) \cap (x' - r, x' + r) = \emptyset$, so $(x - r, x + r) \times \{y\}$ and $(x' - r, x' + r) \times \{y\}$ are disjoint neighborhoods for (x, y) and (x', y), respectively. □

We prove a result on convergence of sequences and the relative topology.

Exercise 3.8 Let A be a subset of a topological space X. Let $(a_n) \subset A$ be a sequence and $a \in A$. Prove that $(a_n) \to a$ in X if and only if $(a_n) \to a$ in A. Give a topological space where this result fails for the topology γ_A defined in (2.6).

Solution Assume first that $(a_n) \to a$ in X. Let U' be a neighborhood of a in $\tau_{|A}$. Then $U' = U \cap A$ for some neighborhood U of a in X. By hypothesis, there is $\nu \in \mathbb{N}$ such that $a_n \in U$ whenever $n \geq \nu$. Thus $a_n \in U \cap A = U'$ for all $n \geq \nu$.

Conversely, suppose $(a_n) \to a$ in A. If $U \in \mathscr{U}_a$, then $U \cap A$ is a neighborhood in A. Then there is $\nu \in \mathbb{N}$ such that $a_n \in U \cap A$ for all $n \geq \nu$; in particular, $a_n \in U$.

For the counterexample, let $A = [0, 1]$. First we see that a basis of neighborhoods of $x = 0$ is $\sigma_0 = \{A\}$. In fact, A is the only element of γ_A containing 0. To prove this, if $O \in \tau_u$, $0 \in O$ and $O \subset [0, 1]$, then there are $a, b \in \mathbb{R}$ such that $0 \in (a, b) \subset [0, 1]$ which is absurd. Likewise, $\sigma_1 = \{A\}$ is a basis of neighborhoods of $x = 1$. As a consequence, every sequence of A converges to $x = 0$ and to $x = 1$ in the topology γ_A, which does not happen with the Euclidean topology $\tau_{|A}$ of A. □

In the following two exercises we compute the interior and closure in the relative topology of subsets of the Euclidean line.

Exercise 3.9 Consider $A = [0, 1] \cup (2, 3) \cup \{5\}$ as a topological subspace of the Euclidean real line (Fig. 3.5).

1. Find the interior and closure of $\{5\}$ in A.
2. Find the interior and closure of the interval $(2, 3)$.
3. Find the closure of $[0, 1)$ in A.
4. Determine if $[0, \frac{1}{2}]$ is a neighborhood of $x = 0$ in A.

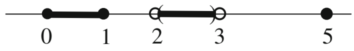


Fig. 3.5 The set of Exercise 3.9

Solution

1. The set $\{5\}$ is open because $\{5\} = (4, 6) \cap A$ and $(4, 6)$ is open in $\mathbb{R}$. It is also closed because $\{5\} = [4, 6] \cap A$ and $[4, 6]$ is closed in $\mathbb{R}$. Thus $\operatorname{int}^A(\{5\}) = \{5\} = \overline{\{5\}}^A$.
2. The set $(2, 3)$ is open in $\mathbb{R}$ and $(2, 3) \subset A$, so it is also open in A. It is also a closed set in A because $(2, 3) = [2, 3] \cap A$ and $[2, 3]$ is closed in $\mathbb{R}$. Then $\operatorname{int}^A((2, 3)) = (2, 3) = \overline{(2, 3)}^A$.
3. For the closure of $[0, 1)$ in A, utilize Proposition 3.3.7. Since $\overline{[0, 1)} = [0, 1]$ in $\mathbb{R}$, the closure in A is $\overline{[0, 1]}^A = [0, 1] \cap A = [0, 1]$.
4. The set $[0, \frac{1}{2}]$ is a neighborhood of $x = 0$, because $[0, \frac{1}{2}) = (-1, \frac{1}{2}) \cap A$ is open in A and $0 \in [0, \frac{1}{2}) \subset [0, \frac{1}{2}]$. □

Exercise 3.10 Let $X = [0, 2) \cup (2, 4]$ be as a topological subspace of the Euclidean real line. Find the interior, closure, exterior and boundary of the following subsets: $[0, 1)$, $(1, 2) \cup (2, 3)$ and $(2, 3]$.

Solution Denote by a super index the operations on X.

1. Let $A = [0, 1)$. Since $A = (-1, 1) \cap X$, the set A is open in X, so its interior is A. For the closure, $\overline{A}^X = \overline{A} \cap X = [0, 1] \cap X = [0, 1]$ and consequently,
$$\operatorname{Bd}^X(A) = \overline{A}^X \setminus \operatorname{int}^X(A) = \{1\}, \quad \operatorname{ext}^X(A) = (1, 2) \cup (2, 4].$$
2. Let $B = (1, 2) \cup (2, 3)$. This set is open in $\mathbb{R}$, so open in X. On the other hand, since $\overline{B} = [1, 3]$, then $\overline{B}^X = [1, 2) \cup (2, 3]$. Thus
$$\operatorname{Bd}^X(B) = \overline{B}^X \setminus \operatorname{int}^X(B) = \{1, 3\}, \quad \operatorname{ext}^X(B) = [0, 1) \cup (3, 4].$$
3. Let $C = (2, 3]$. The set $(2, 3)$ is open in $\mathbb{R}$ and included in X, so it is open in X. Since $\operatorname{int}^X(C)$ is the largest open set contained in X, we have $(2, 3) \subset \operatorname{int}^X(C)$. We prove that 3 is not an interior point. On the contrary, there would be $r > 0$ such that $(3 - r, 3 + r) \cap X \subset (2, 3]$. This, however, is absurd since $3 + r/2 \in (3 - r, 3 + r)$ but it does not belong to $(2, 3]$. Thus $\operatorname{int}^X(C) = (2, 3)$.

   Since $C = [2, 3] \cap X$, the set C is closed in X, so $\overline{C}^X = (2, 3]$. Therefore,
$$\operatorname{Bd}^X(C) = \{3\}, \quad \operatorname{ext}^X(C) = [0, 2) \cup (3, 4].$$

□

The next exercise is especially useful when we calculate the interior and closure in a space where the topology is finer (or coarser) than another topology on the same set and we know the interior and closure in the latter topology. The super index j will indicate the corresponding topology τ_j, $j = 1, 2$.

Exercise 3.11 Let τ_1 and τ_2 be two topologies on a set X. Prove that the following statements are equivalent:

1. τ_1 is finer than τ_2.
2. For every subset A of X, $\overline{A}^1 \subset \overline{A}^2$.
3. For every subset A of X, $\overset{\circ}{A}^2 \subset \overset{\circ}{A}^1$.

As an application, find the interior and closure of $(0, 1]$ in the Sorgenfrey topology.

Solution We prove the equivalence (1)$\Leftrightarrow$ (2).

1. Suppose that τ_1 is finer than τ_2. We see the inclusion $\overline{A}^1 \subset \overline{A}^2$. Let $x \in \overline{A}^1$ and we prove that $x \in \overline{A}^2$. Let $O \in \tau_2$ be an open set such that $x \in O$. Since $\tau_1 \subset \tau_2$, O is open in τ_1, hence $O \cap A \neq \emptyset$ because $x \in \overline{A}^1$.
2. Suppose $\overline{A}^1 \subset \overline{A}^2$ for all $A \subset X$. To establish that τ_1 is finer that τ_2, it suffices to prove that every closed in (X, τ_2) is closed in (X, τ_1). Let $F \in \mathscr{F}_2$. Its closure in (X, τ_1) satisfies $\overline{F}^1 \subset \overline{F}^2 = F$, where we use the hypothesis in the first inclusion, and that F is closed in (X, τ_2) in the last identity. This proves that $\overline{F}^1 \subset F$, hence $\overline{F}^1 = F$ and F is closed in (X, τ_1).

   To prove the equivalence (2)$\Leftrightarrow$(3), utilize the identity $\text{int}(X \setminus A) = X \setminus \overline{A}$. Then

$$\overset{\circ}{A}^2 = \text{int}^2(X \setminus (X \setminus A)) = X \setminus \overline{X \setminus A}^2.$$

Then (2) $\Rightarrow$ (3) is true because

$$\overset{\circ}{A}^2 \subset X \setminus \overline{X \setminus A}^1 = \text{int}^1(X \setminus (X \setminus A)) = \overset{\circ}{A}^1.$$

The implication (3)$\Rightarrow$(2) is similar.

For the last part, we use that $\tau_u \subset \tau_S$. Then we utilize that we know the interior and closure in $(\mathbb{R}, \tau_u)$, namely, $\overset{\circ}{(0, 1]} = (0, 1)$ and $\overline{(0, 1]} = [0, 1]$. Thus we deduce that $(0, 1) \subset \overset{\circ}{(0, 1]}^S$ and $\overline{(0, 1]}^S \subset [0, 1]$. All we have to check is if $x = 1$ is interior and if $x = 0$ is adherent. For $x = 1$, there are no intervals $[1, y)$ included in $(0, 1]$. For $x = 0$, any interval $[0, y)$ intersects $(0, 1]$. Thus $\overset{\circ}{(0, 1]}^S = (0, 1)$ and $\overline{(0, 1]}^S = [0, 1]$. □

Exercise 3.12 In the scattered line (Exercise 2.21), find efficient bases of neighborhoods and the interior and closure of $[0, 1]$ and $[0, \sqrt{2})$.

Solution Recall that the open sets in this topology are the sets $O \cup A$, where $O \in \tau_u$ and A is a subset of the irrational numbers $\mathbb{R} - \mathbb{Q}$.

1. For bases of neighborhoods, we distinguish if the number is rational or not.
   (a) Let $x \in \mathbb{Q}$ be given. We prove that a basis of neighborhoods of x is $\beta_x = \{(x-r, x+r) : r > 0\}$, the standard basis for the Euclidean topology. The set $(x-r, x+r)$ is open in τ taking $A = \emptyset$. Let $O \cup A$ be an open set containing x, where $A \subset \mathbb{R} - \mathbb{Q}$ in particular, $x \in O$ because $x \in \mathbb{Q}$. Then there is $r > 0$ such that $x \in (x-r, x+r) \subset O$, so $x \in (x-r, x+r) \subset O \cup A$.
   (b) If x is irrational, we show that $\beta_x = \{\{x\}\}$ is a basis of neighborhood. First, the set $\{x\}$ is an open set in τ by choosing $O = \emptyset$ and $A = \{x\}$. Finally it is obvious that every neighborhood of x contains $\{x\}$.
2. For the interior and closure of a set, we use that τ is finer than the Euclidean topology and Exercise 3.11. The interior of $[0, 1]$ contains the interval $(0, 1)$. It remains to see if 0 and 1 are interior points. Because the basis of neighborhoods for both (rational) numbers agrees with that of the Euclidean topology, we deduce that 0 and 1 are not interior points. Thus $\mathrm{int}([0, 1]) = (0, 1)$.
   For the closure, $[0, 1]$ is closed in the usual topology. Since $\tau_u \subset \tau$ it is also closed in τ. This proves that the closure is $[0, 1]$.
3. For the interior of $[0, \sqrt{2})$, arguing as above it follows that its interior is $(0, \sqrt{2})$. For the closure, the set $[0, \sqrt{2}]$ is closed in τ_u. Thus $\overline{[0, \sqrt{2})} \subset [0, \sqrt{2}]$. It remains to see if $\sqrt{2}$ is an adherent point or not. The neighborhood $\{\sqrt{2}\}$ does not intersect $[0, \sqrt{2})$, so $\sqrt{2}$ is not an adherent point. Thus $\overline{[0, \sqrt{2})} = [0, \sqrt{2})$. Notice that we could have seen that the complement of $[0, \sqrt{2})$ is open. Indeed, the complement is $(-\infty, 0) \cup [\sqrt{2}, \infty)$. The set $(-\infty, 0)$ is open in τ_u, so also in τ, and $[\sqrt{2}, \infty)$ is open because it coincides with $(\sqrt{2}, \infty) \cup \{\sqrt{2}\}$. □

Exercise 3.13 Consider on $\mathbb{N}$ the topologies of Exercise 2.2.

1. Find the interior and closure of $C = \{4n : n \in \mathbb{N}\}$ in $X = \{2n : n \in \mathbb{N}\}$.
2. Find the interior and closure of $D = \{2, 4\}$ in $Y = \{2, 3, 4\}$.

Solution

1. (a) The intersection of A_n with X is $A_n \cap X = \{2, 4, \ldots, 2m\}$, where $2m$ is the closest even natural number to n with $2m \leq n$. Then every open set in X contains the number 2. Since 2 belongs to every open set in $\tau_{|X}$ and $2 \notin C$, the only open set in $\tau_{|X}$ included in C is $\emptyset$. This proves that $\overset{\circ}{C} = \emptyset$.
   For the closure, use the identity $\overline{C}^X = \overline{C} \cap X$. The nontrivial closed sets in $\mathbb{N}$ are the sets B_n. Among them, the smallest one that contains C is B_4, so $\overline{C} = B_4$. Thus $\overline{C}^X = B_4 \cap X = \{2n : n \geq 2\} = X \setminus \{2\}$.
   (b) We compute the interior and closure for the topology τ'. Now the open sets in $\tau'_{|X}$ are the sets $B_n \cap X$. This set is precisely $B_m \cap X$, where m is the nearest even number that is bigger or equal to n. Because in $B_m \cap X$ there are always numbers $2k$ that are not of the form $4n$, there are no nontrivial open sets contained in C. This proves $\overset{\circ}{C} = \emptyset$.

As above, $\overline{C}^X = \overline{C} \cap X$ and we compute $\overline{C}$ in the topology τ'. The nontrivial closed sets in this topology are the sets A_n; in particular, there is no A_n containing C. This proves that $\overline{C} = \mathbb{N}$, so $\overline{C}^X = X$.

2. Since Y is finite, it is easy to calculate its relative topology in both topologies,

$$\tau_{|Y} = \{\{2\}, \{2, 3\}, Y, \emptyset\}, \quad \tau'_{|Y} = \{\{3, 4\}, \{4\}, Y, \emptyset\}.$$

Thus the closed sets are

$$\mathscr{F}_{|Y} = \{\{4\}, \{3, 4\}, Y, \emptyset\}, \quad \mathscr{F}'_{|Y} = \{\{2\}, \{2, 3\}, Y, \emptyset\}.$$

(a) The interior of D in $\tau_{|Y}$ is the largest open set in Y contained in D, so the interior is $\{2\}$. For the closure, the smallest closed set that contains D is Y.
(b) For the topology $\tau'_{|Y}$, the interior of C is $\{4\}$ and its closure is Y again. □

Exercise 3.14 In the topology τ defined in Exercise 2.24, find efficient bases of neighborhoods. Calculate the interior and closure of $A = [2, \infty)$ and $B = (-\infty, 2]$.

Solution A basis of neighborhoods of x is formed by all open sets containing x, but a smaller basis is $\beta_x = \{U_{x,n} : n \in \mathbb{N}\}$. Recall that $\tau \subset \tau_u$.

1. Since the interior of A in τ_u is $(2, \infty)$, by Exercise 3.11 we have $\text{int}([2, \infty)) \subset (2, \infty)$. If $x \in (2, \infty)$ and in view that this set is open in the usual topology, let $n \in \mathbb{N}$ such that $(x - \frac{1}{n}, x + \frac{1}{n}) \subset (2, \infty)$. If we choose $n > 2$, then $(n, \infty) \subset (2, \infty)$, hence $U_{x,n} \subset (2, \infty) \subset [2, \infty)$. Thus $\text{int}([2, \infty)) = (2, \infty)$.
We compute the closure of A. Any basic neighborhood $U_{x,n}$ intersects $[2, \infty)$ because

$$U_{x,n} \cap [2, \infty) = \left((x - \frac{1}{n}, x + \frac{1}{n}) \cup (n, \infty)\right) \bigcap [2, \infty) \supset (n, \infty) \cap [2, \infty) \neq \emptyset.$$

Thus $\overline{[2, \infty)} = \mathbb{R}$.

2. Since no interval (n, ∞) is included in $(-\infty, 2]$, there are no basic neighborhoods $U_{x,n}$ inside B. Thus $\overset{\circ}{B} = \emptyset$. We have seen that $(2, \infty)$ is open, so B is closed. □

We establish a relationship between the boundary of a set and the boundary of its interior and of its closure.

Exercise 3.15 Let A be a subset in a space X. Prove that $\text{Bd}(\overset{\circ}{A}) \subset \text{Bd}(A)$ and $\text{Bd}(\overline{A}) \subset \text{Bd}(A)$. Give examples that these inclusions are proper in general.

Solution The proof of the inclusions can be performed point-by-point, or using properties of the interior and closure of a set. Let us choose the second option and use Proposition 3.3.7.

1. We have $\mathrm{Bd}(\mathring{A}) = \overline{\mathring{A}} \cap \overline{X \setminus \mathring{A}}$. First, $\mathring{A} \subset A$, so $\overline{\mathring{A}} \subset \overline{A}$. On the other hand, $X \setminus \mathring{A}$ is closed, so $\overline{X \setminus \mathring{A}} = X \setminus \mathring{A} = \overline{X \setminus A}$, where we apply (4) of Proposition 3.3.7. Therefore,

$$\mathrm{Bd}(\mathring{A}) \subset \overline{A} \cap (X \setminus \mathring{A}) = \overline{A} \cap \overline{X \setminus A} = \mathrm{Bd}(A).$$

2. For the boundary of $\overline{A}$, $\mathrm{Bd}(\overline{A}) = \overline{\overline{A}} \cap \overline{X \setminus \overline{A}} = \overline{A} \cap \overline{X \setminus A}$. Thus $\mathrm{Bd}(\overline{A}) \subset \overline{A} \cap \overline{X \setminus A} = \mathrm{Bd}(A)$.
3. For the examples, consider the Euclidean real line. For the interior, let $A = \{0\} \cup (1, 2)$. Then $\mathring{A} = (1, 2)$ and $\mathrm{Bd}(\mathring{A}) = \{1, 2\}$. On the other hand, $\mathrm{Bd}(A) = \{0, 1, 2\}$.
   The example for the closure is $A = [0, 1) \cup (1, 2]$. Then $\overline{A} = [0, 2]$ and $\mathrm{Bd}(\overline{A}) = \{0, 2\}$. But $\mathrm{Bd}(A) = \{0, 1, 2\}$. □

Exercise 3.16 Prove that the following collections β_x for each $x \in \mathbb{R}$ define bases of neighborhoods for a topology τ on $\mathbb{R}$. If $x \neq 0$, β_x is the standard basis for the usual topology. If $x = 0$, let $\beta_0 = \{V_{n,\varepsilon} : n \in \mathbb{N}, \varepsilon > 0\}$, where

$$V_{n,\varepsilon} = (-\infty, -n) \cup (-\varepsilon, \varepsilon) \cup (n, \infty).$$

See Fig. 3.6. This space is called the *looped line*. Compare with the usual topology. Find the interior and closure of $A = (1, \infty)$ and $B = (-1, 1)$. Prove that if $(x_n) \to \infty$ in the usual topology, then $(x_n) \to 0$ in τ.

Solution In testing the hypotheses of Theorem 3.2.4, the reader may think that it suffices to check the properties only at $x = 0$ because for the other points we already know that β_x is a basis of neighborhoods. However, (iii) of the theorem involves other points different from x, so Theorem 3.2.4 must be verified for all points.

1. (a) Case $x \neq 0$. Properties (i) and (ii) are true because β_x is the basis in the usual topology. For (iii), let $V = (x - r, x + r) \in \beta_x$. Since $x \neq 0$, let $s > 0$, with $s < r$, such that $0 \notin U = (x - s, x + s) \subset V$. For this U and because β_x is a basis of neighborhoods in the usual topology, there is $W \in \beta_x$ such that for each $y \in W$, there is $V_y \in \beta_y$ with $V_y \subset U$. This is possible because $0 \notin U$. Since $U \subset V$, we have proved (iii).
   (b) Case $x = 0$. Property (i) is immediate. For (ii), it is clear that $V_{n,\varepsilon} \cap V_{m,\delta} = V_{k,\eta}$, where $k = \max\{n, m\}$, $\eta = \min\{\varepsilon, \delta\}$. We prove (iii). Let $V_{n,\varepsilon} \in \beta_0$. We claim that $V_0 = V_{n,\varepsilon}$ is the neighborhood V_0 of Theorem 3.2.4. Let $y \in V_0$,

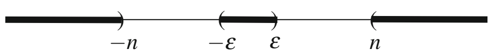


Fig. 3.6 The basic neighborhoods $V_{n,\varepsilon}$ of Exercise 3.16

$y \neq 0$ (if $y = 0$ the next argument is trivial). As $V_{n,\varepsilon}$ is open in the usual topology and β_y is a basis of neighborhoods for the usual topology, there is $V_y \in \beta_y$ such that $V_y \subset V$. This completes the proof.

2. Taking into account that $V_{n,\varepsilon}$ of β_0 are open in the usual topology, the topology τ is included in the Euclidean one. However, there are open sets in the usual topology that are not open in τ. For instance, $A = (-1, 1)$ is open in the usual topology but not in τ because $0 \in A$. However, there are no basic neighborhoods $V_{n,\varepsilon}$ contained in $(-1, 1)$ (note that all $V_{n,\varepsilon}$ are not bounded sets).
3. The points of A are interior points with the usual topology. Since these points are different from 0, they are also interior for τ. This proves that $\overset{\circ}{A} = A$. For the closure, those points different from 0 that are adherent with the usual topology, also are for τ. Thus $[1, \infty) \subset \overline{A}$. For $x = 0$, every basic neighborhood $V_{n,\varepsilon}$ intersects A, so 0 is also an adherent point. We conclude that $\overline{A} = \{0\} \cup [1, \infty)$.
4. The set B is open in the usual topology, hence all its points distinct from 0 are interior points in τ. For $x = 0$, there are no $V_{n,\varepsilon}$ included in B because $V_{n,\varepsilon}$ is not bounded and B is. This proves that $\overset{\circ}{B} = (-1, 0) \cup (0, 1)$.
   For the closure, the interval $[-1, 1]$ is closed in the usual topology. In consequence, all points of $[-1, 1]$ are adherent in τ because $\tau \subset \tau_u$. The remaining points $|x| > 1$ are not adherent because the basis of neighborhoods agrees with that of the usual topology. Thus $\overline{B} = [-1, 1]$. One easily sees that if $A \subset \mathbb{R}$ is a bounded set, its interior coincides with the interior in the usual topology except that $x = 0$ cannot be an interior point. Similarly, if A is not bounded, then its closure is equal to that in the usual topology together with the point $x = 0$.
5. Recall that in the usual topology, the expression $(x_n) \to \infty$ is equivalent to saying that (x_n) is not bounded. Let $V_{n,\varepsilon}$ be a basic neighborhood of $x = 0$. Since (x_n) is not bounded, given $n > 0$, let $\nu \in \mathbb{N}$ such that $|x_m| > n$ for all $m \geq \nu$. Then $x_m \in (-\infty, -n) \cup (n, \infty) \subset V_{n,\varepsilon}$ for all $m \geq \nu$. □

Exercise 3.17 Let $\{A, B\}$ be a partition of a set X and consider the topology $\tau = \{\emptyset, X, A, B\}$. If $p \in A$, find the interior and closure of $C = \{p\}$ and $D = \{p\} \cup B$.

Solution Notice that the family of closed sets coincides with the topology.

1. For C, the largest open set included in C is $\{p\}$ if $A = \{p\}$ or it is $\emptyset$ if A contains more than one element. Thus we distinguish two cases:

   (a) Case $A = \{p\}$. Then $\overset{\circ}{C} = \{p\}$. Since A is also closed, $\overline{C} = \{p\}$.

   (b) Case $A \neq \{p\}$. Then $\overset{\circ}{C} = \emptyset$ and $\overline{C} = \{A\}$, because it is the smallest closed set containing $\{p\}$.

2. For D, the case that $A = \{p\}$ is trivial because then $D = X$. Thus assume that A has more than one element.

   (a) The only open sets included in D are $\emptyset$ and B, so $\overset{\circ}{D} = B$.

   (b) On the other hand, the only closed set containing D is X, so $\overline{D} = X$. □

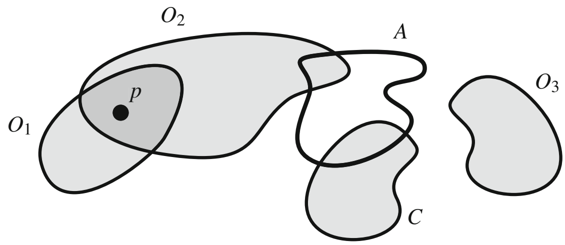


Fig. 3.7 Exercise 3.18. The sets O's are open and C is not

Exercise 3.18 Let X be a set, let $A \subset X$ and fix $p \in X$ with $p \notin A$. Define a topology τ by declaring that a set O is open if it satisfies the following property:

$$\text{if } p \notin O \text{ then } O \cap A = \emptyset.$$

See Fig. 3.7. Prove that τ is a topology and describe the family of closed sets. Find bases of neighborhoods. In the particular case $X = \mathbb{R}$, $A = [2, 3]$ and $p = 0$, find the interior and closure of $[0, 1)$.

Solution From the definition of τ, if $G \subset X$ such that $p \in G$, then $G \in \tau$.

1. It is obvious that the trivial sets belong to τ. Let $\{O_i : i \in I\} \subset \tau$. Suppose $p \notin \bigcup_{i\in I} O_i$. Then $p \notin O_i$ for all $i \in I$. Since $O_i \in \tau$, we have $O_i \cap A = \emptyset$ for all $i \in I$ and it follows that $(\bigcup_{i\in I} O_i) \cap A = \bigcup_{i\in I}(O_i \cap A) = \emptyset$, so $\bigcup_{i\in I} O_i \in \tau$. For the intersection, let $O_1, O_2 \in \tau$. Suppose that $p \notin O_1 \cap O_2$. Then p does not belong to O_1 or O_2. If, say $p \notin O_1$, then $O_1 \cap A = \emptyset$ because $O_1 \in \tau$. Thus $(O_1 \cap O_2) \cap A = \emptyset$, hence $O_1 \cap O_2 \in \tau$.
2. A set $F \subset X$ is closed if $X \setminus F$ is open. This means that if $p \notin X \setminus F$, then $(X \setminus F) \cap A = \emptyset$. By contraposition, F is closed if it satisfies the following property: if $p \in F$ then $A \subset F$. Thus

$$\mathscr{F} = \{F \subset X : p \notin F\} \cup \{F \subset X : A \cup \{p\} \subset F\}.$$

   In the same fashion, the topology is

$$\tau = \{O \subset X : p \in O\} \cup \{O \subset X : O \subset X \setminus (A \cup \{p\})\}.$$

3. We know that if $x \notin A$, then $\{x\}$ is open. Thus a basis of neighborhoods is $\beta_x = \{\{x\}\}$. If $x \in A$, the smallest open set containing x is $A \cup \{p\}$, so $\beta_x = \{A \cup \{p\}\}$ is a basis of neighborhoods of x.
4. The interval $[0, 1)$ is open in τ because $0 \in (0, 1)$. Thus $\text{int}([0, 1)) = [0, 1)$. For the closure and by (2), the closed sets of the first type do not contain $[0, 1)$. Among the sets of second type, the smallest set containing $[0, 1)$ is $[0, 1) \cup [2, 3]$, and $\overline{[0, 1)} = [0, 1) \cup [2, 3]$. □

Exercise 3.19 Find the interior and closure of the coordinate axes of $\mathbb{R}^2$.

Solution Let $A = (\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})$. In $\mathbb{R}^2$ there are two standard bases of neighborhoods for (x, y),

$$\gamma_{(x,y)} = \{(x - r, x + r) \times (y - r, y + r) : r > 0\},$$
$$\beta_{(x,y)} = \{B_r(x, y) : r > 0\}.$$

To show that the two bases are equally valid, we work with both ones.

1. We prove that $\overset{\circ}{A} = \emptyset$ because it is clear that there are no squares included in A. Certainly, let $(x, y) \in A$. Suppose $y = 0$ (similar if $x = 0$). If $(x, 0)$ is an interior point of A, then there is $r > 0$ such that

$$U = (x - r, x + r) \times (-r, r) \subset A,$$

   which is not possible. Indeed, if $x \neq 0$, the point $(x, r/2) \in U$ but $(x, r/2) \notin A$. If $x = 0$, the point $(r/2, r/2) \in U$ but $(r/2, r/2) \notin A$.
2. For the closure, we show that A is closed. In other words, we prove that for a point $(x, y) \notin A$, it is possible to find a ball centered at this point not intersecting A. For that (x, y), let $r = \min\{|x|, |y|\} > 0$, which is positive because $(x, y) \neq (0, 0)$. We prove that $B_r(x, y) \cap A = \emptyset$. If there is a point (a, b) in this intersection, this point belongs to the x-axis or y-axis. We do the argument for points $(a, 0)$ (analogously for $(0, b)$). As $(a, 0) \in B_r(x, y)$, it follows that $(x - a)^2 + y^2 < r^2$, hence $y^2 < r^2$. This is absurd by definition of r. □

Exercise 3.20 In $(\mathbb{R}, \tau_S)$, find the interior and closure of the intervals $(0, 1)$, $[0, 1)$, $(0, 1]$ and $[0, 1]$.

Solution We exploit that the Sorgenfrey topology is finer than the Euclidean topology. Recall that a basis of neighborhoods of x is $\beta_x^S = \{[x, y) : y > x\}$.

1. The interval $(0, 1)$ is open for the Euclidean topology, so also in τ_S. Thus it coincides with its interior.
   We see that $\overline{(0, 1)} = [0, 1)$. It is clear that $x = 0$ is adherent because every element $[0, y)$ of its basis of neighborhoods intersects $(0, 1)$. The point $x = 1$ is not adherent because $[1, 2) \in \beta_1^S$ and does not intersect $(0, 1)$. The remaining points are not adherents because they are not with the Euclidean topology.
2. The interval $[0, 1)$ is open in the Sorgenfrey line because it is a member of the basis, so it coincides with its interior.
   It is also a closed set because its complement $(-\infty, 0) \cup [1, \infty)$ is the union of two open sets: $(-\infty, 0)$ is open in the Euclidean topology, so also in τ_S; the interval $[1, \infty)$ can be expressed as union of open sets of τ_S, $[1, \infty) = \bigcup_{n \in \mathbb{N}} [1, n + 1)$.
3. We see that $\text{int}((0, 1]) = (0, 1)$. All numbers of $(0, 1)$ are interior points because they are interior in the Euclidean topology. However, $x = 1$ is not because no basic neighborhood $[1, y)$ is included in $(0, 1]$.

For the closure, the numbers $x > 1$ and $x < 0$ are not adherent because they are not with the Euclidean topology. It remains to study the point $x = 0$. Since every basic neighborhood $[0, y)$ intersects $(0, 1]$, the point $x = 0$ is adherent. Therefore, $\overline{(0, 1]} = [0, 1]$.

4. The set $[0, 1)$ is an open set included in $[0, 1]$. On the other hand, $x = 1$ is not interior because no basic neighborhood $[1, y)$ is included in $[0, 1]$. This proves that $\operatorname{int}([0, 1]) = [0, 1)$.
   The set $[0, 1)$ is closed because its complement $(-\infty, 0) \cup (1, \infty)$ is union of two open sets in the Euclidean topology, so open in the Sorgenfrey topology. □

In the next exercises the bases of neighborhoods are formed by one element.

Exercise 3.21 Define a topology τ on $\mathbb{N}$ by declaring a set $O \subset \mathbb{N}$ to be open if and only if satisfies the following property:

$$2n \in O \text{ if an only if } 2n - 1 \in O.$$

This topology is called the *Hjalmar Ekdal topology*.

1. Prove that τ is a topology.
2. Find a basis for the topology and a basis of neighborhoods.
3. Find the interior and closure of $A = \{4, 5, 6\}$ and $B = \{4, 5, 6, 7\}$.

Solution By definition of τ, if an even number belongs to the open set, then the previous number belongs to the open set, and if an odd number belongs to the open set, then the next number belongs to the open set.

1. It is obvious that $\emptyset$ and $\mathbb{N}$ belong to τ. Let $\{O_i : i \in I\}$ be a family of elements of τ and let $2n \in \bigcup_{i \in I} O_i$. Then there is $i_0 \in I$ such that $2n \in O_{i_0}$. Since $O_{i_0} \in \tau$, we have $2n - 1 \in O_{i_0}$, so $2n - 1 \in \bigcup_{i \in I} O_i$. A similar argument applies if $2n - 1 \in \bigcup_{i \in I} O_i$.
   Let $O_1, O_2 \in \tau$ and $2n \in O_1 \cap O_2$. Then $2n \in O_1$, so $2n - 1 \in O_1$. Analogously, $2n \in O_2$, hence $2n - 1 \in O_2$. This proves that $2n - 1 \in O_1 \cap O_2$. A similar argument can be performed for $2n - 1$. This reasoning applies equally well to an arbitrary number of open sets, so arbitrary intersection of open sets is open.
2. Note that if an open set is finite, its cardinality is even. For the basis of neighborhoods of $m \in \mathbb{N}$, if $m = 2n$, then $U_{2n} = \{2n-1, 2n\}$ is the smallest open set containing $2n$, so $\beta_{2n} = \{U_{2n}\}$ is a basis of neighborhoods. If $m = 2n - 1$, the smallest open set containing $2n - 1$ is $U_{2n-1} = \{2n - 1, 2n\}$, so $\{U_{2n-1}\}$ is a basis of neighborhoods. Observe that $U_{2n} = U_{2n-1}$.
   Therefore, the family $\beta = \{U_{2n}, U_{2n-1} : n \in \mathbb{N}\}$ is a basis for the topology ((2) of Proposition 3.2.2).
3. Since A and B are finite, so too are $\overset{\circ}{A}$ and $\overset{\circ}{B}$ and it is immediate that $\overset{\circ}{A} = \{5, 6\} = \overset{\circ}{B} = \{5, 6\}$.
4. For the closures of A and B, because U_n contains only two consecutive numbers, any number whose distance from the set is greater than 2 is not adherent. For A, it is enough to check for 2, 3, 7 and 8. Then it is clear that only $x = 3$ is adherent,

so $\overline{A} = A \cup \{3\}$. For B, we follow the same argument. Among the numbers 2, 3, 8 and 9, only U_3 and U_8 intersect B hence $\overline{B} = B \cup \{3, 8\}$. □

Exercise 3.22 In the included point topology and in excluded point topology, find the interior, exterior, boundary and closure of any set.

Solution Let $A \subset X$. We distinguish if p is an element of A or not. The trivial cases $A = \emptyset$ and $A = X$ are discarded.

1. Included point topology.
   (a) Case $p \in A$. Then A is open and $X \setminus A$ cannot contain any open set. Thus $\overset{\circ}{A} = A$, $\mathrm{ext}(A) = \emptyset$, Bd $= X \setminus A$ and $\overline{A} = X$.
   (b) Case $p \notin A$. Then $p \in X \setminus A$, so by the previous item, $\overset{\circ}{A} = \emptyset$, $\mathrm{ext}(A) = X \setminus A$, $\mathrm{Bd}(A) = A$ and $\overline{A} = A$.
2. Excluded point topology.
   (a) Case $p \in A$. Then A is closed and $X \setminus A$ is open, so $\overline{A} = A$ and $\mathrm{ext}(A) = X \setminus A$. The largest open set included in A is $A \setminus \{p\}$, so $\overset{\circ}{A} = A \setminus \{p\}$ and $\mathrm{Bd}(A) = \{p\}$.
   (b) Case $p \notin A$. Then A is open and the largest open set included in $X \setminus A$ is $X \setminus \{A \cup \{p\}\}$. Thus $\overset{\circ}{A} = A$, $\mathrm{ext}(A) = X \setminus \{A \cup \{p\}\}$, $\mathrm{Bd}(A) = \{p\}$ and $\overline{A} = A \cup \{p\}$. □

Exercise 3.23 Find the points of accumulation and isolated points in the included point topology and in excluded point topology.

Solution For τ_{in} a basis of neighborhoods is $\beta_x = \{U_x\}$, where $U_x = \{p, x\}$, and for τ_{ex}, $\gamma_x = \{V_x\}$, with $V_x = \{x\}$ if $x \neq p$ and $V_p = X$.

1. For the included point topology, we see that all points different from p are points of accumulation of X and p is the only isolated point. If $x \neq p$,
$$(U_x \setminus \{x\}) \cap X = \{p\} \cap X = \{p\} \neq \emptyset,$$
so x is a point of accumulation. On the other hand, for p, $(U_p \setminus \{p\}) \cap X = \emptyset$. This proves that p is an isolated point.
2. In the excluded point topology all points are isolated points of X except p, which is a point of accumulation. If $x \neq p$ then $(V_x \setminus \{x\}) \cap X = \emptyset$. By contrast, $(V_p \setminus \{p\}) \cap X \neq \emptyset$, proving that p is a point of accumulation. Here we are assuming that X has more than one point, otherwise that point will be isolated.

□

In the next exercise the computations of the interior and closure of a subset are easy because we have explicit descriptions of all open sets.

Exercise 3.24 On $\mathbb{R}$, consider the topology
$$\tau = \{\emptyset, \mathbb{R}\} \cup \{(-a, a) : a \in \mathbb{R}, a > 1\}.$$

See Exercise 2.20. Find an efficient basis of neighborhoods for each point. Find the interior and closure of the intervals $(2, 3)$, $(-1, 1)$ and $[-1, 1]$.

Solution

1. Let $x \in \mathbb{R}$ be given. A basis of neighborhoods is formed for those members of τ that contain $x \in \mathbb{R}$. Then $\beta_x = \{(-a, a) : a > 1, |x| < a\}$. Note that the intervals $(-a, a)$ are ordered by inclusion according the value a. If we fix a value $a_0 > 1$ such that $x \in (-a_0, a_0)$, then
$$\beta_x = \{(-a, a) : |x| < a < a_0\}$$
is a basis of neighborhoods of x. Observe that if $|x| < 1$, the condition $x \in (-a, a)$ is always valid for all $a > 1$, so if $x \in (-1, 1)$ then $\beta_x = \{(-1, 1)\}$ is a basis of neighborhoods.
2. For the computation of the interior and closure we will use Propositions 3.3.2 and 3.3.8. Recall that
$$\mathscr{F} = \{\emptyset, \mathbb{R}\} \cup \{(-\infty, -a] \cup [a, \infty) : a > 1\}.$$
Since there are no intervals $(-a, a)$ with $a > 1$ included in any of the three intervals, the interior is empty. For the closure,
$$\overline{(2, 3)} = (-\infty, -2] \cup [2, \infty), \quad \overline{(-1, 1)} = \overline{[-1, 1]} = \mathbb{R}.$$

□

Exercise 3.25 In the topology of the divisors (Exercise 2.15), find the interior and closure of $A = \{2n : n \in \mathbb{N}\}$ and $B = \{2, 4, 5, 6\}$.

Solution A basis of neighborhoods of n is $\beta_n = \{U_n = \{\text{divisors of } n\}\}$ (Example 3.5).

1. If $2n \in \overset{\circ}{A}$ then $U_{2n} \subset A$. Then all divisors of $2n$ are even numbers. Therefore, $2n$ is of the form 2^m for some $m \in \mathbb{N}$ proving $\overset{\circ}{A} = \{2^m : m \in \mathbb{N}\}$.
For the closure, we examine whether or not the numbers $2n + 1$ are adherent. Since $2n + 1$ is an odd number, the divisors also are odd numbers, so U_{2n+1} does not intersect A, and we deduce that $2n + 1$ is not an adherent point of A. Thus $\overline{A} = A$ and A is a closed set.
2. Since B is finite, we can calculate the interior points point-by-point. We have $U_2 = \{2\}$, $U_4 = \{2, 4\}$, $U_5 = \{5\}$ and $U_6 = \{2, 3, 6\}$. Because all are included in A except U_6, we deduce $\overset{\circ}{B} = \{2, 4, 5\}$.
For the closure, we examine the intersection $B \cap U_n$. If $2 \in B \cap U_n$, we have $2|n$ and n is a multiple of 2. The same argument for 4, 5 or 6, yields that n is multiple of these numbers. Thus $\overline{B}$ is composed of all multiples of 2, 4, 5 and 6, or equivalently, all multiples of 2, 5 and 6. □

We turn our attention to the sum topology defined in Exercise 2.8.

Exercise 3.26 Consider $X_1 = \mathbb{Q}$ and $X_2 = \mathbb{R} - \mathbb{Q}$ with the usual topologies τ_1 and τ_2, respectively. We endow $\mathbb{R} = X_1 \oplus X_2$ with the sum topology $\tau = \tau_1 + \tau_2$. Find the interior and closure of $\mathbb{Q}$ and $[0, 1]$. Determine the convergent sequences.

Solution

1. As $\mathbb{Q}$ is an open set in $\mathbb{Q}$, it follows that $\mathbb{Q}$ is an element of the basis β for the sum topology, so $\mathbb{Q}$ is open in τ. We see that $\overline{\mathbb{Q}} = \mathbb{Q}$, so irrationals are not adherent points of $\mathbb{Q}$. Let $x \in \mathbb{R} - \mathbb{Q}$ and consider any element $B \in \beta$ such that $x \in B$. As $x \in \mathbb{R} - \mathbb{Q}$, then $B \in \tau_2$, so $B \cap \mathbb{Q} = \emptyset$ because $B \subset \mathbb{R} - \mathbb{Q}$.
2. We see that $\operatorname{int}([0, 1]) = (0, 1)$. The interval $(0, 1)$ can be written as

$$(0, 1) = ((0, 1) \cap \mathbb{Q}) \cup ((0, 1) \cap (\mathbb{R} - \mathbb{Q})) \in \tau_1 + \tau_2.$$

   Thus $(0, 1)$ is open in τ. We show that $x = 0$ and $x = 1$ are not interior points of $[0, 1]$. It will be shown for $x = 0$ and a similar argument holds for $x = 1$. If $x = 0$ is an interior point, there is $B \in \beta$ such that $0 \in B \subset [0, 1]$. Since $0 \in \mathbb{Q}$, then $B \in \tau_1$. Now, by definition of the relative topology on $\mathbb{Q}$ and because $0 \in B$, there is $r > 0$ such that

$$0 \in (-r, r) \cap \mathbb{Q} \subset B.$$

   Thus $(-r, r) \cap \mathbb{Q} \subset [0, 1]$, which is not possible because $(-r, 0) \not\subset [0, 1]$. Finally, we prove that $\overline{[0, 1]} = [0, 1]$. Let $x \notin [0, 1]$. Suppose without loss of generality that $x > 1$. Then there exists $r > 0$ such that $(x-r, x+r) \cap [0, 1] = \emptyset$. If $x \in \mathbb{Q}$ (similarly if $x \in \mathbb{R} - \mathbb{Q}$),

$$((x - r, x + r) \cap \mathbb{Q}) \cap [0, 1] = \emptyset.$$

   But $(x - r, x + r) \cap \mathbb{Q}$ is a neighborhood of x in τ_1, in particular of $\tau_1 + \tau_2$. Since this neighborhood of x does not intersect $[0, 1]$, x is not an adherent point.
3. Let $(x_n) \to x$. As a first case, assume $x \in \mathbb{Q}$. A basis of neighborhoods of x in τ is the collection of elements of $\beta = \tau_1 \cup \tau_2$ containing x. Notice that there are no elements of τ_2 containing x because the sets of τ_2 are subsets of $\mathbb{R} - \mathbb{Q}$. Then a basis of neighborhoods of x is formed by the elements of τ_1 that contain x; in particular, $\beta_x = \{(x - r, x + r) \cap \mathbb{Q} : r > 0\}$. If $(x_n) \to x$, then for $r > 0$ there is $\nu \in \mathbb{N}$ such that $x_n \in (x - r, x + r) \cap \mathbb{Q}$ for $n \geq \nu$. This implies that from the term ν of the sequence, the sequence is included in $\mathbb{Q}$ and converges to x in the usual topology. We have proved that sequences converging to a rational number $x \in \mathbb{Q}$ are the sequences where all but finitely many terms are rationals that converge to x in the usual topology.
   With a parallel argument, sequences convergent to $x \in \mathbb{R} - \mathbb{Q}$ are the sequences where all but finitely many terms are irrationals that converge to x in the usual topology.
   Since the Euclidean topology is coarser than the sum topology τ (Exercise 2.9), a convergent sequence in τ is also convergent in τ_u. But the converse fails in

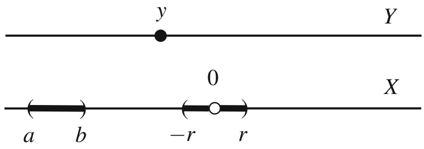


Fig. 3.8 The basic neighborhoods for the space of Exercise 3.27. The interval (a, b) is a neighborhood in X and $\{y\} \cup B_r$ is a neighborhood in Y

general. For example, $(\frac{1}{\sqrt{2n}}) \to 0$ and $((\frac{n+1}{n})^n) \to e$ in the Euclidean topology but not in the sum topology τ.
As a final observation, this result extends to the general case. Let $\{X_i : i \in I\}$ be a partition of a space X. A convergent sequence $(x_n) \to x$ in the sum topology is a convergent sequence in (X, τ) satisfying the following property: if X_{i_0} is the subset such that $x \in X_{i_0}$, then all but finitely many terms of (x_n) belong to X_{i_0} and $(x_n) \to x$ in the space X_{i_0}. $\square$

Exercise 3.27 Let $X = \mathbb{R} \setminus \{0\}$, $Y = \mathbb{R}$ and the disjoint union $Z = X \cup Y$. For $r > 0$, let $B_r = \{x \in X : x \in (-r, 0) \cup (0, r)\}$. Let us assign for each element of Z a collection of subsets as follows. If $x \in X$, β_x consists of all Euclidean balls of X centered at x. For each $y \in Y$, $\beta_y = \{\{y\} \cup B_r : r > 0\}$ (Fig. 3.8). Prove that all β_x and β_y are bases of neighborhoods for a topology on Z. Prove that the relative topology on X is Euclidean but the relative topology on Y is discrete.

Solution

1. We check the hypotheses of Theorem 3.2.4. Property (i) is immediate and (ii) is obvious for the elements of β_x. For β_y,

$$(\{y\} \cup B_{r_1}) \cap (\{y\} \cup B_{r_2}) = \{y\} \cup B_r,$$

where $r = \min\{r_1, r_2\}$. Property (iii) follows because the elements of β_x are included in X and (iii) holds for the Euclidean topology on X. Consider the elements of β_y. Following the notation of Theorem 3.2.4, let $V = \{y\} \cup B_r$. Let $V_0 = V$ and $z \in V_0$. The case $z = y$ is immediate taking $V_z = V$. If $z \in B_r$ then $z \in X$. In such a case, because B_r is open in the usual topology, let $r_0 > 0$ such that $\{x \in X : x \in (z - r_0, z + r_0)\} \subset (-r, 0) \cup (0, r)$. Thus $\{x \in X : x \in (z - r_0, z + r_0)\}$ is a Euclidean ball of X centered at x and included in V_0.
2. Intersecting β_x with X and because the elements of β_x are included in X, we obtain β_x again. This proves that the relative topology is Euclidean.
3. Since $B_r \subset X$, the intersection of an element of β_y with Y is $\{y\}$. This implies that singletons of Y are open sets, so the relative topology is discrete. $\square$

Exercise 3.28 For each $x \in \mathbb{R}$ and $a < x < b$, define $B^x_{a,b} = (-\infty, a] \cup \{x\} \cup [b, \infty)$ and let $\beta_x = \{B^x_{a,b} : a < x < b, a, b \in \mathbb{R}\}$. Prove that the collection $\{\beta_x : x \in \mathbb{R}\}$ defines a topology on $\mathbb{R}$ where β_x is a basis of neighborhoods of x. Prove that $B^x_{a,b}$ is open and find the interior and closure of $\mathbb{Z}$.

Solution

1. Let us check the properties of Theorem 3.2.4. It is clear that (i) is fulfilled and (ii) as well because the intersection of two elements of β_x is another element of β_x,

$$B^x_{a,b} \cap B^x_{c,d} = (-\infty, \min\{a, c\}] \cup \{x\} \cup [\max\{b, d\}, \infty).$$

For (iii), we follow the same notation of Theorem 3.2.4. Let $V = B^x_{a,b}$ and let $V_0 = V$. Let $y \in V_0$ be given. Assume $y \neq x$ because otherwise, $V_x = V$. Without loss of generality, assume $y \in [b, \infty)$. Then

$$B^y_{a,y+1} = (-\infty, a] \cup \{y\} \cup [y + 1, \infty)$$

is included in V.
2. For notational convenience, let $A = (-\infty, a] \cup \{x\} \cup [b, \infty)$ with $a < x < b$. We see that any point of A is interior; for $x \in A$, it is clear that $x \in B^x_{a,b} \subset A$. Suppose that $x \in (-\infty, a]$ (the case where $x \in [b, \infty)$ is similar). We check that $B^x_{x-1,b} \subset A$. We have $B^x_{x-1,b} = (-\infty, x - 1] \cup \{x\} \cup [b, \infty)$ and because $x \leq a$, then $(-\infty, x - 1,] \subset (-\infty, a - 1] \subset (-\infty, a] \subset A$.
3. It is clear that any there are no $B^x_{a,b}$ included in $\mathbb{Z}$ because $\mathbb{Z}$ cannot contain intervals $(-\infty, a]$. Thus $\overset{\circ}{\mathbb{Z}} = \emptyset$. It is also clear that any $B^x_{a,b}$ intersects $\mathbb{Z}$, so $\overline{\mathbb{Z}} = \mathbb{R}$. □

Exercise 3.29 Let A be a nontrivial subset of a set X. Define

$$\tau = \{X\} \cup \mathbf{P}(A) = \{X\} \cup \{\text{subsets of } A\}.$$

Prove that τ is a topology on X and describe the collection of closed sets. Find a basis of neighborhoods for every $x \in X$. If $X = \mathbb{R}$ and $A = (0, 2)$, find the interior and closure of the interval $(1, 3)$.

Solution

1. The proof that τ is a topology is obvious because the properties of the union and intersection hold in the family $\mathbf{P}(A)$ of all subsets of A.
2. Let F be a closed set of X. Then $X \setminus F$ is open. If $X \setminus F = X$ then $F = \emptyset$. Otherwise, $X \setminus F \subset A$ or, equivalently, $X \setminus A \subset F$. Thus the collection of closed sets is

$$\mathscr{F} = \{\emptyset\} \cup \{F \subset X : X \setminus A \subset F\}.$$

3. Let $x \in X$. If $x \in A$ then $\{x\} \subset A$, so $\{x\}$ is open in X. Thus $\beta_x = \{\{x\}\}$ is a basis of neighborhoods because $\{x\}$ is the smallest subset containing x. If $x \notin A$, the only open set containing x is X itself, so $\beta_x = \{X\}$.

4. From the description of the open sets and closed sets, we deduce that the largest open set included in $(1, 3)$ is $(1, 2)$, so $\operatorname{int}((1, 3)) = (1, 2)$. For the closure, the closed sets are those subsets F such that $F \supset \mathbb{R} \setminus A = (-\infty, 0] \cup [2, \infty)$. Thus the smallest closed set containing $(1, 3)$ is $(-\infty, 0] \cup (1, \infty)$. $\square$

Exercise 3.30 Find bases of neighborhoods in the space of Exercise 2.13. Find the interior and closure of $A = [0, 1]$ and $B = \{0\}$.

Solution A basis for τ is $\beta = \{\{x\} : x \neq 0\} \cup \{(-1, 1)\}$, hence by (3.1), we can obtain bases of neighborhoods. But it is possible to find the smallest basis of neighborhoods, namely,

$$\beta_x = \begin{cases} \{x\}, & x \neq 0 \\ \{(-1, 1)\}, & x = 0. \end{cases}$$

1. The set $(0, 1]$ is open, so $(0, 1] \subset \overset{\circ}{A}$. The point $x = 0$ is not interior because its basic neighborhood $(-1, 1)$ is not included in A. Thus $\overset{\circ}{A} = (0, 1]$. The set A is closed because its complement $[-1, 0)$ is open, since $0 \notin [-1, 0)$. This concludes that $\overline{A} = A$.
2. For the interior of B, we see if $x = 0$ is interior or not. As its basic neighborhood $(-1, 1)$ is not contained in B, $\overset{\circ}{B} = \emptyset$. For the closure, let $x \notin B$; that is, $x \neq 0$. Its basic neighborhood $\{x\}$ does not intersect B, so $x \notin \overline{B}$ and thus $\overline{B} = B = \{0\}$. $\square$

Exercise 3.31 Let $A = \{-1/n : n \in \mathbb{N}\} \cup \{0\}$. Prove that the relative topology on A from the Euclidean one is not discrete, but it is discrete from the Sorgenfrey topology. Consider the exercise by replacing A with $B = \{1/n : n \in \mathbb{N}\} \cup \{0\}$.

Solution

1. The set $\{0\}$ is not an open set of A in the Euclidean topology for if $x = 0$ is an interior point, there would be $r > 0$ such that $0 \in (-r, r) \cap A \subset \{0\}$. This is clearly not the case because the interval $(-r, r)$ contains infinite numbers $-1/n$.

   We see that $\tau_{S|A}$ is discrete. The set $\{0\}$ is open because the open set $O = [0, 1) \in \tau_S$ satisfies $O \cap A = \{0\}$. For the other points $-1/n$ of A, let $O_n = [-\frac{1}{n}, -\frac{1}{n+1}) \in \tau_S$. Then $\{-1/n\} = O_n \cap A$.
2. With the usual topology and by Example 2.27, the relative topology of B is not discrete. For the Sorgenfrey topology, the set $\{0\}$ is not open for if $x = 0$ is an interior point in the relative topology, there would be $r > 0$ such that $0 \in [0, r) \cap B \subset \{0\}$, which is not possible. Thus the relative topology on B is not discrete.

   In fact, the relative topologies on B coincide for both topologies because the intersections of the bases β_u and β_S with B coincide. Another way is using bases of neighborhoods. For numbers $\frac{1}{n}$, if we intersect sufficiently small open intervals centered at $\frac{1}{n}$ with B, we obtain $\{\frac{1}{n}\}$ and this also applies to the Sorgenfrey topology for the intersection of $[\frac{1}{n}, \frac{1}{n} + r)$ with B for r sufficiently small. For $x = 0$, bases of neighborhoods for τ_u and τ_S are intervals $(-r, r)$ and $[0, r)$, respectively. Intersecting with B in order to get bases of neighborhoods in the relative topologies, we see that $(-r, r) \cap B = [0, r) \cap B$. □

Exercise 3.32 Consider on $\mathbb{N}$ the topology defined in Exercise 2.16. Find efficient bases of neighborhoods for each number.

Solution Let $2n-1$ be an odd number. Then $U_{2n-1} = \{2n-1\}$ is open; in particular, it is the smallest neighborhood of $2n-1$ and this proves that $\beta_{2n-1} = \{U_{2n-1}\}$ is a basis of neighborhoods.

Let $2n$ be an even number and consider $U_{2n} = \{2n-1, 2n, 2n+1\}$. Then U_{2n} is open, so it is a neighborhood of $2n$. We see that U_{2n} is the smallest open set containing $2n$, hence $\beta_{2n} = \{U_{2n}\}$ is a basis of neighborhoods of $2n$. By Exercise 2.16, there are no open sets formed by only two consecutive numbers. Let $O \in \tau$ with $2n \in O$. Then necessarily $2n+1 \in O$ by definition of τ. If $2n-1 \notin O$ then $2n-2$ and $2n$ do not belong to O by definition of the topology. Since this is not possible because $2n \in O$, we deduce that $2n-1 \in O$. Thus we have proved that $U_{2n} \subset O$. □

Exercise 3.33 Define a topology on $\mathbb{R}^2$ describing a basis of neighborhoods for each point. If $(x, y) \in \mathbb{R}^2$, the basis of neighborhoods is $\beta_{(x,y)} = \{A_{xy}\}$, where $A_{xy} = [x, \infty) \times [y, \infty)$. Prove that $\beta_{(x,y)}$ defines a topology on $\mathbb{R}^2$. Prove that $\beta_{(x,y)}$ is also a basis of neighborhoods of the topology generated by $\beta_r \times \beta_r$. Find the interior and closure of the coordinate axes and of $[0, 1] \times \mathbb{R}$.

Solution

1. Let us check the hypotheses of Theorem 3.2.4. Since $\beta_{(x,y)}$ has only one element, (iii) is the only property to test. Using the notation of the theorem, let $A_{xy} \in \beta_{(x,y)}$ and take $V_0 = A_{xy}$. Let $(a, b) \in V_0$ be given. Then $x \leq a$ and $y \leq b$ so $[a, \infty) \subset [x, \infty)$ and $[b, \infty) \subset [y, \infty)$. This implies that $A_{ab} \subset A_{xy}$ as desired.
2. Here $\beta_r \times \beta_r = \{[x, \infty) \times [y, \infty) : x, y \in \mathbb{R}\}$. First, it is clear that A_{xy} are open sets because they are basic elements, so A_{xy} is a neighborhood of (x, y). Now let $O \in \tau(\beta_r \times \beta_r)$ such that $(x, y) \in O$. Then there is $[a, \infty) \times [b, \infty) \in \beta_r \times \beta_r$ such that $(x, y) \in [a, \infty) \times [b, \infty) \subset O$. Then $x \in [a, \infty)$ and $y \in [b, \infty)$. This implies $a \leq x$ and $b \leq y$, so $A_{xy} \subset [a, \infty) \times [b, \infty)$, hence included in O.
3. Let $A = (\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})$ be the coordinate axes. It is clear that no element A_{xy} is included in A, so $\overset{\circ}{A} = \emptyset$.
   For the closure, if $x, y > 0$ then $A_{xy} = [x, \infty) \times [y, \infty)$ does not intersect A, so (x, y) is not an adherent point. However, the remaining points are adherent. Indeed, if $x \leq 0$, then the intersection $([x, \infty) \times [y, \infty)) \cap A$ contains the point $(0, y)$. If $x > 0$ and $y \leq 0$, then $A_{xy} \cap A$ contains the point $(x, 0)$. This proves definitively that $\overline{A} = \mathbb{R}^2 \setminus ((0, \infty) \times (0, \infty))$.
4. A similar argument as before proves $\overset{\circ}{B} = \emptyset$. We compute its closure. Let $x > 1$. Then $[x, \infty) \times [y, \infty)$ does not intersect B, so (x, y) is not adherent. Again, we

see that the rest of points are adherent of B. Let (x, y) with $x < 0$ (if $x \in [0, 1]$ then the point is contained in B, so it is adherent). Then it is clear that $(0, y) \in A_{xy} \cap B$. This proves that $\overline{B} = (-\infty, 1] \times \mathbb{R}$. □

Exercise 3.34 Let X be a set such that for each $x \in X$, we assign one set A_x with the property that if $y \in A_x$, then $A_y \subset A_x$. If $\beta_x = \{A_x\}$, prove that the collection of all β_x defines a topology on X where β_x is a basis of neighborhoods of x. Prove that the sets A_x are open sets. If $X = \mathbb{R}^2$, apply to the cases $A_{(x,y)} = \mathbb{R} \times \{y\}$ and $A_{(x,y)} = \mathbb{R} \times [y, \infty)$.

Solution

1. We check the hypotheses of Theorem 3.2.4. Properties (i) and (ii) are clear because there is only one element in β_x. For (iii), we follow the same notation. Given $V = A_x$, take $V_0 = A_x$. If $y \in V_0$, by hypothesis, $A_y \subset A_x$, as desired.
2. The set A_x is open if all its points are interior. Let $y \in A_x$ be given. By hypothesis, $A_y \subset A_x$ and $A_y \in \mathscr{U}_y$, so y is an interior point of A_x.
3. If $A_{(x,y)} = \mathbb{R} \times \{y\}$ and $(z, t) \in A_{(x,y)}$, then $t = y$. Then $A_{(z,t)} = \mathbb{R} \times \{t\} = \mathbb{R} \times \{y\} = A_{(x,y)}$. Following the arguments as in Exercise 3.33 and recalling also Exercise 2.1, the topology is generated by $\beta_T \times \beta_D = \{\mathbb{R} \times \{y\} : y \in \mathbb{R}\}$.
4. If $A_{(x,y)} = \mathbb{R} \times [y, \infty)$ and $(z, t) \in A_{(x,y)}$, then $t \in [y, \infty)$. In particular, $[t, \infty) \subset [y, \infty)$ and consequently, $A_{(z,t)} \subset A_{(x,y)}$. It is easy to deduce that a basis for the topology is $\beta_T \times \beta_r = \{\mathbb{R} \times [y, \infty) : y \in \mathbb{R}\}$. □

Exercise 3.35 (Moore plane) On $X = \mathbb{R} \times [0, \infty)$, define a topology τ using bases of neighborhoods. Let $(x, y) \in X$. If $y > 0$, $\beta_{(x,y)}$ is formed by all balls centered at (x, y) and included in X. If $y = 0$,

$$\beta_{(x,0)} = \{B_r(x, r) \cup \{(x, 0)\} : r > 0\}.$$

See Fig. 3.9. This space is called the *Moore plane*. Find the interior of $A = \mathbb{R} \times \{0\}$. Show that the relative topology is discrete on A and Euclidean on $X \setminus A$.

Solution

1. We check the properties of Theorem 3.2.4. If (x, y) is a point where $y \neq 0$, then all properties are fulfilled by two facts. First, because $\beta_{(x,y)}$ consists of all

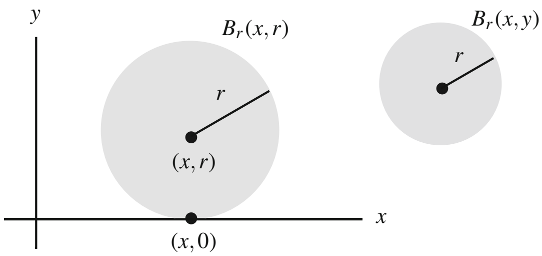


Fig. 3.9 The Moore plane

Euclidean balls centered at (x, y) with radius sufficiently small. Second, because the bases of neighborhoods of all points of these balls are again Euclidean balls since the second coordinate of all these points is positive. Thus, it remains to verify the properties of Theorem 3.2.4 for the points $(x, 0)$. For (ii), the intersection of two elements of $\beta_{(x,0)}$ is another,

$$(B_r(x,r) \cup \{(x,0)\}) \cap (B_s(x,s) \cup \{(x,0)\}) = B_\varepsilon(x,\varepsilon) \cup \{(x,0)\},$$

where $\varepsilon = \min\{r, s\}$. We check (iii). Let $V = B_r(x,r) \cup \{(x,0)\}$ and take $V_0 = V$. Let $(a,b) \in V$. The case $(a,b) = (x,0)$ is trivial, so assume $b \neq 0$. Then $(a,b) \in B_r(x,r)$. Since $B_r(x,r)$ is open in the usual topology, there is $\varepsilon > 0$ such that $B_\varepsilon(a,b) \subset B_r(x,r)$, so $B_\varepsilon(a,b) \in \beta_{(a,b)}$ and it is included in V.

2. For any $(x,0) \in \mathbb{R} \times \{0\}$, there are no sets $B_r(x,r) \cup \{(x,0)\}$ included in A, so its interior is $\emptyset$.
3. A basis of neighborhoods of $(x,0)$ in $\tau_{|A}$ is intersection of the elements of $\beta_{(x,0)}$ with A. Since $(B_r(x,r) \cup \{(x,0)\}) \cap A = \{(x,0)\}$, then the basis coincides with that of the discrete topology.
4. For the points of $X \setminus A$, since the elements of $\beta_{(x,y)}$ are included in $X \setminus A$, when intersecting with $X \setminus A$ we get the same neighborhoods. But these neighborhoods generate the usual topology. □

3.6 Suggested Exercises

3.6.1 On $X = \big(\mathbb{R} \times \{0\}\big) \cup \big([0,\infty) \times \{1\}\big) \subset \mathbb{R}^2$, define a topology using bases of neighborhoods as follows. For points distinct from $(0, 1)$, we take a basis of neighborhoods for the usual topology on $\mathbb{R}^2$. If the point is $(0, 1)$,

$$\beta_{(0,1)} = \Big\{\big((a,0) \times \{0\}\big) \cup \big([0,b) \times \{1\}\big) : a < b, a, b \in \mathbb{R}\Big\}.$$

See Fig. 3.10. Show that this defines a non-Hausdorff topology on X. Find the convergent sequences.

3.6.2 If $p, q \in X$ are fixed, define $\beta_x = \{\{x, p, q\}\}$ for each $x \in X$. Show that the collection of all β_x determine bases of neighborhoods for some topology on X.

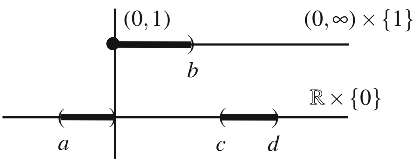


Fig. 3.10 The basic neighborhoods for the space of Exercise 3.6.1. The interval (c, d) belongs to the usual topology

3.6.3 Let ω be an object, $\omega \notin \mathbb{R}$ and $X = \mathbb{R} \cup \{\omega\}$. Prove that

$$\beta = \{(a, b) : a, b \in \mathbb{R}\} \cup \{(\mathbb{R} \setminus [a, b]) \cup \{\omega\} : a, b \in \mathbb{R}\}$$

defines a basis for a topology on X. Prove that the relative topology on $\mathbb{R}$ is Euclidean. Find the interior and closure of $[0, 1] \cup \{\omega\}$, $[0, \infty) \cup \{\omega\}$ and $\{\omega\}$.

3.6.4 Prove that $\tau = \{\emptyset, X, \{a\}, \{a, b\}\}$ is a topology on $X = \{a, b, c, d\}$. Find bases of neighborhoods and the interior and closure of $\{a, d\}$ and $\{b, c, d\}$.

3.6.5 Find the interior and closure of any subset in the topology defined in Exercise 2.12.

3.6.6 Find the interior and closure of a straight line of $\mathbb{R}^n$.

3.6.7 Fix a neighborhood V of a point x. Show that the collection of all neighborhoods of x included in V is a basis of neighborhoods of x.

3.6.8 Let X be a space and $A \subset X$. The set A is said to be *dense* if $\overline{A} = X$. Show that this is equivalent to each nonempty open subset O of X satisfying $O \cap A \neq \emptyset$. Show that in the cofinite topology, every open set is dense.

3.6.9 Prove that $\mathbb{Q}$ is dense in $\mathbb{R}$. Show that this is equivalent to for each $x \in \mathbb{R}$ and $\varepsilon > 0$, there is $q \in \mathbb{Q}$ such that $|x - q| < \varepsilon$. Show that the sets $\mathbb{R} - \mathbb{Q}$ and $\{\sqrt{2} + q : q \in \mathbb{Q}\}$ are also dense.

3.6.10 Prove that $\mathbb{Q}$ and $\mathbb{R} - \mathbb{Q}$ are dense in $(\mathbb{R}, \tau_S)$ and in $(\mathbb{R}, \tau_r)$.

3.6.11 Show that $\mathrm{Bd}(A \cup B) \subset \mathrm{Bd}(A) \cup \mathrm{Bd}(B)$. If $A \subset B$, find an example that shows that $\mathrm{Bd}(A) \not\subset \mathrm{Bd}(B)$.

3.6.12 If $B \subset A$, prove that $\overline{A} \setminus \overline{B} \subset \overline{A \setminus B}$ and give an example that the inclusion is proper.

3.6.13 Find the interior and closure of all the intervals in the right order topology on $\mathbb{R}$.

3.6.14 Let β_x and γ_x be two bases of neighborhoods of x. Show that $\{U \cap V : U \in \beta_x, V \in \gamma_x\}$ is a basis of neighborhoods of x. Give, however, an example where $\beta_x \cap \gamma_x$ is not a basis of neighborhoods.

3.6.15 Let $A \subset X$ and $a \in \overset{\circ}{A}$. Show that there is a common basis of neighborhoods of a for the topologies γ_A, $\tau_{|A}$ and τ.

3.6.16 Prove $\mathrm{int}(\overline{\mathrm{int}(\overline{A})}) = \mathrm{int}(\overline{A})$.

3.6.17 Find the interior and boundary of $[0, 1] \times \{0\}$ in $\mathbb{R}^2$.

3.6.18 Let A and B be two subsets of a space X. If $A \cup B = X$, prove $\overline{A} \cup \overset{\circ}{B} = X$. Analogously, prove that if $A \cap B = \emptyset$ then $\overline{A} \cap \overset{\circ}{B} = \emptyset$.

3.6.19 Show that a space is Hausdorff if and only if every point is the intersection of the closures of all its neighborhoods.

3.6.20 Let $A \subset \mathbb{R}$ be a set bounded from above (resp. from below) and let $\alpha = \sup(A)$ (resp. $\beta = \inf(A)$). Show that $\alpha \in \overline{A}$ (resp. $\beta \in \overline{A}$). Give an example of a bounded set A such that $\alpha, \beta \in A$ but A is not closed. Investigate the same problem in the Sorgenfrey topology and the right order topology.

Chapter 4
Metric Spaces

Metric spaces are a remarkable family of topological spaces and play a fundamental role in many fields of mathematics, especially in mathematical analysis, and thus it is worth dedicating a chapter to their study. A metric space is simply a set equipped with a notion of distance between points, and consequently, a way to indicate proximity between them. An example of a metric space is the Euclidean space, where the notion of distance is consistent with our sense of proximity.

We will topologize a metric space in much the same way that we did for $\mathbb{R}^n$. First, define the balls in the space and later, we prove that the collection of all balls forms a basis for a topology. For computational as well as for theoretical purposes, we shall describe all the topological concepts in terms of balls.

4.1 Distance and Metric Spaces

Definition 4.1.1 A function $d\colon X \times X \to \mathbb{R}$ is said to be a *distance* on X is it satisfies the following properties:

(i) Positive definiteness: $d(x, y) \geq 0$ for all $x, y \in X$. In addition, $d(x, y) = 0$ if and only if $x = y$;
(ii) Symmetry: $d(x, y) = d(y, x)$ for all $x, y \in X$;
(iii) Triangle inequality: $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in X$.

The set X together with the distance d, written (X, d), is called a *metric space*.

Example 4.1 The *Euclidean distance*, or *usual distance*, on $\mathbb{R}^n$ is

$$d_u(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2},$$

where $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$. ◊

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_4

Example 4.2 In any set X, the *discrete distance* is defined by

$$d(x, y) = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y. \end{cases} \tag{4.1}$$

The space (X, d) is called the *discrete metric space*. To check the triangle inequality $d(x, y) \leq d(x, z) + d(y, z)$, note that each one these three numbers is 0 or 1. If $x \neq z$, then $d(x, z) = 1$ and the inequality holds trivially. If $x = z$, the inequality is $d(x, y) \leq d(y, x)$, which is also true ♢

Example 4.3 Let (X, d) be a metric space and $A \subset X$. Define on A the *induced distance* by

$$d_A(x, y) = d(x, y), \quad x, y \in A,$$

and (A, d_A) is called a *metric subspace*. We write simply d if A is understood. Let us observe that d_A and d are different because the domains are distinct. ♢

Example 4.4 Let (V, g) be a Euclidean metric vector space where g is a positive definite metric (see Sect. 2.3). The *metric distance* is defined by

$$d(u, v) = |u - v| = \sqrt{g(u - v, u - v)}, \quad u, v \in V.$$

Recall that the triangle inequality is a consequence of the Cauchy-Schwarz inequality. A particular example is $\mathbb{R}^n$ with the Euclidean metric. ♢

Example 4.5 Let $\mathsf{M}_n(\mathbb{R})$ be the vector space of squared matrices of order n of real numbers. This space can be endowed with the structure of a Euclidean metric vector space as follows. The sum of two matrices and the product by scalars are defined by

$$A + B = (a_{ij} + b_{ij}), \quad \lambda \cdot A = (\lambda a_{ij}),$$

where $A = (a_{ij})$, $B = (b_{ij})$ and $\lambda \in \mathbb{R}$. Define the metric

$$g(A, B) = \text{trace } (AB^t).$$

Here B^t is the transpose matrix of B (the (i, j)-element of B^t is b_{ji}) and the trace is the sum of the elements of its principal diagonal. An elementary computation gives

$$g(A, B) = \sum_{i,j=1}^{n} a_{ij} b_{ij}.$$

The metric g can be expressed as follows. Consider a matrix $A = (a_{ij})$; take the first row as an n-tupla of $\mathbb{R}^n$, followed by the second row and so forth, until obtaining a vector of $\mathbb{R}^{n^2}$. The mapping

$$A \in \mathsf{M}_n(\mathbb{R}) \mapsto (a_{11}, \dots, a_{1n}, a_{21}, a_{22}, \dots, a_{2n}, \dots, a_{n1}, \dots, a_{nn}) \in \mathbb{R}^{n^2} \tag{4.2}$$

is an isomorphism between the vector space $\mathsf{M}_n(\mathbb{R})$ and $\mathbb{R}^{n^2}$. Then $g(A, B) = \sum_{i,j=1}^{n} a_{ij} b_{ij}$ is nothing more than the Euclidean product of A and B regarded as vectors of $\mathbb{R}^{n^2}$. The distance between A and B is

$$d(A, B) = \sqrt{g(A - B, A - B)} = \sqrt{\sum_{i,j=1}^{n} (a_{ij} - b_{ij})^2}.$$

Thus $\mathsf{M}_n(\mathbb{R})$ is merely the Euclidean space $\mathbb{R}^{n^2}$ according to (4.2). $\Diamond$

Example 4.6 On $\mathbb{R}^n$, define two distances d_t and d_∞ by

$$d_t(x, y) = \sum_{i=1}^{n} |x_i - y_i|,$$

$$d_\infty(x, y) = \max\{|x_1 - y_1|, \dots, |x_n - y_n|\}.$$

The relation of d_t and d_∞ with the Euclidean distance d_u is shown in Fig. 4.1. It is clear that $d_\infty \le d_u \le d_t$. The distance d_t is called the *taxi distance*. The reason is the following. Think of a taxi that travels between two points of a city (a subset of the Euclidean plane). A city is made by streets that meet perpendicularly. Then the taxi describes a path formed by orthogonal segments and the length is the distance d_t. The positive definiteness and symmetry properties are trivial for d_t and d_∞. The triangle inequality for d_t is clear because if $x, y, z \in \mathbb{R}^n$, then

$$|x_i - y_i| \le |x_i - z_i| + |z_i - y_i| \tag{4.3}$$

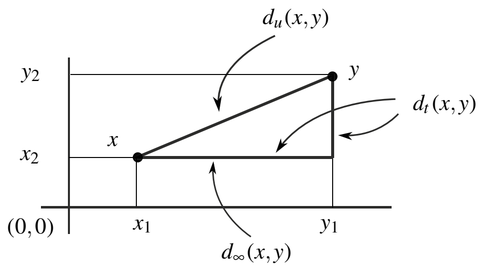


Fig. 4.1 Comparison of the distances d_u, d_t and d_∞ in $\mathbb{R}^2$. The Euclidean distance d_u corresponds with the hypotenuse of the right triangle determined by x and y, d_t is the sum of the two legs and d_∞ is the largest leg

for all $1 \le i \le n$, hence we deduce $d_t(x, y) \le d_t(x, z) + d_t(z, y)$. For d_∞, we can bound the right side (4.3) from above to obtain $|x_i - y_i| \le d_\infty(x, z) + d_\infty(z, y)$. Since this holds for all $1 \le i \le n$, then $d_\infty(x, y) \le d_\infty(x, z) + d_\infty(z, y)$. $\diamondsuit$

Example 4.7 A *normed space* is a real vector space X furnished with a map $\|\cdot\| : X \to \mathbb{R}$, called a *norm*, with the following properties:

(i) $\|x\| \ge 0$ for all $x \in X$ and $\|x\| = 0$ if and only if $x = 0$.
(ii) $\|\lambda x\| = |\lambda|\|x\|$ for all $x \in X$ and $\lambda \in \mathbb{R}$ (homogeneity).
(iii) $\|x + y\| \le \|x\| + \|y\|$ for all $x, y \in X$ (triangle inequality).

Here 0 denotes the unit element of the vector space. The distance associated to the norm is defined by $d(x, y) = \|x - y\|$; in particular $\|x\| = d(x, 0)$. To check the triangle inequality for d, we have from (iii) that

$$d(x, z) = \|(x - y) + (y - z)\| \le \|x - y\| + \|y - z\| = d(x, y) + d(y, z).$$

Normed spaces constitute a large family of metric spaces that appear in different fields of mathematics. On $\mathbb{R}^n$, the Euclidean norm is the modulus of a vector. But on the underlying set $\mathbb{R}^n$ there are another norms, as for example,

$$\|x\|_t = \sum_{i=1}^{n} |x_i|, \quad \|x\|_\infty = \max\{|x_i| : 1 \le i \le n\}.$$

The distances for these norms are d_t and d_∞, respectively, but the norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ do not come from a positive definite metric. The reason is the following. In any Euclidean vector space (V, g), the metric g can be expressed in terms of the norm because $\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2g(x, y)$ implies

$$g(x, y) = \frac{1}{2}\left(\|x + y\|^2 - \|x\|^2 - \|y\|^2\right).$$

Since $g(\lambda x, y) = \lambda g(x, y)$ for all $x, y \in \mathbb{R}^n$, $\lambda \in \mathbb{R}$, if $\|\cdot\|_t$ comes from a metric g, we would have

$$\|\lambda x + y\|^2 - \|y\|^2 = \lambda(\|x + y\|^2 - \|y\|^2).$$

If $x = (1, 0, \ldots, 0)$ and $y = (0, \ldots, 0, 1)$, then the above identity says $|\lambda| = \lambda$ for all $\lambda \in \mathbb{R}$, which is not true.

We summarize with the next chain of implications:

$$\text{Euclidean vector spaces} \Rightarrow \text{normed spaces}$$
$$\Rightarrow \text{metric spaces} \Rightarrow \text{topological spaces}.$$

$\diamondsuit$

Example 4.8 Normed spaces have, in general, infinite dimension. We present one such space. Let X be the set of sequences of real numbers in which all terms but finitely many are zero, where the sum of elements of X and the product by scalars are standardly defined. We show that the dimension of X is infinite. Suppose, on the contrary, that $B = \{e_1, \dots, e_k\}$ is a basis of X. For each sequence e_j, let $m_j \in \mathbb{N}$ be the last position of the sequence e_j whose m_jth element is not zero. Let n_0 be a natural number such that $n_0 > m_j$ for all $1 \leq j \leq k$. Then (x_n) consisting of 0 in all terms except in the n_0th one that is 1, cannot be expressed as a linear combination of the basis B. On the other hand,

$$\|x\| = \sup\{|x_n| : n \in \mathbb{N}\}$$

defines a norm on X. This supremum exists because all numbers $|x_n|$ are 0, except a finite set. $\diamondsuit$

Definition 4.1.2 A *ball* of radius $r > 0$ with center $x \in X$ is defined by

$$B_r(x) = \{y \in X : d(x, y) < r\}.$$

As a consequence, if $A \subset X$ and $x \in A$, then the ball $B_r^A(x)$ in (A, d_A) is $B_r^A(x) = B_r(x) \cap A$.

Example 4.9 We illustrate the balls on $\mathbb{R}^2$ for the distances d_u, d_t and d_∞ (in $\mathbb{R}$ the three distances coincide). For each one, let us express the balls of radius r centered at the origin $p = (0, 0)$ (Fig. 4.2).

1. For d_u,

$$B_r(p) = \{(x, y) \in \mathbb{R}^2 : \sqrt{x^2 + y^2} < r\}$$

is merely a round disc centered at p with radius r whose border is not included.

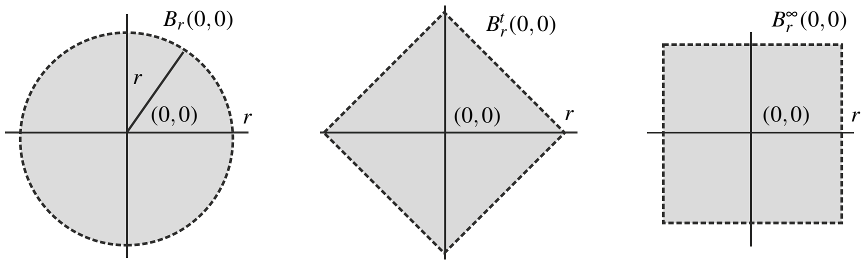


Fig. 4.2 The balls in Example 4.9. In the three cases, the border is not included in the ball

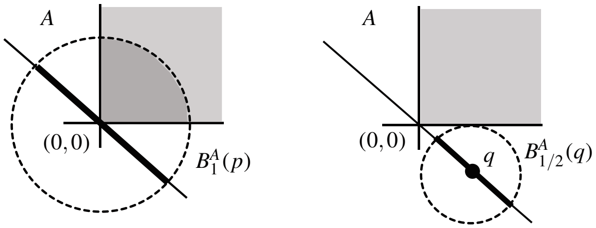


Fig. 4.3 Balls are not "round" in the set A of Example 4.10

2. For d_t,

$$B_r^t(p) = \{(x, y) \in \mathbb{R}^2 : |x| + |y| < r\}$$

is a square centered at the origin whose vertices are the points $(\pm r, 0)$ and $(0, \pm r)$.
3. For d_∞,

$$B_r^\infty(p) = \{(x, y) \in \mathbb{R}^2 : \max\{|x|, |y|\} < r\}$$

is a square again but now the vertices are the points $(\pm r, \pm r)$.

$\diamondsuit$

Example 4.10 Consider the Euclidean distance induced on

$$A = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\} \cup \{(x, -x) \in \mathbb{R}^2 : x \in \mathbb{R}\}.$$

Take $r = 1$ and $p = (0, 0)$. Then the ball $B_r^A(p)$ appears depicted in Fig. 4.3, left. This ball is a segment of length 2 in the straight line of equation $y = -x$ together with part of the first quadrant. If $r = \frac{1}{2}$ and $q = (\frac{1}{2}, -\frac{1}{2})$, the ball $B_r^A(q)$ is a segment of length 1 in the straight line $y = -x$. $\diamondsuit$

Example 4.11 Let $A = [0, 1] \cup (2, 4] \cup \{5\}$ be equipped with the Euclidean distance and consider three balls with identical radius $r = 1$ but different centers:

$$B_1^A(0) = [0, 1), \quad B_1^A(3) = (2, 4), \quad B_1^A(5) = \{5\}.$$

1. The ball $B_1^A(0)$ is an interval that is not centered at $x = 0$, whereas $B_1^A(5)$ is one point.
2. The balls $B_1^A(0)$ and $B_1^A(3)$ have equal radius but the length of $B_1^A(3)$ is double that of $B_1^A(0)$.
3. Finally, the ball $B_2^A(0) = [0, 1]$ with double the radius of $B_1^A(0)$ has equal length. Let us also observe that $B_6^A(0) = B_{100}^A(0) = A$.

$\diamondsuit$

Definition 4.1.3 A metric space is said to be *bounded* if it is included in a ball. A subset of a metric space is said to be bounded if, as a metric subspace, it is bounded.

4.2 Topology on a Metric Space

As in Euclidean space, we topologize a metric space using the collection of all balls as a basis for the topology.

Theorem 4.2.1 *In a metric space* (X, d)*, the collection of all balls*

$$\beta = \{B_r(x) : x \in X, r > 0\}$$

is a basis for a topology τ_d *called the metric topology determined by* d*.*

It is clear that the collection of balls $\beta_x = \{B_r(x) : r > 0\}$ centered at x is a basis of neighborhoods of x. We can also reduce the number of balls even further, as asserted in the next result.

Proposition 4.2.2 *Let* X *be a metric space and* $x \in X$*. The next collections of balls are bases of neighborhoods of* x*:* $\{B_r(x) : r \in \mathbb{Q}^+\}$, $\{B_r(x) : r \in (\mathbb{R} - \mathbb{Q})^+\}$, $\{B_r(x) : 0 < r < r_0\}$ *for a given* $r_0 > 0$ *and* $\{B_{r_n}(x) : n \in \mathbb{N}\}$ *if* (r_n) *a sequence of positive numbers such that* $(r_n) \to 0$.

Example 4.12 We prove that the metric topology τ_d induced by the discrete distance (Example 4.2) coincides with the discrete topology τ_D. It suffices that all singletons are open in τ_d. But $B_1(x) = \{x\}$, so $\{x\}$ is open.

In fact, we can replace the number 1 in the definition of the discrete distance by another positive number. Certainly, if $\lambda > 0$,

$$d_\lambda(x, y) = \begin{cases} 0, & \text{if } x = y \\ \lambda, & \text{if } x \neq y \end{cases}$$

is a distance defined on X and the metric topology induced by d_λ is the discrete topology again because $B_\lambda(x) = \{x\}$ is open for all $x \in X$. $\diamondsuit$

In this example we observe that all d_λ lead to identical metric topologies. An interesting question is when, given a point-set X endowed with two distances, do the corresponding metric topologies coincide.

Definition 4.2.3 Two distances on a given set are *equivalent* if the metric topologies coincide.

Theorem 4.2.4 *If* d_1 *and* d_2 *are two distances on a set* X*, then* $\tau_1 \subset \tau_2$ *if and only if for every* $x \in X$ *and* $B_r^1(x)$*, there is* $s > 0$ *such that* $B_s^2(x) \subset B_r^1(x)$*. As a consequence, if there are* $k, m > 0$ *such that* $d_1 \leq k \cdot d_2$ *and* $d_2 \leq m \cdot d_1$*, then* d_1 *and* d_2 *are equivalent.*

Example 4.13 We see that the distances d_u, d_t and d_∞ of $\mathbb{R}^n$ are all equivalent. Consider $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$ and let $a_i = |x_i - y_i|$. Then

$$d_u(x, y) = \sqrt{\sum_{i=1}^{n} a_i^2}, \quad d_t(x, y) = \sum_{i=1}^{n} a_i, \quad d_\infty(x, y) = \max\{a_i : 1 \le i \le n\}.$$

Since $a_i \ge 0$, we infer

$$\max\{a_i : 1 \le i \le n\} \le \sqrt{\sum_{i=1}^{n} a_i^2} \le \sum_{i=1}^{n} a_i \le n \cdot \max\{a_i : 1 \le i \le n\};$$

that is, $d_\infty \le d_u \le d_t \le n \cdot d_\infty$ and we conclude using Theorem 4.2.4. ♢

Example 4.14 In the previous example, the balls with the three distances are different so the bases. A bit different occurs with the discrete distances of Example 4.12. Let d_1 and d_2 be the distances for $\lambda = 1$ and $\lambda = 2$ respectively, which are equivalent. Here the balls are different, for example $B_2^1(x) = X$ and $B_2^2(x) = \{x\}$. However, the bases β_1 and β_2 for both topologies coincide, $\beta_1 = \beta_2 = \{X\} \cup \{\{x\} : x \in X\}$. ♢

Example 4.15 The converse of the last part of theorem is false in general and it may occur that $\tau_1 \subset \tau_2$ but not an inequality of type $d_1 \le k \cdot d_2$. For example, let $X = \mathbb{R}$ endowed with the Euclidean distance d_u and the discrete distance d. By Example 4.12, the metric topology τ_d is discrete so $\tau_u \subset \tau_d$. However, there is no $k > 0$ such that $d_u \le k \cdot d$; on the contrary, for $x = 2k$ and $y = 0$ we would have

$$2k = d_u(2k, 0) \le k \cdot d(2k, 0) = k \cdot 1 = k,$$

which is not possible. ♢

Proposition 4.2.2 asserts that one can choose a basis with a countable number of elements.

Definition 4.2.5 A topological space satisfies the *first axiom of countability* (or is said to be *first countable*) if there is a countable basis of neighborhoods for all points.

Consequently, *every metric space is first countable*. Analogously to the first axiom of countability, a topological space satisfies the *second axiom of countability* (or it is *second countable*) if there is a countable basis for the topology. It is immediate that second countability implies first countability.

Example 4.16 The real line $\mathbb{R}$ with the discrete distance is not second countable because $\beta = \{\{x\} : x \in \mathbb{R}\}$ is the smallest basis. ♢

Theorem 4.2.6 *Euclidean spaces are second countable. In consequence, every subset of Euclidean space is second countable.*

Proof We prove that the countable collection of balls

$$\beta = \{B_r(q) : r \in \mathbb{Q}^+, q \in \mathbb{Q}^n\}$$

is a basis for the Euclidean topology. Let O be an open set and let $x \in O$. By Proposition 4.2.2, there is $r \in \mathbb{Q}^+$ such that $x \in B_r(x) \subset O$. Let $x = (x_1, \ldots, x_n)$. For each $1 \leq i \leq n$, let $q_i \in \mathbb{Q}$ be a rational number such that $|x_i - q_i| < r/(2\sqrt{n})$ and let $q = (q_1, \ldots, q_n) \in \mathbb{Q}^n$. Note that

$$|x - q| = \sqrt{\sum_{i=1}^{n} (x_i - q_i)^2} < \sqrt{\frac{r^2}{4}} = \frac{r}{2}.$$

We claim that $x \in B_{r/2}(q) \subset B_r(x) \subset O$, which completes the proof. It is obvious that $x \in B_{r/2}(q)$. For the inclusion $B_{r/2}(q) \subset B_r(x)$, if $z \in B_{r/2}(q)$,

$$|x - z| \leq |x - q| + |q - z| < \frac{r}{2} + \frac{r}{2} = r.$$

The last part is a consequence of (2) of Theorem 2.5.2. □

For topological spaces in general, a property $\mathscr{P}$ is said to be *hereditary* if whenever a topological space satisfies $\mathscr{P}$, then every topological subspace also satisfies $\mathscr{P}$. Therefore, first and second countability are hereditary properties.

We relate the notions of the relative topology and metric spaces. Consider a metric space (X, d) with its metric topology τ_d. On a subset $A \subset X$, there are defined two topologies. One of them is the relative topology from τ_d and denoted by $\tau_{d|A}$. The second one is the metric topology τ_{d_A} determined by the induced distance d_A defined in Example 4.3.

Proposition 4.2.7 *If A is a subset of a metric space (X, d), then $\tau_{d|A} = \tau_{d_A}$.*

We conclude this section by asking if there are topological spaces whose topologies are not the metric topologies. The question can be formulated as follows.

> **Problem of metrization.** *Given a topological space (X, τ), does a distance d exist on X such that the metric topology τ_d coincides with τ?*

A topological space is said to be *metrizable* if its topology is a metric topology. Examples of metrizable topologies are the Euclidean topology, discrete topology (Example 4.2) and relative topology (Proposition 4.2.7). In order to exhibit an example of a non-metrizable space, we work with the following argument. Metric spaces, as topological spaces, fulfill some topological properties that topological spaces in general may not. For example, a metric space is first countable and consequently, a topological space that is not first countable is not metrizable. We look back to the cofinite topology.

Example 4.17 We show that $(\mathbb{R}, \tau_{CF})$ is not first countable. Suppose that $x \in \mathbb{R}$ has a countable basis of neighborhoods $\beta_x = \{U_n : n \in J\}$, where J is a countable set of indices. Since every neighborhood is an open set, then $U_n = \mathbb{R} \setminus F_n$ and $F_n \subset \mathbb{R}$ is finite with $x \notin F_n$. Then $\bigcup_{n \in J} F_n$ is countable because it is the countable union of finite sets (in case that J is finite, this set is finite as well). Thus $\bigcup_{n \in J} F_n \neq \mathbb{R}$ because $\mathbb{R}$ is not countable. Let $y \in \mathbb{R}$ distinct of x and $y \notin F_n$ for all $n \in J$. Then $U = \mathbb{R} \setminus \{y\}$ is a neighborhood of x, so there is $U_n \in \beta_x$ such that $U_n \subset \mathbb{R} \setminus \{y\}$. Taking complements, $y \in F_n$, obtaining a contradiction. ♢

Proposition 4.2.8 *Metric spaces are Hausdorff.*

Example 4.18 With the aid of this proposition, we show again that the cofinite topology is not metrizable by demonstrating that any two neighborhoods of two points must intersect. As a convention in this book, we say that *two sets intersect* if their intersection is non-void. Let $U \in \mathscr{U}_x$ and $V \in \mathscr{U}_y$ and assume $U \cap V = \emptyset$. Then

$$X = X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$$

is finite because it is union of two finite sets, but this is not possible since X is infinite. Other non-Hausdorff topologies are the trivial topology, included point topology and right order topology on $\mathbb{R}$ (all these spaces are first countable). ♢

4.3 Interior and Closure in a Metric Space

We learned that the balls centered at a point is a basis of neighborhoods. This permits us to characterize the interior and closure of a set in terms of balls.

Proposition 4.3.1 *Let X be a metric space, $A \subset X$ and $x \in X$.*

1. *$x \in \mathring{A}$ if and only if there is $r > 0$ such that $B_r(x) \subset A$.*
2. *$x \in \overline{A}$ if and only if $B_r(x) \cap A \neq \emptyset$ for all $r > 0$.*

Given a ball $B_r(x)$, the reader may think, at least intuitively, of the set consisting of all points at distance r from x as the "border" of the ball. Also, that this border coincides with the boundary of the ball (in addition, it looks like that plain language confirms this idea). First we need the following concept. Let (X, d) be a metric space, $a \in X$ and $r > 0$. The *sphere of radius r and center a* is defined by

$$\mathbb{S}_r(a) = \{x \in X : d(x, a) = r\}.$$

So, the space can be separated in three disjoint sets, points whose distance to a is less, equal and bigger than r. The next result reflects our intuition in $\mathbb{R}^n$.

Theorem 4.3.2 *In Euclidean space $\mathbb{R}^n$, we have the following identities:*

1. $\operatorname{int}(B_r(a)) = B_r(a) = \{x \in \mathbb{R}^n : d(x, a) < r\}$.
2. $\operatorname{Bd}(B_r(a)) = \mathbb{S}_r(a) = \{x \in \mathbb{R}^n : d(x, a) = r\}$.
3. $\operatorname{ext}(B_r(a)) = \{x \in \mathbb{R}^n : d(x, a) > r\}$.

Proof The first identity is trivial because any ball is an open set. The equality for the exterior is a consequence if we prove the set-theoretical identity $\mathrm{Bd}(B_r(a)) = \mathbb{S}_r(a)$. We do this by double inclusion.

1. We prove $\mathbb{S}_r(a) \subset \mathrm{Bd}(B_r(a))$. Let $x \in \mathbb{S}_r(a)$ and let $B_\delta(x)$ be a ball of the basis of neighborhoods whose radius may be assumed to be $\delta \leq r$ (Proposition 4.2.2). We must prove that $B_\delta(x) \cap B_r(a) \neq \emptyset$ and $B_r(x) \cap (\mathbb{R}^n \setminus B_r(a)) \neq \emptyset$. The latter is immediate because x is an element of this intersection since $d(x, a) = r$. For the first intersection, we prove that $y = x - \delta(x - a)/(2r)$ belongs to this intersection. First,
$$|y - a| = \left|(1 - \tfrac{\delta}{2r})(x - a)\right| = r - \tfrac{\delta}{2} < r,$$
so $y \in B_r(a)$. Also $|y - x| = \delta/2 < \delta$, hence $y \in B_\delta(x)$.
2. We see that $\mathrm{Bd}(B_r(a)) \subset \mathbb{S}_r(a)$. Let $x \in \mathrm{Bd}(B_r(a))$ and assume $|x - a| \neq r$. If $|x - a| < r$ then $x \in B_r(a) = \mathrm{int}(B_r(a))$, which is not possible because x is a boundary point. Suppose $|x - a| > r$. Let $\delta = |x - a| - r > 0$. If $z \in B_\delta(x) \cap B_r(a)$, the triangle inequality leads to
$$|x - a| \leq |x - z| + |z - a| < \delta + r = |x - a|,$$
which is a contradiction. Thus $B_\delta(x) \cap B_r(a) = \emptyset$, which is absurd because x is a boundary point of $B_r(a)$.

$\square$

As the reader can suspect, the above theorem is not true in all metric spaces. Theorem 4.3.2 uses the affine structure of $\mathbb{R}^n$ because the proof of the inclusion $\mathbb{S}_r(a) \subset \mathrm{Bd}(B_r(a))$ is based on the sums of vectors and products by scalars. However, the inclusion $\mathrm{Bd}(B_r(a)) \subset \mathbb{S}_r(a)$ holds in any metric space, hence

$$\mathrm{Bd}(B_r(a)) \subset \mathbb{S}_r(a), \quad \{x \in X : d(x, a) > r\} \subset \mathrm{ext}(B_r(a)). \tag{4.4}$$

The next two examples show that these inclusions may be proper.

Example 4.19 Consider the subset of $\mathbb{R}^2$ given by

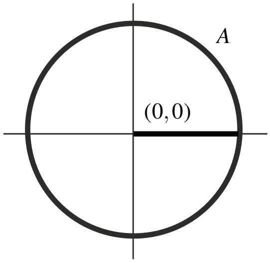
$$A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} \cup \{(x, 0) \in \mathbb{R}^2 : 0 \leq x \leq 1\}.$$

See Fig. 4.4. Let $a = (0, 0)$ and $r = 1$. Then

$$B_1^A(a) = B_1(a) \cap A = \{(x, 0) \in \mathbb{R}^2 : 0 \leq x < 1\}.$$

We see that $\mathrm{Bd}(B_1^A(a)) = \{(1, 0)\}$ and thus $\mathrm{ext}(B_1^A(a)) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} \setminus \{(1, 0)\}$. Because the boundary points are not interior, $\mathrm{Bd}(B_1^A)(a) \subset \{(x, y) \in$

Fig. 4.4 The set A of Example 4.19



$\mathbb{R}^2 : x^2 + y^2 = 1\}$. The point $p = (1, 0)$, which does not belong to $B_1^A(a)$, is a boundary point because for any $r > 0$,

$$\begin{aligned} B_r^A(p) \cap B_1^A(a) &= \{(x, y) \in \mathbb{R}^2 : 0 \leq x < 1, 1 - x < r\} \\ &= \{(x, 0) \in \mathbb{R}^2 : 1 - r < x < 1\} \end{aligned}$$

is non-void. We see that the rest of the points are not boundary points. Let $(x, y) \in \mathbb{R}^2$, $x^2 + y^2 = 1$ and $(x, y) \neq p$. It is clear that if $x \leq 0$, $B_1^A((x, y)) \cap B_1^A(a) = \emptyset$ and if $x > 0$, $B_{|y|}^A((x, y)) \cap B_1^A(a) = \emptyset$.

Finally, $\{(x, y) \in A : d((x, y), a) > 1\} = \emptyset$ and

$$\mathbb{S}_1^A(a) = \mathbb{S}_1(a) \cap A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

This gives proper inclusions in (4.4). $\lozenge$

Example 4.20 Let $A = [0, 1] \cup (2, 4] \cup \{5\}$ be equipped with the Euclidean distance and let drop the index ^{A} on the subspace (A, d_u).

1. Take $a = 0$ and $r = 1$. Then $B_r(a) = [0, 1)$. Now $\mathbb{S}_r(a) = \{1\}$ and $\mathrm{Bd}(B_r(a)) = \emptyset$.
2. Take $a = 0$ and $r = 3/2$. Now $B_r(a) = [0, 1]$ and $\mathbb{S}_r(a) = \mathrm{Bd}(B_r(a)) = \emptyset$.
3. Let $a = 5$ and $r = 1$, then $B_r(a) = \{5\}$, $\mathrm{Bd}(B_r(a)) = \emptyset$ and $\mathbb{S}_r(a) = \{4\}$.
4. If $a = 5$, $r = 2$, then $B_r(a) = \{5\} \cup (3, 4]$ and $\mathrm{Bd}(B_r(a)) = \mathbb{S}_r(a) = \{3\}$. $\lozenge$

4.4 Convergence of Sequences in a Metric Space

The convergence of sequences in metric spaces is discussed in this section. First, metric spaces are Hausdorff and consequently, the limit of a convergent sequence is unique. On the other hand, because the balls centered at a point is a basis of neighborhoods, the following characterization of the convergence is immediate.

Proposition 4.4.1 *In a metric space, $(x_n) \to x$ if and only if $(d(x_n, x)) \to 0$, or equivalently,*

$$\text{for all } \epsilon > 0 \text{ there is } \nu \in \mathbb{N} \text{ such that if } n \geq \nu \text{ then } d(x_n, x) < \epsilon.$$

Theorem 4.4.2 *Let X be a metric space, $A \subset X$ and $x \in X$.*

1. *$x \in \overset{\circ}{A}$ if and only if for each sequence (x_n) converging to x, there is $\nu \in \mathbb{N}$ such that $x_n \in A$ for $n \geq \nu$.*
2. *$x \in \overline{A}$ if and only if there is a sequence $(a_n) \subset A$ such that $(a_n) \to x$.*
3. *$x \in \mathrm{ext}(A)$ if and only if for each sequence (x_n) converging to x, there is $\nu \in \mathbb{N}$ such that $x_n \in X \setminus A$ for $n \geq \nu$.*
4. *$x \in \mathrm{Bd}(A)$ if and only if there are sequences in A and in $X \setminus A$ converging to x.*

As a consequence, if x is a point of accumulation of a sequence (x_n), then there is a subsequence of (x_n) that converges to x.

Since open sets are characterized by sequences, we can assert

Metric topologies are characterized in terms of convergence of sequences.

We will say that *the topology is sequentially characterized*. As expected, the convergence of sequences in Euclidean space $\mathbb{R}^m$ can be reduced to considering convergence of real numbers.

Proposition 4.4.3 *Let $(x_n) \subset \mathbb{R}^m$ and $x \in \mathbb{R}^m$ and denote $x_n = (x_1^n, \ldots, x_m^n)$ and $x = (x_1, \ldots, x_m)$. Then $(x_n) \to x$ if and only if $(x_j^n) \to x_j$ for all $j = 1, \ldots, m$.*

The next result is a consequence of Proposition 4.4.3 and the properties of the convergent sequences of $\mathbb{R}$.

Corollary 4.4.4 *In $\mathbb{R}^m$, assume $(x_n) \to x$ and $(y_n) \to y$.*

1. *For $\lambda, \mu \in \mathbb{R}$, we have $(\lambda x_n + \mu y_n) \to \lambda x + \mu y$.*
2. *If $(y_n) \subset \mathbb{R}$, then $(y_n x_n) \to yx$.*
3. *If $(y_n) \subset \mathbb{R}$ and $y \neq 0$, then $(x_n/y_n) \to x/y$.*

The following result is also expected for the Euclidean topology. As it is usual, we will identify any pair $(x, y) \in R^p \times \mathbb{R}^q$ with a $(p+q)$-tupla in $\mathbb{R}^{p+q}$ simply by $(x, y) \mapsto (x_1, \ldots, x_p, y_1, \ldots, y_q)$, regarding the Cartesian product $\mathbb{R}^p \times \mathbb{R}^q$ as $\mathbb{R}^{p+q}$.

Corollary 4.4.5 *Let $A \subset \mathbb{R}^p$ and $B \subset \mathbb{R}^q$. Then A and B are open (resp. closed) sets if and only if $A \times B$ is open (resp. closed) in $\mathbb{R}^{p+q}$.*

As a consequence, a basis of the Euclidean topology on $\mathbb{R}^n$ is $\beta = \{O_1 \times \ldots \times O_n : O_i \in \tau_u\}$, where τ_u is the Euclidean topology of $\mathbb{R}$.

Example 4.21 Define the *topologist's sine* as

$$X = \left\{(x, y) \in \mathbb{R}^2 : x > 0,\ y = \sin\left(\tfrac{1}{x}\right)\right\} \cup \{(0, 0)\}.$$

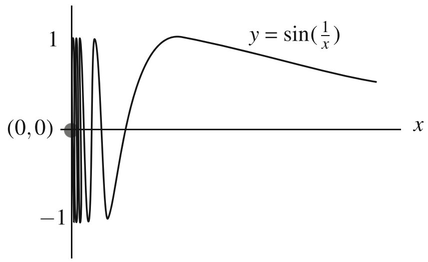


Fig. 4.5 The topologist's sine

See Fig. 4.5. Notice that $X = G(f) \cup \{(0, 0)\}$, where $G(f)$ is the graph of the continuous function

$$f\colon (0, \infty) \to \mathbb{R}, \quad f(x) = \sin\left(\tfrac{1}{x}\right).$$

We prove that $(0, 0)$ is an adherent point of $G(f)$. For showing this, it suffices to see that $\left((\frac{1}{n\pi}, \sin(n\pi))\right) = \left((\frac{1}{n\pi}, 0)\right) \to (0, 0)$. In fact, the closure of $G(f)$ is $\overline{G(f)} = G(f) \cup (\{0\} \times [-1, 1])$. To check this identity by a double inclusion, first note that the points of $\{0\} \times [-1, 1]$ are adherent. So, if $y_0 \in [-1, 1]$, let $x_0 \in \mathbb{R}$ be any real number such that $\sin x_0 = y_0$. Then $\left((\frac{1}{x_0+2\pi n}, \sin(x_0 + 2\pi n))\right) \to (0, y_0)$. On the other hand, if $(x, y) \in \overline{G(f)}$, then there is $((x_n, y_n)) \subset G(f)$ such that $((x_n, y_n)) \to (x, y)$. Since $y_n = \sin(1/x_n)$, then $|y_n| \le 1$; in particular, letting $n \to \infty$ in this inequality, we deduce $y \in [-1, 1]$. Furthermore, $(x_n) \to x$ and as $x_n \in (0, \infty)$, then $x \in [0, \infty)$. If $x > 0$, then $(y_n) \to \sin(1/x)$, so $(x, y) = (x, \sin(1/x)) \in G(f)$. If $x = 0$ then $(x, y) = (0, y) \in \{0\} \times [-1, 1]$. ♢

This section concludes by characterizing an adherent point in a metric space.

Definition 4.4.6 If A is a subset of a metric space X and $x \in X$, the *distance from* x *to* A is defined by

$$d(x, A) = \inf\{d(x, a) : a \in A\}. \tag{4.5}$$

This infimum exists because the set of real numbers $\{d(x, a) : a \in A\}$ is bounded from below by 0. Thus if $x \in A$ and because the number $d(x, x) = 0$ belongs to the set $\{d(x, a) : a \in A\}$, then $d(x, A) = 0$. However, we see that there are more points that do not belong to A at distance equal to 0.

Example 4.22 Let $X = \mathbb{R}$ with the Euclidean distance and $A = (0, 1)$.

1. We see that $d(0, A) = 0$. Since $1/n \in A$, the numbers $d(1/n, 0) = 1/n$ are included in the set $\{d(0, a) : a \in A\}$. As the infimum of $\{1/n : n \in \mathbb{N}\}$ is 0, then $d(0, A) = 0$. Here we are using that if $A, B \subset \mathbb{R}$, $A \subset B$ and B is bounded from below, then $\inf(A) \ge \inf(B)$.

2. We see $d(2, A) = 1$. Taking the elements of A of type $1 - 1/n$ for $n > 1$, we have $d(2, 1 - 1/n) = 1 + 1/n$. Then the set $\{d(x, a) : a \in A\}$ contains the set $\{1 + 1/n : n \in \mathbb{N}\}$, whose infimum is 1. This yields $d(2, A) \leq 1$. On the other hand, for any $a \in A$, $d(2, a) = |2 - a| = 2 - a > 1$ because $a < 1$. Thus $1 \leq d(2, A)$, concluding $d(2, A) = 1$. ♢

Proposition 4.4.7 *Let A be a subset of a metric space X and let $x \in X$. Then $x \in \overline{A}$ if and only if $d(x, A) = 0$.*

Proof

1. Let $x \in \overline{A}$. Thanks to Theorem 4.4.2, there is $(a_n) \subset A$ such that $(a_n) \to x$. As $\{d(a_n, x) : n \in \mathbb{N}\} \subset \{d(a, x) : a \in A\}$,

$$\inf\{d(a_n, x) : n \in \mathbb{N}\} \geq \inf\{d(x, a) : a \in A\}.$$

Since $(d(a_n, x)) \to 0$, the infimum of the set in the left-hand side is 0, proving that the infimum in the right-hand side is also 0, so $d(x, A) = 0$.

2. Suppose $d(x, A) = 0$ and we see that $B_r(x) \cap A \neq \emptyset$ for all $r > 0$. As

$$0 = d(x, A) = \inf\{d(x, a) : a \in A\} < r,$$

the definition of infimum implies that there is $a \in A$ such that $0 \leq d(x, a) < r$. Then $a \in B_r(x)$ so $B_r(x) \cap A \neq \emptyset$.

□

The characterization of adherent points in terms of distances to the set provides similar characterizations for the other types of points. Using Proposition 3.3.2, we have:

1. $x \in \mathring{A}$ if and only if $d(x, X \setminus A) > 0$,
2. $x \in \mathrm{Bd}(A)$ if and only if $d(x, A) = d(x, X \setminus A) = 0$ and
3. $x \in \mathrm{ext}(A)$ if and only if $d(x, A) > 0$.

4.5 Worked Exercises

We know that the convergent sequences in the discrete topology are the eventually constant sequences. We revisit this result using the discrete distance.

Exercise 4.1 Find the convergent sequences in the discrete metric space.

Solution Let $(x_n) \to x$ be a convergent sequence in the discrete metric space. Let $\epsilon = 1$. By definition of convergence, there is $\nu \in \mathbb{N}$ such that $d(x_n, x) < 1$ for $n \geq \nu$ and by definition of d, $d(x_n, x) = 0$, so $x_n = x$ for all $n \geq \nu$. Thus the convergent sequences in this space are the eventually constant sequences. □

Exercise 4.2 Prove that the topology of a finite metric space is discrete.

Solution If X has only one element, the result is trivial. Suppose $X = \{x_1, \ldots, x_n\}$, $n \geq 2$ and let

$$r = \min\{d(x_i, x_j) : i \neq j, 1 \leq i, j \leq n\}.$$

Note that this number is positive. The balls $B_{r/2}(x_i)$ are open, but $B_{r/2}(x_i) = \{x_i\}$ by definition of r. Thus every singleton is open, so the metric topology is discrete. □

We give two equivalent distances that do not satisfy the last hypothesis of Theorem 4.2.4.

Exercise 4.3 Let $X = \{1/n : n \in \mathbb{N}\} \cup \mathbb{N}$ and consider on X the Euclidean distance d_u and the discrete distance d. Prove that the two distances are equivalent but there are no $k, m > 0$ such that $d_u \leq k \cdot d$ and $d \leq m \cdot d_u$.

Solution The topology determined by the discrete distance is discrete. For the usual topology on X, we see that the topology is discrete again. For showing this, we see that singletons are open in X. If $n \neq 1$, we have $\{\frac{1}{n}\} = (\frac{1}{n+1}, \frac{1}{n+1}) \cap X$; if $n = 1$, $\{1\} = (\frac{1}{2}, 2) \cap X$ and if $n > 1$, $\{n\} = (n-1, n+1) \cap X$.

1. Suppose that there is $k > 0$ such that $d_u \leq k \cdot d$. Let $x = E(k+1)$ and $y = E(3k+3)$, where $E(k)$ stands for the integer part of k. Then $d(x, y) = 1$, so $d_u(x, y) \leq k \cdot d(x, y) = k$. On the other hand, using $E(x+n) = E(x) + n$ and $E(nx) \geq nE(x)$ for $x \in \mathbb{R}$ and $n \in \mathbb{N}$, we have

$$\begin{aligned} k \geq d_u(x, y) &= E(3k+3) - E(k+1) = E(3k) - E(k) + 2 \\ &\geq 3E(k) - E(k) + 2 = 2E(k) + 2. \end{aligned}$$

   From this inequality and by definition of $E(k)$, we deduce that $E(k) \geq 2E(k)+2$, which is a contradiction.
2. Suppose that there is an inequality $d \leq m \cdot d_u$ for some $m > 0$. Let n be a natural number such that $m < n(n+1)$ and consider $x = 1/(n+1)$ and $y = 1/n$. Then $d(x, y) = 1$ and the contradiction arrives because

$$1 = d(x, y) \leq m \cdot d_u(x, y) = m \left| \frac{1}{n+1} - \frac{1}{n} \right| = \frac{m}{n(n+1)} < 1.$$

□

In the following exercises we work with the space of real continuous functions defined on $[0, 1]$, a well known set in analysis.

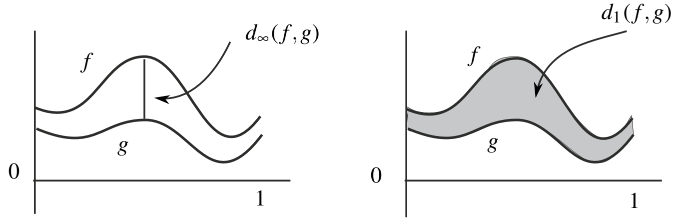


Fig. 4.6 Distances d_∞ and d_1 on $\mathsf{C}([0, 1])$. The distance d_∞ is the maximum distance between $f(x)$ and $g(x)$ (left). The distance $d_1(f, g)$ is the area of the shaded region between the graphs of f and g (right)

Exercise 4.4 Let $\mathsf{C}([0, 1])$ be the space of continuous, real-valued functions in $I = [0, 1]$. Prove that the following functions d_∞ and d_1 are distances:

$$d_\infty(f, g) = \max\{|f(x) - g(x)| : x \in [0, 1]\},$$

$$d_1(f, g) = \int_0^1 |f(x) - g(x)|\, dx.$$

The distance d_∞ is called the *uniform distance* (Fig. 4.6).

Solution

1. The distance d_∞ is well defined because the function $|f - g|$ attains a maximum since $|f - g|$ is a continuous function defined on a bounded closed interval. Moreover, if $d_\infty(f, g) = 0$ then $|f(x) - g(x)| = 0 \leq d_\infty(f, g)$ for all $x \in [0, 1]$, so $f = g$. It remains to prove that d_∞ satisfies the triangle inequality. Let f, g and h be three continuous functions in $[0, 1]$. If $y \in [0, 1]$,

$$|f(y) - g(y)| \leq |f(y) - h(y)| + |h(y) - g(y)|.$$

   Thus

$$|f(y) - g(y)| \leq \max_{x \in I} |f(x) - h(x)| + \max_{x \in I} |h(x) - g(x)| = d_\infty(f, h) + d_\infty(h, g).$$

   So we have found an upper bound for $|f(y) - g(y)|$ for all $y \in I$, deducing

$$\max_{y \in I} |f(y) - g(y)| \leq d_\infty(f, h) + d_\infty(h, g).$$

2. Observe that $d_1(f, g)$ is well defined because the integral is defined for a continuous function, namely, the function $|f - g|$, in the closed bounded interval $[0, 1]$. If $d_1(f, g) = 0$ then $\int_0^1 |f(x) - g(x)|\, dx = 0$. Since $|f(x) - g(x)| \geq 0$, we must have $f(x) - g(x) = 0$ for all $x \in I$, proving that $f = g$. For the triangle

inequality,

$$|f(x) - g(x)| \le |f(x) - h(x)| + |h(x) - g(x)|, \quad x \in I.$$

Integrating in [0, 1],

$$\begin{aligned} d_1(f, g) &= \int_0^1 |f(x) - g(x)|\, dx \le \int_0^1 (|f(x) - h(x)| + |h(x) - g(x)|)\, dx \\ &= \int_0^1 |f(x) - h(x)|\, dx + \int_0^1 |h(x) - g(x)|\, dx = d_1(f, h) + d_1(h, g). \end{aligned}$$

The set $\mathsf{C}([0, 1])$ is a vector space with the usual sum of functions and product of a function by scalars. Its dimension is infinite because, for example, the functions $x \mapsto e^{nx}$, $n \in \mathbb{N}$, are linearly independent. The above two metric spaces are normed spaces where the norms are, respectively,

$$\|f\|_\infty = \max_{x\in I} |f(x)|, \quad \|f\|_1 = \int_0^1 |f(x)|\, dx.$$

□

Exercise 4.5 (Continuation) Describe the balls in $\mathsf{C}([0, 1])$ for the distance d_∞. Prove that $A = \{f \in \mathsf{C}([0, 1]) : f(0) = 0\}$ is a closed set.

Solution

1. Let $f \in \mathsf{C}([0, 1])$ and $r > 0$. If $g \in B_r^\infty(f)$ then $\max\{|g(x) - f(x)| : x \in I\} < r$; in particular, $|g(x) - f(x)| < r$ for all $x \in I$. This may be rewritten as $f(x)-r < g(x) < f(x)+r$ for all $x \in I$. Thus the graph of g lies in the region of $\mathbb{R}^2$ bounded by the graphs of $f(x)-r$ and $f(x)+r$. See the shaded region in Fig. 4.7.

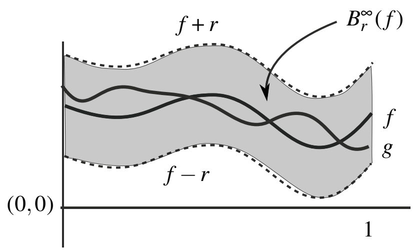


Fig. 4.7 The ball $B_r(f)$ of radius $r > 0$ centered at f in the metric space $(\mathsf{C}([0, 1]), d_\infty)$

2. We prove that A is closed by the sequential characterization. Let $(f_n) \subset A$ and $(f_n) \to f$. Then $d_\infty(f_n, f) \to 0$; in particular,

$$|f_n(0) - f(0)| \leq \max\{|f_n(x) - f(x)| : x \in [0,1]\} = d_\infty(f_n, f).$$

As $f_n(0) = 0$, then $|f(0)| \leq d_\infty(f_n, f)$. Letting $n \to \infty$, we deduce that $f(0) = 0$, concluding $f \in A$. □

Exercise 4.6 In $\mathsf{C}([0,1])$, prove that τ_{d_∞} is finer than τ_{d_1} but $\tau_{d_\infty} \neq \tau_{d_1}$.

Solution

1. We see that $\tau_{d_1} \subset \tau_{d_\infty}$. If $f, g \in \mathsf{C}([0,1])$ then $|f(x) - g(x)| \leq d_1(f, g)$ for all $x \in I$. Thus

$$\begin{aligned} d_1(f,g) &= \int_0^1 |f(x) - g(x)|\, dx \leq \int_0^1 d_\infty(f,g)\, dx \\ &= d_\infty(f,g) \int_0^1 1 \cdot dx = d_\infty(f,g). \end{aligned}$$

Then $d_1 \leq d_\infty$ and this proves that $\tau_{d_1} \subset \tau_{d_\infty}$ by Theorem 4.2.4.

2. Take $g = 0$ and we prove that the ball $B_1^\infty(g)$ is not open in τ_{d_1}. Since $\{B_r^1(g) : 0 < r < 1\}$ is a basis of neighborhoods of g for the distance d_1, we show that $B_r^1(g) \not\subset B_1^\infty(g)$ for all $r < 1$. This is enough according to Proposition 4.2.2. For each $r > 0$, define the continuous function

$$f_r(x) = \begin{cases} 0, & \text{if } 0 \leq x \leq 1 - \frac{r}{2} \\ \frac{4}{r}(x-1) + 2, & \text{if } 1 - \frac{r}{2} \leq x \leq 1. \end{cases}$$

See the graph of f_r in Fig. 4.8. Then $f_r \in B_r^1(g)$ for all $0 < r < 1$ because

$$d_1(f_r, g) = \int_0^1 |f_r(x) - 0|\, dx = \int_{1-r/2}^1 \left(\frac{4}{r}(x-1) + 2\right) dx = \frac{r}{2} < r.$$

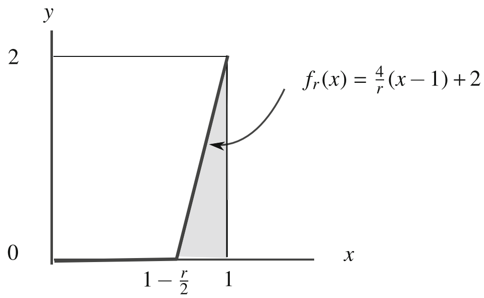


Fig. 4.8 The graph of the function f_r. The area of the shaded region is the distance d_1 between f_r and the function $g = 0$. However, for d_∞, this distance is the maximum of $|f_r(x) - 0|$ which is 2

But $f_r \notin B_1^\infty(g)$ since

$$d_\infty(f_r, g) = \max_{x\in[0,1]} |f_r(x) - 0| = f_r(1) = 2.$$

□

Exercise 4.7 In the space $X = \mathsf{C}([0, 1])$, consider

$$A = \{f \in X : f(x) > 0, \text{ for all } x \in I\},$$
$$B = \{f \in X : f(x) \geq 0, \text{ for all } x \in I\}.$$

Prove that A is open and that B is closed in (X, d_∞), but A is not open in (X, d_1).

Solution

1. Let $f \in A$ and we find $r > 0$ such that $B_r^\infty(f) \subset A$. Since f is positive in the closed interval $[0, 1]$, the number $r = \min\{f(x) : x \in [0, 1]\}$ is positive and we prove that $B_r^\infty(f) \subset A$. If $g \in B_r^\infty(f)$ then $|f(x) - g(x)| < r$ for all $x \in I$, in particular, $f(x) < r + g(x)$. Hence $g(x) > f(x) - r \geq 0$ for all $x \in I$, proving that $g \in A$.
2. We prove that $X \setminus B$ is open. This set can be described as

$$X \setminus B = \{f \in X : f(x) < 0 \text{ at some points } x \in I\}.$$

Let $f \in X \setminus B$ be given. Take $x_0 \in I$ such that $f(x_0) < 0$. Letting $\epsilon = -f(x_0)$, we see that $B_\epsilon^\infty(f) \subset X \setminus B$. If $g \in B_\epsilon^\infty(f)$, then $|g(x) - f(x)| < \epsilon$ for all $x \in I$. For the particular point $x = x_0$, we have $g(x_0) - f(x_0) < \epsilon = -f(x_0)$, hence $g(x_0) < 0$, so $g \in X \setminus B$.
3. The proof is by contradiction. Let f be the constant function $f(x) = 1$, $x \in I$. Since A is open, there is $\epsilon > 0$ such that $B_\epsilon^1(f) \subset A$. The plan is to find a function g that takes negative values at some points but that its integral is close to the integral of f. Let δ fix a positive number such that $\delta < \{\epsilon/2, 1/2\}$. Define

$$g(x) = \begin{cases} \dfrac{1}{\delta}(x - \delta), & x \in [0, 2\delta] \\ 1, & x \in [2\delta, 1]. \end{cases}$$

This function is continuous and $g \notin A$ because $g(0) = -1$. However, the contradiction arrives because $g \in B_\epsilon^1(f)$, namely,

$$d_1(f, g) = \int_0^1 |1 - g(x)|\, dx = \int_0^{2\delta} |1 - g(x)|\, dx = 2\delta < \epsilon.$$

Note that B is closed in (X, d_1) because $\tau_{d_1} \subset \tau_{d_\infty}$ (Exercise 3.11). □

We revisit the spheres $\mathbb{S}_r(x)$ defined in metric spaces.

Exercise 4.8 Let X be a metric space, $x \in X$ and $r > 0$. Prove that

$$V_r(x) = \{y \in X : d(x, y) \leq r\} = B_r(x) \cup \mathbb{S}_r(x).$$

is closed.

Solution Let $y \in \overline{V_r(x)}$. Then there is $(x_n) \subset V_r(x)$ such that $(x_n) \to y$. From the triangle inequality

$$d(x, y) \leq d(x, x_n) + d(x_n, y) \leq r + d(x_n, y).$$

Letting $n \to \infty$, $d(x, y) \leq r + 0 = r$ and this gives $y \in V_r(x)$. □

The graph of a mapping is a remarkable set associated to it. We turn our attention to some topological properties of this set if the function is continuous. Since the same effort is done in arbitrary dimension, we consider functions defined on $\mathbb{R}^m$. Here we assume that the reader is familiarized with the continuity of real-valued functions defined on $\mathbb{R}^m$.

Exercise 4.9 The *graph* of a function $f : \mathbb{R}^m \to \mathbb{R}$ is defined by

$$G(f) = \{(x, y) \in \mathbb{R}^m \times \mathbb{R} = \mathbb{R}^{m+1} : y = f(x)\} = \{(x, f(x)) : x \in \mathbb{R}^m\}.$$

Suppose that f is continuous. Find the interior and closure of $G(f)$ in $\mathbb{R}^{m+1}$.

Solution Let us use the sequential characterization of the Euclidean topology.

1. Let $(x, y) \in \text{int}(G(f))$. Since $((x, y + 1/n)) \to (x, y)$, there is $\nu \in \mathbb{N}$ such that $(x, y + 1/n) \in G(f)$ for $n \geq \nu$. Then $y + 1/n = f(x)$, which contradicts that $y = f(x)$ because $(x, y) \in G(f)$. This proves $\text{int}(G(f)) = \emptyset$. Note that we have not used that f is continuous.
2. We see that $G(f)$ is closed. Let $(x, y) \in \overline{G(f)}$ be given. Then there is $((x_n, y_n)) \subset G(f)$ converging to (x, y). In particular, $(x_n) \to x$ and $(y_n) \to y$. Since f is continuous, $(f(x_n)) \to f(x)$. Because $(x_n, y_n) \in G(f)$, then $y_n = f(x_n)$, so taking limits, $(f(x_n)) = (y_n) \to y$. By uniqueness of limits, $y = f(x)$ and this shows that $(x, y) \in G(f)$.

We mention some observations.

(i) If f is not continuous, this graph may not be closed. For instance, the graph of

$$f(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0 \end{cases}$$

is $((0, \infty) \times \{1\}) \cup ((-\infty, 0] \times \{-1\})$. This set is not closed because $(0, 1) \notin G(f)$ but $(0, 1) \in \overline{G(f)}$ because the sequence $((1/n, 1)) \subset G(f)$ converges to $(0, 1)$.

(ii) One can consider continuous functions $f : \mathbb{R}^m \to \mathbb{R}^k$, obtaining again that the graph of f, $G(f) = \{(x, y) \in \mathbb{R}^m \times \mathbb{R}^k = \mathbb{R}^{m+k} : y = f(x)\}$, is closed in $\mathbb{R}^{m+k}$.

(iii) The arguments are equally valid for functions $f : A \subset \mathbb{R}^m \to \mathbb{R}^k$. Thus the graph of f defined by $G(f) = \{(x, y) \in A \times \mathbb{R}^k = \mathbb{R}^{m+k} : y = f(x)\}$ is closed in $A \times \mathbb{R}^k$. Indeed, now $(x, y) \in \overline{G(f)}$, where the closure is considered in $A \times \mathbb{R}^k$. But $A \times \mathbb{R}^k$ is a metric subspace of $\mathbb{R}^m \times \mathbb{R}^k = \mathbb{R}^{m+k}$, so there is a sequence $((x_n, y_n)) \subset G(f)$ converging to (x, y). The rest of the reasoning is the same.

(iv) As a consequence of the previous item, if $A \subset \mathbb{R}^m$ is a closed set, then $G(f)$ is closed in $\mathbb{R}^{m+k}$ because $G(f)$ is closed in $A \times \mathbb{R}^k$ and $A \times \mathbb{R}^k$ is closed in $\mathbb{R}^{m+k}$ (Corollary 4.4.5). □

Exercise 4.10 Consider the quadrics of $\mathbb{R}^3$:

1. The *ellipsoid*: $\{(x, y, z) \in \mathbb{R}^3 : x^2/a^2 + y^2/b^2 + z^2/c^2 = 1\}$ and $a, b, c \neq 0$. If $a = b = c$, we obtain a sphere of radius $|a|$ centered at the origin.
2. The *paraboloid*: $\{(x, y, z) \in \mathbb{R}^3 : z = x^2 + y^2\}$.
3. The *ruled hyperboloid*: $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = 1\}$.
4. The *hyperboloid of two sheets*: $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = -1\}$.
5. The *cone*: $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$.
6. The *hyperbolic paraboloid*: $\{(x, y, z) \in \mathbb{R}^3 : z = x^2 - y^2\}$.
7. The *cylinder*: $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$.

See Figs. 4.9 and 4.10. Show that all these subsets are closed in $\mathbb{R}^3$.

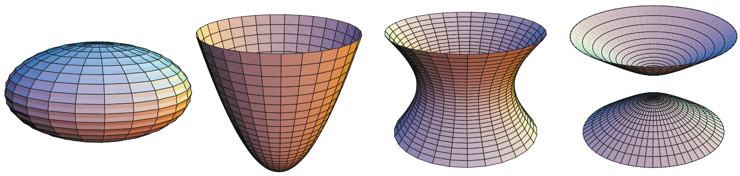


Fig. 4.9 From left to right: an ellipsoid, a paraboloid, a ruled hyperboloid and a hyperboloid of two sheets

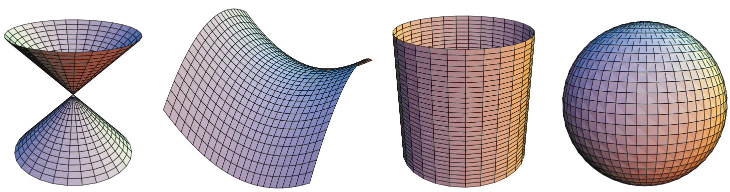


Fig. 4.10 From left to right: a cone and a hyperbolic paraboloid, a cylinder and a sphere

Solution The proof can be done with the sequential characterization of adherent points and using the properties of the convergent sequences of $\mathbb{R}$. The exercise can be also solved thanks to the foregoing one. For example, the paraboloid is the graph of $f(x, y) = x^2 + y^2$ and the hyperbolic paraboloid of $f(x, y) = x^2 - y^2$, being continuous. The cone is the union of two graphs of continuous functions, $f(x, y) = \sqrt{x^2 + y^2}$ and $g(x, y) = -\sqrt{x^2 + y^2}$, so the cone is closed.

The hyperboloid of two sheets is different because in this case, the set is union of two graphs of continuous functions defined on subsets of $\mathbb{R}^2$. Let

$$f, g\colon \mathbb{R}^2 \setminus B_1(0,0) \to \mathbb{R}, \quad f(x, y) = \sqrt{x^2 + y^2 - 1},\ g(x, y) = -\sqrt{x^2 + y^2 - 1}.$$

Then $G(f)$ and $G(g)$ are closed in $\mathbb{R}^2 \setminus B_1(0,0)$ by the last observations in the previous exercise, but as $\mathbb{R}^2 \setminus B_1(0,0)$ is closed in $\mathbb{R}^2$, then $G(f)$ and $G(g)$ are closed in $\mathbb{R}^3$.

For the ellipsoid, the set is the union of two graphs defined on $A = \{(x, y) \in \mathbb{R}^2 : x^2/a^2 + y^2/b^2 \leq 1\}$, where the functions are

$$f, g\colon A \to \mathbb{R}, \quad f(x, y) = c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}},\ g(x, y) = -c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}.$$

The exercise concludes if we prove that A is closed in $\mathbb{R}^2$. If $(x, y) \in \overline{A}$, then there is $((x_n, y_n)) \subset A$ that converges to (x, y). Thus $(x_n) \to x$ and $(y_n) \to y$. Since $x_n^2/a^2 + y_n^2/b^2 \leq 1$, letting $n \to \infty$, we get $x^2/a^2 + y^2/b^2 \leq 1$, proving that $(x, y) \in A$.

Finally, the cylinder is again union of two graphs of functions defined on closed sets, but now the functions are defined on the xz-plane. So

$$f, g\colon C \to \mathbb{R}, \quad f(x, z) = \sqrt{1 - x^2},\ g(x, y) = -\sqrt{1 - x^2},$$

where $C = \{(x, z) \in \mathbb{R}^2 : -1 \leq x \leq 1\} = [-1, 1] \times \mathbb{R}$ and this set is closed. □

Exercise 4.11 For each $n, m \in \mathbb{N}$, let $a_{n,m} = (\frac{1}{n}, \frac{1}{m}) \in \mathbb{R}^2$ and $A = \{a_{n,m} : n, m \in \mathbb{N}\}$. See Fig. 4.11. Prove that all points of A are isolated and find the accumulation points.

Solution

1. Let $a_{n,m} \in A$ be given. Consider the distance from $a_{n,m}$ to the points of A around $a_{n,m}$. The smallest distance from $a_{n,m}$ to the remaining points of A is attained at the points $a_{n+1,m}$ or $a_{n,m+1}$, that are situated to the left or below $a_{n,m}$. Hence let

$$r = \min\left\{\frac{1}{n} - \frac{1}{n+1}, \frac{1}{m} - \frac{1}{m+1}\right\}.$$

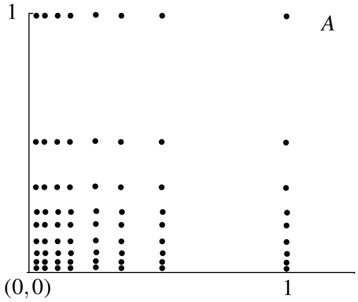


Fig. 4.11 The set A of Exercise 4.11

We see that $B_r(a_{n,m}) \cap A = \{a_{n,m}\}$, proving that $a_{n,m}$ is isolated. Suppose $a_{p,q} \in B_r(a_{n,m})$. Then $|a_{p,q} - a_{n,m}| < r$; in particular, $|1/n - 1/p| < r$ and $|1/q - 1/m| < r$. From these inequalities and the value of r, $p < n+1$ and $q < m+1$. If $a_{p,q} \neq a_{n,m}$, suppose $p \neq n$. Then $p < n-1$. In such a case,

$$\left|\frac{1}{n} - \frac{1}{p}\right| = \frac{1}{n-1} - \frac{1}{n} < r \leq \frac{1}{n} - \frac{1}{n+1},$$

which is not possible.

2. The accumulation points of A coincide with the set $\overline{A} \setminus \operatorname{Ais}(A) = \overline{A} \setminus A$. By the sequential characterization of adherent points, these points are limits of convergent sequences of A. Then the points $(0, \frac{1}{n})$, $(\frac{1}{n}, 0)$ and $(0, 0)$ are adherent points of A because

$$\left(0, \frac{1}{n}\right) = \lim_{m\to\infty}\left(\frac{1}{m}, \frac{1}{n}\right), \ \left(\frac{1}{n}, 0\right) = \lim_{m\to\infty}\left(\frac{1}{n}, \frac{1}{m}\right), \ (0,0) = \lim_{m\to\infty}\left(\frac{1}{m}, \frac{1}{m}\right).$$

We see that there are no other adherent points. Let $(x, y) \in \overline{A}$ and $(x, y) \notin A$. Then there is a sequence $(a_{n_k,m_k}) \subset A$ converging to (x, y); in particular, $(1/n_k) \to x$ and $(1/m_k) \to y$. If $\{n_k : k \in \mathbb{N}\}$ and $\{m_k : k \in \mathbb{N}\}$ are finite, then (x, y) is of the form $a_{n,m} \in A$, which is not possible. If $\{n_k : k \in \mathbb{N}\}$ is infinite and $\{m_k : k \in \mathbb{N}\}$ is finite, then $(1/n_k) \to 0$, so $(x, y) = (0, 1/m)$. In case that $\{n_k : k \in \mathbb{N}\}$ is finite and $\{m_k : k \in \mathbb{N}\}$ is infinite, we have that $(x, y) = (1/n, 0)$. Finally, if $\{n_k : k \in \mathbb{N}\}$ and $\{m_k : k \in \mathbb{N}\}$ are infinite, then the limit is $(0, 0)$. □

Over the course of the next three exercises we introduce new metric spaces. In the first one, we define a curious distance on $\mathbb{R}^n$ where all singletons, with the exception of the origin, are open and the balls centered at the origin are the Euclidean balls.

Exercise 4.12 On $\mathbb{R}^n$, define $d(x, y) = |x| + |y|$ if $x \neq y$ and $d(x, x) = 0$. Prove that d is a distance on $\mathbb{R}^n$, called the *post office distance* (Fig. 4.12). Prove that

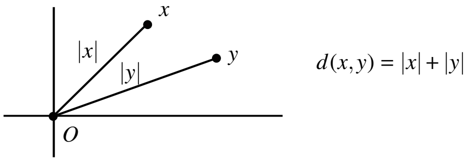


Fig. 4.12 The post office distance

all points other than the origin O are open and describe the balls centered at O. Compare with the Euclidean topology and characterize the convergent sequences.

The number $d(x, y)$ measures the Euclidean distance from x to the origin O of $\mathbb{R}^n$ plus the Euclidean distance from O to y. In some sense, one can think that going from x to y, one must go across O. In other words, the post office distance summarizes the well-know phrase that "all roads lead to Rome".

Solution

1. It is clear that if $x \neq y$, then $d(x, y) = |x| + |y| \neq 0$. The symmetry is obvious, so let us show the triangle inequality. Let $x, y, z \in \mathbb{R}^n$ and we show that

$$d(x, y) \leq d(x, z) + d(z, y).$$

   We have $d(x, z) + d(z, y) = |x| + 2|z| + |y| \geq |x| + |y| = d(x, y)$.
2. Let $x \neq O$ and consider the radius $r = |x|$. We see that $B_r(x) = \{x\}$. In effect, if $y \in B_r(x)$, $y \neq x$, then $d(x, y) < r$ can be rewritten as $|x| + |y| < |x|$, so $|y| < 0$, which is a contradiction. This proves that $B_r(x)$ only contains the point x and $\{x\}$ is open.
3. Let $r > 0$ and $x \in B_r(O)$, $x \neq O$. Then $d(x, O) = |x| + |O| = |x|$. Thus $B_r(x)$ coincides with the Euclidean ball of radius r centered at O.
4. We see that the metric topology is finer than the Euclidean one. Comparing with the bases of neighborhoods of all balls centered at a point (Hausdorff criterion), Euclidean balls centered at O coincide with the ones of the distance d. For the other balls, if $y \in B_r^{d_u}(x)$, $x \neq O$, then $B_r(y) = \{y\} \subset B_r^{d_u}(x)$.
5. Let $(x_n) \to x$. If $x \neq O$, consider the neighborhood $\{x\}$. Because the sequence must belong to this neighborhood after a certain term of the sequence, we conclude that the sequence is constant and equal to x. If $x = O$ then the balls of O agree with those of Euclidean topology, so the convergence coincides with the Euclidean one. We summarize by saying that the convergent sequences are the eventually constant sequences and the convergent sequences in the Euclidean topology whose limit is O. $\square$

Exercise 4.13 Prove that

$$d(x, y) = \begin{cases} |x - y|, & x, y \neq 0 \\ \max\{1, |y|\}, & x = 0 \\ \max\{1, |x|\}, & y = 0 \\ 0, & x = y = 0. \end{cases}$$

is a distance on $\mathbb{R}$ and compare its metric topology with the Euclidean topology. Describe the balls and find the interior and closure of $\{0\}$ and $(0, 1)$.

Solution

1. All properties to be a distance are trivial except the triangle inequality. Let $x, y \in \mathbb{R}$ and let $z \in \mathbb{R}$ be a third point. It is clear that if all of them are distinct from 0, the triangle inequality coincides with that of the Euclidean distance. Thus we discuss the cases such that some of them are 0.

   (a) Case $x = 0$. The job is to show that $d(0, y) \leq d(0, z)+d(z, y)$. Since $y \neq 0$, this inequality is $\max\{1, |y|\} \leq \max\{1, |z|\} + |y - z|$. However,

   $$|y| \leq |y - z| + |z| \leq |y - z| + \max\{1, |z|\} = d(y, z) + d(0, z),$$

   and

   $$1 \leq \max\{1, |z|\} = d(0, z) \leq d(0, z) + d(y, z).$$

   From both inequalities, $d(0, y) \leq d(0, z) + d(z, y)$.

   (b) Case $x \neq 0$. If $y = 0$, the argument is similar to the one above. Thus assume $y \neq 0$. The case to study is when $z = 0$. Then we must show $|x - y| \leq \max\{1, |x|\} + \max\{1, |y|\}$. But this is clear because $|x - y| \leq |x| + |y|$.

   Since $|x - y| \leq d(x, y)$, the metric topology τ_d is finer than the Euclidean one.

2. The inequality $|x-y| \leq d(x, y)$ says that $B_r^d(x) \subset (x-r, x+r)$. We distinguish two cases.

   (a) Case $x = 0$. If $r > 1$, we see that $B_r^d(0) = (-r, r)$. Certainly, if $d(0, y) < r$ then $|y| \leq \max\{1, |y|\} < r$. If $r \leq 1$, we prove that $B_r^d(0) = \{0\}$; if $d(0, y) < r \leq 1$ then $\max\{1, |y|\} < 1$, which is not possible. This implies $y = 0$.

   (b) Case $x \neq 0$. Let $y \in B_r^d(x)$. If $y = 0$, $\max\{1, |x|\} < r$, so $|x| < r$ and $1 < r$. If $y \neq 0$ then $d(x, y) = |x - y| < r$. Thus

   $$B_r^d(x) = \begin{cases} (x - r, x + r) \setminus \{0\}, & |x| < r \leq 1 \\ (x - r, x + r), & \text{otherwise.} \end{cases}$$

   As a result, taking radii less than 1, $\{\{0\}\}$ is a basis of neighborhoods of $x = 0$ and $\{(x - r, x + r) : 0 < r < |x|\}$ is a basis of neighborhoods for $x \neq 0$. Another consequence is that we have a full description of the topology τ_d, namely,

   $$\tau_d = \tau_u \cup \{O \cup \{0\} : O \in \tau_u\}.$$

3. The inclusion $\tau_u \subset \tau_d$ implies that if $A \subset \mathbb{R}$, then $\operatorname{int}(A) \subset \operatorname{int}^d(A)$ and $\overline{A}^d \subset \overline{A}$ by Exercise 3.11. In particular, $\overline{\{0\}}^d = \{0\}$ and $\operatorname{int}^d(0,1) = (0,1)$. Since $\{0\}$ is a neighborhood of $x = 0$, then $\operatorname{int}^d(\{0\}) = \{0\}$. For the closure, $\overline{(0,1)}^d \subset [0,1]$. As $(0,1) \subset \overline{(0,1)}^d$, it remains to check if 0 and 1 are adherent points. Since $\{0\}$ is a neighborhood of $x = 0$, then $x = 0$ is not an adherent point. For $x = 1$, the balls $B_r^d(1)$ intersect $(0,1)$, so it is an adherent point. Hence $\overline{(0,1)}^d = (0,1]$. □

Exercise 4.14 On $\mathbb{R}^2$, define $d\colon \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ by

$$d((x_1,y_1),(x_2,y_2)) = \begin{cases} \max\{1, |y_1 - y_2|\}, & x_1 \neq x_2 \\ |y_1 - y_2|, & \text{otherwise.} \end{cases}$$

Prove that d is a metric and that a basis of neighborhoods at (x,y) is $\beta_{(x,y)} = \{\{x\} \times (y-r, y+r) : r > 0\}$. Prove that $\tau_D \times \tau_u$ is a basis for the topology.

Solution

1. It is obvious that $d \geq 0$. The symmetry property is also trivial. For the positive definiteness, suppose $d((x_1,y_1),(x_2,y_2)) = 0$. The case $x_1 \neq x_2$ cannot occur. Thus $x_1 = x_2$. In such a case, $|y_1 - y_2| = 0$, so $y_1 = y_2$. This proves that both points coincide. For the triangle inequality, we wish to show that

$$d((x_1,y_1),(x_2,y_2)) \leq d((x_1,y_1),(x_3,y_3)) + d((x_3,y_3),(x_2,y_2)). \tag{4.6}$$

We distinguish cases:

(i) Case $x_1 \neq x_2$. Again, there are two alternatives. The first case is $x_3 = x_1$ (a similar argument if $x_3 = x_2$). Now (4.6) is

$$\max\{1, |y_1 - y_2|\} \leq |y_1 - y_3| + \max\{1, |y_2 - y_3|\}.$$

But this is a consequence of

$$|y_1 - y_2| \leq |y_1 - y_3| + |y_3 - y_2| \leq |y_1 - y_3| + \max\{1, |y_2 - y_3|\} \tag{4.7}$$

and

$$1 \leq \max\{1, |y_2 - y_3|\} \leq |y_1 - y_3| + \max\{1, |y_2 - y_3|\}. \tag{4.8}$$

The other cases are $x_3 \neq x_1$ and $x_3 \neq x_2$. Now the inequality (4.6) is

$$\max\{1, |y_1 - y_2|\} \leq \max\{1, |y_1 - y_3|\} + \max\{1, |y_2 - y_3|\}.$$

Following with (4.7) and (4.8), we have respectively,

$$|y_1 - y_2| \leq \max\{1, |y_1 - y_3|\} + \max\{1, |y_2 - y_3|\}$$

and

$$1 \leq |y_1 - y_3| + \max\{1, |y_2 - y_3| \leq \max\{1, |y_1 - y_3|\} + \max\{1, |y_2 - y_3|\}.$$

This proves (4.6).

(ii) Case $x_1 = x_2$. If $x_3 = x_1$, the inequality (4.6) is

$$|y_1 - y_2| \leq |y_1 - y_3| + |y_2 - y_3|,$$

which is obvious. If $x_3 \neq x_1$, the inequality (4.6) is

$$|y_1 - y_2| \leq \max\{1, |y_1 - y_3|\} + \max\{1, |y_2 - y_3|\},$$

which is true by the triangle inequality of the Euclidean distance.

2. By Proposition 4.2.2, a basis of neighborhoods of (x, y) is $\gamma_{(x,y)} = \{B_r((x, y)) : 0 < r < 1\}$. If $d((x, y), (z, t)) < r < 1$ then necessarily $x = z$. In such a case,

$$B_r((x, y)) = \{(x, t) \in \mathbb{R}^2 : |y - t| < r\} = \{x\} \times (y - r, y + r).$$

It is clear that if $\gamma_{(x,y)}$ is a basis of neighborhoods and because any element of $\beta_{(x,y)}$ contains another of $\gamma_{(x,y)}$, then $\beta_{(x,y)}$ is a basis of neighborhoods.

3. For the last item, we see that any element of $\tau_D \times \tau_u$ is open in τ_d. But this is immediate, for if $(x, y) \in O \times O' \in \tau_D \times \tau_u$, then $(x, y) \in \{x\} \times B_r(y) \subset O \times O'$, where $B_r(y)$ is a ball contained in O'. Finally, if $G \in \tau_d$ and $(x, y) \in G$, then there is $0 < r < 1$ such that $(x, y) \in \{x\} \times B_r(y) \subset G$ and $\{x\} \times B_r(y) \in \tau_D \times \tau_u$. □

Exercise 4.15 Using the sequential characterization, prove that $A = \{(x, y) \in \mathbb{R}^2 : y > x + 1\}$ is open in $\mathbb{R}^2$ and that $\overline{A} = \{(x, y) \in \mathbb{R}^2 : y \geq x + 1\}$.

Solution

1. Let $(x, y) \in A$ and we prove that $(x, y) \in \mathring{A}$. Let $((x_n, y_n)) \to (x, y)$. Then $(x_n) \to x$ and $(y_n) \to y$ (Proposition 4.4.3). Thus $(y_n - x_n - 1) \to y - x - 1$ (Corollary 4.4.4). As $y - x - 1 > 0$, then $y - x - 1 \in (0, \infty)$. Since $(0, \infty)$ is open, there exists $\nu \in \mathbb{N}$ such that $y_n - x_n - 1 \in (0, \infty)$ for $n \geq \nu$, proving that $(x_n, y_n) \in A$ for $n \geq \nu$.
2. Let $(x, y) \in \overline{A}$. By Theorem 4.4.2, there is $((x_n, y_n)) \subset A$ such that $((x_n, y_n)) \to (x, y)$; in particular, $(x_n) \to x$ and $(y_n) \to y$. Since $(x_n, y_n) \in A$, then $y_n - x_n - 1 > 0$. Taking limits and by elementary calculus (or Exercise 4.6.10), we obtain $y - x - 1 \geq 0$. This proves that $\overline{A} \subset \{(x, y) \in \mathbb{R}^2 : y \geq x + 1\}$. Another way is the following. Let $z_n = y_n - x_n - 1$ and $(z_n) \to z = y - x - 1$. Then

$x \in \overline{\{z_n : n \in \mathbb{N}\}}$. Since $\{z_n : n \in \mathbb{N}\} \subset (0, \infty)$, then $z \in \overline{(0, \infty)} = [0, \infty)$, so $z \geq 0$. Because

$$\{(x, y) \in \mathbb{R}^2 : y \geq x + 1\} = A \cup \{(x, y) \in \mathbb{R}^2 : y = x + 1\},$$

it remains to show that if (x, y) satisfies $y = x + 1$, then (x, y) is an adherent point of A. But this is clear because $((x, x + 1 + 1/n)) \subset A$ and converges to (x, y). □

Exercise 4.16 Let $\{F_k : k \in \mathbb{Z}\}$ be a family of closed sets of $\mathbb{R}$ such that $F_k \subset (k, k + 1)$ for all $k \in \mathbb{Z}$. Prove that $\bigcup_{k\in\mathbb{Z}} F_k$ is a closed set.

Solution Let us use the sequential characterization of adherent points. Let $x \in \overline{\bigcup_{k\in\mathbb{Z}} F_k}$ and $(x_n) \subset \bigcup_{k\in\mathbb{Z}} F_k$ such that $(x_n) \to x$. We distinguish two alternatives.

1. Case $x \notin \mathbb{Z}$. Then there is $m \in \mathbb{Z}$ such that $x \in (m, m + 1)$. Because this interval is open there is $\nu \in \mathbb{N}$ such that $x_n \in (m, m + 1)$ whenever $n \geq \nu$. Then necessarily, $x_n \in F_m$. Using that F_m is closed, we have $x \in F_m$, so $x \in \bigcup_{k\in\mathbb{Z}} F_k$.
2. Case $x \in \mathbb{Z}$. Let $x = m$. Then $x \in (m - 1, m + 1)$. Again, there is $\nu \in \mathbb{N}$ such that $x_n \in (m - 1, m + 1)$ for all $n \geq \nu$. Thus $x_n \in F_{m-1} \cup F_m$. Because this union is finite, $F_{m-1} \cup F_m$ is closed and $x \in F_{m-1} \cup F_m$ again, hence x belongs to the union.

We can directly solve the exercise with the second argument regardless of whether x is an integer or not, by choosing an interval around x so the sequence is contained in two (it suffices a finite number) consecutive intervals $(n, n + 1)$. □

Exercise 4.17 Prove that if $A \subset \mathbb{R}$ is bounded, then $\overline{A}$ is bounded. In such a case, prove that $\sup(A) = \sup(\overline{A})$ and $\inf(A) = \inf(\overline{A})$.

Solution

1. Denote $\alpha = \sup(A)$ and $\alpha' = \sup(\overline{A})$. Let $x \in \overline{A}$ and let $(a_n) \subset A$ be a sequence converging to x. From the triangle inequality,

$$x = x - a_n + a_n \leq |x - a_n| + a_n \leq |x - a_n| + \alpha.$$

   Letting $n \to \infty$, we deduce $x \leq \alpha$. Since α is an upper bound for $\overline{A}$, then $\alpha' \leq \alpha$. Because $A \subset \overline{A}$ then $\alpha \leq \alpha'$, obtaining $\alpha = \alpha'$.
2. Let $\beta = \inf(A)$ and $\beta' = \inf(\overline{A})$. Let $x \in \overline{A}$ and $(a_n) \subset A$ be a sequence such that $(a_n) \to x$. Then

$$x = x - a_n + a_n \geq -|x - a_n| + a_n \geq -|x - a_n| + \beta.$$

   Letting $n \to \infty$, $x \geq \beta$, hence $\beta' \geq \beta$. Since $A \subset \overline{A}$, $\beta \geq \beta'$, so $\beta = \beta'$. □

Exercise 4.18 Study Exercise 4.17 by replacing the closure with interior.

Solution The inclusion $\mathring{A} \subset A$ proves that if A is bounded, then $\mathring{A}$ is bounded as well. By the way, $\sup(\mathring{A}) \leq \sup(A)$ and $\inf(A) \leq \inf(\mathring{A})$. We see examples where we have strict inequalities.

One example is $A = (0, 1) \cup \{-1, 2\}$. Then $\mathring{A} = (0, 1)$ and its supremum is 1 and its infimum is 0. However, $\sup(A) = 2$ and $\inf(A) = -1$. Another example is $A = \{0\}$. Then $\sup(A) = \inf(A) = 0$. On the other hand, $\mathring{A} = \emptyset$ hence $\sup(\emptyset) = -\infty$ and $\inf(\emptyset) = +\infty$. Observe that for the empty set $\emptyset$, we have $\sup(\emptyset) = -\infty$ and $\inf(\emptyset) = +\infty$. For the remaining bounded subsets of $\mathbb{R}$, the inequality $\inf(A) \leq \sup(A)$ holds. □

We investigate the same problem of Exercise 4.17 in different topologies.

Exercise 4.19 In the Sorgenfrey line and in the right order topology, determine if $\sup(A) = \sup(\overline{A})$ and $\inf(A) = \inf(\overline{A})$.

Solution

1. We prove that both identities hold in the Sorgenfrey topology, such as occurs for the Euclidean topology. We follow the notation of Exercise 4.17. By contradiction, suppose $\alpha < \alpha'$. By definition of α', there is $x \in \overline{A}$ such that $\alpha < x \leq \alpha'$. Consider the neighborhood $[x, x + 1)$. Then $[x, x + 1) \cap A \neq \emptyset$. If a is a point of this intersection, then $x \leq a$, so $\alpha < a$, which is a contradiction.

   The proof for the infimum is also by contradiction again. If $\beta' < \beta$, by definition of infimum, there is $x \in \overline{A}$ such that $\beta' \leq x < \beta$. Let $[x, \beta)$ and $a \in [x, \beta) \cap A$. Then $a < \beta$, a contradiction.
2. In the right order topology τ_r, we see that $\alpha = \alpha'$. If this were not the case, then $\alpha < \alpha'$. Let $x \in \overline{A}$ such that $\alpha < x \leq \alpha'$ by definition of α'. The neighborhood $[x, \infty)$ intersects A, so let $a \in [x, \infty) \cap A$; in particular, $x \leq a$, hence $\alpha < a$, which is a contradiction.

   For the infimum the identity $\beta = \beta'$ does not hold in general. In fact, for a bounded set A from below, the set $\overline{A}$ may fail to be bounded from below as happens with $A = \{0\}$, where $\overline{A} = (-\infty, 0]$. □

We have seen that metric spaces are first countable by taking a countable basis of balls whose radii are decreasing and countable. This can be generalized in any first countable space in two directions, first proving that for any basis of neighborhoods there is a subcollection that is countable and second, preserving the sense of order by inclusions.

Exercise 4.20 Let X be a first countable space. If β_x is a basis of neighborhoods of x, prove that there is a countable collection β'_x of elements of β_x that is a basis of neighborhoods. Moreover, the elements of β'_x can be chosen so $U_{n+1} \subset U_n$ for all $U_n \in \beta'_x$.

Solution Let γ_x be a countable basis of neighborhoods of x. If γ_x is finite, then any basis of neighborhoods of x is finite. Assume that $\gamma_x = \{V_n : n \in \mathbb{N}\}$ is infinite. Consider

$$\Gamma = \{(V_i, V_j) \in \gamma_x \times \gamma_x : \text{there is } U \in \beta_x,\ V_i \subset U \subset V_j\}.$$

The cardinalitu of Γ is less than that of $\gamma_x \times \gamma_x$, so Γ is countable. For each $(V_i, V_j) \in \Gamma$, fix $U_{ij} \in \beta_x$ such that $V_i \subset U_{ij} \subset V_j$ and let $\beta'_x = \{U_{ij} : (V_i, V_j) \in \Gamma\}$. We see that β'_x is a basis of neighborhoods of x. Let $U \in \mathscr{U}_x$. Since γ_x is a basis of neighborhoods of x, there is $V_i \in \gamma_x$ such that $V_i \subset U$. Using that β_x and γ_x are bases of neighborhoods, there are $V \in \beta_x$ and $V_j \in \gamma_x$ such that $V_j \subset V \subset V_i$. Thus $(V_i, V_j) \in \Gamma$. Then $U_{ij} \subset V_j \subset U$, concluding the proof.

For the second part, suppose that $\beta_x = \{U_n : n \in J\}$ is a basis of neighborhoods of x where $J = \{1, \ldots, m\}$ if β_x is finite or $J = \mathbb{N}$ if β_x is infinite. Define

$$V_n = U_1 \cap \ldots \cap U_n$$

for all $n \in J$. We show that $\beta''_x = \{V_n : n \in J\}$ is a basis of neighborhoods of x with the property $V_{n+1} \subset V_n$ for all $n \in J$. Clearly V_n is a neighborhood of x and $V_{n+1} \subset V_n$. Since β_x is basis of neighborhoods, if U a neighborhood of x, then there is $n \in J$ such that $U_n \subset U$; in particular, $V_n = U_1 \cap \ldots \cap U_n \subset U_n \subset U$. □

Consider a second countable space. Although a basis for the topology may not be countable, we prove that we can select a countable basis from any basis.

Exercise 4.21 If β is a basis for the topology in a second countable space, prove that there is a countable subcollection $\beta' \subset \beta$ such that β' is a basis for the topology.

Proof If there is a finite basis for the topology, the result is trivial because the topology is also finite. Thus, assume that the space possesses bases with countably infinite elements. Fix a countable basis for the topology, $\gamma = \{V_n : n \in \mathbb{N}\}$. Consider the collection of pairs

$$\Gamma = \{(V_i, V_j) \in \gamma \times \gamma : \text{there is } B \in \beta \text{ such that } V_i \subset B \subset V_j\}.$$

We check that this family of sets is non-void. Given $V_j \in \gamma$ and $x \in V_j$, there is $B \in \beta$ such that $x \in B \subset V_j$. Using again that γ is a basis and because $x \in B$, there is $V_i \in \gamma$ with $x \in V_i \subset B$. This implies that $(V_i, V_j) \in \Gamma$.

We have $\operatorname{card}(\Gamma) \leq \operatorname{card}(\gamma \times \gamma) = \operatorname{card}(\gamma)$, where the last identity is because γ is countable infinite hence Γ is countable. For each $(V_i, V_j) \in \Gamma$, fix an element $B_{ij} \in \beta$ such that $V_i \subset B_{ij} \subset V_j$ and consider the family of open sets

$$\beta' = \{B_{ij} : (V_i, V_j) \in \Gamma\}.$$

Let us observe that $\operatorname{card}(\beta') \leq \operatorname{card}(\Gamma) \leq \operatorname{card}(\gamma)$. Thus the proof concludes if we prove that β' is a basis for the topology. Let O be an open set and $x \in O$. Since γ is a basis, there is $V_j \in \gamma$ such that $x \in V_j \subset O$. Likewise, as $x \in V_j$ and β is a basis, there is $B \in \beta$ such that $x \in B \subset V_j$. Again, there is $V_i \in \gamma$ such that $x \in V_i \subset B$. Therefore there exist $V_i, V_j \in \gamma$ and $B \in \beta$ such that

$$x \in V_i \subset B \subset V_j \subset O,$$

so $(V_i, V_j) \in \Gamma$. For this pair of open sets, consider the corresponding element $B_{ij} \in \beta'$. Then $V_i \subset B_{ij} \subset V_j$; in particular, $x \in B_{ij} \subset O$. □

In the next exercise we define a distance on the Cartesian product of two metric spaces and related to the distances of each one of the factors.

Exercise 4.22 Let (X_1, d_1) and (X_2, d_2) be two metric spaces. On $X_1 \times X_2$, define the product distance d by

$$d((x_1, x_2), (y_1, y_2)) = d_1(x_1, y_1) + d_2(x_2, y_2).$$

Prove that d is a distance. Study d in the cases $X_1 = X_2 = \mathbb{R}$, $X_1 = X_2 = \mathbb{R}^2$ and $X_1 = \mathbb{R}$ and $X_2 = \mathbb{R}^2$, where all distances d_i are Euclidean.

Solution The fact that $d \geq 0$ and the symmetry of d are trivial. Suppose $d((x_1, y_1), (x_2, y_2)) = 0$. Then $d_1(x_1, x_2) + d_2(y_1, y_2) = 0$. As $d_1, d_2 \geq 0$, we deduce $d_1(x_1, x_2) = d_2(x_2, y_2) = 0$, obtaining $x_1 = x_2$ and $y_1 = y_2$.

We prove the triangle inequality. Let $x = (x_1, y_1)$, $y = (x_2, y_2)$ and $z = (x_3, y_3)$. By means of the triangle inequalities for the distances d_1 and d_2,

$$\begin{aligned} d(x, y) &= d_1(x_1, x_2) + d_2(y_1, y_2) \\ &\leq d_1(x_1, x_3) + d_1(x_3, x_2) + d_2(y_1, y_3) + d_2(y_3, y_2) \\ &= d(x, z) + d(z, y) \end{aligned}$$

In the following, we identify $\mathbb{R}^n \times \mathbb{R}^m$ with $\mathbb{R}^{n+m}$ as usual.

1. If $X_1 = X_2 = \mathbb{R}$, the product distance is $d(x, y) = |x_1 - y_1| + |x_2 - y_2|$ and this tells that d coincides with the taxi distance d_t.
2. Now let $X_1 = X_2 = \mathbb{R}^2$ and $x = ((x_1, x_2), (x_3, x_4))$ and $y = ((y_1, y_2), (y_3, y_4))$. Then

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} + \sqrt{(x_3 - y_3)^2 + (x_4 - y_4)^2}.$$

   Then d is neither the Euclidean distance, nor the taxi distance d_t nor d_∞.
3. If $X_1 = \mathbb{R}$ and $X_2 = \mathbb{R}^2$, then

$$d(x, y) = |x_1 - y_1| + \sqrt{(x_2 - y_2)^2 + (x_3 - y_3)^2}.$$

   Again, this distance is not the Euclidean distance, and neither is it the taxis distance nor d_∞. □

After the definition of the product distance and with the identification $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$, the next exercise is expected.

Exercise 4.23 (Continuation) Consider on $\mathbb{R}^n$ and $\mathbb{R}^m$ the Euclidean distances. Prove that the product distance on $\mathbb{R}^{n+m}$ is equivalent to the Euclidean distance.

Solution To be precise with the notations, denote by d_k the Euclidean distance on $\mathbb{R}^k$ and denote by d the product distance of d_n by d_m. Writing $x, y \in \mathbb{R}^{n+m}$ as $x = (x^1, x^2)$, $y = (y^1, y^2) \in \mathbb{R}^n \times \mathbb{R}^m$, we have

$$d_n(x^1, y^1) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}, \quad d_m(x^2, y^2) = \sqrt{\sum_{i=1}^{m}(x_{n+i} - y_{n+i})^2},$$

$$d(x, y) = d_n(x^1, y^1) + d_m(x^2, y^2), \quad d_{n+m}(x, y) = \sqrt{\sum_{i=1}^{n+m}(x_i - y_i)^2}.$$

Since $d_n(x^1, y^1) \leq d_{n+m}(x, y)$ and $d_m(x^2, y^2) \leq d_{n+m}(x, y)$, then

$$d(x, y) \leq 2 \cdot d_{n+m}(x, y).$$

On the other hand, we have $d_{n+m} \leq d$ because

$$\begin{aligned} d(x, y)^2 &= d_n(x^1, y^1)^2 + d_m(x^2, y^2)^2 + 2d_n(x^1, y^1)d_m(x^2, y^2) \\ &\geq d_n(x^1, y^1)^2 + d_m(x^2, y^2)^2 = d_{n+m}(x, y)^2. \end{aligned}$$

We conclude using Theorem 4.2.4. □

We obtain equivalent distances to the product distance.

Exercise 4.24 Following with Exercise 4.22 and its notation, define

$$d'((x_1, x_2), (y_1, y_2)) = \max\{d_1(x_1, y_1), d_2(x_2, y_2)\},$$

$$\bar{d}((x_1, x_2), (y_1, y_2)) = \sqrt{d_1(x_1, y_1)^2 + d_2(x_2, y_2)^2}.$$

Show that d' and $\bar{d}$ are distances and both are equivalent to the product distance.

Solution

1. The symmetry property for the two distances is obvious. Because $d_i(x_i, y_i) \leq d'((x_1, x_2), (y_1, y_2))$ for $1 \leq i \leq n$, if $d'((x_1, x_2), (y_1, y_2)) = 0$, then $d_1(x_1, y_1) \leq 0$ and $d_2(x_2, y_2) \leq 0$, obtaining that both points coincide. For the distance $\bar{d}$, the argument is similar because $d_1(x_1, y_1) \leq \bar{d}((x_1, x_2), (y_1, y_2))$ and $d_2(x_2, y_2) \leq \bar{d}((x_1, x_2), (y_1, y_2))$.

We prove the triangle inequality for d'. For (z_1, z_2), we have

$$\begin{aligned} d_1(x_1, y_1) &\le d_1(x_1, z_1) + d_1(z_1, y_1) \\ &\le \max\{d_1(x_1, z_1), d_2(x_2, z_2)\} + \max\{d_1(z_1, y_1), d_2(z_2, y_2)\} \\ &= d'((x_1, x_2), (z_1, z_2)) + d'((z_1, z_2), (y_1, y_2)). \end{aligned}$$

For $\bar{d}$, we must check

$$\bar{d}((x_1, x_2), (y_1, y_2)) \le \bar{d}((x_1, x_2), (z_1, z_2)) + \bar{d}((y_1, y_2), (z_1, z_2)),$$

or equivalently,

$$\begin{aligned} d_1(x_1, y_1)^2 + d_2(x_2, y_2)^2 &\le d_1(x_1, z_1)^2 + d_2(x_2, z_2)^2 \\ &+ d_1(y_1, z_1)^2 + d_2(y_2, z_2)^2 \\ &+ 2\sqrt{d_1(x_1, z_1)^2 + d_2(x_2, z_2)^2}\sqrt{d_1(y_1, z_1)^2 + d_2(y_2, z_2)^2}. \end{aligned}$$

With the notation

$$a_i = d_i(x_i, y_i), \quad b_i = d_i(x_i, z_i), \quad c_i = d_i(z_i, y_i),$$

we must prove

$$a_1^2 + a_2^2 \le b_1^2 + b_2^2 + c_1^2 + c_2^2 + 2\sqrt{b_1^2 + b_2^2}\sqrt{c_1^2 + c_2^2}.$$

From the triangle inequality on d_1 and d_2, we have $a_i \le b_i + c_i$. Then

$$a_1^2 + a_2^2 \le (b_1 + c_1)^2 + (b_2 + c_2)^2 = b_1^2 + b_2^2 + c_1^2 + c_2^2 + 2b_1c_1 + 2b_2c_2.$$

Thus it suffices to check

$$b_1c_1 + b_2c_2 \le \sqrt{b_1^2 + b_2^2}\sqrt{c_1^2 + c_2^2},$$

which is immediate simply by squaring.

2. We prove the equivalence of the three distances. But it is clear that $d' \le d \le \bar{d} \le 2d'$, which is enough by Theorem 4.2.4. □

We give a method of constructing equivalent distances to the Euclidean distance.

Exercise 4.25 Let $f : \mathbb{R} \to \mathbb{R}$ be a function. For $x, y \in \mathbb{R}$, define

$$d(x, y) = \sqrt{(x - y)^2 + (f(x) - f(y))^2}.$$

Prove that d is a distance on $\mathbb{R}$ and that d is equivalent to the Euclidean distance if and only if f is continuous.

Solution

1. The fact that $d \geq 0$ and the symmetry of d are obvious. If $d(x, y) = 0$ then $(x - y)^2 + (f(x) - f(y))^2 = 0$, so $x - y = 0$. Finally, we see the triangle inequality. Let $x, y, z \in \mathbb{R}$ be given. Since $d(x, y)$ is the Euclidean distance on $\mathbb{R}^2$ between the points $(x, f(x))$ and $(y, f(y))$, the triangle inequality for d is obviously a consequence of the Euclidean distance of $\mathbb{R}^2$.
2. Apply Theorem 4.2.4 to verify that the two distances are equivalent. Since $d_u \leq d$, if $(x_n) \to x$ for d, then $(x_n) \to x$ for d_u. This is independent of whether or not f is continuous.

   (a) Assume that d_u and d are equivalent. To see that f is continuous we must prove that if $|x_n - x| \to 0$, then $|f(x_n) - f(x)| \to 0$. As $(d(x_n, x)) \to 0$,

   $$|f(x_n) - f(x)| \leq \sqrt{(x_n - x)^2 + (f(x_n) - f(x))^2} = d(x_n, x) \to 0.$$

   (b) Assume that f is continuous. We show that if $(x_n) \to x$ for d_u, then $(x_n) \to x$ for d. But if $(x_n) \to x$ for d_u, by continuity of f, $|f(x_n) - f(x)| \to 0$, so by definition of d, $d(x_n, x) \to 0$.

This exercise generalizes to $\mathbb{R}^n$ by replacing f by a continuous function $f : \mathbb{R}^n \to \mathbb{R}$ and d by $d(x, y) = \sqrt{|x - y|^2 + (f(x) - f(y))^2}$. □

We prove that any distance is equivalent to a bounded distance.

Exercise 4.26 In a metric space (X, d), prove that

$$d'(x, y) = \min\{d(x, y), 1\}, \quad x, y \in X,$$

is an equivalent distance to d. If $X = \mathbb{R}$ and d is the usual distance, prove that there is not $k > 0$ such that $k \cdot d_u \leq d'$.

Solution

1. We prove that d' is a distance. The fact that $d' \geq 0$ and the symmetry of d are clear. Suppose that $d'(x, y) = 0$. Then $d(x, y) = 0$, so $x = y$. We see the triangle inequality. Let $x, y, z \in X$ and set $a = d(x, y)$, $b = d(y, z)$ and $c = d(x, z)$. We wish to show

   $$\min\{c, 1\} \leq \min\{a, 1\} + \min\{b, 1\}. \tag{4.9}$$

   The triangle inequality for d yields $c \leq a + b$. If one of the two numbers on the right-hand side of (4.9) is 1, then the inequality is true because $\min\{c, 1\} \leq 1$. Thus the remaining case is $a+b$ in the right-hand side of (4.9). Then $\min\{c, 1\} \leq c \leq a + b$, proving (4.9).

2. We see that d and d' are equivalent. From the definition, we have $d' \leq d$, hence every convergent sequence in (X, d) is also in (X, d'). Conversely, suppose that $(x_n) \to x$ for d'. Then $(d'(x_n, x)) \to 0$. From a certain term of this sequence of real numbers, the elements of the sequence are strictly less than 1, so $d'(x_n, x) = d(x_n, x)$. In consequence, $d(x_n, x) \to 0$.
3. Assume to the contrary that there is $k > 0$ such that $k \cdot d_u \leq d'$. Then $k \cdot d_u \leq 1$ and $k \cdot d_u(0, n) \leq 1$ for all $n \in \mathbb{N}$; that is, $kn \leq 1$ and that is not possible. □

It was proved in Example 2.6 that the intersection of two topologies is a topology, but this fails for union of topologies. It is natural to ask if the intersection and the union of metric topologies (if it is a topology) are metrizable.

Exercise 4.27 Let d and d' be two distances on a set X. Prove that $d_M(x, y) = \max\{d(x, y), d'(x, y)\}$ is a distance on X. Prove that $B_r^M(x) = B_r(x) \cap B'_r(x)$ and deduce that the metric topology τ_{d_M} satisfies $\tau_d \cup \tau_{d'} \subset \tau_{d_M}$. Find an example where this inclusion is proper.

Solution

1. The symmetry of d_M is obvious and if $d_M(x, y) = 0$, then $d(x, y) \leq 0$ and $d'(x, y) \leq 0$. Then $d(x, y) = 0$, so $x = y$. For the triangle inequality,

$$d(x, y) \leq d(x, z) + d(z, y) \leq d_M(x, z) + d_M(z, y).$$

Likewise, $d'(x, y) \leq d_M(x, z) + d_M(z, y)$. Thus

$$\max\{d(x, y), d'(x, y)\} \leq d_M(x, z) + d_M(z, y),$$

which means $d_M(x, y) \leq d_M(x, z) + d_M(z, y)$.
2. We prove the set-theoretical identity $B_r^M(x) = B_r(x) \cap B'_r(x)$ by double inclusion. Let $y \in B_r^M(x)$. Then $d_M(x, y) < r$. Since $d \leq d_M$, we have $d(x, y) < r$, so $y \in B_r(x)$. A similar argument yields $y \in B'_r(x)$. Now let $y \in B_r(x) \cap B'_r(x)$. Then $d(x, y) < r$ and $d'(x, y) < r$, so $d_M(x, y) < r$ and consequently, $y \in B_r^M(x)$.
3. Since $d \leq d_M$ and $d' \leq d_M$, by Theorem 4.2.4, $\tau_d \subset \tau_{d_M}$ and $\tau_{d'} \subset \tau_{d_M}$. Thus $\tau_d \cup \tau_{d'} \subset \tau_{d_M}$.
4. Consider on $\mathbb{R}^2$ the post distance d and the distance d' defined in Exercise 4.13. Choosing radii sufficiently small in both metric spaces, $B_r(x) = \{x\}$ for $x \neq 0$ and $B'_r(0) = \{0\}$. Thus $B_r(x) \cap B'_r(x) = \{x\}$ for every $x \in \mathbb{R}^2$, hence τ_{d_M} is the discrete topology.

   In this example, we see that the union $\tau_d \cup \tau_{d'}$ is not a topology. The set $\{(0, 0)\}$ is open in $\tau_{d'}$ so it is an element of $\tau_d \cup \tau_{d'}$. Similarly, $\{(1, 1)\} \in \tau_d \subset \tau_d \cup \tau_{d'}$. If $\tau_d \cup \tau_{d'}$ is a topology, then $\{(0, 0), (1, 1)\} \in \tau_d \cup \tau_{d'}$. However, this set is neither open in τ_d ($(0, 0)$ is not an interior point) nor open in $\tau_{d'}$ ($(1, 1)$ is not an interior point). Thus the union of two metric topologies is not a topology in general. □

Exercise 4.28 Let d and d' be two distances on a set X. Prove that $d_m(x, y) = \min\{d(x, y), d'(x, y)\}$ is not, in general, a distance on X. In case that d_m is a distance, prove that $B_r^m(x) = B_r(x) \cup B_r'(x)$ and deduce that the metric topology τ_{d_m} satisfies $\tau_{d_m} \subset \tau_d \cap \tau_{d'}$. Find an example where this inclusion is proper.

Solution

1. We show an example where the triangle inequality fails for d_m. For the Euclidean distance d on $\mathbb{R}^2$, if a, b, c are three positive numbers such that $a \leq b + c$, $b \leq a + c$ and $c \leq a + b$, then there are $x, y, z \in \mathbb{R}^2$ such that $d(x, y) = a$, $d(y, z) = b$ and $d(y, z) = c$. Consider the triples of numbers $(7, 6, 1.1)$ and $(8, 5.8, 3)$, and let (x, y, z) and (x', y', z') be the triples of points of $\mathbb{R}^2$ such that
$$d(x, y) = 7, \quad d(x, z) = 6, \quad d(y, z) = 1.1,$$
$$d(x', y') = 8, \quad d(x', z') = 5.8, \quad d(y', z') = 3.$$
On the other hand, let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be any bijection such that $f(x') = x$, $f(y') = y$ and $f(z') = z$ and consider on $\mathbb{R}^2$ the distance $d_f(p, q) = d'(f^{-1}(p), f^{-1}(q))$. Then
$$d_f(x, y) = 8, \quad d_f(x, z) = 5.8, \quad d_f(y, z) = 3.$$
Thus $d_m(x, y) = 7$, $d_m(x, z) = 5.8$ and $d_m(z, y) = 1.1$, but $d_m(x, y) > d_m(x, z) + d(z, y)$.
2. We prove $B_r^m(x) = B_r(x) \cup B_r'(x)$ by double inclusion. Let $y \in B_r^m(x)$. Then $d_m(x, y) < r$; that is, $\min\{d(x, y), d'(x, y)\} < r$. If, say $\min\{d(x, y), d'(x, y)\} = d(x, y)$, then $d(x, y) < r$ proving that $y \in B_r(x) \subset B_r(x) \cup B_r'(x)$. A similar argument can be established if $\min\{d(x, y), d'(x, y)\} = d'(x, y)$.

   Let $y \in B_r(x) \cup B_r'(x)$. Without loss of generality, we may assume that $y \in B_r(x)$. Then $d(x, y) < r$ and as $d_m \leq d$, then $d_m(x, y) < r$, proving $y \in B_r^m(x)$.
3. Since $d_m \leq d$ and $d_m \leq d'$, by Theorem 4.2.4, $\tau_{d_m} \subset \tau_d \cap \tau_{d'}$. We observe that $\tau_d \cap \tau_{d'}$ is a topology on X.
4. Let us show a space where $\tau_d \cap \tau_{d'}$ is not Hausdorff and thus its topology cannot be a metric topology. Let $X = \{1/n : n \in \mathbb{N}\} \cup \{0, 2\}$ and define on X two distances d and d'. The distance d is the Euclidean distance. The distance d' agrees with d but interchanging the roles of $x = 0$ and $x = 2$. To be precise, let $f : X \to X$ be the bijection that fixes the numbers $1/n$ and $f(0) = 2$ and $f(2) = 0$. Consider d' the distance $d'(x, y) = d(f^{-1}(x), f^{-1}(y))$. We see that the topology $\tau_d \cap \tau_{d'}$ is not Hausdorff because it is not possible to separate $x = 0$ from $x = 2$. If this were not the case, there are disjoint neighborhoods U and V of 0 and 2, respectively. Since $U \in \tau_d$ and $(1/n) \to 0$, then there is $\nu_1 \in \mathbb{N}$ such that $1/n \in U$ for $n \geq \nu_1$. Let $r > 0$ such that $B_r^{d'}(2) \subset V$. As $d'(2, 1/n) = d(0, 1/n) = 1/n \to 0$, there is $\nu_2 \in \mathbb{N}$ such that $1/n \in B_r^{d'}(2)$ for $n \geq \nu_2$. Thus, $1/n \in U \cap V$ for $n \geq \max\{\nu_1, \nu_2\}$, which is not possible. □

We rediscover two notions from calculus. Let X be a set, (Y, d') a metric space and $f : X \to Y$. A sequence of functions $\{f_n : X \to Y\}$ is said to be *pointwise convergent* to f if $(f_n(x)) \to f(x)$ for all $x \in X$. Equivalently, given and arbitrary $x \in X$, for all $\epsilon > 0$, there is $\nu_x \in \mathbb{N}$ such that if $n \geq \nu_x$, $d'(f_n(x), f(x)) < \epsilon$. Similarly, $(f_n) \to f$ *uniformly converges* if for all $\epsilon > 0$, there is $\nu \in \mathbb{N}$ such that $d'(f_n(x), f(x)) < \epsilon$ for all $x \in X$. The name "uniformly" is because the number ν does not depend on x, but only on ϵ.

Exercise 4.29 Let (Y, d') be a metric space. For a set X, consider

$$\mathsf{B}(X, Y) = \{f : X \to Y : f(X) \text{ is a bounded set on } (Y, d')\}.$$

Define on $\mathsf{B}(X, Y)$ the sup distance by

$$d_\infty(f, g) = \sup\{d'(f(x), g(y)) : x \in X\}.$$

Prove that d_∞ is a distance. If (f_n) is a sequence in $\mathsf{B}(X, Y)$, prove that $(f_n) \to f$ if and only if $(f_n) \to f$ uniformly.

Solution The property that $f(X)$ is a bounded set is equivalent to f being a bounded function. Observe that the notation d_∞ has been employed for $\mathsf{C}([0, 1])$ in Exercise 4.4. Note also that $\mathsf{C}([0, 1]) \subset \mathsf{B}([0, 1], \mathbb{R})$ and that these distances coincide in this set because every continuous function in $[0, 1]$ attains a maximum.

Concerning the sup distance, we make two comments. First, there is no distance defined on the domain X. And second, we have $d_\infty(f, g) < \infty$ because if $f(X) \subset B_{r_1}(f(x_0))$ and $g(X) \subset B_{r_2}(g(x_0))$ and $x_0 \in X$ a fixed point, the triangle inequality yields

$$\begin{aligned} d'(f(x), g(x)) &\leq d'(f(x), f(x_0)) + d'(f(x_0), g(x_0)) + d'(g(x), g(x_0)) \\ &\leq r_1 + r_2 + d'(f(x_0), g(x_0)), \end{aligned}$$

for any $x \in X$. This assures that $\{d'(f(x), g(x)) : x \in X\}$ is bounded from above.

The properties of d_∞ as a distance are all trivial because the arguments are essentially as in Exercise 4.4. We proceed to show the equivalence of the convergence of sequences in this space with the notion of uniform convergence.

(i) Assume that $(f_n) \to f$ with the sup distance. Let $\epsilon > 0$ be given. Then there is $\nu \in \mathbb{N}$ such that $d_\infty(f_n, f) < \epsilon$ whenever $n \geq \nu$. For all these numbers n,

$$\sup\{d'(f_n(x), f(x)) : x \in X\} < \epsilon;$$

in particular, $d'(f_n(x), f(x)) < \epsilon$ for all $x \in X$. This is precisely the notion of uniform convergence.

(ii) Reciprocally, suppose that $(f_n) \to f$ uniformly and we prove the convergence $(f_n) \to f$ for d_∞. Let $\epsilon > 0$ be given. For $\epsilon/2$ and by the uniform convergence, there is $\nu \in \mathbb{N}$ such that $d'(f_n(x), f(x)) < \epsilon/2$ for all $n \geq \nu$ and for all $x \in X$.

Thus, for $n \geq \nu$

$$\sup\{d'(f_n(x), f(x)) : x \in X\} \leq \epsilon/2;$$

in particular, $d_\infty(f_n, f) < \epsilon$ for $n \geq \nu$. □

4.6 Suggested Exercises

4.6.1 In a metric space (X, d), prove that

$$d'(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

is a bounded equivalent distance to d.

4.6.2 Let $f : [0, \infty) \to [0, \infty)$ be a non-decreasing function such that $f(t) = 0$ if and only if $t = 0$, $\lim_{t\to 0} f(t) = 0$ and $f(s+t) \leq f(s)+f(t)$ for all $s, t \in [0, \infty)$. If (X, d) is a metric space, prove that $d'(x, y) = f(d(x, y))$ defines a distance on X equivalent to d. Check that the distances of Exercises 4.26 and 4.6.1 are derived with this method. Describe the distance for $f(t) = \lambda t$ and $f(t) = t^\lambda$ for $\lambda > 0$.

4.6.3 Define on $\mathbb{R}^2$

$$d((x_1, y_1), (x_2, y_2)) = \begin{cases} |y_1 - y_2|, & \text{if } x_1 = x_2 \\ |y_1| + |x_1 - x_2| + |y_2|, & x_1 \neq x_2. \end{cases}$$

See Fig. 4.13. Show that d is a distance and compare with the Euclidean distance.

4.6.4 Prove that

$$d(x, y) = \begin{cases} |x - y|, & \text{if } x - y \in \mathbb{Q} \\ 1 + |x - y|, & \text{if } x - y \in \mathbb{R} - \mathbb{Q} \end{cases}$$

is a distance on $\mathbb{R}$. Determine the convergent sequences whose limit is $x = 0$.

4.6.5 In any metric space, prove the *quadrangle inequality*

$$|d(x, y) - d(z, w)| \leq d(x, z) + d(y, w).$$

As a consequence, if $(x_n) \to x$ and $(y_n) \to y$, show that $(d(x_n, y_n)) \to d(x, y)$.

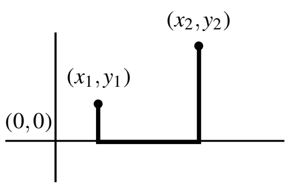


Fig. 4.13 The distance of Exercise 4.6.3

4.6.6 In a metric space, prove that there is not a ball of radius 2 contained strictly in a ball of radius 1, possibly with different centers.

4.6.7 If A and B are two subsets of a metric space, the *distance between A and B* is defined by

$$d(A, B) = \inf\{d(a, b) : a \in A, b \in B\}.$$

Show that $d(A, B) = \inf\{d(a, B) : a \in A\} = \inf\{d(b, A) : b \in B\}$. Find a space and subsets A and B such that $d(A, B) = 0$ but $A \cap B = \emptyset$.

4.6.8 Let (X, d) be a metric space where $d \leq 1$. Let $\omega \notin X$ and consider $X^* = X \cup \{\omega\}$. On X^*, define

$$d^*(x, y) = \begin{cases} d(x, y), & x, y \in X \\ 1, & x = \omega, y \in X \\ 1, & x \in X, y = \omega \\ 0, & x = y = \omega. \end{cases}$$

Show that d^* is a distance on X^*. Show that the relative topology on X is the metric topology of d. Prove that $\{\omega\}$ is open and closed.

4.6.9 Using Theorem 4.4.2, find the interior and closure of each one of the following subsets of $\mathbb{R}^2$: $\mathbb{R}\times\{0\}$, $\mathbb{R}\times(\mathbb{R}\backslash\{0\})$, $\mathbb{Q}\times\mathbb{R}$, $\{(x, y) \in \mathbb{R}^2 : 0 < x^2+y^2 \leq 1\}$ and $\{(x, y) \in \mathbb{R}^2 : x \neq 0, y \leq 1/x\}$.

4.6.10 Let $(x_n) \to x$ and $(y_n) \to y$ be two convergent sequences in $(\mathbb{R}, \tau_u)$. If $x_n \geq y_n$ for all $n \in \mathbb{N}$, prove that $x \geq y$. Investigate the problem replacing the Euclidean topology by the Sorgenfrey topology and right order topology.

4.6.11 Define on $\mathbb{R}^n$ a distance d as follows. If x and y are situated on the same straight line through the origin, the distance is the Euclidean distance; otherwise, the distance is the post office distance (Exercise 4.12). Show that d is a distance and describe the balls in this metric space. Compare with the Euclidean distance and post office distance.

4.6.12 If $A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$, find $d((1, 1), A)$ for the distances d_u, d_t and d_∞.

4.6.13 The *diameter* of a metric space X, denoted diam (X), is defined by

$$\operatorname{diam}(X) = \sup\{d(x, y) : x, y \in X\}.$$

This number can be infinite, but it follows that the diameter is finite if and only if X is bounded. Let A and B two bounded subsets of a metric space.

(i) If $A \cap B = \emptyset$, prove that $\operatorname{diam}(A \cup B) \leq \operatorname{diam}(A) + \operatorname{diam}(B)$.
(ii) If $A \cap B \neq \emptyset$, prove $\operatorname{diam}(A \cup B) \leq \operatorname{diam}(A) + \operatorname{diam}(B) + d(A, B)$.
(iii) Prove that $\operatorname{diam}(\overline{A}) = \operatorname{diam}(A)$.

4.6.14 Prove that there are no surjective maps $f : \mathbb{R} \to \mathbb{R}^2$ that preserve the Euclidean distances, so that $d'(f(x), f(y)) = d(x, y)$ for all $x, y \in \mathbb{R}$. [Hint: apply the Cauchy-Schwarz inequality.]

4.6.15 If $(x_n) \to x$ in a metric space, show that $A = \{x_n : n \in \mathbb{N}\} \cup \{x\}$ is closed and bounded.

4.6.16 With the notation of Exercise 4.8, prove

1. $V_r(x) = \bigcap_{s>r} B_s(x) = \bigcap_{n\in\mathbb{N}} B_{r+1/n}(x)$.
2. $\{x\} = \bigcap_{s>0} B_s(x) = \bigcap_{n\in\mathbb{N}} B_{1/n}(x)$.
3. $B_r(x) = \bigcup_{s<r} V_s(x) = \bigcup_{n\in\mathbb{N}} V_{r-1/n}(x)$.
4. $B_r(x) \subset \operatorname{int}(V_r(x))$.

4.6.17 In a metric space, show that $x \in A'$ if and only if there is $(a_n) \subset A$ of different terms that converges to x.

4.6.18 An isometry of the Euclidean real line is a bijection $f : \mathbb{R} \to \mathbb{R}$ such that $|f(x) - f(y)| = |x - y|$ for all $x, y \in \mathbb{R}$. Given $a \in \mathbb{R}$, prove that there are exactly two isometries that fix a.

Chapter 5
Continuity

In the previous chapters, we introduced the concept of topological space, as well as a variety of notions related to "proximity". Going forward, we shall study the other pillar in point-set topology, those transformations between topological spaces that map the notions of proximity from one space into another. These transformations are the continuous maps. Recall the first sentence in the introduction, *topology studies continuous maps*, which indicates that topological spaces and continuous maps go hand in hand.

Surely the reader has already worked in calculus with the continuity of functions and is familiar with the prominent role that this concept plays in analysis. Turning to the topological setting, we now are tasked with defining the continuity of a mapping in a topological space. The concept of a neighborhood can help to make intuitively clear the notion of continuity of a map at a point. Once we have defined a continuous map, our task is to determine how these maps behave with respect to the topological concepts defined in earlier chapters.

5.1 Continuous Mappings

The continuity of a map $f: X \to Y$ at a point x should reflect that $f(z)$ is arbitrarily close to $f(x)$ if z is sufficiently close to x. As mentioned before, our motivation comes from the familiar definition of continuity of a real-valued function that appears in calculus and given in terms of ϵ's and δ's. In that setting, a function $f: \mathbb{R} \to \mathbb{R}$ is continuous at $x \in \mathbb{R}$ if

$$\forall \epsilon > 0, \exists \delta > 0 : |z - x| < \delta \Rightarrow |f(z) - f(x)| < \epsilon. \tag{5.1}$$

We rewrite this expression in language of topological spaces. First, we avoid employing the distance, that is particular of $\mathbb{R}$ and metric spaces in general. So

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_5

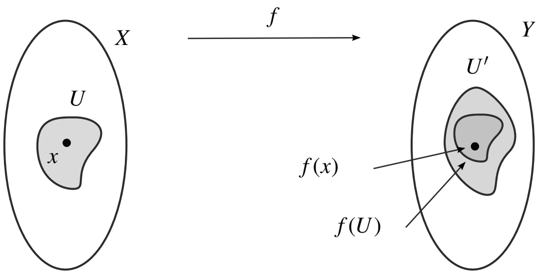


Fig. 5.1 The continuity of f at x says that for every neighborhood U' of $f(x)$, there is a neighborhood U of x such that $f(U) \subset U'$

we may write $|z - x| < \delta$ as $z \in (x - \delta, x + \delta) = B_\delta(x)$. Likewise, we replace $|f(z) - f(x)| < \epsilon$ by $f(z) \in B_\epsilon(f(x))$. The expression (5.1) is now

$$\text{for all } \epsilon > 0, \text{ there is } \delta > 0 \text{ such that } f(B_\delta(x)) \subset B_\epsilon(f(x)). \tag{5.2}$$

The last inequality in (5.1) has been replaced in (5.2) by an inclusion of sets. Since every neighborhood contains a ball, we see the next definition of continuity is then motivated (Fig. 5.1).

Definition 5.1.1 A mapping $f: X \to Y$ is said to be *continuous* at $x \in X$ if for every $U' \in \mathscr{U}'_{f(x)}$, there is $U \in \mathscr{U}_x$ such that $f(U) \subset U'$. If f is continuous at each point of X, we say that f is *continuous on* X, or simply *continuous*. If f is not continuous at a point x, we say that f is *discontinuous* at x.

Example 5.1 Consider on $\mathbb{R}$ the topologies τ_0 and τ_1 of the included point topology for the particular points 0 and 1, respectively. Let

$$f: (\mathbb{R}, \tau_0) \to (\mathbb{R}, \tau_1), \quad f(x) = x^2.$$

A basis of neighborhoods of x for τ_0 is $\beta_x^0 = \{U_x = \{0, x\}\}$ and for τ_1, $\beta_x^1 = \{V_x = \{1, x\}\}$. We first study the continuity at $x = 0$ and $x = 2$ and later, at every point.

1. We prove that f is continuous at $x = 0$. Since $f(0) = 0$, let U' be a neighborhood of 0 in $(\mathbb{R}, \tau_1)$. Then $V_0 \subset U'$. It is clear that $f(U_0) = \{f(0)\} = \{0\} \subset V_0 = \{0, 1\}$, so $f(U_0) \subset U'$. This proves that f is continuous at $x = 0$.
2. We see that f is not continuous at $x = 2$. Now $f(2) = 4$. Take $V_4 = \{1, 4\}$ as a neighborhood of $f(2)$. If f is continuous at $x = 2$, there is $U \in \mathscr{U}_2$ such that $f(U) \subset V_4$, and because $U_2 \subset U$, then $f(U_2) \subset V_4$. However, $f(U_2) = \{0, 4\}$, so $f(U_2) \not\subset V_4$.
3. We study the continuity of f at any $x \in \mathbb{R}$. Since in each neighborhood there is a basic neighborhood, the continuity of f at x is equivalent to testing the inclusion $f(U_x) \subset V_{f(x)}$, or in other words,

$$\{f(0) = 0, f(x)\} \subset \{1, f(x)\}.$$

This inclusion occurs if and only if $f(x) = 0$. Definitively, f is continuous at $x = 0$ and discontinuous at every other point. $\Diamond$

Proposition 5.1.2 *Consider β and β' two bases in X and Y respectively, and let β_x and β'_y be bases of neighborhoods for $x \in X$ and $y \in Y$, respectively. Let $f\colon X \to Y$ be a mapping. Then the following statements are equivalent:*

(i) *The map f is continuous at x.*
(ii) *For every open O' of Y such that $f(x) \in O'$, there is an open set O in X such that $x \in O$ and $f(O) \subset O'$.*
(iii) *For every $B' \in \beta'$ such that $f(x) \in B'$, there is $B \in \beta$ such that $x \in B$ and $f(B) \subset B'$.*
(iv) *For every $V' \in \beta'_{f(x)}$, there is $V \in \beta_x$ such that $f(V) \subset V'$.*
(v) *For every $V' \in \beta'_{f(x)}$, $f^{-1}(V')$ is a neighborhood of x.*

In particular, if X and Y are metric spaces, then f is continuous at x if and only if (5.2) *holds.*

Example 5.2 We prove that constant maps are continuous. Let $f(x) = y_0$ be a constant map, where $y_0 \in Y$ is a given fixed point. Then the preimage of any open set $O' \subset Y$ is open because

$$f^{-1}(O') = \begin{cases} \emptyset, & \text{if } y_0 \notin O' \\ X, & \text{if } y_0 \in O'. \end{cases}$$

$\Diamond$

Example 5.3 A map $f\colon (X, d) \to (Y, d')$ between two metric spaces is said to be *Lipschitz* if there is $K \geq 0$ such that $d'(f(x), f(z)) \leq Kd(x, z)$ for all $x, z \in X$. We see that f is continuous on X. The case $K = 0$ is trivial because f is constant. Assume $K > 0$. Let $x \in X$ and let $\epsilon > 0$ be a given but arbitrary positive number. Choose $\delta = \epsilon/K$. If $d(z, x) < \delta$, then $d'(f(z), f(x)) \leq Kd(z, x) < K \cdot \delta = \epsilon$. $\Diamond$

We investigate the continuity of a map over its entire domain. First we take a quick pause to recall various definitions and properties from set theory. Given a mapping $f\colon X \to Y$ and $B \subset Y$, the *preimage* of B (also called inverse image) is defined by $f^{-1}(B) = \{x \in X : f(x) \in B\}$. Note that f^{-1} is not the inverse of f, because we do not require that f is injective or surjective (see also Exercise 2.7.6). Among the relevant properties, we point out that $f(f^{-1}(B)) \subset B$ and $A \subset f^{-1}(f(A))$ for $B \subset Y$ and $A \subset X$. The preimage under a map behaves as expected with respect to the operations of sets. So $f^{-1}(\emptyset) = \emptyset$, $f^{-1}(Y) = X$, $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$ and

$$f^{-1}\Big(\bigcup_{i\in I} B_i\Big) = \bigcup_{i\in I} f^{-1}(B_i), \quad f^{-1}\Big(\bigcap_{i\in I} B_i\Big) = \bigcap_{i\in I} f^{-1}(B_i).$$

The statements in the next theorem are quite striking in its elegance and simplicity.

Theorem 5.1.3 *If $f: X \to Y$, then the following are equivalent:*

1. *The map f is continuous on X.*
2. *$f^{-1}(O')$ is open for every open set O' of Y.*
3. *$f^{-1}(F')$ is closed for every closed set F' of Y.*
4. *$f(\overline{A}) \subset \overline{f(A)}$ for every $A \subset X$.*
5. *For a basis β' for Y, $f^{-1}(B')$ is open in X for every $B' \in \beta'$.*

Example 5.4 Recall that in set theory, the *identity map* on a set X is the mapping $\mathbb{1}_X : X \to X$ defined by $\mathbb{1}_X(x) = x$ for all $x \in X$. As a consequence of (ii) of Theorem 5.1.3, the identity map $\mathbb{1}_X : (X, \tau_1) \to (X, \tau_2)$ is continuous if and only if τ_1 is finer than τ_2. ◊

A practical problem in studying the continuity of a mapping is that we may find an open set whose preimage is not open, so the map is not continuous. However, this does not tell us at which points the map is continuous or discontinuous. A simple criterion of discontinuity is the following.

Proposition 5.1.4 *Let $f: X \to Y$ be a map and let $O' \subset Y$ be an open set such that $f^{-1}(O')$ is not open in X. Then f is discontinuous at all points of $f^{-1}(O') \setminus$* $\mathrm{int}(f^{-1}(O'))$.

Example 5.5 We prove that the function $f(x) = x^2$ is continuous with the Euclidean topology. Apply (5) of Theorem 5.1.3. Let $\beta_u = \{(a, b) : a, b \in \mathbb{R}\}$ be the standard basis for τ_u. Then

$$f^{-1}((a,b)) = \begin{cases} (-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b}), & \text{if } 0 \leq a \\ (-\sqrt{b}, \sqrt{b}), & \text{if } a < 0 < b \\ \emptyset, & \text{if } b \leq 0 \end{cases}$$

and thus f is continuous. Let us notice the cleanness of the proof compared to the arguments using the definition (5.1) of calculus. ◊

Example 5.6 Consider the discrete space (X, τ_D). Then for every space (Y, τ'), any map $f: (X, \tau_D) \to (Y, \tau')$ is continuous because the preimage of every open set in Y is open in X since it is a subset of X. In fact, this property characterizes the discrete topology: *If (X, τ) is a space such that any mapping from (X, τ) onto another space is continuous, then τ is the discrete topology*. The proof of the result is as follows. Because the property holds for any map, we consider the identity where the codomain is (X, τ_D), $\mathbb{1}_X : (X, \tau) \to (X, \tau_D)$. By hypothesis, this map is continuous, so Example 5.4 implies that $\tau_D \subset \tau$. Since the other inclusion is always true, we conclude $\tau = \tau_D$.

Analogously, *every mapping whose codomain is the trivial space is continuous and this property characterizes the trivial topology*. ◊

In calculus, it is well known that the continuity on the Euclidean real line is characterized by convergence of sequences. Similarly, a map $f: (X, d) \to (Y, d')$ between two metric spaces is continuous at x if and only if for every $(x_n) \to x$, we have $(f(x_n)) \to f(x)$.

However, in a topological space we only have a partial result in the following sense.

Proposition 5.1.5 *If $f: X \to Y$ is continuous at $x \in X$, then for all $(x_n) \to x$, we have $(f(x_n)) \to f(x)$.*

A map f is said to be *sequentially continuous* if for every convergent sequence $(x_n) \to x$, it holds that $(f(x_n)) \to f(x)$. The converse in Proposition 5.1.5 fails in general.

Example 5.7 Let $\mathbb{R}$ be endowed with the co-countable topology τ_{CC} and consider the identity map $\mathbb{1}_{\mathbb{R}} : (\mathbb{R}, \tau_{CC}) \to (\mathbb{R}, \tau_u)$. We see that $\mathbb{1}_{\mathbb{R}}$ is sequentially continuous. Let $(x_n) \to x$ in $(\mathbb{R}, \tau_{CC})$. We learned in Exercise 3.4 that (x_n) is eventually constant. Thus $(\mathbb{1}_{\mathbb{R}}(x_n)) = (x_n)$ is eventually constant, so convergent to x. However, $\mathbb{1}_{\mathbb{R}}$ is not continuous because $(0, 1)$ is open in $(\mathbb{R}, \tau_u)$ but its preimage under $\mathbb{1}_{\mathbb{R}}$ is $(0, 1)$, which is not open in $(\mathbb{R}, \tau_{CC})$. ◊

5.2 Properties of Continuous Maps

We relate the continuity of mappings with various concepts of set theory such as the composition and restrictions of set-theoretic maps.

Proposition 5.2.1

1. *The identity map $\mathbb{1}_X : (X, \tau) \to (X, \tau)$ is continuous.*
2. *If $f: X \to Y$ is continuous at x and $g: Y \to Z$ is continuous at $g(x)$, then $g \circ f: X \to Z$ is continuous at x. As a consequence, composition of continuous maps is continuous.*

We work with the concepts of continuity and inclusion maps. We present the setting. In set theory, if A is a subset of X, the *inclusion map* is defined by

$$i: A \hookrightarrow X, \quad i(a) = a.$$

Making the natural assumption that A supports the relative topology, the next result demonstrates that the inclusion map is continuous.

Proposition 5.2.2 *The inclusion map $i: (A, \tau_{|A}) \to (X, \tau)$ is continuous. Furthermore, $\tau_{|A}$ is the coarsest topology on A that makes continuous the inclusion mapping.*

The following context refers to the restriction of maps. From set theory, if $f: X \to Y$ is a mapping and $A \subset X$, the set-theoretic *restriction of f to A* is defined by

$$f_{|A} \colon A \to Y, \quad f_{|A}(a) = f(a).$$

By contrast, it is not natural to restrict a map to a subset of the codomain; if, for instance, $B \subset Y \setminus \mathrm{Im}(f)$, the expression $f \colon X \to B$ does not make sense. A way to overcome this problem is to require that $\mathrm{Im}(f) \subset B$. In such a case, the restriction of f to B is the map $f \colon X \to B$ defined for each $x \in X$ as $f(x)$ (note that in this situation we keep identical symbol for the map and its restriction).

Theorem 5.2.3 *Let (X, τ) and (Y, τ') be two spaces, $A \subset X$, $B \subset Y$ and $f \colon (X, \tau) \to (Y, \tau')$ continuous.*

1. *The restriction map $f_{|A} \colon (A, \tau_{|A}) \to (Y, \tau')$ is continuous. In fact, it suffices that f is continuous only at those points of A.*
2. *If $Im(f) \subset B$ then $f \colon (X, \tau) \to (B, \tau'_{|B})$ is continuous.*
3. *The mapping $f_{|A} \colon (A, \tau_{|A}) \to (f(A), \tau'_{|f(A)})$ is continuous.*

Of course, continuity on A does not imply continuity on X as we exhibit in the next example.

Example 5.8 Consider the function

$$f \colon \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} 0, & \text{if } x \in \mathbb{Q} \\ 1, & \text{if } x \in \mathbb{R} - \mathbb{Q}. \end{cases}$$

Then $f_{|\mathbb{Q}} \colon \mathbb{Q} \to \mathbb{R}$ is continuous because it is constant, but $f \colon \mathbb{R} \to \mathbb{R}$ is discontinuous at every point of $\mathbb{Q}$; if $(x_n) \subset \mathbb{R} - \mathbb{Q}$ is a sequence of irrationals converging to $x \in \mathbb{Q}$ then $f(x_n) = 1 \to 1 \neq f(x) = 0$. Notice that $i \colon \mathbb{Q} \hookrightarrow \mathbb{R}$ and $f \circ i \colon \mathbb{Q} \to \mathbb{R}$ are both continuous but f is discontinuous at every rational. This example shows that it may occur that $f \colon X \to Y$ and $g \circ f \colon X \to Z$ are continuous but not $g \colon Y \to Z$. $\diamondsuit$

Motivated by this example, it is reasonable to believe that continuity is a *local concept*, so it is enough to consider the behaviour of the mapping in a neighborhood of the point.

Proposition 5.2.4 *If $f \colon X \to Y$ and $x \in X$, then the following are all equivalent:*

1. *The map f is continuous at x.*
2. *For every neighborhood U of x, the mapping $f_{|U} \colon U \to Y$ is continuous at x.*
3. *There is a neighborhood U of x such that $f_{|U} \colon U \to Y$ is continuous at x.*
4. *There is an open set O such that $x \in O$ and $f_{|O} \colon O \to Y$ is continuous at x.*

We close this section with a useful result that guarantees the continuity of a map that is piecewise defined.

Question: *Suppose $X = A \cup B$ and $f \colon X \to Y$ a map such that $f_{|A}$ and $f_{|B}$ are continuous. Under what conditions on A and B is f continuous?*

Theorem 5.2.5 (Pasting Lemma) *Let $f\colon X \to Y$ and suppose $X = A \cup B$, where A and B are closed sets (or both are open). If $f_{|A}$ and $f_{|B}$ are continuous, then f is continuous.*

If A and B are open, we can directly apply (4) of Proposition 5.2.4, so this theorem is interesting when A and B are closed. Moreover, if A and B form a partition by closed sets of X, then they are also open sets. It is clear that Theorem 5.2.5 extends to an arbitrary number of open sets, or a finite number of closed sets.

Example 5.9 Define

$$f\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}, \qquad f(x) = \begin{cases} -1, & \text{if } x \in (-\infty, 0) \\ 1, & \text{if } x \in (0, \infty). \end{cases}$$

Then f is continuous because $(-\infty, 0)$ and $(0, \infty)$ are open in $\mathbb{R} \setminus \{0\}$ and the restriction of f into each subset is a constant function. Notice that this function has a jump and one may think that f is not continuous. Observe that the domain of f is not $\mathbb{R}$. $\Diamond$

Example 5.10 Let $X = \mathbb{R} \times \{0, 1\}$ and let $f\colon X \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} 0, & y = 0 \\ 1, & y = 1. \end{cases}$$

Consider $A = \mathbb{R} \times \{0\}$ and $B = \mathbb{R} \times \{1\}$. Then the restrictions of f to A and B are continuous because these restrictions are constant. Thus f is continuous if A and B are open in X (or both are closed). Observe that $\{A, B\}$ is a partition of X, hence if they are open, so too are they closed and *vice-versa*. However, it is much easier to show that they are closed than open. So A and B are closed in $\mathbb{R}^2$ because are the product of two closed sets (Corollary 4.4.5), hence also closed in X. Notice that A and B are not open in $\mathbb{R}^2$, although they are open in X. $\Diamond$

5.3 Continuous Maps on Euclidean Spaces

It is well known from calculus that the sum or product of continuous real-valued functions of a real variable is continuous. The goal of this section is to extend this type of result for arbitrary topological spaces.

Let us recall some definitions of set theory. For two set-theoretic maps $f, g\colon X \to \mathbb{R}^n$, the *sum* of f and g is defined as

$$f + g\colon X \to \mathbb{R}^n \quad (f + g)(x) = f(x) + g(x).$$

If $\lambda \in \mathbb{R}$, the *product* of λ by f is the mapping defined as

$$\lambda \cdot f : X \to \mathbb{R}^n \quad (\lambda \cdot f)(x) = \lambda f(x).$$

One can also define the *subtraction map* $f - g : X \to \mathbb{R}^n$. In the particular case $g : X \to \mathbb{R}$, the *product* of g and f is defined by

$$g \cdot f : X \to \mathbb{R}^n \quad (g \cdot f)(x) = g(x)f(x),$$

and similarly, the *quotient* $f/g : X \to \mathbb{R}^n$ at those points $x \in X$ where $g(x) \neq 0$.

We turn to the continuity of these maps when X is a topological space, where $\mathbb{R}^n$ is equipped with the Euclidean topology. As a first step we will consider $\mathbb{R}$ as the codomain and, later, the general case $\mathbb{R}^n$.

Proposition 5.3.1 *Let $f, g : X \to \mathbb{R}$ and $\lambda \in \mathbb{R}$. If f, g are continuous at $x_0 \in X$, then $f + g$, λf, $f - g$, fg and f/g (if $g(x_0) \neq 0$) are continuous at x_0.*

The next step substitutes $\mathbb{R}^n$ by $\mathbb{R}$ for the codomain. The idea is to write a map $f : X \to \mathbb{R}^n$ in coordinates and then, to carry the question for maps whose codomain is $\mathbb{R}$. For this reason, we need consider the projections of $\mathbb{R}^n$. The following concept is not topological and stems from set theory. For each $1 \leq i \leq n$, the *ith projection* of $\mathbb{R}^n$ is defined by

$$p_i : \mathbb{R}^n \to \mathbb{R}, \quad p_i(x_1, \ldots, x_n) = x_i.$$

If we consider the usual topologies, the projections are continuous.

Proposition 5.3.2 *The projections of the Euclidean space are continuous.*

Proof Let $i \in \{1, \ldots, n\}$ and we prove that p_i is continuous at every $a \in \mathbb{R}^n$. Let $a = (a_1, \ldots, a_n) \in \mathbb{R}^n$ and $a_i = p_i(a_1, \ldots, a_n)$. Then

$$|p_i(x) - p_i(a)| = |x_i - a_i| \leq \sqrt{\sum_{j=1}^{n} (x_j - a_j)^2} = |x - a|,$$

and this proves that p_i is a Lipschitz map with $K = 1$ (Example 5.3). □

One might also prove the continuity of p_i using Theorem 5.1.3. Indeed, consider the usual basis β_u for the usual topology. We prove the continuity for p_1 (similar for the other projections). See Fig. 5.2. Then

$$p_1^{-1}((a, b)) = (a, b) \times \mathbb{R} \times \overset{n-1}{\smile\ldots} \times \mathbb{R} = (a, b) \times \mathbb{R}^{n-1}.$$

But the Cartesian product of open sets of Euclidean spaces is open (Corollary 4.4.5).

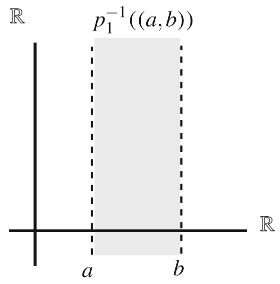


Fig. 5.2 The set $p_1^{-1}((a, b)) = (a, b) \times \mathbb{R}$ is open in $\mathbb{R}^2$

We discuss how projections play their role in the study of the continuity of maps whose codomain is $\mathbb{R}^n$. If $f: X \to \mathbb{R}^n$, the *ith coordinate function* of f, $1 \le i \le n$, is defined by $f_i: X \to \mathbb{R}$, $f_i = p_i \circ f$. Therefore $f(x)$ writes as $f(x) = (f_1(x), \dots, f_n(x))$.

Proposition 5.3.3 *The map $f: X \to \mathbb{R}^n$ is continuous if and only if all its coordinate functions are continuous. As a consequence, if $f, g: X \to \mathbb{R}^n$ are continuous, then $\lambda f + \mu g$ is continuous, with $\lambda, \mu \in \mathbb{R}$. If $h: X \to \mathbb{R}$ is continuous, then $hf: X \to \mathbb{R}^n$ is continuous and f/h is continuous at those points $x \in X$ where $h(x) \neq 0$.*

The previous result shows that the Euclidean topology on $\mathbb{R}^n$ plays well with the vector structure of $\mathbb{R}^n$. But what exactly is behind all this? This nice relationship is based on that *the addition of vectors and product by scalars are continuous maps considering the Euclidean topology* as we see in the following example.

Example 5.11 Consider the sum of vectors as the map

$$\phi: \mathbb{R}^n \times \mathbb{R}^n = \mathbb{R}^{2n} \to \mathbb{R}^n, \quad \phi(x, y) = x + y.$$

We prove that ϕ is continuous. First, we need to indicate which is the topology on $\mathbb{R}^n \times \mathbb{R}^n$. The usual topology on $\mathbb{R}^n$ is metric and it is natural to consider on the Cartesian product $\mathbb{R}^n \times \mathbb{R}^n$ the product distance (Exercise 4.22). It was shown in Exercise 4.23 that this distance is equivalent to the Euclidean distance of $\mathbb{R}^{2n}$. We check that ϕ is continuous by composing with the projections. Indeed, let p_i and q_j be the projections on $\mathbb{R}^n$ and $\mathbb{R}^{2n}$ respectively. Then $p_i(\phi(x, y)) = x_i + y_i$, so $p_i \circ \phi = q_i + q_{n+i}$ for all $1 \le i \le n$. The map

$$\psi: \mathbb{R} \times \mathbb{R}^n = \mathbb{R}^{n+1} \to \mathbb{R}^n, \quad \psi(\lambda, x) = \lambda \cdot x$$

is consequently continuous, for if q_i are the projections on $\mathbb{R}^{n+1}$, then $p_i \circ \psi = q_1 \cdot q_{1+i}$ which is continuous. $\Diamond$

Theorem 5.1.3 provides a technique to show that a set is open (or closed) if this set can be expressed as the preimage of an open (or closed) set under a continuous map.

Example 5.12 A *hyperplane* of $\mathbb{R}^n$ is a subset that can be written as

$$H = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : a_1x_1 + \ldots + a_nx_n + b = 0\},$$

where $a_i, b \in \mathbb{R}$, with some $a_i \neq 0$. We prove that any hyperplane is a closed set in $\mathbb{R}^n$. For this purpose, define

$$f : \mathbb{R}^n \to \mathbb{R}, \quad f(x_1, \ldots, x_n) = a_1x_1 + \ldots + a_nx_n + b.$$

This mapping is continuous because $f = \sum_{i=1}^n a_i p_i + b$ is a linear combination of the projections p_i of $\mathbb{R}^n$ and the constant map $g(x) = b$ (Proposition 5.3.3). Since $H = f^{-1}(\{0\})$ and $\{0\}$ is closed in $\mathbb{R}$, we deduce that H is closed in $\mathbb{R}^n$.

As a consequence, every (affine) subspace of arbitrary dimension is also closed in $\mathbb{R}^n$ because every subspace of dimension k is the intersection of $n - k$ hyperplanes. Observe that a subspace $A \subset \mathbb{R}^n$ of dimension k can also be expressed as $A = p + \vec{A} = \{p + x : x \in \vec{A}\}$, where $p \in \mathbb{R}^n$ and $\vec{A}$ is the vector subspace associated to A. Another consequence is that the half spaces determined by H,

$$\{(x_1, \ldots, x_n) \in \mathbb{R}^n : a_1x_1 + \ldots + a_nx_n + b > 0\},$$

$$\{(x_1, \ldots, x_n) \in \mathbb{R}^n : a_1x_1 + \ldots + a_nx_n + b < 0\},$$

are open sets in $\mathbb{R}^n$ because they are $f^{-1}((0, \infty))$ and $f^{-1}((-\infty, 0))$, respectively. $\Diamond$

The second example considers the sphere in Euclidean space. The concept of a sphere appeared in Chap. 4 in the context of metric spaces, where the notation employed was $\mathbb{S}_r(a)$. Now we particularize to Euclidean spaces.

Example 5.13 The n-dimensional *sphere* of radius $r > 0$ and center $a \in \mathbb{R}^{n+1}$ is defined by

$$\mathbb{S}_r^n(a) = \left\{x = (x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : |x - a| = r\right\}.$$

We prove that $\mathbb{S}_r^n(a)$ is closed in $\mathbb{R}^{n+1}$. Define

$$f : \mathbb{R}^{n+1} \to \mathbb{R}, \quad f(x) = |x - a|^2 - r^2 = \sum_{i=1}^{n+1}(x_i - a_i)^2 - r^2.$$

This map can be expressed as

$$f = \sum_{i=1}^{n+1}(p_i - a_i)^2 - r^2.$$

By Proposition 5.3.3, f is continuous. Finally, $\mathbb{S}^n_r(a) = f^{-1}(\{0\})$ proving that $\mathbb{S}^n_r(a)$ is closed. Henceforth, we denote by $\mathbb{S}^n$ (resp. $\mathbb{S}^n_r$) the n-sphere of radius 1 (resp. $r > 0$) and center the origin of $\mathbb{R}^{n+1}$. For $n = 1$, $\mathbb{S}^1$ is also called a *circle*.

We point out that $\mathrm{int}(\mathbb{S}^n_r(a)) = \emptyset$. The reason is that no ball of $\mathbb{R}^{n+1}$ is included in the sphere $\mathbb{S}^n_r(a)$, as the reader can easily verify. ◊

The reader should be warned about possible confusing arising from the use of plain language. For example, the expression $\mathbb{S}^1 \subset \mathbb{S}^2$ is meaningless because $\mathbb{S}^1 \subset \mathbb{R}^2$ and $\mathbb{S}^2 \subset \mathbb{R}^3$. A circle in $\mathbb{R}^3$ is the locus of points in $\mathbb{R}^3$ situated in a plane and equidistant from a point of this plane. Thus this circle is the intersection of a sphere $\mathbb{S}^2_r(a)$ with a plane of $\mathbb{R}^3$ and consequently, it is a closed set because it is the intersection of two closed sets.

Example 5.14 Let $a \in \mathbb{R}^3$, $r > 0$ and $v \in \mathbb{R}^3$ with $|v| = 1$. The *cylinder* of radius r and axis v through a is defined by

$$C_r(v; a) = \{x \in \mathbb{R}^3 : |x - a|^2 - \langle x - a, v\rangle^2 = r^2\}.$$

One can check that $C_r(v; a)$ is the locus of the points whose distance to the straight line through a and direction v is r. This line is called the rotation axis of the cylinder. For example, if $r = 1$, $a = (0, 0, 0)$ and $v = (0, 0, 1)$, then

$$C_1(v; a) = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 = 1\} = \mathbb{S}^1 \times \mathbb{R}.$$

See Fig. 5.3. This particular cylinder is a Cartesian product of two sets, but this does not hold for any cylinder in general. We prove that any cylinder is a closed set of $\mathbb{R}^3$. Let $v = (v_1, v_2, v_3)$ and $a = (a_1, a_2, a_3)$. By an elementary computation, $(x_1, x_2, x_3) \in C_r(v; a)$ if and only if

$$\sum_{i=1}^{3}(x_i - a_i)^2 - (\sum_{i=1}^{3} v_i(x_i - a_i))^2 = r^2.$$

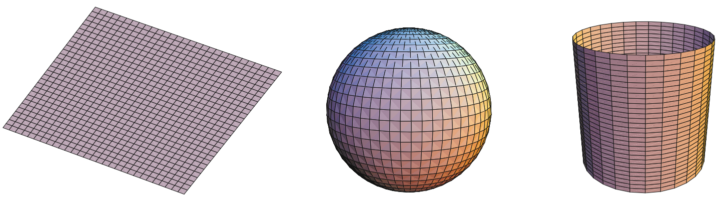


Fig. 5.3 A plane, a sphere and a cylinder are closed sets in $\mathbb{R}^3$

Define the function

$$f : \mathbb{R}^3 \to \mathbb{R}, \quad f(x_1, x_2, x_3) = \sum_{i=1}^{3}(x_i - a_i)^2 - (\sum_{i=1}^{3} v_i(x_i - a_i))^2 - r^2.$$

Then f is continuous, because written in terms of the projections p_i of $\mathbb{R}^3$,

$$f = \sum_{i=1}^{3}(p_i - a_i)^2 - (\sum_{i=1}^{3} v_i(p_i - a_i))^2 - r^2.$$

Since $C_r(v; a) = f^{-1}(\{0\})$, we deduce that $C_r(v; a)$ is closed. $\diamondsuit$

5.4 Worked Exercises

In many of the subsequent exercises, we investigate the continuity of known functions defined on $\mathbb{R}$ but equipped with a non-Euclidean topology.

Exercise 5.1 Consider on $\mathbb{R}$ the included point topology τ_{in} for $p = 1$ and the excluded point topology τ_{ex} for $q = 0$. Determine the continuity of

$$f : (\mathbb{R}, \tau_{in}) \to (\mathbb{R}, \tau_{ex}), \quad f(x) = x^2.$$

Solution A basis of neighborhoods for τ_{in} is $\beta_x = \{U_x = \{1, x\}\}$ and for τ_{ex} is $\gamma_x = \{V_x\}$, where $V_x = \{x\}$ if $x \neq 0$ and $V_0 = \mathbb{R}$. Then f is continuous at x if and only if $f(U_x) \subset V_{f(x)}$. We distinguish cases depending if $f(x) = 0$ or $f(x) \neq 0$, or equivalently, if $x = 0$ of $x \neq 0$.

1. Case $x = 0$. Then $f(U_0) \subset V_0 = \mathbb{R}$, so f is continuous at $x = 0$.
2. Case $x \neq 0$. We check the inclusion $f(\{1, x\}) \subset \{f(x)\}$. This occurs if and only if $f(1) = f(x)$, which yields $x^2 = 1$. Thus f is continuous at $x = -1$ and $x = 1$.

Definitively, f is continuous only at the points 0, -1 and 1. $\square$

Exercise 5.2 Define

$$f : (\mathbb{R}, \tau_1) \to (\mathbb{R}, \tau_2), \quad f(x) = \begin{cases} 0, & \text{if } x \in \mathbb{Q} \\ 1, & \text{if } x \in \mathbb{R} - \mathbb{Q}, \end{cases}$$

where τ_1 and τ_2 are the included point topologies for $p = 1$ and $q = 2$, respectively. Determine at which points f is continuous. Do the same replacing τ_2 by the right order topology.

Solution

1. A basis of neighborhoods for τ_1 is $\beta_x^1 = \{U_x = \{1, x\}\}$ and for τ_2 is $\beta_x^2 = \{V_x = \{2, x\}\}$. Thus f is continuous at x if and only if $f(U_x) \subset V_{f(x)}$, which writes as
$$\{0, f(x)\} \subset \{2, f(x)\}.$$
This occurs if $f(x) = 0$. Thus f is continuous only at rationals.
2. On the right order topology, a basis of neighborhoods of x is $\gamma_x = \{[x, \infty)\}$. Thus f is continuous at x if
$$\{0, f(x)\} \subset [f(x), \infty).$$
This inclusion holds if and only if $0 \in [f(x), \infty)$. Since the value for $f(x)$ is 0 and 1, we deduce that the function f is continuous only at the points where $f(0) = 0$, in other words, at rationals. □

Exercise 5.3 Characterize the continuous maps from a space with the included point topology into other space with the excluded point topology.

Solution Let (X, τ_{in}) and (Y, τ_{ex}) be two spaces with the included point topology for $p \in X$ and the excluded point topology for $q \in Y$. A basis of neighborhoods of $x \in X$ is $\beta_x = \{U_x = \{x, p\}\}$ and for $y \in Y$, $\beta_y = \{V_y\}$, where $V_y = \{y\}$ if $y \neq q$ and $V_q = Y$. Let $f : X \to Y$ be a map and $x \in X$.

1. Case $f(x) \neq q$. Then f is continuous at x if and only if $\{f(p), f(x)\} \subset \{f(x)\}$, or equivalently, if $f(x) = f(p)$.
2. Case $f(x) = q$. It is trivial that $f(U_x) \subset V_{f(x)} = Y$. Thus f is always continuous at those points x such that $f(x) = q$.

Therefore, the continuous maps are the constant maps and the maps whose image is formed by exactly two points $\{q, q_2\}$ subject to the condition that $f(p) = q_2$. An example is the following. Let $X = Y = \mathbb{R}$, $p = 1$, $q = 2$ and
$$f(x) = \begin{cases} 4, & x \leq 3 \\ 2, & x > 3. \end{cases}$$
□

Exercise 5.4 Prove that the function
$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} -1, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ 1, & \text{if } x > 0 \end{cases}$$
is not continuous.

Solution Using the basis β_u of the bounded open intervals, it is obvious that $f^{-1}((-\frac{1}{2}, \frac{1}{2})) = \{0\}$, which is not an open set.

We have proved that f is not continuous; now we determine the precise points at which it is not continuous. With the aid of Proposition 5.1.4, we see that f is discontinuous at $\{0\} \setminus \text{int}(\{0\}) = \{0\}$. Let also observe that there are another open intervals whose inverse image are open sets, as for example, any interval $(0, a)$ with $a > 1$, because $f^{-1}((0, a)) = (0, \infty)$.

At points distinct from 0, f is continuous. This is a consequence of Proposition 5.2.4 because $f_{|(0,\infty)}$ and $f_{|(-\infty,0)}$ are constant and $(0, \infty)$ and $(-\infty, 0)$ are open sets. □

Exercise 5.5 Let $f, g\colon (\mathbb{R}, \tau_{CF}) \to (\mathbb{R}, \tau_{CF})$ be the functions $f(x) = x$ and

$$g(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Prove that f and g are continuous but not the product function $h = f \cdot g$.

Solution The function f is the identity in $(\mathbb{R}, \tau_{CF})$, so it is continuous. On the other hand, we see that g is continuous if $g^{-1}(F)$ is finite for every finite set $F \subset \mathbb{R}$. Since g is bijective, with $g^{-1} = g$, the function g^{-1} maps finite sets to finite sets, proving that g is continuous. On the other hand, the product function h is

$$h(x) = \begin{cases} 1, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

This function is not continuous because $F = \{1\}$ is closed but $h^{-1}(F) = \mathbb{R} \setminus \{0\}$ is not because it is infinite. □

Exercise 5.6 Determine the continuous maps from $(\mathbb{R}, \tau_{CF})$ to $(\mathbb{R}, \tau_u)$.

Solution Let $f\colon (\mathbb{R}, \tau_{CF}) \to (\mathbb{R}, \tau_u)$ be a continuous map. Suppose that $x, y \in \text{Im}(f)$, which we may assume $x < y$. Let c be a real number such that $x < c < y$. Consider the open sets in $(\mathbb{R}, \tau_u)$ defined by $O_1 = (-\infty, c)$ and $O_2 = (c, \infty)$. Both sets are disjoint and $x \in O_1$ and $y \in O_2$. Since f is continuous, $f^{-1}(O_1)$ and $f^{-1}(O_2)$ are open in the cofinite topology. Moreover, they must be nonempty and disjoint because $O_1 \cap O_2 = \emptyset$. However, this contradicts that two nontrivial open sets in the cofinite topology must intersect.

This contradiction arises because we assumed that the image of f has at least two elements. Consequently, if f is continuous, f must be a constant map. Thus the only continuous maps are the constant maps. □

Exercise 5.7 If (X, τ_r) is the right order topology, study which mappings from (X, τ_r) into (X, τ_r) are continuous.

Solution Recall that $\beta_x = \{[x, \rightarrow [\}$ is a basis of neighborhoods of x. Since this basis has only one element, $f : (X, \tau_r) \to (X, \tau_r)$ is continuous at x if and only if

$$f([x, \rightarrow [) \subset [f(x), \rightarrow [$$

for every $x \in X$. This inclusion says that for each $y \in X$ with $y \in [x, \rightarrow [$, we have $f(y) \in [f(x), \rightarrow [$, or equivalently, if $x \leq y$ then $f(x) \leq f(y)$. As a conclusion, continuous maps are increasing. An example of a continuous map when $X = \mathbb{R}$ is

$$f : (\mathbb{R}, \tau_r) \to (\mathbb{R}, \tau_r), \quad f(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ 1, & \text{if } x > 0 \end{cases}$$

□

Exercise 5.8 Define the function

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ 1, & \text{if } x > 0. \end{cases}$$

Determine the continuity of f considering the topologies τ_r and τ_u in the domain and codomain in all its possibilities. See Fig. 5.4, left.

Solution If both spaces have the Euclidean topology, the function is discontinuous only at $x = 0$. If they support the right order topology, then f is continuous by the preceding exercise because f is increasing. Consider the remaining cases.

1. Case $f : (\mathbb{R}, \tau_u) \to (\mathbb{R}, \tau_r)$. We study continuity of f by means of the preimages of a basis for τ_r. Using the basis $\beta_r = \{[x, \infty) : x \in \mathbb{R}\}$ for τ_r,

$$f^{-1}([x, \infty)) = \begin{cases} \emptyset, & \text{if } x > 1 \\ (0, \infty), & \text{if } 0 < x < 1 \\ \mathbb{R}, & \text{if } x \leq 0. \end{cases}$$

   Since all the three above sets are open in τ_u, we deduce that f is continuous.
2. Case $f : (\mathbb{R}, \tau_r) \to (\mathbb{R}, \tau_u)$. This function is not continuous because for $O' = (-\frac{1}{2}, \frac{1}{2}) \in \tau_u$, we have $f^{-1}(O') = (-\infty, 0]$ which it is not open in τ_r. We discuss in which points f is continuous. On τ_r, consider the basis

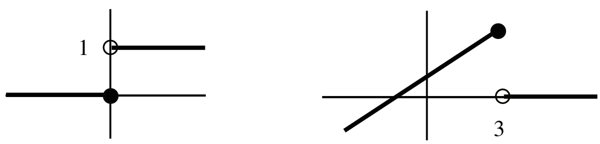


Fig. 5.4 The functions f of Exercise 5.8 (left) and that of Exercise 5.9 (right)

of neighborhoods $\{[x, \infty)\}$ and on the Euclidean topology, the basis $\gamma_x = \{(x - r, x + r) : r > 0\}$.

(a) Let $x \leq 0$. We prove that f is discontinuous at x. Now $f(x) = 0$. Consider the neighborhood $(-\frac{1}{2}, \frac{1}{2})$ of 0. However,

$$f([x, \infty)) = \{0, 1\} \not\subset (-\tfrac{1}{2}, \tfrac{1}{2}).$$

(b) Let $x > 0$. Now $f(x) = 1$. Then f is continuous at x because for all $r > 0$, $f([x, \infty)) = \{1\} \subset (1 - r, 1 + r)$. □

Exercise 5.9 Define

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} x + 2, & x \leq 3 \\ 0, & x > 3. \end{cases}$$

See Fig. 5.4, right. Determine the continuity of f considering the Sorgenfrey topology and Euclidean topology on the domain and codomain in all its possibilities.

Solution It is clear that if both spaces are endowed with the Euclidean topology, then f is continuous except at $x = 3$. A basis of neighborhoods in the Sorgenfrey topology is $\beta_x^S = \{[x, y) : y > x\}$.

1. Case $f : (\mathbb{R}, \tau_u) \to (\mathbb{R}, \tau_S)$.

 (a) Let $x > 3$. Then $f(x) = 0$ and replace β_0^S with the basis $\gamma_0 = \{[0, r) : 0 < r < 1\}$. Then $f^{-1}([0, r)) = (3, \infty) \cup [-2, -2 + r)$. Then $(3, \infty)$ is a neighborhood of x and $f((3, \infty)) = \{0\} \subset [0, r)$. Thus f is continuous at x.
 (b) Let $x = 3$. As $f(3) = 5$, given $[5, y)$, $f^{-1}([5, y)) = \{3\}$ that is not a neighborhood of $x = 3$. Thus f is discontinuous at $x = 3$.
 (c) Let $x < 3$. Since $f(x) = x + 2$, we have

 $$f^{-1}([x + 2, 5)) = \begin{cases} [x, 3), & -2 < x < 3 \\ [x, \infty), & x \leq -2, \end{cases}$$

 which is not a neighborhood of x in the Euclidean topology. Thus f is not continuous at x.

 As a conclusion, f is continuous only at points of $(3, \infty)$.
2. Case $f : (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_S)$.

 (a) Case $x > 3$. We may modify the arguments as follows. On the neighborhood $[x, \infty)$ of x in τ_S, $f_{|[x,\infty)}$ is constant, so continuous on $[x, \infty)$. This proves that f is continuous at x.
 (b) Case $x = 3$. The function is not continuous because $f(3) = 5$ and taking $[5, 6)$ as a neighborhood of 5, we have $f^{-1}([5, 6)) = \{3\}$ that is not a neighborhood of $x = 3$ in τ_S.

(c) Case $x < 3$. If $r > 0$ is sufficiently small so $r < 3 - x$,

$$f^{-1}([x+2, x+2+r)) = \begin{cases} [x, x+r), & x \geq -2 \\ [x, x+r) \cup (3, \infty), & x \leq -2 \end{cases}$$

is a neighborhood of x in τ_S.

The function is discontinuous only at $x = 3$.

3. Case $f\colon (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_u)$. The study can be done point-by-point or using that $\tau_u \subset \tau_S$ in the codomain, that is, $\mathscr{U}_x \subset \mathscr{U}_x^S$. This means that if $f\colon (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_S)$ is continuous at x, then $f\colon (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_u)$ is continuous at x. In such a case, it only remains to determine what happens at $x = 3$. Now $f(3) = 5$. The function f is not continuous because $f^{-1}((4, 6)) = (2, 3]$ which is not a neighborhood of $x = 3$. We have proved that f is discontinuous only at $x = 3$.

□

The above exercise is peculiar because one of the two topologies is finer than the other one. Also, if $\tau_1 \subset \tau_2$, it is clear that if $f\colon (X, \tau_1) \to Y$ is continuous, then it is also $f\colon (X, \tau_2) \to Y$. A similar discussion can be done for two comparable topologies in the codomain.

In the setting of metric spaces, various maps were defined involving the distance of the space. We investigate the continuity of these maps. The first example is the distance $d\colon X \times X \to \mathbb{R}$. First we topologize $X \times X$. As in $\mathbb{R}^n \times \mathbb{R}^n$, it is natural to consider on $X \times X$ the product distance

$$d'((x_1, x_2), (y_1, y_2)) = d(x_1, y_1) + d(x_2, y_2)$$

defined in Exercise 4.22.

Exercise 5.10 The distance $d\colon (X \times X, d') \to \mathbb{R}$ is continuous. As a consequence, the functions $f_x\colon X \to \mathbb{R}$ defined by $f_x(y) = d(x, y)$ and the norm $\|\cdot\|$ on a normed space are continuous.

Solution Let $((x_n, y_n)) \to (x, y)$ in $X \times X$ for the distance d' and we prove that $(d(x_n, y_n)) \to d(x, y)$, or equivalently, $|d(x_n, y_n) - d(x, y)| \to 0$. As a consequence of the so-called quadrangle inequality (Exercise 4.6.5), we have

$$|d(x_n, y_n) - d(x, y)| \leq d(x_n, x) + d(y_n, y) = d'((x_n, y_n), (x, y)) \to 0.$$

For the function f_x, if $(x_n) \to y$, by the quadrangle inequality again

$$|f_x(z) - f_x(y)| = |d(x, z) - d(x, y)| \leq d(z, y),$$

proving that f_x is Lipschitz with $K = 1$. Finally, the norm function $\|\cdot\|\colon X \to \mathbb{R}$ defined on a normed space is simply $x \mapsto d(x, 0) = f_0(x)$.

Another way to prove that f_x is continuous is to observe that f_x is composition of two continuous maps. Indeed, $f_x = d \circ \phi$, where $\phi\colon (X, d) \to (X \times X, d')$ is defined by $\phi(y) = (x, y)$. The map ϕ is continuous because it is Lipschitz with $K = 1$.

We study the function $d(x, A)$ used in Proposition 4.4.7 to characterize adherent points.

Exercise 5.11 If $A \subset X$, the function $f\colon X \to \mathbb{R}$, $f(x) = d(x, A)$, is continuous.

Solution Let $x, x' \in X$. If $a \in A$ is arbitrary, then $d(x, A) \le d(x, a) \le d(x, x') + d(x', a)$. Thus $d(x, A) - d(x, x') \le d(x', a)$. Because this holds for any $a \in A$, using the definition of $d(x, A)$, we deduce $d(x, A) - d(x, x') \le d(x', A)$. Hence $f(x) - f(x') \le d(x, x')$. Similarly, $f(x') - f(x) \le d(x, x')$. This proves that $|f(x) - f(x')| \le d(x, x')$ and consequently f is Lipschitz with $K = 1$.

Exercise 5.12 Prove that the upper hemisphere of $\mathbb{S}^n$ defined by

$$\mathbb{S}^n_+ = \{(x_1, \dots, x_{n+1}) \in \mathbb{S}^n : x_{n+1} > 0\}$$

is open in $\mathbb{S}^n$.

Solution We offer three proofs.

1. It suffices to observe that $\mathbb{S}^n_+ = \mathbb{S}^n \cap H^+$, where

$$H^+ = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : x_{n+1} > 0\}$$

   and H^+ is open in $\mathbb{R}^{n+1}$ (Example 5.12).
2. The restriction map $p_{n+1|\mathbb{S}^n}\colon \mathbb{S}^n \to \mathbb{R}$ of the projection $p_{n+1}\colon \mathbb{R}^{n+1} \to \mathbb{R}$ is continuous. Then $\mathbb{S}^n_+ = (p_{n+1|\mathbb{S}^n})^{-1}((0, \infty))$ is open in $\mathbb{S}^n$.
3. Use sequential characterization of interior points. Let $x \in \mathbb{S}^n_+$ and $(x^m) \subset \mathbb{S}^n$ be a sequence such that $(x^m) \to x$. Since the topology on $\mathbb{S}^n$ is the metric topology of the Euclidean distance, if $x^m = (x^m_1, \dots, x^m_{n+1})$ and $x = (x_1, \dots, x_{n+1})$, we deduce $(x^m_k) \to x_k$ for all $1 \le k \le n + 1$. In the particular case $k = n + 1$, $(x^m_{n+1}) \to x_{n+1}$. As $x \in \mathbb{S}^n_+$, we must have $x_{n+1} > 0$. This tells us that x_{n+1} belongs to the open set $(0, \infty)$. From the sequential characterization of interior points in $(\mathbb{R}, \tau_u)$, there is $\nu \in \mathbb{N}$ such that $x^m_{n+1} > 0$ for all $m \ge \nu$. This implies that $x^m \in \mathbb{S}^n_+$ wherein $m \ge \nu$, as we want to prove. □

Exercise 5.13 We return to Exercise 4.15 and we present an alternative proof that $A = \{(x, y) \in \mathbb{R}^2 : y > x+1\}$ is open in $\mathbb{R}^2$ and that $\overline{A} = \{(x, y) \in \mathbb{R}^2 : y \ge x+1\}$.

Solution Define the function

$$f\colon \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = y - x - 1.$$

Then f is continuous because $f = p_2 - p_1 - 1$, where p_1 and p_2 are the projections of $\mathbb{R}^2$. It is clear that $A = f^{-1}((0, \infty))$ and as a consequence, A is open in $\mathbb{R}^2$. Comparing with Exercise 4.15, this proof is more simple.

For the computation of $\overline{A}$, we simplify the proof of Exercise 4.15 in the following sense. Using the above function f, the set

$$B = \{(x, y) \in \mathbb{R}^2 : y \geq x + 1\} = f^{-1}([0, \infty))$$

is closed in $\mathbb{R}^2$. Since $A \subset B$, we have $\overline{A} \subset \overline{B} = B$. Because

$$B = A \cup f^{-1}(\{0\}) = A \cup \{(x, y) \in \mathbb{R}^2 : y = x + 1\},$$

it remains to decide if the points (x, y) that satisfy $y - x - 1 = 0$ are adherent or not. Now we follow the final steps of Exercise 4.15. □

Exercise 5.14 Let $\mathbb{N}$ be the set of natural numbers and $\omega \notin \mathbb{N}$. Define a topology on $\mathbb{N}^* = \mathbb{N} \cup \{\omega\}$ by

$$\tau^* = \mathbf{P}(\mathbb{N}) \cup \{\mathbb{N}^* \setminus A : \ A \subset \mathbb{N} \text{ is finite}\}.$$

Prove that τ^* is a topology and that $\tau^*_{|\mathbb{N}}$ is discrete. Let X be a topological space, $(x_n) \subset X$ a sequence and $x \in X$. Define

$$f : (\mathbb{N}^*, \tau^*) \to X, \quad \begin{cases} f(n) = x_n \\ f(\omega) = x. \end{cases}$$

Prove that f is continuous at ω if and only if $(x_n) \to x$.

Solution

1. Let us check that τ^* is a topology on $\mathbb{N}^*$. First, $\emptyset \in \mathbf{P}(\mathbb{N})$ and $\mathbb{N}^* = \mathbb{N}^* \setminus \emptyset$. There are two types of open sets. The first ones belong to the discrete topology of $\mathbb{N}$ and the second ones are similar to open sets of the cofinite topology on $\mathbb{N}^*$. Thus the axioms of being a topology in each type are fulfilled. For unions, if we take elements of the two types, then the union is an element of the second type. However, the intersection of two elements of both types is an element of the first type.
2. If $n \in \mathbb{N}$, then $\{n\}$ is an open set of the first type. Then all singletons of $\mathbb{N}$ are open and the topology on $\mathbb{N}$ is discrete.
3. As a previous step, we see that $(n) \to \omega$ in the space $(\mathbb{N}^*, \tau^*)$. Certainly, let $O \in \tau^*$ be an arbitrary open set with $\omega \in O$. By definition of τ^*, $O = \mathbb{N}^* \setminus A$, where $A \subset \mathbb{N}$ is finite. Since A is finite, let $\nu \in \mathbb{N}$ be a fixed number such that $\nu > a$ for all $a \in A$. Then $n \in O$ whenever $n \geq \nu$.

   We proceed to prove the equivalence of the exercise.

   (a) Since $(n) \to \omega$, we have $(f(n)) \to f(\omega)$ by continuity, so $(x_n) \to x$.

(b) For the converse, since $f(\omega) = x$, let O' be an open set in X such that $x \in O'$. Since $(x_n) \to x$, there is $\nu \in \mathbb{N}$ such that $x_n \in O'$ for all $n \geq \nu$. Let $A = \{1, \dots, \nu\}$. This set is finite, so $\mathbb{N}^* \setminus A$ is open in τ^* and contains ω. If $n \notin A$, then $n > \nu$, so $f(n) = x_n \in O'$. This proves that $f(\mathbb{N}^* \setminus A) \subset O'$. □

Exercise 5.15 Consider on $\mathbb{N}$ the topology defined in Exercise 3.21. Determine the continuity of the functions $f, g\colon \mathbb{N} \to \mathbb{N}$ defined by

$$f(n) = \begin{cases} 1, & n \leq 4 \\ 5, & n > 4. \end{cases}, \quad g(n) = \begin{cases} 1, & n < 4 \\ 5, & n \geq 4. \end{cases}$$

Solution A basis of neighborhoods for $2n$ is $\{U_{2n} = \{2n-1, 2n\}\}$ and for $2n-1$, $\{U_{2n-1} = \{2n-1, 2n\}\}$. Because the bases possess only one element, the continuity of f at n reduces to proving the inclusions $f(U_n) \subset U_{f(n)}$ and $g(U_n) \subset U_{g(n)}$.

1. It is clear that if $n \leq 3$ or $n \geq 5$, the function f is constant on U_n, so f is continuous at n (Proposition 5.2.4). It remains the case $n = 4$, where we need to check the inclusion $f(U_4) \subset U_1$. This is inclusion is $f(\{3, 4\}) \subset \{1, 2\}$, which is true. Thus f is continuous at $n = 4$.
2. The function g is constant on U_n for $n \leq 2$ and $n \geq 5$, so g is continuous in these points. It remains to study the continuity at $n = 3$ and $n = 4$. The inclusion $g(U_3) \subset U_1$ should be $g(\{3, 4\}) = \{1, 5\} \subset \{1, 2\}$, which is not true. At $n = 4$, $g(U_4) = g(\{3, 4\}) = \{1, 5\}$, which is not included in $U_{g(4)} = U_5 = \{5, 6\}$. Thus g is discontinuous only at 3 and 4. □

Exercise 5.16 Consider on $\mathbb{N} \setminus \{1\}$ the topology τ of divisors (Exercise 2.15). Characterize the continuous maps $f\colon (\mathbb{N}, \tau) \to (\mathbb{N}, \tau)$. Apply to the function $f\colon \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x + 1$ studying continuity at the prime numbers, at the even numbers and at all numbers up to 10.

Solution A basis of neighborhoods is $\beta_n = \{U_n\}$, where U_n is the set of all divisors of n (Example 3.5). Since there is only one element in the basis, f is continuous at n if and only if $f(U_n) \subset U_{f(n)}$. This inclusion means that if $m|n$, then $f(m)|f(n)$; that is, f is continuous if and only if f preserves the property of be divisor.

For $f(x) = x+1$, we verify this property, or equivalently, the inclusion $f(U_n) \subset U_{n+1}$.

1. Let n be a prime number. Then $U_n = \{n\}$, so $f(U_n) = \{n+1\}$. It is clear that $n + 1 \in U_{n+1}$, so f is continuous on the prime numbers.
2. We prove that f is not continuous at any even number except at 2. Let $n = 2m > 2$. Then $m|n$ and if f were continuous at n, then $m + 1|2m + 1$. In such a case, there is $k \in \mathbb{N}$, $k \geq 2$, such that $k(m + 1) = 2m + 1$. Hence $km + k = 2m + 1$, so $(k - 2)m = 1 - k$, which is absurd. For $n = 2$, $f(U_2) = f(\{2\}) = \{3\} = U_3$.
3. By the preceding items, it remains to study the case $n = 9$. However, $3|9$ but $f(3) = 4$ is not a divisor of $f(9) = 10$. Thus f is continuous at 2, 3, 5 and 7 □

We revisit a result of calculus that states that the limit of a uniformly convergent sequence of continuous functions is continuous. We prove it for bounded functions.

Exercise 5.17 Let (X, τ) be a space, let (Y, d') be a metric space and consider the set consisting of the bounded continuous maps from X into Y,

$$\mathsf{B}_c(X, Y) = \{f : X \to Y : f \text{ is continuous and bounded}\}.$$

As subspace of $\mathsf{B}(X, Y)$, let d_∞ be the sup distance of Exercise 4.29 induced on $\mathsf{B}_c(X, Y)$. Prove that $\mathsf{B}_c(X, Y)$ is closed in $\mathsf{B}(X, Y)$.

Solution Use the sequential characterization of adherent points. Let $(f_n) \subset \mathsf{B}_c(X, Y)$ and $g \in \mathsf{B}(X, Y)$ such that $(f_n) \to g$ and we must show that $g \in \mathsf{B}_c(X, Y)$, or in other words, the continuity of g. Let $x_0 \in X$ and $\epsilon > 0$ be a given but arbitrary positive number. From the triangle inequality, for any $x \in X$ and $n \in \mathbb{N}$,

$$d'(g(x), g(x_0)) \leq d'(g(x), f_n(x)) + d'(f_n(x), f_n(x_0)) + d'(f_n(x_0), g(x_0)). \tag{5.3}$$

Since $(d_\infty(f_n, g)) \to 0$, given $\epsilon/3 > 0$, there is $\nu \in \mathbb{N}$ such that $d_\infty(f_n, g) < \epsilon/3$ whenever $n \geq \nu$. This implies

$$d'(f_n(x), g(x)) \leq \max\{d'(f_n(t), g(t)) : t \in X\} = d_\infty(f_n, g) < \frac{\epsilon}{3}$$

for all $x \in X$. On the other hand, f_n is continuous at x_0, so there is $U \in \mathscr{U}_{x_0}$ such that $d'(f_n(x), f_n(x_0)) < \epsilon/3$ for all $x \in U$. Using these inequalities in (5.3), we have $d'(g(x), g(x_0)) < \epsilon$ for all $x \in U$. This proves the continuity of g at x_0.

We revisit the aforementioned result of calculus. Let $X = [a, b]$ and $Y = \mathbb{R}$. Suppose that (f_n) is a sequence of real-valued continuous functions defined on $[a, b]$; in particular, all f_n are bounded because they are defined on a bounded closed interval. If $(f_n) \to g$ uniformly then $(f_n) \to g$ with the sup distance by Exercise 4.29, but we have proved that $g \in \mathsf{B}_c(X, Y)$, hence g is continuous. □

In the context of metric spaces, the next exercise shows that we can change the metric on the domain of a continuous function by another one equivalent and transforming the map to be uniformly continuous.

Exercise 5.18 Let $f : (X, d) \to (Y, d')$ be a continuous map between two metric spaces and define a distance d^* on X by

$$d^*(x_1, x_2) = d(x_1, x_2) + d'(f(x_1), f(x_2)).$$

Prove that d^* is an equivalent distance to d and that $f : (X, d^*) \to (Y, d')$ is uniformly continuous.

Solution

1. It is clear that $d^* \geq 0$ and that it is symmetric. Since $d \leq d^*$, if $d^*(x_1, x_2) = 0$, we have $d(x_1, x_2) = 0$, so $x_1 = x_2$. The triangle inequality is immediate from the corresponding ones of d and d' because for $x_1, x_2, x_3 \in X$,

$$\begin{aligned} d^*(x_1, x_2) &= d(x_1, x_2) + d'(f(x_1), f(x_2)) \\ &\leq d(x_1, x_3) + d(x_3, x_2) + d'(f(x_1), f(x_3)) + d'(f(x_3), f(x_2)) \\ &= d^*(x_1, x_3) + d^*(x_2, x_3). \end{aligned}$$

2. For the equivalence of the two distances, we see that $(x_n) \to x$ for d if and only if $(x_n) \to x$ for d^*. Inequality $d \leq d^*$ implies that if $(x_n) \to x$ for d^* then $(x_n) \to x$ for d. Reciprocally, suppose $(x_n) \to x$ for d. Since f is continuous, we have $(f(x_n)) \to f(x)$ for d'. Then

$$d^*(x_n, x) = d(x_n, x) + d'(f(x_n), f(x)) \to 0 + 0 = 0.$$

3. We see that $f : (X, d^*) \to (Y, d')$ is uniformly continuous. Let $\epsilon > 0$ be given. Take $\delta = \epsilon$. If $d^*(x_1, x_2) < \delta$ then $d(x_1, x_2) + d'(f(x_1), f(x_2)) < \delta = \epsilon$, in particular, $d'(f(x_n), f(x)) < \epsilon$. $\square$

Exercise 5.19 Let $f, g : X \to Y$ be two continuous maps. If Y is Hausdorff, prove that $A = \{x \in X : f(x) = g(x)\}$ is a closed set of X. Thus if A is dense then $f = g$. Apply to prove that if $f : \mathbb{R} \to \mathbb{R}$ is continuous function such that

$$f(x + y) = f(x)f(y) \tag{5.4}$$

for all $x, y \in \mathbb{R}$, then $f = 0$ or there is $a > 0$ such that $f(x) = a^x$.

Solution

1. We see that the complement of A is open. Let $x \notin A$. Then $f(x) \neq g(x)$. Since Y is Hausdorff, there are disjoint neighborhoods U' and V' of $f(x)$ and $g(x)$, respectively. By continuity of f and g, there are $U, V \in \mathscr{U}_x$ such that $f(U) \subset U'$ and $g(V) \subset V'$. Then $U \cap V$ is a neighborhood of x which does not intersect A for if $z \in (U \cap V) \cap A$, then $f(z) = g(z) \in f(U) \cap g(V) \subset U' \cap V'$, a contradiction.
2. The proof consists of demonstrating that f coincides with the zero function or a function of type a^x at all rationals. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, part (1) establishes the required identity. Let us first notice that f is nonnegative because

$$f(x) = f\left(\frac{x}{2} + \frac{x}{2}\right) = f\left(\frac{x}{2}\right)^2 \geq 0.$$

We distinguish two cases.

(a) There is $x_0 \in \mathbb{R}$ such that $f(x_0) = 0$. Then $f = 0$ because for all $x \in \mathbb{R}$ and by (5.4),

$$f(x) = f(x - x_0 + x_0) = f(x - x_0) f(x_0) = 0.$$

(b) An alternative is that $f(x) > 0$ for all $x \in \mathbb{R}$. Set $a = f(1)$ and define the function $g(x) = a^x$. We show that $f(x) = g(x)$ for all $x \in \mathbb{Q}$ and this concludes the proof.

The relation (5.4) yields $f(0) = f(0+0) = f(0)^2$, so $f(0) = 1$. Thus $f(0) = g(0)$. On the other hand, if $x \in \mathbb{R}$, (5.4) implies

$$f(x) f(-x) = f(x - x) = f(0) = 1,$$

hence $f(-x) = 1/f(x)$. We begin with the naturals. Let $n \in \mathbb{N}$ be given. Then (5.4) yields

$$f(n) = f(1 + \overset{n}{\smile\!\!\ldots} + 1) = f(1)^n = a^n = g(n).$$

If $n \in \mathbb{Z}^-$, then

$$f(n) = \frac{1}{f(-n)} = \frac{1}{a^{-n}} = a^n = g(n).$$

Let us observe that we have proved $f = g$ in $\mathbb{Z}$. However, $\mathbb{Z}$ is not dense in $\mathbb{R}$, so we must go one step further. If $n \in \mathbb{N}$, then

$$a = f(1) = f\left(\tfrac{1}{n} + \overset{n}{\smile\!\!\ldots} + \tfrac{1}{n}\right) = f\left(\tfrac{1}{n}\right)^n,$$

so we obtain $f(1/n) = a^{1/n} = g(1/n)$. Analogously, $f(1/n) = g(1/n)$ if $n \in \mathbb{Z}^-$. Now let $q \in \mathbb{Q}$ write as $q = m/n$, where $m > 0$. Then

$$f(q) = f\left(\tfrac{1}{n} + \overset{m}{\smile\!\!\ldots} + \tfrac{1}{n}\right) = f\left(\tfrac{1}{n}\right)^m = (a^{1/n})^m = a^q = g(q).$$

This proves definitively that $f(x) = g(x)$ for all $x \in \mathbb{Q}$. □

As was announced, we characterize the relative topology by a universal property.

Exercise 5.20 Let (X, τ) be a topological space and let A be a subset of X. Prove that the relative topology on A is the only topology on A with the following property: for any topological space (Y, τ') and any map $f : Y \to A$, we have f is continuous if and only if $i \circ f : Y \to X$ is continuous where $i : A \to X$ is the inclusion.

Solution First we prove that $\tau_{|A}$ satisfies the property. It is clear that if $f : Y \to A$ is continuous, then $i \circ f$ is continuous. Suppose that $i \circ f$ is continuous and we see the continuity of f. Let $O \cap A \in \tau_{|A}$, where $O \in \tau$. We see that $f^{-1}(O \cap A)$ is open in Y by proving that $f^{-1}(O \cap A) = (i \circ f)^{-1}(O)$, hence, $f^{-1}(O \cap A)$ is open because $i \circ f$ is continuous. For the equality,

$$(i \circ f)^{-1}(O) = f^{-1}(i^{-1}(O)) = f^{-1}(O \cap A).$$

It remains to prove uniqueness. Let $\tilde{\tau}$ be a topology on A with the same property that $\tau_{|A}$. Since the property holds for any space (Y, τ'), we choose $(Y, \tau') = (A, \tau_{|A})$ and $f = \mathbb{1}_A\colon (A, \tau_{|A}) \to (A, \tilde{\tau})$. The composition $i \circ f$ is the inclusion $(A, \tau_{|A}) \hookrightarrow (X, \tau)$, which is continuous. By hypothesis, $\mathbb{1}_A$ is continuous, so $\tilde{\tau} \subset \tau_{|A}$. Since the relative topology $\tau_{|A}$ is the coarsest topology on A that makes continuous the inclusion, then $\tau_{|A} \subset \tilde{\tau}$. Thus we conclude $\tau_{|A} = \tilde{\tau}$. □

Exercise 5.21 On $\mathbb{R}$, consider the topology

$$\tau = \{(-a, a) : a > 1, a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}.$$

Determine the continuity of $f(x) = x^2$, when we alternate in the domain and codomain the topology τ and the Euclidean topology τ_u.

Solution

1. If both topologies are Euclidean, then f is continuous.
2. Since $f\colon (\mathbb{R}, \tau_u) \to (\mathbb{R}, \tau_u)$ is continuous, replacing the codomain by $(\mathbb{R}, \tau)$, we derive the continuity of $f\colon (\mathbb{R}, \tau_u) \to (\mathbb{R}, \tau)$ because $\tau \subset \tau_u$.
3. Let $f\colon (\mathbb{R}, \tau) \to (\mathbb{R}, \tau)$. For any $(-a, a) \in \tau$, $f^{-1}((-a, a)) = (-\sqrt{a}, \sqrt{a})$ where $\sqrt{a} > 1$. This proves that f is continuous.
4. We see that $f\colon (\mathbb{R}, \tau) \to (\mathbb{R}, \tau_u)$ is discontinuous at all points. We show it by two manners.

   (a) Point-by-point. In the Euclidean topology, consider open intervals centered at a point as a basis of neighborhoods. We distinguish two cases:

      (i) Case $x = 0$. Then $f(0) = 0$ and consider for $f(0) = 0$, the basis $\beta_0 = \{(-r, r) : 0 < r < 1\}$. Given $r > 0$, if f is continuous at 0, there is $a > 1$ such that $f((-a, a)) \subset (-r, r)$. Rewriting as $[0, a^2) \subset (-r, r)$, we see this is not possible because $a > 1$ and $r < 1$.

      (ii) Case $x \neq 0$. For $f(x) = x^2$, consider the basis $\beta_{x^2} = \{(x^2 - r, x^2 + r) : 0 < r < x^2\}$. If f is continuous at x then there is $a > 1$ such that $x \in (-a, a)$ and $f((-a, a)) \subset (x^2 - r, x^2 + r)$. Now the inclusion is $[0, a^2) \subset (x^2 - r, x^2 + r)$, which is not possible because $0 \notin (x^2 - r, x^2 + r)$.

   (b) We see that the preimages of all elements of a basis for the Euclidean topology are not open sets in $(\mathbb{R}, \tau)$. With the standard basis for τ_u,

$$f^{-1}((a, b)) = \begin{cases} \emptyset, & b \leq 0 \\ (-\sqrt{b}, \sqrt{b}), & a < 0 < b \\ (-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b}), & 0 \leq a < b. \end{cases}$$

Consider the intervals (a, b) with $a < 0 < b \leq 1$ and with $0 \leq a < b$. The preimages are the intervals $(-\sqrt{b}, \sqrt{b})$ and $(-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b})$ respectively. These sets are not open in τ. In order to use Proposition 5.1.4, compute the interior of both sets. Since there are no sets $(-c, c)$, with $c > 1$ included in them, the interior is empty in both cases. By Proposition 5.1.4, f is discontinuous at all points of $(-\sqrt{b}, \sqrt{b})$ if $0 < b \leq 1$ and of $(-\sqrt{b}, -\sqrt{a}) \cup (\sqrt{a}, \sqrt{b})$ if $0 \leq a < b$. Taking $b = 1$ in the first case and $a \to 0$, $b \to \infty$ in the second one, we deduce that f is discontinuous at $(-1, 1)$ and in $\mathbb{R} \setminus \{0\}$, respectively, so f is discontinuous at all real numbers. □

Exercise 5.22 Determine the continuity of the linear maps

$$f\colon (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_S), \quad f(x) = \lambda x.$$

Solution

1. Case $\lambda > 0$. If $[a, b)$ is a basic element of τ_S, we have $f^{-1}([a, b)) = [\frac{a}{\lambda}, \frac{b}{\lambda})$ because f is increasing. This implies that f is continuous.
2. Case $\lambda = 0$. Now $f = 0$ is continuous because f is constant.
3. Case $\lambda < 0$. Now $f^{-1}([a, b)) = (\frac{b}{\lambda}, \frac{a}{\lambda}]$ is not open in τ_S. This implies that f is not continuous. Furthermore, by Proposition 5.1.4, f is discontinuous at points $(\frac{b}{\lambda}, \frac{a}{\lambda}] \setminus \mathrm{int}((\frac{b}{\lambda}, \frac{a}{\lambda}])$. The interior of $(\frac{b}{\lambda}, \frac{a}{\lambda}]$ is $(\frac{b}{\lambda}, \frac{a}{\lambda})$. Thus Proposition 5.1.4 assures that f is discontinuous at a/λ. Taking all real numbers a, the set of all a/λ covers all reals, so f is discontinuous at all points. □

We prove again that the distances defined in Exercise 4.4 are not equivalent.

Exercise 5.23 Let $X = \mathsf{C}([0, 1])$ and define

$$F\colon X \to \mathbb{R}, \quad F(f) = \max_{x \in I} |f(x)|.$$

Prove that $F\colon (X, d_\infty) \to \mathbb{R}$ is continuous but $F\colon (X, d_1) \to \mathbb{R}$ is discontinuous at the null function $g = 0$.

Solution

1. Continuity of $F\colon (X, d_\infty) \to \mathbb{R}$ is a direct consequence of Exercise 5.10 because $F(f) = d_\infty(f, 0)$.
2. We see that $F\colon (X, d) \to \mathbb{R}$ is not continuous at g finding a sequence (f_n) with $(f_n) \to g$ for the distance d_1, but the sequence $(F(f_n))$ does not converge to $F(g)$. Here we use the functions that appeared in Exercise 4.4, letting $r = 1/n$. For every $n \in \mathbb{N}$ define

$$f_n(x) = \begin{cases} 0, & \text{if } x \in [0, 1 - \frac{1}{2n}] \\ 4n(x-1) + 2, & \text{if } x \in [1 - \frac{1}{2n}, 1]. \end{cases}$$

Then

$$d_1(f_n, g) = \int_0^1 |f_n(x)|\,dx \;=\; \frac{1}{2n} \to 0,$$

obtaining $(f_n) \to g$ with the distance d_1. On the other hand,

$$F(f_n) = \max_{x\in I} |f_n(x)| = f_n(1) = 2,$$

that is, $(F(f_n)) \to 2$. However, $F(g) = 0 \neq \lim_{n\to\infty} F(f_n)$. □

Exercise 5.24 Consider on $\mathbb{R}$ the topology τ defined in Exercise 3.16. Prove that

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

is continuous at $x = 0$.

Solution First, we prove that

$$\gamma_0 = \left\{ U_n = (-\infty, -n) \cup \left(\tfrac{1}{n}, \tfrac{1}{n}\right) \cup (n, \infty) : n \in \mathbb{N} \right\}$$

is a basis of neighborhoods of 0. Because $U_n \in \beta_0$, the set U_n is a neighborhood of 0. On the other hand, given $V_{n,\epsilon}$, let $m \in \mathbb{N}$ sufficiently large satisfying $m > n$ and $1/m < \epsilon$. Then $U_m \subset V_{n,\epsilon}$ because $(m, \infty) \subset (n, \infty)$ and $(-1/n, 1/n) \subset (-1/\epsilon, 1/\epsilon)$.

We prove the continuity of f at 0. Note that $f(0) = 0$, so let $U_n \in \gamma_0$. The proof is completed if we see that $f(U_n) = U_n$ for all $n \in \mathbb{N}$. But this is clear because

$$f((n, \infty)) = \left(0, \tfrac{1}{n}\right),$$

$$f((-\infty, -n)) = \left(-\tfrac{1}{n}, 0\right),$$

$$f\left((-\tfrac{1}{n}, \tfrac{1}{n})\right) = \{0\} \cup (-\infty, -n) \cup (n, \infty).$$

□

It is known from calculus that continuity is preserved by the absolute value and the maximum and minimum of functions. We extend this result to topological spaces.

Exercise 5.25 Let $f, g : X \to \mathbb{R}$ be two functions defined on a space X. Define the functions $\max(f, g), \min(f, g) : X \to \mathbb{R}$ by

$$\max(f, g)(x) = \max\{f(x), g(x)\}, \quad \min(f, g)(x) = \min\{f(x), g(x)\}.$$

If f and g are continuous, prove that $\max(f, g)$ and $\min(f, g)$ are continuous.

Solution In terms of the functions f and g,

$$\max(f,g) = \frac{f+g+|f+g|}{2}, \quad \min(f,g) = \frac{f+g-|f-g|}{2}.$$

Using Proposition 5.3.3, it suffices to verify that the modulus of a continuous function is continuous. Let $f\colon X \to \mathbb{R}$ be a continuous function and let $|f|\colon X \to \mathbb{R}$ be the function $|f|(x) = |f(x)|$. Let $x_0 \in X$ be given. We have three alternatives:

1. Case $f(x_0) > 0$. Consider the neighborhood $(0, f(x_0)+1)$ of $f(x_0)$. Since f is continuous, there is $U \in \mathscr{U}_{x_0}$ such that $f(U) \subset (0, f(x_0)+1)$; in particular, $f(x) > 0$ all $x \in U$. Thus $|f| = f$ in U. This implies that the restriction of the function $|f|$ to U coincides with $f_{|U}$, so $|f|$ is continuous at x_0.
2. Case $f(x_0) < 0$. It is analogous to the case $f(x_0) > 0$.
3. Case $f(x_0) = 0$. Now $|f|(x_0) = 0$. Let $\epsilon > 0$ be an arbitrary positive number and let $(-\epsilon, \epsilon)$ be the corresponding interval around $f(x_0)$. Since f is continuous, there is $U \in \mathscr{U}_{x_0}$ such that $|f(x) - f(x_0)| < \epsilon$ for all $x \in U$, that is, $|f(x)| < \epsilon$. In terms of $|f|$, this inequality is

$$||f|(x) - |f|(x_0)| < \epsilon \quad \text{for all } x \in U,$$

proving the continuity of $|f|$ at x_0.

Notice that the argument in (3) applies equally well to (1) after some modifications. Suppose that $f(x_0) > 0$ and let $\epsilon > 0$. Let $\epsilon' > 0$ such that $\epsilon' < \min\{\epsilon, f(x_0)\}$. From continuity of f, there is $U \in \mathscr{U}_{x_0}$ such that $|f(x) - f(x_0)| < \epsilon'$. Then $-\epsilon' < f(x) - f(x_0)$ so $f(x) > f(x_0) - \epsilon' > 0$ for all $x \in U$. This implies $|f|(x) = f(x)$, hence $||f|(x) - |f|(x_0)| = |f(x) - f(x_0)| < \epsilon' < \epsilon$ as claimed.

□

Exercise 5.26 Let $X = \mathsf{C}(\mathbb{R}, \mathbb{R})$ be the set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$. For each $f \in X$, let

$$B_f = \{(x, y) \in \mathbb{R}^2 : y > f(x)\}.$$

Prove that $\beta = \{B_f : f \in X\}$ is a basis for a topology on $\mathbb{R}^2$ and compare with the usual topology.

Solution

1. It is clear that $\mathbb{R}^2 = \bigcup_{f \in X} B_f$ for if $(a, b) \in \mathbb{R}^2$, the constant function $f(x) = b - 1$ belongs to X and $(a, b) \in B_f$. We see that the intersection of two members of β is another of β because

$$B_f \cap B_g = B_{\max(f,g)}$$

and $\max(f, g)$ is continuous by the foregoing exercise. In establishing the above identity, let $(x, y) \in B_f \cap B_g$. Then $y > f(x)$ and $y > g(x)$, so $y > \max(f, g)(x)$. This proves that $(x, y) \in B_{\max(f,g)}$. Conversely, let $(x, y) \in B_{\max(f,g)}$. Then

$$y > \max(f, g)(x) = \max\{f(x), g(x)\}.$$

Thus $y > f(x)$ and $y > g(x)$, which means that $(x, y) \in B_f \cap B_g$.

2. We prove that the usual topology is finer than $\tau(\beta)$. This is because each set B_f is open in the usual topology since $B_f = h^{-1}((0, \infty))$ and $h\colon \mathbb{R}^2 \to \mathbb{R}$ is the continuous function $h(x, y) = y - f(x)$.

   However, the topology $\tau(\beta)$ is not the Euclidean one. For example, a ball, which is open in τ_u, is not the union of elements of β because a ball is a bounded set, but all elements of β are not bounded.

   Two observations. The continuity of the functions has been used to ensure that $\max\{f, g\}$ is continuous. If we replace X with the set Y of all functions $f : \mathbb{R} \to \mathbb{R}$, then $\gamma = \{B_f : f \in Y\}$ is trivially a basis for some topology. The main point in the present exercise is that we can substitute Y by the smaller family X and the corresponding family β follows being a basis for some topology (different than $\tau(\gamma)$). The second observation is that we can replace X with a smaller family Z of continuous functions with the only requisite that $\max\{f, g\} \in Z$. An example is $Z = \{f_a : a \in \mathbb{R}\}$ where $f_a(x) = x + a$ (see also Exercise 2.18). □

Exercise 5.27 Prove that the planar region bounded by a triangle without its borders is an open set.

Solution A triangle is determined by three non-collinear points. The straight-line defined by two points is a set that can be written as $\{(x, y) \in \mathbb{R}^2 : ax + by + c = 0\}$, where a, b and c are three constants. Thus the region determined by the three points is the intersection of three half planes of $\mathbb{R}^2$ of type $A = \{(x, y) \in \mathbb{R}^2 : ax+by+c < 0\}$ and each of these sets is open in $\mathbb{R}^2$ by Example 5.12 (Fig. 5.5). Thus the region is open.

The result can be generalized in different ways.

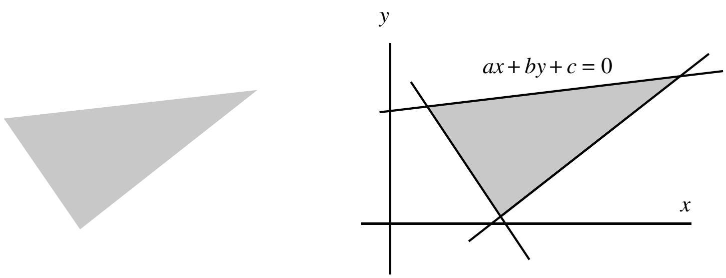


Fig. 5.5 A triangle is determined by the intersection of three half planes

(i) If we consider the triangle together with the three borders, then the set is closed. Now the set is the intersection of three closed sets of type $ax + by + c \leq 0$.
(ii) Angular sectors, the intersection of two sets $ax + by + c < 0$, are open sets.
(iii) Regions bounded by convex polygons of the plane. Again they are open because it is the intersection of a finite number of half planes. □

Exercise 5.28 Consider τ the nested interval topology (Exercise 2.25) and

$$f : ((0, 1), \tau) \to ((0, 1), \tau), \quad f(x) = x^2.$$

Determine the continuity of the points of the interval $(0, \frac{1}{2}]$ and at $x = 0.7$.

Solution The preimage of any open set $(0, 1 - \frac{1}{n})$ is $(0, \sqrt{1 - \frac{1}{n}})$. However, the number $\sqrt{1 - \frac{1}{n}}$ is not of type $1 - \frac{1}{m}$. Thus the preimage of any open set is not open and this implies that f is not continuous.

We see that any point has a basis of neighborhoods formed by only one element. This is a consequence that among all open sets containing x, there is a smallest one under the inclusion of subsets, so this open set constitutes a basis of neighborhoods of x. To see this, note that the open sets containing x are intervals $(0, 1 - \frac{1}{n})$ where $x < 1 - \frac{1}{n}$. Since $\{n \in \mathbb{N} : x < 1 - \frac{1}{n}\}$ is bounded from below, let n_x be the minimum of this set and consequently, $U_x = (0, 1 - 1/n_x)$ is the smallest neighborhood containing x. This proves that $\beta_x = \{U_x\}$ is a basis of neighborhoods of x.

Consequently f is continuous at x if and only $f(U_x) \subset U_{x^2}$. Furthermore, since f is increasing, this inclusion holds if and only if $f(n_x) \leq n_{x^2}$, that is, $(n_x)^2 \leq n_{x^2}$. Let us utilize this basis to study the continuity of f.

1. Let $x \in (0, \frac{1}{2}]$. The image of x is $f(x) = x^2$ with $x^2 \leq \frac{1}{4}$. Then $U_{x^2} = (0, \frac{1}{2})$. For x, we have $U_x = (0, \frac{1}{2})$ if $x < \frac{1}{2}$ and $U_x = (0, \frac{2}{3})$ if $x = \frac{1}{2}$. In the first case, $f(U_x) = (0, \frac{1}{4})$ and in the second one, $f(U_x) = (0, \frac{4}{9})$. Both intervals are included in $(0, \frac{1}{2})$ proving that f is continuous in $(0, \frac{1}{2}]$.
2. For $x = 0.7$, $U_x = (0, \frac{3}{4})$. The image of x is 0.49, so $U_{x^2} = (0, \frac{1}{2})$. Finally, $f(\frac{3}{4}) \sim 0.56 > \frac{1}{2}$, so $f(U_x) \not\subset U_{x^2}$. Here $n_x = \frac{3}{4}$ and $n_{x^2} = 1/2$ but $(n_x)^2 > n_{x^2}$.

As a final observation, we can use Proposition 5.1.4 to discover the points where f is discontinuous. For the first open set, $f^{-1}((0, \frac{1}{2})) = (0, \frac{1}{\sqrt{2}})$ whose interior is $(0, \frac{2}{3})$. Thus f is discontinuous at $[\frac{2}{3}, \frac{1}{\sqrt{2}})$. For $(0, \frac{2}{3})$, we have $f^{-1}((0, \frac{2}{3})) = (0, \sqrt{\frac{2}{3}})$ whose interior is $(0, \frac{4}{5})$. Thus f is discontinuous at $[\frac{4}{5}, \sqrt{\frac{2}{3}})$ and so forth. The reader is invited to investigate that as $x \to 1$, there are sub-intervals of $(0, 1)$ where f is continuous and others that not. □

5.5 Suggested Exercises

5.5.1 Determine the continuity of $f, g\colon (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_S)$, where $f(x) = x^3$ and $g(x) = \sin(x)$.

5.5.2 Let $X = \{a, b\}$, $\tau_1 = \{\emptyset, X, \{a\}\}$ and $\tau_2 = \{\emptyset, X, \{b\}\}$. Find all continuous maps from (X, τ_i) into (X, τ_j) for all possibilities $1 \le i, j \le 2$.

5.5.3 Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x - 1$. Determine the continuity by taking in the domain and codomain the topologies of Example 2.12.

5.5.4 Determine the continuity of $f\colon \mathbb{N} \to \mathbb{N}$ defined by $f(x) = x^2$, where the domain and codomain are the spaces of Exercise 2.2. Change f by $g(x) = x + 1$.

5.5.5 Consider on $\mathbb{R}^2$ the included point topology τ_{in} for $p = (0, 0)$. Find the points where the projection $p\colon (\mathbb{R}^2, \tau_{in}) \to (\mathbb{R}, \tau_u)$, $p(x, y) = x$, is continuous.

5.5.6 Prove that the projections $p_i\colon (\mathbb{R}^2, \tau(\beta_1 \times \beta_2)) \to (\mathbb{R}, \tau(\beta_i))$, $i = 1, 2$, are continuous, where β_i is any of the topologies defined in Exercise 2.1.

5.5.7 Let $f\colon (X, d) \to (Y, d')$ be a map between two metric spaces. Show that f is uniformly continuous if and only if for all sequences (x_n) and (z_n) in X such that $d(x_n, z_n) \to 0$, it holds $d'(f(x_n), f(z_n)) \to 0$.

5.5.8 Let (X, d) be a metric space. Using Exercise 5.11, show that $O \subset X$ is open if and only there is an open $G \subset \mathbb{R}$ and a continuous function $f\colon X \to \mathbb{R}$ such that $O = f^{-1}(G)$.

5.5.9 Let X be a metric space and $f\colon X \to X$ continuous. Prove that $A = \{x \in X : f(x) = x\}$ is closed. Find a non-metric space where this result is false.

5.5.10 Let d be the distance defined in Exercise 4.6.4. Show that

$$f\colon (\mathbb{R}, d) \to (\mathbb{R}, d), \qquad f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \in \mathbb{R} - \mathbb{Q} \end{cases}$$

is continuous at $x = 0$.

5.5.11 Consider on $X = \{a, b, c, d\}$ the topology $\tau = \{\emptyset, X, \{a\}, \{b\}, \{a, b\}, \{b, c, d\}\}$. Define $f\colon X \to X$ by $f(a) = b$, $f(b) = d$, $f(c) = b$ and $f(d) = c$. Determine the continuity of f.

5.5.12 Determine the continuity of $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = \sin(x)$, considering in the domain and codomain the Euclidean and right order topology in all its possibilities.

5.5.13 (Generalization of the Pasting Lemma) Let $X = \bigcup_{i \in I} A_i$ where A_i are open sets and let $f\colon X \to Y$. If $f_{|A_i}$ is continuous, prove that f is continuous for all $i \in I$. Show that the result remains valid if we replace the open sets by a finite number of closed sets. For closed sets, find an example where the result fails if I is infinite.

5.5.14 If $A \subset X$, study the continuity of the *characteristic function* of A:

$$\chi_A\colon X \to \mathbb{R}, \qquad \chi_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A. \end{cases}$$

5.5.15 Prove that $f\colon X \to Y$ is continuous if and only if it satisfies one of the following properties:

(i) $f^{-1}(\text{int}\,(B)) \subset \text{int}\,(f^{-1}(B))$ for all $B \subset Y$.
(ii) $f^{-1}(\text{ext}(B)) \subset \text{ext}(f^{-1}(B))$ for all $B \subset Y$.
(iii) $\text{Bd}(f^{-1}(B)) \subset f^{-1}(\text{Bd}(B))$ for all $B \subset Y$.

5.5.16 Given a collection $\{(X_i, \tau_i) : i \in I\}$ of pairwise disjoint spaces, prove that the topological sum $\sum_{i\in I} \tau_i$ is the finest topology on $X = \bigcup_{i\in I} X_i$ for which all the inclusions $(X_i, \tau_i) \hookrightarrow X$ are continuous.

5.5.17 On $\mathbb{N}$ consider the topology $\tau = \{\emptyset, \mathbb{N}\} \cup \{A_n : n \in \mathbb{N}\}$ where $A_n = \{1, 2, \ldots, n\}$. Determine the continuous maps from $(\mathbb{N}, \tau)$ to $(\mathbb{N}, \tau)$.

5.5.18 Prove that the interior of an affine subspace of $\mathbb{R}^n$, a n-sphere of $\mathbb{R}^{n+1}$ and a cylinder of $\mathbb{R}^3$, is empty.

5.5.19 Prove that the regions bounded by polygons of the plane are open sets.

Chapter 6
Homeomorphisms and Topological Invariants

In this chapter we define the stretching and contracting deformations that we discusses early in this book. These deformations transform a topological space into an equivalent topological space, such that both spaces are indistinguishable from a topological point of view. Deformations are mappings between topological spaces that are bijective and continuous as well as their inverses. These mappings are the so-called *homeomorphisms*.

It is at this point when it makes sense to say that a topologist considers objects “made of play dough” because such objects will remain identical after this type of deformation. Although standards of mathematical rigor require that there must be no reliance whatsoever on figures, the reader is encouraged to draw pictures throughout that illustrate the transformations between spaces.

On the other hand, we will see the important role that the homeomorphisms play in the classification of topological spaces. From now until the end of the book, we will take on this task of classification, showing various tools and techniques to carry it out.

6.1 Homeomorphisms

Definition 6.1.1 A map $f: X \to Y$ is called a *homeomorphism* if it is bijective and f and f^{-1} are continuous.

Example 6.1 The *graph* of a function $f: \mathbb{R}^n \to \mathbb{R}$ is defined by

$$G(f) = \{(x, f(x)) \in \mathbb{R}^{n+1} : x \in \mathbb{R}^n\}.$$

We prove that if f is *continuous*, then there is a homeomorphism between $G(f)$ and $\mathbb{R}^n$, the domain of f. The idea of the homeomorphism is intuitive. The mapping

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_6

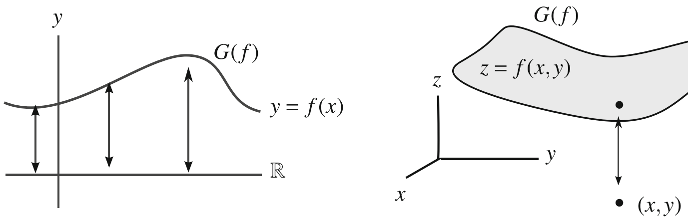


Fig. 6.1 The projection of the points of $G(f)$ onto $\mathbb{R}^n$ establishes a homeomorphism between both spaces: case $n = 1$ (left) and $n = 2$ (right)

is merely the projection $G(f)$ onto the hyperplane of equation $x_{n+1} = 0$ as Fig. 6.1 shows. The point $(x, f(x))$ is mapped into $(x, 0)$, which belongs to $\mathbb{R}^{n+1}$, next, we map $(x, 0)$ to $x \in \mathbb{R}^n$. We cannot say that $G(f)$ is "flattening" into $\mathbb{R}^n$ because $G(f)$ is included in a different Euclidean space since $G(f) \subset \mathbb{R}^{n+1}$. Define

$$\phi\colon G(f) \to \mathbb{R}^n, \quad \phi(x, f(x)) = x.$$

For continuity, notice that ϕ is the restriction to $G(f)$ of

$$\Phi\colon \mathbb{R}^{n+1} \to \mathbb{R}^n, \quad \Phi(x_1, \ldots, x_{n+1}) = (x_1, \ldots, x_n).$$

The map Φ is continuous because composing with the projections p_i' of $\mathbb{R}^n$, $1 \le i \le n$, we obtain $p_i' \circ \Phi = p_i$, where p_i are the projections of $\mathbb{R}^{n+1}$. Thus $\phi = \Phi_{|G(f)}$ is continuous. On the other hand, the inverse of ϕ is

$$\phi^{-1}\colon \mathbb{R}^n \to G(f), \quad \phi^{-1}(x) = (x, f(x)).$$

Continuity of ϕ^{-1} follows by observing that ϕ^{-1} is the restriction of

$$\psi\colon \mathbb{R}^n \to \mathbb{R}^{n+1}, \quad \psi(x) = (x, f(x)).$$

Since the codomain of ψ is $\mathbb{R}^{n+1}$, ψ is continuous if $p_i \circ \psi$ is continuous for every $1 \le i \le n+1$ (Proposition 5.3.3). But now

$$\begin{cases} p_i \circ \psi = p_i', & 1 \le i \le n \\ p_{n+1} \circ \psi = f. \end{cases}$$

Each one of these maps is continuous, where we use the continuity of f. Finally, the range of ψ is $G(f)$ and restricting ψ to $G(f)$, we get the initial map ϕ^{-1}. $\Diamond$

We show that homeomorphisms carry any topological object from one space into another.

Theorem 6.1.2 *If $f : X \to Y$ is a bijection, then the following are all equivalent:*

1. *f is a homeomorphism.*
2. *f is continuous and f carries open sets of X into open sets of Y.*
3. *f is continuous and f carries closed sets of X into closed sets of Y.*
4. *$f(\overline{A}) = \overline{f(A)}$ for all $A \subset X$.*
5. *f is continuous and f maps neighborhoods of X into neighborhoods of Y.*

Similarly homeomorphisms preserve the interior, exterior and boundary (Exercise 6.6.1).

Theorem 6.1.3 *If $f : (X, \tau) \to (Y, \tau')$ is a bijection, then the following are all equivalent:*

1. *f is a homeomorphism.*
2. *$\tau' = f(\tau) = \{f(O) : O \in \tau\}$.*
3. *$\mathscr{F}' = f(\mathscr{F}) = \{f(F) : F \in \mathscr{F}\}$.*
4. *$f(\beta)$ is a basis for τ' for every basis β for τ.*
5. *$\mathscr{U}'_{f(x)} = \{f(U) : U \in \mathscr{U}_x\}$ for all $x \in X$.*
6. *$f(\beta_x)$ is a basis of neighborhoods of $f(x)$ for every $x \in X$ and basis of neighborhoods β_x of x.*

Example 6.2 Consider the Sierpinski space (X, τ), with $X = \{a, b\}$ and $\tau = \{\emptyset, X, \{a\}\}$. We prove that there is a homeomorphism from (X, τ) into (X, τ'), where $\tau' = \{\emptyset, X, \{b\}\}$. To show this, it suffices to define

$$f : (X, \tau) \to (X, \tau'), \qquad \begin{cases} a \longmapsto b \\ b \longmapsto a. \end{cases}$$

Then f is bijective with $f^{-1} = f$ and it is obvious that $\tau' = f(\tau)$. $\Diamond$

Example 6.3 Let τ_S be the Sorgenfrey topology and let τ_S^l be the topology generated by $\beta_S^l = \{(a, b] : a, b \in \mathbb{R}\}$. Then

$$f : (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_S^l), \qquad f(x) = -x$$

is a homeomorphism. Indeed, $f^{-1} = f$ and it is clear that $f(\beta_S) = \beta_S^l$. $\Diamond$

We study the behavior of a homeomorphism regarding the restrictions of maps.

Proposition 6.1.4 *If $f : (X, \tau) \to (Y, \tau')$ is a homeomorphism and $A \subset X$, then $f_{|A} : (A, \tau_{|A}) \to (f(A), \tau'_{|f(A)})$ is a homeomorphism.*

This proposition says a little more. If we are looking for a homeomorphism between two topological spaces A and B, it may sometimes be easier to view A and B as subspaces of other homeomorphic space X and Y such that $B = f(A)$.

We have discussed that homeomorphisms classify topological spaces. We make precise this idea of classification.

Definition 6.1.5 A topological space X is said to be *homeomorphic* to Y if there is a homeomorphism from X into Y. We write $X \cong Y$.

Homeomorphisms establish an equivalence relation in the set of all topological spaces. This is because: the identity map; the inverse of a homeomorphism is a homeomorphism and; the composition of two homeomorphisms is a homeomorphism. Moreover, the set $\mathsf{H}(X, \tau)$ of all homeomorphisms of (X, τ) to itself is a group where the operation is the composition of homeomorphisms.

Once we have established that $\cong$ is an equivalence relation, if X is homeomorphic to Y, we say simply that *X and Y are homeomorphic*. We are in position to formulate the problem of classification in topology.

> **Problem of classification.** *Given two spaces, decide whether they are homeomorphic.*

Example 6.4 Although $G(f) \subset \mathbb{R}^{n+1}$, the homeomorphism between $G(f)$ and $\mathbb{R}^n$ cannot be restriction of another between $\mathbb{R}^{n+1}$ and $\mathbb{R}^n$ because $\mathbb{R}^{n+1} \ncong \mathbb{R}^n$. The non-existence of homeomorphisms between $\mathbb{R}^{n+1}$ and $\mathbb{R}^n$ is particular of the next result.

Theorem 6.1.6 *The Euclidean spaces $\mathbb{R}^n$ and $\mathbb{R}^m$ are homeomorphic if and only if $n = m$.*

This theorem is expected but difficult to prove because it requires hard techniques that go beyond those in this book. We will be able to prove the theorem only for the values $n = 1$ and $n = 2$!.

1. Coming back to Example 6.1, one can view the codomain $\mathbb{R}^n$ of ϕ as the hyperplane $A = \{(x, 0) : x \in \mathbb{R}^n\}$. We show that there is a family of homeomorphisms from $\mathbb{R}^{n+1}$ to itself carrying $G(f)$ into A. The idea is to consider the segment joining every point $(x, f(x)) \in G(f)$ with $(x, 0)$ and parametrized by $\{(x, f(x)) + t((x, f(x)) - (x, 0)) : t \in [0, 1]\}$. Then we can deform $G(f)$ into A along these segments by the map $\phi_t \colon G(f) \to \mathbb{R}^{n+1}$ given by
$$\phi_t(x, f(x)) = (1 - t)(x, f(x)) + t(x, 0) = (x, (1 - t)f(x)),$$
viewing t as a real parameter. If $A_t = \phi_t(G(f))$ and arguing as in Example 6.1, $\phi_t \colon G(f) \to A_t$ is a homeomorphism. Furthermore, for $t = 0$, $\phi_0 \colon G(f) \to A_0$ is the identity and for $t = 1$, $\phi_1 \colon G(f) \to A$ is the homeomorphism ϕ. It is now that the reader can see the family of homeomorphisms $\{\phi_t : t \in [0, 1]\}$ as a deformation between $G(f)$ and A.
2. There is a homeomorphism $\Theta \colon \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ that maps $G(f)$ into the hyperplane A, namely,
$$\Theta(x, y) = (x, y - f(x)).$$

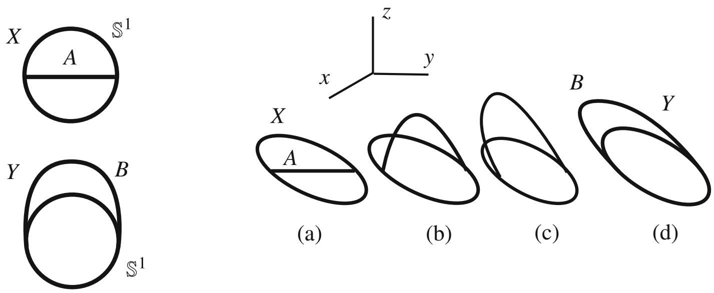


Fig. 6.2 The sets X and Y are homeomorphic but they are not restrictions of a homeomorphism of $\mathbb{R}^2$ (left), but yes of $\mathbb{R}^3$ (right)

The inverse of Θ is $\Theta^{-1}(x, y) = (x, y + f(x))$ and restricting Θ to $G(f)$,

$$\Theta(G(f)) = \{\Theta(x, f(x)) : x \in \mathbb{R}^n\} = \{(x, 0) : x \in \mathbb{R}^n\} = A.$$

$\diamondsuit$

Continuing with this discussion, even if two homeomorphic sets are included in the same Euclidean space $\mathbb{R}^n$, it may happen that the homeomorphism is not restriction of another of $\mathbb{R}^n$ to itself. An example appears in Fig. 6.2, left. It is clear that $X = \mathbb{S}^1 \cup A$ and $Y = \mathbb{S}^1 \cup B$ are homeomorphic. Also it is true that there is an homeomorphism between X and Y carrying A into B (Exercise 6.17). However, there is no homeomorphism from $\mathbb{R}^2$ to itself mapping X into Y and A going to B. This fact is not trivial and requires other techniques that will appear in Chap. 8. But if we see X and Y as subsets of Euclidean space $\mathbb{R}^3$ (for example, inserted in the plane of equation $z = 0$), one can find a family of homeomorphisms in $\mathbb{R}^3$ mapping X into Y. Indeed, from the position in the plane $z = 0$ (a), pick up the piece A and stretch it upwards (b); then continue to stretch it up and drop it on the xy-plane (c) until we get to the final position Y (d). See Fig. 6.2, right.

We illustrate the problem of classification in a simple topological space such as the Euclidean real line. Let us turn our attention to the intervals of $\mathbb{R}$.

Theorem 6.1.7 *Any two open intervals are homeomorphic.*

Proof There are four types of open intervals in $\mathbb{R}$: bounded open intervals, intervals (a, ∞), $(-\infty, a)$ and finally the real line $\mathbb{R}$. We proceed by considering different cases.

1. If $a, b \in \mathbb{R}$, we show that $(a, b) \cong (0, 1)$. The idea is simply to stretch the interval and move it to the correct position. Stretching is made by dilations $x \mapsto \lambda x$ and the displacements by translations $x \mapsto x + \mu$. Thus the required map writes as $f(x) = \lambda x + \mu$. If we impose that $f(a) = 0$ and $f(b) = 1$ to obtain the values λ and μ, the expression for f is

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \frac{x - a}{b - a}. \tag{6.1}$$

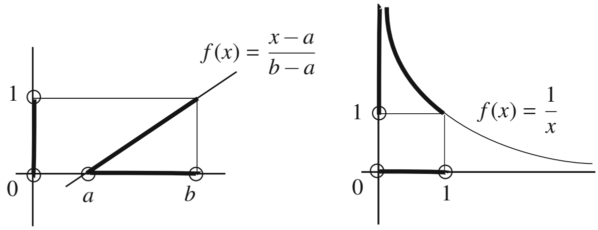


Fig. 6.3 A homeomorphism between the intervals (a, b) and $(0, 1)$ (left) and a homeomorphism between $(0, 1)$ and $(1, \infty)$ (right). The circles indicate that the endpoints of the intervals are not included

It is clear that f is continuous as well as its inverse $f^{-1}(x) = (b - a)x + a$. Thus f is a homeomorphism. If we see that $f((a, b)) = (0, 1)$, Proposition 6.1.4 implies that (a, b) is homeomorphic to $(0, 1)$ (Fig. 6.3, left). Notice that $f(a) = 0$ and $f(b) = 1$ does not assure the identity $f((a, b)) = (0, 1)$. For an explicit proof, the best way to check the set-theoretical identity is by double inclusion. For instance, if $x \in (a, b)$ then $a < x < b$, so $0 < x - a < b - a$. Dividing by $b - a$, we find $0 < f(x) < 1$, hence $f(x) \in (0, 1)$, proving $f((a, b)) \subset (0, 1)$. The other inclusion is analogous.

2. Any two intervals (a, ∞) and (b, ∞) are homeomorphic. Again the idea is using a function $f(x) = \lambda x + \mu$ with $f(a) = b$ and $f(\infty) = \infty$, obtaining

$$f\colon \mathbb{R} \to \mathbb{R}, \quad f(x) = x + b - a.$$

This map is continuous as well as its inverse $f^{-1}(x) = x - b + a$. As $f((a, \infty)) = (b, \infty)$, we deduce that $(a, \infty) \cong (b, \infty)$.

3. Similarly, two intervals of the type $(-\infty, b)$ are homeomorphic.
4. The interval $(1, \infty)$ is homeomorphic to $(-\infty, -1)$ because

$$f\colon \mathbb{R} \to \mathbb{R}, \quad f(x) = -x \tag{6.2}$$

is a homeomorphism and it is clear that $f((1, \infty)) = (-\infty, -1)$.

5. The interval $(0, 1)$ is homeomorphic to $(1, \infty)$ by means of

$$f\colon (0, \infty) \to (0, \infty), \quad f(x) = \frac{1}{x}. \tag{6.3}$$

Here $f^{-1} = f$ and $f((0, 1)) = (1, \infty)$ (Fig. 6.3, right).

6. Finally, the mapping

$$f\colon \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad f(x) = \tan(x)$$

is a homeomorphism, where $f^{-1}(x) = \arctan(x)$. Thus $\mathbb{R} \cong (-\pi/2, \pi/2)$.

$\square$

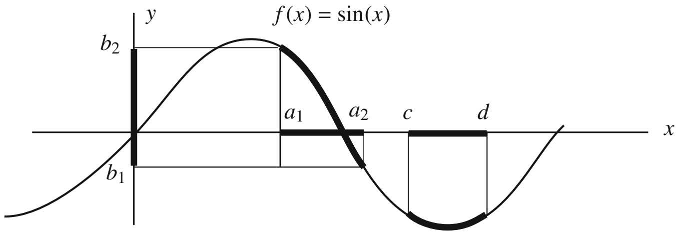


Fig. 6.4 In small domains, the mapping $f(x) = \sin(x)$ is a homeomorphism. For instance, between the open intervals (a_1, a_2) and (b_1, b_2). However, f is not injective in (c, d)

If we examine the above proof, the homeomorphism defined in (6.1) can be restricted to bounded closed intervals. Also, the homeomorphisms (6.1), (6.2) and (6.3), can be restricted to unbounded closed intervals and open-closed intervals.

Corollary 6.1.8

1. *Any two bounded closed intervals with more than one point are homeomorphic.*
2. *All intervals of type* $[a, \infty)$, $[a, b)$, $(-\infty, a]$ *and* $(a, b]$ *are homeomorphic.*

Example 6.5 The function $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = \sin(x)$, is continuous and it is not bijective. But we can restrict f to smaller domains in order to achieve that f is bijective. Figure 6.4 shows that f is a homeomorphism between the intervals (a_1, a_2) and (b_1, b_2) where the inverse is the arc sine function. $\Diamond$

Example 6.6 There are many homeomorphisms between a bounded open interval and $\mathbb{R}$. For example, $h(x) = x/(1+|x|)$ maps homeomorphically $\mathbb{R}$ into $(-1, 1)$ and its inverse is $h^{-1}(x) = x/(1-|x|)$. In Exercise 6.22, we provide a procedure to build homeomorphisms between bounded open intervals and $\mathbb{R}$. Another homeomorphism between $\mathbb{R}$ and $\mathbb{R}$ is

$$f\colon \mathbb{R} \to \mathbb{R}, \qquad \begin{cases} x + \frac{\pi}{2} - 1, & x \leq -\frac{\pi}{2} \\ \sin x, & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ x^2 + 1 - \frac{\pi^2}{4}, & x \geq \frac{\pi}{2}, \end{cases}$$

where f is piecewise defined. $\Diamond$

Discarding singletons (closed intervals with one element), consider the following division of intervals of $\mathbb{R}$:

1. Open intervals: intervals of type (a, b), (a, ∞), $(-\infty, a)$ and $\mathbb{R}$.
2. Closed bounded intervals: intervals of type $[a, b]$ with $a < b$.
3. Open-closed intervals, closed-open intervals and unbounded closed intervals: interval of type $(a, b]$, $[a, b)$, $(-\infty, a]$ and $[a, \infty)$.

We denote by X_1, X_2 and X_3 the sets of each of the above types of intervals, respectively. Theorem 6.1.7 and Corollary 6.1.8 have demonstrated that two intervals belonging to the same set X_i are homeomorphic. The proof that intervals of different X_i are not homeomorphic requires the intermediate value theorem from calculus. This theorem states that if $f: [a, b] \to \mathbb{R}$ is continuous and if $f(a) < 0 < f(b)$, then there is $x \in (a, b)$ such that $f(x) = 0$.

Theorem 6.1.9 *The topological classification of intervals of* $\mathbb{R}$ *is the following: (a) open intervals; (b) closed bounded intervals; (c) open-closed intervals, closed-open intervals and unbounded closed intervals; and (d) singletons.*

Proof We prove that $(0, 1) \in X_1$ is not homeomorphic to $[0, 1) \in X_3$. By contradiction, let $f: [0, 1) \to (0, 1)$ be a homeomorphism and let $x_0 = f(0)$. Then $f: [0, 1] \setminus \{0\} = (0, 1) \to (0, 1) \setminus \{f(0)\} = (0, 1) \setminus \{x_0\}$ is a homeomorphism. Let $a', b' \in (0, 1)$ such that $a' < x_0 < b'$ and let $a = f^{-1}(a')$, $b = f^{-1}(b')$. Without loss of generality, assume $a < b$. Then $f: [a, b] \to \mathbb{R}$ is continuous with $f(a) < x_0 < f(b)$. By the intermediate value theorem, there is $x \in [a, b]$ such that $f(x) = x_0$. This is a contradiction because $f(0) = x_0$ and f is injective on $[0, 1)$.

With similar arguments, $(0, 1) \in X_1$ is not homeomorphic to $[0, 1] \in X_2$ and $[0, 1) \in X_3$ is not homeomorphic to $[0, 1] \in X_2$. □

6.2 Topological Invariants

As a first approach, proving that two spaces are homeomorphic reduces to constructing an explicit homeomorphism. However, it is a problem of a completely different nature to decide if two spaces are not homeomorphic. How can we prove that there does not exist a homeomorphism between two given topological spaces?

Example 6.7 We see that $(\mathbb{R}, \tau_D) \not\cong (\mathbb{R}, \tau_u)$. Suppose that $f: (\mathbb{R}, \tau_D) \to (\mathbb{R}, \tau_u)$ is a homeomorphism. Since $\{x\}$ is a neighborhood of x in $(\mathbb{R}, \tau_D)$, we have that $\{f(x)\}$ is a neighborhood of $f(x)$ in $(\mathbb{R}, \tau_u)$, which is not possible. ♢

In this example, the property "the singletons are open sets" is preserved by homeomorphisms which allows us to conclude that $(\mathbb{R}, \tau_D)$ and $(\mathbb{R}, \tau_u)$ are not homeomorphic. This can be extended to other "properties" and it only remains to make precise which properties we are talking about. We formalize this idea with the next concept that will accompany us throughout this book.

Definition 6.2.1 A property $\mathscr{P}$ is said to be a *topological invariant* if whenever a space X satisfies property $\mathscr{P}$, then every homeomorphic space to X satisfies $\mathscr{P}$.

From this point of view,

Topology studies topological invariants that allow for the classification of topological spaces.

An inconvenience of the concept of topological invariant is that it can occur that two non-homeomorphic spaces satisfy a specific topological invariant. For instance, the spaces $\mathbb{R}^3$ and $\mathbb{R}^4$ will satisfy all the topological invariants that will appear in this book! However, they are not homeomorphic.

Topological spaces can be grouped into equivalence classes under the homeomorphism relation. By contrast, topological invariants separate spaces into only two groups: those that satisfy the invariant and those that do not. So, if $\mathscr{P}$ is a topological invariant and $\sim_{\mathscr{P}}$ is the equivalence relation

$$(X, \tau) \sim_{\mathscr{P}} (Y, \tau') \text{ if } (X, \tau) \text{ and } (Y, \tau') \text{ satisfy } \mathscr{P},$$

then there are only two equivalence classes. On the other hand, it is clear that

$$(X, \tau) \cong (Y, \tau') \Rightarrow (X, \tau) \sim_{\mathscr{P}} (Y, \tau'),$$

but the converse does not necessarily hold. This implies that the equivalence classes of $\sim_{\mathscr{P}}$ is formed by a union of equivalence classes under the homeomorphism relation. This idea is visualized in Fig. 6.5. On the left, we show the equivalence classes by $\cong$ where each point indicates a space and all spaces homeomorphic to it. In Fig. 6.5, middle and right, the classification is done with two different topological invariants $\mathscr{P}_1$ and $\mathscr{P}_2$. For example, think of two spaces X and Y. These spaces fulfill the property $\mathscr{P}_1$ but this does not ensure that they are homeomorphic. If we find another topological invariant $\mathscr{P}_2$ such that X satisfies it but not Y, then necessarily $X \ncong Y$.

Example 6.8 The Hausdorff property (Definition 3.4.3) is a topological invariant because homeomorphisms map disjoint neighborhoods to disjoint neighborhoods. With this property we see that $(\mathbb{R}, \tau_r) \ncong (\mathbb{R}, \tau_u)$. The right order topology is not Hausdorff because any two open sets intersect. However, the usual topology is Hausdorff because it is a metric topology. ♢

Example 6.9 We say that a space X satisfies the property $\mathscr{M}$ if and only if every continuous function $f : X \to (\mathbb{R}, \tau_u)$ attains a maximum. We prove that $\mathscr{M}$ is a

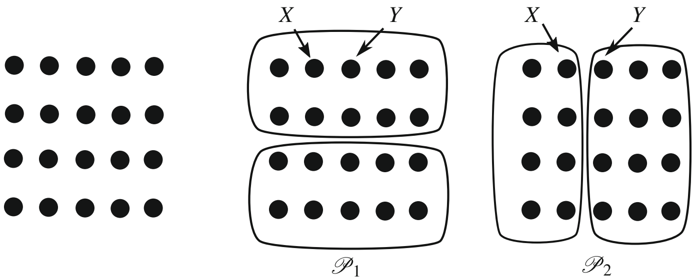


Fig. 6.5 Topological spaces put in groups according to the equivalence classes by the relation be homeomorphic (left), the topological invariant $\mathscr{P}_1$ (middle) and the topological invariant $\mathscr{P}_2$ (right)

topological invariant. For this purpose, suppose that X satisfies $\mathscr{M}$. Let $\phi\colon X \to Y$ a homeomorphism and we show that Y satisfies $\mathscr{M}$ too. Let $g\colon Y \to \mathbb{R}$ be a continuous function. Since $g \circ \phi$ defined on X is continuous, there exists an $x_0 \in X$ such that $(g \circ \phi)(x) \leq (g \circ \phi)(x_0)$ for all $x \in X$. Then g attains a maximum at $\phi(x_0)$.

This topological invariant distinguishes again intervals of $\mathbb{R}$. A known result of calculus asserts that every continuous function defined on a bounded closed interval attains a maximum, so $[0, 1]$ satisfies $\mathscr{M}$. However, the interval $(0, 1)$ does not satisfy $\mathscr{M}$ as the function $f(x) = 1/x$ shows. As a consequence, every interval of the family X_1 is not homeomorphic to an interval of X_2. Likewise, every interval of X_2 is not homeomorphic to an interval of X_3 because $(0, 1] \in X_3$ does not satisfy $\mathscr{M}$ using the same function f. ♢

Example 6.10 We prove that a topological space which is homeomorphic to a discrete space is, indeed, a discrete space. Certainly, if $f\colon (X, \tau_D) \to (Y, \tau')$ is a homeomorphism, then any singleton $\{y\}$ of (Y, τ') is open because $f^{-1}(\{y\})$ is open in (X, τ_D), $\{y\} = f(f^{-1}(\{y\}))$ and f carries open sets in open sets. This proves that τ' is the discrete topology. A similar result can be derived for the trivial and cofinite topologies (Exercise 6.6.4). ♢

Example 6.11 If a space is homeomorphic to the Euclidean real line $(\mathbb{R}, \tau_u)$, is this space the Euclidean real line? The answer is negative because we know that $(0, 1)$ is homeomorphic to $\mathbb{R}$. Another example is the following. Define the bijective map

$$f\colon \mathbb{R} \to \mathbb{R}, \qquad f(x) = \begin{cases} x, & \text{if } x \neq 0, 2 \\ 2, & \text{if } x = 0 \\ 0, & \text{if } x = 2. \end{cases}$$

Consider in the domain of f the usual topology τ_u and in the codomain the topology $\tau_u(f)$ formed by the images of all open sets of τ_u via f (Exercise 2.7.6). As a consequence, $f\colon (\mathbb{R}, \tau_u) \to (\mathbb{R}, \tau_u(f))$ is a homeomorphism. However, the topology $\tau_u(f)$ does not coincide with τ_u. For instance,

$$f((-1, 1)) = (-1, 0) \cup (0, 1) \cup \{2\}$$

is open in $\tau_u(f)$ but not in τ_u. Thus $\tau_u \neq \tau_u(f)$. ♢

6.3 Construction of Homeomorphisms in Euclidean Spaces

This section is devoted constructing explicit homeomorphisms between subsets of Euclidean spaces. The goal is to formalize, if possible, our intuition of deformations.

Proposition 6.3.1 *Every affine map is continuous. As a consequence, affinities are homeomorphisms. In particular, linear isomorphisms and rigid motions (such as translations and dilations) are homeomorphisms.*

Proof If $f : \mathbb{R}^n \to \mathbb{R}^m$ is an affine map, then f can be expressed by $f(x) = Ax+b$, where $A \in \mathsf{M}_{m\times n}(\mathbb{R})$, $x \in \mathsf{M}_{n\times 1}(\mathbb{R})$ and $b \in \mathsf{M}_{m\times 1}(\mathbb{R})$, or equivalently

$$f(x_1,\ldots,x_n) = A\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = \left(\sum_{j=1}^{n} a_{1j}x_j + b_1, \ldots, \sum_{j=1}^{n} a_{mj}x_j + b_m\right).$$

Here $A = (a_{ij})$ and $b = (b_j)$. We prove the continuity of f by checking that $p'_i \circ f$ are continuous, where p'_i are the projections of $\mathbb{R}^m$. From the expression of f,

$$(p'_i \circ f)(x_1,\ldots,x_n) = \sum_{j=1}^{n} a_{ij}x_j + b_i = (\sum_{j=1}^{n} a_{ij}p_j + b_i)(x_1,\ldots,x_n),$$

where now p_i are the projection of $\mathbb{R}^n$. Thus $p'_i \circ f = \sum_{j=1}^{n} a_{ij}p_j + b_i$ is continuous for all $1 \le i \le m$.

An *affinity* $f : \mathbb{R}^n \to \mathbb{R}^n$ is an affine bijection, or equivalently, the matrix A is regular. The inverse of an affinity is another affine map, hence a homeomorphism. □

Affinities reflect the intuitive idea of stretching and shrinking. This was the case of the functions $f(x) = ax + b$ utilized in the proof of Theorem 6.1.7. For example, the affinity $f(x, y) = (3x, y)$ deforms the plane $\mathbb{R}^2$ along the x-direction stretching to the triple but preserving along the y-direction, as Fig. 6.6 shows.

Example 6.12 We see that any two n-dimensional spheres are homeomorphic.

Following the notation employed in Example 5.13, we prove that $\mathbb{S}^n_r(a)$ is homeomorphic to $\mathbb{S}^n$. The idea is simple. First we translate $\mathbb{S}^n_r(a)$ such that it is centered at the origin and next, we stretch it until its radius is 1 (Fig. 6.7). Let $\Phi = H \circ T$ where T is the translation $T(x) = x - a$ and H is the dilation $H(x) = x/r$. Then Φ is composition of two homeomorphisms and $\Phi(\mathbb{S}^n_r(a)) = \mathbb{S}^n$.

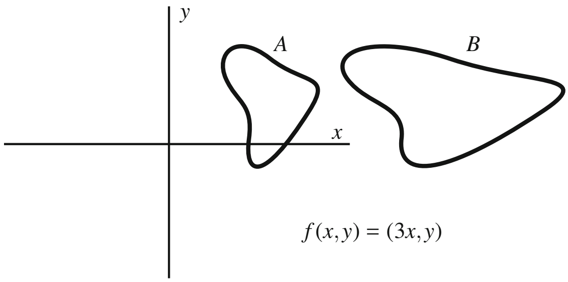


Fig. 6.6 By using the affinity $f(x, y) = (3x, y)$, the shape B is obtained by stretching A along the x-direction

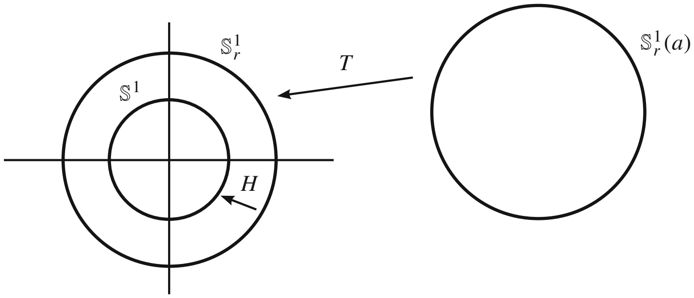


Fig. 6.7 The circles $\mathbb{S}^1_r(a)$, $T(\mathbb{S}^1_r(a)) = \mathbb{S}^1_r$ and $H(\mathbb{S}^1_r) = \mathbb{S}^1$ are all homeomorphic

Indeed, if $x \in \mathbb{S}^n_r(a)$,

$$|\Phi(x)| = |H(x-a)| = \frac{|x-a|}{r} = 1,$$

so $\Phi(x) \in \mathbb{S}^n_r(a)$. This proves that $\phi(\mathbb{S}^n_r(a)) \subset \mathbb{S}^n$. Similarly, $\Phi^{-1}(\mathbb{S}^n) \subset \mathbb{S}^n_r(a)$. $\Diamond$

Example 6.13 We prove that any two cylinders of $\mathbb{R}^3$ are homeomorphic.

Using the notation of Example 5.14, we prove that $C_r(v; a)$ is homeomorphic to $\mathbb{S}^1 \times \mathbb{R}$. The only difference with the case of spheres is that a rigid motion may be needed so that the axes of the two cylinders coincide. The translation of $\mathbb{R}^3$ given by $T(x) = x - a$ does not change the radius of the cylinder. Since $T(a) = 0$, we have $T(C_r(v; a)) = C_r(v; 0)$. We change this cylinder into another one whose axis is $e_3 = (0, 0, 1)$ using a linear isometry $\phi\colon \mathbb{R}^3 \to \mathbb{R}^3$ such that $\phi(v) = e_3$. Indeed, as ϕ preserves the Euclidean metric $\langle , \rangle$,

$$|\phi(x)|^2 - \langle \phi(x), \phi(v)\rangle^2 = |x|^2 - \langle p, e_3\rangle^2.$$

This implies $\phi(C_r(v; 0)) \subset C_r(e_3; 0)$. Again, the reverse inclusion is immediate. Finally, a dilation H of ratio $1/r$ maps $C_r(e_3; 0)$ to $C_1(e_3; 0) = \mathbb{S}^1 \times \mathbb{R}$. $\Diamond$

We formalize the definition of a segment in Euclidean space. If $x, y \in \mathbb{R}^n$, the *segment* determined by x and y is the set of $\mathbb{R}^n$, written $[x, y]$, defined by

$$[x, y] = \{x + t(y - x) : t \in [0, 1]\}.$$

The segment $[x, y]$ is simply the piece of the straight line in $\mathbb{R}^n$ determined by x and y that lies between both points. If $n = 1$, the segment $[x, y]$ is the closed interval $[x, y]$.

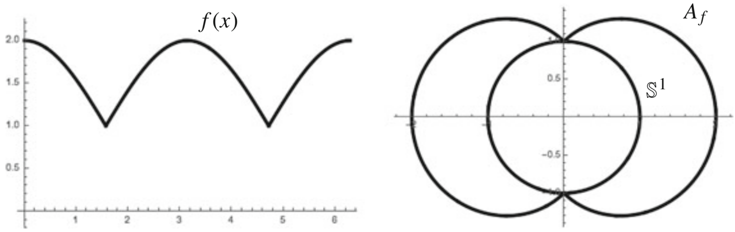


Fig. 6.8 The graph of $f(x, y) = 1 + |x|$ when f winds one time along $\mathbb{S}^1$ starting at $(1, 0)$ (from $t = 0$) until returning at the point at $t = 2\pi$ (left). The set A_f and the circle $\mathbb{S}^1$ (right)

Example 6.14 Let $f : \mathbb{S}^n \to \mathbb{R}$ be a positive function and define

$$A_f = \{f(p)p : p \in \mathbb{S}^n\}.$$

We prove that *if f is continuous then A_f is homeomorphic to $\mathbb{S}^n$*. Geometrically, the set A_f is constructed as follows. For each $p \in \mathbb{S}^n$, let $\{\lambda p : \lambda > 0\}$ be the ray from the origin of $\mathbb{R}^{n+1}$ through the point p. In this ray, consider the point λp of modulus $f(p)$, hence $\lambda = |\lambda p| = f(p)$ and all these points form the set A_f. As we vary p along $\mathbb{S}^n$, the modulus varies.

For example, consider $f : \mathbb{S}^1 \to \mathbb{R}$ defined by $f(x, y) = 1 + |x|$, whose graph is depicted in Fig. 6.8. As (x, y) moves along $\mathbb{S}^1$, the range of f goes from 1 to 2. In Fig. 6.8, left, we show the graph of f when f winds one time around $\mathbb{S}^1$. If we parametrize $\mathbb{S}^1$ by $\{(\cos t, \sin t) : t \in [0, 2\pi]\}$, then $f(x, y) = 1 + |\cos t|$. Depicted on the right is the corresponding set A_f. As a consequence,

1. The set A_f is characterized by the property that each ray emanating from the origin meets A_f at only one point.
2. The set C_f enclosed by A_f is

$$C_f = \bigcup_{p \in \mathbb{S}^n} [0, f(p)p] = \{x \in \mathbb{R}^{n+1} \setminus \{0\} : |x| \leq f(\tfrac{x}{|x|})\} \cup \{0\},$$

   that is, formed by all segments joining the origin $0 \in \mathbb{R}^{n+1}$ with the point $f(p)p$.

The homeomorphism assigns to each point $p \in \mathbb{S}^n$ the point of A_f obtained by intersection of the ray determined by p. Define

$$\phi : \mathbb{S}^n \to A_f, \quad \phi(p) = f(p)p.$$

The inverse map ϕ^{-1} carries each point $q \in A_f$ to its projection onto $\mathbb{S}^n$ through the half-line $\{\lambda q : \lambda > 0\}$. Since $\phi^{-1}(q)$ is proportional to q and of modulus 1, then $\phi^{-1}(q) = q/|q|$. We see that ϕ is continuous. The map ϕ is simply the restriction to A_f of $\Phi : \mathbb{S}^n \to \mathbb{R}^{n+1}$ defined by $\Phi(p) = f(p)p$. The map Φ is

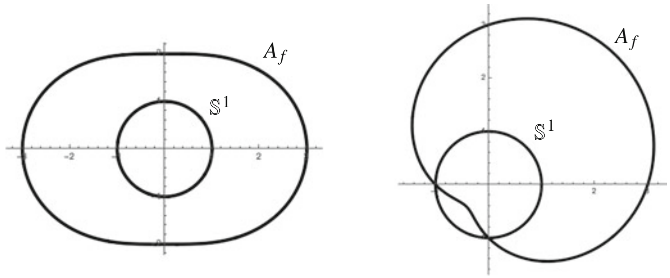


Fig. 6.9 Other choices for f in $\mathbb{R}^2$: $f(x, y) = 2 + x^2$ (left) and $f(x, y) = 2 + x + y$ (right)

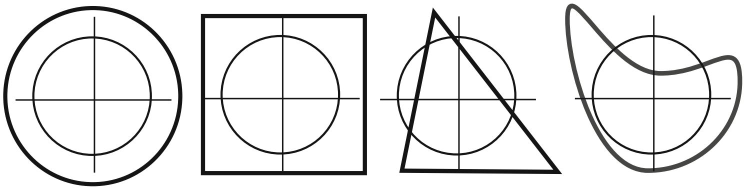


Fig. 6.10 A circle, a square, a triangle and a closed curve are examples of sets A_f in $\mathbb{R}^2$ defined for suitable continuous functions $f : \mathbb{S}^1 \to \mathbb{R}$. These sets are all homeomorphic

continuous because it is the product of the inclusion $i : \mathbb{S}^n \hookrightarrow \mathbb{R}^{n+1}$ by the function $f : \mathbb{S}^n \to \mathbb{R}$. With respect to ϕ^{-1}, we note that ϕ^{-1} is the restriction to A_f of

$$\Psi : \mathbb{R}^{n+1} \setminus \{0)\} \to \mathbb{R}^{n+1} \setminus \{0)\}, \quad \Psi(q) = \frac{q}{|q|},$$

that is continuous. Another pictures of sets A_f for different choices of f appear in Figs. 6.9 (for explicit functions f) and 6.10 (without to explicit f). ◊

In order to facilitate the next arguments, we extend Proposition 5.3.3.

Proposition 6.3.2 *Let $f : X \to Y$ be a map where $Y \subset \mathbb{R}^n$ is endowed with the usual topology. Denote by p'_i the projections of $\mathbb{R}^n$ and let $q_i = p'_{i|Y}$, $1 \le i \le n$. Then f is continuous if and only if $q_i \circ f$ is continuous for every $1 \le i \le n$.*

In the remainder of this section, we show:

1. A ball of $\mathbb{R}^n$ is homeomorphic to $\mathbb{R}^n$.
2. A m-dimensional affine subspace is homeomorphic to $\mathbb{R}^m$.
3. An annulus is homeomorphic to a cylinder.
4. A punctured sphere is homeomorphic to a Euclidean space.

Example 6.15 Every ball in Euclidean space $\mathbb{R}^n$ is homeomorphic to $\mathbb{R}^n$. As a consequence, any two balls of $\mathbb{R}^n$ are homeomorphic. It will be crucial that the ball is immersed in Euclidean space, where stretching transformations are admissible.

If $n = 1$, the ball $B_r(a)$ is the open interval $(a - r, a + r)$ and it is known that this interval is homeomorphic to $\mathbb{R}$ (Theorem 6.1.7).

In a first stage we prove that every ball $B_r(a)$ is homeomorphic to $B_1(0)$, where $0 \in \mathbb{R}^n$ is the origin of coordinates. The idea is simple. We dilate $B_1(0)$ until that the radius is r and then, translate until that its center coincides with a, like we made in Example 6.12. Define

$$F : \mathbb{R}^n \to \mathbb{R}^n, \quad F = T \circ H,$$

where $H(x) = rx$ is the dilation of ratio $r > 0$ and center $0 \in \mathbb{R}^n$ and $T(x) = x + a$ is the translation of translation vector a. By Proposition 6.3.1, F is a homeomorphism. If we see that $F(B_1(0)) = B_r(a)$, then Proposition 6.1.4 gives that $B_1(0)) \cong B_r(a)$. The set-theoretical identity is checked by double inclusion. If $x \in B_1(0)$,

$$|F(x) - a| = |(rx + a) - a| = r|x| < r \cdot 1 = r,$$

so $F(x) \in B_r(a)$. The other inclusion is analogous observing that $F^{-1} = H^{-1} \circ T^{-1}$.

We proceed to construct a homeomorphism $F : B_1(0) \to \mathbb{R}^n$. For each $x \in B_1(0)$, consider the segment through x from the origin to the intersection with the boundary of $B_1(0)$, namely, $x/|x|$. This segment is mapped to the line $\{\lambda x : \lambda > 0\}$ by stretching the modulus of the vectors. Let $h : [0, 1) \to [0, \infty)$ be a homeomorphism with $h(0) = 0$ (Theorem 6.1.7). For x, consider its modulus $|x|$, next $h(|x|) \in (0, \infty)$, which will be the modulus of $F(x) = \lambda x$. So $|\lambda x| = h(|x|)$, hence $\lambda = h(|x|)/|x|$. Thus, define

$$F : B_1(0) \to \mathbb{R}^n, \quad F(x) = \begin{cases} h(|x|)\frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0. \end{cases} \tag{6.4}$$

For the inverse, let $h(|x|)\frac{x}{|x|} = y$. Then $h(|x|) = |y|$, so $|x| = h^{-1}(|y|)$. Thus, if $y \neq 0$,

$$x = \frac{|x|}{h(|x|)} y = \frac{h^{-1}(|y|)}{|y|} y = F^{-1}(y),$$

and $F^{-1}(0) = 0$. For the continuity of F at the points $x \neq 0$, we use Proposition 5.3.3 with the inclusion of $B_1(0) \hookrightarrow \mathbb{R}^n$, the modulus $x \mapsto |x|$ and h. Analogously, F^{-1} is continuous. At $x = 0$, if $(x_n) \to 0$ with $x_n \neq 0$, $|F(x_n) - F(0)| = h(|x_n|) \to 0$ because $\lim_{t \to 0} h(t) = 0$. The argument for F^{-1} is analogous because $\lim_{t \to 0} h^{-1}(t) = 0$. $\diamondsuit$

Notice that no explicit description of the homeomorphism h is needed. Anyway, an example of such h is $h(t) = t/(1-|t|)$ where $h^{-1}(t) = t/(1+|t|)$. Then for $x, y \neq 0$, the expressions of F and F^{-1} are

$$F(x) = \frac{x}{1-|x|}, \quad F^{-1}(y) = \frac{y}{1+|y|}.$$

The above examples illustrate that we can modify the statement of a topological problem for another more manageable one. In the above example, $B_r(p)$ is replaced by $B_1(0)$. Of course the change is done by means of homeomorphisms, so the topological properties remain unchanged.

Example 6.15 does not hold for arbitrary metric spaces in general because we have used translations and dilations which are particular to $\mathbb{R}^n$.

Example 6.16 Let X be a set (with more than two elements) equipped with the discrete distance. Then $B_1(x) = \{x\}$ and $B_2(x) = X$ and both sets are not homeomorphic.

Even the conclusion of Example 6.15 is not applicable to subsets of $\mathbb{R}^n$ with the Euclidean distance. For example, in $X = \{0\} \cup [2, 6]$ as subset of $\mathbb{R}$, the following three balls have identical radius $r = 1$, but neither is homeomorphic to the other:

$$B_1(0) = \{0\}, \quad B_1(2) = [2, 3), \quad B_1(4) = (3, 5).$$

Even balls with the same center are not homeomorphic, as for example,

$$B_1(6) = (5, 6], \quad B_5(6) = [2, 6], \quad B_7(6) = \{0\} \cup [2, 6].$$

$\diamondsuit$

The next result is also intuitive since tells us that an affine subspace is homeomorphic to the Euclidean space of the same dimension. If we think of a straight line within the Euclidean plane $\mathbb{R}^2$, we may simply assume that the line is one of the coordinate axes, say the x-axis $\mathbb{R} \times \{0\}$ and we remove the "0" from the second coordinate.

Example 6.17 If $A \subset \mathbb{R}^n$ is a m-dimensional affine subspace, then $A \cong \mathbb{R}^m$.

We may rewrite the subspace A as $A = p + \mathbf{A}$, where $p \in \mathbb{R}^n$ and $\mathbf{A} \subset \mathbb{R}^n$ is the vector subspace of dimension m associated to A. Suppose $m < n$ (the case $m = n$ is trivial). We need a manageable affine subspace homeomorphic to A, exactly $\mathbb{R}^m \times \{0\}$, with $0 \in \mathbb{R}^{n-m}$.

1. Consider the translation $T(x) = x - p$. Then $T(A) = \mathbf{A}$ so $A \cong \mathbf{A}$. This reduces the proof to vector subspaces of $\mathbb{R}^n$. See Fig. 6.11.
2. Suppose that $\mathbf{A}$ is spanned by m vectors, $\mathbf{A} = L(\{v_1, \dots, v_m\})$. This notation for the subspace spanned by a set of vectors will be used throughout the book. Let $B = \{e_1, \dots, e_n\}$ be the usual vector basis of $\mathbb{R}^n$, $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, \dots, 0)$ and so on until $e_n = (0, 0, \dots, 1)$. In view that $B' = \{v_1, \dots, v_m\}$

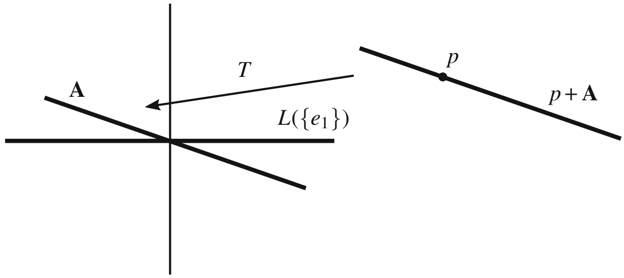


Fig. 6.11 Two straight lines are homeomorphic

are linearly independent, we extend to a basis $\{v_1, \ldots, v_m, v_{m+1}, \ldots, v_n\}$ of $\mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}^n$ be the linear isomorphism such that $f(v_i) = e_i$ for $1 \le i \le n$. Thus f is a homeomorphism and by linearity,

$$\mathbf{A} \cong f(\mathbf{A}) = L(\{f(v_1), \ldots, f(v_m)\}) = L(\{e_1, \ldots, e_m\}).$$

3. We prove that $X = L(\{e_1, \ldots, e_m\}) \cong \mathbb{R}^m$. Notice that X is

$$X = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_{m+1} = 0, \ldots, x_n = 0\} = \mathbb{R}^m \times \{0\},$$

where $\{0\} \in R^{n-m}$. Define

$$g : \mathbb{R}^m \to \mathbb{R}^n, \quad g(x_1, \ldots, x_m) = (x_1, \ldots, x_m, 0, \ldots, 0).$$

Then g is injective, continuous and $g(\mathbb{R}^m) = X$. Thus $\phi = g_{|\mathbb{R}^m} : \mathbb{R}^m \to X$ is bijective, continuous and its inverse is

$$\phi^{-1} : X \to \mathbb{R}^m, \quad \phi^{-1}(x_1, \ldots, x_m, 0, \ldots, 0) = (x_1, \ldots, x_m).$$

Now ϕ^{-1} is continuous because it is the restriction to X of

$$\Phi : \mathbb{R}^n \to \mathbb{R}^n, \quad \Phi(x_1, \ldots, x_n) = (x_1, \ldots, x_m).$$

This concludes that $\phi : \mathbb{R}^m \to X$ is a homeomorphism.

$\Diamond$

We proceed by establishing more homeomorphisms between subsets of Euclidean spaces. If $0 \le r < R \le \infty$, the *circular annulus* (or simply annulus) centered at (x_0, y_0) having inner radius r and outer radius R is defined by

$$A = \{(x, y) \in \mathbb{R}^2 : r < \sqrt{(x - x_0)^2 + (y - y_0)^2} < R\}.$$

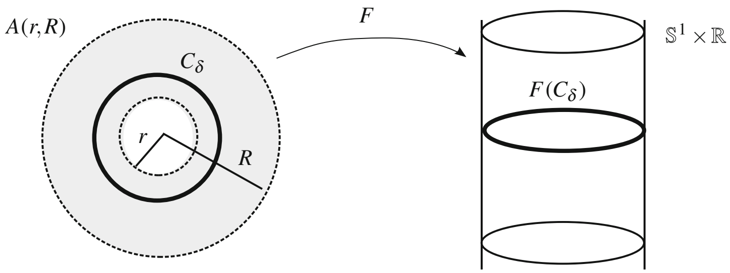


Fig. 6.12 A homeomorphism between the annulus $A(r, R)$ and $\mathbb{S}^1 \times \mathbb{R}$

Up to a translation of $\mathbb{R}^2$, assume that the center is the origin $(0, 0)$,

$$A(r, R) = \{(x, y) \in \mathbb{R}^2 : r < \sqrt{x^2 + y^2} < R\}.$$

Two interesting annuli are $A(r, \infty) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 > r^2\}$ and the punctured plane $A(0, \infty) = \mathbb{R}^2 \setminus \{(0, 0)\}$.

Example 6.18 Any circular annulus is homeomorphic to the cylinder $\mathbb{S}^1 \times \mathbb{R}$.

We visualize the homeomorphism in Fig. 6.12. We will map every concentric circle $C_\delta \subset A(r, R)$ of radius $\delta > 0$ into a horizontal circle of the cylinder such that as δ goes from r to R, the circle C_δ will be located at a height varying from $-\infty$ to ∞. The key is that the interval (r, R) representing all the radii of the annulus is homeomorphic to $(-\infty, \infty) = \mathbb{R}$ which indicates all the heights in the cylinder.

Let $h\colon (r, R) \to \mathbb{R}$ be a homeomorphism. We position C_δ, $r < \delta < R$, in the corresponding horizontal circle in the cylinder at the height $h(\delta)$. If $(x, y) \in C_\delta$, the first two coordinates in the cylinder are proportional to (x, y) and of modulus 1. Thus define

$$F\colon A(r, R) \to \mathbb{S}^1 \times \mathbb{R}, \quad F(x, y) = \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, h(\sqrt{x^2 + y^2}) \right).$$

Continuity of F is proved with the aid of Proposition 6.3.2. So it is clear that

$$q_1 \circ F = \frac{p_1}{\sqrt{p_1^2 + p_2^2}}, \quad q_2 \circ F = \frac{p_2}{\sqrt{p_1^2 + p_2^2}}, \quad q_3 \circ F = h \circ \sqrt{p_1^2 + p_2^2},$$

where p_i and q_i are the restrictions of the corresponding projections of $\mathbb{R}^2$ to $A(r, R)$ and of $\mathbb{R}^3$ to $\mathbb{S}^1 \times \mathbb{R}$, respectively. To get F^{-1}, from $F^{-1}(x, y, z) = (a, b)$, we can obtain the variables a and b in terms of x, y and z. Indeed,

$$x = \frac{a}{\sqrt{a^2 + b^2}}, \quad y = \frac{b}{\sqrt{a^2 + b^2}}, \quad z = h(\sqrt{a^2 + b^2}),$$

and from the third equation, $\sqrt{a^2+b^2} = h^{-1}(z)$, so $a = xh^{-1}(z)$ and $b = yh^{-1}(z)$. Thus $F^{-1}(x, y, z) = (xh^{-1}(z), yh^{-1}(z))$. Continuity of F^{-1} is clear because

$$p_1 \circ F^{-1} = q_1 \cdot (h^{-1} \circ q_3), \quad p_2 \circ F^{-1} = q_2 \cdot (h^{-1} \circ q_3).$$

$\Diamond$

In the construction of F, it has not been necessary to specify the expression of h, in fact, we are not even utilizing that the circles are located at heights varying from $-\infty$ to ∞ as we go from r to R in the annulus. In fact, we do not have evidence that for an arbitrary homeomorphism between (r, R) and $\mathbb{R}$, the endpoints of (r, R) go (in the limit) to the "endpoints" ∞ and $-\infty$. Nevertheless, it will be proved in Chap. 8 that *every homeomorphism between two intervals of* $\mathbb{R}$ *is strictly increasing or strictly decreasing* and, as a consequence, the endpoints are mapped to the endpoints.

Example 6.18 generalizes to arbitrary dimensions. So let $A(r, R) = \{p \in \mathbb{R}^n : r < |p| < R\}$. Then $A(r, R) \cong \mathbb{S}^{n-1} \times \mathbb{R}$ and the homeomorphism is

$$F : A(r, R) \to \mathbb{S}^{n-1} \times \mathbb{R}, \quad F(p) = \left(\frac{p}{|p|}, h(|p|)\right).$$

A second extension is to include the circles C_r and C_R of the boundary of $A(r, R)$. For instance, an annulus $r \leq \sqrt{x^2+y^2} < R$ for $r > 0$ is homeomorphic to $\mathbb{S}^1 \times [r, R)$ (Exercise 6.6.6).

Example 6.19 When the annulus is the punctured plane, F is simply the *polar coordinates*. To be precise, for $r = 0$ and $R = \infty$, take h the identity map. Then $F : \mathbb{R}^2 \setminus \{0\} \to \mathbb{S}^1 \times \mathbb{R}$ is

$$F(x, y) = \left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}}, \sqrt{x^2+y^2}\right).$$

The usual notation for polar coordinates is $r = \sqrt{x^2+y^2}$ and $\theta \in [0, 2\pi)$ is the unique number such that

$$\cos\theta = \frac{x}{\sqrt{x^2+y^2}}, \quad \sin\theta = \frac{y}{\sqrt{x^2+y^2}}.$$

Then the polar coordinates of (x, y) are (θ, r), where $F(x, y) = (\cos\theta, \sin\theta, r)$. $\Diamond$

The next example tells us that the punctured sphere is homeomorphic to the Euclidean space of the same dimension.

Example 6.20 If $p \in \mathbb{S}^n$, then $\mathbb{S}^n \setminus \{p\}$ is homeomorphic to $\mathbb{R}^n$.

We follow the strategy of previous examples and consider more manageable sets. If $p, q \in \mathbb{S}^n$, we prove that $\mathbb{S}^n \setminus \{p\} \cong \mathbb{S}^n \setminus \{q\}$. To accomplish this, let $\{p, v_1, \ldots, v_n\}$ and $\{q, u_1, \ldots, u_n\}$ be two orthonormal bases of $\mathbb{R}^n$ respectively

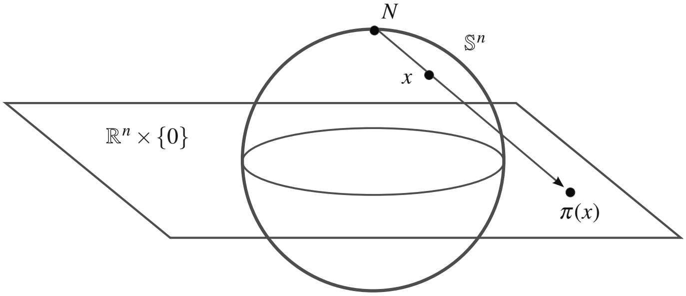


Fig. 6.13 The stereographic projection π from $\mathbb{S}^n \setminus \{N\}$ into $\mathbb{R}^n$

containing p and q. Let $\phi\colon \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be the linear isometry such that $\phi(p) = q$ and $\phi(v_i) = u_i$ for all $1 \le i \le n$. Then ϕ preserves modulus, so the restriction of ϕ to $\mathbb{S}^n$ leads to a homeomorphism $\phi\colon \mathbb{S}^n \to \mathbb{S}^n$. Since $\phi(p) = q$,

$$\mathbb{S}^n \setminus \{p\} \cong \phi(\mathbb{S}^n \setminus \{p\}) = \phi(\mathbb{S}^n) \setminus \{\phi(p)\} = \mathbb{S}^n \setminus \{q\}.$$

After this, let us pick the point p to be the *north pole* $N = (0, \ldots, 0, 1)$ of $\mathbb{S}^n$. We proceed to define a homeomorphism $\pi : \mathbb{S}^n \setminus \{N\} \to \mathbb{R}^n$. The idea appears in Fig. 6.13. The map π carries each point $x \in \mathbb{S}^n \setminus \{N\}$ to the point obtained by the intersection of the hyperplane $x_{n+1} = 0$ with the straight line that passes through N and x. Finally, the hyperplane $x_{n+1} = 0$ is identified with $\mathbb{R}^n$ (Example 6.17). Analytically the ray from N through $x \in \mathbb{S}^n$ can be written as $N + \lambda(x - N)$, with $\lambda > 0$. If we impose that the $(n + 1)$th coordinate is 0, then $1 + \lambda(x_{n+1} - 1) = 0$, hence $\lambda = 1/(1 - x_{n+1})$. Thus

$$\pi : \mathbb{S}^n \setminus \{N\} \to \mathbb{R}^n, \quad \pi(x_1, \ldots, x_{n+1}) = \left(\frac{x_1}{1 - x_{n+1}}, \ldots, \frac{x_n}{1 - x_{n+1}}\right). \tag{6.5}$$

This mapping is called the *stereographic projection*. Likewise the inverse map $\pi^{-1}\colon \mathbb{R}^n \to \mathbb{S}^n \setminus \{N\}$ is

$$\pi^{-1}(x_1, \ldots, x_n) = \left(\frac{2x_1}{1 + |x|^2}, \ldots, \frac{2x_n}{1 + |x|^2}, \frac{|x|^2 - 1}{1 + |x|^2}\right). \tag{6.6}$$

The proof that π and π^{-1} are continuous follows by arguing as in previous examples and we do it only for π. Consider $p_i'\colon \mathbb{R}^n \to \mathbb{R}$ and $p_i\colon \mathbb{R}^{n+1} \to \mathbb{R}$ the corresponding projections. If $q_i = (p_i)_{|\mathbb{S}^n \setminus \{N\}}$, then

$$p_i' \circ \pi = \frac{q_i}{1 - q_{n+1}}, \quad 1 \le i \le n,$$

proving that π is continuous. $\Diamond$

Remark 6.3.3 When $n = 1$, the inverse of the stereographic projection is

$$\pi^{-1}\colon \mathbb{R} \to \mathbb{S}^1 \setminus \{(0,1)\}, \quad \pi^{-1}(t) = \left(\frac{2t}{1+t^2}, \frac{t^2-1}{1+t^2}\right).$$

Since $\pi^{-1}(t) \in \mathbb{S}^1$,

$$\left(\frac{2t}{1+t^2}\right)^2 + \left(\frac{t^2-1}{1+t^2}\right)^2 = 1,$$

so $(2t)^2 + (t^2-1)^2 = (1+t^2)^2$. This map provides many *Pythagorean triples*. A Pythagorean triple is formed by three integers x, y and z such that $x^2 + y^2 = z^2$. Choosing $c \in \mathbb{N}$, let $x = 2c$, $y = c^2 - 1$ and $z = 1 + c^2$. Several examples are $(4, 3, 5)$, $(6, 8, 10)$ and $(200, 9999, 10001)$. The value c may be rational, as for example, $c = 3/2$ where $a = 12/13$, $b = 5/13$ and the Pythagorean triple is $(12, 5, 13)$.

Remark 6.3.4 The sphere $\mathbb{S}^2$ is a model of the Earth's surface and $\mathbb{R}^2$ is a map (a drawing of the Earth on a sheet of paper). Then the stereographic projection gives a one-to-one representation of the Earth that preserves proximity. However, by a simple computation, π does not preserve the distances between two points, that is, there is no a constant $\lambda > 0$ (the so-called scale) such that $|\pi(x) - \pi(y)| = \lambda|x - y|$ for all $x, y \in \mathbb{S}^2$. In fact, a result in mathematics (that is not topological) affirms that it is not possible to represent the Earth on a map preserving, up to a scale, the distances.

6.4 Embeddings and Open Maps

Something weaker than homeomorphisms are those maps that immerse one space within another one, identifying the domain space homeomorphically with its image.

Definition 6.4.1 A map $f\colon X \to Y$ is said to be an *embedding* if $f\colon X \to f(X)$ is a homeomorphism. In such a case, we say that X *is embedded* in Y by f.

As a consequence, the space X can be regarded as a subset of Y equipped with the relative topology. An example of an embedding is any homeomorphism. The composition of embeddings is an embedding and the restriction of an embedding to a subspace of the domain is an embedding again.

Proposition 6.4.2

1. *Let $f\colon X \to Y$ be an embedding. If $\phi\colon Z \to X$ and $\psi\colon Y \to W$ are homeomorphisms, then $f \circ \phi$ and $\psi \circ f$ are embeddings.*
2. *Let (X, τ) be a space and let $A \subset X$ a subspace supporting a topology τ'. Then the inclusion $i\colon (A, \tau') \to (X, \tau)$ is an embedding if and only if $\tau' = \tau_{|A}$.*

Example 6.21 If $n \leq m$, then $\mathbb{R}^n$ can be embedded in $\mathbb{R}^m$ and $\mathbb{S}^n$ in $\mathbb{S}^m$.

An embedding of $\mathbb{R}^n$ into $\mathbb{R}^m$ is the map

$$g\colon \mathbb{R}^n \to \mathbb{R}^m, \quad g(x_1, \dots, x_n) = (x_1, \dots, x_n, 0, \dots, 0)$$

defined in (3) of Example 6.17. Another embedding is letting $\mathbb{R}^m = \mathbb{R}^{n+k}$ and let $y = (y_1, \dots, y_k)$ be an arbitrary but fixed point of $\mathbb{R}^k$. Then

$$\phi\colon \mathbb{R}^n \to \mathbb{R}^{n+k}, \quad \phi(x_1, \dots, x_n) = (x_1, \dots, x_n, y_1, \dots, y_k)$$

is an embedding and consequently, $\mathbb{R}^n \cong \mathbb{R}^n \times \{y\}$. The inverse of $\phi\colon \mathbb{R}^n \to \phi(\mathbb{R}^n)$ is the restriction to $\phi(\mathbb{R}^n)$ of $\psi(x_1, \dots, x_m) = (x_1, \dots, x_n)$. For spheres, the above map $g\colon \mathbb{R}^{n+1} \to \mathbb{R}^{m+1}$ carries $\mathbb{S}^n$ homeomorphically to a subset of $\mathbb{S}^m$, so $g_{|\mathbb{S}^n}\colon \mathbb{S}^n \to \mathbb{S}^m$ is an embedding. $\Diamond$

Example 6.22

1. The interval $[0, 1]$ embeds in $(0, 1)$. The idea is that $[0, 1] \cong [\frac{1}{4}, \frac{1}{2}]$ and $[\frac{1}{4}, \frac{1}{2}] \hookrightarrow (0, 1)$ is an embedding.
2. The interval $[0, 1]$ embeds in $(0, 1)$ and $(0, 1)$ embeds in $[0, 1]$ with the inclusion, but $[0, 1]$ and $(0, 1)$ are not homeomorphic.
3. Let $n < m$. The Euclidean space $\mathbb{R}^n$ embeds in $\mathbb{R}^m$ but $\mathbb{R}^m$ cannot be embedded in $\mathbb{R}^n$ by Theorem 6.1.6.

$\Diamond$

It is clear that Euclidean spaces are special topological spaces and possess many properties. It would be desirable that a space could be embedded in some Euclidean space, hence the following problem arises.

The Embedding problem. *If X is a topological space, find, if possible, an embedding from X into a Euclidean space $\mathbb{R}^n$ for some $n \in \mathbb{N}$. In such a case, find the smallest number n.*

Consequently, a non-metrizable space cannot be embedded in a Euclidean space because the relative topology of Euclidean topology is metric. If a space cannot be embedded in a Euclidean space of dimension n, then it cannot be embedded in a space of smaller dimension either (Example 6.21).

Definition 6.4.3 A mapping $f\colon X \to Y$ is called *open* (resp. *closed*) if $f(O)$ is open for every open set O of X (resp. $f(F)$ closed for every closed set F of X).

Using this notion and Theorem 5.1.3, we see that a bijection is a homeomorphism if and only if is continuous and open (or it is continuous and closed). Also open maps carry neighborhoods onto neighborhoods.

It is clear that f is open if for every $A \subset X$, $f(\overset{\circ}{A}) \subset \text{int}(f(A))$. Likewise, a mapping $f\colon X \to Y$ is closed if and only if $\overline{f(A)} \subset f(\overline{A})$ for all $A \subset X$. Significant examples of open maps are the projections of Euclidean spaces.

Example 6.23 We see that the projections $p_i : \mathbb{R}^n \to \mathbb{R}$, $1 \le i \le n$, are open but not closed.

To prover that p_i is open, it is enough to show that if β is a basis for $\mathbb{R}^n$ then $f(B)$ is open for all $B \in \beta$. Consider the standard basis for the Euclidean space $\mathbb{R}^n$ given in Definition 2.3.2. Then clearly $p_i((a_1, b_1) \times \ldots \times (a_n, b_n)) = (a_i, b_i)$ that is open in $\mathbb{R}$.

We prove that projections are not closed. Consider just the first projection p_1, as the other cases are analogous (the case $n = 1$ is discarded because p_1 is the identity). Let $A = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : \Pi_{i=1}^n x_i = 1\}$. This set is closed because $A = f^{-1}(\{0\})$, where $f(x_1, \ldots, x_n) = \Pi_{i=1}^n x_i - 1$ and f is continuous. On the other hand, the set $p_1(A)$ coincides with $\mathbb{R} \setminus \{0\}$ which is not closed in $\mathbb{R}$. In effect, the inclusion $p_1(A) \subset \mathbb{R} \setminus \{0\}$ is obvious because $x_1 \neq 0$ for all $x \in A$. For the inclusion $\mathbb{R} \setminus \{0\} \subset p_1(A)$, if $t \neq 0$ then $x = (t, 1/t, 1, \ldots, 1) \in A$ and $p_1(x) = t$. ♢

We show an open and closed mapping that is not continuous.

Example 6.24 Define the function $f : [0, 2] \to \{0, 1\}$ by

$$f(x) = \begin{cases} 1, & \text{if } x \in [0, 1] \\ 0, & \text{if } x \in (1, 2], \end{cases}$$

The map f is obviously open and closed because the topology on $\{0, 1\}$ is discrete, so every subset is open and closed. However, f is discontinuous at $x = 1$. ♢

6.5 Worked Exercises

In the next three exercises, we answer various questions that were formulated in Example 2.1 of Chap. 2.

Exercise 6.1 Let X be a set and $p, q \in X$. For the included point topologies τ_p and τ_q, prove that $(X, \tau_p) \cong (X, \tau_q)$ but $\tau_p \neq \tau_q$.

Solution It is obvious that $\{p\} \in \tau_p$ but $\{p\} \notin \tau_q$, so the topologies are different. We prove that (X, τ_p) is homeomorphic to (X, τ_q) by way of the map

$$f : (X, \tau_p) \to (X, \tau_q), \qquad \begin{cases} x \longmapsto x, & x \neq p, q \\ p \longmapsto q, & \\ q \longmapsto p. & \end{cases}$$

The map f is bijective and its inverse is f again. We prove that f is continuous. Let $O' \in \tau_q$. Then $q \in O'$. As a consequence, it follows from $f(p) = q$ that $p \in f^{-1}(O')$, hence $f^{-1}(O') \in \tau_p$. This proves continuity of f. Similarly

$f^{-1}\colon (X, \tau_q) \to (X, \tau_p)$ is continuous. As a consequence, *every space homeomorphic to an included point topology is an included point topology.* □

Exercise 6.2 Let X be a set and $p, q \in X$. For the excluded point topologies τ^p and τ^q, prove that $(X, \tau^p) \cong (X, \tau^q)$ but $\tau^p \neq \tau^q$.

Solution Now $\{q\} \in \tau^p$ but $\{q\} \notin \tau^q$, so the topologies are different. For the homeomorphism, we use again

$$f\colon (X, \tau^p) \to (X, \tau^q), \qquad \begin{cases} x \longmapsto x, & x \neq p, q \\ p \longmapsto q, & \\ q \longmapsto p. & \end{cases}$$

To prove that $f\colon (X, \tau^p) \to (X, \tau^q)$ is a homeomorphism, we see that f maps bases to bases. Let $\beta^p = \{\{x\} : x \neq p\} \cup \{X\}$ be a basis por τ^p and we see that $f(\beta^p) = \beta^q$, where $\beta^q = \{\{x\} : x \neq q\} \cup \{X\}$ is a basis for τ^q. It is clear that for the singletons $\{x\}$ with $x \neq p$, we have

$$\{f(x)\} = \begin{cases} \{x\}, & x \neq q \\ \{p\}, & x = q, \end{cases}$$

so $\{f(x)\} \in \beta^q$. Finally $X \in \beta^p$ and $f(X) = X \in \beta^q$. We conclude that *every space homeomorphic to an excluded point topology is an excluded point topology.* □

Exercise 6.3 Prove that the included point topology is not homeomorphic to the excluded point topology.

Solution We discard the situation in which the spaces have one or two elements because in such a case, both topologies are homeomorphic. Let $f\colon (X, \tau_{ex}) \to (Y, \tau_{in})$ be a homeomorphism where $p \in X$ and $q \in Y$ are the particular points. Let $x \neq p$ be a point of X such that $f(x) \neq q$. A basis of neighborhoods of x is $\beta_x = \{\{x\}\}$. Since f is a homeomorphism, $\{f(x)\}$ is a neighborhood of $f(x)$. But in τ_{in}, singletons are neighborhoods only at the case $\{q\}$. This contradiction completes the exercise. □

Exercise 6.4 Consider on $\mathbb{R}$ the right order topology τ_r and left order topology τ_l. Prove $(\mathbb{R}, \tau_r) \cong (\mathbb{R}, \tau_l)$, but $\tau_r \neq \tau_l$.

Solution A basis for τ_r is $\beta_r = \{[x, \infty) : x \in \mathbb{R}\}$ and a basis for τ_l is $\beta_l = \{(-\infty, x] : x \in \mathbb{R}\}$. See Example 2.12. We know explicitly the topologies,

$$\tau_r = \{\emptyset, \mathbb{R}\} \cup \beta_r \cup \{(x, \infty) : x \in \mathbb{R}\}, \quad \tau_l = \{\emptyset, \mathbb{R}\} \cup \beta_l \cup \{(-\infty, x) : x \in \mathbb{R}\}.$$

Thus $\tau_r \neq \tau_l$. We find a homeomorphism between these spaces. Define

$$f\colon (\mathbb{R}, \tau_r) \to (\mathbb{R}, \tau_l), \quad f(x) = -x.$$

This map is bijective and its inverse is $f^{-1} = f$. It is clear that $f(\beta_r) = \beta_l$, proving that f is a homeomorphism. □

We present in subsequent exercises new topological invariants.

Exercise 6.5 Prove that the second axiom of countability is a topological invariant. Use this property to see that the Euclidean real line is not homeomorphic to the included point topology $(\mathbb{R}, \tau_{in})$.

Solution Let X be a space with a countable basis β and $f: X \to Y$ a homeomorphism. Then it is clear that $f(\beta)$ is basis for τ' by Theorem 6.1.3 and that $f(\beta)$ is countable because f is a bijection. Thus Y is also second countable.

By Theorem 4.2.6, the Euclidean real line has a countable basis. Suppose that $p \in \mathbb{R}$ is the particular point that defines the included point topology τ_{in}. It was proven in Example 2.9 that any basis for τ_{in} must contain at least the collection of subsets $\{\{p, x\} : x \in \mathbb{R}\}$. Since this set is uncountable, any basis for τ_{in} is uncountable.

There is also a simple way to prove that these spaces are not homeomorphic. If $f: (\mathbb{R}, \tau_{in}) \to (\mathbb{R}, \tau_u)$ is a homeomorphism, then $\{p\} \in \tau_{in}$ but $f(\{p\}) = \{f(p)\} \notin \tau_u$. □

Exercise 6.6 Let $\mathsf{H}(X, \tau)$ be the set of all homeomorphisms of a space (X, τ) to itself and let $\mathsf{H}(X, \tau)$ be endowed with a group structure with the composition of maps as group operation. If (Y, τ') is homeomorphic to (X, τ), prove that $\mathsf{H}(Y, \tau')$ is isomorphic to $\mathsf{H}(X, \tau)$. Find two non-homeomorphic spaces whose groups of homeomorphisms are isomorphic.

Solution

1. Let $f: X \to Y$ be a homeomorphism and define

$$\phi: \mathsf{H}(X, \tau) \to \mathsf{H}(Y, \tau'), \quad \phi(g) = f \circ g \circ f^{-1}.$$

The map ϕ is well defined because $\phi(g)$ is composition of three homeomorphisms. The inverse is $\phi^{-1}(g) = f^{-1} \circ g \circ f$. Finally, ϕ is a homomorphism of groups because

$$\begin{aligned}\phi(g \circ h) &= f \circ g \circ h \circ f^{-1} = f \circ g \circ f^{-1} \circ f \circ h \circ f^{-1} \\ &= (f \circ g \circ f^{-1}) \circ (f \circ h \circ f^{-1}) = \phi(g) \circ \phi(h).\end{aligned}$$

2. For an infinite set X, consider the discrete topology τ_D and the cofinite topology τ_{CF}. It was proved in Example 5.6 that every map from (X, τ_D) into (X, τ_D) is continuous. Thus $\mathsf{H}(X, \tau_D)$ coincides with the set $\mathrm{Bij}(X)$ of all bijections from X to itself.

   We claim that $\mathsf{H}(X, \tau_{CF})$ also coincides with $\mathrm{Bij}(X)$. Let $f: (X, \tau_{CF}) \to (X, \tau_{CF})$ be a bijection. Then the image of every finite set is finite, proving that f carries closed sets onto closed sets and analogously for f^{-1}. Thus f is

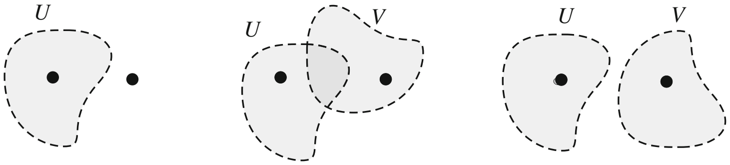


Fig. 6.14 The T_0-property (left), the T_1-property (middle) and the T_2-property (right)

a homeomorphism by Theorem 6.1.3 and consequently, $\mathsf{H}(X, \tau_{CF}) = \mathrm{Bij}(X)$. Definitively, the sets $\mathsf{H}(X, \tau_D)$ and $\mathsf{H}(X, \tau_{CF})$ are actually identical.

However, (X, τ_D) and (X, τ_{CF}) are not homeomorphic because any bijection between them maps singletons into singletons, but singletons are open in (X, τ_D) but not in (X, τ_{CF}). □

The Hausdorff property is one of many so-called *separation axioms* in topology. This refers to certain properties that allow us to separate, by means of open sets, certain types of subsets in a topological space. We give two new axioms of separation (Fig. 6.14).

Exercise 6.7 A space X is said to satisfy the T_0-property if for each $x_1 \neq x_2 \in X$, there is $U_1 \in \mathscr{U}_{x_1}$ such that $x_2 \notin U_1$, or there is $U_2 \in \mathscr{U}_{x_2}$ such that $x_1 \notin U_2$. Prove that the T_0-property is a topological invariant. Find two non-homeomorphic spaces that satisfy the T_0-property.

Solution

1. Let $f : X \to Y$ be a homeomorphism, where X is a T_0-space. Let y_1 and y_2 be two points in Y. In view that f is bijective, let $x_i \in X$ such that $f(x_i) = y_i$, $i = 1, 2$ and obviously, $x_1 \neq x_2$. Since X satisfies the T_0-property, there is, say $U \in \mathscr{U}_{x_1}$ such that $x_2 \notin U$. Using again that f is bijective, $y_2 \notin f(U)$ and $f(U)$ is a neighborhood of y_1 because f is a homeomorphism.
2. Consider on $\mathbb{R}$ the discrete topology τ_D and the usual topology τ_u. We see that they satisfy the T_0-property. Let $x \neq y$. For τ_D, take $U = \{x\}$ and for τ_u, let $U = (x - r, x + r)$, where $r < |x - y|$. In both cases, $y \notin U$. However, these spaces are not homeomorphic because singletons in $(\mathbb{R}, \tau_u)$ are not open. □

Exercise 6.8 A space X is said to satisfy the T_1-property if for each $x_1 \neq x_2 \in X$, there is $U_1 \in \mathscr{U}_{x_1}$ such that $x_2 \notin U_1$ and there is $U_2 \in \mathscr{U}_{x_2}$ such that $x_1 \notin U_2$. Prove that a space is T_1 if and only if any singleton is closed. Find examples of a T_0-space that is not T_1 and T_1-space that is not T_2.

Solution The T_1-property is a topological invariant because homeomorphisms preserve neighborhoods. Also, we have the following chain of implications:

$$\text{metric spaces} \Rightarrow T_2 \Rightarrow T_1 \Rightarrow T_0.$$

1. Assume that the space is T_1. If $\{x\}$ is not closed, there is $y \in \overline{\{x\}}$ such that $x \neq y$. Since the space is T_1, there is $V \in \mathscr{U}_y$ such that $x \notin V$. Thus $\{x\} \cap V = \emptyset$, which is contradictory since y is an adherent point of $\{x\}$.

   Suppose that every singleton is closed. Let $x, y \in X$ be given. In view that $y \notin \{x\} = \overline{\{x\}}$, there is $V \in \mathscr{U}_y$ such that $\{x\} \cap V = \emptyset$, which means that $x \notin V$. Analogously, there is $U \in \mathscr{U}_x$ such that $y \notin U$.
2. Let $X = \{a, b\}$ with the Sierpinski topology $\tau = \{\emptyset, X, \{a\}\}$. This space is T_0 because $\{a\}$ is a neighborhood of a and $b \notin \{a\}$, but it is not T_1 because $\{a\}$ is not closed.
3. It was proven in Chap. 4 that the cofinite topology is not T_2. However, it is T_1 because any singleton is finite, hence closed. □

Exercise 6.9 Prove that the space $(\mathbb{N}^*, \tau^*)$ defined in Exercise 5.14 is homeomorphic to $X = \{0\} \cup \{1/n : n \in \mathbb{N}\}$.

Solution We prove that

$$f : \mathbb{N}^* \to X, \qquad \begin{cases} f(n) = \frac{1}{n} \\ f(\omega) = 0 \end{cases}$$

is a homeomorphism. It is elementary that f is bijective. For the continuity of f, let $n \in \mathbb{N}$. The singletons $\{n\}$ are neighborhoods of n and this also applies to the points of $X \setminus \{0\}$. As $f : \{n\} \to \{f(n)\}$ is continuous because it is constant and $\{n\}$ is a neighborhood of n in $\mathbb{N}^*$, item (2) of Proposition 5.2.4 assures the continuity of $f : \mathbb{N}^* \to X$ at n. We check the continuity of f at p. Let $G \subset \mathbb{R}$ be an open set such that $f(\omega) = 0 \in G$ and let $\nu \in \mathbb{N}$ such that $1/n \in G$ for $n \geq \nu$. Then $O = \mathbb{N}^* \setminus \{1, \ldots, \nu\}$ is open in $\mathbb{N}^*$, $p \in O$ and clearly $f(O) = \{\omega\} \cup \{1/n : n > \nu\} \subset G$.

Finally, we see that f is open. Let $O \in \tau^*$. If O is open in $\mathbb{N}$, then it is merely a subset of $\mathbb{N}$, so $f(O) \subset X \setminus \{0\}$. As $X \setminus \{0\}$ has the discrete topology, $f(O)$ is open in $X \setminus \{x\}$. Since $X \setminus \{0\}$ is open in X, $f(O)$ is open in X. Note that $\{0\}$ is closed in $\mathbb{R}$, so also in X. If $O = \mathbb{N}^* \setminus A$, where A is a finite subset of $\mathbb{N}$, then $f(O) = X \setminus f(A)$. The set $f(A)$ is finite, so $f(A)$ is closed in $\mathbb{R}$; in particular, it is closed in X. Thus its complement $f(O)$ is open. □

In the next exercise, we add to the Euclidean real line $\mathbb{R}$ the "endpoints" $+\infty$ and $-\infty$ and define a topology such that the new space is homeomorphic to $[-1, 1]$.

Exercise 6.10 Let ω^- and ω^+ be two objects that are not real numbers. Let $\mathbb{R}^* = \mathbb{R} \cup \{\omega^-, \omega^+\}$ and define the topology τ generated by the basis

$$\beta = \beta_u \cup \{\{\omega^-\} \cup (-\infty, a) : a \in \mathbb{R}\} \cup \{(a, \infty) \cup \{\omega^+\} : a \in \mathbb{R}\},$$

where β_u is the standard basis of τ_u. Prove that the relative topology on $\mathbb{R}$ is Euclidean and that $(\mathbb{R}^*, \tau)$ is homeomorphic to $[-1, 1]$.

Solution Notice that β satisfies the properties of being a basis for a topology because the intersection of two members of β is another member of β. Specifically, if both members belong to each one of the three types of elements of β, then the intersection is an element of the same type. If the intersection is between two elements of different types, the intersection is an element of β_u, hence of β.

1. We prove that the relative topology on $\mathbb{R}$ coincides with the Euclidean topology. The intersection of an element of the first type of β with $\mathbb{R}$ is the same element, which belongs to β_u. If we intersect elements of β of the second or third type with $\mathbb{R}$, the resulting sets are intervals of type $(-\infty, a)$ and of type (a, ∞), which are open in τ_u. Definitively, the intersection of all members of β with $\mathbb{R}$ provides a collection of open sets in the usual topology that includes a basis of this topology. This gives a basis for the Euclidean topology.
2. We give the idea of the homeomorphism. We stretch the real line $\mathbb{R} = (-\infty, \infty)$ to the open interval $(-1, 1)$. If we think of the endpoints "$-\infty$" and "$+\infty$" of $\mathbb{R}$ as being glued along this deformation, one can picture that "$-\infty$" goes to -1 and "$+\infty$" into 1. Under this deformation, the open sets (a, ∞) are mapped to intervals of type $(b, 1)$ in $(-1, 1)$. If the point ω^+ of $\mathbb{R}^*$, identified with $+\infty$, is glued in the stretching process. Then the point ω^+ will map onto 1, obtaining in the codomain an interval $(b, 1]$, which is open in $[-1, 1]$. This suggests that the process is continuous. Therefore, define

$$f\colon \mathbb{R}^* \to [-1, 1], \quad f(x) = \begin{cases} h(x) = \dfrac{x}{1+|x|}, & \text{if } x \in \mathbb{R} \\ -1, & \text{if } x = \omega^- \\ 1, & \text{if } x = \omega^+. \end{cases}$$

Then f is, clearly, bijective. We see that f is open. Using that h is increasing, for each $B \in \beta$,

$$f(B) = \begin{cases} [-1, h(a)), & \text{if } B = \{\omega^-\} \cup (-\infty, a) \\ (h(a), h(b)), & \text{if } B = (a, b) \\ (h(a), 1], & \text{if } B = (a, \infty) \cup \{\omega^+\}. \end{cases}$$

In all cases, the image is open in $[-1, 1]$. We check that f^{-1} is also open. Consider the basis for $[-1, 1]$ obtained by the intersection of the usual basis of the Euclidean real line with $[-1, 1]$:

$$\beta' = \{(a, b) : -1 < a < b < 1\} \cup \{[-1, a) : a < 1\} \cup \{(a, 1] : a > -1\}.$$

Then

$$f^{-1}(B') = \begin{cases} \{\omega^-\} \cup (-\infty, h^{-1}(a)), & \text{if } B' = [-1, a) \\ (h^{-1}(a), h^{-1}(b)), & \text{if } B' = (a, b) \\ (h^{-1}(a), \infty) \cup \{\omega^+\}, & \text{if } B' = (a, 1], \end{cases}$$

proving that f^{-1} is continuous. □

The following exercise works with the map $f(t) = (\cos t, \sin t)$. This map rolls $\mathbb{R}$ along $\mathbb{S}^1$ in such a way that as we run an interval of length 2π in $\mathbb{R}$, its image winds one time on $\mathbb{S}^1$.

Exercise 6.11 Let

$$f : \mathbb{R} \to \mathbb{S}^1, \quad f(t) = (\cos t, \sin t).$$

Prove that for each $t \in \mathbb{R}$, there is a neighborhood U of t and other V of $f(t)$ such that $f : U \to V$ is a homeomorphism. Prove that f is open but not closed. As a consequence, show that $f : (a, a+2\pi) \to f((a, a+2\pi))$ is a homeomorphism for every $a \in \mathbb{R}$. As a consequence, the punctured circle is homeomorphic to $\mathbb{R}$.

Solution

1. We first solve the exercise when $t = 0$ and next, after a translation in $\mathbb{R}$ and a rotation in $\mathbb{S}^1$, we extend it to every point $t \in \mathbb{R}$. Let $t = 0$ with $f(0) = (1, 0)$. Let $U_0 = (-\pi/2, \pi/2)$ and $V_0 = f(U_0) = \{(\cos t, \sin t) : -\pi/2 < t < \pi/2\}$. This set is open in $\mathbb{S}^1$ because V_0 is the right hemicircle of $\mathbb{S}^1$, $V_0 = \mathbb{S}^1 \cap \{(x, y) \in \mathbb{R}^2 : x > 0\}$ and $\{(x, y) \in \mathbb{R}^2 : x > 0\}$ is open in $\mathbb{R}^2$. The inverse map is $f^{-1}(x, y) = \text{arc}\tan(y/x)$. The maps f and f^{-1} are continuous.

   Now let t_0 be arbitrary and define $U = T(U_0)$, where $T(t) = t + t_0$. Then T is a homeomorphism and U is neighborhood of t_0. For $f(t_0)$, let $V = f(U)$. Then

$$\begin{aligned} V &= f(\{t + t_0 : t \in U_0\}) = \{(\cos(t + t_0), \sin(t + t_0)) : t \in U_0\} \\ &= \left\{ \begin{pmatrix} \cos t_0 & -\sin t_0 \\ \sin t_0 & \cos t_0 \end{pmatrix} \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} : t \in U_0 \right\} \\ &= \left\{ \begin{pmatrix} \cos t_0 & -\sin t_0 \\ \sin t_0 & \cos t_0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} : (x, y) \in V_0 \right\} = \phi(V_0), \end{aligned}$$

   where $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ is the rotation

$$\phi(x, y) = \begin{pmatrix} \cos t_0 & -\sin t_0 \\ \sin t_0 & \cos t_0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

   Since ϕ is a homeomorphism and $\phi(\mathbb{S}^1) = \mathbb{S}^1$, then $V = \phi(V_0)$ is a neighborhood of $\phi(1, 0) = f(t_0)$. We prove that $f_{|U} = \phi \circ f_{|U_0} \circ T^{-1}$. If $t \in U$,

$$\phi \circ f_{|U_0} \circ T^{-1}(t) = \phi(\cos(t - t_0), \sin(t - t_0)) = (\cos t, \sin t) = f(t).$$

   Finally, since ϕ is a homeomorphism, then $f_{|U} : U \to f(U) = V$ is also a homeomorphism.

2. On $\mathbb{R}$, consider the basis β of open intervals of length less than π. If $B \in \beta$, there is a certain interval $(t - \pi/2, t + \pi/2)$ such that $B \subset (t - \pi/2, t + \pi/2)$. We have defined a homeomorphism $f : (t - \pi/2, t + \pi/2) \to V$, where V is open in $\mathbb{S}^1$. Since B is open in $(t - \pi/2, t + \pi/2)$ and $f : (t - \pi/2, t + \pi/2) \to V$ is open, we deduce that $f(B)$ is open in V, so also open in $\mathbb{S}^1$.
3. The map f is not closed. Consider $F = \left\{2\pi n + \frac{1}{n} : n \in \mathbb{N}\right\}$. This set is closed in $\mathbb{R}$ because its complement

$$\mathbb{R} \setminus F = \bigcup_{n \in \mathbb{N}} \left(2\pi n + \tfrac{1}{n}, 2\pi(n+1) + \tfrac{1}{n+1}\right) \bigcup (-\infty, 2\pi + 1)$$

is open in $\mathbb{R}$. The image of F under f is

$$f(F) = \left\{f\left(2\pi n + \tfrac{1}{n}\right) : n \in \mathbb{N}\right\} = \left\{f\left(\tfrac{1}{n}\right) : n \in \mathbb{N}\right\}.$$

Since $(1/n) \to 0$ and f is continuous, we have $(f(1/n)) \to f(0) = (1, 0)$. However, $(1, 0) \notin f(F)$ as indicated in Fig. 6.15.
4. In order to make precise the notation, we let g denote the restriction map $g : (a, a + 2\pi) \to g((a, a + 2\pi))$. The map g is bijective and also continuous because it is the restriction of the initial map f. We prove that g^{-1} is continuous. Let $t \in (a, a + 2\pi)$. By (1), there are neighborhoods U and V of t and $g(t)$, respectively, such that $g^{-1} : V \to U$ is a homeomorphism and V is an open set of $\mathbb{S}^1$. On the other hand, $g((a, a + 2\pi))$ is the punctured circle $\mathbb{S}^1 \setminus \{f(a)\}$, which is also open in $\mathbb{S}^1$. Thus the restriction

$$f^{-1} : V \cap f((a, a + 2\pi)) \to U \cap (a, a + 2\pi)$$

is a homeomorphism, where $V \cap f((a, a+2\pi))$ is an open set of $f((a, a+2\pi))$. Thus g^{-1} coincides with f^{-1}, so g^{-1} is continuous at $f(t)$. □

In Example 6.1 we have seen that $G(f) \cong \mathbb{R}^n$ for a continuous function f defined on $\mathbb{R}^n$. We extend this result for functions defined on subsets of $\mathbb{R}^n$.

Exercise 6.12 If $f : A \subset \mathbb{R}^n \to \mathbb{R}$ is a continuous function, prove that $G(f) \cong A$.

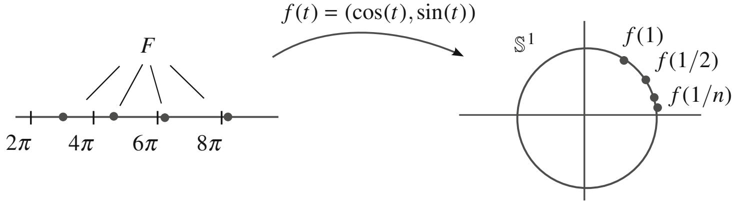


Fig. 6.15 The set F is closed in $\mathbb{R}$ but $f(F)$ is not in $\mathbb{S}^1$

Solution The proof is similar to that of Example 6.1. Define

$$\phi\colon A \to G(f), \quad \phi(x) = (x, f(x)).$$

This map is the restriction of $\Phi\colon A \to \mathbb{R}^{n+1}$ defined by $\Phi(x) = (x, f(x))$. We see that Φ is continuous by composing with the projections p_i of $\mathbb{R}^{n+1}$,

$$p_i \circ \Phi\colon A \to \mathbb{R}, \quad (p_i \circ \Phi)(x) = \begin{cases} x_i, & 1 \le i \le n \\ f(x), & i = n+1. \end{cases}$$

Thus $p_i \circ \Phi = q_{i|A}$, $1 \le i \le n$, where $q_i\colon \mathbb{R}^n \to \mathbb{R}$ is the corresponding ith projection and $p_{n+1} \circ \Phi = f$ is continuous by the hypothesis on f. This completes the proof of the continuity of Φ. As a consequence, since $\Phi(A) = G(f)$, the map $\phi\colon A \to G(f)$ is the restriction of Φ to its image, hence is continuous. We prove that

$$\phi^{-1}\colon G(f) \to A, \quad \phi^{-1}(x, f(x)) = x,$$

is continuous. Consider the projections $q_{i|A}$, $1 \le i \le n$. Then we have $(q_{i|A} \circ \phi^{-1})(x, f(x)) = x_i$, hence $q_{i|A} \circ \phi^{-1} = p_{i|G(f)}$ for $1 \le i \le n$, as all functions are continuous. □

Exercise 6.13 Find a homeomorphism between $X = \{(0, y) : y \in \mathbb{R}\}$ and $Y = \{(x, x^2) : -1 < x < 1\}$.

Solution The space X is homeomorphic to $\mathbb{R}$ because it is the graph of the continuous function $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = 0$, but viewing the graph on the y-line, $G(f) = \{(x, y) \in \mathbb{R}^2 : x = f(y)\} = \{(0, y) : y \in \mathbb{R}\}$. It can be demonstrated as in Example 6.1 that $G(f)$ is homeomorphic to $\mathbb{R}$.

On the other hand, Y is the graph of the function $g\colon (-1, 1) \to \mathbb{R}$, $g(x) = x^2$. By Exercise 6.12, the space Y is homeomorphic to $(-1, 1)$. Finally, we know that $\mathbb{R}$ is homeomorphic to $(-1, 1)$. The explicit homeomorphism is the composition of the above homeomorphisms. Let $h\colon \mathbb{R} \to (-1, 1)$ be the homeomorphism $h(x) = x/(1 + |x|)$. The order is

$$\begin{aligned} &X \to \mathbb{R}, && (0, y) \mapsto y, \\ &\mathbb{R} \to (-1, 1), && x \mapsto h(x), \\ &(-1, 1) \to Y, && x \mapsto (x, x^2). \end{aligned}$$

Then the homeomorphism that we are looking for is

$$F\colon X \to Y, \quad F(0, y) = (h(y), h(y)^2) = \left(\frac{y}{1 + |y|}, \frac{y^2}{(1 + |y|)^2}\right).$$

□

We show an open, closed map that is not continuous (see also Example 6.24).

Exercise 6.14 Let $X = [1, \infty)$ and consider the bijective map

$$f : X \to \mathbb{S}^1, \quad f(x) = (\cos(\tfrac{2\pi}{x}), \sin(\tfrac{2\pi}{x}))$$

and the usual topology on $\mathbb{S}^1$. Denote by τ the direct image topology on X by f^{-1} (Exercise 2.7.6). Prove that $\mathbb{1}_X : (X, \tau) \to (X, \tau_u)$ is open, closed but it is not continuous.

Solution A first observation is that we have used the bijection $x \mapsto 1/x$ between $[1, \infty)$ and $(0, 1]$, so the image of f is $\mathbb{S}^1$, f being one-to-one. Thus $f : (X, \tau) \to \mathbb{S}^1$ is bijective. The open sets of τ are simply the image under f^{-1} that of $\mathbb{S}^1$. By definition of f (see also Exercise 6.11), if we take balls of $\mathbb{S}^1$ of radius sufficiently small, the images of these balls are open intervals in $(1, \infty)$ except if the ball contains the point $(1, 0) \in \mathbb{S}^1$. Note that $f(1, 0) = (1, 0)$. In such a case, the image under f^{-1} is a set of type $[1, a) \cup (b, \infty)$ with $a < b$. Definitively, a basis for τ is

$$\beta = \{(a, b) : a, b \in \mathbb{R}, a \geq 1\} \cup \{[1, a) \cup (b, \infty) : a < b\}.$$

The image of the elements of β under $\mathbb{1}_X$ are the same elements. Because all of them are open in the Euclidean topology, the identity is an open map. As a consequence, $\mathbb{1}_X$ is also closed because if $F \in \mathscr{F}$, then $X \setminus \mathbb{1}_X(F) = X \setminus F$ and $X \setminus F$ is open in the Euclidean topology.

Finally, $\mathbb{1}_X$ is not continuous because $\mathbb{1}_X^{-1}([1, 2)) = [1, 2)$. If $[1, 2)$ is open in τ, then 1 would be an interior point so there is $B \in \beta$ such that $1 \in B \subset [1, 2]$. But necessarily, B is of type $[1, a) \cup (b, \infty)$, a contradiction. $\square$

Exercise 6.15 Find a homeomorphism between the cylinder $\mathbb{S}^1 \times \mathbb{R}$ and the ruled hyperboloid

$$H = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = 1\}.$$

Solution We sketch the homeomorphism. The deformation maps each horizontal circle C of the cylinder $\mathbb{S}^1 \times \mathbb{R}$ to the corresponding circle in the ruled hyperboloid with equal height as appears in Fig. 6.16. To accomplish this, we make a convenient dilation on the plane containing each circle C of $\mathbb{S}^1 \times \mathbb{R}$. For a point $(x, y, z) \in C$, the corresponding point in the circle of H writes as $(\lambda x, \lambda y, z)$, with $\lambda > 0$. The value of λ is derived imposing that $(\lambda x, \lambda y, z)$ belongs to H, or equivalently, $\lambda^2 x^2 + \lambda^2 y^2 - z^2 = 1$. Since $x^2 + y^2 = 1$, we get $\lambda = \sqrt{1 + z^2}$.

The homeomorphism is defined by

$$f : \mathbb{S}^1 \times \mathbb{R} \to H, \quad f(x, y, z) = \left(x\sqrt{1 + z^2}, y\sqrt{1 + z^2}, z\right).$$

For the inverse we can do an analogous argument by contracting each circle of H to the corresponding one in the cylinder at equal height. So, if $(x, y, z) \in H$, the

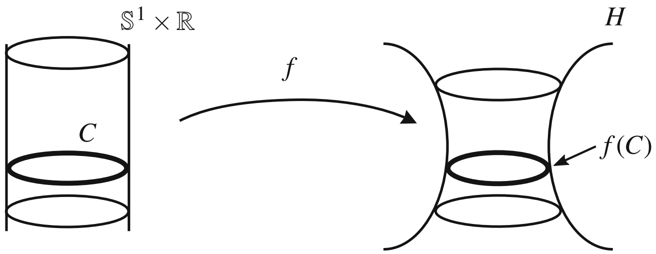


Fig. 6.16 The sets $\mathbb{S}^1 \times \mathbb{R}$ and H of Exercise 6.15

corresponding point in $\mathbb{S}^1 \times \mathbb{R}$ has equal height z and we dilate (x, y) to be of modulus 1, so $(x, y)/\sqrt{x^2 + y^2}$. Thus

$$f^{-1}\colon H \to \mathbb{S}^1 \times \mathbb{R}, \quad f^{-1}(x, y, z) = \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, z \right).$$

The map f is continuous by composing with the projections. We include the details for f and f^{-1} is similar. Let p_i be the projections of $\mathbb{R}^3$ and $q_i = p_{i|H}$ the restrictions on H. It is trivial that

$$q_1 \circ f = h_1 \cdot \sqrt{1 + h_3^2}, \quad q_2 \circ f = h_2 \cdot \sqrt{1 + h_3^2}, \quad q_3 \circ f = h_3,$$

where $h_i = p_{i|C}$, for $1 \leq i \leq 3$ and the three functions are continuous.

This exercise can be generalized to arbitrary dimensions by proving that $H = \{x \in \mathbb{R}^{n+1} : \sum_{i=1}^{n} x_i^2 - x_{n+1}^2 = 1\}$ is homeomorphic to $\mathbb{S}^{n-1} \times \mathbb{R}$. □

We establish under what circumstances two spaces are homeomorphic when each one is defined by pieces and we have homeomorphisms between the pieces.

Exercise 6.16 Let X and Y be two spaces given by partitions of two open sets $X = A \cup B$ and $Y = A' \cup B'$. If $A \cong A'$ and $B \cong B'$, prove that $X \cong Y$. Apply this result to prove that $X = A \cup B$ is homeomorphic to $Y = A' \cup B'$, where

$$A = \{(x, y) \in \mathbb{R}^2 : y = x^2\}, \quad B = \{(x, y) \in \mathbb{R}^2 : y = x^2 + 1\},$$

$$A' = \mathbb{R} \times \{0\}, \quad B' = \mathbb{R} \times \{1\}.$$

Solution Let $\phi\colon A \to A'$ and $\varphi\colon B \to B'$ be the homeomorphisms given in the hypothesis and define

$$F\colon X \to Y, \quad F(x) = \begin{cases} \phi(x), & x \in A \\ \varphi(x), & x \in B. \end{cases}$$

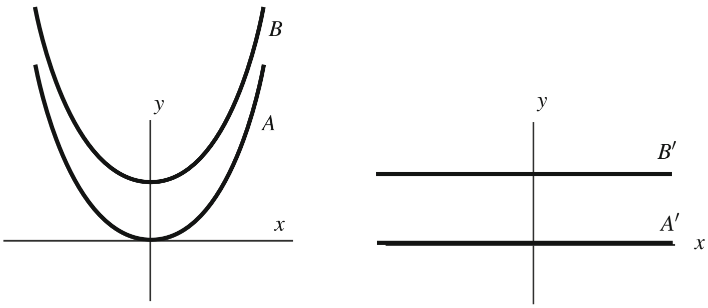


Fig. 6.17 The sets X and Y of Exercise 6.16

This map is well defined because $A \cap B = \emptyset$, F is bijective and its inverse is

$$F^{-1}\colon Y \to X, \quad F^{-1}(y) = \begin{cases} \phi^{-1}(y), & y \in A' \\ \varphi^{-1}(y), & y \in B'. \end{cases}$$

Continuity of F and F^{-1} is assured by the Pasting Lemma. The result is equally valid if the partitions are formed by closed sets.

The set X is formed by two disjoint parabolas and Y by two parallel horizontal lines of $\mathbb{R}^2$ as are depicted in Fig. 6.17. Each one of the parabolas is homeomorphic to $\mathbb{R}$ because it is the graph of a continuous function defined on $\mathbb{R}$. Analogously, each straight line is also homeomorphic to $\mathbb{R}$. Thus the homeomorphism maps each parabola into each one of the two lines of Y. It remains to show that A and B are open (or closed) in X and A' and B' in Y. In fact, the four sets are closed in $\mathbb{R}^2$, so are closed in the respective relative topologies. For A, we have $A = w^{-1}(\{0\})$, where $w\colon \mathbb{R}^2 \to \mathbb{R}$ is $w(x, y) = y - x^2$. In the same fashion, B is closed. For A' and B', we know that $A' = p_2^{-1}(\{0\})$ and $B' = p_2^{-1}(\{1\})$, where $p_2\colon \mathbb{R}^2 \to \mathbb{R}$ is the second projection. Another argument is that the four sets are graphs of continuous functions defined on $\mathbb{R}$ and it was proved that graphs are closed in $\mathbb{R}^2$. Here A, B, A' and B' are the graphs of the functions x^2, $x^2 + 1$, 0 and 1, respectively. Also A' and B' are closed because they are Cartesian products of closed sets.

We conclude with some observations.

1. The difference to prove that A is closed instead of open is that A is closed in $\mathbb{R}^2$. However, A is not open in $\mathbb{R}^2$. Regardless, the proof that A is open in X is not difficult if we consider the continuous map $q(x) = x^2 + 1/2$. The sets $O_1 = \{(x, y) : y < q(x)\}$ and $O_2 = \{(x, y) : y > q(x)\}$ are open in $\mathbb{R}^2$ and $A = O_1 \cap X$ and $B = O_2 \cap X$, proving that A and B are open in X. Analogously the sets $O_1' = \{(x, y) : y < 1/2\}$ and $O_2' = \{(x, y) : y > 1/2\}$ are open in $\mathbb{R}^2$ and $A' = O_1' \cap Y$ and $B' = O_2' \cap Y$, proving that A' and B' are open in Y.
2. In this exercise we have employed the Pasting Lemma. It is clear that the result can be generalized replacing (in the domain and in the codomain) two open sets by an arbitrary number of open sets or by a finite number of closed sets.

3. The result fails if the sets are not disjoint. Right now the proof does not work because it may occur that the map F is not well defined on $A \cap B$. For instance, let $X = [0, 2) \cup (1, 3]$ and $Y = (0, 2] \cup (1, 3]$. Here we have four open-closed intervals, so all are homeomorphic. However, $X = [0, 3]$ and $Y = (0, 3]$ are not homeomorphic. □

We prove that the sets of Fig. 6.2, left, are homeomorphic.

Exercise 6.17 Prove that there is a homeomorphisms between X and Y, where

$$X = \mathbb{S}^1 \cup \{(x, 0) \in \mathbb{R}^2 : -1 \leq x \leq 1\},$$

$$Y = \mathbb{S}^1 \cup \{(x, y) \in \mathbb{R}^2 : -1 \leq x \leq 1, y = 2\sqrt{1 - x^2}\}$$

and such that $\{(x, 0) \in \mathbb{R}^2 : -1 \leq x \leq 1\}$ goes to $\cup\{(x, y) \in \mathbb{R}^2 : -1 \leq x \leq 1, y = 2\sqrt{1 - x^2}\}$.

Solution Denote

$$A = \{(x, 0) \in \mathbb{R}^2 : -1 \leq x \leq 1\}, \quad B = \{(x, y) \in \mathbb{R}^2 : -1 \leq x \leq 1, y = 2\sqrt{1 - x^2}\}.$$

The set A is homeomorphic to B since they are graphs of two continuous functions defined on the domain $[-1, 1]$, $f(x) = 0$ and $g(x) = 2\sqrt{1 - x^2}$, respectively. Thus the homeomorphism is simply the composition of $(x, 0) \mapsto x$ and $x \mapsto (x, 2\sqrt{1 - x^2})$:

$$h : A \to B, \quad h(x, 0) = (x, 2\sqrt{1 - x^2}).$$

We cannot directly utilize the above exercise because $\mathbb{S}^1$ and A are not a partition of X (nor $\mathbb{S}^1$ and B of Y). However, the homeomorphism that we are looking for follows the same idea. Observe that $\mathbb{S}^1 \cap A = \mathbb{S}^1 \cap B = \{(-1, 0), (1, 0)\}$ and that $\mathbb{S}^1$ is closed in X and Y because they are closed in $\mathbb{R}^2$. We see that A and B are closed in X and Y respectively, because they are closed in $\mathbb{R}^2$. Indeed, $A = [-1, 1] \times \{0\}$ is the product of two closed sets of $\mathbb{R}$ (Corollary 4.4.5), so closed in $\mathbb{R}^2$. The set B is $q^{-1}(\{0\})$, where $q : [-1, 1] \times \mathbb{R} \to \mathbb{R}$ is defined by $q(x, y) = y - 2\sqrt{1 - x^2}$. Then B is closed in $[-1, 1] \times \mathbb{R}$, but the latter is closed in $\mathbb{R}^2$ because it is the product of two closed sets of $\mathbb{R}$, so closed in $\mathbb{R}^2$.

Finally, the homeomorphism that we are looking for is

$$\phi : X \to Y, \quad \phi(p) = \begin{cases} p, & p \in \mathbb{S}^1 \\ h(p), & p \in A. \end{cases}$$

It is clear that the inverse is

$$\phi^{-1} : Y \to X, \quad \phi(p) = \begin{cases} p, & p \in \mathbb{S}^1 \\ h^{-1}(p), & p \in B. \end{cases}$$

The maps ϕ and ϕ^{-1} are well defined at $\mathbb{S}^1 \cap A$ and $\mathbb{S}^1 \cap B$ respectively, because $h(-1, 0) = (-1, 0)$ and $h(1, 0) = (1, 0)$. On the other hand, ϕ and ϕ^{-1} are continuous by the Pasting Lemma. □

It has been emphasized that a homeomorphism between subsets of Euclidean space is not necessarily the restriction of a homeomorphism between Euclidean spaces. In the next example both subsets belong to the Euclidean real line $\mathbb{R}$.

Exercise 6.18 Prove that $X = \mathbb{R} \setminus \{0\}$ is homeomorphic to $Y = \mathbb{R} \setminus [-1, 1]$ but there is no homeomorphism $\phi \colon \mathbb{R} \to \mathbb{R}$ such that $\phi(X) = Y$.

Solution The set X is $(-\infty, 0) \cup (0, \infty)$ and Y is $(-\infty, -1) \cup (1, \infty)$. We apply the foregoing Exercise 6.16. The four intervals are open intervals, so open sets in X and Y. On the other hand, open intervals are all homeomorphic; in particular, $(-\infty, 0) \cong (-\infty, -1)$ and $(0, \infty) \cong (1, \infty)$. It follows that $X \cong Y$.

Assume that there is a homeomorphism $\phi \colon \mathbb{R} \to \mathbb{R}$ such that $\phi(X) = Y$. Restricting to $\mathbb{R} \setminus X$, we deduce that $\mathbb{R} \setminus X \cong \mathbb{R} \setminus \phi(X) = \mathbb{R} \setminus Y$. However, $\mathbb{R} \setminus X = \{0\}$ and $\mathbb{R} \setminus Y = [-1, 1]$, which is not possible because they have distinct cardinalities. □

Exercise 6.19 Prove that each pair of the next spaces are homeomorphic.

1. $C = \{(x, y) \in \mathbb{R}^2 : r^2 < x^2 + y^2 < R^2\}$ and $H = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = 1\}$.
2. $\mathbb{S}^2 \setminus \{(0, 0, 1), (0, 0, -1)\}$ and $\mathbb{S}^1 \times \mathbb{R}$.
3. $\mathbb{S}^1 \times \mathbb{R}$ and $A = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2, z > 0\}$.

Solution

1. The set C is the circular annulus $A(r, R)$, that is homeomorphic to the cylinder $\mathbb{S}^1 \times \mathbb{R}$. In Exercise 6.15, it was proved that the ruled hyperboloid H is homeomorphic to $\mathbb{S}^1 \times \mathbb{R}$.
2. The punctured sphere $\mathbb{S}^2 \setminus \{N\}$ is homeomorphic to $\mathbb{R}^2$ thanks to the stereographic projection. Thus $\mathbb{S}^2$ with two removed points is homeomorphic to the punctured plane which is a circular annulus, hence homeomorphic to the cylinder $\mathbb{S}^1 \times \mathbb{R}$.
3. The set A is the graph of the function
$$f \colon \mathbb{R}^2 \setminus \{(0, 0)\} \to \mathbb{R}, \quad f(x, y) = \sqrt{x^2 + y^2}.$$
Since f is continuous, A is homeomorphic to $\mathbb{R}^2 \setminus \{(0, 0)\}$. Finally, we know that the annulus $\mathbb{R}^2 \setminus \{(0, 0)\}$ is homeomorphic to the cylinder $\mathbb{S}^1 \times \mathbb{R}$. □

The following result is expected and tells us that if we prescribe two finite sets of $\mathbb{R}$ with equal number of elements, there is a homeomorphism from $\mathbb{R}$ to itself that maps one of these sets to the other.

Exercise 6.20 Let $A = \{a_1, \ldots, a_n\}$ and $B = \{b_1, \ldots b_n\}$ be two finite sets of $\mathbb{R}$. Find a homeomorphism $f : \mathbb{R} \to \mathbb{R}$ such that $f(a_i) = b_i$ for all $i = 1, \ldots, n$.

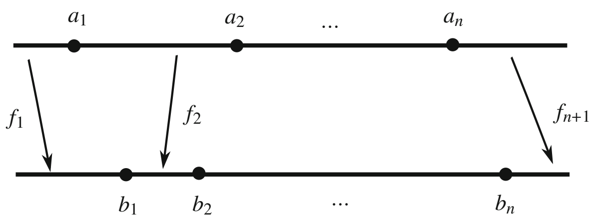


Fig. 6.18 The homeomorphism of Exercise 6.20

Solution The idea of the homeomorphism is clear (Fig. 6.18). The sets A and B separate $\mathbb{R}$ in $n+1$ closed intervals, which two are not bounded and correspond with the end intervals and the remaining intervals are bounded. Since all unbounded closed intervals are homeomorphic and any two bounded closed intervals are homeomorphic too, the homeomorphism will map each interval that is divided $\mathbb{R}$ by A in the corresponding interval (in the same order) determined by B in $\mathbb{R}$.

For simplicity, we assume that the numbers of A and B are ordered, $a_1 < a_2 < \ldots < a_n$ and $b_1 < b_2 < \ldots < b_n$. Let $f_1 \colon (-\infty, a_1] \to (-\infty, b_1]$ be a homeomorphism such that $f_1(a_1) = b_1$. For $i = 2, \ldots, n$, let $f_i \colon [a_{i-1}, a_i] \to [b_{i-1}, b_i]$ be a homeomorphism such that $f(a_i) = b_i$. Finally, let $f_{n+1} \colon [a_n, \infty) \to [b_n, \infty)$ be a homeomorphism with $f_{n+1}(a_n) = b_n$. Define

$$F \colon \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} f_1(x), & x \in (-\infty, a_1] \\ f_i(x), & x \in [a_{i-1}, a_i],\ 2 \le i \le n \\ f_{n+1}(x), & x \in [a_n, \infty). \end{cases}$$

The map f is well defined because $f_i(a_i) = f_{i+1}(a_{i-1}) = b_i$ for $1 \le i \le n$. Clearly F is a bijection. Each one of the subintervals of the domain is closed and the number of such intervals is finite. Further, F is a homeomorphism in each one of them, so F is a homeomorphism: see observation (2) of Exercise 6.16.

Notice that the extension of this exercise for infinite sets fails in general, even assuming that A and B are countable. For example, the same construction fails to map $\mathbb{N}$ into $\mathbb{Z}$. Even more, there is no homeomorphism $f \colon \mathbb{R} \to \mathbb{R}$ such that $f(\mathbb{N}) = \mathbb{Z}$ because any homeomorphism from $\mathbb{R}$ to itself must be increasing or decreasing (this will be proved in Chap. 8). Then $f(1)$ is necessarily the minimum or the maximum of $f(\mathbb{N}) = \mathbb{Z}$, a contradiction. Also, there is no homeomorphism $f \colon \mathbb{R} \to \mathbb{R}$ such that $f(\mathbb{Z}) = \mathbb{Q}$. If, for instance, f is increasing, then $f(0) < f(1)$ and thus there are no rationals between $f(0)$ and $f(1)$, a contradiction. □

We extend the foregoing exercise to $\mathbb{R}^2$.

Exercise 6.21 Let $A = \{p_1, \ldots, p_n\}$ and $B = \{q_1, \ldots q_n\}$ be two finite sets of $\mathbb{R}^2$. Find a homeomorphism from $\mathbb{R}^2$ to $\mathbb{R}^2$ mapping A into B. In consequence, $\mathbb{R}^2 \setminus A$ is homeomorphic to $\mathbb{R}^2 \setminus B$.

Solution It suffices to prescribe B to be included in the x-axis. Let $B = \{(1, 0), \ldots, (n, 0)\}$. Let $p_i = (x_i, y_i)$, $1 \le i \le n$. We distinguish two cases.

1. As a first step, assume that all x_i are pairwise distinct. Let $f\colon \mathbb{R} \to \mathbb{R}$ be any continuous function such that $f(x_i) = y_i$, $1 \le i \le n$. Then

$$F\colon \mathbb{R}^2 \to \mathbb{R}^2, \quad F(x, y) = (x, y - f(x))$$

is a homeomorphism. This map coincides with Θ which appeared in the remarks after Theorem 6.1.6. Moreover, $F(p_i) = (x_i, 0)$, $1 \le i \le n$. On the other hand, by Exercise 6.20, there is a homeomorphism $g\colon \mathbb{R} \to \mathbb{R}$ such that $g(x_k) = k$ for $k = 1, \ldots, n$. The map

$$G\colon \mathbb{R}^2 \to \mathbb{R}^2, \quad G(x, y) = (g(x), y).$$

is a homeomorphism and its inverse is $G^{-1}(x, y) = (g^{-1}(x), y)$. Moreover, $G(x_k, 0) = (k, 0)$, for $k = 1, \ldots, n$. Finally, the homeomorphism that we are looking for is the composition $G \circ F\colon \mathbb{R}^2 \to \mathbb{R}^2$ and clearly $G \circ F$ maps A to B.
2. The proof is complete once we prove that it is possible to find a homeomorphism from $\mathbb{R}^2$ into $\mathbb{R}^2$ mapping all points p_k to others where the x-coordinates are all distinct. Let $R_\theta\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the rotation of angle θ defined by

$$R_\theta(x, y) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta).$$

We claim that there is $\theta \in [0, 2\pi)$ such that the x-coordinates of all points $\{R_\theta(p_k) : 1 \le k \le n\}$ are pairwise distinct. Let us check first that the claim concludes the proof. Consider the homeomorphism R_θ that maps A in $R_\theta(A)$. The points of $R_\theta(A)$ are under the hypothesis of the preceding case, so there is a homeomorphism $\psi\colon \mathbb{R}^2 \to \mathbb{R}^2$ that maps $R_\theta(A)$ onto B. Finally, the desired homeomorphism is $\psi \circ R_\theta$.

The proof of the claim is as follows. The x-coordinates of all points $R_\theta(p_k)$ form the set

$$\{x_1\cos\theta - y_1\sin\theta, \ldots, x_n\cos\theta - y_n\sin\theta\}.$$

Suppose that for all $\theta \in [0, 2\pi)$, there is always a pair of the above numbers that are equal. Thus there are two indices $i, j \in \{1, \ldots, n\}$ and an infinite set $J \subset [0, 2\pi)$ such that

$$x_i\cos\theta - y_i\sin\theta = x_j\cos\theta - y_j\sin\theta$$

for all $\theta \in J$. In other words, each pair $(\cos\theta, \sin\theta)$, $\theta \in J$, is a solution of the linear equation $(x_i - x_j)a - (y_i - y_j)b = 0$. Since the vectors $(\cos\theta, \sin\theta)$ are not proportional as they are an infinite number of them, the only possibility is that $x_i - x_j = y_i - y_j = 0$. Thus $(x_i, y_i) = (x_j, y_j)$, which is contradictory. □

We give a method of constructing homeomorphisms between the open interval $(-1, 1)$ and $\mathbb{R}$ that generalizes the function h of Example 6.15.

Exercise 6.22 Let $f : [-1, 1] \to \mathbb{R}$ be a continuous function such that $f(1) = f(-1) = 0$, f is positive in $(-1, 1)$ and f satisfies the following property:

$$\text{if } (t, f(t)) = \lambda(s, f(s)) \text{ for some } \lambda > 0, \text{ then } t = s. \tag{6.7}$$

The condition (6.7) implies that every ray from the origin of $\mathbb{R}^2$ meets the graph of f exactly once. Let L be the horizontal line of equation $y = 1$. For each $t \in (-1, 1)$, the ray emanating from the origin $(0, 0)$ through the point $(t, f(t))$ intersects L at some point which we denote by $(h(t), 1)$. Prove that $h : (-1, 1) \to \mathbb{R}$ is a homeomorphism such that $h(0) = 0$ and $\lim_{t\to\pm1} h(t) = \pm\infty$. If, in addition, f is an even function, then h is odd.

Solution The construction of h appears in Fig. 6.19, where the function f is even.

We proceed to calculate the value of h. The ray crossing $(t, f(t))$ is $\lambda(t, f(t))$, $\lambda > 0$ and the intersection with the line $y = 1$ occurs when $\lambda f(t) = 1$, so $\lambda = 1/f(t)$. The x-coordinate of the intersection point is $\lambda t = t/f(t)$. Thus

$$h : (-1, 1) \to \mathbb{R}, \quad h(t) = \frac{t}{f(t)}.$$

This function is continuous because it is the quotient of continuous functions. Notice that $\lim_{t\to\pm1} h(t) = \pm\infty$. We see that h is bijective and h^{-1} is continuous. For injectivity, if $h(t) = h(s)$ then $t/f(t) = s/f(s)$, which implies $tf(s) = sf(t)$, or equivalently,

$$(s, f(s)) = \left(\frac{tf(s)}{f(t)}, f(s)\right) = \frac{f(s)}{f(t)}(t, f(t)).$$

Since $f(s)/f(t) > 0$, the hypothesis (6.7) yields $s = t$.

We prove that h is surjective. Let $y \in \mathbb{R}$. If $y = 0$, it is clear that $h(0) = y$. Suppose that $y > 0$ (a parallel argument if $y < 0$). The ray crossing the point $(y, 1)$

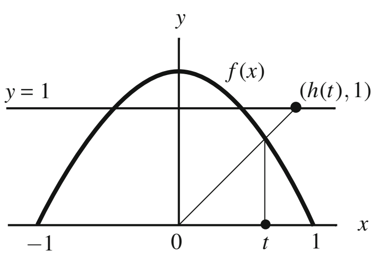


Fig. 6.19 The homeomorphism h between $(-1, 1)$ and $\mathbb{R}$ of Exercise 6.22

is $\lambda(y, 1)$. Thus we find λ such that $\lambda(y, 1) \in G(f)$ so $\lambda = f(\lambda y)$. In such a case, the preimage of y by h is the point λy. Consider

$$g\colon [0, \tfrac{1}{y}) \to \mathbb{R}, \quad g(t) = t - f(ty).$$

This function is continuous and $g(0) = -f(0) < 0$ and $\lim_{t\to 1/y} g(t) = 1/y > 0$. By the intermediate value theorem, there is λ such that $g(\lambda) = 0$ and thus $\lambda = f(\lambda y)$, as we want to prove.

Finally, we prove that h^{-1} is continuous. Note that we do not have an explicit formula for h^{-1}. Let $(y_n) \to y$ and we show that $(h^{-1}(y_n)) \to h^{-1}(y)$. Set $t_n = h^{-1}(y_n)$ and $t = h^{-1}(y)$. By definition of h,

$$\frac{t_n}{f(t_n)} \to \frac{t}{f(t)}.$$

Suppose to the contrary that (t_n) does not converge to t. Then there is an $\epsilon_0 > 0$ and a subsequence $(t_{\sigma(n)})$ of (t_n) such that $|t_{\sigma(n)} - t| \geq \epsilon_0$. Since $(t_{\sigma(n)})$ is bounded because $|t_{\sigma(n)}| \leq 1$, a result of calculus states the existence of a subsequence convergent to some $\bar{t} \in [-1, 1]$. Denote this subsequence as $(t_{\sigma(n)})$ again. Since $y_{\sigma(n)} = h(t_{\sigma(n)}) \to y = h(t)$, the sequence $(t_{\sigma(n)})$ does not converge to ± 1, otherwise, $y_n \to \infty$. Thus $\bar{t} \neq \pm 1$ and $h(\bar{t}) = h(t)$. Since h is injective, $\bar{t} = t$ but this is a contradiction because $|t_{\sigma(n)} - t| \geq \epsilon_0$.

If f is even then $h(-t) = -t/f(-t) = -t/f(t) = -h(t)$, so h is odd.

We conclude by giving some explicit choices for f, then computing directly the inverse functions of h. For these f, we leave it as exercise for the reader to check (6.7).

1. $f(x) = 1 - |x|$. Then $h(t) = t/(1 - |t|)$ and $h^{-1}(t) = t/(1 + |t|)$. This is the homeomorphism of Example 6.15.
2. $f(x) = 1 - x^2$. Then $h(t) = t/(1 - t^2)$ and $h^{-1}(t) = (-1 + \sqrt{1 + 4t^2})/(2t)$.
3. $f(x) = \sqrt{1 - x^2}$. Then $h(t) = t/\sqrt{1 - t^2}$ and $h^{-1}(t) = t/\sqrt{1 + t^2}$. □

Exercise 6.23 Prove that segments of $\mathbb{R}^n$ are homeomorphic to $[0, 1]$.

Solution Let $x, y \in \mathbb{R}^n$, $x \neq y$. We see that the restriction of

$$\alpha\colon [0, 1] \to \mathbb{R}^n, \quad \alpha(t) = x + t(y - x)$$

to its image is a homeomorphism. It is clear that α is continuous and we find an explicit expression of $\alpha^{-1}\colon \alpha([x, y]) \to [x, y]$. If $z = x + t(y - x)$, then $\alpha^{-1}(z) = t$. Using the Euclidean product $\langle , \rangle$ of $\mathbb{R}^n$, we have

$$\langle z - x, y - x\rangle = t\langle y - x, y - x\rangle = t|y - x|^2.$$

From here, we get t which yields the expression of α^{-1},

$$\alpha^{-1}(z) = \frac{\langle z - x, y - x\rangle}{|y - x|^2}.$$

Then α^{-1} is continuous for if p_1 and p_2 are the projections of $\mathbb{R}^2$ and letting $x = (x_1, x_2)$, $y = (y_1, y_2)$, then

$$\alpha^{-1} = \frac{1}{|y - x|^2}\left((y_1 - x_1)(p_1 - x_1) + (y_2 - x_2)(p_2 - x_2)\right).$$

Another way to prove that $\alpha\colon [x, y] \to \alpha([x, y])$ is a homeomorphism goes back to Example 6.17 where was proven that every straight line of $\mathbb{R}^n$ is homeomorphic to $\mathbb{R}$. Precisely in the last part of the proof of that theorem, we took $e_1 = (1, 0, \ldots, 0)$ and the map $g\colon \mathbb{R} \to \mathbb{R}^n$ given by $g(t) = (t, 0, \ldots, 0)$ is a homeomorphism between $\mathbb{R}$ and the straight line $L(\{e_1\})$. Note that $g([0, 1]) = [0, 1] \times \{0\} \subset \mathbb{R} \times \mathbb{R}^{n-1}$. For the case of the segment $[x, y]$ considered here, let the affinity

$$\phi\colon \mathbb{R}^n \to \mathbb{R}^n, \quad \phi(p) = \Phi(p) + x,$$

where the associated isomorphism Φ satisfies $\Phi(e_1) = y - x$. Then

$$\phi(te_1) = t\Phi(e_1) + x = t(y - x) + x.$$

Thus $\phi(\{te_1 : t \in [0, 1]\}) = [x, y]$. It follows that α is a homeomorphism because α is composition of two homeomorphisms, $\alpha = \phi_{|g([0,1])} \circ g_{|[0,1]}$. □

Exercise 6.24 Let $(x_0, y_0) \in \mathbb{R}^2$ and $a, b > 0$. Prove that the ellipse

$$E = \left\{(x, y) \in \mathbb{R}^2 : \frac{(x - x_0)^2}{a^2} + \frac{(y - y_0)^2}{b^2} = 1\right\}$$

is homeomorphic to $\mathbb{S}^1$.

Solution By an affinity ϕ, we can map the circle $\mathbb{S}^1$ in the ellipse E,

$$\phi\colon \mathbb{R}^2 \to \mathbb{R}^2, \quad \phi(x, y) = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = (ax + x_0, by + y_0).$$

If $(x, y) \in \mathbb{S}^1$, then clearly $\phi(x, y) \in E$, proving that $\phi(\mathbb{S}^1) \subset E$. Working with the inverse of ϕ, whose expression is $\phi^{-1}(x, y) = ((x - x_0)/a, (y - y_0)/b)$, we obtain $\phi^{-1}(E) \subset \mathbb{S}^1$. Therefore, $\phi_{|\mathbb{S}^1}\colon \mathbb{S}^1 \to E$ is a homeomorphism.

The result extends trivially to arbitrary dimensions proving that

$$\mathbb{S}^n \cong \left\{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} : \sum_{i=1}^{n+1} \frac{(x_i - c_i)^2}{a_i^2} = 1\right\},$$

where $a_i > 0$, $a_i, c_i \in \mathbb{R}$. □

We prove that $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^n$ for $n > 1$ using the intermediate value theorem. The result is a particular case of Theorem 6.1.6.

Exercise 6.25 Prove that $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^n$ for $n > 1$.

Solution The proof proceeds by contradiction. Assume that $\phi\colon \mathbb{R}^n \to \mathbb{R}$ is a homeomorphism and let $p \in \mathbb{R}^n$ be the point such that $\phi(p) = 0$. Then $\mathbb{R}^n \setminus \{p\}$ is homeomorphic to $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$. We see that this is not possible.

Let $a, b \in \mathbb{R}^n$ such that $\phi(a) = 1$ and $\phi(b) = -1$ and consider the segment $[a, b] \subset \mathbb{R}^n$. We distinguish two alternatives.

1. Case $p \notin [a, b]$ (Fig. 6.20, left). Then $[a, b] \subset \mathbb{R}^n \setminus \{p\}$. Define

   $$f\colon [0, 1] \to \mathbb{R}, \quad f(t) = \phi(a + t(b - a)).$$

   This function is continuous and $f(0) = \phi(a) = 1$ and $f(1) = \phi(b) = -1$. By the intermediate value theorem, there is $t_0 \in [0, 1]$ such that $f(t_0) = 0$, that is, $\phi(a + t_0(b - a)) = 0$. Since ϕ is injective, $a + t_0(b - a) = p$, which is contradictory with that $p \notin [a, b]$.
2. Case $p \in [a, b]$ (Fig. 6.20, right). We replace the segment $[a, b]$ by union of two segments by means of a third point $c \in \mathbb{R}^n$ that is not collinear with a and b. This can done because the dimension of $\mathbb{R}^n$ is bigger than 1. We analyze this in detail. Since the points a, b and p are collinear, consider a point $c \in \mathbb{R}^n$ that does not belong to the straight line that contains a and b. Let $X = [a, c] \cup [c, b]$. Note that $p \notin X$. We can parametrize X by

   $$X = \{a + t(c - a) : t \in [0, 1]\} \cup \{c + (t - 1)(b - c) : t \in [1, 2]\}$$

   and define

   $$f\colon [0, 2] \to \mathbb{R}, \quad f(t) = \begin{cases} \phi(a + t(c - a)), & t \in [0, 1] \\ \phi(c + (t - 1)(b - c)), & t \in [1, 2]. \end{cases}$$

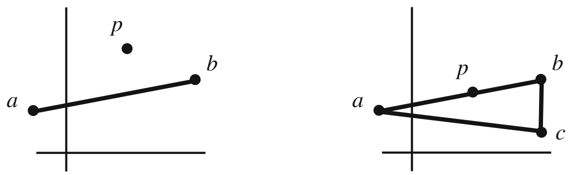


Fig. 6.20 Exercise 6.25. Cases $p \notin [a, b]$ (left) and $p \in [a, b]$ (right)

The map f is well defined at $t = 1$ and is continuous by the Pasting Lemma. The final argument is like part (1). Certainly, $f(0) = 1$, $f(2) = -1$ and we obtain a contradiction again by using the intermediate value theorem. □

The following homeomorphism illustrates a situation where the intuition is clear but the explicit expression of the map is not so easy.

Exercise 6.26 Prove that the unit *disc* $\mathbb{D}^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ is homeomorphic to the unit square $K = \{(x, y) \in \mathbb{R}^2 : |x| \leq 1, |y| \leq 1\}$.

Solution The homeomorphism consists of expanding the disc from the origin (Fig. 6.21). So, if $(x, y) \in \mathbb{D}^2$ is different from the origin, consider the ray $L = \{\lambda(x, y) : \lambda > 0\}$. This ray intersects the border of K at a point. Suppose that this occurs in the right vertical side $x = 1$. Then $\lambda = 1/x$ and the intersection point is $(1, y/x)$. We expand the segment $[(0, 0), (x, y)/\sqrt{x^2 + y^2})]$ into $[(0, 0), (1, y/x)]$. To accomplish this, define the dilation

$$h\colon (0, 1] \to (0, \tfrac{\sqrt{x^2+y^2}}{x}], \quad h(t) = \frac{\sqrt{x^2 + y^2}}{x}\, t.$$

Given a point of L, consider its modulus $\sqrt{x^2 + y^2}$ and pick up the point of modulus $h(\sqrt{x^2 + y^2})$ contained in the ray L. Then $F\colon \mathbb{D}^2 \to K$ is

$$F(x, y) = h(\sqrt{x^2 + y^2})\frac{(x, y)}{\sqrt{x^2 + y^2}} = \frac{\sqrt{x^2 + y^2}}{x}(x, y).$$

The argument is equally valid for the remaining the points, where the only care to take is to distinguish in what part of the border of K, the ray L intersects it. Following this motivation, the explicit formula for F is

$$F\colon \mathbb{D}^2 \to K, \quad F(x, y) = \begin{cases} \dfrac{\sqrt{x^2 + y^2}}{\max\{|x|, |y|\}}(x, y), & (x, y) \neq (0, 0) \\ (0, 0), & (x, y) = (0, 0). \end{cases}$$

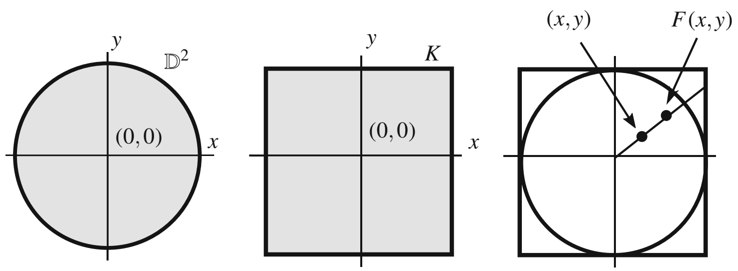


Fig. 6.21 The sets $\mathbb{D}^2$ and K of Exercise 6.26 and the homeomorphism $f\colon \mathbb{D}^2 \to K$

The argument to deduce the expression of the inverse of F is similar, obtaining

$$F^{-1}\colon K \to \mathbb{D}^2, \quad F^{-1}(x, y) = \begin{cases} \dfrac{\max\{|x|, |y|\}}{\sqrt{x^2 + y^2}}(x, y), & (x, y) \neq (0, 0) \\ (0, 0), & (x, y) = (0, 0). \end{cases}$$

In the open set $\mathbb{D}^2 \setminus \{(0, 0)\}$, the composition of F with the projections is continuous. Analogously, F^{-1} is continuous in the open $K \setminus \{(0, 0)\}$. It remains to study the continuity at $(0, 0)$. Notice that $f(0, 0) = F^{-1}(0, 0) = (0, 0)$. Let $((x_n, y_n)) \to (0, 0)$, or equivalently, $d_\infty((x_n, y_n), (0, 0)) \to 0$. For the map F, if $((x_n, y_n)) \to (0, 0)$, then

$$|F(x_n, y_n)| = \frac{x_n^2 + y_n^2}{\max\{|x_n|, |y_n|\}} \leq \frac{2(\max\{|x_n|, |y_n|\})^2}{\max\{|x_n|, |y_n|\}} = 2\max\{|x_n|, |y_n|\} \to 0,$$

$$|F^{-1}(x_n, y_n)| = \max\{|x_n|, |y_n|\} \to 0.$$

□

The next exercise is expected but difficult to formalize. It establishes that if we prescribe two points of the unit ball of $\mathbb{R}^2$, we can transform $\mathbb{D}^2$ to itself by a homeomorphism carrying a point to the other one and preserving $\mathbb{S}^1$.

Exercise 6.27 Let $\mathbb{D}^2$ be the unit disc and let $p, q \in B_1(0)$. Find a homeomorphism $f\colon \mathbb{D}^2 \to \mathbb{D}^2$ such that $f(p) = q$ and $f_{|\mathbb{S}^1} = \mathbb{1}_{\mathbb{S}^1}$.

Solution It suffices to fix one of the two points p and q. Thus, suppose $p = 0$. Any point $x \in \mathbb{D}^2$, with the exception of the origin, has a representation in polar coordinates $(|x|, x/|x|) \in (0, 1] \times \mathbb{S}^1$ (Example 6.19). Let $x \in \mathbb{D}^2$ be a point distinct from the origin and consider the ray from x to its intersection with $\mathbb{S}^1$ at $x/|x|$. The segment $[0, \frac{x}{|x|}]$ is mapped to the segment $[\frac{x}{|x|}, q]$ by means of a dilation between the modulus of their points. Here we use that $q \notin \mathbb{S}^1$ because this last segment could be reduced to one point. This dilation is

$$h\colon [0, 1] \to \left[0, \left|\tfrac{x}{|x|} - q\right|\right], \quad h(t) = \left|\frac{x}{|x|} - q\right| t.$$

Finally, we translate the point to q. Then x is mapped into the point of modulus $h(|x|)$ in the segment $[q, x/|x|]$ as indicated in Fig. 6.22. Thus define

$$f\colon \mathbb{D}^2 \to \mathbb{D}^2, \quad f(x) = \begin{cases} h(|x|)\dfrac{\frac{x}{|x|} - q}{|\frac{x}{|x|} - q|} + q = x + (1 - |x|)q, & x \neq 0 \\ q, & x = 0. \end{cases}$$

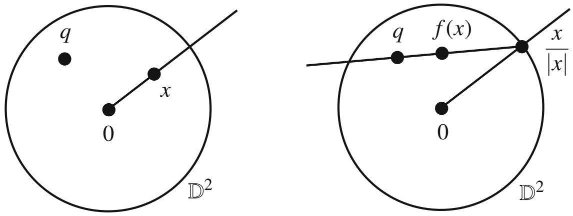


Fig. 6.22 The homeomorphism of Exercise 6.27

Notice that $|f(x)| \leq 1$ because

$$\begin{aligned}|f(x)|^2 &= |x + (1-|x|)q|^2 = |x|^2 + (1-|x|)^2|q|^2 + 2(1-|x|)\langle q, x\rangle \\ &\leq |x|^2 + |q|^2(1-|x|)^2 + 2(1-|x|)|q||x| = (|q|(1-|x|) + |x|)^2 \\ &\leq ((1-|x|) + |x|)^2 = 1,\end{aligned}$$

where have used $\langle q, x\rangle \leq |q||x|$ by the Cauchy-Schwarz inequality and because $|q| < 1$. Also it is obvious that $f(0) = q$ and that if $x \in \mathbb{S}^1$, then $f(x) = x$.

The continuity at the points different from 0 is clear. At $x = 0$, if $(x_n) \to 0$,

$$|f(x_n) - f(0)| = |f(x_n) - q| = |x_n - |x_n|q| \leq |x_n| + |x_n||q| \to 0$$

because $|x_n| \to 0$.

For the inverse map, we will obtain the expression for x in terms of y from the identity $x + (1-|x|)q = y$. Let us write $x - |x|q = y - q$. Multiplying with the Euclidean product by q and x and after some manipulations,

$$(1-|q|^2)|x|^2 - 2|x|\langle y-q, q\rangle + |y-q|^2 = 0.$$

Using that $|q| < 1$, the value of $|x|$ is

$$|x| = \frac{1}{1-|q|^2}\left(\langle y-q, q\rangle + \sqrt{\langle y-q, q\rangle^2 + |y-q|^2(1-|q|^2)}\right).$$

If $\phi(y)$ denotes the right-hand side of the above expression, then

$$f^{-1}\colon \mathbb{D}^2 \to \mathbb{D}^2, \quad f^{-1}(y) = y - (1-\phi(y))q.$$

This proves that f^{-1} is continuous because ϕ is continuous and, as a consequence, f is a homeomorphism. As a last remark, notice that this exercise proves that *given* $p, q \in B_1(0)$, *there is a homeomorphism from* $B_1(0)$ *to itself mapping* p *into* q. □

The next exercise is surprising because it asserts that every homeomorphism between two circles of $\mathbb{R}^2$ can be extended to a homeomorphism of $\mathbb{R}^2$ to itself.

Exercise 6.28 Let C_1 and C_2 two circles of $\mathbb{R}^2$ and $f: C_1 \to C_2$ a homeomorphism. Prove that there is an homeomorphism $F: \mathbb{R}^2 \to \mathbb{R}^2$ such that $F_{|C_1} = f$.

Solution Without loss of generality, we assume that $C_1 = \mathbb{S}^1$. Let $C_2 = \mathbb{S}^1_r(c)$ and let $f: \mathbb{S}^1 \to \mathbb{S}^1_r(c)$ be a homeomorphism. The idea of the homeomorphism F is the following. Let $p \in \mathbb{R}^2$ and consider the ray $\overline{Op}$ from the origin O through p. This ray intersects $\mathbb{S}^1$ at $p/|p|$. Consider $f(p/|p|) \in \mathbb{S}^1_r(c)$, which together with c defines a ray. Then $F(p)$ is the point in this ray whose modulus is of p after a dilation so $\mathbb{S}^1$ converts into $\mathbb{S}^1_r(c)$. This dilation maps the point p to the corresponding point on the ray $\{c + \lambda(f(p/|p|) - c) : \lambda \geq 0\}$. Thus

$$F: \mathbb{R}^2 \to \mathbb{R}^2, \quad F(p) = \begin{cases} c + |p|(f(\frac{p}{|p|}) - c), & p \neq O \\ c, & p = O. \end{cases}$$

If $p \in \mathbb{S}^1$, then $|p| = 1$, so $F(p) = c + (f(p) - c) = f(p)$. This yields $F_{|\mathbb{S}^1} = f$ as required. The continuity of F in the open set $\mathbb{R}^2 \setminus \{O\}$ is clear. We study the case when F is continuous at O. Since $F(O) = c$, if $(p_n) \to O$,

$$|F(p_n) - F(O)| = |p_n||f(\tfrac{p_n}{|p_n|}) - c| = r|p_n| \to 0.$$

It remains to prove that F is bijective and that F^{-1} is continuous. The construction of F^{-1} is analogous. For $p \in \mathbb{R}^2$, consider the point $c + \lambda(p - c)$ such that $|\lambda(p - c)| = r$. Then $\lambda = r/|p - c|$. Now, let $f^{-1}(c + \frac{r(p-c)}{|p-c|}) \in \mathbb{S}^1$. Then

$$F^{-1}: \mathbb{R}^2 \to \mathbb{R}^2, \quad F^{-1}(p) = \begin{cases} \frac{|p-c|}{r} f^{-1}(c + \frac{r(p-c}{|p-c|}), & p \neq c \\ O, & p = c. \end{cases}$$

For the continuity of F^{-1} at c, let $(p_n) \to c$. Since $F^{-1}(c) = O$, we have

$$|F^{-1}(p_n) - F^{-1}(c)| = \left| \frac{|p_n - c|}{r} f^{-1}(c + \frac{r(p_n - c)}{|p_n - c|}) \right| = \frac{|p_n - c|}{r} \to 0$$

as $(p_n) \to c$. □

Again, the next exercise is expected, but difficult to formalize.

Exercise 6.29 Prove that $[0, 1) \times [0, 1)$ is homeomorphic to $[0, 1] \times [0, 1)$.

Solution Note first that these sets are homeomorphic by a simple look at their pictures (Fig. 6.23). Let us observe that the second factors in both sets coincide, but the first ones are not homeomorphic.

Let $X = [0, 1) \times [0, 1)$. First we show that X is homeomorphic to the triangle T determined by the vertices $(0, 0)$, $(0, 1)$ and $(1, 0)$,

$$T = \{(x, y) \in \mathbb{R}^2 : 0 \leq x < 1, 0 \leq y < -x + 1\}.$$

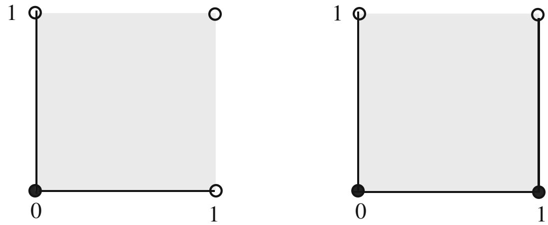


Fig. 6.23 The sets $[0, 1) \times [0, 1)$ and $[0, 1] \times [0, 1)$ of Exercise 6.29

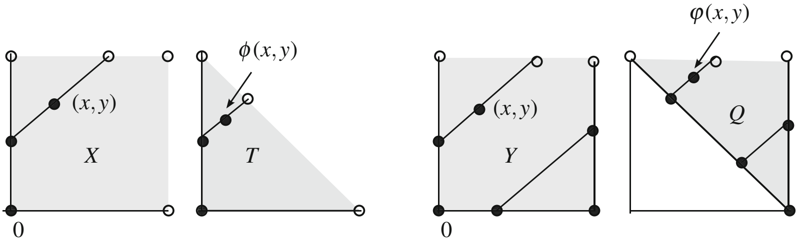


Fig. 6.24 The sets X and T are homeomorphic (left) and Y and Q are homeomorphic

For the homeomorphism between X and T, consider the segments defined by the border of X with the lines $y = x + k$. On each on these lines, the segment can be regarded as an interval $[z, t)$. Then we shrink the segment in the line to be included in T. This is done by a dilation (Fig. 6.24, left). We include the details.

Let (a, b) be a point of X and consider the line $y = x+b-a$. This line intersects the side $\{0\} \times [0, 1)$ or the side $[0, 1) \times \{0\}$. Assume the first case. The point of intersection with the y-line is $(0, b - a)$. We dilate the segment $[(0, b - a), (1 - b+a, 1)]$ in the segment $[(0, b-a), (\frac{a-b+1}{2}, \frac{1+b-a}{2})]$. Notice that the length of the second segment is the half than the first one. Thus

$$(a, b) \mapsto (0, b - a) + \frac{1}{2}((a, b) - (0, b - a)) = \left(\frac{a}{2}, b - \frac{a}{2}\right).$$

A parallel procedure can be performed with the side $[0, 1) \times \{0\}$, obtaining that the point (a, b) is mapped into $(a - \frac{b}{2}, \frac{b}{2})$. Thus

$$\phi\colon X \to T, \quad \phi(x, y) = \begin{cases} \left(\dfrac{x}{2}, \dfrac{-x + 2y}{2}\right), & x \le y \\ \left(\dfrac{2x - y}{2}, \dfrac{y}{2}\right), & x \ge y. \end{cases}$$

This map is continuous by the Pasting Lemma since $\{(x, y) \in \mathbb{R}^2 : x \le y\}$ and $\{(x, y) \in \mathbb{R}^2 : x \ge y\}$ are closed in $\mathbb{R}^2$, hence closed in X.

Take $Y = [0, 1] \times [0, 1)$. A similar argument proves that Y is homeomorphic to the triangle Q determined by the points $(1, 0)$, $(0, 1)$ and $(1, 1)$,

$$Q = \{(x, y) \in \mathbb{R}^2 : y \geq -x + 1, 0 < x \leq 1, y < 1\}.$$

See Fig. 6.24, right. Here the main point is that the lines $y = x + k$ for $k \geq 0$ intersect Y on closed-open segments and this holds on Q as well. For the line with $k < 0$, the intersections on Y and on Q are closed segments. The map is

$$\varphi\colon Y \to Q, \quad \phi(x, y) = \begin{cases} \left(\dfrac{1 + 2x - y}{2}, \dfrac{1 + y}{2}\right), & x \leq y \\ \left(\dfrac{1 + x}{2}, \dfrac{1 - x + 2y}{2}\right), & x \geq y. \end{cases}$$

The next step is transforming Q in the triangle P determined by the points $(1, 0)$, $(0, 1)$ and $(2, 1)$,

$$P = \{(x, y) \in \mathbb{R}^2 : y < 1, y \geq -x + 1, y \geq x - 1\}.$$

The plan is considering the horizontal lines $y = k$ that intersect Q and dilating until they are included in P. These segments in Q are closed and similarly its image on P is closed. Let $(a, b) \in Q$ and the line $y = b$. The intersection with the borders of Q are the points $(1 - b, b)$ and $(1, b)$. The segment $[(1 - b, b), (1, b)]$ is multiplied by 2 in the x-direction (Fig. 6.25). The point (a, b) is mapped into

$$(1 - b, b) + 2\,((a, b) - (1 - b, b)) = (2a + b - 1, b),$$

and the homeomorphism is

$$\psi\colon Q \to P, \quad \psi(x, y) = (2x + y - 1, y).$$

The final step in this exercise is to prove that P is homeomorphic to T. But now this is clear because both spaces are the regions determined by two consecutive

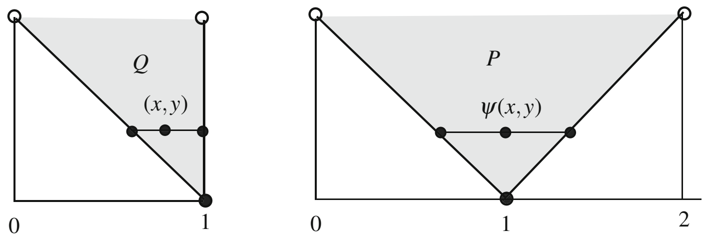


Fig. 6.25 The sets Q and P are homeomorphic

sides of a square (with different length, $\sqrt{2}$ for P and 1 for T) and the diagonal that joins the endpoints of the two sides. The reader is invited to find explicitly this homeomorphism since it is obvious that after applying a dilation of P, we transform P so the sides are of length 1 and after a rotation, the image of P coincides with T. □

It is a rather startling exercise that may clash with one's intuition to exhibit a space where an open subset is homeomorphic to another subset which is neither open nor closed.

Exercise 6.30 Let $X = [-1, 1] \cup \{1 + 1/n : n \in \mathbb{N}\}$ and consider the subsets of X given by $A = [-1, 0)$ and $B = (0, 1]$. Prove that A and B are homeomorphic, A is open in X but B is neither open nor closed in X.

Solution It is clear that A and B are homeomorphic by Corollary 6.1.8. On the other hand A is open in X because $A = (-\infty, 0) \cap X$.

We see that B is not open in X. If B were open in X then $x = 1$ would be an interior point. However, $(1 + \frac{1}{n}) \to 1$ and the elements of the sequence do not belong to B.

We show that B is not closed in X. In this case, we see that 0, which does not belong to B, is an adherent point of B. This is clear because the sequence $(\frac{1}{n})$, which is contained in B, converges to 0. In fact, the closure of B in X is $\overline{B}^X = \overline{B} \cap X = [0, 1] \cap X = [0, 1]$, where $\overline{B}$ is the closure of B as subspace of $\mathbb{R}$. □

6.6 Suggested Exercises

6.6.1 Let $f : X \to Y$ be a bijection. Show that f is a homeomorphism if and only if one of the following conditions holds:

1. $\text{int}(f(A)) = f(\text{int}(A))$ for all $A \subset X$.
2. $\text{ext}(f(A)) = f(\text{ext}(A))$ for all $A \subset X$.
3. $\text{Bd}(f(A)) = f(\text{Bd}(A))$ for all $A \subset X$.

6.6.2 Let $X = \{a, b, c\}$ and $\tau_1 = \{\emptyset, X, \{a\}, \{a, b\}\}$, $\tau_2 = \{\emptyset, X, \{b\}, \{b, c\}\}$ and $\tau_3 = \{\emptyset, X, \{a\}, \{b, c\}\}$. Show that (X, τ_1) is homeomorphic to (X, τ_2) but not to (X, τ_3). How many non-homeomorphic topologies do exist on X?

6.6.3 Prove that metrizability is a topological invariant.

6.6.4 Let (Y, τ') be a space homeomorphic to a trivial space (X, τ_T). Show that τ' is the trivial topology on Y. Obtain the analogous result for the cofinite topology.

6.6.5 Find a homeomorphism between $\mathbb{R}^2$ and $\{(x, y) \in \mathbb{R}^2 : y > 0\}$.

6.6.6 If $0 < r < R$, define the closed annulus as $C = \{(x, y) \in \mathbb{R}^2 : r^2 \leq x^2 + y^2 \leq R^2\}$. Prove that C is homeomorphic to $\mathbb{S}^1 \times [0, 1]$. Obtain the analogous results replacing in the definition of C $\leq$ by $<$ everywhere.

6.6.7 Construct an explicit homeomorphism between each pair of the next spaces.

1. $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1, z \geq 1\}$ and $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1, z \leq 0\}$.
2. (Cylinders with different axes) $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$ and $\{(x, y, z) \in \mathbb{R}^3 : x^2 + z^2 = 1\}$.
3. The interval $[0, 1)$ and $\{(x, y, z) \in \mathbb{R}^3 : -1 < y \leq 0, x = z = 0\}$.
4. The graphs of $\sin x$ and $\cos x$.
5. $\mathbb{N}$ and $\mathbb{Z}$.
6. (Two planes of Euclidean spaces) $\{(x, y, z) \in \mathbb{R}^3 : x - y = 0\}$ and $\{(x, y, z) \in \mathbb{R}^3 : x - y - z = 0\}$.

6.6.8 Construct an explicit homeomorphism between $\mathbb{R}$ and the following subsets of $\mathbb{R}^2$:

1. $(0, 1) \times \{0\}$.
2. The bridge $(\{0, 1\} \times (0, 1]) \cup ([0, 1] \times \{1\})$.
3. The spiral $\{e^t(\cos t, \sin t) : t \in (0, 4\pi)\}$.

6.6.9 Prove that all bounded open intervals in the Sorgenfrey line are homeomorphic. Determine if the same occurs for the other types of intervals. Is $[0, \infty)$ homeomorphic to $(-\infty, 0]$?

6.6.10 Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be two continuous functions such that $f(x) < g(x)$ for all $x \in \mathbb{R}$. Show that $\{(x, y) \in \mathbb{R}^2 : f(x) \leq y \leq g(x)\}$ is homeomorphic to $\mathbb{R} \times [0, 1]$.

6.6.11 Prove that $\mathbb{S}^1$ satisfies the property $\mathscr{M}$ of Example 6.9 but show that $\mathbb{S}^1$ is not homeomorphic to $[0, 1]$.

6.6.12 Determine if $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, is open and closed. Consider the problem replacing the usual topology by the Sorgenfrey topology.

6.6.13 A space X is said to have the *fixed-point property* if every continuous map $f\colon X \to X$ has a fixed point, a point $x_0 \in X$ such that $f(x_0) = x_0$. Prove that the fixed-point property is a topological invariant.

6.6.14 Let $f\colon \mathbb{R} \to \mathbb{S}^1$, $f(t) = (\cos t, \sin t)$ Prove that if we restrict to $[0, 2\pi]$, then f is closed but it is not open. If we restrict to $[0, 2\pi)$, prove that f is bijective, continuous but not a homeomorphism.

6.6.15 Determine if $f\colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 1/(1 + x^2)$ is open and closed. Do the same with the function $f(x) = \sin x$.

6.6.16 If $A \subset X$, show that the inclusion $i\colon A \hookrightarrow X$ is open (resp. closed) if and only if A is an open (resp. closed) set in X.

6.6.17 Consider on $\mathbb{R}$ the topology $\tau(\beta_{sr})$ of Example 2.12. Let

$$f\colon ((0, \infty), \tau(\beta_{sr})_{|(0,\infty)}) \to (\mathbb{R}, \tau(\beta_{sr})), \qquad f(x) = \operatorname{arc\,tan}(x).$$

Show that f is an embedding but $f((0, \infty))$ is not open.

6.6.18 For each basis $B = \{e_1, \ldots, e_n\}$ for $\mathbb{R}^n$, define the linear isomorphism

$$f_B\colon \mathbb{R}^n \to \mathbb{R}^n, \quad f_B(x_1, \ldots, x_n) = x_1 e_1 + \ldots + x_n e_n.$$

Prove that the direct image topology $\tau_u(f_B)$ coincides with τ_u.

6.6.19 In $\mathbb{R}^3$, let $\{L_1, L_2\}$ and $\{R_1, R_2\}$ be two pairs of disjoint straight lines. Prove that $L_1 \cup L_2 \cong R_1 \cup R_2$.

6.6.20 With the notation $\mathbb{S}^1_r(a)$ for circles of $\mathbb{R}^2$, prove:

1. $\mathbb{S}^1_1(-1, 0) \cup \mathbb{S}^1_2(3, 0) \cong \mathbb{S}^1_3(-4, 0) \cup \mathbb{S}^1_4(6, 0)$.
2. $\mathbb{S}^1_1(-1, 0) \cup \mathbb{S}^1_1(1, 0) \cong \mathbb{S}^1_2(-2, 0) \cup \mathbb{S}^1_2(2, 0)$.

Chapter 7
Product Topology

The subject of this chapter is the construction of a topology on the Cartesian product of two spaces. The problem is the following:

> **Question:** *Given two topological spaces* (X_1, τ_1) *and* (X_2, τ_2)*, in what sense may we define a topology on* $X_1 \times X_2$ *that has some relation with the topologies* τ_1 *and* τ_2*?*

This topology will be called the *product topology*. Since our model will be the Euclidean plane $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ regarded as the Cartesian product of $(\mathbb{R}, \tau_u)$ with itself, a list of properties should be transferred to the product topology.

1. The projections $p_i : X_1 \times X_2 \to X_i$, $p_i(x_1, x_2) = x_i$, $i = 1, 2$, are continuous.
2. Continuity of a mapping $f : Y \to X_1 \times X_2$ is equivalent to the continuity of the coordinate maps $f_i = p_i \circ f$, $i = 1, 2$.
3. If $X_1 \cong Y_1$ and $X_2 \cong Y_2$ then $X_1 \times X_2 \cong Y_1 \times Y_2$.
4. If we identify $\mathbb{R}^n \times \mathbb{R}^m$ with $\mathbb{R}^{n+m}$ as point sets, it is desirable that the product space of $\mathbb{R}^n$ with $\mathbb{R}^m$ is the Euclidean space $\mathbb{R}^{n+m}$.

Once we have defined the product topology, there is work ahead to deduce which properties of the product space can be inherited from each one of the factors. The last part of the chapter concerns the generalization of the product space to the case where the number of factors is infinite.

7.1 The Product Topology

We recall the Euclidean topology on $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ defined in Chap. 2. A basis for this topology is formed by all the Cartesian products of bounded open intervals (Fig. 7.1). Open intervals form a basis for the Euclidean topology, so we can consider Cartesian products of bases of X_1 and X_2.

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_7

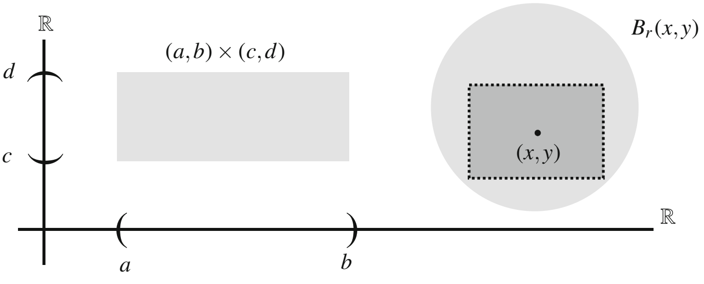


Fig. 7.1 The family of rectangles $(a, b) \times (c, d)$ constitutes a basis for the Euclidean topology on $\mathbb{R}^2$ (left). Balls of $\mathbb{R}^2$ are open sets (right)

Definition 7.1.1 Let (X_1, τ_1) and (X_2, τ_2) be two spaces. The *product topology* is defined as the topology on $X_1 \times X_2$ that has as basis for the topology

$$\tau_1 \times \tau_2 = \{O_1 \times O_2 : O_1 \in \tau_1, O_2 \in \tau_2\}.$$

Abusing slightly the notation, we still use $\tau_1 \times \tau_2$ to denote the product topology. Let observe that $\tau_1 \times \tau_2$ obeys the property (ii*) of Theorem 2.2.3 because

$$(O_1 \times O_2) \cap (O_1' \times O_2') = (O_1 \cap O_1') \times (O_2 \cap O_2') \in \tau_1 \times \tau_2.$$

Proposition 7.1.2

1. *The Cartesian product of bases is a basis for the product topology.*
2. *The Cartesian product of bases of neighborhoods is a basis of neighborhoods in the product topology. In particular, the Cartesian product of neighborhoods is a basis of neighborhoods in the product topology.*
3. *Let X_1 and X_2 be two spaces and $A_i \subset X_i$, $i = 1, 2$. Then*

   (a) $\operatorname{int}(A_1 \times A_2) = \overset{\circ}{A_1} \times \overset{\circ}{A_2}$.
   (b) $\overline{A_1 \times A_2} = \overline{A_1} \times \overline{A_2}$.
   (c) $\operatorname{Bd}(A_1 \times A_2) = (\operatorname{Bd}(A_1) \times \overline{A_2}) \cup (\overline{A_1} \times \operatorname{Bd}(A_2))$.

   As a consequence, the Cartesian product of closed sets is closed.

Example 7.1

1. Consider (X, τ_D) and (Y, τ_D') two discrete spaces. Let $x \in X$ and $y \in Y$. Due to that $\{x\}$ and $\{y\}$ are open in X and Y, respectively, $\{x\} \times \{y\} = \{(x, y)\}$ is open in the product topology. Since singletons are open in $\tau_D \times \tau_D'$, this topology is discrete. Consequently, *the topological product of two discrete spaces is the discrete topology on the Cartesian product.*
2. Similarly, if (X, τ_T) and (Y, τ_T') are two trivial spaces, then $\tau_T \times \tau_T' = \{\emptyset, X \times Y\}$. So, *the topological product of two trivial spaces is the trivial topology on the Cartesian product.* ◊

The remaining the section deals with the relationship between the product topology and the following:

1. Euclidean topology.
2. Metric spaces.
3. Relative topologies.

First we analyze the Euclidean topology in detail. The point set $\mathbb{R}^k$, with $k > 1$, can be written as different Cartesian products $\mathbb{R}^k = \mathbb{R}^n \times \mathbb{R}^m$, where n and m are natural numbers with the only requirement being that $k = n + m$. Here we are identifying an ordered pair of $\mathbb{R}^n \times \mathbb{R}^m$ with a $(n+m)$-tupla of $\mathbb{R}^{n+m}$. For example, $\mathbb{R}^4 = \mathbb{R} \times \mathbb{R}^3 = \mathbb{R}^2 \times \mathbb{R}^2$. Consider the Euclidean topology in each one of the factors and we see that the product topology is the Euclidean topology on $\mathbb{R}^k$. We let τ_u^n stand for the Euclidean topology on $\mathbb{R}^n$.

Theorem 7.1.3 *Consider the Euclidean spaces* $(\mathbb{R}^n, \tau_u^n)$ *and* $(\mathbb{R}^m, \tau_u^m)$. *Then*

$$\tau_u^n \times \tau_u^m = \tau_u^{n+m}.$$

This result can be rephrased by saying that *the product topology of Euclidean spaces is the Euclidean topology on the Cartesian product.*

Proof A basis for τ_u^n is $\beta_u^n = \{\Pi_{i=1}^n (a_i, b_i) : a_i, b_i \in \mathbb{R}, 1 \leq i \leq n\}$. Since

$$\Pi_{i=1}^n (a_i, b_i) \times \Pi_{i=1}^m (a_{n+i}, b_{n+i}) = \Pi_{i=1}^{n+m} (a_i, b_i),$$

then $\beta_u^n \times \beta_u^m = \beta_u^{n+m}$ and (1) of Proposition 7.1.2 concludes the proof. □

Now we can show an example where an open (closed) set in a product topology is not the Cartesian product of open (closed) set in the factors.

Example 7.2 Consider the Euclidean plane $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ with the product topology $\tau_u^1 \times \tau_u^1$. From the theorem, this topology is simply the Euclidean topology τ_u^2 of $\mathbb{R}^2$. The ball $B_1(0) \subset \mathbb{R}^2$ is open in $\mathbb{R}^2$ and its closure, $\overline{B_1(0)}$, is closed. However, $B_1(0)$ is neither the Cartesian product of open sets of $(\mathbb{R}, \tau_u)$, nor $\overline{B_1(0)}$ is the product of closed sets of $(\mathbb{R}, \tau_u)$. In fact, these sets are not the Cartesian product two sets of $\mathbb{R}$. For closed sets, another example is $F = \{(x, x) \in \mathbb{R}^2 : x \in \mathbb{R}\}$, where it was proved in Exercise 4.9 that it is closed in $\mathbb{R}^2$. However, F is not the product of two sets of $\mathbb{R}$. ◊

Consider the product of two metric spaces. Let (X_1, d_1) and (X_2, d_2) be two metric spaces and let $(X_1 \times X_2, \tau_{d_1} \times \tau_{d_2})$ be the product space, where τ_{d_i} are the metric topologies, $i = 1, 2$. Let us use the product distance d defined in Exercise 4.22,

$$d((x_1, x_2), (y_1, y_2)) = d_1(x_1, y_1) + d_2(x_2, y_2).$$

The next result was proved for the Euclidean distance (Exercise 4.23) and the proof is similar.

Theorem 7.1.4 *The product topology of two metric spaces is the metric topology determined by the product distance.*

We conclude with the product topology of two relative topologies. Let (X_1, τ_1) and (X_2, τ_2) be two spaces and $A_i \subset X_i$, $i = 1, 2$. On $A_1 \times A_2$ there are two topologies. First, the topological product of the relative topologies $\tau_{1|A_1} \times \tau_{2|A_2}$. The other topology is the relative topology obtained from the topological product $X_1 \times X_2$ and denoted by $(\tau_1 \times \tau_2)_{|A_1 \times A_2}$. As expected, both topologies coincide.

Theorem 7.1.5 *Following the above notation, we have $\tau_{1|A_1} \times \tau_{2|A_2} = (\tau_1 \times \tau_2)_{|A_1 \times A_2}$. As a consequence, if $f : X_1 \times X_2 \to Y$ is continuous, then*

$$f_{|A_1 \times A_2} : (A_1 \times A_2, \tau_{1|A_1} \times \tau_{2|A_2}) \to Y$$

is continuous.

7.2 Product Topology and Continuity

Following the definition of the product topology, we turn our attention to the continuity of mappings having a product space as domain or codomain. As was promised in the introduction of the chapter, we begin with the projections. Recall that the concept of projection is not topological, but stems from set theory. Specifically, if X_1 and X_2 are two point sets, the set-theoretic *projections* are defined by

$$p_i : X_1 \times X_2 \to X_i, \quad p_i(x_1, x_2) = x_i, \ i = 1, 2.$$

Theorem 7.2.1

1. *The projections $p_i : (X_1 \times X_2, \tau_1 \times \tau_2) \to (X_i, \tau_i)$, $i = 1, 2$, are continuous and open.*
2. *A mapping $f : X \to X_1 \times X_2$ is continuous if and only if $p_i \circ f$ are continuous, $i = 1, 2$.*

We construct set-theoretic maps from other maps that have a product space as their codomain. The next setting comes from set theory and for that reason, we bring out from the topological result that we will formulate later. If $f_i : X \to X_i, i = 1, 2$, are two set-theoretic maps, the *evaluation* of f_1 and f_2 is defined by (Fig. 7.2)

$$e(f_1, f_2) : X \to X_1 \times X_2, \quad e(f_1, f_2)(x) = (f_1(x), f_2(x)).$$

$$\begin{tikzcd}
& X_1 & \\
X \arrow[ur, "f_1"] \arrow[rr, "e(f_1,f_2)"] \arrow[dr, "f_2"'] & & X_1 \times X_2 \arrow[ul, "p_1"'] \arrow[dl, "p_2"] \\
& X_2 &
\end{tikzcd}$$

Fig. 7.2 The evaluation map $e(f_1, f_2)$ of f_1 and f_2

Corollary 7.2.2 *If* $f_i \colon X \to X_i$*,* $i = 1, 2$*, are continuous, then* $e(f_1, f_2)$ *is continuous.*

Example 7.3 We show three examples of evaluation maps whose interest resides in their images.

1. Fix $y_0 \in Y$ and consider the constant map $f \colon X \to Y$, $f(x) = y_0$. Then

$$e(\mathbb{1}_X, f) \colon X \to X \times Y, \quad e(\mathbb{1}_X, f)(x) = (x, y_0)$$

is continuous. The image of $e(\mathbb{1}_X, f)$ is $X \times \{y_0\}$.

2. If $f \colon X \to Y$ is continuous, then

$$e(\mathbb{1}_X, f) \colon X \to X \times Y, \quad e(\mathbb{1}_X, f)(x) = (x, f(x))$$

is continuous. Now the image of $e(\mathbb{1}_X, f)$ is the graph of f.

3. Consider the identity $\mathbb{1}_X \colon X \to X$. Then

$$e(\mathbb{1}_X, \mathbb{1}_X) \colon X \to X \times X, \quad e(\mathbb{1}_X, \mathbb{1}_X)(x) = (x, x)$$

is continuous and its image is $\Delta = \{(x, x) : x \in X\}$, called the *diagonal* of $X \times X$. $\Diamond$

Now we consider mappings whose domain is a product space. We analyze the problem in the next natural setting. Consider a map $f \colon X_1 \times X_2 \to Y$ and fix variables in the domain as follows. For each $x_2 \in X_2$, define

$$f_{x_2} \colon X_1 \to Y, \quad f_{x_2}(x_1) = f(x_1, x_2).$$

Likewise, for each $x_1 \in X_1$, define

$$f^{x_1} \colon X_2 \to Y, \quad f^{x_1}(x_2) = f(x_1, x_2).$$

These maps simply regard the map f as a map of one variable. So, f_{x_2} depends on the first variable of f and we have a collection $\{f_{x_2} : x_2 \in X_2\}$ of maps parametrized by the second variable.

Corollary 7.2.3 *If $f : X_1 \times X_2 \to Y$ is continuous, then f_{x_2} and f^{x_1} are continuous for all $x_i \in X_i$, $i = 1, 2$.*

Regarding the converse of the above result, it may occur that *all* mappings f_{x_2} and f^{x_1} are continuous, but not the very mapping f.

Example 7.4 Define

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0). \end{cases} \tag{7.1}$$

For the functions $f_y : \mathbb{R} \to \mathbb{R}$, $f_0 = 0$, so it is continuous. If $y \neq 0$,

$$f_y : \mathbb{R} \to \mathbb{R}, \quad f_y(x) = \frac{xy}{x^2 + y^2},$$

which is also continuous because it is the quotient of polynomial functions on x. Note that the denominator of f_y is nowhere zero because $y \neq 0$. Likewise, all functions f^x are continuous. However, f is not continuous at $(0, 0)$ because $((\frac{1}{n}, \frac{1}{n})) \to (0, 0)$, but $(f(\frac{1}{n}, \frac{1}{n})) = (\frac{1}{2}) \to \frac{1}{2} \neq f(0, 0) = 0$. $\Diamond$

We proceed with the construction of new mappings from others. The next definition is again set-theoretic. If $f_i : X_i \to Y_i$, $i = 1, 2$, are two set-theoretic maps, the *product map* is defined by

$$f_1 \times f_2 : X_1 \times X_2 \to Y_1 \times Y_2, \quad (f_1 \times f_2)(x_1, x_2) = (f_1(x_1), f_2(x_2)).$$

After this definition, we present the topological result: see Fig. 7.3.

Corollary 7.2.4 *The map $f_1 \times f_2$ is continuous (resp. homeomorphism) if and only $f_i : X_i \to Y_i$, $i = 1, 2$, are continuous (resp. homeomorphisms). As a consequence, if $X_1 \cong Y_1$ and $X_2 \cong Y_2$, then $X_1 \times X_2 \cong Y_1 \times Y_2$.*

We use the evaluation map to obtain partial answers of the Embedding Problem.

Fig. 7.3 The mappings f_1, f_2 and $f_1 \times f_2$. This diagram is commutative

$$\begin{tikzcd} X_1 \times X_2 \arrow[r, "f_1 \times f_2"] \arrow[d, "p_i"'] & Y_1 \times Y_2 \arrow[d, "p'_i"] \\ X_i \arrow[r, "f_i"] & Y_i \end{tikzcd}$$

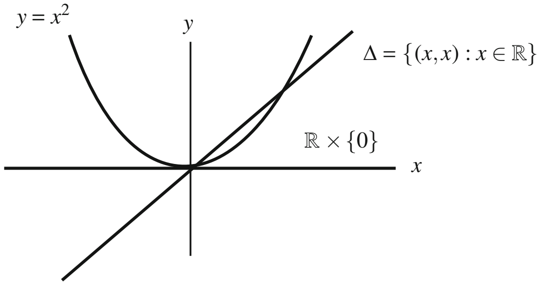


Fig. 7.4 Three embeddings of the Euclidean line $\mathbb{R}$ into $\mathbb{R}^2$ as graphs of functions

Theorem 7.2.5 *If* $f: X \to Y$ *is continuous, then the evaluation map* $e(\mathbb{1}_X, f): X \to X \times Y$ *is an embedding. In particular,*

1. *If* p *is an object then* $X \cong X \times \{p\}$.
2. *The space* X *is homeomorphic to the diagonal* Δ.

Conversely, if $e(\mathbb{1}_X, f) : X \to X \times Y$ *is an embedding, then* f *is continuous.*

Example 7.5 Figure 7.4 shows three embeddings of $\mathbb{R}$ in $\mathbb{R}^2$: as the graph of a continuous function, as a product space $\mathbb{R} \times \{a\}$ for $a \in \mathbb{R}$ and as the diagonal Δ of $\mathbb{R}^2$.

$\Diamond$

We close this section by considering the Embedding Problem for a famous set in topology, also in mathematics, which deserves to have its own space in this book.

Example 7.6 The 2-dimensional *torus* $\mathbb{T}^2$ is defined as the product space

$$\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1 = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1^2 + x_2^2 = x_3^2 + x_4^2 = 1\}, \tag{7.2}$$

where each one of the factors $\mathbb{S}^1$ is endowed by the Euclidean topology.

By Theorems 7.1.3 and 7.1.5, this product space coincides with the topological subspace $\mathbb{S}^1 \times \mathbb{S}^1$ as subset of $\mathbb{R}^4$. Moreover, the inclusion $i: \mathbb{S}^1 \times \mathbb{S}^1 \hookrightarrow \mathbb{R}^4$ is an embedding. However, we are able to prove that *the torus* $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ *can be embedded in* $\mathbb{R}^3$, in fact, that $\mathbb{T}^2$ embeds in $\mathbb{R}^3$ as the surface of a doughnut.

We shall show that $\mathbb{S}^1 \times \mathbb{S}^1$ is homeomorphic to the set $T \subset \mathbb{R}^3$ of Fig. 7.5, left. This set is obtained by rotating a circle contained in the xz-plane with respect to the z-axis. The set T is part of a class of subsets of $\mathbb{R}^3$ called *surfaces of revolution* (spheres and cylinders are other examples).

We first need to have a manageable expression for T. Rotations of $\mathbb{R}^3$ with respect to the z-axis and angle $\theta \in \mathbb{R}$ are the mappings

$$R_\theta : \mathbb{R}^3 \to \mathbb{R}^3, \quad R_\theta(x, y, z) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

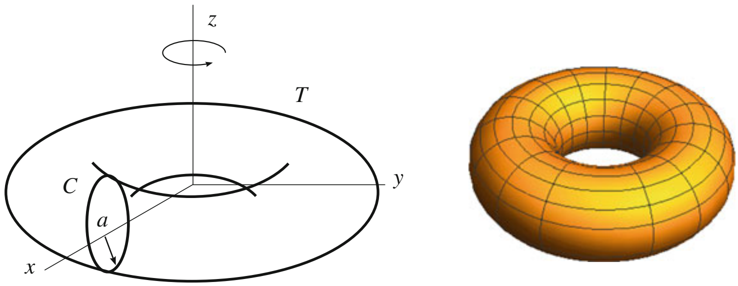


Fig. 7.5 The torus $\mathbb{S}^1 \times \mathbb{S}^1$ embedded in Euclidean space $\mathbb{R}^3$ as the surface of revolution T (left). The set T according the parametrization (7.3) for the values $a = 4$ and $r = 2$ (right)

The set $G = \{R_\theta : \theta \in \mathbb{R}\}$ of all rotations about the z-axis is a group isomorphic to $\mathbb{Z}$ with the composition as group operation because $R_{\theta_1} \circ R_{\theta_2} = R_{\theta_1+\theta_2}$. In the xz-plane, let C be the circle of radius r and center $(a, 0, 0)$ with $a > r$. This circle can be expressed by

$$C = \{(x, 0, z) \in \mathbb{R}^3 : (x - a)^2 + z^2 = r^2\}.$$

Example 6.12 asserts that $C \cong \mathbb{S}^1$. The set T is obtained by rotating C with respect to the z-axis, $T = \{R_\theta(C) : \theta \in \mathbb{R}\}$. From the implicit equation of C, we have $x - a = r \cos t$ and $z = r \sin t$ for some $t \in \mathbb{R}$ and an arbitrary point of C can be expressed as $(a + r \cos t, 0, r \sin t)$. If we rotate all these points by all R_θ, then

$$\begin{aligned} T &= \{R_\theta(a + r \cos t, 0, r \sin t) : \theta, t \in \mathbb{R}\} \\ &= \{((a + r \cos t) \cos\theta, (a + r \cos t) \sin\theta, r \sin t) : \theta, t \in \mathbb{R}\}. \end{aligned} \tag{7.3}$$

The requirement $a > r$ guarantees that C does not meet the z-axis and consequently, T has no self-intersections. The set T is simply the surface of a doughnut as shows Fig. 7.5, right. The points of T can be written as the zeroes of a continuous function,

$$T = \{(x, y, z) \in \mathbb{R}^3 : (\sqrt{x^2 + y^2} - a)^2 + z^2 = r^2\}. \tag{7.4}$$

This identity between these point sets is proved by a double inclusion. The details are left to the reader. As a consequence, T is a closed subset of $\mathbb{R}^3$. If $(x, y) = (\cos t, \sin t)$ and $(x', y') = (\cos\theta, \sin\theta)$, the expression (7.3) enables us to define

$$f : \mathbb{S}^1 \times \mathbb{S}^1 \to T, \quad f(x, y, x', y') = ((a + rx)x', (a + rx)y', ry).$$

We prove that f is a homeomorphism. Let p_i' and q_j be the restrictions of the projections of $\mathbb{R}^3$ to T and of $\mathbb{R}^4$ to $\mathbb{S}^1 \times \mathbb{S}^1$, respectively. Then

$$p_1' \circ f = (a + r \cdot q_1) \cdot q_3, \quad p_2' \circ f = (a + r \cdot q_1) \cdot q_4, \quad p_3' \circ f = r \cdot q_2.$$

The inverse of f is calculated as usual. From $f(x, y, x', y') = (x_1, x_2, x_3)$, one gets the expressions of x_1, x_2 and x_3 in terms of x, y, x' and y', obtaining

$$f^{-1}(x_1, x_2, x_3) = \left(\frac{\sqrt{x_1^2 + x_2^2} - a}{r}, \frac{x_3}{r}, \frac{x_1}{\sqrt{x_1^2 + x_2^2}}, \frac{x_2}{\sqrt{x_1^2 + x_2^2}} \right).$$

The terms $x_1^2 + x_2^2$ in the denominators cannot be 0 because if $x_1^2 + x_2^2 = 0$, substituting in (7.4), we obtain $a^2 + z^2 = r^2$, hence $a \leq r$ which is not possible. Continuity of f^{-1} follows in much the same way,

$$q_1 \circ f^{-1} = \frac{\sqrt{p_1'^2 + p_2'^2} - a}{r}, \; q_2 \circ f^{-1} = \frac{p_3'}{r},$$
$$q_3 \circ f^{-1} = \frac{p_1'}{\sqrt{p_1'^2 + p_2'^2}}, \quad q_4 \circ f^{-1} = \frac{p_2'}{\sqrt{p_1'^2 + p_2'^2}}.$$

Finally, the composition of the homeomorphism f and the inclusion of $i : T \hookrightarrow \mathbb{R}^3$ establishes the desired embedding of $\mathbb{T}^2$ into $\mathbb{R}^3$ whose image is T. $\Diamond$

In Exercise 7.23, we will give another embedding of the torus in $\mathbb{R}^3$ using the stereographic projection from $\mathbb{S}^3$ to $\mathbb{R}^3$.

7.3 Product Topology: The Infinite Case

So far we have defined the topological product of two spaces. A similar procedure extends to a finite number of topological spaces $\{(X_i, \tau_i) : 1 \leq i \leq n\}$. The purpose of this section is to generalize the notion of a topological product to an arbitrary number of spaces. The plan consists of characterizing the product topology of two spaces in terms of a special subbasis. This characterization is the key for the definition of the product topology in the general case.

Proposition 7.3.1 *Let (X_1, τ_1) and (X_2, τ_2) be two spaces.*

1. *The collection $\mathscr{S} = \{p_i^{-1}(O_i) : O_i \in \tau_i, i = 1, 2\}$ is a subbasis for $\tau_1 \times \tau_2$.*
2. *The product topology $\tau_1 \times \tau_2$ is the coarsest topology on $X_1 \times X_2$ that makes continuous the projections.*

We are ready to generalize the notion of a product topology to an arbitrary family of spaces. First we recall from set theory the notion of a Cartesian product of an indexed family of sets $\{X_i : i \in I\}$, where the index set I is any set.

Definition 7.3.2 Let $\{X_i : i \in I\}$ be a collection of sets. The set-theoretical *Cartesian product* of $\{X_i : i \in I\}$ is defined by

$$\Pi_{i\in I} X_i = \left\{x : I \to \bigcup\nolimits_{i\in I} X_i : x \text{ is a map and } x(i) \in X_i \text{ for all } i \in I\right\}.$$

Each element x is written by (x_i) where $x_i = x(i) \in X_i$. For each $j \in I$, the *projection* on X_j is defined by

$$p_j : \Pi_{i\in I} X_i \to X_j, \quad p_j((x_i)) = x_j.$$

Three remarks are necessary to point out.

1. The sets X_i are not empty if and only if the Cartesian product $\Pi_{i\in I} X_i$ is nonvoid. This statement is a consequence of the Axiom of Choice of set theory which asserts that *given a collection of nonempty sets* $\{X_i : i \in I\}$, *there exists a mapping* $f : I \to \bigcup_{i\in I} X_i$ *such that* $f(i) \in X_i$ *for all* $i \in I$. This map f allows us to choose one element of X_i, hence the Cartesian product is a nonempty set for a family $\{X_i : i \in I\}$ of nonempty sets.
2. If I is finite then we have the common definition of Cartesian product. For example, if I is a set of indices with two elements, say $I = \{1, 2\}$, we endow I with an order relation. A map $x : I \to X_1 \cup X_2$ is known by its image $(x(1), x(2))$ and we identify by the pair $(x(1), x(2))$ in the same order as in I. In such a case, $\Pi_{i=1}^2 X_i$ is the usual Cartesian product $X_1 \times X_2$.
3. If $X_i = X_j = X$ for all i, j, the Cartesian product writes as X^I and it is merely the set of mappings from I onto X. A special case occurs when $I = \mathbb{N}$, obtaining the set, written X^ω, of all sequences of X. In the same manner, $\mathbb{R}^\mathbb{R}$ is the set of real-valued functions of real variable.

Definition 7.3.3 Let $\{(X_i, \tau_i) : i \in I\}$ be a collection of spaces. The *product topology* on $\Pi_{i\in I} X_i$, denoted $\Pi_{i\in I} \tau_i$, is the topology that has as a subbasis

$$\mathscr{S} = \{p_i^{-1}(O_i) : O_i \in \tau_i, i \in I\}.$$

We shall call $(\Pi_{i\in I} X_i, \Pi_{i\in I} \tau_i)$ the *topological product* of $\{(X_i, \tau_i) : i \in I\}$.

We describe the basic open sets. An element of $\mathscr{S}$, say $p_j^{-1}(O_j)$, is

$$p_j^{-1}(O_j) = \Pi_{i\in I} G_i, \text{ where } G_i = \begin{cases} X_i, & i \neq j \\ O_j, & i = j. \end{cases}$$

Thus an element of $\beta(\mathscr{S})$ is

$$p_{i_1}^{-1}(O_{i_1}) \cap \ldots \cap p_{i_k}^{-1}(O_{i_k}) = \Pi_{i\in I} G_i, \text{ where } G_i = \begin{cases} X_i, & i \neq i_j \\ O_{i_j}, & i = i_j. \end{cases}$$

Definitively, the basis for the product topology is

$$\beta(\mathscr{S}) = \{\Pi_{i\in I} O_i : O_i \in \tau_i, \text{ such that for all but a finite number of } i,\ O_i = X_i\}.$$

In essence, any basic open set fills almost all the product space and it is in this sense that we say that the product topology contains few open sets. In case of two spaces, it is obvious that $\beta(\mathscr{S})$ coincides with $\tau_1 \times \tau_2$. The next result is immediate.

Proposition 7.3.4

1. *If β_i is a basis for X_i for all $i \in I$, then*

   $$\Pi_{i\in I}\beta_i = \{\Pi_{i\in I} B_i : B_i \in \beta_i, \text{ and all } B_i = X_i \text{ but for a finite number of indices}\}$$

   is a basis for the product topology.
2. *If for each $i \in I$, $\beta^i_{x_i}$ is a basis of neighborhoods of $x_i \in X_i$, then*

   $$\Pi_{i\in I}\beta^i_{x_i} = \{\Pi_i V_i : V_i \in \beta^i_{x_i}, \text{ and all } V_i = X_i \text{ but for a finite number of indices}\}$$

   is a basis of neighborhoods of (x_i) in the product topology.

Theorems 7.1.5 and 7.2.1 and Proposition 7.3.1 can be extended to the product of infinite spaces. We conclude with the box topology. A basis of this topology is simply to consider the product of open sets of each one of the factors. As we will show in successive exercises, we will see that this topology is not useful.

Definition 7.3.5 Given a collection of spaces $\{(X_i, \tau_i) : i \in I\}$, the *box topology* on $\Pi_{i\in I} X_i$ is the topology, denoted τ_b, whose basis is

$$\beta_b = \{\Pi_{i\in I} O_i : O_i \in \tau_i, i \in I\}.$$

1. The family β_b is a basis for a topology because the intersection of two members of β_b is other of β_b.
2. The projections $p_j : (\Pi_{i\in I} X_i, \tau_b) \to (X_j, \tau_j)$ are continuous.
3. If I is finite, the box topology τ_b coincides with the product topology.

7.4 Worked Exercises

Exercise 7.1 Consider the product space $(\mathbb{R} \times \mathbb{R}, \tau_u \times \tau_r)$, where τ_u and τ_r are the usual topology and the right order topology, respectively. Find the interior and closure of the square $A = \{(x, y) \in \mathbb{R}^2 : |x| < 1, |y| < 1\}$.

Solution We present two ways to solve the exercise.

1. Point-by-point. A basis of neighborhoods for (x, y) is

   $$\beta^u_x \times \beta^r_y = \{(x - \varepsilon, x + \varepsilon) \times [y, \infty) : \varepsilon > 0\}.$$

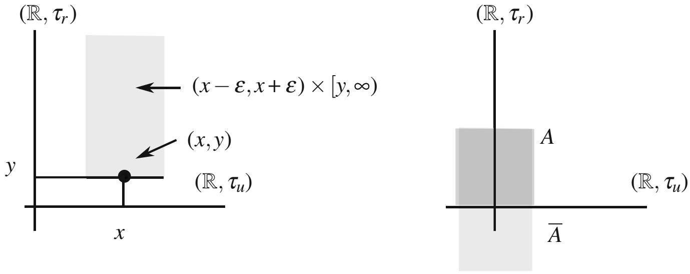


Fig. 7.6 A basis of neighborhood of (x, y) in $(\mathbb{R}^2, \tau_u \times \tau_r)$ (left) and the closure of A (right)

See Fig. 7.6, left. Since all these sets are not bounded, there are no elements of $\beta_x^u \times \beta_y^r$ included in A. Consequently, $\mathrm{int}(A) = \emptyset$. For the closure, and motivated by the shape of the basic neighborhoods, we claim that

$$\overline{A} = [-1, 1] \times (-\infty, 1).$$

See Fig. 7.6, right. Certainly, let (x, y) such that $|x| \leq 1$, $y < 1$ and let $\varepsilon > 0$ be given. Consider the point (z, t), where $z \in (x - \varepsilon, x + \varepsilon) \cap [-1, 1]$ and $t \in (\max\{y, 0\}, 1)$. It is clear that $(z, t) \in A \cap ((x - \varepsilon, x + \varepsilon) \times [y, \infty))$, proving that $(x, y) \in \overline{A}$.
We see that the remaining points are not adherent. For the points where $y \geq 1$, the neighborhoods $(x - \varepsilon, x + \varepsilon) \times [y, \infty)$ do not intersect A because

$$((x - \varepsilon, x + \varepsilon) \times [y, \infty)) \cap A = ((x - \varepsilon, x + \varepsilon) \cap (-1, 1)) \times ([y, \infty) \cap (-1, 1))$$

and $[y, \infty) \cap (-1, 1) = \emptyset$. Finally, let (x, y) with $|x| > 1$. Suppose $x > 1$ (the argument is similar if $x < -1$). Let $\varepsilon_0 > 0$ such that $1 + \varepsilon_0 < x$. Then the neighborhood $(x - \varepsilon_0, x + \varepsilon_0) \times [y, \infty)$ does not intersect A.

2. A second proof uses the property that A is a Cartesian product of two sets, $A = (-1, 1) \times (-1, 1)$. In such a case, its interior and closure is the Cartesian product of interiors and closures of each one of its factors. Then

$$\mathrm{int}(A) = \mathrm{int}_u(-1, 1) \times \mathrm{int}_r(-1, 1) = (-1, 1) \times \emptyset = \emptyset.$$

The interior of $(-1, 1)$ in τ_r is $\emptyset$ since $\emptyset$ is the only open set in the right order topology contained in $(-1, 1)$. For the closure, taking into account that the closure of $(-1, 1)$ in the right order topology is $(-\infty, 1)$, we conclude that

$$\overline{A} = \overline{(1, 1)}^u \times \overline{(1, 1)}^r = [-1, 1] \times (-\infty, 1).$$

$\square$

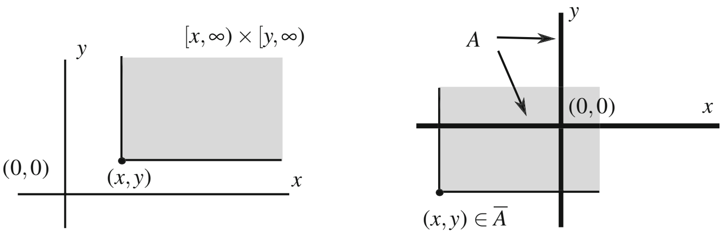


Fig. 7.7 A basic neighborhood and the closure of A of Exercise 7.2

Exercise 7.2 Find the interior and closure of the coordinate axes

$$A = \{(x, 0) : x \in \mathbb{R}\} \cup \{(0, x) : x \in \mathbb{R}\}$$

in the product topology $(\mathbb{R} \times \mathbb{R}, \tau_r \times \tau_r)$.

Solution A basis of neighborhoods of (x, y) in the product topology $\tau_r \times \tau_r$ is $\beta_{(x,y)} = \{[x, \infty) \times [y, \infty)\}$. See Fig. 7.7, left.

1. It is clear that there are no sets $[x, \infty) \times [y, \infty)$ included in A. This proves that $\text{int}(A) = \emptyset$.
2. For the closure of A, we study which points of $\mathbb{R}^2 \setminus A$ are adherent. From the shape of A and of the basic neighborhoods, we show that $\overline{A} = \mathbb{R}^2 \setminus \{(x, y) \in \mathbb{R}^2 : x, y > 0\}$. Certainly, if $x, y > 0$, the point (x, y) is not adherent because $([x, \infty) \times [y, \infty)) \cap A = \emptyset$. Indeed, otherwise there would be $(a, 0)$ (or $(0, b)$) such that $(a, 0) \in [x, \infty) \times [y, \infty)$. Thus $0 \in [y, \infty)$, so $y \leq 0$, which is contradictory.
   Now let $(x, y) \in \mathbb{R}^2$ such that $x \leq 0$ (resp. $y \leq 0$). Then $[x, \infty) \times [y, \infty)$ intersects A because the point $(0, y)$ (resp. $(x, 0)$) belongs to the basic neighborhood and to A. See Fig. 7.7, right. □

Exercise 7.3 Prove that, in general, the product space of two cofinite topologies is not the cofinite topology on the Cartesian product.

Solution Consider two sets X and Y endowed with the cofinite topology τ_{CF} and τ'_{CF}, respectively and the product topology $\tau_{CF} \times \tau'_{CF}$ on $X \times Y$. On the other hand, we stand for $\widetilde{\tau}_{CF}$ the cofinite topology on $X \times Y$. We show that

$$\tau'_{CF} \subsetneq \tau_{CF} \times \tau_{CF}.$$

1. Let $O = X \times Y \setminus \{(x_1, y_1), \ldots, (x_n, y_n)\} \in \widetilde{\tau}_{CF}$ and we prove that O is open in the product topology $\tau_{CF} \times \tau_{CF}$. Observe that O is not necessarily a basic open set of $\tau_{CF} \times \tau'_{CF}$. In fact, O is not a Cartesian product. For example, if $X = Y = \mathbb{R}$, the set $O = \mathbb{R}^2 \setminus \{(0, 0)\}$ is not a set of type $A \times B$.

We see that O is a neighborhood of all its points in the product topology. Let $(x, y) \in O$ and we shall find $O_1 \in \tau_{CF}$ and $O_2 \in \tau'_{CF}$, such that $(x, y) \in O_1 \times O_2 \subset O$. We distinguish three cases.

(a) If $x \neq x_i$ and $y \neq y_j$ for all i and j, it is enough to choose the open sets $O_1 = X \setminus \{x_1, \ldots, x_n\}$ and $O_2 = Y \setminus \{y_1, \ldots, y_n\}$.

(b) Suppose $x = x_i$ for some i. In order to simplify the reasoning, we may assume that $x = x_1$, hence $y \neq y_1$. We distinguish two cases.

  (i) Case $y \neq y_j$ for all j. Take $O_1 = X \setminus \{x_2, \ldots, x_n\}$ and $O_2 = Y \setminus \{y_1, \ldots, y_n\}$.

  (ii) Case $y = y_j$ for some $j \in \{2, \ldots, n\}$. Let $O_1 = X \setminus \{x_2, \ldots, x_n\}$ again and $O_2 = Y \setminus \{y_1, \ldots, y_{j-1}, y_{j+1}, \ldots, y_n\}$.

(c) The case $y = y_j$ for some j is analogous interchanging the roles of x and y.

2. We see that $\tau_{CF} \times \tau'_{CF}$ is not included in $\widetilde{\tau}_{CF}$. Fix $y_0 \in Y$. The set $X \times (Y \setminus \{y_0\})$ is open in $\tau_{CF} \times \tau'_{CF}$ because it is the product of two open sets. However, it is not open in $\widetilde{\tau}_{CF}$ because its complement $X \times \{y_0\}$ is not finite.

Thus *the topological product of two cofinite topologies is finer than the cofinite topology on the Cartesian product.* □

Exercise 7.4 Determine if the topological product of two included topologies is the included topology on the Cartesian product.

Solution Let X and Y be two sets, $p \in X$, $q \in Y$ and τ_p and τ_q the corresponding included point topologies. On $X \times Y$, we consider the included point topology $\tau_{(p,q)}$. We prove that

$$\tau_p \times \tau_q \subset \tau_{(p,q)}$$

and we show an example where this inclusion is proper.

1. If $O \in \tau_p$ and $O' \in \tau_q$ then $O \times O'$ contains the point (p, q), which implies that $O \times O'$ is open in $\tau_{(p,q)}$. This proves that all members of the basis for the product topology are open in $\tau_{(p,q)}$, hence the same occurs for the topology $\tau_p \times \tau_q$.
2. The inclusion $\tau_{(p,q)} \subset \tau_p \times \tau_q$ does not hold in general. An example is $X = \{a, b\}$, $X = Y$ and $p = q = b$. Let $A = \{(b, b), (a, a)\}$. This set contains (b, b), so $A \in \tau_{(b,b)}$. However, A is not open in $\tau_b \times \tau_b$. To do this, since $(a, a) \in A$, there would be $O, O' \in \tau_b$ such that $(a, a) \in O \times O' \subset A$. Since $a \in O$ and $a \in O'$, we have $O = O' = \{a, b\}$ so, $O \times O' = X \times X$, but $X \times X$ is not included in A.

Definitively, *the included point topology on the Cartesian product is finer than the topological product of two included point topologies.* □

Exercise 7.5 Prove that $A = \{(x, y) \in \mathbb{R}^2 : x \in (0, 1)\}$ is homeomorphic to $\mathbb{R}^2$.

Solution This is an elementary exercise because $A = (0,1) \times \mathbb{R}$. Since $(0,1) \cong \mathbb{R}$, Corollary 7.2.4 implies $A \cong \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. However, the question of this example is to emphasize the topologies that are present in each one of these spaces. We make precise the notation and the topologies. Let τ_u^1 and τ_u^2 denote the usual topology on $\mathbb{R}$ and $\mathbb{R}^2$, respectively. Corollary 7.2.4 asserts that

$$\left((0,1) \times \mathbb{R}, \tau^1_{u\,|(0,1)} \times \tau_u^1\right) \cong (\mathbb{R} \times \mathbb{R}, \tau_u^1 \times \tau_u^1).$$

On the other hand, Theorem 7.1.5 implies that $(A, \tau^1_{u\,|(0,1)} \times \tau_u^1) = (A, (\tau_u^1 \times \tau_u^1)_{|A})$ and Theorem 7.1.3 yields $\tau_u^1 \times \tau_u^1 = \tau_u^2$. Thus $(A, \tau^2_{u\,|A}) \cong (\mathbb{R}^2, \tau_u^2)$, which it is just that we wanted to prove. □

The next exercise follows the footsteps of similar results that characterize a topology in terms of the continuity of some maps (relative topology and product topology). In the present case, we prove that the distance is continuous and this characterizes, in some sense, the metric topology.

Exercise 7.6 Let (X, d) be a metric space and let τ be a topology on X. If $d : (X \times X, \tau \times \tau) \to \mathbb{R}$ is continuous, prove that τ is finer than the metric topology of (X, d). This characterizes the metric topology as the coarsest topology on X that makes continuous the distance function d.

Solution First, we see that $d : (X \times X, \tau_d \times \tau_d) \to \mathbb{R}$ is continuous. But Theorem 7.1.4 asserts that $\tau_d \times \tau_d$ is the metric topology of the product distance and Exercise 5.10 that d is continuous.

We show that $\tau_d \subset \tau$. A basis for the metric topology is formed by all balls. Thus it suffices to show that every ball $B_x(r)$ is an element of τ. If $d : (X \times X, \tau \times \tau) \to \mathbb{R}$ is continuous, its restriction $d : (\{x\} \times X, \tau_T \times \tau) \to \mathbb{R}$ is continuous. On the other hand, the map

$$\phi : (X, \tau) \to (\{x\} \times X, \tau_T \times \tau), \quad \phi(y) = (x, y)$$

is a homeomorphism because $\phi = e(\mathbb{1}_X, y)$. Definitively,

$$f : (X, \tau) \to \mathbb{R}, \quad f(y) = (d \circ \phi)(y) = d(x, y)$$

is continuous. Thus the preimage of the open interval $(-1, r)$ under f is open in (X, τ). But this set is

$$f^{-1}((-1, r)) = \{y \in X : d(x, y) < r\} = B_r(x),$$

proving that $B_r(x) \in \tau$. □

Motivated by the definition of the product topology, let $\{(X_i, \tau_i) : i \in I\}$ be a collection of spaces and let X be a set and consider a family of maps $\{f_i : X \to X_i : i \in I\}$. The *initial topology* on X induced by $\{f_i\}_{i \in I}$ is defined as the topology on

X, written $\tau(f_i)$, where

$$\mathscr{S} = \{f_i^{-1}(O_i) : O_i \in \tau_i, i \in I\}$$

is a subbasis. As a consequence, the product topology is the initial topology on the Cartesian product $\Pi_{i \in I} X_i$ induced by the projections $\{p_i\}_{i \in I}$. We prove that the metric topology is an initial topology.

Exercise 7.7 Let (X, d) be a metric space. For each $x \in X$, let $f_x : X \to (\mathbb{R}, \tau_u)$ be the function defined by $f_x(y) = d(x, y)$. Prove that the initial topology on X determined by the collection of maps $\{f_x : x \in X\}$ is the metric topology τ_d.

Solution A subbasis for the initial topology is $\mathscr{S} = \{f_x^{-1}(O) : x \in X, O \in \tau_u\}$. We calculate the preimages of the open intervals of $\mathbb{R}$. For $(a, b) \in \tau_u$,

$$f_x^{-1}((a, b)) = \{y \in X : a < d(x, y) < b\}.$$

Thus if $a < 0$ and $b > 0$, the set $f_x^{-1}((a, b))$ is the ball $B_b(x)$. This implies the inclusion $\tau_d \subset \tau$. Note that the subbasis contains more elements, for example, if $a \geq 0$, or taking the preimages of other open sets of $(\mathbb{R}, \tau_u)$.

We prove the inclusion $\tau \subset \tau_d$. It suffices to check that every element of the subbasis for the initial topology is open in the metric topology. Let $f_x^{-1}(O) \in \mathscr{S}$ and we see that all its points are interior in the metric topology. Let $y \in f_x^{-1}(O)$ be an arbitrary point. Since O is open in $\mathbb{R}$ and $f_x(y) = d(x, y) \in O$, there is $r > 0$ such that $(d(x, y) - r, d(x, y) + r) \subset O$. We prove that $B_r(y) \subset f_x^{-1}(O)$ and consequently, y is an interior point. For $z \in B_r(y)$, $f_x(z) = d(x, z)$ satisfies

$$|f_x(z) - d(x, y)| = |d(x, z) - d(x, y)| \leq d(y, z) < r,$$

by the quadrangle inequality, proving that $f_x(z) \in (d(x, y) - r, d(x, y) + r) \subset O$. $\square$

Exercise 7.8 Define $f : \mathbb{R} \to (\mathbb{R}, \tau_{CF})$ by $f(x) = E(x)$, where $E(x)$ is the *integer part* of x (Fig. 7.8). Characterize the open and the closed sets for the initial topology τ on $\mathbb{R}$. Find the interior and closure of $A = [-\frac{1}{2}, \frac{3}{2}]$ and of $\mathbb{Z}$.

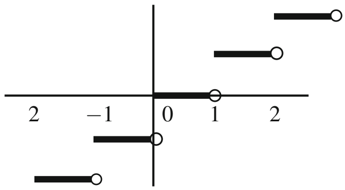


Fig. 7.8 The integer part function

Solution

1. If $a \notin \mathbb{Z}$, $E^{-1}(\{a\}) = \emptyset$, but if $a \in \mathbb{Z}$, $E^{-1}(\{a\}) = [a, a+1)$. Because $\{a\}$ is closed in τ_{CF}, then $[a, a+1)$ is closed in the initial topology τ.
Let $F = \{a_1, \ldots, a_m\}$ be an arbitrary closed set in the cofinite topology with $a_i \in \mathbb{R}$. Since
$$E^{-1}(F) = E^{-1}(\{a_1, \ldots, a_m\}) = \bigcup_{i=1}^{m} E^{-1}(\{a_i\}) = \bigcup\nolimits_{k=1}^{n} [a_{i_k}, a_{i_k} + 1),$$
where $a_k \in \mathbb{Z}$ and $n \leq m$. We have proved that the collection of such sets is the family of closed sets in the initial topology τ. Consequently, the open sets in $(\mathbb{R}, \tau)$ are the complements of sets $\bigcup_{k=1}^{n} [a_{i_k}, a_{i_k} + 1)$; that is, the trivial sets and sets
$$(-\infty, x_{i_1}) \cup [x_{i_2}, x_{i_2} + 1) \cup \ldots \cup [x_{i_n}, x_{i_{n+1}}) \cup [x_{i_{n+1}}, \infty), \tag{7.5}$$
where $x_1 < \ldots < x_{n+1}$, $x_k \in \mathbb{Z}$ and $n \in \mathbb{N}$.
2. It is clear that there are no open sets included in A because every open set in this topology is not bounded. The closure of A is the smallest closed set containing A, so $\overline{A} = [-1, 2)$.
3. The interior of $\mathbb{Z}$ is $\emptyset$ because there are no intervals included in $\mathbb{Z}$. Finally $\overline{\mathbb{Z}} = \mathbb{R}$ because any open set of (7.5) contains infinite integers thanks to the intervals $(-\infty, x_{i_1})$ and $[x_{i_{n+1}}, \infty)$. □

Over the course of the next exercises, we study the box topology in comparison with the product topology. First we show that the box topology τ_b does not satisfy (2) of Theorem 7.2.1.

Exercise 7.9 Consider $\mathbb{R}^\omega$, the set of sequences of real numbers. Define
$$f : \mathbb{R} \to \mathbb{R}^\omega, \quad f(x)(n) = x,$$
that is, $f(x)$ is a constant sequence where all terms are x, $(x)_n = x$. Prove that $f : (\mathbb{R}, \tau_u) \to (\mathbb{R}^\omega, \Pi_{n\in\mathbb{N}}\tau_u)$ is continuous but not $f : (\mathbb{R}, \tau_u) \to (\mathbb{R}^\omega, \tau_b)$.

Solution

1. Since the codomain of $f : (\mathbb{R}, \tau_u) \to (\mathbb{R}^\omega, \Pi_{n\in\mathbb{N}}\tau_u)$ is a product space, f is continuous if all compositions with the projections are continuous. If $p_n : \mathbb{R}^\omega \to \mathbb{R}$ is the nth projection then $p_n \circ f(x) = p_n((x)_n) = x$. This proves that $p_n \circ f = \mathbb{1}_{\mathbb{R}}$, hence is continuous.
2. We see that $f : (\mathbb{R}, \tau_u) \to (\mathbb{R}^\omega, \tau_b)$ is not continuous at $x = 0$. The image of $x = 0$ under f is the constant map $h_0 \in \mathbb{R}^\omega$ defined by $h_0(n) = 0$ for $n \in \mathbb{N}$. Let $O = \Pi_{n\in\mathbb{N}}(-\frac{1}{n}, \frac{1}{n})$, which is a neighborhood of h_0 in the box topology τ_b. If f

were continuous at 0, there would be a $r > 0$ such that $f((-r,r)) \subset O$. Using the nth projection in this inclusion,

$$p_n \circ f((-r,r)) \subset p_n(O) = \left(-\frac{1}{n}, \frac{1}{n}\right)$$

for all $n \in \mathbb{N}$. This, however, is false. Certainly, $r/2 \in (-r,r)$ and $f(r/2)$ is the constant sequence where all its terms are equal to $r/2$, hence

$$p_n \circ f(\frac{r}{2}) = \frac{r}{2} \in \left(-\frac{1}{n}, \frac{1}{n}\right),$$

which is not valid for all $n \in \mathbb{N}$. As a result of this example, consider $(1/n) \to 0$. The sequence $(f(1/n))$ does not converge to $f(0)$ in the box topology. Indeed, $f(0) = (0, 0, \dots, 0, \dots)$ is the zero sequence and $(f(1/n))$ is

$$\begin{aligned} f(1) &= (1, 1, 1, \dots, 1, \dots) \\ f(\tfrac{1}{2}) &= (\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \dots, \tfrac{1}{2}, \dots) \\ f(\tfrac{1}{3}) &= (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, \dots, \tfrac{1}{3}, \dots) \\ \vdots &= \vdots \end{aligned}$$

which converges to $f(0)$ in the product topology but not in the box topology. □

Exercise 7.10 Prove that the box topology $(\mathbb{R}^\omega, \tau_b)$ is not metrizable.

Solution We give a subset A of $\mathbb{R}^\omega$ and $x_0 \in \overline{A}$ which cannot be characterized by sequences: this is necessary in a metric space (Theorem 4.4.2). Let $A = \{(x_n) : x_n > 0, n \in \mathbb{N}\}$ be the set formed for all sequences whose terms are all positive and let x_0 be the zero constant sequence $x_0 = (0)$.

1. We prove that $x_0 \in \overline{A}$. Given $\varepsilon_n > 0$, let $V = \Pi_{n\in\mathbb{N}}(-\varepsilon_n, \varepsilon_n)$ be an arbitrary basic neighborhood of x_0 in the box topology. Then the sequence $(\varepsilon_n/2)$ belongs to A and V, hence $A \cap V \neq \emptyset$.
2. If the space were metrizable, there would be $(a_m) \subset A$ such that $(a_m) \to x_0$ in the box topology. We denote $a_m = (a_n^m)$. Consider the neighborhood of $x_0 \in \mathbb{R}^\omega$ defined by $U = \Pi_{n\in\mathbb{N}}(-a_n^n, a_n^n)$. Note that $a_n^n > 0$ because $a_m \in A$. If $(a_m) \to x_0$, there is $\nu \in \mathbb{N}$ such that $a_m \in U$ for all $m \geq \nu$. Then $p_m((a_m)) \in p_m(U)$, which forces that the mth element a_m^m of (a_m) belongs to $(-a_m^m, a_m^m)$, a contradiction. □

We investigate if the interior of a Cartesian product of subsets is the Cartesian product of their interiors.

Exercise 7.11 Let $\{X_i : i \in I\}$ be a family of spaces and $A_i \subset X_i$, $i \in I$. In the product topology, prove

$$\text{int}(\Pi_{i\in I} A_i) \subset \Pi_{i\in I} \overset{\circ}{A_i} .$$

However, equality does not hold in general as shows the space $\mathbb{R}^\omega$ of sequences of real numbers and $A_n = (0, 1) \subset \mathbb{R}$.

Solution

1. Let $(x_i) \in \text{int}(\Pi_{i\in I} A_i)$ be given. Then there is a basic open set $\Pi_{i\in I} O_i$ such that $(x_i) \in \Pi_{i\in I} O_i \subset \Pi_{i\in I} A_i$. Here O_i is open in X_i and $O_i = X_i$ for all but finitely many indices $J \subset I$. From the inclusion $O_i \subset X_i$, we deduce that $X_i = A_i$ for all $i \in I \setminus J$, in particular $\overset{\circ}{A_i}= A_i = X_i$. For the indices of J, $x_j \in O_j \subset A_j$ and this implies $x_j \in\overset{\circ}{A}_j$. Definitively, $(x_i) \in \Pi_{i\in I} \overset{\circ}{A_i}$.
2. Observe that $\Pi_{n\in\mathbb{N}} A_n$ consists of the sequences whose terms belong to $(0, 1)$. We prove that $\text{int}(\Pi_{n\in\mathbb{N}} A_n) = \emptyset$. Notice that $\Pi_{n\in\mathbb{N}} \overset{\circ}{A_n}= \Pi_{n\in\mathbb{N}}(0, 1)$. If the interior were not empty, the set $\Pi_{n\in\mathbb{N}} A_n$ would contain inside a basic open set $\Pi_{n\in\mathbb{N}} O_n$ which is not possible because $O_n = \mathbb{R}$ for all but finitely many indices. □

Exercise 7.12 Let $\{X_i : i \in I\}$ be a family of spaces and $A_i \subset X_i$, $i \in I$. In the box topology, prove

$$\text{int}\,(\Pi_{i\in I} A_i) = \Pi_{i\in I} \overset{\circ}{A_i} .$$

Solution Since each $\overset{\circ}{A_i}$ is open in X_i, $\Pi_{i\in I} \overset{\circ}{A_i}$ is open in the box topology. Because $\overset{\circ}{A_i}\subset A_i$, we have the inclusion $\Pi_{i\in I} \overset{\circ}{A_i}\subset \text{int}\,(\Pi_{i\in I} A_i)$.

Conversely, let $(x_i) \in \text{int}\,(\Pi_{i\in I} A_i)$. By definition of the box topology, there is a basic open set $\Pi_{i\in I} O_i$, with O_i open in X_i, such that $(x_i) \in \Pi_{i\in I} O_i \subset \Pi_{i\in I} A_i$. Then $x_i \in O_i \subset A_i$, so $x_i \in\overset{\circ}{A_i}$ for all $i \in I$. Therefore, $(x_i) \in \Pi_{i\in I} \overset{\circ}{A_i}$. □

Exercise 7.13 In the product space $(\mathbb{R} \times \mathbb{R}, \tau_u \times \tau_{CF})$, find the interior and closure of $A = \mathbb{R} \times (0, 1)$ and $B = \{(x, x) \in \mathbb{R}^2 : x \in \mathbb{R}\}$. Determine if

$$f : (\mathbb{R}^2, \tau_u \times \tau_u) \to (\mathbb{R}^2, \tau_u \times \tau_{CF}), \quad f(x, y) = (y, x)$$

is continuous and open.

Solution

1. Taking into account that A is a Cartesian product of two sets,

$$\overset{\circ}{A}=\overset{\circ}{\mathbb{R}}^u \times\text{int}^{CF}((0, 1)) = \mathbb{R} \times \emptyset = \emptyset, \quad \overline{A} = \overline{\mathbb{R}}^u \times \overline{(0, 1)}^{CF} = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2.$$

The set B is not, by contrast, a Cartesian product. If $(x, x) \in \overset{\circ}{B}$, then there is a basic neighborhood of (x, x) included in B, that is,

$$(x, x) \in (x - r, x + r) \times (\mathbb{R} \setminus F) \subset B,$$

where $r > 0$ and $F \subset \mathbb{R}$ is a finite set. Let $y \in \mathbb{R} \setminus F$ be given with $x \neq y$. Then

$$(x, y) \in (x - r, x + r) \times (\mathbb{R} \setminus F),$$

but $(x, y) \notin B$. This shows $\overset{\circ}{B} = \emptyset$. We claim that $\overline{B} = \mathbb{R}^2$. Let $(x, y) \in \mathbb{R}^2$ and $(x - r, x + r) \times (\mathbb{R} \setminus F)$ a basic neighborhood of (x, y). We see that $(x - r, x + r) \times (\mathbb{R} \setminus F)$ intersects B. In view that F is finite, $(x - r, x + r) \cap (\mathbb{R} \setminus F) \neq \emptyset$. Then if z belongs to this intersection, we have $(z, z) \in B$.

2. We prove that f is continuous by studying the composition with the projections p_1 and p_2. With the first projection,

$$p_1 \circ f : (\mathbb{R}^2, \tau_u) \to (\mathbb{R}, \tau_u), \quad (x, y) \longmapsto y$$

is clearly continuous. The composition with the second projection is

$$p_2 \circ f : (\mathbb{R}^2, \tau_u) \to (\mathbb{R}, \tau_{CF}), \quad (x, y) \longmapsto x.$$

We see that this map is continuous showing that the preimage of a closed set is closed. If $F \subset \mathbb{R}$ is finite, $(p_2 \circ f)^{-1}(F) = F \times \mathbb{R}$, which is closed in the usual topology because it is the product of two closed sets. In contrast, f is not open. For instance, $(0, 1) \times \mathbb{R}$ is open $(\mathbb{R}^2, \tau_u \times \tau_{CF})$ and its image under f is $\mathbb{R} \times (0, 1)$. Since this set is a Cartesian product, it would be open if each factor is open, but $(0, 1) \notin \tau_{CF}$. □

Exercise 7.14 Prove that the relative topology on $A = \{(x, -x) : x \in \mathbb{R}\}$ as subspace of $(\mathbb{R} \times \mathbb{R}, \tau_S \times \tau_S)$ is discrete. Consider the functions

$$f(x) = x + 2, \quad g(x) = x^2.$$

Determine if the graphs $G(f)$ and $G(g)$, as subspaces of $(\mathbb{R} \times \mathbb{R}, \tau_S \times \tau_S)$, are homeomorphic to $(\mathbb{R}, \tau_S)$.

Solution A basis of neighborhoods of (x, y) in $\tau_S \times \tau_S$ is

$$\beta_{(x,y)} = \{([x, z) \times [y, t)) : z, t \in \mathbb{R}\}.$$

1. It suffices to see that every singleton of A is open in A. Let $(x, -x) \in A$. Then

$$\{(x, -x)\} = ([x, x + 1) \times [-x, -x + 1)) \cap A \in (\tau_S \times \tau_S)_{|A}.$$

See Fig. 7.9, left.

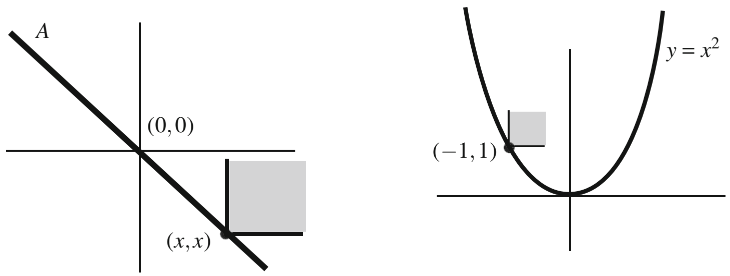


Fig. 7.9 The set A inherits the discrete topology from the product space $(\mathbb{R} \times \mathbb{R}, \tau_S \times \tau_S)$ (left). In the graph of $g(x) = x^2$, there are singletons that are open sets in the relative topology (right)

2. We prove that $G(f) \cong (\mathbb{R}, \tau_S)$. The function $f : (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_S)$ is continuous because $f^{-1}([x, y)) = [x - 2, y - 2)$ for all $x, y \in \mathbb{R}$. Thus $G(f) \cong (\mathbb{R}, \tau_S)$ by Corollary 7.2.5.
3. The function g is not continuous because $g^{-1}([1, 4)) = (-2, -1] \cup [1, 2)$ which is not open since $x = -1$ is not an interior point. Thus Theorem 7.2.5 implies that $\phi : (\mathbb{R}, \tau_S) \to G(g)$, $\phi(x) = (x, g(x))$, is not a homeomorphism. This, however, does not imply that $G(g)$ is not homeomorphic to $(\mathbb{R}, \tau_S)$. We prove that $G(g)$ has singletons that are open, which cannot happen in the Sorgenfrey topology and, consequently, $G(g) \ncong (\mathbb{R}, \tau_S)$. For example, $\{(-1, 1)\} \subset G(g)$ is open because

$$\{(-1, 1)\} = ([-1, 0) \times [1, 2)) \cap G(g).$$

See Fig. 7.9, right. Indeed, if $(z, z^2) \in [-1, 0) \times [1, 2)$, then $z \in [-1, 0)$ implies $z^2 \in (0, 1]$. Since also $z^2 \in [1, 2)$, then $z = -1$, so $(z, z^2) = (-1, 1)$. □

Exercise 7.15 Let $A = \{(x, x) : x \in \mathbb{R}\}$ be a subspace of $(\mathbb{R} \times \mathbb{R}, \tau_S \times \tau_r)$ and define $\phi : A \to \mathbb{R}$ by $\phi(x, x) = x$. Prove that $\phi : A \to (\mathbb{R}, \tau_S)$ is a homeomorphism, but not $\phi : A \to (\mathbb{R}, \tau_r)$. Prove that A is not homeomorphic to $(\mathbb{R}, \tau_r)$.

Solution

1. We see that

$$\phi : (A, (\tau_S \times \tau_r)_{|A}) \to (\mathbb{R}, \tau_S), \quad \phi(x, x) = x$$

is a homeomorphism. As in the above exercise, ϕ is the restriction to A of the first projection of $(\mathbb{R} \times \mathbb{R}, \tau_S \times \tau_r)$, so ϕ is continuous. We see that ϕ is open proving that the image of a basis of neighborhoods are neighborhoods of the image. A basis of neighborhoods of (x, x) in $(\tau_S \times \tau_r)_{|A}$ is

$$\beta^A_{(x,x)} = \{([x, y) \times [x, \infty)) \cap A : y \in \mathbb{R}\}.$$

Note that $([x, y) \times [x, y)) \cap A = \{(t, t) : x \le t < y\}$. Thus

$$\phi(([x, y) \times [x, \infty)) \cap A) = \{\phi(t, t) : x \le t < y\} = \{t : x \le t < y\} = [x, y)$$

and $[x, y)$ is a neighborhood of $\phi(x, x) = x$ in $(\mathbb{R}, \tau_S)$.

2. The map $\phi\colon A \to (\mathbb{R}, \tau_r)$ is continuous because it is the restriction to A of the second projection $p_2\colon (\mathbb{R} \times \mathbb{R}, \tau_S \times \tau_r) \to (\mathbb{R}, \tau_r)$. However, ϕ is not open because $\phi([x, y) \times [x, \infty)) \cap A) = [x, y)$ is not open in τ_r.
3. From part (2), if $A \cong (\mathbb{R}, \tau_r)$, then $(\mathbb{R}, \tau_S) \cong (\mathbb{R}, \tau_r)$. But this is not possible because in $(\mathbb{R}, \tau_r)$ there are bases of neighborhoods consisting of one element ($\beta_x^r = \{[x, \infty)\}$), which does not occur in the Sorgenfrey topology.

The point (1) is also a consequence of (1) of Corollary 7.2.5 (not of (2)!) because $\mathbb{1}_{\mathbb{R}}\colon (\mathbb{R}, \tau_S) \to (\mathbb{R}, \tau_r)$ is continuous since $\tau_r \subset \tau_S$. □

Exercise 7.16 On $\mathbb{R}^2$, define the *lexicographic order* by

$$(x, y) \le (x', y') \quad \text{if and only if} \quad \begin{cases} x < x', \text{ or} \\ x = x',\ y \le y'. \end{cases}$$

Consider the order topology τ_o, called the *topology of the lexicographic order*. Prove that this topology coincides with $\tau_D \times \tau_u$. This topology appeared in Exercise 4.14 and it was proven that it is metrizable.

Solution The term lexicographic comes from the method to order the words of a dictionary. Given two different words, one firstly compares the first letter and one orders according to the alphabet. In case that the first letters of both words coincide, one compares the second letter and repeats the procedure. This process finishes because the number of letters of each word is finite.

The reader should pay attention on the two uses of (x, y) as a point of $\mathbb{R}^2$ and as an open interval of $\mathbb{R}$.

1. We prove that $\tau_D \times \tau_u$ is included in the lexicographic order topology τ_o. A basis for the topology $\tau_D \times \tau_u$ is

$$\{\{x\} \times (a, b) : x \in \mathbb{R}, a, b \in \mathbb{R}\}.$$

It suffices to see that every element $\{x\} \times (a, b)$ is open in τ_o. Let $(x, y) \in \{x\} \times (a, b)$ and we show that

$$(x, y) \in](x, a), (a, b)[\subset \{x\} \times (a, b).$$

It is clear that $(x, y) \in](x, a), (a, b)[$. Let $(z, t) \in](x, a), (a, b)[$ be given. From $(x, a) < (z, t)$ and $(z, t) < (x, b)$, we deduce $x = z$ and $a < t < b$, so $(z, t) \in \{x\} \times (a, b)$.
2. We show that every basic element $](a, b), (c, d)[$ of τ_o is open in $\tau_D \times \tau_u$. Let $(x, y) \in](a, b), (c, d)[$. We discuss case-by-case according the inequalities

$(a, b) < (x, y)$ and $(x, y) < (c, d)$. From $(a, b) < (x, y)$, we distinguish two alternatives.

(a) Case $a < x$. Since $(x, y) < (c, d)$, we separate two cases.

(i) If $x < c$, then it holds the inclusion $\{x\} \times (y-1, y+1) \subset](a, b), (c, d)[$ by comparing only the first coordinate.

(ii) If $x = c$, we have $y < d$. We show that $\{x\} \times (y-1, d) \subset](a, b), (c, d)[$. Indeed, if $(x, t) \in \{x\} \times (y-1, d)$ then $(a, b) < (x, t)$ because $a < x$ and $(x, t) < (x, d)$ because $t < d$.

(b) Case $a = x$. Then $b < y$.

(i) If $x < c$, we prove that $\{x\} \times (b, y+1) \subset](a, b), (c, d)[$. Certainly, if $(x, t) \in \{x\} \times (b, y+1)$ then $b < t < y+1$. Thus $(a, b) < (x, t)$ because $b < t$ (now $a = x$) and $(x, t) < (c, d)$ because $x < c$.

(ii) If $x = c$, we see that $\{x\} \times (b, d) \subset](a, b), (c, d)[$. In effect, if $(x, t) \in \{x\} \times (b, d)$ then $b < t < d$. Now $(a, b) < (x, t)$ because $b < t$ again (now $a = x$) and $(x, t) < (c, d)$ because $x < d$. □

Exercise 7.17 If $X = [0, 1] \times [0, 1]$, prove that the relative topology on X as a subspace of $(\mathbb{R}^2, \tau_o)$ does not coincide with the lexicographic order topology on X.

Solution Here let τ^o stand for the order topology on X. Consider the points $p = (\frac{1}{2}, \frac{1}{2})$ and $q = (\frac{1}{2}, 2)$. Let $A =]p, q[\cap X$, where $]p, q[$ denotes the basic open in τ_o, which can be expressed by

$$A = \left\{(x, y) \in X : \tfrac{1}{2} \leq x \leq \tfrac{1}{2}, \tfrac{1}{2} < y < 2\right\} = \left\{\tfrac{1}{2}\right\} \times \left(\tfrac{1}{2}, 1\right].$$

Then A is open in the relative topology $\tau_{o|X}$. We see that A is not open in (X, τ^o). In fact, we show that $(\frac{1}{2}, 1)$ is not an interior point of A in (X, τ^o). Otherwise, there would exist a basic open $](a, b), (c, d)[$ in τ^o such that

$$\left(\tfrac{1}{2}, 1\right) \in](a, b), (c, d)[\subset \left\{\tfrac{1}{2}\right\} \times \left(\tfrac{1}{2}, 1\right].$$

Thus $\frac{1}{2} \leq a$. From this inclusion and $(a, b) < (\frac{1}{2}, 1)$, we deduce $a = \frac{1}{2}$. On the other hand, if $c = \frac{1}{2}$, we would have $1 < d$ because $(\frac{1}{2}, 1) < (c, d)$, which is absurd since $(c, d) \in X$. This proves that $c > \frac{1}{2}$. Finally, we see that

$$](\tfrac{1}{2}, b), (c, d)[\not\subset \left\{\tfrac{1}{2}\right\} \times \left(\tfrac{1}{2}, 1\right].$$

For any $r \in (\frac{1}{2}, c)$, we have $(r, 1) \notin \{\frac{1}{2}\} \times (\frac{1}{2}, 1]$, but $(r, 1) \in](\frac{1}{2}, b), (c, d)[$ because $\frac{1}{2} < r$. □

By Theorem 7.1.4, the product of two metric spaces is metrizable. We extend this result for a countable number of metric spaces.

Exercise 7.18 If $\{(X_n, d_n) : n \in \mathbb{N}\}$ is a countable family of metric spaces, prove that $\Pi_{n\in\mathbb{N}} X_n$ is metrizable.

Solution In view of Example 4.6.1, we can assume that all distances d_n are bounded by 1. Define a distance d on $\Pi_{n\in\mathbb{N}} X_n$ by

$$d((x_n), (y_n)) = \sum_{k=1}^{\infty} \frac{d_k(x_k, y_k)}{2^k}.$$

Notice that d is well defined because

$$\sum_{k=1}^{\infty} \frac{d_k(x_k, y_k)}{2^k} < \sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$$

We see that d is a distance on $\Pi_{n\in\mathbb{N}} X_n$. All the properties are obvious except the triangle inequality. Let $(x_n), (y_n) \in \Pi_{n\in\mathbb{N}} X_n$. If (z_n) is another element, the triangle inequality for each distance d_k yields $d_k(x_k, y_k) \le d_k(x_k, z_k) + d_k(z_k, y_k)$ and thus

$$\frac{d_k(x_k, y_k)}{2^k} \le \frac{d_k(x_k, z_k)}{2^k} + \frac{d_k(z_k, y_k)}{2^k}.$$

Taking the sum of the series, we derive the triangle inequality for d.

We use the Hausdorff criterion to compare τ_d and $\Pi_{n\in\mathbb{N}} \tau_{d_n}$. Consider the balls as basis of neighborhoods in τ_d and the product of balls of d_n in $\Pi_{n\in\mathbb{N}} \tau_{d_n}$. Let $(x_n) \in \Pi_{n\in\mathbb{N}} X_n$.

1. Let U be a basic neighborhood of (x_n) in $\Pi_{n\in\mathbb{N}} \tau_{d_n}$. This means that there are a finite set of indices $\{i_1, \ldots, i_m\}$ such that

$$U = \Pi_{n\in\mathbb{N}} U_n, \quad U_n = \begin{cases} B^{i_k}_{\varepsilon_{i_k}}(x_{i_k}), & n \in \{i_1, \ldots, i_m\} \\ X_n, & n \notin \{i_1, \ldots, i_m\}. \end{cases}$$

Let $\varepsilon = \min\{\frac{\varepsilon_{i_1}}{2^{i_1}}, \ldots, \frac{\varepsilon_{i_m}}{2^{i_m}}\}$. We see that $B^d_\varepsilon((x_n)) \subset U$. Let $(y_n) \in B^d_\varepsilon((x_n))$. We only need to verify that $y_{i_n} \in B^{i_n}_{\varepsilon_{i_n}}(x_{i_n})$ for $n \in \{i_1, \ldots, i_m\}$. By definition of d,

$$\frac{d_{i_n}(x_{i_n}, y_{i_n})}{2^{i_n}} \le \sum_{k=1}^{\infty} \frac{d_k(x_k, y_k)}{2^k} < \varepsilon \le \frac{\varepsilon_{i_n}}{2^{i_n}},$$

in particular, $d_{i_n}(x_{i_n}, y_{i_n}) < \varepsilon_{i_n}$ and this proves that $y_{i_n} \in B^{i_n}_{\varepsilon_{i_n}}(x_{i_n})$.

2. Let $B^d_\varepsilon((x_n))$ be a ball of radius ε and centered at (x_n) in τ_d. Since the series $\sum_{k\geq 1} 1/2^k$ is convergent, let $N \in \mathbb{N}$ such that

$$\sum_{i=N+1}^{\infty} \frac{1}{2^i} < \frac{\varepsilon}{2}.$$

We prove the inclusion

$$B^1_{\varepsilon/2N}(x_1) \times \ldots B^N_{\varepsilon/2N}(x_n) \times \left(\Pi_{i\geq N+1} X_i\right) \subset B^d_\varepsilon((x_n)).$$

Let $(y_n) \in B^1_{\varepsilon/2N}(x_1) \times \ldots B^N_{\varepsilon/2N}(x_n) \times \left(\Pi_{i\geq N+1} X_i\right)$. Then

$$\begin{aligned} d((y_n),(x_n)) &= \sum_{k=1}^{\infty} \frac{d_k(x_k,y_k)}{2^k} = \sum_{k=1}^{N} \frac{d_k(x_k,y_k)}{2^k} + \sum_{k=N+1}^{\infty} \frac{d_k(x_k,y_k)}{2^k} \\ &< \sum_{k=1}^{N} \frac{\varepsilon}{2N} + \sum_{k=N+1}^{\infty} \frac{1}{2^k} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

□

We give a characterization of the product topology by a universal property.

Exercise 7.19 Let (X_i, τ_i) be two topological spaces, $i = 1, 2$. Then the product topology $\tau_1 \times \tau_2$ is the only topology on $X_1 \times X_2$ with the following property: for any topological space (Y, τ') and $f : Y \to X_1 \times X_2$, we have f is continuous if and only if $p_i \circ f$ are continuous, $i = 1, 2$.

Solution First, notice that the product topology satisfies the given property by Theorem 7.2.1. Let $\tilde{\tau}$ be a topology on $X_1 \times X_2$ with the same property and we see that $\tilde{\tau} = \tau_1 \times \tau_2$. As in Exercise 5.20, the result is obtained by two particular choices of (Y, τ'). First, let $(Y, \tau') = (X_1 \times X_2, \tilde{\tau})$ and f the identity from $(X_1 \times X_2, \tilde{\tau})$ itself. Then f is continuous, so by hypothesis, $p_i \circ f = p_i$ are continuous. By the characterization of Proposition 7.3.1, $\tau_1 \times \tau_2 \subset \tilde{\tau}$.

Now, let $(Y, \tau') = (X_1 \times X_2, \tau_1 \times \tau_2)$ and let f be the identity map of the point set $X_1 \times X_2$. Then $p_i \circ f = p_i$ are continuous. By hypothesis, $\mathbb{1}_{X_1\times X_2} : (X_1 \times X_2, \tau_1 \times \tau_2) \to (X_1 \times X_2, \tilde{\tau})$ is continuous, so $\tilde{\tau} \subset \tau_1 \times \tau_2$. □

The following two exercises give a relationship between the product topology and the box topology with the pointwise convergence and the uniform convergence of sequences of functions of a real variable.

Exercise 7.20 Let $\mathbb{R}^{\mathbb{R}}$ be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$ endowed with the product topology assuming on $\mathbb{R}$ the usual topology τ_u. Let $(f_n)_{n\in\mathbb{N}} \subset \mathbb{R}^{\mathbb{R}}$ and $f \in \mathbb{R}^{\mathbb{R}}$. Prove that $(f_n) \to f$ in the product topology if and only if $(f_n) \to$

f pointwise (in the sense of calculus). Compare this statement with the uniform convergence defined in Exercise 4.29.

Solution

1. Suppose $(f_n) \to f$ pointwise and we show that $(f_n) \to f$ in the product topology. Let $\Pi_{x\in\mathbb{R}} U_x$ be a basic neighborhood of f; that is, there is $m \in \mathbb{N}$ and $x_1, \ldots, x_m \in \mathbb{R}$, $\varepsilon_1 > 0, \ldots, \varepsilon_m > 0$ such that

$$U_x = \begin{cases} (f(x_i) - \varepsilon_i, f(x_i) + \varepsilon_i), & x = x_i, 1 \le i \le m \\ \mathbb{R}, & x \neq x_i, \text{ for all } i = 1, \ldots, m. \end{cases}$$

   Fix $i = 1, \ldots, m$. Since $(f_n(x_i)) \to f(x_i)$, there is $\nu_i \in \mathbb{N}$ such that

$$|f_n(x_i) - f(x_i)| < \varepsilon_i$$

   for all $n \ge \nu_i$. Let $\nu = \max\{\nu_1, \ldots, \nu_m\}$ and we prove that if $n \ge \nu$ then $f_n \in \Pi_{x\in\mathbb{R}} U_x$. If $x \neq x_i$, there is nothing to prove because $U_x = \mathbb{R}$. If $x = x_i$ for some $i \in \{1, \ldots, m\}$, then the inequality $|f_n(x_i) - f(x_i)| < \varepsilon_i$ implies that $f_n(x_i) \in (f(x_i) - \varepsilon_i, f(x_i) + \varepsilon_i) = U_{x_i}$.
2. Suppose that $(f_n) \to f$ in the product topology. We see that $(f_n(x)) \to f(x)$ for all $x \in \mathbb{R}$. Let x_0 be an arbitrary but fixed real number and we show that $(f_n(x_0)) \to f(x_0)$. Let $\varepsilon > 0$ be given and define

$$\Pi_{x\in\mathbb{R}} U_x, \quad U_x = \begin{cases} \mathbb{R}, & x \neq x_0 \\ (f(x_0) - \varepsilon, f(x_0) + \varepsilon), & x = x_0. \end{cases}$$

   This set is a neighborhood of f in the product topology. From the convergence $(f_n) \to f$, there is $\nu \in \mathbb{N}$ such that $f_n \in \Pi_{x\in\mathbb{R}} U_x$ for all $n \ge \nu$. For the case $x = x_0$, we have $f_n(x_0) \in U_{x_0} = (f(x_0) - \varepsilon, f(x_0) + \varepsilon)$, or equivalently, $|f_n(x_0) - f(x_0)| < \varepsilon$, completing the proof. □

Exercise 7.21 In the space of functions $\mathbb{R}^{\mathbb{R}}$, prove that if $(f_n) \to f$ in the box topology, then $(f_n) \to f$ uniformly. Find an example where the converse is false.

Solution

1. Suppose that $(f_n) \to f$ in the box topology. Let $\varepsilon > 0$ be given. Consider the basic neighborhood of f in the box topology defined by $\Pi_{x\in\mathbb{R}}(f(x) - \varepsilon, f(x) + \varepsilon)$. From the convergence in τ_b, there is $\nu \in \mathbb{N}$ such that $f_n \in \Pi_{x\in\mathbb{R}}(f(x) - \varepsilon, f(x) + \varepsilon)$ for all $n \ge \nu$ and $x \in \mathbb{R}$. Then $f_n(x) \in (f(x) - \varepsilon, f(x) + \varepsilon)$, so $|f_n(x) - f(x)| < \varepsilon$ for all $n \ge \nu$ and $x \in \mathbb{R}$. This proves the uniform convergence.
2. For each $n \in \mathbb{N}$, define

$$f_n(x) = \frac{1}{n(1 + x^2)}.$$

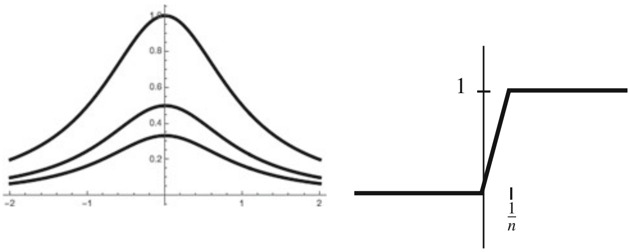


Fig. 7.10 The functions f_n of Exercise 7.21, for $n = 1, 2, 3$ (left) and that of Exercise 7.22 (right)

See Fig. 7.10, left. We show that $(f_n) \to 0$ uniformly but not in the box topology. For uniform convergence (Exercise 4.29),

$$d_\infty(f_n, 0) = \sup_{x\in\mathbb{R}} \frac{1}{n(1+x^2)} = \frac{1}{n}.$$

Since $(1/n) \to 0$, we have $d_\infty(f_n, 0) \to 0$. Now we prove that (f_n) does not converge to 0 in the box topology. Let

$$O = \Pi_{x\in\mathbb{R}} U_x, \quad U_x = \begin{cases} (-1, 1), & x = 0 \\ (-x^2, x^2), & x \neq 0. \end{cases}$$

If $(f_n) \to 0$ in the box topology, there is $\nu \in \mathbb{N}$ such that $f_n \in O$ for $n \geq \nu$. Thus if $x \neq 0$ then $f_n(x) \in U_x$, or in other words,

$$\frac{1}{n(1+x^2)} \in (-x^2, x^2)$$

for $n \geq \nu$. For $n = \nu$, we have $1/(\nu(1 + x^2)) < x^2$. Letting $x \to 0$, we get $1/\nu \leq 0$, a contradiction. □

In Exercise 5.17, it was proved that the limit of uniform sequences of continuous functions is continuous. This is not valid for the pointwise convergence.

Exercise 7.22 Prove that the space $\mathsf{B}_c(\mathbb{R}, \mathbb{R})$ of all bounded and continuous functions as a subspace of the product space of $\mathbb{R}^{\mathbb{R}}$ is not closed.

Solution For each $n \in \mathbb{N}$, define

$$f_n : \mathbb{R} \to \mathbb{R}, \qquad \begin{cases} 0, & x \leq 0 \\ nx, & x \in [0, \frac{1}{n}] \\ 1, & x \geq 1/n. \end{cases}$$

See Fig. 7.10, right. It is clear that $f_n \in \mathsf{B}_c(\mathbb{R}, \mathbb{R})$. We prove that $(f_n) \to f$ in $\mathbb{R}^{\mathbb{R}}$, where

$$f : \mathbb{R} \to \mathbb{R}, \quad \begin{cases} 0, & x \leq 0 \\ 1, & x > 0, \end{cases}$$

but f is not continuous. Since the convergence in the product topology is the pointwise convergence, we see that $(f_n(x)) \to f(x)$ for all $x \in \mathbb{R}$. If $x \leq 0$ then $f_n(x) = 0 = f(0)$, so it is obvious that $(f_n(x)) \to f(0)$. If $x > 0$ then $f_n(x) = 1$ for all $n \geq n_0$, where n_0 is any natural number such that $1/n_0 < x$. Thus, up to a finite number of terms of the sequence, $(f_n(x)) = (1)$ which converges to $1 = f(1)$. □

We prove the announced result that the torus $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ can be embedded in $\mathbb{R}^3$ by employing the stereographic projection $\pi : \mathbb{S}^3 \setminus \{N\} \to \mathbb{R}^3$. The idea is the following. Although $\mathbb{T}^2$ is not included in $\mathbb{S}^3 \setminus \{N\}$, it is contained in the sphere $\mathbb{S}^3_{\sqrt{2}}$ of radius $\sqrt{2}$. By a dilation of $\mathbb{R}^4$, we insert $\mathbb{T}^2$ in $\mathbb{S}^3$ and next, we map to $\mathbb{R}^3$ via π.

Exercise 7.23 Prove that the torus $\mathbb{T}^2$ is homeomorphic to $\mathbb{S}^1_r \times \mathbb{S}^1_r$. Use the stereographic projection $\pi : \mathbb{S}^3 \to \mathbb{R}^3$ to find an embedding of $\mathbb{T}^2$ in $\mathbb{R}^3$.

Solution All circles of $\mathbb{R}^2$ are homeomorphic, in particular $\mathbb{S}^1 \cong \mathbb{S}^1_r$. Using Corollary 7.2.4, we deduce that $\mathbb{T}^2$ is homeomorphic to $\mathbb{S}^1_r \times \mathbb{S}^1_r$. Even more, since $\mathbb{S}^1 \cong \mathbb{S}^1_r$ by the dilation $h(x, y) = r(x, y)$, then

$$f = h \times h : \mathbb{T}^2 \to \mathbb{S}^1_r \times \mathbb{S}^1_r, \quad f(x, y, z, t) = r(x, y, z, t)$$

is an explicit homeomorphism.

For the second part, we choose $r = 1/\sqrt{2}$. Then $\mathbb{S}^1_r \times \mathbb{S}^1_r$ is a subset of the unit sphere $\mathbb{S}^3$: if $(x, y, z, t) \in \mathbb{S}^1_r \times \mathbb{S}^1_r$, then $|(x, y, z, t)|^2 = x^2 + y^2 + z^2 + t^2 = 2r^2 = 1$. By composing the above f with the stereographic projection π, we derive the desired embedding of $\mathbb{T}^2$ into $\mathbb{R}^3$. The only thing to check is that the north pole $N = (0, 0, 0, 1)$ does not belong to $\mathbb{S}^1_r \times \mathbb{S}^1_r$, which is immediate.

A curiosity: We can show that the image of $\mathbb{T}^2$ by the embedding $\pi \circ f$ is precisely the surface of revolution T given in (7.4). To do this, choose $r = 1$ and $a = \sqrt{2}$ in (7.4). By definition of the stereographic projection,

$$(\pi \circ f)(x, y, z, t) = \left(\frac{rx}{1 - rt}, \frac{ry}{1 - rt}, \frac{rz}{1 - rt} \right).$$

Using $x^2+y^2=1$ and $z^2+t^2=1$, and from the implicit equation in (7.4),

$$\left(\sqrt{(\tfrac{rx}{1-rt})^2+(\tfrac{ry}{1-rt})^2}-a\right)^2+(\tfrac{rz}{1-rt})^2=\left(\sqrt{\tfrac{r^2}{(1-rt)^2}}-\tfrac{1}{r}\right)^2+\left(\tfrac{rz}{1-rt}\right)^2$$

$$=\left(\tfrac{r}{1-rt}-\tfrac{1}{r}\right)^2+\left(\tfrac{rz}{1-rt}\right)^2=\frac{(r^2-1+rt)^2+r^4(1-t^2)}{r^2(1-rt)^2}=1$$

because $r=1/\sqrt{2}$. □

We see that the product of Hausdorff spaces is Hausdorff. In general, a property $\mathscr{P}$ is said to be *productive* if whenever two spaces satisfy the property $\mathscr{P}$, then its topological product satisfies the property $\mathscr{P}$.

Exercise 7.24 Let $\{X_i : i \in I\}$ be a collection of topological spaces. Show that $\Pi_{i\in I}X_i$ is Hausdorff if and only if X_i is Hausdorff for all $i \in I$.

Solution If $\Pi_{i\in I}X_i$ is Hausdorff then each factor is Hausdorff because the projections are open so they carry neighborhoods to neighborhoods. Thus, fixed $j \in I$, if $a, b \in X_j$ are two points, consider in $\Pi_{i\in I}X_i$ any two points $x = (x_i)$, $y = (y_i)$ such that $x_j = a$ and $y_j = b$. Then $x \neq y$ and let W_x and W_y be two disjoint neighborhoods of x and y respectively. Then $p_j(W_x)$ and $p_j(W_y)$ are disjoint open neighborhoods of a and b respectively.

For the converse, let $x, y \in \Pi_{i\in I}X_i$ where $x \neq y$. Let $x = (x_i)$, $y = (y_i)$. In particular, there is $j \in I$ such that $x_j \neq y_j$. Since X_j is Hausdorff, there are disjoint neighborhoods U and V of x_j and y_j, respectively. Let us construct $W_x = \Pi_{i\in I}U_i$ and $W_y = \Pi_{i\in I}V_i$ where all U_i and V_i coincide with X_i, with the exception $U_j = U$ and $V_j = V$. Then W_x and W_y are disjoint neighborhoods of x and y, respectively. □

Exercise 7.25 Prove that a space Y is Hausdorff if and only if its diagonal Δ' is closed in $Y \times Y$. Deduce that if $f : X \to Y$ is continuous and Y is Hausdorff, then the graph of f is closed in $X \times Y$.

Solution

1. Assume that Y is Hausdorff. If Δ' is the diagonal of Y, we see that $Y \times Y \setminus \Delta'$ is open. Let $(y_1, y_2) \notin \Delta'$. Since $y_1 \neq y_2$ and Y is Hausdorff, there are disjoint open sets O_1 and O_2 such that $y_i \in O_i$, $i = 1, 2$. We prove that $O_1 \times O_2 \subset Y \times Y \setminus \Delta'$. Otherwise, there is $(y, y) \in \Delta' \cap (O_1 \cap O_2)$, in particular, $y \in O_1 \cap O_2$, a contradiction.
   Assume that Δ' is closed. Let $y_1 \neq y_2$. Then $(y_1, y_2) \notin \Delta'$. Since $Y \times Y \setminus \Delta'$ is open, there is a basic open $O_1 \times O_2 \in \tau' \times \tau'$ such that $(y_1, y_2) \in O_1 \times O_2 \subset Y \times Y \setminus \Delta'$. Thus $O_1 \cap O_2 = \emptyset$.
2. Define

$$\phi : X \times Y \to Y \times Y, \quad \phi(x, y) = (f(x), y).$$

In order to prove that ϕ is continuous, let p_i and q_i be the projections of $Y \times Y$ and $X \times Y$, respectively. Then $p_1 \circ \phi = f \circ q_1$ and $p_2 \circ \phi = q_2$, hence ϕ is continuous, in particular, $\phi^{-1}(\Delta')$ is closed. But

$$\phi^{-1}(\Delta') = \{(x, y) \in X \times Y : f(x) = y\} = G(f).$$

□

The last exercises introduce the concept of a topological group. The idea generalizes the property established in Sect. 5.3 when it was proved that the addition of vectors, regarded as a map $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{2n}$, is continuous.

Exercise 7.26 Let $(G, *)$ be a group and let τ be a topology on G. We say that G is a *topological group* if the group operation φ and the inversion θ are continuous maps,

$$\varphi\colon (G \times G, \tau \times \tau) \to (G, \tau), \quad \varphi(x, y) = x * y,$$

$$\theta\colon (G, \tau) \to (G, \tau), \quad \theta(x) = x^{-1}.$$

For notational convenience, we write xy instead of $x * y$. Prove:

1. Continuity of the mappings φ and θ are equivalent to the continuity of $\psi\colon G \times G \to G$ defined by $\psi(x, y) = xy^{-1}$.
2. Define the left translations and right translation with respect to $a \in G$ by

$$L_a\colon G \to G, \quad L_a(x) = ax,$$

$$R_a\colon G \to G, \quad R_a(x) = xa,$$

respectively. Prove that the translations are homeomorphisms.
3. If O_1 and O_2 are two open sets of G, show that $O_1 O_2 = \{xy : x \in O_1, y \in O_2\}$ is open in G.

Solution

1. Define the map

$$F\colon G \times G \to G \times G, \quad F(x, y) = (x, y^{-1}).$$

Then F is bijective, with $F^{-1} = F$. Furthermore, $\psi = \varphi \circ F$. To prove the necessity, suppose that φ and θ are continuous. Then F is continuous because composing with the projections p_1 and p_2 of $G \times G$, one gets $p_1 \circ F = p_1$ and $p_2 \circ F = \theta \circ p_2$ both of which are continuous. We conclude that $\psi = \varphi \circ F$ is continuous.

For the reverse implication, we first see that θ is continuous. Define

$$K\colon G \to G \times G, \quad K(x) = (e, x).$$

Then K is continuous because $p_1 \circ K$ is constant and $p_2 \circ K = \mathbb{1}_G$. Since $\theta = \psi \circ K$, then θ is continuous. Having proved that θ is continuous, the above map F is continuous and consequently, φ is too because $\varphi = \psi \circ F$.

2. We solve the exercise for the left translations and analogously the argument holds for right translations. The map $f\colon G \to G \times G$ defined by $f(x) = (a, x)$ is continuous because $p_1 \circ f$ is constant and $p_2 \circ f = \mathbb{1}_G$. Then L_a is continuous because $L_a = \varphi \circ f$. On the other hand, the inverse of L_a is the left translation $L_{a^{-1}}$, so it is continuous again.
3. It suffices to observe that $O_1 O_2 = \bigcup_{x \in O_1} L_x(O_2)$ is a union of open sets. □

Exercise 7.27 If $H < G$ is a subgroup of a topological group G, then H is a topological group. Furthermore,

1. The closure $\overline{H}$ is a topological subgroup of G.
2. If $H \lhd G$ is a normal subgroup of G, then $\overline{H}$ too.
3. If H is open, then it is also closed.

Solution To facilitate the notation, let $\widetilde{\varphi}\colon H \times H \to H$ and $\widetilde{\theta}\colon H \to H$ be the group operation and the inversion on H. The codomain of these mappings is H because H is a subgroup of G. Since $\widetilde{\varphi}$ and $\widetilde{\theta}$ are simply the restrictions of φ to $H \times H$ and of θ to H, we conclude that they are continuous.

1. Proving that $\overline{H}$ is a group is equivalent to showing that $\psi(\overline{H} \times \overline{H}) \subset \overline{H}$. Since ψ is continuous and H is a subgroup, Theorem 5.1.3 and Proposition 7.1.2 imply

$$\psi(\overline{H} \times \overline{H}) = \psi(\overline{H \times H}) \subset \overline{\psi(H \times H)} \subset \overline{H}.$$

2. The map $\eta_a\colon G \to G$ defined by $\eta_a(x) = axa^{-1}$ is a homeomorphism because $\eta_a = L_a \circ R_{a^{-1}}$. Using that $aHa^{-1} \subset H$ because H is normal,

$$a\overline{H}a^{-1} = \eta_a(\overline{H}) = \overline{\eta_a(H)} = \overline{aHa^{-1}} \subset \overline{H},$$

and this proves that $\overline{H}$ is a normal subgroup of G.
3. Define on G the equivalence relation $a \sim b$ if $b^{-1}a \in H$. The equivalence class $[a]$ of $a \in G$ is

$$\begin{aligned}[a] &= \{b \in H : b^{-1}a = h \text{ for some } h \in H\} = \{ah^{-1} : h \in H\} \\ &= \{ah : h \in H\} = aH = L_a(H).\end{aligned}$$

This relation provides a partition on G in equivalence classes where the class of the unit element e is H. Thus

$$G = \bigcup_{a \in G} aH = H \bigcup \left(\bigcup_{a \notin H} aH \right).$$

Since H is open, the set $L_a(H) = aH$ is also open and consequently $\bigcup_{a \notin H} aH$ is open. Since $H = G \setminus \bigcup_{a \notin H} aH$, we conclude that H is closed. □

Remark 7.4.1 The last argument deserves to be highlighted. In topological spaces, if $\{A_i : i \in I\}$ is a partition of a space by open sets, then all A_i are closed because any A_i is the complement of the union of the remaining elements of the partition. This does not hold if we exchange open sets for closed sets. A simple example is $\mathbb{R}$ and the partition $\mathbb{R} = \bigcup_{x \in \mathbb{R}} \{x\}$.

Besides the Euclidean space $\mathbb{R}^n$ (Example 5.11), we present two new topological groups, the punctured complex plane and the general linear group.

Exercise 7.28 The set of nonzero complex numbers $\mathbb{C}^* = \{z \in \mathbb{C} : z \neq 0\}$ is a topological group where the group operation is the multiplication of complex numbers and the topology is the usual one. As a consequence, the unit circle $\mathbb{S}^1 \subset \mathbb{C}^*$ is a topological subgroup.

Solution As a topological space, $\mathbb{C}^*$ coincides with $\mathbb{R}^2 \setminus \{(0, 0)\}$ and this is because we consider the usual topology on $\mathbb{C}^*$. In order to prove that the multiplication of complex numbers, denoted $\cdot$, is continuous, we write the operation rather explicitly. If $z = x + iy \equiv (x, y)$ and $z' = x' + iy' \equiv (x', y')$ then

$$\varphi(z, z') = z \cdot z' = (x + iy) \cdot (x' + iy') = (xx' - yy', xy' + yx').$$

Then $\varphi \colon \mathbb{C}^* \times \mathbb{C}^* \to \mathbb{C}^* \hookrightarrow \mathbb{R}^2$ is continuous because the compositions with the projections p_i of $\mathbb{R}^2$ yield

$$p_1 \circ \varphi = q_1 q_3 - q_2 q_4, \quad p_2 \circ \varphi = q_1 q_4 + q_2 q_3,$$

where q_i are the restrictions of the projections of $\mathbb{R}^4$ to $\mathbb{C}^* \times \mathbb{C}^*$. Similarly, the inverse of complex numbers

$$\theta(z) = \frac{1}{z} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$$

is continuous because

$$p_1 \circ \theta = \frac{p_1}{p_1^2 + p_2^2}, \quad p_2 \circ \theta = \frac{-p_2}{p_1^2 + p_2^2}.$$

□

The next topological group is based on spaces of matrices. Denote by $\mathsf{M}_{n\times m}(\mathbb{R})$ the space of matrices of n rows and m columns of real numbers. If $n = m$, we write $\mathsf{gl}(n, \mathbb{R})$. The set $\mathsf{M}_{n\times m}(\mathbb{R})$ is a group with respect to the addition of matrices. Using Exercise 2.7.6, consider on $\mathsf{M}_{n\times m}(\mathbb{R}) \to \mathbb{R}^{nm}$ the direct image topology identifying $\mathsf{M}_{n\times m}(\mathbb{R})$ with $\mathbb{R}^{nm}$ as usual (see (4.2)). As $\mathbb{R}^{nm}$ is a topological group, then $\mathsf{M}_{n\times m}(\mathbb{R})$ too. We also know that the usual topology on $\mathsf{gl}(n, \mathbb{R})$ is determined by the Euclidean metric $g(A, B) = \operatorname{trace}(AB^t)$ (Example 4.5).

Exercise 7.29 The general linear group defined by

$$\mathsf{GL}(n, \mathbb{R}) = \{A \in \mathsf{gl}(n, \mathbb{R}) : \det(A) \neq 0\}$$

is a topological group where the group operation is the multiplication of matrices. Prove that $\mathsf{GL}(n, \mathbb{R})$ is open in $\mathsf{gl}(n, \mathbb{R})$.

Solution We show that the multiplication of matrices is continuous in $\mathsf{gl}(n, \mathbb{R})$. Denote by a_{ij} (or A_{ij}) the (ij)-element of a matrix $A = (a_{ij})$. The multiplication is the mapping

$$P\colon \mathsf{gl}(n, \mathbb{R}) \times \mathsf{gl}(n, \mathbb{R}) \to \mathsf{gl}(n, \mathbb{R}), \quad P(A, B) = AB. \tag{7.6}$$

Let p_{ij} be the projections of $\mathsf{gl}(n, \mathbb{R}) = \mathbb{R}^{n^2}$ thanks to the identification (4.2). Then

$$\begin{aligned} p_{ij} \circ P(A, B) = (AB)_{ij} &= \sum_{k=1}^{n} a_{ik}b_{kj} = \sum_{k=1}^{n} p_{ik}(A)p_{kj}(B) \\ &= \sum_{k=1}^{n} (p_{ik} \circ \pi_1)(p_{kj} \circ \pi_2)(A, B). \end{aligned}$$

Here π_1 and π_2 stand for the corresponding projections of the product space $\mathsf{gl}(n, \mathbb{R}) \times \mathsf{gl}(n, \mathbb{R})$. Thus $p_{ij} \circ P$ are all continuous. We prove that the inversion

$$\theta\colon \mathsf{GL}(n, \mathbb{R}) \to \mathsf{GL}(n, \mathbb{R}), \quad \theta(A) = A^{-1}$$

is continuous. We carry out it in a sequence of steps.

1. First, we see that the determinant $\det\colon \mathsf{gl}(n, \mathbb{R}) \to \mathbb{R}$ is continuous. Recall that the determinant of $A = (a_{ij})$ is

$$\det(A) = \sum_{\sigma \in S_n} (-1)^{sgn(\sigma)} a_{1\sigma(1)} \dots a_{n\sigma(n)},$$

   where S_n is the symmetric group of order n and $sgn(\sigma)$ is the signature of the permutation $\sigma \in S_n$. Then

$$\det(A) = \sum_{\sigma \in S_n} (-1)^{sgn(\sigma)} (p_{1\sigma(1)} \dots p_{n\sigma(n)})(A),$$

and the map det can be expressed as

$$\det = \sum_{\sigma \in S_n} (-1)^{sgn(\sigma)} p_{1\sigma(1)} \cdots p_{n\sigma(n)}.$$

This proves that det is continuous on $\mathsf{gl}(n, \mathbb{R})$.

2. The transpose map $t\colon \mathsf{gl}(n, \mathbb{R}) \to \mathsf{gl}(n, \mathbb{R})$ defined by $t(A) = A^t$, is continuous because $p_{ij} \circ t = p_{ji}$.
3. The adjoint map $\mathrm{adj}\colon \mathsf{gl}(n, \mathbb{R}) \to \mathsf{gl}(n, \mathbb{R})$ that assigns each matrix A to its adjoint A^{ad} is continuous. The argument is similar to that of the determinant, by observing simply that if $i < j$, we have

$$\begin{aligned} p_{ij}(\mathrm{adj}(A)) &= (-1)^{i+j} \det(A^{ad}_{ij}) \\ &= (-1)^{i+j} \sum_{\tau \in S_{n-1}} a_{1\tau(1)} \ldots a_{i-1\tau(i-1)} a_{i+1\tau(i)} \ldots a_{j\tau(j-1)} a_{j+1\tau(j+1)} \ldots a_{n\tau(n)}. \end{aligned}$$

Finally, θ is continuous because

$$\theta = \frac{1}{\det}(t \circ \mathrm{adj}),$$

where all these maps are the restrictions to $\mathsf{GL}(n, \mathbb{R})$.

Finally, $\mathsf{GL}(n, \mathbb{R})$ is open in $\mathsf{gl}(n, \mathbb{R})$ because $\mathsf{GL}(n, \mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$ and $\mathbb{R} \setminus \{0\}$ is open. □

7.5 Suggested Exercises

7.5.1 Determine if the topological product of two excluded point topologies is comparable with the excluded point topology on the Cartesian product.

7.5.2 Let $A \subset X_1 \times X_2$ be an open (resp. closed) set. For each $x_1 \in X_1$, define $A_{x_1} = \{x_2 \in X_2 : (x_1, x_2) \in A\}$. Show that A_{x_1} is an open (resp. closed) set of X_2 for all $x_1 \in X_1$. Give examples of subsets of $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ such that A is not open (resp. closed) but A_x is open (resp. closed) for all $x \in \mathbb{R}$.

7.5.3 Determine if the properties T_0, T_1, T_2, first countable and second countable are productive.

7.5.4 Let $X = \{x_1, \ldots, x_n\}$ be a finite set equipped with the discrete topology and let $\{f_i\colon Y \to Y : 1 \le i \le n\}$ be a collection of continuous maps. Show that $\phi\colon X \times Y \to Y$ defined by $\phi(x_i, y) = f_i(y)$ is continuous.

7.5.5 If $f_i\colon X_i \to Y_i$, $i = 1, 2$, are open maps, prove that $f_1 \times f_2$ is open and determine if the converse is also true. Consider the same exercise replacing open by closed.

7.5.6 If (X, τ_{in}) is the included point topology for $p \in X$, study if projections from $(X \times X, \tau_{in} \times \tau_{in})$ onto (X, τ_{in}) are closed. Determine the continuity of

$$f\colon (X \times X, \tau_{in} \times \tau_{in}) \to (X, \tau_{in}), \quad f(x, y) = \begin{cases} x, & \text{if } x, y \neq p \\ p, & \text{otherwise.} \end{cases}$$

7.5.7 Let $\{X_i : i \in I\}$ be a collection of spaces and let $A_i \subset X_i$ for all $i \in I$. Determine if it is true the identity $\Pi_{i\in I}\overline{A_i} = \overline{\Pi_{i\in I}A_i}$ in the product topology and in the box topology.

7.5.8 Prove that $\mathbb{R}^n \setminus \{0\}$ is homeomorphic to $\mathbb{S}^{n-1} \times (0, \infty)$.

7.5.9 If $f\colon \mathbb{R}^n \to \mathbb{R}$ is continuous, prove that $A = \{(x, y) \in \mathbb{R}^n \times \mathbb{R} : y < f(x)\}$ and $B = \{(x, y) \in \mathbb{R}^n \times \mathbb{R} : y > f(x)\}$ are homeomorphic to $\mathbb{R}^n \times (0, \infty)$.

7.5.10 Prove that $\mathbb{R}$ endowed with the initial topology for the mappings

$$f_1, f_2\colon \mathbb{R} \to (\mathbb{R}, \tau_S), \quad f_1(x) = x + 1,\ f_2(x) = -x,$$

has discrete topology.

7.5.11 Prove that $F_1 = \mathbb{N}$ and $F_2 = \{-n + \frac{1}{n} : n \in \mathbb{N}, n \geq 2\}$ are closed in $\mathbb{R}$, but $F_1 + F_2$ is not.

7.5.12 Prove that $F_1 = \{(x, e^x) : x \in \mathbb{R}\}$ and $F_2 = \mathbb{R} \times \{0\}$ are closed in $\mathbb{R}^2$ but $F_1 + F_2$ is not. [Hint: show that every point $(x, 0)$ is adherent of $F_1 + F_2$ but does not belong to $F_1 + F_2$.]

7.5.13 Using Exercise 7.25, show that a union of two closed Hausdorff subspaces is Hausdorff. Give an example showing that the result fails if we change "closed" to "open".

7.5.14 Determine if arbitrary products of Hausdorff spaces are Hausdorff with the box topology.

7.5.15 In a topological group G, denote by $\mathscr{U}$ the collection of the neighborhoods of the unit element $e \in G$. Prove:

1. For each $U \in \mathscr{U}$, there is $V \in \mathscr{U}$ such that $VV \subset U$.
2. For each $U \in \mathscr{U}$, there is $V \in \mathscr{U}$ such that $V^{-1} \subset U$.
3. For each $U \in \mathscr{U}$, there is $V \in \mathscr{U}$ such that $VV^{-1} \subset U$.
4. For each $U \in \mathscr{U}$ and for each $a \in G$, there is $V \in \mathscr{U}$ such that $aVa^{-1} \subset U$.
5. For each $U \in \mathscr{U}$, there is $V \in \mathscr{U}$ with $V = V^{-1}$, such that $V \subset U$.

The notations V^{-1} and VV^{-1} are in the sense of groups. So, $V^{-1} = \{x^{-1} : x \in V\}$ and $VV^{-1} = \{xy^{-1} : x, y \in V\}$.

Chapter 8
Connectedness

Thus far, we have introduced the notion of topological spaces together with appropriate mappings between topological spaces, the continuous maps. Armed with the concepts of homeomorphism and topological invariants, we may now tackle the problem of classifying the spaces. In this chapter, we investigate one of the most fundamental invariants in topology, connectedness. In plain words, a connected space is simply a space made up of a unique piece. The simplicity of this idea does not undermine its importance and it appears in many fields of mathematics. Connectedness is employed as a powerful tool in the classification of topological spaces. A classic example is the problem of classification of Euclidean spaces (Theorem 6.1.6); we shall prove that for $n > 1$, the Euclidean real line $\mathbb{R}$ is not homeomorphic to another Euclidean space $\mathbb{R}^n$.

8.1 Connected Spaces and Examples

Before formulating the definition of a connected space, we try to understand what it means to have a space made up of a "single piece". From the point of view of set theory, every set X can be expressed as the union of two disjoint subsets by taking any subset A and writing $X = A \cup (X \setminus A)$. If now X is a topological space, it seems plausible to impose some topological assumptions on the elements of this partition and it is natural that A and $X \setminus A$ must be open sets.

Definition 8.1.1 A space X is said to be *connected* if the only partition of X by open sets is the trivial partition $\{\emptyset, X\}$. A subset A of X is called connected if A is connected as a topological subspace of X.

Henceforth, we say *open partition* to mean a partition by open sets. On the other hand, a space X is *disconnected* if there are two disjoint nontrivial open subsets A and B such that $X = A \cup B$. In this case, we say that $\{A, B\}$ is an *open separation*

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_8

of X. Because the complement of an open set is closed, this is equivalent to saying that a space is connected if the only partition by closed sets is trivial. Clearly, connectedness is a topological invariant.

Proposition 8.1.2 *The following statements are all equivalent:*

1. *The space is connected.*
2. *Trivial subsets are the only subsets that are open and closed.*
3. *Every continuous map from the space into a discrete space is constant.*
4. *There are no surjective continuous maps from the space onto* $\{0, 1\}$.

Example 8.1

1. The trivial topology is connected because the only open sets are the trivial ones.
2. A discrete space with more than one element is not connected because if $x \in X$, the singleton $\{x\}$ is open and closed.
3. A space with the property that any two nontrivial open sets intersect is connected (the same occurs replacing open by closed). Indeed, if $X = O_1 \cup O_2$ is an open partition then $O_1 \cap O_2 = \emptyset$, so O_1 and O_2 must be the trivial sets. Examples of such topologies are the Sierpinski topology, cofinite topology, included point topology and the right order topology.
4. The excluded point topology is connected because if $\{A, B\}$ is an open partition, one of these open sets, say A, must contain the distinguished point of the topology. By definition of the topology, A must be the entire space, so $B = \emptyset$.
5. The set of rational numbers $\mathbb{Q}$ is disconnected since $\{\mathbb{Q} \cap (-\infty, \sqrt{2}), \mathbb{Q} \cap (\sqrt{2}, \infty)\}$ is an open separation of $\mathbb{Q}$. In the same fashion, the set of irrationals $\mathbb{R} - \mathbb{Q}$ is disconnected because an open separation is $\{(\mathbb{R} - \mathbb{Q}) \cap (-\infty, 0), (\mathbb{R} - \mathbb{Q}) \cap (0, \infty)\}$.
6. The hyperboloid of two sheets $H = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 = -1\}$ is disconnected because
$$H^+ = \{(x, y, z) \in H : z > 0\}, \quad H^- = \{(x, y, z) \in H : z < 0\}$$
is an open separation of H (Fig. 8.1).
7. The space of regular matrices $\mathsf{GL}(n, \mathbb{R})$ defined in Exercise 7.29 is disconnected. Certainly, the determinant map $A \longmapsto \det(A)$ is continuous. Since $\det(A) \neq 0$

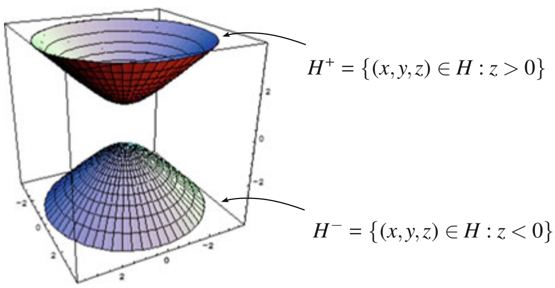


Fig. 8.1 The hyperboloid of two sheets is disconnected

for all $A \in \mathsf{GL}(n, \mathbb{R})$, an open separation of $\mathsf{GL}(n, \mathbb{R})$ is

$$\mathsf{GL}^+(n, \mathbb{R}) = \det^{-1}((0, \infty)), \quad \mathsf{GL}^-(n, \mathbb{R}) = \det^{-1}((-\infty, 0)).$$

8. The sum topology is not connected. If $X = \bigcup_{i \in I} X_i$, then each X_i is open because belongs to the basis for the topology. Since all X_i are open, all X_i are also closed (Remark 7.4.1). $\lozenge$

The first interesting space to study is the Euclidean real line. The next theorem can be considered the "crown jewel" of this chapter because it is the door to showing that many other subsets of Euclidean spaces are connected. We prove that intervals are the only connected sets of $\mathbb{R}$. The proof is not trivial and it uses fundamental properties of the real numbers.

Theorem 8.1.3 *Intervals are the only connected subsets of the Euclidean real line* $\mathbb{R}$.

Proof First we prove that any connected subset A of $\mathbb{R}$ must be an interval. Indeed, if A is not an interval, there are $a, b \in A$ and $c \notin A$ such that $a < c < b$. Then $\{A \cap (-\infty, c), A \cap (c, \infty)\}$ is an open separation of A. This argument was employed in Example 8.1 to prove that $\mathbb{Q}$ is disconnected (taking $c = \sqrt{2}$).

The reverse implication is proved by contradiction. Let I be an interval and let A and B be an open separation of I. Let $a \in A$ and $b \in B$ and without loss of generality, we may assume $a < b$. Define the set

$$C = \{x \in \mathbb{R} : [a, x) \subset A\}.$$

We will arrive to a contradiction after the following steps.

1. The set C is nonempty. Since $a \in A$ and A is open in I, there is an $r > 0$ such that $(a - r, a + r) \cap I \subset A$. Choose $r > 0$ small enough so $a + r < b$. Because I is an interval, $a + r \in I$ so $[a, a + r] \subset I$ and

$$[a, a + r) = [a, a + r) \cap I \subset (a - r, a + r) \cap I \subset A.$$

   This proves that $a + r \in C$, hence $C \neq \emptyset$.
2. The number b is an upper bound of C. Otherwise, there would be $x \in C$ such that $b < x$. Since $[a, x) \subset A$ and $a < b < x$, then $b \in A$, which is not possible.
3. Let $\alpha = \sup(C)$. Then $\alpha \in I$ because $a \le \alpha \le b$ and I is an interval.
4. We prove that $\alpha \in \overline{A}$, where $\overline{A}$ the closure of A in $\mathbb{R}$. This is achieved if for all $r > 0$, we have $(\alpha - r, \alpha + r) \cap A \neq \emptyset$. Let $r_0 > 0$ such that $0 < r_0 < r$ and $a + r_0 < \alpha$. By definition of supremum, there is $x \in C$ such that $\alpha - r_0 < x \le \alpha$. Since $[a, x) \subset A$ because $x \in C$, then $\alpha - r_0 \in A$, hence $\alpha - r_0 \in (\alpha - r, \alpha + r) \cap A$.
5. We prove that $\alpha \in A$. By the two preceding items and Proposition 3.3.7,

$$\alpha \in \overline{A} \cap I = \overline{A}^I = A,$$

   where the last equality occurs because A is closed in I.

We may now conclude the proof. Since $\alpha \leq b$ and $b \notin A$, then $\alpha < b$. Thus $a < \alpha < b$ and because A is open in I, let $\epsilon > 0$ such that

$$(\alpha - \epsilon, \alpha + \epsilon) \cap I \subset A.$$

Let ϵ be small enough so $a < \alpha - \epsilon < \alpha + \epsilon < b$. Since $a, b \in I$ and I is an interval, then $(\alpha - \epsilon, \alpha + \epsilon) \subset I$ and thus $(\alpha - \epsilon, \alpha + \epsilon) \subset A$. By definition of supremum, there is $c \in C$ such that $\alpha - \epsilon < c \leq \alpha$. Then

$$[a, \alpha + \epsilon) = [a, c) \cup (\alpha - \epsilon, \alpha + \epsilon) \subset A.$$

This proves $\alpha + \epsilon \in C$ but it contradicts that α is the supremum of C. □

Corollary 8.1.4

1. *The Euclidean real line is connected.*
2. *If $A \subset \mathbb{R}$, the intervals included in A are the only connected subsets of A.*

We offer three applications of Theorem 8.1.3.

1. We revisit the topological classification of the intervals of $\mathbb{R}$.
2. We exhibit many examples of connected subsets of $\mathbb{R}^n$.
3. We prove the celebrated theorems of intermediate value, Bolzano and fixed-point.

First we return to the topological classification of all intervals of $\mathbb{R}$ given in Chap. 6, giving a new proof.

Theorem 8.1.5 *Every interval $I \subset \mathbb{R}$ is homeomorphic to one of the following types:*

1. $\{0\}$, *if I has only one point.*
2. $[0, 1]$, *if $I = [a, b]$ for $a < b$.*
3. $[0, 1)$, *if $I = [a, b)$, $[a, \infty)$, $(-\infty, a]$ or $(a, b]$.*
4. $(0, 1)$, *if I is an open interval or $\mathbb{R}$.*

Moreover, intervals of different classes are not homeomorphic.

Proof It was proved in Theorem 6.1.7 and its corollaries that any two intervals of each class are homeomorphic. We use connectedness to prove that two intervals of different classes are not homeomorphic. Suppose by contradiction that $[0, 1)$ is homeomorphic to $(0, 1)$ and let $f : [0, 1) \to (0, 1)$ be a homeomorphism. Delete the endpoint 0 in $[0, 1)$ and its image under f. Then $[0, 1) \setminus \{0\} = (0, 1)$ is homeomorphic to $f([0, 1]) \setminus \{f(0)\} = (0, 1) \setminus \{f(0)\}$. However, $(0, 1)$ is connected but $(0, 1) \setminus \{f(0)\}$ is not because it is not an interval.

A similar argument proves that $[0, 1]$ is not homeomorphic to $(0, 1)$ and $[0, 1)$. In effect, if $f : [0, 1] \to (0, 1)$ is a homeomorphism, then $(0, 1] \cong (0, 1) \setminus \{f(0)\}$. This is not possible because the first space is connected but the second one is not, since it is not an interval. Similarly, if $f : [0, 1] \to [0, 1)$ is a homeomorphism, then

$$[0, 1] \setminus \{0, 1\} = (0, 1) \cong [0, 1) \setminus \{f(0), f(1)\},$$

obtaining a contradiction again because $(0, 1)$ is connected and $[0, 1) \setminus \{f(0), f(1)\}$ is not an interval. $\square$

As a consequence, if $f : [a, b] \to [c, d]$ is a homeomorphism, then $f(\{a, b\}) = \{c, d\}$. Analogously, if $f : [a, b) \to [c, d)$ is a homeomorphism, then $f(a) = c$.

We see that connectedness is a topological invariant; furthermore, connectedness is preserved under continuous maps.

Theorem 8.1.6 *If $f : X \to Y$ is continuous and X is connected, then $f(X)$ is connected.*

This result provides a strategy to prove that a space is connected: it suffices to show that the space is the image of a connected space under a continuous map.

Example 8.2 The circle $\mathbb{S}^1$ is connected. Furthermore, $\mathbb{S}^1$ and $\mathbb{R}$ are not homeomorphic.

Indeed, the circle $\mathbb{S}^1$ is connected because $\mathbb{S}^1 = p(\mathbb{R})$, where $p : \mathbb{R} \to \mathbb{S}^1$ is defined by $p(t) = (\cos t, \sin t)$. Now suppose that $\phi : \mathbb{S}^1 \to \mathbb{R}$ is a homeomorphism. If $x \in \mathbb{S}^1$ then $\mathbb{S}^1 \setminus \{x\}$ is homeomorphic to $\mathbb{R} \setminus \{\phi(x)\}$. However, a punctured circle is homeomorphic to $\mathbb{R}$ (Example 6.20) which it is connected, but $\mathbb{R} \setminus \{\phi(x)\}$ is not because it is not an interval. $\diamondsuit$

Example 8.3 Segments of $\mathbb{R}^n$ are connected. Following the notation of Exercise 6.23, the segment $[x, y]$ is $\alpha([0, 1])$. $\diamondsuit$

Theorem 8.1.7 *Let X be a connected space and let $f : X \to \mathbb{R}$ be a continuous map.*

1. *If $y_1, y_2 \in Im(f)$, then $[y_1, y_2] \subset Im(f)$ (intermediate value).*
2. *If $a, b \in X$ such that $f(a)f(b) < 0$, then there is $x_0 \in X$ such that $f(x_0) = 0$ (Bolzano).*
3. *If $f : [0, 1] \to [0, 1]$ is continuous then there is $x_0 \in [0, 1]$ such that $f(x_0) = x_0$ (fixed-point).*

Proof

1. Let $x_i \in X$ such that $f(x_i) = y_i$, $i = 1, 2$, and we may assume $y_1 \le y_2$. As $f(X)$ is connected, the set $f(X)$ is an interval. Since this interval contains y_1 and a y_2, it must contain the closed interval $[y_1, y_2]$ proving that $[y_1, y_2] \subset f(X)$.
2. Nothing is lost in assuming that $f(a) < 0 < f(b)$. By the intermediate value theorem, the interval $[f(a), f(b)] \subset \mathrm{Im}(f)$, so $0 \in f(X)$.
3. The function $h : [0, 1] \to [0, 1]$ defined by $h(x) = f(x) - x$ is continuous, $h(0) = f(0) \ge 0$ and $h(1) = f(1) - 1 \le 0$. See Fig. 8.2. By Bolzano's theorem, there is $x_0 \in [0, 1]$ such that $h(x_0) = 0$, thus $f(x_0) = x_0$.

$\square$

The above fixed-point theorem is the simplest version of a large family of similar results with profound implications in different fields of mathematics. Besides the interval $[0, 1]$, other space with this property is the n-dimensional *unit disc* $\mathbb{D}^n =$

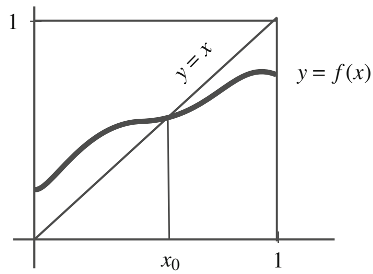


Fig. 8.2 The fixed-point theorem

$\{x \in \mathbb{R}^n : |x| \le 1\}$. However, the needed techniques are extremely sophisticated and we will be only able to prove the case $n = 2$ in Exercise 11.14.

The following result is not trivial and requires of connectedness.

Theorem 8.1.8 (Invariance of the Domain) *Let $A \subset \mathbb{R}$ be an open set. If $f : A \to \mathbb{R}$ is an embedding and A is connected, then $f(A)$ is an open set in $\mathbb{R}$.*

Proof As a first step, assume that A is a bounded open interval (a, b). Then $f(A)$ is connected, so it is an interval. Since $A \cong f(A)$, Theorem 8.1.5 implies that $f(A)$ is an open interval.

Now consider the general case that A an open set in $\mathbb{R}$. Then A is a union of bounded open intervals $A = \bigcup_{j\in J} I_j$ and

$$f(A) = f(\bigcup_{j\in J} I_j) = \bigcup_{j\in J} f(I_j).$$

Since the restriction $f_{|I_j} : I_j \to \mathbb{R}$ is an embedding, the first part of the proof leads to that $f(I_j)$ is an open interval, hence $f(A)$ is open. □

One cannot argue that $f(A)$ is open by saying that a homeomorphism maps open sets to open sets. In the present situation, the homeomorphism is $f : A \to f(A)$ but the hypothesis says that A is open *in* $\mathbb{R}$.

8.2 Further Properties of Connectedness

Although the idea behind the notion of connectedness of a space is very intuitive, it is, rather involved to decide if a space is connected in practice. In this section, we provide new techniques for determining whether a space is connected. The idea behind of the following result is that the union of a collection of connected subsets where all them overlap, is connected as may be seen pictorially in Fig. 8.3.

Theorem 8.2.1 *If one of the following statements holds, then X is connected.*

(i) There is a collection $\{X_i\}_{i\in I}$ of connected subsets of X such that $X = \bigcup_{i\in I} X_i$ and $\bigcap_{i\in I} X_i \neq \emptyset$.

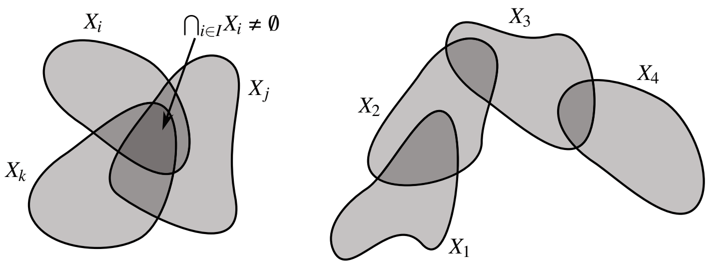


Fig. 8.3 Theorem 8.2.1, cases (i) (left) and (iv) (right)

(ii) There is $x_0 \in X$ such that for every $x \in X$, there is a connected subset containing x_0 and x.
(iii) For every $x, y \in X$, there is a connected subset containing x and y.
(iv) There is a countable collection $\{X_n : n \in \mathbb{N}\}$ of connected subsets of X such that $X = \bigcup_{n\in\mathbb{N}} X_n$ and $X_n \cap X_{n+1} \neq \emptyset$ for all $n \in \mathbb{N}$.

Example 8.4 Theorem 8.2.1 allows us to prove that intervals of $\mathbb{R}$ are connected if we only know that $[0, 1]$ is connected. Certainly, the intervals $(0, 1)$ and $[0, 1)$ can be expressed as union of bounded closed intervals with nonempty intersection, namely,

$$(0,1) = \bigcup_{n\geq 2} [\tfrac{1}{n}, 1 - \tfrac{1}{n}], \quad [0,1) = \bigcup_{n\geq 2} [0, 1 - \tfrac{1}{n}).$$

$\lozenge$

We illustrate how Theorem 8.2.1 may be employed. First, we recall two affine concepts of $\mathbb{R}^n$. A subset A of $\mathbb{R}^n$ is called *star-shaped* with respect to a point $x_0 \in A$ if for each $x \in A$, the segment $[x_0, x]$ is included in A. The set A is said to be *convex* if for each $x, y \in A$, the segment $[x, y]$ is included in A. Therefore every convex set is star-shaped with respect to any point of the set.

Example 8.5 We see that star-shaped and convex sets of $\mathbb{R}^n$ are connected, in particular, $\mathbb{R}^n$ is connected. If A is a star-shaped set with respect to x_0, the segments $[x_0, x]$ are included in A for all $x \in A$ and we apply (ii) of Th. 8.2.1. $\lozenge$

Example 8.6 For $n \geq 2$, the set $\mathbb{R}^n \setminus \{p\}$ is connected for all $p \in \mathbb{R}^n$. Indeed, after a translation of $\mathbb{R}^n$, we may assume that p is the origin 0 of $\mathbb{R}^n$. The connectedness of the punctured space $\mathbb{R}^n \setminus \{0\}$ is shown using (iii) of Theorem 8.2.1. Let $x, y \in \mathbb{R}^n \setminus \{0\}$.

1. If $[x, y] \subset \mathbb{R}^n \setminus \{0\}$, then $[x, y]$ is the connected set containing both points.
2. Suppose that $0 \in [x, y]$; in particular, 0, x and y are collinear. Pick a point $z \in \mathbb{R}^n$ that does not belong to the straight line containing $[x, y]$ which can be found because the dimension of $\mathbb{R}^n$ is bigger or equal than 2. Thus $[x, z]$ and $[z, y]$

are contained in $\mathbb{R}^n \setminus \{0\}$. Since these segments are connected, $[x, z] \cup [z, y]$ is connected.

$\Diamond$

We give a partial classification of Euclidean spaces (Theorem 6.1.6).

Theorem 8.2.2 *1. The Euclidean line $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^n$ for $n \geq 2$.*
2. The circle $\mathbb{S}^1$ is not homeomorphic to $\mathbb{S}^n$ for $n \geq 2$.

Proof

1. If $f: \mathbb{R} \to \mathbb{R}^n$ is a homeomorphism then $\mathbb{R} \setminus \{0\}$ is homeomorphic to $\mathbb{R}^n \setminus \{f(0)\}$. By the foregoing corollary, $\mathbb{R}^n \setminus \{f(0)\}$ is connected, but $\mathbb{R} \setminus \{0\}$ is not connected because it is not an interval.
2. If $f: \mathbb{S}^1 \to \mathbb{S}^n$ is a homeomorphism and $p \in \mathbb{S}^1$, then $\mathbb{S}^1 \setminus \{p\}$ is homeomorphic to $\mathbb{S}^n \setminus \{f(p)\}$. But $\mathbb{S}^1 \setminus \{p\} \cong \mathbb{R}$ and $\mathbb{S}^n \setminus \{f(p)\} \cong \mathbb{R}^n$.

$\square$

The next result reflects the intuitive idea that if one adds adherent points to a connected set, the new set remains connected.

Theorem 8.2.3 *Let X be a space and let A be a connected subset of X. If $B \subset X$ satisfies*

$$A \subset B \subset \overline{A},$$

then B is connected. In particular, $\overline{A}$ is connected.

Proof We employ Proposition 8.1.2. Let $f: B \to \{0, 1\}$ be a continuous map. Then $f_{|A}: A \to \{0, 1\}$ is continuous. Since A is connected, we deduce that $f(A) \neq \{0, 1\}$. The closure of A in B is $\overline{A}^B = \overline{A} \cap B = B$ because $B \subset \overline{A}$. Using Theorem 5.1.3 with the map $f: B \to \{0, 1\}$,

$$f(B) = f(\overline{A}^B) \subset \overline{f(A)} = f(A).$$

The last identity occurs because $\{0, 1\}$ is discrete, so every subset is closed. Consequently, $f(B) \subset f(A)$ and f is not surjective. $\square$

Example 8.7 The topologist's sine is connected. Following the notation of Example 4.21, $X = G(f) \cup \{(0, 0)\}$ and we know that $G(f) \cong (0, \infty)$ is connected. Since $(0, 0) \in \overline{G(f)}$, then X is connected. Analogously, $\overline{G(f)} = G(f) \cup (\{0\} \times [-1, 1])$ is also connected. $\Diamond$

Example 8.8 The sphere $\mathbb{S}^n$ is connected. For this, let $A = \mathbb{S}^n \setminus \{N\}$ where N is the north pole. Then A is connected because A is homeomorphic to $\mathbb{R}^n$ (Example 6.20), so $\overline{A}$ is connected. We show that $\overline{A} = \mathbb{S}^n$. It only remains to verify that N is an adherent point of A. But this is true because $((0, \ldots, 0, \cos(t_n), \sin(t_n)) \subset A \to N$ as $(t_n) \nearrow \pi/2$. $\Diamond$

Once we know that $\mathbb{S}^n$ is connected, the next result establishes right away that for any real continuous function on a sphere there are two antipodal points having the same image.

Example 8.9 (Borsuk-Ulam) We prove that if $f\colon \mathbb{S}^n \to \mathbb{R}$ is continuous, then there is $x_0 \in \mathbb{S}^n$ such that $f(x_0) = f(-x_0)$. Define

$$h\colon \mathbb{S}^n \to \mathbb{R}, \quad h(x) = f(x) - f(-x).$$

This map is continuous because $h = f - f \circ A$, where $A = -\mathbb{1}_{\mathbb{S}^n}$. The result is proved if we find a solution of the equation $h(x) = 0$. Fix a point y of $\mathbb{S}^n$. There are two possibilities. If $h(y) = 0$, the point y is the solution that we are looking for. Otherwise, we may assume $h(y) > 0$. Then

$$h(-y) = f(-y) - f(y) = -h(y) < 0.$$

Since $\mathbb{S}^n$ is connected, the Bolzano's theorem asserts the existence of $x_0 \in \mathbb{S}^n$ such that $h(x_0) = 0$ and x_0 is the required solution. $\diamondsuit$

The *antipodal point* of $x \in \mathbb{S}^n$ is $-x$, which also belongs to $\mathbb{S}^n$. It is customary to interpret the theorem of Borsuk-Ulam as follows. On the Earth's surface, consider the function $T = T(x)$ that measures the temperature at each point x of the Earth. The temperature is a continuous function because it does not change abruptly between close points. The Borsuk-Ulam theorem asserts that *there are two antipodal points on the Earth with identical temperature*. Of course, nothing is said about which points these are.

Example 8.10 We show four methods to prove that $\mathbb{S}^n$ is connected.

1. Apply (i) of Theorem 8.2.1. Let $X_1 = \mathbb{S}^n \setminus \{N\}$ and $X_2 = \mathbb{S}^n \setminus \{S\}$. Then X_1 and X_2 are connected (homeomorphic to $\mathbb{R}^n$) such that $\mathbb{S}^n = X_1 \cup X_2$ and $X_1 \cap X_2 \neq \emptyset$.
2. Apply induction on n, the dimension of $\mathbb{S}^n$. For $n = 1$, $\mathbb{S}^1$ is connected by Example 8.2. We prove that $\mathbb{S}^n$ is connected assuming that $\mathbb{S}^{n-1}$ is connected. Use (ii) of Theorem 8.2.1 fixing $x_0 = N$. Let $x \in \mathbb{S}^n$ and consider a vector hyperplane $H \subset \mathbb{R}^{n+1}$ throughout N and x (in case that $x = -N$, there are infinite; if $x \neq -N$, we extend $\{N, x\}$ by n linearly independent vectors $\{N, x, \ldots, x_{n-1}\}$ and let $H = L(\{N, x, \ldots, x_{n-1}\})$). Then $H \cap \mathbb{S}^n$ is a set homeomorphic to $\mathbb{S}^{n-1}$ (Exercise 6.6.15), so it is connected by the induction hypothesis.
3. Apply (ii) of Theorem 8.2.1. Again, let $x_0 = N$ and $x \in \mathbb{S}^n$. Consider a 2-vector plane P containing N and x (if $x = -N$, there are many but if $x \neq N$, then $P = L(\{N, x\})$). The set $P \cap \mathbb{S}^n$ is a circle of $\mathbb{R}^{n+1}$, which is connected.
4. The sphere $\mathbb{S}^n$ is the image under a continuous map of a connected set, such as it was proved for $\mathbb{S}^1$. See the details in Exercise 8.30 for the case $n = 2$. $\diamondsuit$

The following theorem tells us that connectedness is preserved under product.

Theorem 8.2.4 *Two spaces X and Y are connected if and only $X \times Y$ is connected.*

Example 8.11

1. The cylinder $\mathbb{S}^1 \times \mathbb{R}$ and the torus $\mathbb{S}^1 \times \mathbb{S}^1$ are connected. More generally, the product spaces $\mathbb{S}^n \times \mathbb{R}^m$ and $\mathbb{S}^n \times \mathbb{S}^m$ are connected.
2. $\mathbb{R}^n \setminus \{(0, 0)\}$, $n \geq 2$, is connected because it was shown in Example 6.18 that $\mathbb{R}^n \setminus \{0\} \cong \mathbb{S}^{n-1} \times \mathbb{R}$.

It is immediate by induction that Theorem 8.2.4 can be extended for a finite number of spaces, but this is not so clear for an arbitrary number of spaces.

Theorem 8.2.5 *If $\{X_i : i \in I\}$ is a collection of connected spaces, then $\Pi_{i\in I} X_i$ is connected.*

Proof The proof utilizes Theorem 8.2.3 finding a connected subset A whose closure is the whole space $\Pi_{i\in I} X_i$. Fix a point $a = (a_i) \in \Pi_{i\in I} X_i$.

1. For each finite subset of indices $J \subset I$, define $A_J = \{(x_i) : x_i = a_i, \text{ for all } i \notin J\}$. Observe that $a \in A_J$ for all finite sets J. We prove that A_J is connected with the help of (ii) of Theorem 8.2.1. Let $x \in A_J$ be given. Define
$$B = \Pi_{i\in I} Y_i, \quad \text{where } Y_i = \begin{cases} \{a_i\}, & \text{if } i \notin J \\ X_i, & \text{if } i \in J. \end{cases}$$
The set B is homeomorphic to $\Pi_{j\in J} X_j$ (Corollary 7.2.5) which is connected since it is a finite product of connected sets. Moreover $a, x \in B$.
2. Define
$$A = \bigcup \{A_J : J \subset I,\ J \text{ is finite}\}.$$
Using (i) of Theorem 8.2.1, this set is connected because it is the union of the connected spaces A_J. Moreover, $\bigcap_J A_J \neq \emptyset$ because $a \in A_J$ for all J.
3. Finally, we show that $\overline{A} = \Pi_{i\in I} X_i$. Let $x = (x_i) \in \Pi_{i\in I} X_i$ and let U be a basic-neighborhood of x,
$$U = \Pi_{i\in I} U_i, \quad \text{where } U_i = \begin{cases} X_i, & \text{if } i \notin J \\ U_i \in \mathscr{U}^i_{x_i}, & \text{if } i \in J, \end{cases}$$
for a certain finite set of indices $J \subset I$. Then the point $y = (y_i)$ defined by
$$y_i = \begin{cases} a_i, & \text{if } i \notin J \\ x_i, & \text{if } i \in J, \end{cases}$$
belongs to $U \cap A_J$, so $U \cap A \neq \emptyset$.

$\square$

8.3 Connected Components

We have frequently repeated that the idea of a connected space is that it cannot be divided into pieces. A thorough understanding of the notion of connectedness lies in understanding clearly the properties of these pieces.

Definition 8.3.1 Let X be a space and $x \in X$. The *connected component* C_x of x is the union of all connected sets of X containing x,

$$C_x = \bigcup \{C \subset X : C \text{ is connected, } x \in C\}.$$

Notice that $C_x \neq \emptyset$ because C_x contains at least $\{x\}$. The following properties reflect that the connected components are the pieces of a space.

Proposition 8.3.2

1. *The connected component of a point is connected and it is the largest connected set containing x.*
2. *The space is connected if and only if there is a unique connected component.*
3. *The collection of the connected components is a partition of the space.*
4. *Every connected component is a closed set.*

The partition of the connected components defines the equivalence relation

$$x \sim y \text{ if } x \text{ and } y \text{ belong to the same connected component.}$$

In other words, $x \sim y$ if there is a connected set $C \subset X$ such that $x, y \in C$.

Example 8.12 1. Let (X, τ_D) be the discrete space. Since every subset of X has the discrete topology (Example 2.26) and C_x is connected, the set C_x must have only one element, so $C_x = \{x\}$.
2. The components of $\mathbb{R} \setminus \{0\}$ are $(-\infty, 0)$ and $(0, \infty)$ because they are the largest intervals contained in $\mathbb{R} \setminus \{0\}$ (Corollary 8.1.4).
3. Arguing similarly, the connected components of $\mathbb{R} \setminus \mathbb{Z}$ are the intervals $(n, n+1)$, $n \in \mathbb{Z}$.
4. The connected components of $\mathbb{Q}$ are the singletons because they are the only intervals inside $\mathbb{Q}$. The connected components are not open (compare with (4) in Proposition 8.3.2). Besides the discrete topology, the set $\mathbb{Q}$ is another example of a space where the connected components are the singletons. These spaces are called *totally disconnected*.
5. The Cantor set is also totally disconnected. So, if $x \in \mathfrak{C}$ and C_x is the component of x, then C_x is an interval of $\mathbb{R}$ contained in $\mathfrak{C}$. Since $\mathfrak{C}$ cannot contain open intervals, necessarily $C_x = [x, x] = \{x\}$. $\Diamond$

If a space is disconnected, then there is an open separation. Also, the space has more than one connected component. One may hope that the sets that constitute the separation are the family of the connected components. However, this does not hold in general. The following examples are relevant.

Example 8.13

1. In $\mathbb{Q}$, $\{\mathbb{Q} \cap (-\infty, \sqrt{2}), \mathbb{Q} \cap (\sqrt{2}, \infty)\}$ is an open separation but the connected components of $\mathbb{Q}$ are the singletons.
2. Consider $X = \{1, 2, 3, 4\}$ with the discrete topology. The subsets $\{\{1\}, \{2, 3, 4\}\}$ constitute an open separation but $\{2, 3, 4\}$ is not a connected component of X because the components are the singletons.
3. A partition by connected sets of $\mathbb{R}$ is $\{(-\infty, 0], (0, \infty)\}$ but $\mathbb{R}$ is connected.

$\Diamond$

We give a sufficient condition that guarantees that a partition by connected sets coincides with the partition of the connected components.

Proposition 8.3.3 *If $\{A_i : i \in I\}$ is a partition by connected open subsets of a space X, then $\{A_i : i \in I\}$ is the collection of connected components of X.*

Example 8.14 We see that $\mathbb{R}^{n+1} \setminus \mathbb{S}^n$ has exactly two connected components. Indeed,

$$\mathbb{R}^{n+1} \setminus \mathbb{S}^n = \{x \in \mathbb{R}^n : |x| < 1\} \cup \{x \in \mathbb{R}^{n+1} : |x| > 1\}.$$

Both subsets are open in $\mathbb{R}^{n+1}$, so open in $\mathbb{R}^{n+1} \setminus \mathbb{S}^n$. The first set is the ball $B_1(0)$ which is connected because it is convex. The other set is the annulus $A(1, \infty) \subset \mathbb{R}^{n+1}$ which is connected because it is homeomorphic to $\mathbb{S}^n \times \mathbb{R}$ (Example 6.18). Thus $\{x \in \mathbb{R}^n : |x| < 1\}$ and $\{x \in \mathbb{R}^{n+1} : |x| > 1\}$ are the connected components of $\mathbb{R}^{n+1} \setminus \mathbb{S}^n$. $\Diamond$

Since every continuous mapping preserves connected sets, we deduce the following behavior of the connected components by continuous maps.

Proposition 8.3.4

1. *Let $f : X \to Y$ be a continuous map. If $x \in X$ then $f(C_x) \subset C_{f(x)}$. If, in addition, f is a homeomorphism then $f(C_x) = C_{f(x)}$; in particular, the number of connected components is a topological invariant.*
2. *If $(x_1, x_2) \in X_1 \times X_2$, then $C_{(x_1,x_2)} = C_{x_1} \times C_{x_2}$.*

In view of (1) of the above proposition, we can make the next definition.

Definition 8.3.5 Let $n \in \mathbb{N}$. A point x of a space X is called an *intersection point of* n*th order* if $X \setminus \{x\}$ has exactly n connected components.

The order of intersection is preserved by homeomorphisms (Proposition 8.3.4). As a result, the number of all points of nth order coincides in X and in Y.

Example 8.15

1. Every $x \in \mathbb{R}$ is an intersection point of 2nd order.
2. For $n \geq 2$, every point of $\mathbb{R}^n$ is an intersection point of 1st order. In particular, $\mathbb{R}^n$ is not homeomorphic to $\mathbb{R}$, which it is the proof of Theorem 8.2.2.
3. Let $X = [0, 1] \cup \{2\}$. We find the components of $X \setminus \{x\}$ by looking for the largest intervals contained in $X \setminus \{x\}$. A point $x \in (0, 1)$ is of 3rd order because the components of $X \setminus \{x\}$ are $[0, x)$, $(x, 1]$ and $\{2\}$. For $x = 0$, the components of $X \setminus \{0\}$ are $(0, 1]$ and $\{2\}$ and for $x = 1$, the components of $X \setminus \{1\}$ are $[0, 1)$ and $\{2\}$. Finally, $x = 2$ is an intersection point of first order, because $X \setminus \{2\} = [0, 1]$ is connected.
4. Consider $(0, 1) \cup \{2\}$. Then there are no intersection points of order 2, so $(0, 1) \cup \{2\}$ is not homeomorphic to $[0, 1] \cup \{2\}$. ♢

Example 8.16 The circle $\mathbb{S}^1$ is not homeomorphic to any subset of $\mathbb{R}$. Indeed, if $\mathbb{S}^1$ is homeomorphic to a subset A of $\mathbb{R}$, then A must be an interval because $\mathbb{S}^1$ is connected. Every point p of $\mathbb{S}^1$ has order 1 because $\mathbb{S}^1 \setminus \{p\} \cong \mathbb{R}$. But in an interval (except singletons) every point of its interior is an intersection point of 2nd order.

As a consequence, $\mathbb{S}^1$ cannot be embedded in $\mathbb{R}$. Since the inclusion $\mathbb{S}^1 \hookrightarrow \mathbb{R}^2$ is an embedding, we conclude that 2 is the smallest dimension of the Euclidean space where $\mathbb{S}^1$ can be embedded. ♢

8.4 Path Connectedness

An inspection of the arguments employed in many of the previous spaces shows the utility of segments. Of course, segments are connected. Recall that a segment $[x, y] \subset \mathbb{R}^n$ is the image of the map $\alpha\colon \mathbf{I} \to \mathbb{R}^n$ defined by $\alpha(t) = x + t(y - x)$ where $\mathbf{I} = [0, 1]$. The affine structure of $\mathbb{R}^n$ cannot be generalized to an arbitrary topological space but the connectedness of $\alpha(\mathbf{I})$ is only a consequence of the continuity of α and the connectedness of $\mathbf{I}$. This motivates the next definition.

Definition 8.4.1 A *path* in X is a continuous map $\alpha\colon \mathbf{I} \to X$. In this case, we say that α *joins* the beginning point $\alpha(0)$ and the endpoint $\alpha(1)$.

Example 8.17 Comparing with the parametrizations of segments, paths may not be injective and we can go back and forth along the path. For instance, if $x, y \in \mathbb{R}$ and $\alpha(t) = x + t(y - x)$, define

$$\beta\colon [0, 5\pi/2] \to \mathbb{R}, \quad \beta(t) = \alpha(\sin(t)).$$

Then $\beta(0) = x$, $\beta(5\pi/2) = y$ but β is not injective because $\beta(0) = \beta(\pi)$. Furthermore, we have $[x, y] \subset \beta([0, 5\pi/2])$ and this inclusion is proper because $\beta(3\pi/2) = 2x - y \notin [x, y]$. ♢

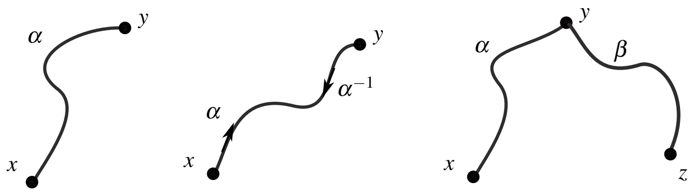


Fig. 8.4 A path between p and 1 (left), a path α and its inverse α^-1 (middle) and the operation $*$ for two paths (right)

A path is connected, or to be more precise, its image $\alpha(\mathbf{I})$ is connected. In some sense, paths replace the role of segments in arbitrary topological spaces (Fig. 8.4, left).

In the notion of a path the map α has an implicit direction because we are going from the point $\alpha(0)$ to $\alpha(1)$ as t goes runs on $\mathbf{I}$. We can reverse the direction going from $\alpha(1)$ to $\alpha(0)$ (Fig. 8.4, middle). To proceed formally, define the *inverse path* of α, denoted α^{-1}, by

$$\alpha^{-1} \colon \mathbf{I} \to X, \quad \alpha^{-1}(t) = \alpha(1-t).$$

The image of the inverse path coincides with of α, $\alpha^{-1}(\mathbf{I}) = \alpha(\mathbf{I})$, but now the curve joins $\alpha(1)$ with $\alpha(0)$. A path in a topological space will not be bijective in general so here α^{-1} is not the inverse as a map. Curiously, in the definition of a segment in $\mathbb{R}^n$, the map $\alpha \colon \mathbf{I} \to \alpha(\mathbf{I})$ is really a homeomorphism (Exercise 6.23).

Another technique of construct paths is to glue two paths together where the endpoint of one path is equal to the beginning point of the other one. Let α, β be two paths such that $\alpha(1) = \beta(0)$. The idea is clear. Start at $\alpha(0)$ running at double speed along α so at $t = 1/2$ we arrive at the point $\alpha(1)$. Since $\alpha(1) = \beta(0)$, we continue through β and running at double speed again, so we finish at $t = 1$ at the point $\beta(1)$. We formalize this idea. Define a new path $\alpha * \beta \colon \mathbf{I} \to X$ by

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t), & \text{if } t \in [0, \frac{1}{2}] \\ \beta(2t-1), & \text{if } t \in [\frac{1}{2}, 1]. \end{cases}$$

Notice that the function $t \mapsto 2t$ for $t \in [0, 1/2]$ runs all the domain $\mathbf{I}$ of α and the second map $t \mapsto 2t - 1$, $t \in [1/2, 1]$, runs the domain $\mathbf{I}$ of β from $t = 1/2$ to $t = 1$ such as indicated in Fig. 8.4, right. The map $\alpha * \beta$ is well defined because α and β coincide at $t = 1/2$. For the continuity of $\alpha * \beta$, we use the Pasting Lemma. First, $[0, 1/2]$ and $[1/2, 1]$ are closed in $[0, 1]$. Now we see that the restrictions of $\alpha * \beta$ in each one of the intervals are continuous. The first restriction is $\alpha * \beta \colon [0, 1/2] \to X$ which is continuous because it is the composition $\alpha \circ \phi$, where $\phi \colon [0, 1/2] \to [0, 1]$ is $\phi(t) = 2t$. In the same fashion, $\alpha * \beta \colon [1/2, 1] \to X$ is continuous.

Proposition 8.4.2 *On a space X, the binary relation*

$$x \sim y \text{ if there is a path joining } x \text{ with } y$$

is an equivalence relation. The equivalence class of $x \in X$, denoted P_x, is called the path-component of x.

Definition 8.4.3 A space is said to be *path connected* if every two points of the space can be joined by a path. Equivalently, a path connected space has a unique path-component.

Proposition 8.4.4

1. *Every path-component is path connected.*
2. *Every path-component is connected. In particular, every path-component is contained in some connected component.*
3. *Every path connected space is connected.*
4. *If $f: X \to Y$ is continuous and $x \in X$, then $f(P_x) \subset P_{f(x)}$. In particular, if X is path connected, then $f(X)$ is path connected.*
5. *Path connectedness is a topological invariant.*

Theorem 8.2.1 has its analogue for path connections, replacing 'connected' by 'path connected'. As with the connected components, the path-component of $x \in X$ is the largest path connected set containing x.

Example 8.18

1. Star-shaped subsets and convex subsets of $\mathbb{R}^n$ are path connected (Fig. 8.5). Here we use segments as paths.
2. The path connected subsets of $\mathbb{R}$ are the intervals. Indeed, a path connected set is connected so it must be an interval. Conversely, an interval of $\mathbb{R}$ is convex, so is path connected by the previous item.
3. A discrete space is not path connected because it is disconnected. Moreover, the path-components are the unitary sets.
4. The set $\mathbb{Q}$ of rationals is not path connected because it is not connected and, again, the path-components are the singletons.
5. The circle $\mathbb{S}^1$ is path connected because $\mathbb{S}^1 = p(\mathbb{R})$, where $p(t) = (\cos t, \sin t)$. Here we use (4) of Proposition 8.4.4.

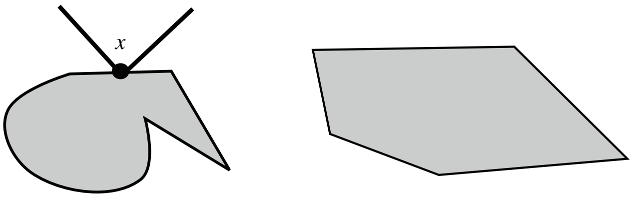


Fig. 8.5 A star-shaped set (left) and a convex set in Euclidean plane

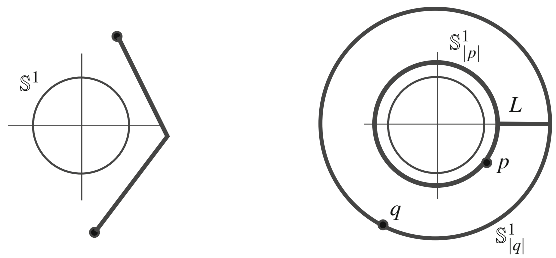


Fig. 8.6 The set $C_2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 > 1\}$ is path connected

6. Spheres are path connected. We use (iii) of the analogous Theorem 8.2.1. So let $x, y \in \mathbb{S}^n$ and let $p \in \mathbb{S}^n$ be a third point. Then $x, y \in \mathbb{S}^n \setminus \{p\} \cong \mathbb{R}^n$ that is path connected.
7. The excluded point topology is path connected. If p is the distinguished point, we show that p can be joined with every point $x \in X$. Define $\alpha\colon \mathbf{I} \to X$ by

$$\alpha(t) = \begin{cases} p, & \text{if } t \in [0, \frac{1}{2}] \\ x, & \text{if } t \in (\frac{1}{2}, 1]. \end{cases}$$

Then α is continuous because for each open set $O \subset X$,

$$\alpha^{-1}(O) = \begin{cases} \emptyset, & p, x \notin O \\ (\frac{1}{2}, 1], & \text{if } p \notin O, x \in O \\ [0, 1], & \text{if } p \in O, \end{cases}$$

since the only open set containing p is X. $\Diamond$

Example 8.19 By Example 8.14, the connected components of $\mathbb{R}^2 \setminus \mathbb{S}^1$ are $C_1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$ and $C_2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 > 1\}$. Also, C_1 and C_2 are path connected. Indeed, first note that C_1 is convex, so it is path connected. The proof concludes if we prove that C_2 is path connected because any path-component is included in a connected component (Proposition 8.4.4).

There are various ways to prove that C_2 is path connected. For example, any two points of C_2 can be joined by a finite number of segments (one followed by the other) in C_2, so (iii) of the analogous Theorem 8.2.1 can be utilized (Fig. 8.6, left). Another argument is the following. If $p, q \in C_2$, let $\mathbb{S}^1_{|p|}$ and $\mathbb{S}^1_{|q|}$ be the circles centered at the origin of radii $|p|$ and $|q|$, respectively. Both circles are contained in C_2, contain p and q and are path connected. However, they are, in general, disjoint. Now consider the segment L joining the points $(|p|, 0)$ and $(|q|, 0)$, which is contained in C_2. The proof concludes if we show that $\mathbb{S}^1_{|p|} \cup L \cup \mathbb{S}^1_{|q|}$ is path connected (Fig. 8.6, right). To accomplish this, apply (iv) of Theorem 8.2.1, noting that $\mathbb{S}^1_{|p|} \cap L = (|p|, 0)$ and $L \cap \mathbb{S}^1_{|q|} = (|q|, 0)$.

$\Diamond$

Example 8.20 The equation

$$e^x(y+1)+xy-1=0$$

has at least three zeroes in $\mathbb{R}^2$. Indeed, consider the continuous function $f(x,y) = e^x(y+1)+xy-1$ and $a=(-1,0)$, $b=(1,0)$. Then $f(a)=e^{-1}-1<0$ and $f(b)=e-1>0$. Consider $c_y=(0,y)$ and the path from a to b whose image is $[a,c_y]\cup[c_y,b]$, or in other words, $\alpha*\beta$, where α and β parametrize $[a,c_y]$ and $[c_y,b]$, respectively. The intersection of all these paths is $\{a,b\}$ and thus, for each $y\in\mathbb{R}$, there is $p_y\in(a,c_y]\cup[c_y,b)$ such that $f(p_y)=0$. $\lozenge$

One should observe that until now, we have not seen a connected space that is not path connected.

Example 8.21 Let $X=A\cup B$ be the topologist's sine, where

$$A=\left\{\left(x,\sin\tfrac{1}{x}\right):x\in(0,\infty)\right\},\qquad B=\{(0,0)\}.$$

The sets A and B are path connected because $A\cong(0,\infty)$ and B is formed by only one point. Now we prove that the points $(0,0)\in B$ and $(1/\pi,0)\in A$ cannot be joined by a path in X and, as a consequence, the sets A and B are the path-components. If $\alpha\colon\mathbf{I}\to X$ is such a path, denote $\alpha(t)=(x(t),y(t))$, where $x(t)$ and $y(t)$ are the coordinates of α. In view that $\alpha(0)=(0,0)$ and $\alpha(1)=(1/\pi,0)$,

$$x(0)=0,\ x(1)=\frac{1}{\pi}.$$

Since $0<2/(3\pi)<1/\pi$, the intermediate theorem for the function $x=x(t)$ implies that there is $t_2\in(0,1)$ such that $x(t_2)=2/3\pi$. Because $x(t_2)\neq0$, we have $\alpha(t_2)\in A$ and by definition of A, $y(t_2)=-1$. We restrict α to the interval $[0,t_2]$ and likewise, we define a decreasing sequence $(t_n)\subset[0,1]$ such that

$$x(t_n)=\frac{2}{(2n-1)\pi}.$$

Thus $y(t_n)=(-1)^{n+1}$. Since (t_n) is decreasing and bounded from below, the sequence (t_n) converges to its infimum $a\in[0,1]$. Since $y=y(t)$ is continuous, $(y(t_n))\to y(a)$ which is not possible because $(y(t_n))$ is not convergent. $\lozenge$

From this example, it is worth noting the following.

1. The closure of a path connected subset may not be path connected, such as A in the topologist's sine because $\overline{A}=X$.
2. A path-component need be neither a closed set nor open. For example, in the topologist's sine, the set A is not closed and B is not open.
3. With a similar argument, the path-components of $Y=A\cup C$, where $C=\{0\}\times[-1,1]$, are A and C.

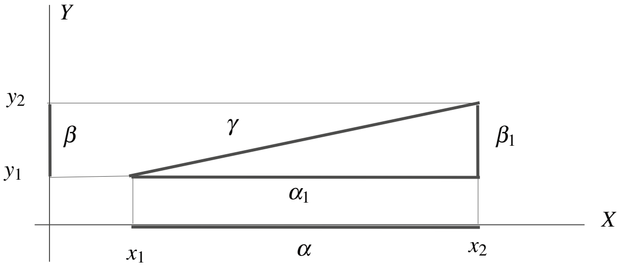


Fig. 8.7 The paths of the proof of Proposition 8.4.6

We give a sufficient condition for a connected space to be path connected.

Theorem 8.4.5 *If every point has a path connected neighborhood, then path-components and connected components coincide. In particular, if every point of a connected space has a path connected neighborhood, then the space is path connected.*

Example 8.22 In an open set of $\mathbb{R}^n$, path-components coincide with connected components. In particular, a connected open set of $\mathbb{R}^n$ is path connected. Indeed, if $A \subset \mathbb{R}^n$ is connected and open and $x \in A$, there is $r > 0$ such that $B_r(x) \subset A$. This ball is a path connected neighborhood of x (homeomorphic to $\mathbb{R}^n$) and the result follows from the theorem.

This result is not applicable in general to closed sets: in Example 8.21, the closure of A is $\overline{A} = A \cup (\{0\} \times [-1, 1])$ which is connected but not path connected. $\diamondsuit$

This section concludes with the proof that path connectedness is a productive property.

Proposition 8.4.6 *Two spaces X and Y are path connected if and only if $X \times Y$ is path connected.*

Proof Using the projections p and p' and (4) of Proposition 8.4.4, if $X \times Y$ is path connected then $X = p(X \times Y)$ and $Y = p'(X \times Y)$ are also path connected.

For the reverse implication, suppose that X and Y are path connected and let $(x_1, y_1), (x_2, y_2) \in X \times Y$. Let α and β be two paths in X and Y respectively joining x_1 with x_2 and y_1 with y_2, respectively. Define

$$\gamma : \mathbf{I} \to X \times Y, \quad \gamma(t) = (\alpha(t), \beta(t)).$$

See Fig. 8.7. Then γ is continuous because $p \circ \gamma = \alpha$ and $p' \circ \gamma = \beta$. Moreover, γ joins (x_1, y_1) with (x_2, y_2).

$\square$

Example 8.23 The cylinder $\mathbb{S}^1 \times \mathbb{R}$, the n-torus $\mathbb{T}^n = \mathbb{S}^1 \times \overset{n}{\cdots} \times \mathbb{S}^1$ and the product spaces $\mathbb{S}^n \times \mathbb{R}^m$ and $\mathbb{S}^n \times \mathbb{S}^m$, are path connected. $\diamondsuit$

Example 8.24 In the proof of Proposition 8.4.6, we find another path between the points (x_1, y_1) and (x_2, y_2). Indeed, define

$$\alpha_1 : \mathbf{I} \to X \times Y,\ \alpha_1(t) = (\alpha(t), y_1),$$

$$\beta_1 : \mathbf{I} \to X \times Y,\ \beta_1(t) = (x_2, \beta(t)),$$

and clearly α_1 and β_1 are continuous. As $\alpha_1(1) = \beta_1(0) = (x_2, y_1)$, the path $\alpha_1 * \beta_1$ joins (x_1, y_1) with (x_2, y_2). In $\mathbb{R}^2$, we can picture the paths γ and $\alpha_1 * \beta_1$, where $X = Y = \mathbb{R}$. As indicated in Fig. 8.7, choose α and β to be the segments

$$\alpha(t) = x_1 + t(x_2 - x_1), \quad \beta(t) = y_1 + t(y_2 - y_1).$$

Then γ is

$$\begin{aligned}\gamma(t) = (\alpha(t), \beta(t)) &= (x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1)) \\ &= (x_1, y_1) + t((x_2, y_2) - (x_1, y_1)).\end{aligned}$$

Thus γ is the segment $[(x_1, y_1), (x_2, y_2)] \subset \mathbb{R}^2$. On the other hand,

$$\alpha_1(t) = (x_1 + t(x_2 - y_2), y_1), \quad \beta_1(t) = (x_2, y_1 + t(y_2 - y_1)),$$

and the image of $\alpha_1 * \beta_1$ is

$$\mathrm{Im}(\alpha_1 * \beta_1) = \mathrm{Im}(\alpha_1) \cup \mathrm{Im}(\beta_1) = [(x_1, y_1), (x_2, y_1)] \cup [(x_2, y_1), (x_2, y_2)],$$

that is, the union of two segments of $\mathbb{R}^2$ whose intersection is (x_2, y_1). $\Diamond$

8.5 Worked Exercises

Exercise 8.1 Find the connected sets of the Sorgenfrey line.

Solution Let $\beta_S = \{[a, b) : a < b, a, b \in \mathbb{R}\}$ be the standard basis for τ_S. Recall that an interval $[a, b)$ is open and closed and thus $(\mathbb{R}, \tau_S)$ is disconnected. As a consequence, the Sorgenfrey line is not homeomorphic to the Euclidean real line.

We study those subsets of $(\mathbb{R}, \tau_S)$ that are connected. First, if A is connected, we show that A must be an interval. The argument follows the same steps as in the Euclidean real line. Certainly, if A is not an interval, there are $a, b \in A$, $a < b$, and $c \notin A$ with $a < c < b$. In view that (c, ∞) and $(-\infty, c)$ are open in τ_S and constitute an open separation of A, namely, $A = (A \cap (-\infty, c)) \cup (A \cap (c, \infty))$, we deduce that A is not connected. In fact, because $\tau_u \subset \tau_S$, if A is not connected in $(\mathbb{R}, \tau_u)$, then it is not connected either in $(\mathbb{R}, \tau_S)$.

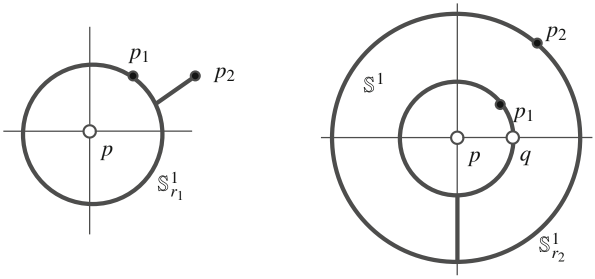


Fig. 8.8 The sets of Exercises 8.2 (left, case $n = 2$) and 8.3 (right, case $n = 2$)

Finally, we check what intervals are connected. Using again that the intervals $[a, b)$ are open and closed, a connected interval cannot contain an interval of this type, hence the only intervals with such a property are the intervals with only one point. Thus *singletons are the only connected subsets of the Sorgenfrey line*. This space is then totally disconnected. □

Exercise 8.2 If $n \geq 2$, prove that the punctured space $\mathbb{R}^n \setminus \{p\}$ is path connected.

Solution We present two proofs. First, $\mathbb{R}^n \setminus \{p\} \cong \mathbb{S}^{n-1} \times \mathbb{R}$ by Exercise 7.5.8, so $\mathbb{R}^n \setminus \{p\}$ is path connected because it is the product of two path connected spaces. The second proof uses (iii) of the analogous Theorem 8.2.1. After a translation of $\mathbb{R}^n$, we may suppose that p is the origin of coordinates. We will find for each $p_1, p_2 \in \mathbb{R}^n \setminus \{0\}$, a path connected set $\Delta_{12} \subset \mathbb{R}^n \setminus \{0\}$ containing p_1 and p_2. Recall that the segment $[p_1, p_2]$ joining both points may be outside $\mathbb{R}^n \setminus \{0\}$ as for example when $p_2 = -p_1$.

Let $r_1 = |p_1|$ and $r_2 = |p_2|$ with $r_1 \leq r_2$ and define

$$\Delta_{12} = [p_2, \frac{r_1}{r_2} p_2] \cup \mathbb{S}^n_{r_1},$$

where $\mathbb{S}^n_{r_1}$ is the sphere of radius r_1 centered at the origin (Fig. 8.8, left). Since the segment $[p_2, \frac{r_1}{r_2} p_2]$ and the sphere $\mathbb{S}^n_{r_1}$ are path connected and

$$[p_2, \frac{r_1}{r_2} p_2] \cap \mathbb{S}^n_{r_1} = \left\{ \frac{r_1}{r_2} p_2 \right\},$$

then Δ_{12} is connected. Observe that $\Delta_{12} \subset \mathbb{R}^n \setminus \{0\}$ because the points of the sphere have modulus r_1 and of the segment are bigger than or equal to $|\frac{r_1}{r_2} p_2| = r_1 > 0$. □

We remove two points from $\mathbb{R}^n$, proving that the resulting set is path connected.

Exercise 8.3 If $n \geq 2$ and $p, q \in \mathbb{R}^n$, prove that $\mathbb{R}^n \setminus \{p, q\}$ is path connected.

Solution We simplify the arguments observing that the points p and q can be fixed initially. This was proved in Exercise 6.21 for a finite number of removed points. Thus we may assume that $p = (0, \ldots, 0)$ and $q = (1, 0, \ldots, 0)$ and we prove that $\mathbb{R}^n \setminus \{p, q\}$ is path connected.

Let p_1 and p_2 be two points of $\mathbb{R}^n \setminus \{p, q\}$ and set $r_i = |p_i|$, $i = 1, 2$. The argument is entirely similar as in the preceding exercise but distinguishing more cases, depending if $r_i = 1$ or $r_i \neq 1$. The value 1 is special because it is the modulus of q.

1. Case $r_i \neq 1$, for $i = 1, 2$. Let $c_i = (0, \dots, 0, -r_i)$, $i = 1, 2$, and

$$\Delta_{12} = \mathbb{S}^{n-1}_{r_1} \cup [c_1, c_2] \cup \mathbb{S}^{n-1}_{r_2}.$$

   The set Δ_{12} is path connected using (iv) of Theorem 8.2.1. In effect, $\mathbb{S}^{n-1}(r_1)$ and the segment $[c_1, c_2]$ are path connected whose intersection is the point c_1. Also $[c_1, c_2]$ and $\mathbb{S}^{n-1}_{r_2}$ are path connected intersecting at the point c_2. Thus the union of the three sets is path connected. Finally, it is clear that $p, q \notin \Delta_{12}$ because the points of $\mathbb{S}^{n-1}_{r_1}$ and $\mathbb{S}^{n-1}_{r_2}$ have modulus r_1 and r_2, respectively, and it is obvious that $p, q \notin [c_1, c_2]$.
2. Case $r_1 = 1$ and $r_2 \neq 1$. Let

$$\Delta_{12} = \mathbb{S}^{n-1} \setminus \{q\} \cup [c_1, c_2] \cup \mathbb{S}^{n-1}_{r_2}.$$

   See Fig. 8.8, right. This set is path connected by (iv) of Theorem 8.2.1 again. Indeed, the three sets are path connected and $\mathbb{S}^{n-1} \setminus \{q\}$ intersects $[c_1, c_2]$ at the point c_1 and the segment $[c_1, c_2]$ intersects $\mathbb{S}^{n-1}_{r_2}$ at c_2. Also, $p, q \notin \mathbb{S}^{n-1} \setminus \{q\}$.
3. Case $r_1 = r_2 = 1$. Let $\Delta_{12} = \mathbb{S}^{n-1} \setminus \{q\}$. This set is homeomorphic to $\mathbb{R}^{n-1}$ which is connected, and clearly $p, q \in \Delta_{12}$. □

We prove that connectedness is not preserved by the box topology.

Exercise 8.4 Prove that the set of sequences of real numbers $\mathbb{R}^\omega$ endowed with the box topology τ_b is not connected.

Solution As a point set, $\mathbb{R}^\omega$ is the product space $\Pi_{n\in\mathbb{N}}\mathbb{R}$, so $\mathbb{R}^\omega$ is connected with the product topology. We prove that $\mathbb{R}^\omega$ with the box topology is not connected because

$$A = \{(x_n) : (x_n) \text{ is bounded}\}, \quad B = \mathbb{R}^\omega \setminus A$$

is an open separation of $\mathbb{R}^\omega$. We show that A and B are open sets in the box topology.

1. The set A is open. We prove that all its points are interior. Let $x = (x_n) \in A$ be given. Because the sequence x is bounded, there is $M > 0$ such that $|x_n| < M$ for all $n \in \mathbb{N}$. Set $U = \Pi_{n\in\mathbb{N}}(x_n - M, x_n + M)$. This set is open in the box topology and contains x, so U is a neighborhood of x. We see that $U \subset A$. If $y = (y_n) \in U$, the sequence y is bounded because using that $y_n \in (x_n - M, x_n + M)$, we derive that

$$|y_n| \leq |x_n| + |y_n - x_n| < |x_n| + M \leq 2M.$$

2. The set B is open. Again, we prove that all its points are interior. If $x = (x_n) \in B$, let $V = \Pi_{n \in \mathbb{N}}(x_n - 1, x_n + 1)$. Then V is open in the box topology and contains x. We prove that $V \subset B$. Let $y = (y_n) \in V$ be given. The sequence y is not bounded, on the contrary, x is also bounded because

$$|x_n| \leq |y_n| + |x_n - y_n| < |y_n| + 1.$$

Since this is not possible, then $y \in B$, so $x \in V \subset B$. □

Exercise 8.5 Prove that connectedness is preserved under continuous maps.

Solution Let $f: X \to Y$ be a continuous map where X is a space. We must show that $f(X)$ is connected. Let $A \subset f(X)$ be an open and closed set. Since $f: X \to f(X)$ is continuous, $f^{-1}(A)$ is open and closed in X. By connectedness, $f^{-1}(A)$ is a trivial subset of X. If $f^{-1}(A) = \emptyset$ then $A = f(f^{-1}(A)) = f(\emptyset) = \emptyset$ and if $f^{-1}(A) = X$ then $A = f(f^{-1}(A)) = f(X)$. In either case, A is a trivial subset of $f(X)$. □

In subsequent exercises, we shall revisit three results using the characterizations of Proposition 8.1.2.

Exercise 8.6 Prove Theorem 8.2.3.

Solution We present two new proofs.

1. (By definition of connectedness.) Let $B = C \cup D$ be an open partition of B. This partition gives another one in A by intersecting with A, $A = (A \cap C) \cup (A \cap D)$. Observe that $A \cap C$ and $A \cap D$ are open in A by the transitivity of relative topologies. Since A is connected, this partition is trivial, so assume $A \cap C = \emptyset$ and $A \cap D = A$. Thus $A \subset D$. Since $D \subset B$, we have $A \subset D \subset B$. In these inclusions, take closures in B yielding

$$\overline{A}^B \subset \overline{D}^B \subset B. \tag{8.1}$$

On the other hand, using Proposition 3.3.7, it follows that $\overline{A}^B = \overline{A} \cap B$. But this intersection is B because $B \subset \overline{A}$. By (8.1), we deduce $B = \overline{D}^B$. Since D is closed (and open) in B, $D = B$ and consequently, $C = \emptyset$.
2. (By Proposition 8.1.2.) Let $C \subset B$ be an open and closed set in B. Then $A \cap C$ is open and closed in A by the transitivity of relative topologies. Since A is connected, then $A \cap C = \emptyset$ or $A \cap C = A$.

   (a) Case $A \cap C = \emptyset$. Then $A \subset B \setminus C$ and taking closures in B,

   $$\overline{A}^B \subset \overline{B \setminus C}^B = B \setminus \text{int}(C)^B = B \setminus C,$$

   because C is open in B. Using that $\overline{A}^B = B = \overline{A} \cap B = B$, the above inclusion reduces into $B \subset B \setminus C$, so $C = \emptyset$.

(b) Case $A \cap C = A$. Then $A \subset C$. In this inclusion, take closures in B again, deducing

$$B = \overline{A}^B \subset \overline{C}^B = C \subset B,$$

where we use that C is closed in B. Thus $C = B$. □

Exercise 8.7 Prove Theorem 8.2.4.

Solution As in the preceding exercise, we present two proofs.

1. (By definition of connectedness.) Let $X \times Y = A \cup B$ be an open partition. Fixing $x \in X$, this partition determines another one in $\{x\} \times Y$,

$$\{x\} \times Y = \Big(A \cap (\{x\} \times Y)\Big) \cup \Big(B \cap (\{x\} \times Y)\Big).$$

The space $\{x\} \times Y \cong Y$ is connected, so the above partition is trivial. If, for instance, $\{x\} \times Y = A \cap (\{x\} \times Y)$, then $\{x\} \times Y \subset A$. If $\{x\} \times Y = B \cap (\{x\} \times Y)$, we deduce $\{x\} \times Y \subset B$. Doing the argument for all $x \in X$, we infer that either $\{x\} \times Y$ is included in A or in B. Suppose that there are $x_1, x_2 \in X$ such $\{x_1\} \times Y \subset A$ and $\{x_2\} \times Y \subset B$. Let $y \in Y$ be an arbitrary but fixed point. The set $X \times \{y\}$ is connected because it is homeomorphic to X. The reasoning as above proves that $X \times \{y\} \subset A$ or $X \times \{y\} \subset B$. However, neither of the two cases is possible because $(x_1, y) \in A$ and $(x_2, y) \in B$. This contradiction proves that $\{x\} \times Y \subset A$ for all $x \in X$ or $\{x\} \times Y \subset B$ for all $x \in X$. If it occurs, say the first case, then

$$X \times Y = \bigcup_{x \in X} \{x\} \times Y \subset A,$$

that is, $X \times Y = A$ and the partition is trivial.
2. (By Proposition 8.1.2.) Let $f: X \times Y \to \{0, 1\}$ be a continuous map. For every $y \in Y$, define

$$f^y: X \to \{0, 1\}, \quad f^y(x) = f(x, y).$$

This map is continuous and because X is connected, then f^y is constant. This implies that for each $y \in Y$, the set $B^y = f^y(X) = \{f(x, y) : x \in X\}$ is $\{0\}$ or $\{1\}$. In a similar way, define for each $x \in X$ the maps

$$f_x: Y \to \{0, 1\}, \quad f_x(y) = f(x, y)$$

and again, each one of these maps is constant because Y is connected. Now, for all $x \in X$, the sets $A_x = f_x(Y) = \{f(x, y) : y \in Y\}$ are formed by one element, namely, $\{0\}$ or $\{1\}$. We claim that all B^y must be $\{0\}$ or all must be $\{1\}$ and this concludes that f is constant. If this were not the case, there are y_1 and y_2 such that $B^{y_1} = \{0\}$ and $B^{y_2} = \{0\}$. Let $x \in X$ be given. Then $f(x, y_1) = 0$ and $f(x, y_2) = 1$, hence $f_x(y_1) = 0$ and $f_x(y_2) = 1$. This contradicts our assumption that $A_x = \{0\}$ or $A_x = \{1\}$. □

Exercise 8.8 Let $X = (\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})$ be the two coordinate axes of $\mathbb{R}^2$. Prove that (0, 0) is an intersection point of 4th order and that the rest have order 2.

Solution We compute the connected components by finding a partition of the space by connected open sets (Proposition 8.3.3). Let $p \in X$ be given.

1. Case $p \neq (0, 0)$. Assume that $p = (x_0, 0)$ with $x_0 > 0$ because for the other possibilities, the arguments are similar. We see that the connected components of $X \setminus \{p\}$ are $A = (x_0, \infty) \times \{0\}$ and $(X \setminus \{p\}) \setminus A$ (Fig. 8.9, left). The set A is connected because it is product space of connected sets. Further,

$$(X \setminus \{p\}) \setminus A = ((-\infty, x_0) \times \{0\}) \bigcup (\{0\} \times \mathbb{R})$$

is connected because it is union of connected sets (topological product of two connected sets) whose intersection is (0, 0) because $x_0 > 0$.

On the other hand, A and $(X \setminus \{p\}) \setminus A$ are open in $X \setminus \{p\}$ because

$$A = \{(x, y) \in \mathbb{R}^2 : x > x_0\} \cap (X \setminus \{p\}),$$

$$(X \setminus \{p\}) \setminus A = \{(x, y) \in \mathbb{R}^2 : x < x_0\} \cap (X \setminus \{p\}).$$

2. Case $p = (0, 0)$. Then $X \setminus \{p\}$ is the union of the four half axes

$$(0, \infty) \times \{0\}, \quad (-\infty, 0) \times \{0\}, \quad \{0\} \times (0, \infty), \quad \{0\} \times (-\infty, 0).$$

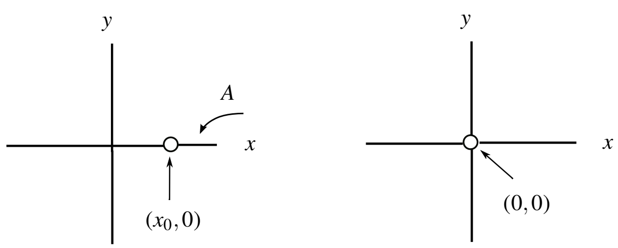


Fig. 8.9 The set $(\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})$ without the point $(x_0, 0)$, $x_0 > 0$, has two connected components, but if we remove the origin, there are four components

We show that they are precisely the four components of $X \setminus \{p\}$. First, all them are connected because they are the product of two connected sets. Also, all them are open sets in X because they are the intersection of open half planes of $\mathbb{R}^2$ with X. For example, the first half axis is

$$(0, \infty) \times \{0\} = \{(x, y) \in \mathbb{R}^2 : x > 0\} \cap (X \setminus \{p\})$$

and a similar argument applies to the other half axes. □

We characterize connectedness of the complement of affine subspaces of Euclidean spaces in terms of its dimension.

Exercise 8.9 Let $H \subset \mathbb{R}^n$ be an affine subspace with $n \geq 2$. Prove that $\mathbb{R}^n \setminus H$ is disconnected if and only if $\dim(H) = n - 1$.

Solution After a homeomorphism of $\mathbb{R}^n$ to itself, we can assume that H is a particular affine subspace. Indeed, as in Example 6.17, if $m = \dim(H)$ we can assume that H is the vector subspace

$$H = \{x = (x_1, \dots, x_n) \in \mathbb{R}^n : x_{m+1} = 0, \dots, x_n = 0\}.$$

1. Case $\dim(H) = n - 1$. Then $H = \{x \in \mathbb{R}^n : x_n = 0\}$ and write

$$\mathbb{R}^n \setminus H = \{x \in \mathbb{R}^n : x_n > 0\} \cup \{x \in \mathbb{R}^n : x_n < 0\}.$$

   Each one of the two sets on the right-hand side are open in $\mathbb{R}^n \setminus H$ because $\{x \in \mathbb{R}^n : x_n < 0\}$ and $\{x \in \mathbb{R}^n : x_n < 0\}$ are open in $\mathbb{R}^n$. This proves that $\mathbb{R}^n \setminus H$ is disconnected. Even more, this partition forms the connected components of $\mathbb{R}^n \setminus H$ because each one of these sets is open and connected. Notice that, for example, $\{x \in \mathbb{R}^n : x_n > 0\} = \mathbb{R}^{n-1} \times (0, \infty)$.
2. Case $\dim(H) < n - 1$. We prove that $\mathbb{R}^n \setminus H$ is path connected. Let $p = (0, \dots, 0, 1) \in \mathbb{R}^n \setminus H$. Use (ii) of the analogous Theorem 8.2.1, showing that for each $x \in \mathbb{R}^n \setminus H$, there is a path connected set containing p and x. This path connected set is either a segment, or a union of two segments. We distinguish two cases depending on the coordinates of x.

   (a) Case $x_i \neq 0$ for some $i \in \{m + 1, \dots, n - 1\}$. Consider the segment $[p, x]$ and we show that every point of $[p, x]$ is included $\mathbb{R}^n \setminus H$. Otherwise, there is $t \in (0, 1)$ such that $p + t(x - p) \in H$. In particular, its ith coordinate is 0, but this coordinate is tx_i, a contradiction.
   (b) Case $x_i = 0$ for all $i \in \{m+1, \dots, n-1\}$. Note that $x_n \neq 0$ (on the contrary, x would belong to H). Let $q = (0, \dots, 0, 1, 0) \notin H$, where the existence of q is assured because $m < n - 1$. Consider $C = [p, q] \cup [q, x]$. This set is

path connected containing p and x. It remains to show that $C \subset \mathbb{R}^n \setminus H$. If $z \in [p, q]$ then $z = p + t(q - p)$ for some $t \in (0, 1)$. Then

$$z = p + t(q - p) = (0, \ldots, 0, t, 1 - t) \notin H.$$

If $z \in [q, x]$ then there is $t \in (0, 1)$ such that

$$z = q + t(x - q) = (tx_1, \ldots, tx_{n-2}, 1 - t, tx_n).$$

If $z \in H$, $tx_n = 1 - t = 0$, so $t = 1$ and $x_n = 0$, a contradiction. □

Exercise 8.10 Determine the connectedness of the space given in Exercise 3.21.

Solution

1. The space is not connected because $\{1, 2\}$ and $\{3, 4, \ldots\}$ is an open partition.
2. We study the connected subspaces of $\mathbb{N}$. Let $A \subset \mathbb{N}$. We prove that A is connected if and only if A is formed by only one element, or A is $\{2n-1, 2n\}$ for a certain $n \in \mathbb{N}$. A first observation is that the relative topology of $\{2n - 1, 2n\}$ is trivial, hence $\{2n - 1, 2n\}$ is connected.

   Reciprocally, let A be a set with more than one element and distinct from $\{2n-1, 2n\}$ for all n. Let $a \in A$ the minimum of A. We have two possibilities.

   (a) Case $a = 2n$ for some $n \in \mathbb{N}$. Then $O_1 = \{1, \ldots, 2n\}$ and $O_2 = \{2n + 1, 2n + 2, \ldots\}$ is an open separation of $\mathbb{N}$ that provides an open separation of A because there are elements in $O_2 \cap A$. This proves that A is not connected.
   (b) Case $a = 2n - 1$ for some $n \in \mathbb{N}$. Take $b \in A$ such that $b > 2n$, which is possible since A has more than two elements and A is distinct of $\{2n-1, 2n\}$. Now $O_1 = \{1, \ldots, 2n\}$ and $O_2 = \{2n+1, 2n+2, \ldots\}$ is an open separation of $\mathbb{N}$ that induces other one of A.

3. We calculate the connected components. By the preceding item, the component of $2n$ is $C_{2n} = \{2n - 1, 2n\}$ and of $2n - 1$, $C_{2n-1} = \{2n - 1, 2n\}$. □

Exercise 8.11 Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous map. Find the connected components of $\mathbb{R}^{n+1} \setminus G(f)$.

Solution In order to facilitate the arguments, we stand for the points of $\mathbb{R}^{n+1}$ as $(x, y) \in \mathbb{R}^n \times \mathbb{R}$. It is clear that $\mathbb{R}^{n+1} \setminus G(f) = A \cup B$, where

$$A = \{(x, y) \in \mathbb{R}^{n+1} : y < f(x)\}, \quad B = \{(x, y) \in \mathbb{R}^{n+1} : y > f(x)\}.$$

We prove that A and B are the connected components. First, they are open in $G(f)$. Certainly, the continuous function

$$F : \mathbb{R}^{n+1} \to \mathbb{R}, \quad F(x, y) = y - f(x)$$

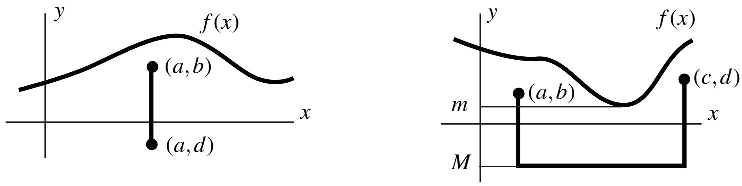


Fig. 8.10 The set $y < f(x)$ of the proof of Exercise 8.11 is connected

implies that $A = F^{-1}((-\infty, 0))$ and $B = F^{-1}((0, \infty))$ are open in $\mathbb{R}^{n+1}$, hence open in $\mathbb{R}^{n+1} \setminus G(f)$. Thus A and B are the connected components if they are connected. We perform the proof for A and the same manner holds for B.

One can invoke Exercise 7.5.9 where was proved that A is homeomorphic to $\mathbb{R}^n \times (0, \infty) \cong \mathbb{R}^{n+1}$. Another way is proving that for any two points of A, there is a connected set containing both points. In fact, this connected set is a finite union of segments of $\mathbb{R}^{n+1}$, where each one intersects the next one (in fact A is path connectedness). Indeed, let $(a, b), (c, d) \in A$, with $b < f(a)$ and $d < f(c)$. There are two cases to discuss.

1. Case $a = c$. It is immediate that the segment $[(a, b), (a, d)]$ is included in A because any point of this segment is of the form $(a, b + t(d - b), t \in [0, 1]$ and $b + t(d - b) \leq \max\{b, d\} < f(a)$. See Fig. 8.10, left.
2. Case $a \neq c$, which we may assume, $a < c$. Consider the restriction of f to $[a, c]$, $f : [a, c] \to \mathbb{R}$. Since the domain is a bounded closed interval, we learned from calculus that f attains a minimum m, $m \leq f(t)$ for all $t \in [a, c]$. Set $M = m - 1$. Then the connected set containing (a, b) and (c, d) is the union of the next three segments:

$$[(a, b), (a, M)] \cup [(a, M), (c, M)] \cup [(c, M), (c, d)].$$

See Fig. 8.10, right. We check that each one of the three segments are included in A. The first segment is constituted by points of type $(a, b+t(M-b))$. Since b and M are strictly less that $f(a)$, the x_{n+1} coordinates of the points of the segment are strictly less or equal than b and $b < f(a)$. A similar argument is equally valid for the third segment $[(c, M), (c, d)]$. The points of the second segment are of the form (t, M) with $t \in [a, c]$. Then

$$f(t) \geq m > m - 1 = M,$$

so $(t, M) \in A$. . □

Exercise 8.12 In a space X, define a binary relation:

$$x \sim y \text{ if there is no open separation } X = A \cup B \text{ with } x \in A \text{ and } y \in B$$

Prove that $\sim$ is an equivalence relation. Each equivalence class is called *quasicomponent*. Prove that the quasicomponent of a point is the intersection of all open and closed subsets that contain the point x. Prove that the connected components are included in the quasicomponents and that a quasicomponent is the union of the connected components of its elements.

Solution

1. The reflexive and symmetry properties are obvious. For the transitive, assume $x \sim y$ and $y \sim z$. Let $X = A \cup B$ be an open separation and suppose that $x \in A$. Because $x \sim y$, then $y \in A$. Using $y \sim z$, then $z \in A$. This proves the transitivity. Denote by Q_x the quasicomponent of x.
2. Let

$$K_x = \bigcap \{G \in \tau \cap \mathscr{F} : x \in G\}.$$

   We see the set-theoretically identity $Q_x = K_x$ by the method of showing that each of the sets is contained in the other. Let $y \in Q_x$. Suppose that there is $G \in \tau \cap \mathscr{F}$, $x \in G$ such that $y \notin G$. Then $\{G, X \setminus G\}$ constitutes an open separation with $x \in G$ and $y \in X \setminus G$. Thus x and y are not related by $\sim$, a contradiction.

   Let $y \in K_x$ be given. If $x \not\sim y$, there there is an open separation $X = A \cup B$ with $x \in A$ and $y \in B$. As a consequence, the set A is open and closed with $x \in A$ so $K_x \subset A$. Since $y \in B$, we obtain a contradiction.
3. We prove the inclusion $C_x \subset Q_x$. Let $y \in C_x$. Let $X = A \cup B$ be an open separation and suppose that, for instance, that $x \in A$. Then $C_x \cap A$ and $C_x \cap B$ is an open separation of C_x. Since C_x is connected, this separation is trivial. In view that $x \in C_x \cap A$, $C_x = C_x \cap A$. This implies, $C_x \subset A$. Thus $y \in A$ hence $x \sim y$.
4. We must see the identity $Q_x = \bigcup_{y \in Q_x} C_y$. The inclusion $Q_x \subset \bigcup_{y \in Q_x} C_y$ is obvious. Let $z \in \bigcup_{y \in Q_x} C_y$ and suppose that $z \in C_y$ for some $y \in Q_x$. Since $z \in C_y \subset Q_y$, then $z \sim y$. Since $y \sim x$, by transitivity we have $z \sim x$, that is $z \in Q_x$. □

We show a space where the connected components do not coincide with the quasicomponents.

Exercise 8.13 Consider $X = \bigcup_{n \in \mathbb{N}} L_n \cup \{(0, 0), (0, 1)\}$, where $L_n = \{1/n\} \times \mathbf{I}$. See Fig. 8.11, left. Find the connected components and quasicomponents.

Solution

1. Each set L_n is connected because L_n is the product of two connected sets. We see that the connected components of X are the sets L_n together with $\{(0, 0)\}$ and $\{(0, 1)\}$. It is clear that L_n is closed in X because L_n is closed in $\mathbb{R}^2$ (product of two closed sets). If $n > 1$, openness of L_n follows because

$$L_n = X \cap \{(x, y) \in \mathbb{R}^2 : \tfrac{1}{n+1} < x < \tfrac{1}{n-1}\}.$$

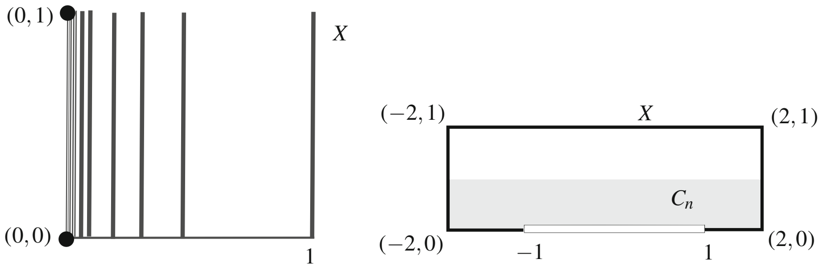


Fig. 8.11 The set X of Exercise 8.13 (left) and that of Exercise 8.14 (right)

For L_1 it is enough to observe that $L_1 = X \cap \{(x, y) \in \mathbb{R}^2 : x > 1/2\}$. Since L_n is closed and open in X, L_n cannot be included in a larger connected set, so L_n is a connected component of X. Finally, the singletons $\{(0, 0)\}$ and $\{(0, 1)\}$ are connected but $\{(0, 0), (0, 1)\}$ is not, because its topology is discrete, so $\{(0, 0)\}$ and $\{(0, 1)\}$ are connected components.

2. Since each L_n is open and closed in X, by (2) of the foregoing exercise, they are quasicomponents. We see that $\{(0, 0), (0, 1)\}$ is also a quasicomponent. Otherwise, if $(0, 0) \not\sim (0, 1)$, then there is an open separation of $X = A \cup B$ with $(0, 0) \in A$ and $(0, 1) \in B$. Since A and B are open sets of X, there is $r > 0$ such that $B_r(0, 0) \cap X \subset A$ and $B_r(0, 1) \cap X \subset B$. If $\frac{1}{n} < r$, then $(\frac{1}{n}, 0) \in B_r(0, 0)$ and $(\frac{1}{n}, 1) \in B_r(0, 1)$. As both points belong to L_n, then $(0, 0) \sim (\frac{1}{n}, 0)$, giving a contradiction.

Exercise 8.14 Consider the set of $\mathbb{R}^2$ given by

$$X = ([-2, 2] \times [0, 1]) \setminus ([-1, 1] \times \{0\}).$$

See Fig. 8.11, right. Let $C_n = \{(x, y) \in X : y < 1/n\}$. Prove that C_n is connected but $\bigcap_{n\in\mathbb{N}} C_n$ is not connected.

Solution

1. The set C_n can be written as

$$C_n = \left([-2, 1) \times [0, \tfrac{1}{n})\right) \cup \left([-2, 2] \times (0, \tfrac{1}{n})\right) \cup \left((1, 2] \times [0, \tfrac{1}{n})\right).$$

Each one of these three sets is connected because it is the topological product of connected sets. The first set intersects the second one and the second one, the third one. Thus the union of all them is connected.

2. Observe that $C_{n+1} \subset C_n$ for all $n \in \mathbb{N}$. It is clear that

$$\bigcap_{n\in\mathbb{N}} C_n = ([-2,-1) \times \{0\}) \cup ((1,2] \times \{0\}) = ([-2,-1) \cup (1,2]) \times \{0\}.$$

Thus this set is homeomorphic to $[-2,-1) \cup (1,2]$ which is not connected. □

The next exercise establishes that every homeomorphism between two intervals of $\mathbb{R}$ is strictly monotone. Although this is expected by the homeomorphisms that appeared in Chap. 6, the following proof appeals directly to the intermediate value theorem.

Exercise 8.15 Prove that every homeomorphism between two intervals is strictly monotone.

Solution Let $f : I \to J$ be an homeomorphism where I and J are two intervals. Let $a, b \in I$. Suppose without loss of generality that $f(a) < f(b)$.

1. We prove that f is increasing in $[a, b]$. Let $c \in (a, b)$ be given. Suppose that $f(c) > f(b)$. The restriction of f to the interval $[a, c]$ is continuous and $f(a) < f(b) < f(c)$. By the intermediate value theorem, there is $x \in (a, c)$ such that $f(x) = f(b)$, which is absurd because f is injective. Analogously, it can be proved that $f(c) < f(a)$ is not possible. Definitively, we have $f(a) < f(c) < f(b)$ for all $c \in (a, b)$. Note that a and b are arbitrary with $f(a) < f(b)$.

   If $x, y \in (a, b)$ with $x < y$, then $f(a) < f(y) < f(b)$ and repeat the above argument but now in the interval $[a, y]$, obtaining $f(a) < f(x) < f(y)$.
2. We prove that f is increasing in I. If this were not the case, there is $x < a$ such that $f(x) > f(a)$ or there is $x > b$ such that $f(x) < f(b)$. We may suppose the first case (the other one is analogous). The function f takes all values of the interval $[f(a), f(b)]$. As $f(x) > f(a)$, let $y \in (a, b)$ such that $f(y) < f(x)$. Then in the interval $[x, y]$, we have $f(x) > f(y)$. We have proved previously that in such a case, the function is strictly decreasing. In particular f is decreasing in $[a, y] \subset [x, y]$, which is contradictory because it was proven in part (1) that f is increasing. □

Exercise 8.16 If $C_n = [0, 1] \times \{\frac{1}{n}\}$, prove that

$$A = (\{0\} \times \mathbb{R}) \cup (\{1\} \times \mathbb{R}) \cup ([0, 1] \times \{0\}) \cup \bigcup_{n\in\mathbb{N}} C_n$$

is connected. See Fig. 8.12, left. Prove that the point $(1/2, 0)$ has no connected neighborhoods contained in the ball of radius $1/2$.

Solution First, we prove that A is connected. Notice that each set C_n is connected because it is the product of two connected spaces. The other pieces of A

$$A_1 = \{0\} \times \mathbb{R}, \quad A_2 = [0, 1] \times \{0\}, \quad A_3 = \{1\} \times \mathbb{R}$$

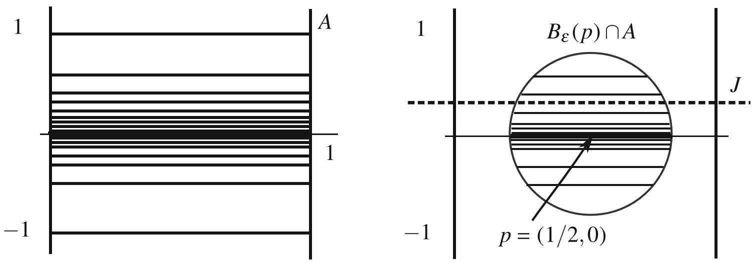


Fig. 8.12 The set A of Exercise 8.16 (left) and the ball $B_\epsilon(p) \cap A$ is not connected (right)

are connected by an analogous argument. We apply Theorem 8.2.1. The reader should note that the intersection of all above sets is empty so it is not possible to use (i) of Theorem 8.2.1. However, $C = A_1 \cup A_2 \cup A_3$ is connected by (iv) of Theorem 8.2.1. The set $C \cup C_n$ is connected because its intersection is non-empty. Finally, $A = \bigcup_{n\in\mathbb{N}}(C \cup C_n)$ is a union of connected sets and their intersection is C. Now we utilize (i) of Theorem 8.2.1.

For the second part, let $p = (\frac{1}{2}, 0)$ and let U be a connected neighborhood included in $B^A_{1/2}(p)$. Observe that the balls centered at p are not connected, as indicated in Fig. 8.12, right. To make this precise, let γ_p be the basis of neighborhoods of p defined by

$$\gamma_p = \{B_\epsilon(p) \cap A : 0 < \epsilon < \tfrac{1}{2}\}.$$

We see that $B_\epsilon(p) \cap A$ is not connected. As $\epsilon < 1/2$, let $n \in \mathbb{N}$ such that $1/n < \epsilon$ and let $r > 0$ with $\frac{1}{n+1} < r < \frac{1}{n}$. Consider the horizontal line $J = \{(x, r) \in \mathbb{R}^2 : x \in \mathbb{R}\}$. Because $\epsilon < 1/2$,

$$B_\epsilon(p) \cap (\{0\} \times \mathbb{R}) = B_\epsilon(p) \cap (\{1\} \times \mathbb{R}) = \emptyset. \tag{8.2}$$

If $(x, y) \in J \cap (B_\epsilon(p) \cap A)$, we have $x \neq 0$ and $x \neq 1$ by (8.2). Then $y = 1/m$ for some $m \in \mathbb{N}$ but this is not possible because $y = r$. This straight line J is used to prove that $B_\epsilon(p) \cap A$ is disconnected because

$$(B_\epsilon(p) \cap A) \cap \{(x, t) \in \mathbb{R}^2 : t > r\}, \quad (B_\epsilon(p) \cap A) \cap \{(x, t) \in \mathbb{R}^2 : t < r\}$$

is an open separation of $B_\epsilon(p) \cap A$. Now the intersection $(B_\epsilon(p) \cap A) \cap \{(x, t) \in \mathbb{R}^2 : t > r\}$ is non-empty because it contains points with $t = 1/n$ with $1/n < \epsilon$.

Let $\epsilon > 0$ with $\epsilon < 1/2$ such that

$$B_\epsilon(p) \cap A \subset U.$$

Since $B_\epsilon(p) \cap A$ is not connected and that $U \subset B^A_{1/2}(p) = B_{1/2}(p) \cap A$, the same horizontal straight line J determines an open separation of U,

$$U \cap \{(x,t) \in \mathbb{R}^2 : t > r\} \quad \text{and} \quad U \cap \{(x,t) \in \mathbb{R}^2 : t < r\},$$

obtaining a contradiction. □

Exercise 8.17 Determine the properties of connectedness of the right order topology $(\mathbb{R}, \tau_r)$.

Solution The topology τ_r is fully determined (Example 2.12),

$$\tau_r = \{\emptyset, \mathbb{R}\} \cup \{[a, \infty) : a \in \mathbb{R}\} \cup \{(a, \infty) : a \in \mathbb{R}\}.$$

We learned in Example 8.1 that this space is connected.

1. Let A be a subspace. We see again that two nontrivial open sets in the relative topology must intersect. Assume that A has at least two points, otherwise, the topology of A is trivial and connected. Let $[a_1, \infty) \cap A$ and $[a_2, \infty) \cap A$ be two open sets in A and without loss of generality, we may assume $a_1 < a_2$. If $x \in [a_2, \infty) \cap A$, then $a_1 \leq a_2 \leq x$, hence $x \in [a_1, \infty) \cap A$. As a consequence, every subset is connected by (3) of Example 8.1.
2. We prove that the space is also path connected. Let $x, y \in \mathbb{R}$ and we find a path joining x with y. Suppose, for instance, that $x < y$ and define

$$\alpha \colon \mathbf{I} \to \mathbb{R}, \quad \alpha(t) = \begin{cases} x, & t = 0 \\ y, & t \in (0, 1]. \end{cases}$$

   Since α is increasing, then α is continuous (Exercise 5.7) and joins x with y. As a result, the spaces defined in Exercise 2.2 are path connected. Recall that $(\mathbb{N}, \tau)$ is a subspace of $(\mathbb{R}, \tau_l)$ and $(\mathbb{R}, \tau_l) \cong (\mathbb{R}, \tau_r)$. □

Exercise 8.18 Determine the properties of connectedness of the included point topology.

Solution Let $p \in X$ be the particular point that defines the topology. We have seen in Example 8.1 that the space is connected.

1. We study which subspaces are connected. Let $A \subset X$ be given. The relative topology on A is known (Example 2.26). If $p \in A$, the relative topology is the included point topology, so A is connected. If $p \notin A$, the relative topology on A is the discrete topology, which is connected only if A has one element. Thus the connected subsets are the singletons and the subsets that contain p.
2. The space is path connected. It suffices to find a path joining p with every point of X. Certainly, let $x \in X$ and define

$$\alpha \colon \mathbf{I} \to X, \quad \alpha(t) = \begin{cases} p, & \text{if } t \in [0, 1) \\ x, & \text{if } t = 1. \end{cases}$$

Then α is continuous for if O is open, then $\alpha^{-1}(O) = [0, 1)$ if $x \notin O$ and $\alpha^{-1}(O) = [0, 1]$ if $x \in O$. □

Exercise 8.19 Determine connectedness and path connectedness of the space of Exercise 3.6.3. Prove that any element of β is connected.

Solution Recall that the relative topology on $\mathbb{R}$ is the usual topology.

1. Let $\mathbb{R}^* = A \cup B$ be an open partition and assume that $\omega \in A$. The open sets A and B define a partition $\{A \cap \mathbb{R}, B \cap \mathbb{R}\}$ for $\mathbb{R}$. Since that $\mathbb{R}$ is connected in $\mathbb{R}^*$ because $\tau^*_{|\mathbb{R}} = \tau_u$, this partition is trivial.

   a. If $A \cap \mathbb{R} = \mathbb{R}$ then $\mathbb{R} \subset A$. As $\omega \in A$, then $A = \mathbb{R} \cup \{\omega\} = \mathbb{R}^*$.
   b. If $A \cap \mathbb{R} = \emptyset$ then $A = \emptyset$ or $A = \{\omega\}$. However, $\{\omega\}$ is not open in τ because any open containing ω must intersect $\mathbb{R}$. Thus $A = \emptyset$.

   In either case, $\{A, B\}$ is trivial, proving that $\mathbb{R}^*$ is connected.
2. We see that $\mathbb{R}^*$ is path connected. Because $\mathbb{R}$ is path connected, we need only find a path joining ω with any real number, say 0. Consider the homeomorphism $f : [0, 1) \to [0, \infty)$ given by $f(x) = x/(1 - x)$. Note that $f(0) = 0$, $\lim_{x \to 1^-} f(x) = \infty$ and that f is strictly increasing. Define

$$\alpha(t) = \begin{cases} f(t), & t \in [0, 1) \\ \omega, & t = 1, \end{cases}$$

   and we see that α is continuous. Let $O \in \beta$. Suppose the case $O = (a, b)$. Since $O \in \tau_u$ and $f : [0, 1) \to [0, \infty)$ is continuous, then $\alpha^{-1}(O) = f^{-1}(O)$ is open in $[0, 1]$. Now suppose $O = \{\omega\} \cup (-\infty, a) \cup (b, \infty)$, $a < b$. The preimage of O depends on the relative position of a and b with respect to 0. We have

$$\alpha^{-1}(O) = \begin{cases} [0, 1], & b < 0, \\ (f^{-1}(b), 1], & a < 0 \leq b \\ [0, f^{-1}(a)) \cup (f^{-1}(b), 1], & a \geq 0. \end{cases}$$

   In either case, $\alpha^{-1}(O)$ is open in the interval $[0, 1]$.
3. The elements of β of type (a, b) are path connected because they are in the usual topology. Let $U = \{\omega\} \cup (-\infty, a) \cup (b, \infty)$ with $a < b$. Let A and B be a nontrivial partition of U by open sets of U and suppose that $\omega \in A$. The above partition determines another one in $C = (-\infty, a) \cup (b, \infty)$ given by $C \cap A$ and $C \cap B$. Because the relative topology on C is Euclidean, the components of C are the open sets $(-\infty, a)$ and (b, ∞). This tells us that $C \cap A$ cannot contain points of $(-\infty, a)$ and (b, ∞) and this holds for $C \cap B$ as well. Thus we can assume that $C \cap A = (-\infty, a)$ and $C \cap B = (b, \infty)$. Since $\omega \in A$ and $A \cap B = \emptyset$, we deduce $A = (-\infty, a) \cup \{\omega\}$ and $B = (b, \infty)$. Since A is open in U and U open

in $\mathbb{R}^*$, A is open in $\mathbb{R}^*$. This, however, is absurd because there would be $c, d \in \mathbb{R}$ such that

$$\omega \in \{\omega\} \cup (-\infty, c) \cup (d, \infty) \subset (-\infty, a) \cup \{\omega\},$$

which is not possible. □

The next result asserts that the complement of a countable subset of $\mathbb{R}^n$, for $n \geq 2$, is path connected. The result is expected if we remove a finite subset, but it not so clear for an infinite countable set. Let us notice that if $n = 1$, the result is not valid.

Exercise 8.20 Let $n \geq 2$. If $A \subset \mathbb{R}^n$ is a countable set, prove that $\mathbb{R}^n \setminus A$ is path connected. As a consequence, $\mathbb{R}^n \setminus \mathbb{Q}^n$ is path connected.

Solution We prove that for any two points $p, q \in \mathbb{R}^n \setminus A$, there is a path connected set containing both points. Let $[p, q]$ be the segment joining p and q and let L be a straight line intersecting $[p, q]$ with $p, q \notin L$. We can do this because $n \geq 2$. For each $x \in L$, define

$$\Delta_x = [p, x] \cup [x, q].$$

This set is path connected because it is union of two segments and $x \in [p, x] \cap [x, q]$. Moreover, $p, q \in \Delta_x$. Observe that Δ_x may intersect A. However, we claim that there is $x \in L$ such that $\Delta_x \subset \mathbb{R}^n \setminus A$ and this completes the exercise. The proof of the claim is by contradiction. Thus suppose $A \cap \Delta_x \neq \emptyset$ for all $x \in L$. For each $x \in L$, let a_x be a fixed element of this intersection. Define

$$\phi \colon L \to A, \quad \phi(x) = a_x.$$

We prove that ϕ is injective. Suppose $\phi(x) = \phi(y)$. Then $a_x = a_y$; in particular, $\Delta_x \cap \Delta_y \neq \emptyset$ and $a_x \in \Delta_x \cap \Delta_y$. If $x \neq y$, this intersection is precisely $\{p, q\}$, which does not belong to A. Consequently, $x = y$.

Because ϕ is injective, the contradiction arrives because L is not countable (bijective with $\mathbb{R}$) and A is countable. □

Exercise 8.21 Let $A_n = [0, 1] \times \{\frac{1}{n}\}$ and $X = \bigcup_{n \in \mathbb{N}} A_n \cup \{(0, 0)\}$. Find the connected components of X.

Solution We prove that the components are the sets A_n and $\{(0, 0)\}$. We begin with A_n. Each A_n is connected because it is the product of two connected sets. Let $p = (x, \frac{1}{n}) \in A_n$. If there is $(a, b) \in C_p \setminus A_n$, then (a, b) is of the form $(a, \frac{1}{m})$ or $(0, 0)$. Assume $n < m$ (the case $m < n$ is similar). In either of the two cases, consider $r > 0$ such that $\frac{1}{n+1} < r < \frac{1}{n}$. Then the sets $O_1 = \{(x, y) \in \mathbb{R}^2 : y > r\}$ and $O_2 = \{(x, y) \in \mathbb{R}^2 : y < r\}$ are open in $\mathbb{R}^2$ and $O_1 \cap C_p$ and $O_2 \cap C_p$ form a separation of C_p by open sets of C_p because $p \in O_1 \cap C_p$ and $(a, b) \in O_2 \cap C_p$. This is contradictory with the connectedness of C_p.

As $(0, 0)$ cannot belong to any of the components A_n, the other component is $\{(0, 0)\}$. □

Exercise 8.22 With the notation of Exercise 6.17, prove that there is no homeomorphism $f : \mathbb{R}^2 \to \mathbb{R}^2$ such that $f_{|X} = \phi$.

Solution The complements of X and Y are formed by three connected components, $\mathbb{R}^2 \setminus X = \{A_1, B_1, C_1\}$ and $\mathbb{R}^2 \setminus Y = \{A_2, B_2, C_2\}$ where A_1 and A_2 are the non-bounded components and C_2 is the component bounded by $\mathbb{S}^1$. Since f is a homeomorphism, f maps connected components to connected components and boundaries to boundaries. Since the boundary of A_1 in $\mathbb{R}^2$ is $\mathbb{S}^1$,

$$\mathrm{Bd}(f(A_1)) = f(\mathrm{Bd}(A_1)) = f(\mathbb{S}^1) = \phi(\mathbb{S}^1) = \mathbb{S}^1.$$

This implies $\phi(A_1) = C_2$. Therefore

$$f(\mathrm{Bd}(B_1) \cap \mathrm{Bd}(C_1)) = \mathrm{Bd}(A_2) \cap \mathrm{Bd}(B_2).$$

But $\mathrm{Bd}(B_1) \cap \mathrm{Bd}(C_1)$ is infinite and $\mathrm{Bd}(A_2) \cap \mathrm{Bd}(B_2)$ are two points. □

Exercise 8.23 For each $q \in \mathbb{Q}$, consider the segment $L_q = [(0, 0), (q, 1)]$ and let $X = \bigcup_{q \in \mathbb{Q}} L_q$. Prove that X is path connected. Find the connected components of $X \setminus \{(0, 0)\}$ and prove that these components are not open in $X \setminus \{(0, 0)\}$.

Solution

1. Each one of the segments L_q is path connected and the intersection of all of them is $(0, 0)$. Thus the space is path connected. Other argument is observing that X is star-shaped from $(0, 0)$.
2. Let $p = (0, 0)$. We prove that the connected components of $X \setminus \{p\}$ are the subsets

$$L_q \setminus \{p\} = ((0, 0), (q, 1)] = \{(1 - t)(0, 0) + t(q, 1) : t \in (0, 1]\}.$$

It is clear that each of these sets $L_q \setminus \{p\}$ is homeomorphic to $(0, 1]$, so it is connected. Let $(q, 1) \in X$ and C_q be its connected component. Thus $L_q \setminus \{p\} \subset C_q$. Suppose that there are points of C_q in distinct segments, say in $L_{q'}$. We find an open separation of L_q as follows. Without loss of generality, assume $q < q'$. Let r be an irrational such that $q < r < q'$ and consider the straight line L of equation $y = x/r$ joining the origin with the point $(r, 1)$. Construct the open sets

$$O_1 = \{(x, y) \in \mathbb{R}^2 : y > x/r\}, \quad O_2 = \{(x, y) \in \mathbb{R}^2 : y < x/r\}.$$

Because $L \cap (X \setminus \{p\}) = \emptyset$, we have the separation of C_q given by $O_1 \cap C_q$ and $O_2 \cap C_q$, where $(q, 1) \in O_1$ and $(q', 1) \in O_2$, but this is contradictory.

3. If $L_q \setminus \{p\}$ is open in $X \setminus \{p\}$, then $(q, 1)$ is an interior point, so there is $r > 0$ such that $B_r(q, 1) \cap (X \setminus \{p\}) \subset L_q \setminus \{p\}$. Let $q' \in \mathbb{Q}$ with $q' \in (q - r, q)$. Then $(q', 1) \in B_r(q, 1)$ but $(q', 1) \notin L_q$, a contradiction. □

Exercise 8.24 Let X be a connected space and $a, b \in X$. If $\{O_i : i \in I\}$ is a collection of open sets with $\bigcup_{i \in I} O_i = X$, prove that there is $n \in \mathbb{N}$ and open sets $O_1, \ldots, O_n$ such that $a \in O_1$, $b \in O_n$ and O_i and O_j are mutually disjoint except if i and j are consecutive, for $i, j \in \{1, \ldots, n\}$.

Solution Consider the set A of all points of X that can be joined with a by means of a finite numbers of open sets of $\{O_i : i \in I\}$, where O_i and O_{i+1} have non-empty intersection for all i but the O_i are mutually disjoint if they are not consecutive. Let us notice that $a \in A$ because a belongs to some O_{i_0}. We prove that A is open and closed. In such a case, because X is connected, $A = X$, showing the result.

1. The set A is open. Let $x \in A$ be given. Then there are open sets $O_1, \ldots, O_n$, with $a \in O_1$, $x \in O_n$ and with the properties previously described. Then all points of O_n satisfy the same property, so $O_n \subset A$. Since $x \in O_n$, this proves that $x \in \mathring{A}$.
2. The set A is closed. Let $x \in \overline{A}$. Let $i_0 \in I$ be an index such that $x \in O_{i_0}$. Since x is an adherent point of A, let $y \in O_{i_0} \cap A$. By definition of A, there are open sets $O_1, \ldots, O_n$ such that $a \in O_1$, $y \in O_n$ and O_i and O_j are mutually disjoint except if they are consecutive. Let $m \in \{1, \ldots, n\}$ be the first number such that $O_m \cap O_{i_0} \neq \emptyset$. Then the chain $O_1, \ldots, O_m, O_{i_0}$ satisfies the desired properties. Observe that $O_{i_0} \cap O_i = \emptyset$ for $i < m$. Since $x \in O_{i_0}$, then $x \in A$. □

It is clear that in Euclidean space $\mathbb{R}^n$ there are non-connected neighborhoods for each point. For example, for $x = (x_1, \ldots, x_n)$, let $U = B_r(x) \cup \{(x_1 + 2r, 0, \ldots, 0)\}$. A bit more difficult is to find a basis of non-connected neighborhoods.

Exercise 8.25 In Euclidean space $\mathbb{R}^n$, find a basis of neighborhoods for each point such that all neighborhoods are not connected.

Solution It suffices to construct such a basis of neighborhoods for the origin $0 = (0, \ldots, 0)$ because by means of translations of $\mathbb{R}^n$, we can carry this basis to every point. Let $\mathbb{N}^* = \mathbb{N} \setminus \{1\}$. For each $n \in \mathbb{N}^*$, define

$$p_n = (\tfrac{1}{n-1}, 0, \ldots, 0), \quad V_n = B_{1/n}(0) \cup \{p_n\}.$$

We prove that $\beta_0 = \{V_n : n \in \mathbb{N}^*\}$ is a basis of neighborhoods of 0 and that all neighborhoods V_n are not connected.

1. It is clear that $V_{n+1} \subset B_{1/n}(0)$ because $B_{1/(n+1)}(0) \subset B_{1/n}(0)$ and $|p_{n+1} - 0| = \frac{1}{n+1} < \frac{1}{n}$.
2. Since $B_{1/n}(0) \subset V_n$, V_n is a neighborhood of 0. Now let U be a neighborhood of 0. Since $\{B_{1/n}(0) : n \in \mathbb{N}^*\}$ is a basis of neighborhoods of 0, there is $n \in \mathbb{N}^*$ such that $B_{1/n}(0) \subset U$. Thus $V_{n+1} \subset U$.

3. The set V_n is not connected. Let $r \in \mathbb{R}$ such that $\frac{1}{n} < r < \frac{1}{n-1}$. Then

$$\{V_n \cap \{x \in \mathbb{R}^n : x_1 < r\}, V_n \cap \{x \in \mathbb{R}^n : x_1 > r\}\}$$

is an open separation of V_n because $B_{1/n}(0) \subset \{x \in \mathbb{R}^n : x_1 < r\}$ and $p_n \in \{x \in \mathbb{R}^n : x_1 > r\}$. $\square$

Exercise 8.26 If $A \subset \mathbb{R}^n$ is a convex set, prove that the space of continuous maps $\mathsf{C}([a,b], A)$ with the sup distance is path connected (Exercise 4.29).

Solution Recall that any continuous function defined on a bounded closed interval $[a,b]$ is bounded, so the sup distance d_∞ is well defined. Fix $a_0 \in A$. We prove that every $f \in \mathsf{C}([a,b], A)$ can be joined through a path with the constant function $g(x) = a_0$, $x \in [a,b]$. Define $\alpha\colon \mathbf{I} \to \mathsf{C}([a,b], A)$ by $\alpha(t)(x) = f(x) + t(a_0 - f(x))$. It is elementary that $\alpha(t)$ is a continuous map and that its image is the segment $[f(x), a_0]$. This set is included in A because $f(x), a_0 \in A$ and A is convex.

On the other hand, $\alpha(0) = f$ and $\alpha(1) = g$ so it remains to prove that α is continuous. Let $t_0 \in \mathbf{I}$ be arbitrary and we see that α is continuous at t_0. For $t \in \mathbf{I}$ and for $x \in [a,b]$, we have

$$|(\alpha(t) - \alpha(t_0))(x)| = |(t - t_0)(a_0 - f(x))| \leq |t - t_0|(|a_0| + M),$$

where $M = \max_{x \in [a,b]} |f(x)|$. Thus

$$d_\infty(\alpha(t), \alpha(t_0)) \leq (|a_0| + M)|t - t_0|$$

and this proves that α is Lipschitz, so continuous. $\square$

Exercise 8.27 Let $f\colon X \to Y$ be an open, surjective, continuous map. Suppose that Y is connected and $f^{-1}(\{y\})$ is connected for every $y \in Y$. Prove that X is connected. Show an example that this result is false if f is not open.

Solution Suppose that X is disconnected and let A, B be an open separation, $X = A \cup B$. Since f is open, then $f(A)$ and $f(B)$ are two open sets of Y. Using that Y is connected, $f(A)$ and $f(B)$ are non trivial, so they must intersect. Let $y \in f(A) \cap f(B)$. Then

$$f^{-1}(\{y\}) = (A \cap f^{-1}(\{y\})) \cup (B \cap f^{-1}(\{y\})).$$

The sets $A \cap f^{-1}(\{y\})$ and $B \cap f^{-1}(\{y\})$ are open in $f^{-1}(\{y\})$ and they are nonempty. Because $f^{-1}(\{y\})$ is connected, they cannot be disjoint. Let $z \in (A \cap f^{-1}(\{y\})) \cap (B \cap f^{-1}(\{y\}))$; in particular, $A \cap B \neq \emptyset$, a contradiction.

The example is the identity $\mathbb{1}_{\mathbb{R}}\colon (\mathbb{R}, \tau_D) \to (\mathbb{R}, \tau_u)$. This map is surjective and continuous because $\tau_u \subset \tau_D$. The space $(\mathbb{R}, \tau_u)$ is connected and $\mathbb{1}^{-1}(\{y\}) = \{y\}$

is connected. The map $\mathbb{1}_{\mathbb{R}}$ is not open, otherwise, $\mathbb{1}_{\mathbb{R}}$ would be a homeomorphism. Finally the domain $(\mathbb{R}, \tau_D)$ is not connected. $\square$

In the next two exercises we work with the concept of a topological group (Exercise 7.26).

Exercise 8.28 Let G be a connected topological group and let e denote the identity. If U is a neighborhood of e with the property that $U^{-1} = U$, prove that G is spanned by U.

Solution Recall that U^{-1} denotes the set of inverses of all elements of U, $U^{-1} = \{x^{-1} : x \in U\}$. In a first step, we replace U by an open neighborhood of e. Let O be an open set such that $e \in O \subset U$. Then $O^{-1} \subset U^{-1} = U$. If $V = O \cap O^{-1}$ then clearly V is open with $V = V^{-1}$ and $e \in V \subset U$.

We proceed to show that the group $\langle V \rangle$ spanned by V is G and consequently, $\langle U \rangle = G$. For each $k \in \mathbb{N}$, let

$$V^k = V \cdot \overset{k}{\overbrace{\ldots}} \cdot V = \{x_1 x_2 \cdots x_k : x_i \in V, i = 1, \ldots, k\}.$$

As a first step, we see that V^k is open for all $k \in \mathbb{N}$. We use induction on k, the exponent of V^k. The case $k = 1$ is clear. Now assume that V^k is open and we show that V^{k+1} is open. If L_x is the left-translation about $x \in G$, then

$$V^{k+1} = V V^k = \bigcup_{x \in V} x V^k = \bigcup_{x \in V} L_x(V^k).$$

Since L_x is a homeomorphism and V^k is open by induction hypothesis, the set $L_x(V^k)$ is open, hence V^{k+1} is open because it is the union of open sets. The subgroup $\langle V \rangle$ is formed for all arbitrary products of all elements of V and their inverses. Thus

$$\langle V \rangle = \bigcup\nolimits_{k=1}^{\infty} V^k.$$

Hence we deduce that $\langle V \rangle$ is open because it is union of open sets. Finally, it was proved in Exercise 7.27 that every open subgroup is also closed and thus $\langle V \rangle$ is also closed. As G is connected and $\langle V \rangle \neq \emptyset$, we conclude $\langle V \rangle = G$. $\square$

Exercise 8.29 Let G be a topological group and denote by G_0 the connected component of the unit element.

1. Prove that the connected component of $x \in G$ is $xG_0 = G_0 x$.
2. Prove that G_0 is a normal subgroup.

Solution

1. The left translations L_x are homeomorphisms, so they map connected components to connected components. Since $L_x(e) = x$ and G_0 is the connected component of e, $L_x(G_0) = xG_0$ is the connected component of x. A parallel

argument applies to the right translation R_x, so $G_0 x$ is the component of x; in particular, $xG_0 = G_0 x$ (observe that G may not be commutative).

2. (a) We see that G_0 is a subgroup. This is equivalent to see that $x^{-1}y \in G_0$ for all $x, y \in G_0$, or equivalently, $G_0^{-1}G_0 \subset G_0$. Note that

$$G_0^{-1}G_0 = \bigcup_{x \in G_0} L_{x^{-1}}(G_0).$$

Since $e = x^{-1}x \in L_{x^{-1}}(G_0)$, $L_{x^{-1}}$ is continuous and G_0 is connected, it follows that $L_{x^{-1}}(G_0) \subset G_0$ for all $x \in G$, hence the result.

(b) Proving that G_0 is a normal subgroup is equivalent to checking that $x^{-1}G_0 x \subset G_0$ for all $x \in G$. Since $e = x^{-1}x \in x^{-1}G_0$,

$$x^{-1}G_0 x = (L_{x^{-1}} \circ R_x)(G_0).$$

As $L_{x^{-1}} \circ R_x$ is a homeomorphism, $L_{x^{-1}} \circ R_x(e) = e$ and G_0 is the connected component of the unit, we deduce that $x^{-1}G_0 x$ is the connected component of e again, that is, $x^{-1}G_0 x = G_0$. □

We present a new proof that $\mathbb{S}^2$ is connected.

Exercise 8.30 Prove that sphere $\mathbb{S}^2$ is connected.

Solution The sphere $\mathbb{S}^2$ is obtained by rotating the semicircle

$$A = \mathbb{S}^2 \cap \{(x, 0, z) : x \geq 0\} = \{(x, 0, z) : x^2 + z^2 = 1, x \geq 0\}$$

about the z-axis. Observe that a similar construction allowed in Example 7.6 to define a torus as a surface of revolution by rotating a circle contained in the xz-plane. The set A can be expressed as $A = \{(\cos t, 0, \sin t) : t \in [-\pi/2, \pi/2]\}$. On the other hand, using the rotations R_θ described in Example 7.6,

$$\mathbb{S}^2 = \{R_\theta(A) : \theta \in \mathbb{R}\} = F([-\tfrac{\pi}{2}, \tfrac{\pi}{2}] \times \mathbb{R})$$

where $F : [-\pi/2, \pi/2] \times \mathbb{R}$ is

$$F(t, \theta) = R_\theta(\cos t, 0, \sin t) = (\cos t \cos\theta, \cos t \sin\theta, \sin t). \tag{8.3}$$

In view that F is continuous and the domain is connected (product of two connected spaces), we conclude that $\mathbb{S}^2$ is connected by Theorem 8.1.6. □

Exercise 8.31 Construct an explicit path joining two points of $\mathbb{S}^n$.

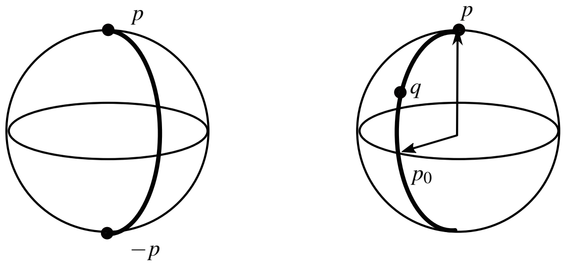


Fig. 8.13 Paths between two points of the sphere $\mathbb{S}^2$

Solution Let $p, q \in \mathbb{S}^n$ be given. A first case is when $q = -p$. Let $p_0 \in \mathbb{S}^n$ be a vector orthogonal to p. Define the continuous map $\alpha(t) = (\cos t)p + (\sin t)p_0$, $t \in [0, \pi]$. Note that

$$|\alpha(t)|^2 = (\cos t)^2 + (\sin t)^2 + 2 \sin t \cos t \langle p, p_0\rangle = (\cos t)^2 + (\sin t)^2 = 1,$$

so $\alpha\colon [0, \pi] \to \mathbb{S}^n$. Further, $\alpha(0) = p$ and $\alpha(\pi) = -p$ (Fig. 8.13, left).

Suppose that $q \neq -p$. In the vector subspace of dimension 2 spanned by p and q, let p_0 be a unitary vector orthogonal to p (there are two of such vectors). This vector can be explicitly calculated as follows. Writing $p_0 = ap + bq$, $a, b \in \mathbb{R}$, then from $\langle p, p_0\rangle = 0$, we deduce $0 = a + b\langle p, q\rangle$. Then $a = -b\langle p, q\rangle$. Moreover, because $|p_0|^2 = 1$, we have a second equation for a and b, namely, $a^2 + b^2 + 2ab\langle p, q\rangle = 1$. Combining both equations, we derive $b^2(1 - \langle p, q\rangle^2) = 1$. The number $\langle p, q\rangle$ is not 1 ($q \neq p$) neither -1 ($q \neq -p$). Hence we deduce b, with two possible values. Let $b = 1/\sqrt{1 - \langle p, q\rangle^2}$ and, consequently, $a = -\langle p, q\rangle/\sqrt{1 - \langle p, q\rangle^2}$. Then

$$p_0 = -\frac{\langle p, q\rangle}{\sqrt{1 - \langle p, q\rangle^2}}p + \frac{1}{\sqrt{1 - \langle p, q\rangle^2}}q.$$

We define the path joining p and q (Fig. 8.13, right). Let

$$\alpha\colon \mathbb{R} \to \mathbb{S}^n, \quad \alpha(t) = (\cos t)p + (\sin t)p_0.$$

It is clear that $|\alpha(t)| = 1$, so $\alpha(t) \in \mathbb{S}^n$ for all $t \in \mathbb{R}$. Also, $\alpha(0) = p$. Thus it suffices to verify that α crosses the point q at some value t_0. This is equivalent to solve on t the equation $(\cos t)p + (\sin t)p_0 = q$. Replacing the value of p_0, we have

$$(\cos t + a \sin t)p + (b \sin t - 1)q = 0.$$

Since p and q are linearly independent, $\cos t + a \sin t = 0$ and $b \sin t - 1 = 0$. Thus

$$\cos t = -\frac{a}{b}, \quad \sin t = \frac{1}{b}. \tag{8.4}$$

From the expressions of a and b, we have $(a/b)^2 + (1/b)^2 = 1$. Thus equations (8.4) have a solution $t_0 \in \mathbb{R}$. In fact, there are many solutions, all of type $t_0 + 2k\pi$, $k \in \mathbb{Z}$. Definitively the path $\alpha\colon [0, t_0] \to \mathbb{S}^n$ joins p with q. □

Exercise 8.32 Let X and Y be two connected spaces and let $A \subset X$, $B \subset Y$ be two nontrivial subsets. Show that $X \times Y \setminus A \times B$ is connected. Deduce that the complement of an affine subspace of dimension m in $\mathbb{R}^n$, with $m < n - 1$, is connected.

Solution

1. By assumption, there are $x_0 \notin A$ and $y_0 \notin B$. Given $(x, y) \in X \times Y \setminus A \times B$, we find a connected set containing (x, y) and (x_0, y_0). Without loss of generality assume that $x \notin A$. Define

$$C = (\{x_0\} \times Y) \cup (X \times \{y_0\}) \cup (\{x\} \times Y).$$

   It is clear that C contains both points. Each of these sets is connected because each one is the product of two connected sets and $(\{x_0\} \times Y) \cap (X \times \{y_0\}) \neq \emptyset$ and $(X \times \{y_0\}) \cap (\{x\} \times Y) \neq \emptyset$. Finally $C \subset X \times Y \setminus A \times B$ because $\{x_0\} \times Y$, $X \times \{y_0\}$ and $\{x\} \times Y$ do not intersect $A \times B$.
2. Let $U \subset \mathbb{R}^n$ be an affine subspace of dimension $m < n - 1$. After applying an affine transformation, we may assume that $U = \mathbb{R}^m \times \{0\}$, where $0 \in \mathbb{R}^{n-m}$. In order to apply (1), write $U = A \times B \subset \mathbb{R}^{m+1} \times \mathbb{R}^{n-m-1}$, where $A = \mathbb{R}^m \times \{0_1\}$ and $B = \{0_2\}$, where $0_1 \in \mathbb{R}$ and $0_2 \in \mathbb{R}^{n-m-1}$ are the zero vectors. Note that $n - m - 1 \geq 1$. Then A and B are nontrivial subsets of $\mathbb{R}^{m+1}$ and $\mathbb{R}^{n-m-1}$, respectively. □

8.6 Suggested Exercises

8.6.1 Let $X = \{a, b, c, d, e\}$ and $\tau = \{X, \emptyset, \{a\}, \{c, d\}, \{a, c, d\}, \{b, c, d, e\}\}$. Prove that τ is a topology on X and that (X, τ) is disconnected. Find the connected components. Determine if $A = \{b, d, e\}$ is connected.

8.6.2 Investigate if $\mathbb{R}^n \setminus (\mathbb{R} - \mathbb{Q})^n$ is connected.

8.6.3 Let X be a space and let $\{X_i\}_{i\in I}$ be a family of connected subsets such that $X = \bigcup_{i\in I} X_i$ and there is $i_0 \in I$ such that $X_{i_0} \cap X_i \neq \emptyset$ for all $i \in I$. Show that X is connected. Is there an analogous result replacing connected by path connected?

8.6.4 Let τ_1 and τ_2 be two topologies on X such that τ_1 is finer than τ_2. Relate all notions of connectedness between (X, τ_1) and (X, τ_2).

8.6.5 Prove that a space X is connected if and only if $\mathrm{Bd}(A) \neq \emptyset$ for all nontrivial subsets A of X.

8.6.6 Let A and C be two subsets of a space X such that C is connected. If $C \cap A \neq \emptyset$ and $C \cap (X \setminus A) \neq \emptyset$, prove that $C \cap \mathrm{Bd}(A) \neq \emptyset$. Apply the argument to the subsets of $\mathbb{R}^2$ given by $A = \{(x, 0) \in \mathbb{R}^2 : -1 \leq x \leq 1\}$ and $C = \{(0, y) \in \mathbb{R}^2 : y \in \mathbb{R}\}$.

8.6.7 Prove that any nonempty, open and closed connected subset of a space is a connected component.

8.6.8 If A and B are two subsets of two connected spaces X and Y respectively, determine if $(A \times Y) \cup (X \times B)$ is connected.

8.6.9 Prove that any space having the fixed-point property is connected.

8.6.10 Let A and B be two connected subsets in a space. If $\overline{A} \cap B \neq \emptyset$, prove that $A \cup B$ is connected.

8.6.11 Give a connected subset in a space whose interior is disconnected.

8.6.12 Let X be the coordinate axes of $\mathbb{R}^2$. Using connectedness, prove that $(0, 0)$ has not a neighborhood homeomorphic to $\mathbb{R}$.

8.6.13 If X is formed by the coordinate axes of $\mathbb{R}^2$ and $f : \mathbb{R} \to X$ is a continuous surjective map, prove that $f^{-1}(\{(0, 0)\})$ contains at least three points.

8.6.14 Prove that the cylinder $\mathbb{S}^1 \times \mathbb{R}$ is a surface of revolution obtained by the rotation about the z-axis of the vertical straight line of equations $x = 1$, $y = 0$. Deduce that the cylinder is connected.

8.6.15 Let A and B be two closed subsets in a space X. If $A \cup B$ and $A \cap B$ are connected, prove that A and B are connected. Give an example showing that the result is not true if one of the subsets is not closed.

8.6.16 Show that $[0, 1] \times [0, 1]$ with the lexicographic topology (Exercise 7.16) is connected but there is no a path between $(0, 0)$ and $(1, 1)$.

8.6.17 In a normed vector space X, prove that the vector subspace $\{\lambda x : \lambda \in \mathbb{R}\}$ is connected for any $x \in X$ and deduce that X is connected.

8.6.18 Prove that the punctured torus and the punctured cylinder are connected. Deduce that $\mathbb{S}^2$, the torus and the cylinder are not homeomorphic to any subset of $\mathbb{R}$.

8.6.19 Let $\omega \notin \mathbb{Q}$. On $X^* = \mathbb{Q} \cup \{\omega\}$ define τ^* consisting of all elements of τ_u and the subsets $A \cup \{\omega\} \subset X^*$ such that $\mathbb{Q} \setminus A$ is finite. Show that τ^* is a topology and study if X^* is connected.

8.6.20 Let X be a connected metric space with at least two elements $x_1, x_2 \in X$. Define the function $f(x) = d(x, x_1)$. Prove that the image of f contains the interval $[0, d(x_1, x_2)]$. Deduce that X is not countable.

Chapter 9
Compactness

Compactness, like connectedness, is a topological invariant with a prominent role in topology. This invariant will be a very useful tool when classifying topological spaces, and it is used in a variety of mathematical problems. A clear example where compactness is employed is in finding local minima (or maxima) of continuous functions. For a simple example, suppose that f is a continuous function of one variable $x \in [a, b]$. It will be proved that the interval $[a, b]$, as a topological space, is compact and that every continuous real-valued function whose domain is compact attains a minimum (Fig. 9.1). We think of $f = f(x)$ as a function that measures the energy at each position x. The local minimum also called equilibrium points are of interest in many contexts of mathematics and physics.

9.1 Compact Spaces and Examples

The idea of compactness is related to the notion of finiteness in the sense that if a space is covered by a union of open sets, it is desirable to remove those sets that are redundant in the covering. If the number of remaining open sets is finite for any covering, the space is said to be compact. To be precise, an *open covering* of a space X is a family $\{O_i : i \in I\}$ of open sets such that $X = \bigcup_{i \in I} O_i$. A *subcovering* of the covering $\{O_i : i \in I\}$ is a subset of this family that is also a covering of X.

Definition 9.1.1 A space is called *compact* if for every open covering there is a finite subcovering. A subset of a space is called compact if it is a compact space as subspace of a topological space.

Since in much of the work we deal with compactness of subspaces, it is desirable to manage open sets of the ambient space instead of the subspace.

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_9

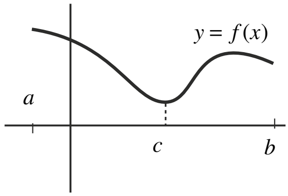


Fig. 9.1 A continuous function defined on the bounded closed interval $[a, b]$ attains a minimum

Proposition 9.1.2 *Let A be a subset of a space X. Then A is compact if and only if for every collection of open sets $\{O_i : i \in I\}$ of X, with $A \subset \bigcup_{i\in I} O_i$, there is $n \in \mathbb{N}$ such that $A \subset O_{i_1} \cup \ldots \cup O_{i_n}$.*

In this context, if $\{O_i : i \in I\}$ are open sets in X with $A \subset \bigcup_{i\in I} O_i$, we also say that $\{O_i : i \in I\}$ is an *open covering* of A.

We establish various characterizations of compactness in terms of closed sets and bases of topologies. The proofs are elementary.

Theorem 9.1.3 *The following are all equivalent for a space X:*

(i) The space X is compact.
(ii) For every family of closed subsets $\{F_i : i \in I\}$ such that $\bigcap_{i\in I} F_i = \emptyset$, there is $n \in \mathbb{N}$ such that $\bigcap_{j=1}^{n} F_{i_j} = \emptyset$.
(iii) If β is a basis for the topology, then for every covering of X by members of β, there is a finite subcovering.

Example 9.1

1. The simplest examples of compact spaces are those ones whose topology is finite. This occurs, for example, if the underlying set X is finite.
2. The Euclidean line $\mathbb{R}$ is not compact because from the open covering $\{(-n, n) : n \in \mathbb{N}\}$, we cannot extract a finite subcovering.
3. An open interval (a, b) is not compact because one cannot find a finite subcovering from the covering $\{(a + \frac{1}{n}, b - \frac{1}{n}) : n \in \mathbb{N}\}$. Moreover $(a, b) \cong \mathbb{R}$ and it is expected that compactness is a topological invariant.
4. The Euclidean plane $\mathbb{R}^2$ is not compact. The argument is entirely similar as in $\mathbb{R}$ by choosing the open covering $\bigcup_{n\in\mathbb{N}}(-n, n) \times (-n, n)$. Analogously, $\mathbb{R}^m$ is not compact.

$\Diamond$

Example 9.2 A compact metric space must be bounded. Indeed, fixing $x \in X$, the collection of balls $\{B_n(x) : n \in \mathbb{N}\}$ is an open covering and by compactness, there are $n_1, \ldots, n_k \in \mathbb{N}$ such that $X = B_{n_1}(x) \cup \ldots \cup B_{n_k}(x)$. If $m = \max\{n_1, \ldots, n_k\}$, then $X = B_m(x)$.

Boundedness is not sufficient to ensure that a metric space is compact, as is shown by $(0, 1)$. Recall that in each (compact or not compact) metric space (X, d) there is an equivalent distance d' that is bounded (Exercise 4.26). However, if the space is compact, all distances that determine the metric topology must be bounded. $\Diamond$

Example 9.3 If $(x_n) \to x$ in a space, the set $A = \{x\} \cup \{x_n : n \in \mathbb{N}\}$ is compact. Indeed, if $A \subset \bigcup_{i \in I} O_i$ is an open covering of A, we fix an O_k containing x. By convergence, there is $\nu \in \mathbb{N}$ such that $x_n \in O_k$ for all $n \geq \nu$. On the other hand, each term x_n of the sequence with $1 \leq n \leq \nu - 1$ is contained in a certain open set O_{i_n}. Then $A \subset O_k \cup O_{i_1} \cup \ldots \cup O_{i_{\nu-1}}$. $\Diamond$

Example 9.4 A discrete space (X, τ_D) is compact if and only if X is finite. Certainly, if X is not finite then $\{\{x\} : x \in X\}$ is an open covering of X from which we cannot take a finite subcovering. $\Diamond$

Example 9.5 The cofinite topology (X, τ_{CF}) is compact. Indeed, if $\{O_i : i \in I\}$ is an open covering of X, we fix an open set O_k of this covering. This open set covers X except a finite number of points $\{x_1, \ldots, x_n\}$. Each one of these points is contained in an open set of the covering which, together with O_k, form a finite subcovering of X. $\Diamond$

Compactness is, of course, a topological invariant, but like connectedness, the property is preserved by continuous maps.

We now turn our attention to the Euclidean line proving that bounded closed intervals are compact.

Theorem 9.1.4 *Every bounded closed interval of* $\mathbb{R}$ *is compact.*

Proof It suffices to consider the interval $[0, 1]$. Let $\{O_i : i \in I\}$ be a covering of $[0, 1]$ by open sets of $\mathbb{R}$ and define

$$A = \{x \in [0, 1] : \text{ there is } n \in \mathbb{N},\ [0, x] \subset O_{i_1} \cup \ldots \cup O_{i_n}\}.$$

The result is proved if we show that $1 \in A$. We see that A is a non-empty closed and open subset in $[0, 1]$ and because $[0, 1]$ is connected, then $A = [0, 1]$, in particular, $1 \in A$. The set A is non-empty because $0 \in A$ since $[0, 0] = \{0\}$ is included in some open set of the covering. On the other hand, it is elementary that

$$\text{if } x \in A \text{ and } y \in [0, 1] \text{ with } y \leq x, \text{ then } y \in A. \tag{9.1}$$

1. The set A is closed in $[0, 1]$. Let $x \in [0, 1]$ be an adherent point of A. Then there is $\{a_n\} \subset A$ such that $(a_n) \to x$. If some a_n satisfies $x \leq a_n$, then $x \in A$ by (9.1). Now assume that $a_n < x$ for all $n \in \mathbb{N}$. Let $k \in I$ be such that $x \in O_k$ and fix k. Then there is $r > 0$ such that $(x - r, x + r) \subset O_k$. For the neighborhood $(x - r, x + r)$ of x and by the convergence $(a_n) \to x$, there is $\nu \in \mathbb{N}$ such that $a_n \in (x - r, x + r)$ for all $n \geq \nu$. Fix a natural number n_0 with $n_0 \geq \nu$. Then

$$[0, x] = [0, a_{n_0}] \cup [a_{n_0}, x] \subset [0, a_{n_0}] \cup (x - r, x + r) \subset [0, a_{n_0}] \cup O_k.$$

   Since $a_{n_0} \in A$, there is $m \in \mathbb{N}$ such that $[0, a_{n_0}] \subset O_{i_1} \cup \ldots \cup O_{i_m}$. Consequently, $[0, x] \subset O_{i_1} \cup \ldots \cup O_{i_m} \cup O_k$, and this proves that $x \in A$.

2. The set A is open in $[0, 1]$. We prove that every point of A is an interior point of A in the relative topology on $[0, 1]$. Let $x \in A$, so $[0, x] \subset O_{i_1} \cup \ldots \cup O_{i_n}$ for some $n \in \mathbb{N}$. Let $i_0 \in I$ be an index such that $x \in O_{i_0}$. Since O_{i_0} is open, there is $r > 0$ with $(x - r, x + r) \subset O_{i_0}$. We proceed to show that

$$(x - r, x + r) \cap [0, 1] \subset A,$$

and this proves that x is an interior point of A. Let $y \in (x - r, x + r) \cap [0, 1]$. Since $x \in A$, if $y \leq x$, then $y \in A$ by (9.1). Now suppose $y > x$. From $[x, y] \subset [0, 1]$, we have

$$\begin{aligned}[0, y] = [0, x] \cup [x, y] &\subset [0, x] \cup (x - r, x + r) \subset [0, x] \cup O_{i_0} \\ &\subset O_{i_1} \cup \ldots \cup O_{i_n} \cup O_{i_0},\end{aligned}$$

and this proves that $y \in A$.

□

Theorem 9.1.4 may be considered equivalent with Theorem 8.1.3. The reason is that using that bounded closed intervals are compact, we will be able to prove that many subsets of $\mathbb{R}^n$ are compact.

Compactness is not a hereditary property. For example $A = \{0\} \cup \{1/n : n \in \mathbb{N}\}$ is compact (Example 9.3) but $B = \{1/n : n \in \mathbb{N}\}$ is not because its topology is discrete and B is infinite. However, compactness is hereditary on closed sets. If the space is Hausdorff, we can reverse that result in the following sense.

Proposition 9.1.5 *Every compact subset of a Hausdorff space is closed. In particular, every compact set in a metric space is bounded and closed.*

Example 9.6 The Hausdorff property cannot be removed in this result. For instance, the cofinite topology is not Hausdorff. This space is compact and any (closed or not closed) subset is compact because the relative topology is cofinite again (Example 2.26). In particular, the complements of singletons are compact but not closed. ♢

Example 9.7 We see that if X is compact, then any continuous function $f : X \to \mathbb{R}$ attains a minimum and a maximum.

The set $f(X)$ is a compact subset of $\mathbb{R}$, so it is bounded, hence there are supremum and infimum of $f(X)$. Both numbers are adherent points of $f(X)$ (Exercise 3.6.20). Since $f(X)$ is closed, both numbers belong to $f(X)$. Consequently, the supremum and infimum are the maximum and minimum, respectively, which may be not necessarily unique.

This result is not applicable if X is not compact. For example, the interval $(-\pi/2, \pi/2)$ is not compact because it is not closed, and the function $f(x) = \tan(x)$ defined on $(-\pi/2, \pi/2)$ is continuous but its image is $\mathbb{R}$. ♢

A further application of Proposition 9.1.5 is a criterion to prove that a map is a homeomorphism.

Theorem 9.1.6 *If $f: X \to Y$ is continuous, X is compact and Y is Hausdorff, then f is closed. Furthermore, if f is one-to-one (resp. injective) then f is a homeomorphism (resp. embedding).*

Example 9.8 In Example 6.23 we have seen that any segment $[x, y] \subset \mathbb{R}^n$, $x \neq y$, is homeomorphic to $[0, 1]$. Taking

$$\alpha: [0, 1] \to [x, y], \quad \alpha(t) = x + t(y - x),$$

then α is continuous and bijective and concluding that α is a homeomorphism because $[0, 1]$ is compact and $[x, y]$ is Hausdorff. $\Diamond$

We conclude this section proving that compactness is a productive property.

Theorem 9.1.7 (Tychonoff) *Two spaces are compact if and only if their product space is compact. Following an argument of induction, the result holds for a finite number of spaces.*

Proof If $X \times Y$ is compact then X and Y are compact because they are the images under the projections of $X \times Y$, that is compact.

Suppose that X and Y are compact and let $X \times Y = \bigcup_{i \in I}(O_i \times V_i)$ be a covering by open sets of the standard basis for the product topology, where O_i and V_i are open in X and of Y, respectively. Since $\{x\} \times Y \cong Y$ (Corollary 7.2.5), then $\{x\} \times Y$ is compact. Thus $\{O_i \times V_i : i \in I\}$ of $X \times Y$ is also an open covering of $\{x\} \times Y$. If we remove those $O_i \times V_i$ such that $x \notin O_i$, the remaining open sets form a covering of $\{x\} \times Y$. Thus there is $n_x \in \mathbb{N}$ such that

$$\{x\} \times Y \subset (O_1^x \times V_1^x) \cup \ldots \cup (O_{n_x}^x \times V_{n_x}^x).$$

Let $O^x = O_1^x \cap \ldots \cap O_{n_x}^x$, which it is an open neighborhood of x. We claim that

$$O^x \times Y \subset (O_1^x \times V_1^x) \cup \ldots \cup (O_{n_x}^x \times V_{n_x}^x).$$

Let $(z, y) \in O^x \times Y$. As $(x, y) \in \{x\} \times Y$, there is $i \in \{1, \ldots, n_x\}$ such that $y \in V_i^x$. Thus $(z, y) \in O^x \times V_i^x \subset O_i^x \times V_i^x$. Once the inclusion is proved, the collection of open sets $\{O^x : x \in X\}$ constitutes a covering of X and by compactness of X, there is $k \in \mathbb{N}$ such that $X = O^{x_1} \cup \ldots \cup O^{x_k}$. We then have

$$X \times Y = (O^{x_1} \cup \ldots \cup O^{x_k}) \times Y = (O^{x_1} \times Y) \cup \ldots \cup (O^{x_k} \times Y)$$

$$\subset \bigcup_{i=1}^{k} \Big((O_i^{x_i} \times V_i^{x_i}) \cup \ldots \cup (O_{n_{x_i}}^{x_i} \times V_{n_{x_i}}^{x_i}) \Big),$$

which is a finite subcovering. $\square$

9.2 Compactness in Euclidean Spaces

In this section, we study compactness in Euclidean spaces. Recall that a compact subset in a metric space must be bounded and closed. In Euclidean space, the converse is true; this is one of the main results of this chapter.

Theorem 9.2.1 (Heine-Borel) *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded with the Euclidean distance.*

Proof We shall prove the reverse implication. Let A be a bounded closed set in $\mathbb{R}^n$. Since A is bounded, there is $M > 0$ such that $|x| \leq M$ for all $x \in A$. If $x = (x_1, \ldots, x_n)$ then $|x_i| \leq |x| \leq M$ for all $x \in A$ and all $1 \leq i \leq n$. Thus

$$A \subset [-M, M] \times \overset{n}{\ldots} \times [-M, M].$$

Since A is closed and the finite Cartesian product of bounded closed intervals is compact by the Tychonoff theorem, A is compact. □

Corollary 9.2.2 *The only compact subsets of $A \subset \mathbb{R}^n$ are the bounded closed sets of $\mathbb{R}^n$ included in A.*

We give some crucial observations regarding the Heine-Borel theorem:

1. The notion of compactness in Euclidean space is now more vague and loses the idea of finiteness. Certainly, compactness mixes two different concepts: closedness, which is topological, and boundedness, which is metric.
2. In the proof of the Heine-Borel theorem, it is crucial that if A is bounded with respect to the Euclidean distance, then A is included in a Cartesian product of bounded closed intervals, and therefore in a compact set.
3. The Heine-Borel theorem fails if the subset is bounded with respect to another distance.
4. The Heine-Borel theorem is not true for subsets of $\mathbb{R}^n$. For example, $A = (0, 1]$ is a bounded closed subset of $(0, 2)$ but it is not compact.

Example 9.9 The sphere $\mathbb{S}^n$, the torus $\mathbb{T}^n = \mathbb{S}^1 \times \overset{n}{\ldots} \times \mathbb{S}^1$ and the Cantor set are compact.

Example 9.10 If $A \subset \mathbb{R}^n$ is bounded, then $\overline{A}$ is compact. This is because the closure of a bounded set is bounded (Exercise 4.17). ♢

The following theorem holds in any topological space and has many applications in analysis.

Theorem 9.2.3 (Bolzano-Weierstrass) *In a compact space, every infinite subset has a point of accumulation. As a consequence, each infinite bounded subset of $\mathbb{R}^n$ has a point of accumulation.*

Proof Suppose the assertion of theorem is false. Let $A \subset X$ be an infinite set without points of accumulation. Then for each $x \in X$ there is an open set O_x such that $(O_x \setminus \{x\}) \cap A = \emptyset$. Since X is compact, taking the open covering $\{O_x : x \in X\}$ of X, there is a finite subcovering $X = O_{x_1} \cup \ldots \cup O_{x_n}$. Then

$$A = X \cap A = (O_{x_1} \cup \ldots \cup O_{x_n}) \cap A = (O_{x_1} \cap A) \cup \ldots \cup (O_{x_n} \cap A).$$

Because $O_{x_i} \cap A$ contains one point at most, then A is finite, a contradiction. □

Corollary 9.2.4 *Every sequence in a compact subset of* $\mathbb{R}^n$ *has a convergent subsequence. As a consequence, every bounded sequence of* $\mathbb{R}^n$ *has a convergent subsequence.*

Proof Let (a_n) be a sequence in a compact set A. If $\{a_n : n \in \mathbb{N}\}$ is finite, then necessarily there is one element in the sequence that appears infinitely repeated. All these elements constitute a convergent subsequence.

Suppose that $\{a_n : n \in \mathbb{N}\}$ is infinite. By the Bolzano-Weierstrass theorem, there is an accumulation point x of the sequence. We shall construct a convergent subsequence as follows. Let $B_1(x)$ be the ball of radius 1 centered at x. Then $B_1(x) \setminus \{x\}$ intersects (a_n) and we select one element in the intersection which is denoted by $a_{\sigma(1)}$. The set $\{a_n : n > \sigma(1)\}$ is infinite. Taking the radius $1/2$, there is $n > \sigma(1)$ such that $a_n \in B_{1/2}(x) \setminus \{x\}$. Let $\sigma(2)$ denote this natural number n. Repeating the argument, we construct a subsequence $(a_{\sigma(n)})$ such that $|a_{\sigma(n)} - x| < 1/n$ for all $n \in \mathbb{N}$. This proves that $(a_{\sigma(n)}) \to x$ and because A is closed, then $x \in A$.

If (x_n) is a bounded sequence of $\mathbb{R}^n$, there is $r > 0$ such that $\{x_n : n \in \mathbb{N}\} \subset \overline{B_r(0)}$. Since $\overline{B_r(0)}$ is compact, we apply the result. □

The next corollary proves to be especially useful in characterizing compact sets of the Euclidean space.

Corollary 9.2.5 *A subset* $A \subset \mathbb{R}^n$ *is compact if and only if every sequence of* A *has a convergent subsequence.*

Proof We need only establish the reverse implication. If A is finite, then A is clearly compact. Assume that A is infinite. Consider a countable basis β for the topology (Theorem 4.2.6). Applying (iii) of Theorem 9.1.3, let $\{O_n : n \in \mathbb{N}\}$ be an open covering of A formed by elements of β. If A is not compact, there is no a finite subcovering from the above covering. Thus for each $n \in \mathbb{N}$, let $a_n \in A$ such that $a_n \notin \bigcup_{k=1}^n O_k$ with the property that $a_n \neq a_k$ for $k = 1, \ldots, n-1$, because $A \setminus \bigcup_{k=1}^n O_k$ is infinite. By hypothesis, the sequence (a_n) has an accumulation point $a \in A$. This is contradictory because the point a belongs to a certain O_m and $a_n \notin O_m$ for $n > m$. □

As a conclusion, in Euclidean spaces there are three equivalent definitions of a compact space. Besides the initial definition regarding finite subcoverings, compact sets are the closed and bounded subsets (Heine-Borel) and those sets where every sequence possesses a convergent subsequence (Bolzano-Weierstrass).

9.3 Worked Exercises

Exercise 9.1 Let $\mathbb{R}$ be the real line equipped with the strictly right topology $\tau_{sr} = \{(x, \infty) : x \in \mathbb{R}\} \cup \{\mathbb{R}, \emptyset\}$. Prove that the space is not compact and investigate which subsets are compact. Prove that the intervals $(-\infty, a]$ are closed but not compact.

Solution

1. The space is not compact because $\{(x, \infty) : x \in \mathbb{R}\}$ is an open covering but there is no finite subcovering: if $\{(x_i, \infty) : 1 \leq i \leq n\}$ covers $\mathbb{R}$, then $\mathbb{R} = \bigcup_{i=1}^{n}(x_i, \infty) = (x, \infty)$, where $x = \min\{x_i : 1 \leq i \leq n\}$, which is not possible.
2. Let $A \subset \mathbb{R}$ be a compact set. Using again the covering $\{(x, \infty) : x \in \mathbb{R}\}$, we deduce that there is $x \in \mathbb{R}$ such that $A \subset (x, \infty)$. This proves that A is bounded from below. Let $\alpha = \inf(A)$. We prove that a subset A is compact if and only if A has a minimum, or equivalently, if $\alpha \in A$.

   Assume that $A \subset \mathbb{R}$ is compact and suppose that $\alpha \notin A$. Consider the collection of open sets $\{(x, \infty) : x > \alpha\}$. We see that $A \subset \bigcup_{x>\alpha}(x, \infty)$. If $a \in A$ then $\alpha \leq a$, in fact $\alpha < a$ because $\alpha \notin A$. By definition of infimum, there is $b \in A$ such that $\alpha \leq b < a$, concluding that $a \in (b, \infty)$. The compactness of A then implies that there is $n \in \mathbb{N}$ such that

$$A \subset (x_1, \infty) \cup \ldots \cup (x_n, \infty).$$

   Thus $A \subset (x_{i_0}, \infty)$ where $x_{i_0} = \min\{x_1, \ldots, x_n\}$. From this inclusion it follows that $x_{i_0} \leq a$ for all $a \in A$, so x_{i_0} is a lower bound of A. Thus $x_{i_0} \leq \alpha$, a contradiction.

   We prove that the subsets having a minimum are compact. Let $A \subset \bigcup_{i \in I}(x_i, \infty)$ and $\alpha = \min(A)$. Since $\alpha \in A$, it follows that $\alpha \in (x_{i_0}, \infty)$ for some $i_0 \in I$. This means that $x_{i_0} < \alpha$, so $x_{i_0} < a$ for all $a \in A$. This proves that $A \subset (x_{i_0}, \infty)$, so we have obtained a finite subcovering.
3. It is clear that an interval $(-\infty, x]$ is closed but not compact because it is not bounded from below. □

Exercise 9.2 Study the compactness properties of the right order topology on $\mathbb{R}$.

Solution

1. The collection of open sets $\{[x, \infty) : x \in \mathbb{R}\}$ is a covering of $\mathbb{R}$. However, there is no finite subcovering because the union of a finite number of intervals $[x, \infty)$ is an interval of type $[a, \infty)$. Thus the space is not compact.
2. By using the above open covering, if $A \subset \mathbb{R}$ is compact, then A is included in some interval $[x, \infty)$, in particular, A is bounded from below. Let $\alpha = \inf(A)$. We see that $A \subset \mathbb{R}$ is compact if and only if $\alpha \in A$.

   Assume that A is compact. It is clear that $A \subset \bigcup_{a \in A}[a, \infty)$. Since A is compact, there is finite subcovering of A,

$$A \subset [a_1, \infty) \cup \ldots \cup [a_n, \infty) = [a_0, \infty),$$

where $a_0 = \min\{a_1, \ldots, a_n\}$. From this inclusion, we deduce $a_0 \leq \alpha$ for all $a \in A$ and consequently $a_0 \leq \alpha$. Since $a_0 \in A$, then $a_0 = \alpha$ proving that $\alpha \in A$.

Assume that A has a minimum and we prove that A is compact. Indeed, assume $A \subset \bigcup_{i \in I}[x_i, \infty)$. Since $\alpha \in A$, let $i_0 \in I$ such that $\alpha \in [x_{i_0}, \infty)$. Then $x_{i_0} \leq \alpha$, so $x_{i_0} \leq a$ for all $a \in A$. This proves that $A \subset [x_{i_0}, \infty)$ and this is a finite subcovering as desired.

In conclusion, compact subspaces are those sets that are bounded from below and have a minimum. □

Exercise 9.3 Study the compactness properties of $(\mathbb{R}, \tau_{CC})$ (Exercise 3.4).

Solution For each $n \in \mathbb{N}$, let $F_n = \{n, n+1, \ldots\}$. Since F_n is a countable set, $\mathbb{R} \setminus F_n$ is open in τ_{CC}. It is clear that $\{\mathbb{R} \setminus F_n : n \in \mathbb{N}\}$ is a covering of $\mathbb{R}$ because

$$\bigcup_{n \in \mathbb{Z}}(\mathbb{R} \setminus F_n) = \mathbb{R} \setminus \bigcap_{n \in \mathbb{N}} F_n = \mathbb{R} \setminus \emptyset = \mathbb{R}.$$

Assume that there is a finite subcovering $\{\mathbb{R} \setminus F_{n_1}, \ldots, \mathbb{R} \setminus F_{n_m}\}$ of $\mathbb{R}$. Then

$$\mathbb{R} = (\mathbb{R} \setminus F_{n_1}) \cup \ldots \cup (\mathbb{R} \setminus F_{n_m}) = \mathbb{R} \setminus (\bigcap_{j=1}^{m} F_{n_j}) = \mathbb{R} \setminus F_{\max\{n_1,\ldots,n_m\}},$$

which is impossible because $F_{\max\{n_1,\ldots,n_m\}} \neq \emptyset$.

We see which subspaces are compact. Recall that the collection of closed sets is formed by the countable subsets of $\mathbb{R}$. Since any subset of a countable set is countable, the relative topology in any subspace is the co-countable topology. The previous argument for $\mathbb{R}$ proves that any uncountable subset of $\mathbb{R}$ is not compact.

Suppose that A is countable. If A is finite, then it is compact. If A is infinite, let $A = \{a_n : n \in \mathbb{N}\}$. We prove that A is not compact. For each $n \in \mathbb{N}$, let $F_n = A \setminus \{a_n\}$. Then F_n is closed in the relative topology of A and $\bigcap_{n \in \mathbb{N}} F_n = \emptyset$. If A is compact, there is $k \in \mathbb{N}$ such that $F_{i_1} \cap \ldots \cap F_{i_k} = \emptyset$. This is a contradiction because $F_{i_1} \cap \ldots \cap F_{i_k} = A \setminus \{a_{i_1}, \ldots, a_{i_k}\} \neq \emptyset$ since A is infinite. □

Exercise 9.4 Study the compactness of the nested interval topology (Exercise 2.25) and characterize the compact subsets.

Solution

1. The space is not compact because $(0, 1) = \bigcup_{n \in \mathbb{N}}(0, 1 - \frac{1}{n})$ and if there is a finite subcovering $\bigcup_{i=1}^{k}(0, 1 - \frac{1}{m_j})$, then
$$(0, 1) = \left(0, 1 - \frac{1}{m_1}\right) \cup \ldots \cup \left(0, 1 - \frac{1}{m_k}\right) = \left(0, 1 - \frac{1}{m}\right),$$
where $m = \max\{m_1, \ldots, m_n\}$, which is absurd.
2. Likewise, if A is a compact subspace and taking the above covering, then A must be included in some interval $(0, 1 - \frac{1}{m})$. Thus $\sup(A) < 1$. We see that the compact subsets are those subsets A such that $\sup(A) < 1$. It remains to prove the backwards implication. Suppose that $\{(0, 1 - 1/m_i) : i \in I\}$ is an open

covering of A. If the set of indices $\{m_i : i \in I\}$ is not bounded from above, then the supremum of $\{1 - 1/m_i : i \in I\}$ is 1, in particular, there is m_{i_0} such that $\sup(A) < 1 - 1/m_{i_0}$. This implies that $A \subset (0, 1 - 1/m_{i_0})$, obtaining the finite subcovering. If $\{m_i : i \in I\}$ is bounded from above, this set is finite and the same occurs for the covering. □

Exercise 9.5 If Y is compact, prove that the projection $p\colon X \times Y \to X$ is closed.

Solution Let F be closed in $X \times Y$. To prove that $p(F)$ is closed, we see that $X \setminus p(F)$ is open, or equivalently, that all its points are interior. Let $x \notin p(F)$. Then $(\{x\} \times Y) \cap F = \emptyset$, so $\{x\} \times Y \subset (X \times Y) \setminus F$. As $(X \times Y) \setminus F$ is open, for each $y \in Y$ there are open sets O_y and O^y of X and Y respectively, such that $(x, y) \in O_y \times O^y \subset (X \times Y) \setminus F$. Since $\{O^y : y \in Y\}$ is an open covering of Y and Y is compact, there is a finite subcovering $Y = O^{y_1} \cup \ldots \cup O^{y_n}$.

Let $U = O_{y_1} \cap \ldots \cap O_{y_n}$, with U a neighborhood of x. If we show that $U \subset X \setminus p(F)$, then x is an interior point of $X \setminus p(F)$. Indeed, if $z \in U \cap p(F)$ then there is $y \in Y$ such that $(z, y) \in F$. Let $j \in \{1, \ldots, n\}$ be an index such that $y \in O^{y_j}$. Then

$$(z, y) \subset U \times O^{y_j} \subset O_{y_j} \times O^{y_j} \subset (X \times Y) \setminus F,$$

a contradiction. □

The following exercise defines *balls of subsets* in metric spaces.

Exercise 9.6 Let (X, d) be a metric space, $A \subset X$ and $r > 0$. A *r-ball* of A in X is defined by

$$U_r(A) = \{x \in X : d(x, A) < r\}.$$

Prove that $U_r(A)$ is an open set containing A. If A is compact and O is an open set with $A \subset O$, prove that there is $r > 0$ such that $A \subset U_r(A) \subset O$.

Solution

1. It is clear that $A \subset U_r(A)$. To establish that $U_r(A)$ is open, let $f\colon X \to \mathbb{R}$ define $f(x) = d(x, A)$. This function is continuous (Exercise 5.11) and $U_r(A) = f^{-1}((-\infty, r))$, proving that $U_r(A)$ is open.
2. The function

$$g\colon A \to \mathbb{R}, \quad g(a) = d(a, X \setminus O)$$

is continuous and defined on a compact set, so g attains a minimum. Let $a_0 \in A$ such that $d(a_0, X \setminus O) \le d(a, X \setminus O)$ for all $a \in A$. Since $X \setminus O$ is a closed set, if $d(a_0, X \setminus O) = 0$, then $a_0 \in X \setminus O$, which is impossible because $A \subset O$. Thus $r = d(a_0, X \setminus O)$ is a positive number. Finally we prove that $U_r(A) \subset O$. Indeed, let $x \in U_r(A)$, that is, $d(x, A) < r$. In particular, $d(x, a_0) < r$. Then $x \in O$ because, otherwise, $x \in X \setminus O$, so $r \le d(a_0, x)$ and this is contradictory. □

Exercise 9.7 Let X be a Hausdorff compact space. Prove that if X has no isolated points, then X is not countable.

Solution Assume X is countable. A first case is when X is finite, $X = \{x_1, \ldots, x_n\}$. For each $i \geq 2$, take disjoint neighborhoods U_i of x_1 and V_i of x_i. Thus $U = U_2 \cap \ldots \cap U_n$ is a neighborhood of x_1 and $(U \setminus \{x_1\}) \cap X = \emptyset$, hence x_1 is an isolated point, a contradiction.

Now assume that X is an infinite countable set. Let $X = \{x_n : n \in \mathbb{N}\}$.

Claim If $x \in X$ and $G \neq \emptyset$ is an open set, then there is a nonempty open set O such that $O \subset G$ and $x \notin \overline{O}$.

Let us prove first that the claim yields the desired contradiction. Taking $G = X$, there is an open set O_1 such that $x_1 \notin \overline{O_1}$. Now take x_2 and the open set O_1. By the claim, let O_2 be an open set such that $O_2 \subset O_1$ and $x_2 \notin \overline{O_2}$. Continuing in this way, we construct a sequence of open sets (O_n) with the property $O_{n+1} \subset O_n$ and $x_n \notin \overline{O_n}$. Consider the family of closed sets $\{\overline{O_n} : n \in \mathbb{N}\}$. We have $\bigcap_{n\in\mathbb{N}} \overline{O_n} = \emptyset$, for if there were a point $x_m \in X$ in the intersection, then $x_m \in \overline{O_m}$, which is not possible. On the other hand, the space X is compact and Theorem 9.1.3 implies that there is $n_0 \in \mathbb{N}$ such that

$$\overline{O_{i_1}} \cap \ldots \cap \overline{O_{i_{n_0}}} = \emptyset.$$

But this intersection is $\overline{O_k}$ where $k = \max\{i_1, \ldots, i_{n_0}\}$, which is nonempty, giving a contradiction.

We now prove the claim. Given the open set G, we can assert that there is $y \in G$ such that $x \neq y$. This is clear if $x \notin G$ because $G \neq \emptyset$. If $x \in G$, then because x is not isolated, we derive that $(G \setminus \{x\}) \cap X \neq \emptyset$ and thus we can take a point y in this intersection. By the Hausdorff property, there are two disjoint open sets W and W' such that $x \in W$ and $y \in W'$. Set $O = W' \cap G$, so $O \subset G$ and $y \in O$. If $x \in \overline{O}$, then $W \cap O = (W \cap W') \cap G \neq \emptyset$, which is not possible. □

Exercise 9.8 Let $f : X \to Y$ be a mapping between two Hausdorff compact spaces. Prove that f is continuous if and only if the graph $G(f)$ of f is closed in $X \times Y$.

Solution Define

$$\phi : X \to X \times Y, \quad \phi(x) = (x, f(x)).$$

Notice that the image of ϕ is $G(f)$. This map is merely the evaluation $e(\mathbb{1}_X, f)$. By Theorem 7.2.5, f is continuous if and only if ϕ is an embedding, that is, $\phi : X \to G(f)$ is a homeomorphism. If f is continuous, then $G(f) \cong X$ is compact. Since $G(f) \subset X \times Y$ and $X \times Y$ is Hausdorff, then $G(f)$ is closed.

Reciprocally, assume that $G(f)$ is closed in $X \times Y$. We prove that f is continuous by showing that the inverse image of any closed set is closed. Let $F' \subset Y$ be a closed set. Since $X \times Y$ is compact and $G(f)$ is closed, $G(f)$ is compact. The restriction $p_{|G(f)} : G(f) \to X$ of the first projection is closed because the domain is compact and the codomain is Hausdorff. On the other hand,

$$f^{-1}(F') = p_{|G(f)}((p_{|G(f)})^{-1}(F')).$$

Since p is continuous, $(p_{|G(f)})^{-1}(F')$ is a closed set in $G(f)$ and thus $f^{-1}(F')$ is closed in X because $p_{|G(f)}$ is a closed map. □

Exercise 9.9 Let (X, d) be a compact metric space. If $f: X \to X$ satisfies

$$d(f(x), f(y)) = d(x, y)$$

for all $x, y \in X$, prove that f is surjective.

Solution Notice that f is injective because if $f(x) = f(y)$, then $d(f(x), f(y) = 0 = d(x, y)$, so $x = y$. Observe that f is continuous because f is Lipschitz.

Assume that f is not surjective and let $x \notin f(X)$. Since $f(X)$ is a compact set in a Hausdorff space, the set $f(X)$ is closed. Thus $d(x, f(X)) > 0$ in view of Proposition 4.4.7. Let $r = d(x, f(X))$. Construct a sequence (x_n) by recurrence,

$$x_1 = x, \quad x_{n+1} = f(x_n).$$

Since (x_n) is included in a compact set, and passing to a subsequence if necessary, we may assume that (x_n) is convergent. By the previous exercise, (x_n) is a Cauchy sequence. Since $x \notin f(X)$, then $d(x_1, x_n) \geq r$ for all $n > 1$. Let $n, m \in \mathbb{N}$ and suppose $n \leq m$. Then

$$d(x_n, x_m) = d(f(x_{n-1}), f(x_{m-1})) = d(x_{n-1}, x_{m-1})) = \ldots = d(x_1, x_{m-n}) \geq r,$$

proving that (x_n) is not Cauchy, a contradiction. As a last remark, notice that the result fails if the space is not compact. For example, $f: (0, \infty) \to (0, \infty)$ defined by $f(x) = x + 1$ preserves distances but it is not surjective. □

Exercise 9.10 Let (X, d) be a compact metric space. If $f: X \to X$ satisfies

$$d(f(x), f(y)) < d(x, y)$$

for all $x, y \in X$, $x \neq y$, then f has a unique fixed point.

Solution We prove the existence of the fixed point. The map f is continuous because it is Lipschitz (with $K = 1$). Define $g: X \to \mathbb{R}$ by $g(x) = d(x, f(x))$. This function g is continuous because $g = d \circ e(\mathbb{1}_X, f)$. Since X is compact, there is $x_0 \in X$ such that $g(x_0) = \min g(X) = m$. If $m > 0$, then $x_0 \neq f(x_0)$, but by the property of f,

$$d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = g(x_0) = m.$$

This contradicts the definition of m. Thus $m = 0$ and $f(x_0) = x_0$. Uniqueness is clear because if $y_0 \in X$ is a distinct fixed point, then

$$d(y_0, x_0) = d(f(y_0), f(x_0)) < d(y_0, x_0),$$

which is absurd. The result fails, in general, if f preserves d. For instance, the rotation $f : \mathbb{S}^1 \to \mathbb{S}^1$ defined by $f(x, y) = (-y, x)$ has no fixed points but $d(f(p), f(q)) = d(p, q)$ for all $p, q \in \mathbb{S}^1$. □

In a Hausdorff space, two points can be separated by open sets. We generalize this result for two disjoint compact sets.

Exercise 9.11 If A and B are two disjoint compact sets in a Hausdorff space, prove there are two disjoint open sets O_1 and O_2 such that $A \subset O_1$ and $B \subset O_2$.

Solution Fix $a \in A$. Because the space is Hausdorff, it follows that for each $b \in B$ there are two disjoint open sets G_b and O_b with $a \in G_b$ and $b \in O_b$. Since $\{O_b : b \in B\}$ is an open covering of B, compactness of B leads to the existence of a finite subcovering $B \subset O_{b_1} \cup \ldots \cup O_{b_n}$ for some $n \in \mathbb{N}$. Then the neighborhood $G^a = G_{b_1} \cap \ldots \cap G_{b_n}$ of a and the open set $O^a = O_{b_1} \cup \ldots O_{b_n}$ are disjoint.

Consider the open covering $\{G^a : a \in A\}$ of A. Again, there is a finite subcovering $A \subset G^{a_1} \cup \ldots \cup G^{a_m}$ with $m \in \mathbb{N}$. Finally, the open sets

$$O_1 = G^{a_1} \cup \ldots \cup G^{a_m}, \quad O_2 = O^{a_1} \cap \ldots \cap O^{a_m}$$

are disjoint (as in the proof of Proposition 9.1.5) and $A \subset O_1$ and $B \subset O_2$. □

Exercise 9.12 Let $X = (\mathbb{R} \times \{0, 1\}, \tau_{CC} \times \tau_T)$, where τ_{CC} and τ_T are the topologies of the countable complements and trivial topology, respectively. Prove that X is not compact but every infinite set has a point of accumulation.

Solution

1. The space X is not compact because the factor $(\mathbb{R}, \tau_{CC})$ is not (Exercise 9.3).
2. Argue by contradiction. Let $A \subset \mathbb{R} \times \{0, 1\}$ be an infinite set and assume that there are no points of accumulation. Let $(x, y) \in A$ be given. Since this point is isolated, there is a countable set F such that $x \in X \setminus F$ and

$$\Big(((\mathbb{R} \setminus F) \times \{0, 1\}) \setminus \{(x, y)\}\Big) \cap A = \emptyset.$$

Then $A \subset (F \times \{0, 1\}) \cup \{(x, y)\}$. As $F \times \{0, 1\} = (F \times \{0\}) \cup (F \times \{1\})$, there are two countable sets F_1 and F_2 of $\mathbb{R}$ such that

$$A = (F_1 \times \{0\}) \cup (F_2 \times \{1\}). \tag{9.2}$$

We prove that $F_1 = F_2$. On the contrary, if there is $x \in F_2$ such that $x \notin F_1$, the point $(x, 0)$ does not belong to A. In view that $(x, 0)$ is not a point of accumulation, there is a countable set F_3 such that $x \notin F_3$ and

$$\Big((\mathbb{R} \setminus F_3) \times \{0, 1\}) \setminus \{(x, 0)\}\Big) \cap A = \emptyset,$$

or equivalently, $A \subset (F_3 \times \{0, 1\}) \cup \{(x, 0)\}$. Since $(x, 1) \in F_2 \times \{1\} \subset A$, we have $x \in F_3$, which is not possible.

From (9.2), we have $A = F_1 \times \{0, 1\}$ where F_1 is a countable set. Let $x \in F_1$. Since $(x, 0)$ is not a point of accumulation, there is a countable set $F' \subset \mathbb{R}$, $x \notin F'$, such that

$$A \subset (F' \times \{0, 1\}) \cup \{(x, 0)\}.$$

As $x \in F_1$, we have $(x, 1) \in A$, in particular, $(x, 1) \in F' \times \{0, 1\}$, hence $x \in F'$, obtaining a contradiction. □

Exercise 9.13 Consider $(\mathbb{R}^*, \tau)$, where $\mathbb{R}^* = \mathbb{R} \cup \{\omega\}$, $\omega \notin \mathbb{R}$ and τ is the topology generated by

$$\beta = \{(a, b) : a, b \in \mathbb{R}\} \cup \{\{\omega\} \cup (-\infty, a) \cup (b, \infty) : a \leq b, a, b \in \mathbb{R}\}.$$

(See Exercise 8.19.) Prove that $(\mathbb{R}^*, \tau)$ is homeomorphic to $\mathbb{S}^1$.

Solution The idea of the homeomorphism is to map the points of $\mathbb{R}$ to the punctured circle $\mathbb{S}^1 \setminus \{N\}$ where $N = (0, 1)$ and the point ω into N. Let $\pi : \mathbb{S}^1 \to \mathbb{R}$ be the stereographic projection whose expression is $\pi(x, y) = x/(1 - y)$. Define

$$\phi \colon \mathbb{S}^1 \to \mathbb{R}^*, \quad \phi(x, y) = \begin{cases} \pi(x, y), & (x, y) \neq N \\ \omega, & (x, y) = N. \end{cases}$$

It is elementary that ϕ is bijective. Since $\mathbb{S}^1$ is compact, ϕ is a homeomorphism if ϕ is continuous and $(\mathbb{R}^*, \tau^*)$ is Hausdorff.

1. The map ϕ is continuous. The relative topology on $\mathbb{R}$ as subspace of $(\mathbb{R}^*, \tau)$ is the usual topology. It is obvious that ϕ is continuous in all points of $\mathbb{S}^1 \setminus \{N\}$. Here use that $\mathbb{S}^1 \setminus \{N\}$ is open in $\mathbb{S}^1$ and Proposition 5.2.4. We prove that ϕ is continuous at N. Since $\phi(N) = \omega$, let $U' = \{\omega\} \cup (-\infty, a) \cup (b, \infty)$ be a basic neighborhood of ω and we will find $\delta > 0$ such if $|(x - y) - N| < \delta$, then $\phi(x, y) \in U'$. Let $r > 0$ such that $(-\infty, -r) \cup (r, \infty) \subset U'$. If $(x, y) \in \mathbb{S}^1$, with $y > 0$, then

$$\pi(x, y) = \frac{x}{1 - y} = \frac{x}{1 - \sqrt{1 - x^2}} = \frac{1 + \sqrt{1 - x^2}}{x} \to \infty$$

as $x \to 0$. Given $r > 0$, let $\delta > 0$ with $\delta < 1$, such that if $|x| < \delta$, $|\pi(x, y)| > r$. Finally, if $|(x, y) - N| < \delta$, then $|x| < \delta$ and $|y - 1| < \delta$. From $|y - 1| < \delta < 1$, we have $y > 0$, and from $|x| < \delta$, $|\pi(x, y)| > r$, that is, $\phi(x, y) \in U'$.

2. The space $(\mathbb{R}^*, \tau)$ is Hausdorff. Let $x, y \in \mathbb{R}^*$. If $x, y \in \mathbb{R}$ then they can be separated with the usual topology, so also in $(\mathbb{R}^*, \tau)$ because the relative topology on $\mathbb{R}$ is the Euclidean topology. For $x \in \mathbb{R}$ and ω, it is enough $U = (x-1, x+1)$ and $V = \{\omega\} \cup (-\infty, x - 1) \cup (x + 1, \infty)$. □

Exercise 9.14 Let (X, τ) be a Hausdorff space and define

$$\beta' = \{\emptyset\} \cup \{O \in \tau : X \setminus O \text{ is compact in } X\}.$$

Prove that β' is a basis for a topology τ' and that $\tau' \subset \tau$. Prove that (X, τ') is compact. If (X, τ) is compact, prove that $\tau' = \tau$.

Solution

1. We show that β' is a basis for a topology. We point out that $X \in \beta'$ because $X \setminus X = \emptyset$ is compact. Now we see that the intersection of two elements of β' is an element of β'. If $B_1, B_2 \in \beta'$ then $X \setminus B_i$ is compact, so $(X \setminus B_1) \cup (X \setminus B_2) = X \setminus (B_1 \cap B_2)$. This set is compact because in a Hausdorff space, the intersection of two compact sets is compact: the intersection of two compact sets is intersection of two closed sets, so it is a closed set contained in a compact set, hence, it is a compact set. This proves that $B_1 \cap B_2 \in \beta'$.
2. By definition of β', is clear that $\beta' \subset \tau$, so this also applies to the topology τ' generated by β'. This yields the inclusion $\tau' \subset \tau$. Observe that $\tau' \neq \tau$ in general. For example, if $X = \mathbb{R}$, the set $(0, \infty)$ is open in τ_u but not in τ'.
3. We prove that (X, τ') is compact. Let $\{O_i : i \in I\}$ be a covering by basic elements of β'. Fix O_{i_0} an element of the covering, $i_0 \in I$. Then $\{O_i : i \in I, i \neq i_0\}$ is a covering of $X \setminus O_{i_0}$ by open sets of τ because $\tau' \subset \tau$. Since $X \setminus O_{i_0}$ is compact, there is a finite subcovering of $X \setminus O_{i_0}$. If we add O_{i_0} to this subcovering, we obtain the finite subcovering of X.
4. If (X, τ) is compact, we see that $\tau' = \tau$. It only remains to check the inclusion $\tau \subset \tau'$. Let $O \in \tau$ be given. Then $X \setminus O$ is closed in a compact space, hence $X \setminus O$ is compact. By definition of β', $O \in \beta' \subset \tau'$, so $O \in \tau'$. □

Exercise 9.15 Study the compactness properties of the included point topology.

Solution Let p be the particular point that defines the topology. For each $x \in X$, let $B_x = \{p, x\}$.

1. The family $\{B_x : x \in X\}$ is a covering by open sets. If X is compact, there is a finite subcovering

$$X = B_{x_1} \cup \ldots \cup B_{x_n} = \{p, x_1, \ldots, x_n\},$$

hence X is finite. Thus the space is compact if and only if it is finite.

2. Let A be a subspace of X. Then the relative topology on A is discrete if $p \notin A$ or it is the included point topology (for identical point p) if $p \in A$. In either case, A is compact if and only if it is finite. □

Exercise 9.16 Study the compactness properties of the excluded point topology.

Solution Let p be the particular point in the space that defines the topology.

1. The space is compact. Certainly, if $\{O_i : i \in I\}$ is an open covering, let $i_0 \in I$ be the index such that $p \in O_{i_0}$. But the only open set containing p is X itself, so $O_{i_0} = X$. Thus $\{O_{i_0}\}$ is the finite subcovering.
2. Let $A \subset X$ be given. If $p \notin A$ then the relative topology is discrete, so A is compact if and only if A is finite. If $p \in A$, the relative topology is the excluded point topology (for the same p), which is compact by item (1). Thus the compact subspaces are the finite sets and sets containing the point p. □

Exercise 9.17 Let X be a Hausdorff compact space, $x \in X$ and C_x its connected component. Prove that C_x coincides with the quasicomponent Q_x (Exercise 8.12).

Solution We learned in Exercise 8.12 the identity

$$Q_x = \bigcap \{O \subset X : x \in O,\ O \text{ is open and closed}\} \tag{9.3}$$

and that $C_x \subset Q_x$. For the inclusion $Q_x \subset C_x$, it suffices to verify that Q_x is connected since C_x is the largest connected set containing x. Notice that $x \in Q_x$ and that Q_x is closed because it is intersection of closed sets.

Let $Q_x = F_1 \cup F_2$ be a separation by two disjoint closed sets F_1 and F_2 of Q_x. Suppose that $x \in F_1$. Because we must prove that Q_x is connected, we need to show that $F_2 = \emptyset$.

Since Q_x is closed, the sets F_1 and F_2 are also closed in X, hence compact. In view of Exercise 9.11, there are disjoint open sets G_1 and G_2 that separate F_1 and F_2. Taking complements in the inclusion $Q_x \subset G_1 \cup G_2$,

$$(X \setminus G_1) \cap (X \setminus G_2) \subset X \setminus Q_x = \bigcup \{X \setminus O : x \in O, O \text{ is open and closed}\}.$$

The set $(X \setminus G_1) \cap (X \setminus G_2)$ is compact because it is closed in X. Thus there is $n \in \mathbb{N}$ such that

$$(X \setminus G_1) \cap (X \setminus G_2) \subset (X \setminus O_1) \cup \ldots \cup (X \setminus O_n) = X \setminus \bigcap_{i=1}^{n} O_i.$$

Let $O = \bigcap_{i=1}^{n} O_i$. Then $F_1 \cup F_2 = Q_x \subset O \subset G_1 \cup G_2$. The set $O \cap G_1$ is open in X. We see that it is also closed in X. The set O is closed because it is intersection of closed sets. Since $G_1 \subset X \setminus G_2$, then $\overline{G_1} \subset X \setminus G_2$ and it follows

$$\overline{O \cap G_1} \subset \overline{O} \cap \overline{G_1} = O \cap \overline{G_1} = O \cap (G_1 \cup G_2) \cap \overline{G_1} = O \cap G_1,$$

giving that $O \cap G_1$ is closed. Because $O \cap G_1$ is open and closed, $O \cap G_1$ is a member of the elements that define Q_x, in particular $Q_x \subset O \cap G_1 \subset G_1$. This yields $Q_x \cap F_2$. Thus $\emptyset = Q_x \cap F_2 = (F_1 \cup F_2) \cap F_2 = F_2$. □

Exercise 9.18 Let X be a Hausdorff connected compact space. Let F be a nontrivial closed set and let C be a connected component of F. Prove that

$$C \cap \mathrm{Bd}(F) \neq \emptyset.$$

Solution Since F is closed, $\mathrm{Bd}(F) \subset F$. The proof proceeds by contradiction. Assume that $C \cap \mathrm{Bd}(F) = \emptyset$. Then $C \subset \mathring{F} \cup \mathrm{ext}(F)$ but as C is a connected component of F, this union is disjoint and $\mathrm{ext}(F) \subset X \setminus F$, we deduce $C \subset \mathring{F}$. We prove that C is closed and open in X. In such a case, $C = X$ because X is connected. Since $C \subset \mathring{F}$, then $F = X$, obtaining a contradiction.

The set C is closed in X because C is closed in F (any connected component is closed) and F is closed in X. We prove that C is open in X showing that all its points are interior. Let $x \in C$ and we see that x is an interior point. The set F is compact and Hausdorff, so $C = Q_x$ by Exercise 9.17. Taking complements in F, and by the identity (9.3),

$$\mathrm{Bd}(F) \subset F \setminus C = \bigcup \{F \setminus O : \ O \text{ is open and closed in } F \text{ with } x \in O\}.$$

Since $\mathrm{Bd}(F)$ is closed and X is compact, we deduce that $\mathrm{Bd}(F)$ is compact. Then there is a finite subcovering

$$\mathrm{Bd}(F) \subset (F \setminus O_1) \cup \ldots \cup (F \setminus O_n) = F \setminus \bigcap_{i=1}^{n} O_i.$$

Let $O = O_1 \cap \ldots \cap O_n$, which is open in F with $x \in O$. Taking complements, it follows $O \subset F \setminus \mathrm{Bd}(F) = \mathring{F}$. As O is open in F, there is an open set G in X such $O = G \cap F$. Then $O = G \cap \mathring{F}$ because $O \subset \mathring{F}$, so O is intersection of two open sets in X, hence open. Finally, $O \subset C$ because all O_i are included in C. Then we have found an open set O of X such that $x \in O \subset C$, proving that $x \in \mathring{C}$. □

We give an equivalent definition of the Cantor set as a product space.

Exercise 9.19 For each $n \in \mathbb{N}$, let $X_n = \{0, 2\}$ be endowed with the discrete topology. Prove that the Cantor set $\mathfrak{C}$ is homeomorphic to $\Pi_{n \in \mathbb{N}} X_n = \{0, 2\}^{\mathbb{N}}$.

Solution Note that $\{0, 2\}^{\mathbb{N}}$ is the set consisting of all sequences where the terms of the sequences are 0 or 2. The crucial issue in this exercise is the following

Claim. Any number of $\mathfrak{C}$ has a unique ternary expansion using only the digits 0 and 2.

We only give an outline of the proof of the claim. Any $x \in [0,1]$ can be expressed as a ternary expansion

$$x = \sum_{n=1}^{\infty} \frac{b_n}{3^n},$$

where $b_n \in \{0,1,2\}$, or equivalently, x is written as $x = 0.b_1b_2b_3\ldots$. This expansion is unique in case that the ternary expansion does not contain the number 1. We say how is derived such representation for the numbers of $\mathfrak{C}$. If $x \in \mathfrak{C}$ then $x \in C_n$ for all n. Each digit $b_n \in \{0,2\}$ indicates in what interval C_{ij} x belongs (see Example 2.19). Recall that in each stage of the process of definition of $\mathfrak{C}$, we divide each closed interval in three subintervals of identical length. The digits 0, 1 and 2 represent that x lies in the left, middle or right interval, respectively. Because all middle intervals are removed, the digit 1 cannot appear in the ternary expansion. If, for instance, $b_1 = 0$ then $x \in C_{11} = [0, \frac{1}{3}]$ and if $b_1 = 2$ then $x \in C_{12} = [\frac{2}{3}, 1]$. If, say $b_1 = 0$, the number b_2 indicates in which of the two intervals after dividing C_{11} the number x is contained. So, if $b_2 = 0$ then $x \in [0, \frac{1}{9}]$ and if $b_2 = 2$, $x \in [\frac{2}{9}, \frac{3}{9}]$. This process follows for each step in the definition of $\mathfrak{C}$.

As a consequence, the *Cantor set is uncountable*. The proof is the following. For each $x \in \mathfrak{C}$, we replace each digit 2 in its ternary expansion with the digit 1. Then we get a number in binary expansion, which belongs to $[0,1]$. This defines a map $h\colon \mathfrak{C} \to [0,1]$. Since the 0 and 1 in all these expressions are arbitrary, we obtain all numbers of $[0,1]$. Hence h is surjective and $\mathfrak{C}$ is not countable.

Once we have described the ternary expansion of all numbers of the Cantor set, it is now clear how one should define the homeomorphism between $\mathfrak{C}$ and X. Define

$$\phi\colon \mathfrak{C} \to X, \quad \phi(x) = (b_1, b_2, b_3, \ldots),$$

where $x = 0.b_1b_2b_3\ldots$ is the ternary expansion of x and, of course, ϕ is bijective. We apply Theorem 9.1.6. The domain is compact because $\mathfrak{C}$ is closed and bounded. The codomain $\{0,2\}^{\mathbb{N}}$ is Hausdorff (Exercise 7.24). Therefore, the proof that ϕ is a homeomorphism comes down to see that ϕ is continuous.

By Exercise 7.18, the space X is metrizable and we need to make explicit the distance. Since each $X_n = \{0,2\}$ is discrete, consider the discrete distance d_n and the metric topology on X is determined by the distance

$$d((x_n),(y_n)) = \sum_{n=1}^{\infty} \frac{d_n(x_n,y_n)}{2^n}.$$

Let $x \in \mathfrak{C}$ and we prove that ϕ is continuous at x. Let $\epsilon > 0$ be given and we will find $\delta > 0$ such that $d(\phi(x),\phi(y)) < \epsilon$ whenever $|x-y| < \delta$. Let $N \in \mathbb{N}$ such that $1/2^N < \epsilon$ and $\delta = 1/3^N$. Let $y \in \mathfrak{C}$ and write $x = (b_1, b_2, b_3, \ldots)$ and

$y = (c_1, c_2, c_3, \ldots)$ for the ternary expansions of x and y. If $|x - y| < \delta$ then

$$|x - y| = \left|\sum_{n=1}^{\infty} \frac{b_n}{3^n} - \sum_{n=1}^{\infty} \frac{c_n}{3^n}\right| = \left|\sum_{n=1}^{\infty} \frac{b_n - c_n}{3^n}\right| < \frac{1}{3^N}.$$

An argument by induction on N, which is not difficult but is a bit tedious, proves that $b_j = c_j$ for all $1 \leq j \leq N$. We leave it as an exercise for the reader to check the details. Since d_n is the discrete distance, then $d_n \leq 1$ and we conclude

$$d(\phi(x), \phi(y)) = \sum_{n=N+1}^{\infty} \frac{d_n(b_n, c_n)}{2^n} \leq \sum_{n=N+1}^{\infty} \frac{1}{2^n} = \frac{1}{2^N} < \epsilon.$$

Finally, we point out that the Cantor set is not discrete (Example 3.13), it is totally disconnected (Example 8.12) and, as we have just showed, it is compact. A result in topology asserts that a totally disconnected compact metric space without isolated points is homeomorphic to the Cantor set. □

Exercise 9.20 Let $X = [0, 1]$, $\omega \notin X$ and $X^* = X \cup \{\omega\}$. On X^* define a topology τ by means of the basis $\beta = \beta_u \cup \{(a, 1) \cup \{\omega\} : 0 < a < 1\}$, where β_u is a basis of the usual topology on X. Prove that X^* is compact but not Hausdorff.

Solution

1. Let β_u be the basis formed by the intersection of all bounded open intervals with $[0, 1]$. We see that β is a basis for a topology proving (ii*) of Theorem 2.2.3. Property (ii*) is clear if the two open sets are of β_u or of the second type. In case of different types,

$$((a, b) \cap X) \cap ((c, 1) \cup \{\omega\}) = ((a, b) \cap (c, 1)) \cap X,$$

and $(a, b) \cap (c, 1) \neq \emptyset$. In particular, $(a, b) \cap (c, 1)$ is a bounded open interval of $\mathbb{R}$. This proves that the intersection is one element of β_u. In view of β, since the elements of the first type do not contain ω, we find that $\beta_\omega = \{(a, 1) \cup \{\omega\} : 0 < a < 1\}$ is a basis of neighborhoods of ω.
2. The relative topology on $[0, 1]$ is Euclidean. This is because intersecting all elements of β with $[0, 1]$, we obtain β_u and all open intervals of type $(a, 1)$.
3. Let $\{B_i : i \in I\}$ be a covering of X^* by members of β. Let $i_0 \in I$ be an index such that $\omega \in O_{i_0}$. Then $O_{i_0} = (a, 1) \cup \{\omega\}$ for some $a < 1$. The complement of O_{i_0} is $X^* \setminus O_{i_0} = [0, a] \cup \{1\}$. Now $[0, a] \cup \{1\}$ is compact in X^* because $\tau_{|X} = \tau_u$, so there is a finite subcovering of $[0, a] \cup \{1\}$. This subcovering together with O_{i_0} is our desired finite subcovering of X^*.
4. The space X^* is not Hausdorff because we cannot separate ω from $x = 1$. Certainly, a basis of neighborhoods of 1 in τ^* coincides with a basis of

neighborhoods in the usual topology, $\beta_1 = \{(b, 1] : b \in (0, 1)\}$. If we consider any element of β_ω, then

$$(b, 1) \cap ((a, 1) \cup \{\omega\}) = (b, 1) \cap (a, 1) \neq \emptyset.$$

□

We show a non-compact space where any infinite subset has points of accumulation.

Exercise 9.21 On $X = \mathbb{R} \setminus \mathbb{Z}$, consider the topology that has as basis $\beta = \{(n, n+1) : n \in \mathbb{Z}\}$. Prove that X is not compact but every subset has an accumulation point. Study which subsets are compact.

Solution Notice that β obeys (ii*) of Theorem 2.2.3 because any two elements of β are disjoint. Moreover, given $x \in X$, a basis of neighborhoods of x is formed by only one element, $\beta_x = \{(E(x), E(x)+1)\}$, where $E(x)$ is the integer part of x. The reason is that the interval $(E(x), E(x)+1)$ is the smallest element of β containing x. Let $U_x = (E(x), E(x)+1)$.

1. For the covering consisting of all elements of β, there is no finite subcovering because $\mathbb{R} \setminus \mathbb{Z}$ would be the union of a finite number of bounded intervals.
2. Let $A \subset X$ and we see that the set of accumulation points A' is not empty. Let $n \in \mathbb{Z}$ such that $A \cap (n, n+1) \neq \emptyset$ and let a be a point of this intersection. We prove that if $x \in (n, n+1)$, with $x \neq a$, then x is an accumulation point of A. It suffices to realize that $U_x = (n, n+1)$ and that $a \in (U_x \setminus \{x\}) \cap A \neq \emptyset$.
3. We examine the compact subspaces. Let $A \subset \mathbb{R} \setminus Z$ and consider the covering of A formed by all elements of β. If A is compact then $A \subset (n_1, n_1+1) \cup \ldots \cup (n_k, n_k+1)$ for some $n_i \in \mathbb{Z}$, in particular, A is bounded. We establish the converse. Let $A \subset X$ be a bounded set and let $\{(n_i, n_i+1) : i \in I\}$ be a covering of A formed by basic open sets. Since A is bounded, there is $M \in \mathbb{N}$ such that $A \subset (-M, M)$. Consider

$$K = \{i \in I : (n_i, n_i+1) \subset (-M, M)\}.$$

Since the number of intervals $(n, n+1)$ included in $(-M, M)$ is finite, K is finite. Let $K = \{i_1, \ldots, i_k\}$. We prove that

$$A \subset (n_{i_1}, n_{i_1}+1) \cup \ldots \cup (n_{i_k}, n_{i_k}+1),$$

obtaining a finite subcovering. Let $a \in A$ be given. If $a \notin (n_{i_j}, n_{i_j}+1)$ for all $i_j \in K$, then $a \in (n_i, n_i+1)$ for some $i \in I$ and $(n_i, n_i+1) \not\subset (-M, M)$. Then $a \notin (-M, M)$, a contradiction. □

Exercise 9.22 Let A and B be two closed compact subsets in a topological group G. Prove that $AB = \{ab : A \in A, b \in B\}$ is a closed set.

Solution We see that $G \setminus AB$ is an open set proving that all its points are interior. Let $x \notin AB$. For each $b \in B$, we have $xb^{-1} \notin A$. Consider the continuous map $\psi\colon G \times G \to G$ defined by $\psi(x, y) = xy^{-1}$ (Exercise 7.26). Since $\psi(x, b) \in G \setminus A$ and $G \setminus A$ is open, there are open neighborhoods U_b and V_b of x and b respectively, such that $\psi(U_b \times V_b) \subset G \setminus A$. This inclusion is written as $U_b V_b^{-1} \subset G \setminus A$. Then $\{V_b : b \in B\}$ is a covering of B and compactness of B implies $B \subset V_{b_1} \cup \ldots \cup V_{b_n}$ for some $n \in \mathbb{N}$.

Finally we prove that the neighborhood $U = U_{b_1} \cap \ldots \cap U_{b_n}$ of x is included in $G \setminus AB$. Suppose this is false and let $z \in U \cap AB$. Then there are $b \in B$ and $a \in A$ such that $z = ab$, which writes as $a = zb^{-1}$. If $b \in V_{b_j}$ for some $j \in \{1, \ldots, n\}$,

$$a = zb^{-1} \in UV_{b_j}^{-1} \subset U_{b_j} V_{b_j}^{-1} \subset G \setminus A,$$

giving a contradiction. □

We work with subsets of the general linear group $\mathsf{GL}(n, \mathbb{R})$ (Exercise 7.29).

Exercise 9.23 Consider the following subgroups of $\mathsf{GL}(n, \mathbb{R})$:

(i) the *orthogonal group* $\mathsf{O}(n) = \{A \in \mathsf{GL}(n, \mathbb{R}) : AA^t = A^t A = I\}$;
(ii) the *special orthogonal subgroup* $\mathsf{SO}(n) = \{A \in \mathsf{O}(n) : \det(A) = 1\}$;
(iii) the *special linear group* $\mathsf{SL}(n, \mathbb{R}) = \{A \in \mathsf{GL}(n, \mathbb{R}) : \det(A) = 1\}$.

Prove that $\mathsf{O}(n)$ and $\mathsf{SO}(n)$ are compact but $\mathsf{SL}(n, \mathbb{R})$ is not if $n \geq 2$.

Solution Recall that $\mathsf{GL}(n, \mathbb{R})$ is a subspace of $\mathsf{gl}(n, \mathbb{R}) = \mathbb{R}^{n^2}$. The Euclidean metric on $\mathsf{gl}(n, \mathbb{R})$ is $\langle A, B\rangle = \operatorname{trace}(AB^t)$, so the modulus of $A \in \mathsf{gl}(n, \mathbb{R}) = \mathbb{R}^{n^2}$ is $|A| = \sqrt{\langle A, A\rangle} = \operatorname{trace}(AA^t)$.

1. We prove that $\mathsf{O}(n)$ is closed and bounded. Define

$$T\colon \mathsf{gl}(n, \mathbb{R}) \to \mathsf{gl}(n, \mathbb{R}), \quad T(A) = AA^t.$$

This map is continuous because composing with the projections p_{ij} of $\mathsf{gl}(n, \mathbb{R})$,

$$(p_{ij} \circ T)(A) = (AA^t)_{ij} = \sum_{k=1}^{n} a_{ik} a_{kj} = \left(\sum_{k=1}^{n} p_{ik} p_{kj}\right)(A).$$

Since $\mathsf{O}(n) = T^{-1}(\{I\})$ and $\{I\} \in \mathbb{R}^{n^2}$ is closed, the orthogonal group is closed. In the same fashion, $\mathsf{SL}(n, \mathbb{R}) = \det^{-1}(\{1\})$, hence it is closed too.

Another way to check that T is continuous is writing $T = P \circ e(\mathbb{1}, t)$, where $\mathbb{1}$ is the identity of $\mathsf{gl}(n, \mathbb{R})$, $t\colon \mathsf{gl}(n, \mathbb{R}) \to \mathsf{gl}(n, \mathbb{R})$ is the transposing map $t(A) = A^t$ and $P\colon \mathsf{gl}(n, \mathbb{R}) \times \mathsf{gl}(n, \mathbb{R}) \to \mathsf{gl}(n, \mathbb{R})$ is the product map of two matrices, $P(A, B) = AB$.

We see that $\mathsf{O}(n)$ is bounded. If $A \in \mathsf{O}(n)$, then

$$|A|^2 = \langle A, A\rangle = \text{trace}(AA^t) = \text{trace}(I) = n.$$

This proves that $|A| = \sqrt{n}$. In particular, $\mathsf{O}(n)$ is a subset of the sphere $\mathbb{S}^{n^2-1}_{\sqrt{n}}$.

2. Closedness of $\mathsf{SO}(n)$ follows because it is the intersection of two closed sets, namely, $\mathsf{O}(n)$ and $\mathsf{SL}(n, \mathbb{R})$. Then $\mathsf{SO}(n)$ is compact because it is closed in the compact set $\mathsf{O}(n)$.
3. The set $\mathsf{SL}(n, \mathbb{R})$ is not bounded if $n \geq 2$ (notice that $\mathsf{SL}(1, \mathbb{R}) = \{-1, 1\}$ which is compact). Indeed, for each $m \in \mathbb{N}$, let $A_m \in \mathsf{SL}(n, \mathbb{R})$ be defined by

$$A_m = \begin{pmatrix} m & 0 & 0 & \ldots & 0 \\ 0 & \frac{1}{m} & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & 0 & \ldots & 1. \end{pmatrix}.$$

Then $|A_m|^2 = m^2 + \frac{1}{m^2} + (m-2) \to \infty$ as $m \to \infty$. $\square$

We prove the known result of Dini on uniform convergence.

Exercise 9.24 Let X be a compact space and let (f_n) be a sequence of continuous functions with $f_n \colon X \to \mathbb{R}$. Prove that if (f_n) converges monotonically pointwise to a continuous function g, then $(f_n) \to g$ with the sup distance d_∞ (Exercise 4.29), that is, the convergence is uniform.

Solution Without loss of generality, assume that $(f_n(x)) \to g(x)$ for all $x \in X$. Since X is compact, every continuous function is bounded; in particular, all functions belong to $\mathsf{B}(X, \mathbb{R})$. Let $\epsilon > 0$ be given and $O_n(\epsilon) = \{x \in X : |f_n(x) - g(x)| < \epsilon\}$. This set is open because $O_n(\epsilon) = (|f_n - g|)^{-1}((-\infty, \epsilon))$ and $|f_n - g|$ is continuous. On the other hand, we see that if $n < m$ then $O_n(\epsilon) \subset O_m(\epsilon)$. Indeed, if $x \in O_n(\epsilon)$ and using that $f_n(x) \leq f_m(x)$, we deduce

$$|f_m(x) - g(x)| = g(x) - f_m(x) \leq g(x) - f_n(x) = |f_n(x) - g(x)| < \epsilon,$$

hence $x \in O_m(\epsilon)$.

We see that $\{O_n(\epsilon) : n \in \mathbb{N}\}$ is a covering of X. Let $x \in X$ be given. Since $(f_n(x)) \to g(x)$, there is $m \in \mathbb{N}$ such that $|f_n(x) - g(x)| < \epsilon$ whenever $n \geq m$. Thus $x \in O_m(\epsilon)$ and this proves that $\bigcup_{n\in\mathbb{N}} O_n(\epsilon) = X$.

Using this covering and that X is compact, there is a finite subcovering $X = O_{n_1}(\epsilon) \cup \ldots \cup O_{n_k}(\epsilon)$. Since the sets $O_n(\epsilon)$ are ordered by inclusions, $X = O_\nu(\epsilon)$, where $\nu = \max\{n_1, \ldots, n_k\}$. Thus $X = O_n(\epsilon)$ whenever $n \geq \nu$. But this identity means that for all $x \in X$, $|f_n(x) - g(x)| < \epsilon$. Because X is compact, the supremum is a maximum, so

$$d_\infty(f_n, g) = \sup_{x \in X} |f_n(x) - g(x)| < \epsilon$$

completing the proof. □

Exercise 9.25 Consider the product space $(\mathbb{R} \times \{0, 1\}, \tau_u \times \tau_T)$. Determine if the following subsets are compact: $([0, 1] \times \{0\}) \cup ((0, 1) \times \{1\})$ and $(0, 1) \times \{0, 1\}$.

Solution

1. Consider the basis of $\mathbb{R} \times \{0, 1\}$ given by $\beta = \{(a, b) \times \{0, 1\} : a, b \in \mathbb{R}\}$ and let

$$([0, 1] \times \{0\}) \cup ((0, 1) \times \{1\}) \subset \bigcup_{i \in I} ((a_i, b_i) \times \{0, 1\}) .$$

Therefore $[0, 1] \times \{0\} \subset \bigcup_{i \in I} (a_i, b_i) \times \{0, 1\}$. By the Tychonoff theorem, $[0, 1] \times \{0\}$ is compact, so there is $n \in \mathbb{N}$ such that $[0, 1] \times \{0\} \subset \bigcup_{j=1}^{n} (a_{i_j}, b_{i_j}) \times \{0\}$. Thus $[0, 1] \subset \bigcup_{j=1}^{n} (a_{i_j}, b_{i_j})$. The same subcovering covers the interval $(0, 1)$. It follows easily that

$$([0, 1] \times \{0\}) \cup ((0, 1) \times \{1\}) \subset \bigcup_{j=1}^{n} (a_{i_j}, b_{i_j}) \times \{0, 1\}.$$

2. The set $(0, 1) \times \{0, 1\}$ is the Cartesian product of two subspaces, but $(0, 1)$ is not compact, hence neither is $(0, 1) \times \{0, 1\}$ by the Tychonoff theorem. □

We offer a construction of an embedding of a non-compact space into a compact space by adding one point to the initial space.

Exercise 9.26 Let (X, τ) be a non-compact space and let ω be an object that does not belong to X. Let $X^* = X \cup \{\omega\}$ and the collection of subsets

$$\tau^* = \tau \cup \{O \subset X^* : X^* \setminus O \text{ is closed and compact in } X\}.$$

1. Prove that τ^* is a topology.
2. Prove that the relative topology on X is τ.
3. Prove that X^* is compact.
4. Prove that X is dense in X^*.

The space X^* is called the *Alexandroff compactification* of X.

Solution The element ω only belongs to the subsets of the second type.

1. We see the conditions on τ^* to be a topology.

   (a) It is clear that $\emptyset \in \tau \subset \tau^*$. Further, since $\{\emptyset\}$ is closed and compact in X, its complement X^* is open in τ^*.

   (b) Let $O_1, O_2 \in \tau^*$. If O_1 and O_2 are of τ, then their intersection belongs to τ and therefore to τ^*. If they are of second type, then $X^* \setminus O_1,\ X^* \setminus O_2$ are closed and compact in X, so their union is a closed compact set in X. The complement of the union is thus open in τ^*. But this complement is

   $$X^* \setminus \big((X^* \setminus O_1) \cup (X^* \setminus O_2)\big) = O_1 \cap O_2.$$

   If $O_1 \in \tau$ and O_2 is of second type, then $\omega \notin O_1 \cap O_2$ and

   $$O_1 \cap O_2 = O_1 \cap (O_2 \setminus \{\omega\}) = O_1 \cap (O_2 \cap X) = O_1 \cap (X \setminus (X^* \setminus O_2))$$

   is open in τ because $X^* \setminus O_2$ is closed.

   (c) Let $\{O_i\}_{i \in I}$ be a collection of sets of τ^*. If $O_i \in \tau$ for all $i \in I$, then its union belongs to τ, hence to τ^*. Suppose that there is $i_0 \in I$ such that $\omega \in O_{i_0}$. Set $J \subset I$ the set of indices such that $\omega \in O_j$ for $j \in J$ and let $K = I \setminus J$. The set $X^* \setminus O_{i_0}$ is closed and compact in (X, τ), hence

   $$\begin{aligned} X^* \setminus \bigcup\nolimits_{i \in I} O_i &= \Big(\bigcap\nolimits_{j \in J} (X^* \setminus O_j)\Big) \bigcap \Big(\bigcap\nolimits_{k \in K} (X^* \setminus O_k)\Big) \\ &= \Big(\bigcap\nolimits_{j \in J} (X^* \setminus O_j)\Big) \bigcap \Big(\bigcap\nolimits_{k \in K} (X \setminus O_k)\Big) \end{aligned}$$

   because $\omega \notin X^* \setminus \bigcup_{i \in I} O_i$ and to make these intersections, we can replace $X^* \setminus O_k$ by $X \setminus O_k$. Then $X^* \setminus \bigcup_{i \in I} O_i$ is closed in X, hence it is compact because it is closed in the compact set $X^* \setminus O_{i_0}$. Thus $\bigcup_{i \in I} O_i \in \tau^*$.

2. We see the set-theoretical identity $\tau = (\tau^*)_{|X}$ by showing that each of the sets is contained in the other. If O is open in τ and as $\tau \subset \tau^*$, then $O = O \cap X \in (\tau^*)_{|X}$. If now $O \in (\tau^*)_{|X}$, there is $O^* \in \tau^*$ such that $O = O^* \cap X$. If $O^* \in \tau$, then $O^* \cap X \in \tau$. The other case is that $\omega \in O^*$. Then $X^* \setminus O^*$ is closed, so

   $$O = (O^* \setminus \{\omega\}) \cap X = X \setminus (X^* \setminus O^*) \in \tau.$$

   Now observe that $O^* \setminus \{\omega\}$ is open in (X, τ).
3. Let $\{O_i : i \in I\}$ be an open covering of X^*. We fix $i_0 \in I$ such that $\omega \in O_{i_0}$. Then $X^* \setminus O_{i_0}$ is compact in (X, τ). Since $(\tau^*)_{|X} = \tau$, the subset $X^* \setminus O_{i_0}$ is compact in X^*. Thus there is $n \in \mathbb{N}$ such that $X^* \setminus O_{i_0} \subset O_{i_1} \cup \ldots \cup O_{i_n}$. Then $\{O_{i_0}, O_{i_1}, \ldots, O_{i_n}\}$ is a finite subcovering of X^*.

4. To prove that X is dense in X^* it suffices to see that $\omega \in \overline{X}$. If $\omega \notin \overline{X}$, there is $O \in \tau^*$ with $\omega \in O$ and $O \cap X = \emptyset$. Then $O = \{\omega\}$. However, $\{\omega\}$ is not open because $X^* \setminus \{\omega\} = X$ is not compact. □

Exercise 9.27 Show that $\mathbb{S}^1$ is the Alexandroff compactification of $\mathbb{R}$.

Solution From Exercise 9.13, it suffices to check that

$$\beta = \{(a, b) : a, b \in \mathbb{R}\} \cup \{\{\omega\} \cup (-\infty, a) \cup (b, \infty) : a \leq b, a, b \in \mathbb{R}\}$$

is a basis for the topology in the Alexandroff compactification. The elements of β are open in τ^* because the intervals (a, b) belong to τ_u and the complements of second type of β are intervals $[a, b]$, which are closed and compact in $\mathbb{R}$.

Now let $O \in \tau^*$ and $x \in O$. If $O \in \tau_u$, then $x \neq \omega$ and it is obvious that there are $a < b$ such that $x \in (a, b) \subset O$. If $O \notin \tau_u$, then $\mathbb{R}^* \setminus O$ is closed and compact in $\mathbb{R}$, or equivalently, $\mathbb{R}^* \setminus O$ is compact. We distinguish two cases:

1. Case $x \in \mathbb{R}$. Since $O \setminus \{\omega\}$ is open in $\mathbb{R}$, there are $a < b$ such that $x \in (a, b) \subset O \setminus \{\omega\} \subset O$.
2. Case $x = \omega$. Since $\mathbb{R}^* \setminus O$ is bounded, let $a \leq b$ such that $\mathbb{R}^* \setminus O \subset [a, b]$. Taking complements, $\omega \in \{\omega\} \cup (-\infty, a) \cup (b, \infty) \subset O$ as desired. □

Exercise 9.28 Prove that $X = [0, 1] \times [0, 1]$, endowed with the order topology from the lexicographic order, is compact.

Solution Consider an open covering of X by basic open sets $](a_i, b_i), (c_i, d_i)[$, $i \in I$. For each $0 < x < 1$, let two basic open sets containing $(x, 0)$ and $(x, 1)$, respectively. If $(x, 0) \in](a_i, b_i), (c_i, d_i)[$, then $(a_i, b_i) < (x, 0)$ which implies $a_i < x$ and from $(x, 0) < (c_i, d_i)$, we have $x < c_i$ or $c_i = x$ and $d_i > 0$. A similar argument applies to $(x, 1)$. Thus, there exists $\epsilon_x > 0$ such that the union of both basic open sets contains the set

$$](x - \epsilon_x, 0), (x + \epsilon_x, 1)[\cap(X \setminus [(x, \epsilon_x), (x, 1 - \epsilon_x)]).$$

On the other hand, $[(x, \epsilon), (x, 1 - \epsilon)] = \{x\} \times [\epsilon, 1 - \epsilon]$ is homeomorphic to $[\epsilon, 1 - \epsilon]$, where we are using that the relative topology on $\{x\} \times [\epsilon, 1 - \epsilon]$ is Euclidean. Thus there is a finite number of elements of the covering whose union covers $[(x, \epsilon), (x, 1 - \epsilon)]$. Definitively, the set $](x - \epsilon_x, 0), (x + \epsilon_x, 1)[$ is covered by a finite number of basic open sets.

The same argument for the basic open sets in the case of the points $(0, 0)$ and $(1, 1)$ establishes that there are $\epsilon_0, \epsilon_1 > 0$ such that two open sets of the covering contain $](0, 0), (0, \epsilon_0)[$ and $](1, 1 - \epsilon), (1, 1)[$.

Consider the open sets of $[0, 1]$ with the usual topology consisting of all $(x - \epsilon_x, x + \epsilon_x)$ together with $[0, \epsilon_0)$ and $(1 - \epsilon_1, 1]$. Since $[0, 1]$ is compact, there is a finite subcovering. Each open set of this subcovering has associated a finite number of open sets of the initial covering of x, the collection of which obviously covers X itself. □

Exercise 9.29 Let $A \times B$ be a compact subset of a product space $X \times Y$. If G is an open set of $X \times Y$ such that $A \times B \subset G$, prove that there are O and O' open sets in X and Y respectively, such that $A \times B \subset O \times O' \subset G$.

Solution Let $a \in A$ be an arbitrary but fixed element of A. For each $b \in B$, we have $(a, b) \in G$ so there are U_a^b and V_b^a open sets of X and Y respectively, such that $(a, b) \in U_a^b \times V_b^a \subset G$. Since $\{V_a^b : b \in B\}$ is an open covering of B, we deduce that $B \subset V_{b_1}^a \cup \ldots \cup V_{b_{n_a}}^a$ for some $n_a \in \mathbb{N}$. Let

$$U_a = U_a^{b_1} \cap \ldots \cap U_a^{b_{n_a}}, \quad V^a = V_{b_1}^a \cup \ldots \cup V_{b_{n_a}}^a.$$

Now $\{U_a : a \in A\}$ is a covering of A so $A \subset U_{a_1} \cup \cdots \cup U_{a_m}$ for some $m \in \mathbb{N}$ because A is compact. If $O = U_{a_1} \cup \ldots \cup U_{a_m}$ and $O' = V^{a_1} \cap \ldots \cap V^{a_m}$, then clearly $A \times B \subset O \times O' \subset G$. □

Exercise 9.30 Let (X, d) be a metric space and let $A \subset X$ be a closed set. Prove that if $x \in X$, then there is $a \in A$ such that $d(x, A) = d(x, a)$. As a consequence, if A and B are two compact subsets of X, then there are $a \in A$ and $b \in B$ such that $d(a, b) = d(A, B)$, where $d(A, B) = \inf\{d(a, b) : a \in A, b \in B\}$.

Solution Recall that $d(x, A)$ was defined in (4.5). Let $(a_n) \subset A$ such that $d(a_n, x) \to d(x, A)$. Then $(d(a_n, x))$ is bounded (Exercise 4.6.15) and this proves that (a_n) is bounded.

1. Assume that $\{a_n : n \in \mathbb{N}\}$ is a finite set. Then $\{d(a_n, x) : n \in \mathbb{N}\}$ is finite so it is immediate that $d(x, A)$ is attained at some point.
2. If $\{a_n : n \in \mathbb{N}\}$ is infinite, the Bolzano-Weierstrass theorem implies that $\{a_n : n \in \mathbb{N}\}$ has a point of accumulation which we call y. Then there is a subsequence $(a_{\sigma(n)})$ converging to y; in particular, $y \in \overline{A}$. Since A is closed, $y \in A$. By continuity of the distance function $z \mapsto d(z, x)$ (Exercise 5.10), we have $(d(a_{\sigma(n)}, x)) \to d(y, x)$, so $d(y, x) = d(x, A)$.

The last part is a consequence of continuity of the distance $d\colon X \times X \to \mathbb{R}$.

This result fails if A is not closed. For instance, if $A = (0, 1)$ and $x = 0$, we have $d(x, A) = 0$, but $d(x, a) = a \neq 0$ for all $a \in A$. □

Exercise 9.31 Use Exercise 9.30 to provide a new proof that the only connected subsets of $\mathbb{R}$ are the intervals assuming that the intervals $[a, b]$ are compact.

Solution By Example 8.4, it suffices to prove that $[0, 1]$ is connected. Let $[0, 1] = A \cup B$ be an open partition; in particular, A and B are closed in $[0, 1]$. Since they are bounded, then A and B are compact. Thanks to Exercise 9.30, there are $a \in A$ and $b \in B$ such that $d(a, b) = d(A, B)$. Since $A \cap B = \emptyset$, this number must be positive (otherwise, $a \in \overline{B} = B$ by Proposition 4.4.7). Take the middle point $c = (a+b)/2$, in particular, $c \in [0, 1]$. Then

$$|c - a| = \left|\frac{a+b}{2} - a\right| = \left|\frac{a-b}{2}\right| < |a - b|.$$

This means that $c \notin B$ and likewise, $c \notin A$. This contradicts that $[0, 1] = A \cup B$. $\square$

We conclude by proving that in a finite dimensional vector space all the distances determined by norms are equivalent.

Exercise 9.32 On a finite dimensional vector space, there is a unique topology that gives a normed space. Equivalently, two different norms determine the same topology.

Solution Let X be a vector space of dimension n and let $B = \{e_1, \ldots, e_n\}$ be a basis. This basis establishes an isomorphism of vector spaces between $\mathbb{R}^n$ and X given by

$$f : \mathbb{R}^n \to X, \quad f(x_1, \ldots, x_n) = \sum_{i=1}^{n} x_i e_i.$$

Define the map $\|\cdot\|_1 : X \to \mathbb{R}$ by

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

where $(x_1, \ldots, x_n) \in \mathbb{R}^n$ are the coordinates of $x \in X$ with respect to B. It is clear that $\|\cdot\|_1$ is a norm on X and that $\|x\|_1 = \|f^{-1}(x)\|_t$, being $\|\cdot\|_t$ the norm on $\mathbb{R}^n$ associated to the taxi distance. Let d_1 be the distance for $\|\cdot\|_1$.

Let $\|\cdot\|$ be another norm on X whose distance is d. We will find two positive constants $k, m > 0$ such that

$$k\|x\|_1 \leq \|x\| \leq m\|x\|_1. \tag{9.4}$$

Once we have established these inequalities, we have the analogous ones $k \cdot d_1 \leq d \leq m \cdot d_1$ and finally apply Theorem 4.2.4 to complete the proof of theorem.

If we replace in (9.4) x by λx for arbitrary $\lambda \in \mathbb{R}$, we obtain (9.4) again by the homogeneity property of the norm. Thus it is enough to see

$$k \leq \|x\| \leq m \tag{9.5}$$

for all vectors x such that $\|x\|_1 = 1$. Let

$$Q = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : \|(x_1, \ldots, x_n)\|_t = 1\}.$$

The set Q is closed because $Q = h^{-1}(\{1\})$, where $h(x_1, \dots, x_n) = \|(x_1, \dots, x_n)\|_t$ is continuous (Exercise 5.10). Also Q is bounded with the Euclidean distance because the Euclidean distance d_u and the taxi distance d_t are related by $d_u \leq \sqrt{n} \cdot d_t$. Consider the restriction $f : Q \to (X, d)$.

(i) The map f is continuous because it is Lipschitz. Indeed, if $a, b \in Q$, we have

$$d(f(a), f(b)) = \|f(a) - f(b)\| = \|\sum_{i=1}^{n} a_i e_i - \sum_{i=1}^{n} b_i e_i\| = \|\sum_{i=1}^{n}(a_i - b_i)e_i\|$$
$$\leq \sum_{i=1}^{n} |a_i - b_i|\|e_i\| \leq M \sum_{i=1}^{n} |a_i - b_i| = M\, d_t(a, b)$$

where $M = \max\{\|e_1\|, \dots, \|e_n\|\}$.

(ii) Thus $\|.\| \circ f : Q \to \mathbb{R}$ attains a minimum $k \geq 0$ and a maximum $m \geq 0$ by Example 9.7. In other words,

$$k \leq \Big\| \sum_{i=1}^{n} x_i e_i \Big\| \leq m \tag{9.6}$$

for all $(x_1, \dots, x_n) \in \mathbb{R}^n$ such that $\sum_{i=1}^{n} |x_i| = 1$.

(iii) We claim that $k > 0$. If $k = 0$, let $z \in Q$ such that $\|f(z)\| = \|z_1 e_1 + \dots + z_n e_n\| = 0$. Then $f(z) = 0$. Since $\{e_1, \dots, e_n\}$ are linearly independent, all z_i are 0. Thus $z = 0$, which is contradictory because $0 \notin Q$.

Having proved that $k > 0$, then (9.6) is equal to (9.5) for all vectors $x = \sum_{i=1}^{n} x_i e_i \in X$ with $\|(x_1, \dots, x_n)\|_t = 1$. □

This exercise does not assert that any two distances on a finite dimensional vector space are equivalent, but only those distances determined by norms. For example, since any norm $\|\cdot\|$ is not bounded by the homogeneity property, any bounded distance is not associated with a norm.

9.4 Suggested Exercises

9.4.1 Find a space and two compact subsets A and B such that $A \cap B$ is not compact. Give an example that the union of two non-compact sets is compact.

9.4.2 Let A and B be two subsets of a metric space such that A is compact. Show there is $a \in A$ such that $d(a, B) = d(A, B)$. If, in addition, B is closed and $A \cap B = \emptyset$, show that $d(A, B) > 0$.

9.4.3 Investigate the compact sets of $(\mathbb{R}, \tau)$ where $\tau = \{\emptyset, \mathbb{R}\} \cup \{[0, a) : a > 0\}$.

9.4.4 Characterize the compact subsets of the space of Exercise 2.13.

9.4.5 In a Hausdorff space, prove that if A is compact then A' is compact.

9.4.6 Give an example of a space and a subset A satisfying:

1. A is not compact and $\overline{A}$ is compact.
2. A is compact and $\overline{A}$ is not compact.
3. A is not compact and $\mathring{A}$ is compact.
4. A is compact and $\mathring{A}$ is not compact.

9.4.7 Prove that there does not exist a continuous function $f: [0, 1] \to \mathbb{R}$ such that $f(x) \in \mathbb{R} - \mathbb{Q}$ if $x \in \mathbb{Q}$ and $f(x) \in \mathbb{Q}$ if $x \in \mathbb{R} - \mathbb{Q}$.

9.4.8 Let X be a compact space and let $f: \mathbb{R} \to X$ be a continuous closed map. Show that $f^{-1}(\{x\})$ is infinite for all $x \in X$.

9.4.9 If $f: X \to Y$ is a continuous map between two metric spaces where X is compact, show that f is uniformly continuous (compare with Exercise 4.29).

9.4.10 Prove that there are no compact neighborhoods in the Sorgenfrey line.

9.4.11 We offer a proof of the compactness of $[0, 1]$ without using connectedness. Let $\{O_i : i \in I\}$ be a covering of $[0, 1]$ by open sets of $\mathbb{R}$ and define

$$A = \{x \in [0, 1] : \text{ there is } n \in \mathbb{N},\ [0, x] \subset O_{i_1} \cup \ldots \cup O_{i_n}\}.$$

Let $\alpha = \sup(A)$. Show that $\alpha > 0$. Next, use arguments as in Theorem 9.1.4 to prove it is not possible that $\alpha < 1$. Thus $\alpha = 1$. Finally, prove that for some $\epsilon > 0$, the interval $[1 - \epsilon, 1]$ is covered by one open set and $[0, 1 - \epsilon]$ by a finite number.

9.4.12 If O is an open set of $\mathbb{D}^2$ such that $\mathbb{S}^1 \subset O$, prove that there exists a positive number $r < 1$ such that $\mathbb{D}^2 = B_r(0) \cup O$.

9.4.13 The *Hawaiian earring* is the set defined by $\bigcup_{n\in\mathbb{N}} A_n$ where $A_n = \mathbb{S}^1_{1/n}(\frac{1}{n}, 0)$. Prove that this space is compact and connected.

9.4.14 Let X be a compact Hausdorff space and let $f: X \to X$ be a continuous map. Define $X_1 = X$ and $X_{n+1} = f(X_n)$. Show that $A = \bigcap_{n\in\mathbb{N}} X_n \neq \emptyset$ and that $f(A) = A$.

9.4.15 Let X and Y be two Hausdorff compact spaces such that there are $x \in X$ and $y \in Y$ with $X \setminus \{x\} \cong Y \setminus \{y\}$. Show that $X \cong Y$.

9.4.16 Prove that $A \subset \mathbb{R}$ is compact if and only if every real-valued function defined and continuous on A has a maximum.

Chapter 10
Quotient Topology

In this chapter we discuss a final method of construction of new topological spaces from old ones using set theory. The set-theoretic concepts are quotient sets and equivalence relations. Consider X a set with an equivalence relation $\sim$. The *equivalence class* of x, denoted $[x]$, is the set of all elements y such that $x \sim y$ and the quotient space $X/\sim$ is the set whose elements are the equivalence classes. Equivalence classes, as subset of X, define a partition of X. Between X and $X/\sim$ there is a natural map $p\colon X \to X/\sim$ called *projection* that assigns to each element $x \in X$ its equivalence class $[x]$. We present the fundamental question of this chapter.

Question: *Given a topological space (X, τ) and an equivalence relation $\sim$, under what conditions may we consider a topology on $X/\sim$ that has some relation with the topology τ?*

A typical equivalence relation is the following. Let A be a subset of X and define on X the equivalence relation

$$x \sim y \text{ if and only if } \begin{cases} x = y, & \text{or} \\ x, y \in A. \end{cases} \tag{10.1}$$

The relation $\sim$ identifies all points of A to the same equivalence class, leaving the remaining elements of X in another equivalence class. Consequently, the equivalence classes are

$$[x] = \begin{cases} \{x\}, & \text{if } x \notin A \\ A, & \text{if } x \in A. \end{cases}$$

For instance, if X is the unit closed disc $\mathbb{D}^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ and $A = \mathbb{S}^1 \subset \mathbb{D}^2$, the equivalence relation identifies all points of $\mathbb{S}^1$ keeping the other points of $\mathbb{D}^2$ as is. Because the quotient topology must preserve proximity of points, it is not difficult to visualize this construction if we view $\mathbb{D}^2$ inserted in the Euclidean space $\mathbb{R}^3$ as the set $\mathbb{D}^2 \times \{0\}$ in the plane of equation $z = 0$. Then we

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_10

fold the disc so that we glue the border of the disc at one point. The final space is, clearly, a round sphere $\mathbb{S}^2$ embedded in $\mathbb{R}^3$.

This chapter concludes a series of constructions of new topological spaces from others using set-theoretical concepts. We review them below.

1. (Subsets, relative topology) In Chap. 2, we established a topological structure on a subset A of a space (X, τ) called the relative topology. Thus the inclusion $i : A \hookrightarrow X$ is an embedding.
2. (Cartesian product, product topology) In Chap. 7, we equipped a topological structure with the Cartesian product $X_1 \times X_2$ of two topological spaces X_1 and X_2. The projections $p_i : X_1 \times X_2 \to X_i$, $i = 1, 2$, are continuous.
3. (Quotient space, quotient topology) In the present chapter, we define a topological structure on the quotient space $X/\sim$ determined by an equivalence relation $\sim$ defined on a given topological space X. The projection $p : X \to X/\sim$ will be continuous.

10.1 Motivation and Definition of the Quotient Topology

The beginning of this section is deliberately kept informal in order to motivate the quotient topology. We start with an example. Let $Q = [0, 1] \times [0, 1]$ be the unit square of $\mathbb{R}^2$. Define an equivalence relation on Q relating each point of the edge $\{0\} \times [0, 1]$ with the corresponding point on $\{1\} \times [0, 1]$ of the same height, and relating the remaining points only to themselves (Fig. 10.1, left). This relation $\sim$ is

$$(t, s) \sim (t', s') \text{ if and only if } \begin{cases} (t, s) = (t', s'), & \text{or} \\ t, t' \in \{0, 1\}, & s = s'. \end{cases} \tag{10.2}$$

The equivalence classes are

$$[(t, s)] = \begin{cases} \{(t, s)\}, & \text{if } t \notin \{0, 1\} \\ \{(1, s), (0, s)\}, & \text{if } t = 0 \text{ or } t = 1. \end{cases}$$

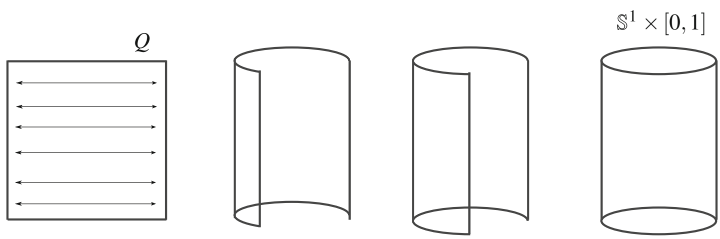


Fig. 10.1 The cylinder as a quotient space of the square $Q = [0, 1] \times [0, 1]$ of $\mathbb{R}^2$

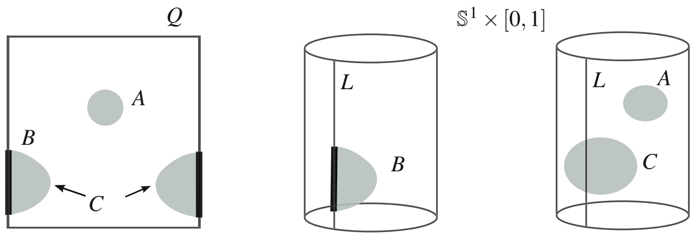


Fig. 10.2 The quotient topology on the quotient space $Q/{\sim}$

If we think of Q as a sheet of paper, the quotient set can be visualized when we fold Q in such a way that we glue each point $(0, s)$ with the corresponding point $(1, s)$ (Fig. 10.1). The resulting object is the cylinder $\mathbb{S}^1 \times [0, 1]$. How can we define a topology on the abstract set $Q/{\sim}$ in such a manner that, after an adequate development of the theory, we can establish that $Q/{\sim}$ is homeomorphic to $\mathbb{S}^1 \times [0, 1]$?

With the above gluing process in mind, it is clear that on the square Q there are two types of open sets. First, there are those open sets A that do not intersect the vertical borders K of Q, where $K = \{0, 1\} \times [0, 1]$. Since the classes of elements $(t, s) \in A$ are the very elements (t, s), these sets A will be an open in the quotient topology. Once we have folded Q to the cylinder $\mathbb{S}^1 \times [0, 1]$, the set A is open in the cylinder such as it is depicted in Fig. 10.2.

The second type of open sets are those subsets that intersect K. Let $B \subset Q$ be such a subset, as depicted in Fig. 10.2. Then B cannot be open in the quotient topology because after folding Q, points of $B \cap K$ are not interior in the cylinder. Moreover, B regarded in the quotient set misses the corresponding points of the right edge (for $s = 1$) of K. Consequently, we must add the points $\{(1, s) : (0, s) \in B\}$ to the equivalence classes. However, $B' = B \cup \{(1, s) : (0, s) \in B\}$ is not open in Q because the points $(1, s)$ are not interior in B'. We add points of Q close to the vertical line $\{(1, s) : (0, s) \in B\}$ to get an open set in Q. This is the case for the set C of Fig. 10.2.

We can also reverse the process from the cylinder and unfold to discover which sets will be open in the quotient topology. In Fig. 10.2 we have indicated with a line L the two edges K glued on the cylinder. Thus C is open in $\mathbb{S}^1 \times [0, 1]$ and after unfolding, C appears as two open subsets in Q described in the preceding paragraph.

Before we define the quotient topology, we need some basic concepts from set theory about equivalence relations. Let X be a set and let $\sim$ be an equivalence relation on X. If $A \subset X$, the *saturation of* A with respect to $\sim$ is defined by

$$R[A] = p^{-1}(p(A)).$$

If $x \in R[A]$, then $p(x) = [x] \in p(A)$, so there is $a \in A$ such that $[x] = [a]$ and this means $x \sim a$. Thus

$$R[A] = \{x \in X : \text{ there is } a \in A,\ x \sim a\} = \bigcup_{a \in A} [a].$$

The saturation of A is nothing but the set of all points of X related with all points of A. The set A is said to be *saturated* if $A = R[A]$. The following properties are trivial.

1. If $x \in X$ then $R[\{x\}] = [x]$ (definition of equivalence class).
2. If $A \subset X$ then $A \subset R[A]$ by the reflexive property.
3. If $A \subset B$ then $R[A] \subset R[B]$. Certainly, if $A \subset B$ then $p(A) \subset p(B)$, so $p^{-1}(p(A)) \subset p^{-1}(p(B))$.
4. The set $R[A]$ is saturated. Indeed, since p is surjective, $p(p^{-1}(C)) = C$ for all $C \subset X/\sim$. Then

$$\begin{aligned} R[R[A]] &= p^{-1}(p(R[A])) = p^{-1}(p(p^{-1}(p(A)))) \\ &= p^{-1}(p \circ p^{-1})(p(A)) = p^{-1}(p(A)) = R[A]. \end{aligned}$$

In the example of Fig. 10.2, the set B is not saturated because its saturation is $R[B] = B \cup \{(1, s) : (0, s) \in B\}$. In contrast, the sets A and C are saturated.

Henceforth, we will assume that X is equipped with a topology τ. The open sets in $X/\sim$ will be those sets that, viewed in X, are open and saturated, as is the case with the sets A and C (but not with B') in Fig. 10.2. The open sets in the quotient topology are obtained by projecting onto $X/\sim$ the saturated open sets in X.

Definition 10.1.1 Let (X, τ) be a topological space and let $\sim$ be an equivalence relation on X. The *quotient topological space* is defined by the space $(X/\sim, \tau/\sim)$, where the topology $\tau/\sim$, called *quotient topology*, is defined by

$$\tau/\sim = \{p(O) : O \in \tau,\ R[O] = O\}. \tag{10.3}$$

Notice that

$$\tau/\sim = \{\widetilde{O} \subset X/\sim : p^{-1}(\widetilde{O}) \in \tau\}. \tag{10.4}$$

As a consequence, the collection of closed sets $\mathscr{F}/\sim$ of the quotient topology is

$$\mathscr{F}/\sim = \{\widetilde{F} \subset X/\sim : p^{-1}(\widetilde{F}) \in \mathscr{F}\} = \{p(F) : F \in \mathscr{F}, R[F] = F\}. \tag{10.5}$$

On the other hand, for each $[x] \in X/\sim$, the collection of neighborhoods of $[x]$ is

$$\mathscr{U}_{[x]} = \{\widetilde{U} \subset X/\sim : p^{-1}(\widetilde{U}) \in \mathscr{U}_x\} = \{p(U) : U \in \mathscr{U}_x,\ R[U] = U\}.$$

Proposition 10.1.2

1. *The projection $p\colon X \to X/{\sim}$ is continuous.*
2. *The quotient topology is the finest topology on the quotient set $X/{\sim}$ that makes continuous the projection.*
3. *The map $f\colon X/{\sim} \to Y$ is continuous if and only if $f \circ p\colon X \to Y$ is continuous.*

This result shows nicely that the construction of the quotient topology is dual to that of the product topology (Theorem 7.2.1 and Proposition 7.3.1).

From the underlying idea of gluing in a quotient space, it is natural that "something" is lost in the process and it seems reasonable to add some topological properties to the equivalence relation. In this sense, recall that a binary relation $\sim$ is actually a subset $\mathscr{R}$ of $X \times X$, where $(x, y) \in \mathscr{R}$ if and only if $x \sim y$, or in other words, $\mathscr{R} = \{(x, y) \in X \times X : x \sim y\}$.

A task in this chapter is to relate the topological properties of X with those in the quotient space $X/{\sim}$ and this can be expressed as follows.

> **Question:** *If X satisfies a topological invariant $\mathscr{P}$, then does $X/{\sim}$ satisfy $\mathscr{P}$?*

Since the projection $p\colon X \to X/{\sim}$ is continuous, then if a space is connected (resp. compact) then every quotient space is connected (resp. compact).

However, the topological properties of X are not carried over the quotient space $X/{\sim}$ in general, nor are topological properties of the quotient space $X/{\sim}$ carried over the space X. To illustrate the latter, it may happen that $X/{\sim}$ is connected (or compact) but X is not. The former is illustrated in the following example.

Example 10.1 By definition of T_1, a quotient space $X/{\sim}$ is T_1 if all singletons $[x]$ are closed. Viewed in X, this is equivalent to all sets $[x]$ are closed in X. We give an example.

Consider the partition of $\mathbb{R}$ given by $\{(-\infty, 0), [0, \infty)\}$ which defines an equivalence relation on $\mathbb{R}$. Then $\mathbb{R}/{\sim}$ is not T_1 because the singleton $\{[-1]\}$, which is $(-\infty, 0)$ in $\mathbb{R}$, is not closed. In this case, the quotient space has only two elements, $X/{\sim} = \{[-1], [0]\}$, so the topology is easy to compute. Besides trivial sets, we have $p^{-1}([-1]) = (-\infty, 0)$ and $p^{-1}([0]) = [0, \infty)$, so $\{[-1]\}$ is the only extra open set and the topology is $\tau_u/{\sim} = \{\emptyset, \mathbb{R}/{\sim}, \{[-1]\}\}$. This space is clearly homeomorphic to the Sierpinski space which is not T_1. $\diamondsuit$

Example 10.2 About the Hausdorff property, we prove that if $p\colon X \to X/{\sim}$ is open and $\mathscr{R}$ is closed in $X \times X$, then $X/{\sim}$ is Hausdorff.

If $[x] \neq [y]$ then $(x, y) \notin \mathscr{R}$. Since $(X \times X) \setminus \mathscr{R}$ is open and (x, y) is an interior point, there are two open sets O_x and O_y such that

$$(x, y) \in O_x \times O_y \subset X \times X \setminus \mathscr{R}.$$

Thus $(O_x \times O_y) \cap \mathscr{R} = \emptyset$. Since p is an open map, the sets $p(O_x)$ and $p(O_y)$ are neighborhoods of $[x]$ and $[y]$, respectively. Finally, $p(O_x)$ and $p(O_y)$ are disjoint because if $[z] \in p(O_x) \cap p(O_y)$, then $z \sim x'$ and $z \sim y'$ for some $x' \in O_x$ and $y' \in O_y$. Thus $x' \sim y'$ and $(x', y') \in O_x \times O_y$, a contradiction. $\diamondsuit$

10.2 Final Topology and Identifications

Once we have defined the quotient topology, the next objective is the construction of homeomorphisms between a quotient space and another space. First we need to extend the notion of quotient topology which will facilitate the arguments. The generalization is clear from the observation that the definition of quotient topology in (10.4) only employs the properties of the inverse image under the projection p. We replace the codomain $X/\sim$ by another space and p by a surjective map (see also Exercise 2.7.6).

Definition 10.2.1 Let (X, τ) be a topological space and let $f: X \to Y$ be a surjective map. The *final topology* on Y by f is defined by

$$\tau(f) = \{O' \subset Y : f^{-1}(O') \in \tau\}.$$

In such a case, $f: (X, \tau) \to (Y, \tau(f))$ is called an *identification*.

Some observations are the following.

1. If f were not surjective, any subset B not intersecting $f(X)$ would satisfy $f^{-1}(B) = \emptyset$ and B would be open in X.
2. If f is a bijection, $\tau(f)$ coincides with the direct image topology (Exercise 2.7.6).
3. A final topology may be the result of the same map and distinct topologies in the domain (see examples in Exercise 10.19).

Now we need conditions that ensure that a mapping is an identification.

Proposition 10.2.2 *Suppose that $f: (X, \tau) \to (Y, \tau')$ is a surjective continuous mapping between two spaces. In any of the following cases, f is an identification:*

(i) f is open.
(ii) f is closed.
(iii) There is a continuous map $g: Y \to X$ such that $f \circ g = \mathbb{1}_Y$. We say that g is a continuous right-inverse of f.

As a consequence of Theorem 9.1.6, a surjective continuous mapping from a compact space onto a Hausdorff space is an identification.

We study how a final topology behaves with respect to the relative topology. Let $f: (X, \tau) \to (Y, \tau(f))$ be an identification and let $B \subset Y$. There are two topologies on B, namely, the relative topology from Y and denoted by $\tau(f)_{|B}$, and the final topology determined by the surjective mapping

$$f: (f^{-1}(B), \tau_{|f^{-1}(B)}) \to B,$$

and denoted by $\tau_{|f^{-1}(B)}(f)$. Because $f\colon (X, \tau) \to (Y, \tau(f))$ is continuous, the restriction map

$$f\colon (f^{-1}(B), \tau_{|f^{-1}(B)}) \to (B, \tau(f)_{|B})$$

is continuous. Since the final topology is the finest one that makes f continuous, we have the inclusion $\tau(f)_{|B} \subset \tau_{|f^{-1}(B)}(f)$. However, the other inclusion does not hold in general. We illustrate an example.

Example 10.3 Let $B = [0, 1] \cap \mathbb{R} - \mathbb{Q}$ and $Y = B \cup \{1\}$. Let τ_u denote the usual topology on any subset of $\mathbb{R}$. Define

$$f\colon [0, 1] \to Y, \quad f(x) = \begin{cases} x, & \text{if } x \in B \\ 1, & \text{if } x \notin B. \end{cases}$$

While $f\colon ([0, 1], \tau_u) \to (Y, \tau_u(f))$ is continuous, $f\colon [0, 1] \to Y$ is not if we consider usual topology on both sets. Notice that $f^{-1}(B) = B$.

1. The set $(0, \frac{1}{2}) \cap B$ is an element of $\tau_{u|f^{-1}(B)}(f)$ because

$$f^{-1}\left((0, \tfrac{1}{2}) \cap B\right) = \left(0, \tfrac{1}{2}\right) \cap B \in \tau_{u|B} = \tau_{u|f^{-1}(B)}.$$

2. However, $(0, \frac{1}{2}) \cap B \notin \tau_u(f)_{|B}$. Otherwise, there would exist $O \in \tau_u(f)$ such that $(0, \frac{1}{2}) \cap B = O \cap B$.
   (a) If $1 \notin O$ then $O \subset B$. But $f^{-1}(O) = O$ is not open in $[0, 1]$ because every open set of $[0, 1]$ must intersect $\mathbb{Q}$.
   (b) If $1 \in O$ then $O = \{1\} \cup ((0, 1/2) \cap B)$. Now

$$f^{-1}(O) = \left((0, \tfrac{1}{2}) \cap B\right) \cup (\mathbb{Q} \cap [0, 1]) = [0, \tfrac{1}{2}] \cup \left((\tfrac{1}{2}, 1] \cap \mathbb{Q}\right)$$

   is not open in $[0, 1]$ because any number of $(\frac{1}{2}, 1) \cap \mathbb{Q}$ is not interior. $\diamondsuit$

Sufficient conditions that ensure the coincidence of both topologies appear in the following result.

Proposition 10.2.3 *Let $B \subset X/\!\sim$ and $A = p^{-1}(B)$. If either B is open (or closed), or p is open (or closed), then*

$$\frac{\tau_{|A}}{\sim} = \left(\frac{\tau}{\sim}\right)_{|B}.$$

10.3 Construction of Homeomorphisms in a Quotient Space

The preceding section gives the necessary tools to define a homeomorphism from a quotient space into another space. We return to the space $Q/\!\sim$ from the beginning of the chapter when we "fold" Q to get the cylinder $\mathbb{S}^1 \times [0, 1]$. The idea is to define a map $f : Q \to \mathbb{S}^1 \times [0, 1]$ such that points which are related via $\sim$ will be mapped under f to the same point of $\mathbb{S}^1 \times [0, 1]$. Then f will induce the homeomorphism between $Q/\!\sim$ and $\mathbb{S}^1 \times [0, 1]$.

First we recall from set theory what is the context. Let $f : X \to Y$ be a map between two sets and define an equivalence relation $\sim_f$ on X by

$$x \sim_f x' \text{ if and only if } f(x) = f(x').$$

Then f induces a map on $X/\!\sim_f$ defined by

$$\widetilde{f} : \frac{X}{\sim_f} \to f(X), \quad \widetilde{f}([x]) = f(x),$$

and $\widetilde{f}$ is bijective. This map $\widetilde{f}$ satisfies $i \circ \widetilde{f} \circ p = f$, where $p : X \to X/\!\sim_f$ is the projection and $i : f(X) \hookrightarrow Y$ is the inclusion. The commutative diagram of Fig. 10.3 helps to visualize the situation. Notice that p is surjective and i is injective.

Suppose now that all above sets are endowed with the structure of a topological space. With all this in mind, it should be plausible that the mapping $\widetilde{f}$ is the candidate to be a homeomorphism between $X/\!\sim_f$ and $f(X)$. The next theorem is the main tool in proving that any quotient space is homeomorphic to another space.

Theorem 10.3.1 *Let $f : X \to Y$ be a map between two spaces and*

$$\widetilde{f} : \frac{X}{\sim_f} \to Y, \quad \widetilde{f}([x]) = f(x).$$

Then $\widetilde{f}$ is a homeomorphism if and only if f is an identification.

Corollary 10.3.2 *Let $f : X \to Y$ be a surjective continuous mapping. If X is compact and Y is Hausdorff, then $X/\!\sim_f$ is homeomorphic to Y.*

Fig. 10.3 The commutative diagram $i \circ \widetilde{f} \circ p = f$

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "p"'] & Y \\ X/\sim_f \arrow[r, "\widetilde{f}"] & f(X) \arrow[u, "i"'] \end{tikzcd}$$

The method of applying the preceding theorem is now clear and simple. Let (X, τ) be a space and an equivalence relation $\sim$ on X. Assume that one thinks that a space Y is homeomorphic to the quotient space $X/\sim$. The strategy is to define a mapping $f\colon \ \to Y$ such that, first, the relation $\sim_f$ coincides with $\sim$ and second, that such a map is an identification. The map f can be thought of a *gluing map* in X, a term that we will employ in this chapter. We illustrate this idea with the following six examples.

Example 10.4 Define the equivalence relation on the closed interval $[0, 1]$ by declaring $0 \sim 1$ and $t \sim s$ if $t = s$ for $t, s \in (0, 1)$. This can be expressed as

$$t \sim s \text{ if and only if } \begin{cases} t = s, & \text{or} \\ t, s \in \{0, 1\}. \end{cases} \tag{10.6}$$

This relation appeared in (10.1). If we think of $[0, 1]$ as made up of rope, then we are gluing the endpoints of the interval and, of course, we get a circle so we suspect that $[0, 1]/\sim$ is homeomorphic to $\mathbb{S}^1$. The idea is to roll the interval $[0, 1]$ along $\mathbb{S}^1$, so the endpoints 0 and 1 coincide at the same point of $\mathbb{S}^1$. The gluing map is

$$f\colon [0, 1] \to \mathbb{S}^1, \quad f(t) = (\cos(2\pi t), \sin(2\pi t)).$$

This map appeared in Exercise 6.11. See Fig. 10.4. Clearly f is surjective and continuous and Corollary 10.3.2 says that f is an identification. Thus $[0, 1]/\sim_f$ is homeomorphic to $\mathbb{S}^1$. It remains to prove that $\sim$ coincides with $\sim_f$.

1. Let $t, s \in [0, 1]$ such that $t \sim s$. If $t, s \in \{0, 1\}$ then $f(t) = f(s) = (1, 0)$, so $t \sim_f s$.
2. Suppose that $t \sim_f s$, that is, $f(t) = f(s)$. Then $\cos(2\pi t) = \cos(2\pi s)$ and $\sin(2\pi t) = \sin(2\pi s)$. Then there is $k \in \mathbb{Z}$ such that $2\pi t - 2\pi s = 2\pi k$. Since $t, s \in [0, 1]$, the number k must be 0, 1 or -1. If $k = 0$ then $t = s$ and if k is 1 or -1, then $t, s \in \{0, 1\}$. In either case, $t \sim s$.

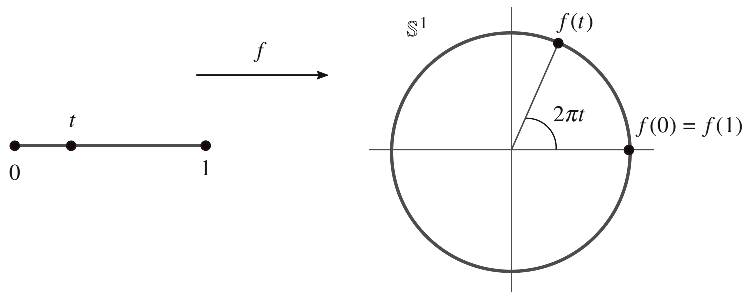


Fig. 10.4 The circle $\mathbb{S}^1$ obtained as a quotient space of $[0, 1]$

With the notation employed at the beginning of this chapter,

$$\frac{[0,1]}{\{0,1\}} \cong \mathbb{S}^1.$$

$\Diamond$

Example 10.5 Consider on $\mathbb{R}$ the equivalence relation defined by

$$t \sim s \text{ if and only if } t - s \in \mathbb{Z}.$$

The equivalence class of $t \in \mathbb{R}$ is $t + \mathbb{Z} = \{t + n : n \in \mathbb{Z}\}$. We need to glue a number $t \in \mathbb{R}$ with all numbers $t + n$ where $n \in \mathbb{Z}$ and furthermore for all $t \in \mathbb{R}$. To accomplish this, we roll $\mathbb{R}$ onto the circle such that we overlap points related to themselves (Fig. 10.5). Again the gluing map is as the preceding example, but now the domain is $\mathbb{R}$. Define

$$f : \mathbb{R} \to \mathbb{S}^1, \quad f(t) = (\cos(2\pi t), \sin(2\pi t)).$$

Then f is surjective and continuous. We proceed to show that f is open, thus concluding that f is an identification. To prove openness of f, it enough that the image of every open interval (a, b) is open in $\mathbb{S}^1$. If $|b-a| > 1$, then $f((a, b)) = \mathbb{S}^1$ which is open in $\mathbb{S}^1$. Assume that $|b - a| \leq 1$. Then $\mathbb{S}^1 \setminus f((a, b)) = f([b, a + 1])$. The set $f([b, a + 1])$ is compact because $[b, a + 1]$ is compact and f is continuous. Since $\mathbb{S}^1$ is Hausdorff, then $f([b, a + 1])$ is closed and this proves definitively that $f((a, b))$ is open. Thus $\mathbb{R}/\sim_f \cong \mathbb{S}^1$. Finally, arguing as in the preceding example, we find that $\sim_f = \sim$ and this concludes that $\mathbb{R}/\sim$ is homeomorphic to $\mathbb{S}^1$. In the literature, the quotient space is denoted by $\mathbb{R}/\mathbb{Z}$ (not to be confused with the relation given in (10.1)). $\Diamond$

We have seen that $\mathbb{S}^1$ can be viewed as a quotient space from both $[0, 1]$ and $\mathbb{R}$, which themselves are not homeomorphic. This shows that for the same quotient space, the underlying sets may not be homeomorphic. Furthermore, $\mathbb{S}^1$ is the quotient space of $\mathbb{R}$ which is not compact, even though $\mathbb{S}^1$ is compact. The next example motivated the definition of the quotient topology.

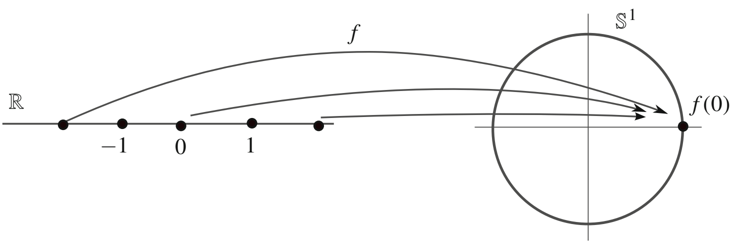


Fig. 10.5 The circle $\mathbb{S}^1$ obtained as a quotient space of $\mathbb{R}$

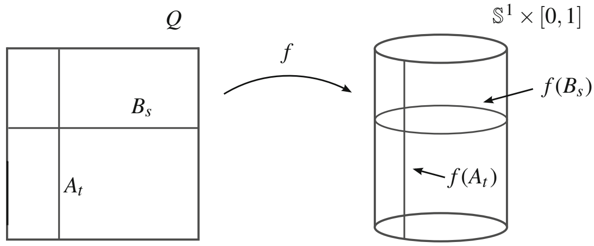


Fig. 10.6 The cylinder $\mathbb{S}^1 \times [0, 1]$ is obtained as a quotient space of the square Q

Example 10.6 Consider the equivalence relation (10.2) and prove that $Q/\sim$ is homeomorphic to the cylinder $\mathbb{S}^1 \times [0, 1]$ by folding Q as was done in Fig. 10.2. As in the foregoing example, the t-coordinate rolls in a circle, whereas we keep the s-coordinate unchanged which corresponds with the z-coordinate of the cylinder. Define

$$f : Q \to \mathbb{S}^1 \times [0, 1], \quad f(t, s) = (\cos(2\pi t), \sin(2\pi t), s).$$

Figure 10.6 illustrates f. The vertical segment $A_t = \{t\} \times [0, 1]$ is mapped to the vertical segment of the cylinder at the point $(\cos(2\pi t), \sin(2\pi t)) \in \mathbb{S}^1$. On the other hand, the horizontal segment $B_s = [0, 1] \times \{s\}$ goes to the circle included in the cylinder at height s. The map f is continuous, surjective and closed because its domain $[0, 1] \times [0, 1]$ is compact and the range $\mathbb{S}^1 \times [0, 1]$ is Hausdorff. Then $Q/\sim_f$ is homeomorphic to $\mathbb{S}^1 \times [0, 1]$. Finally, the proof that $\sim_f$ coincides with $\sim$ follows identical steps as in the above example. $\Diamond$

Example 10.7 Define an equivalence relation on $\mathbb{R}^2$ by

$$(t, s) \sim (t', s') \text{ if and only if } \begin{cases} (t, s) = (t', s'), & \text{or} \\ t - t' \in \mathbb{Z}, & s = s'. \end{cases}$$

We prove that $\mathbb{R}^2/\sim$ is homeomorphic to $\mathbb{S}^1 \times \mathbb{R}$. The gluing map is

$$f : \mathbb{R}^2 \to \mathbb{S}^1 \times [0, 1], \quad f(t, s) = (\cos(2\pi t), \sin(2\pi t), s).$$

It suffices to prove that f is open. The map f can be written as $f = g \times \mathbb{1}_{\mathbb{R}}$ where $g : \mathbb{R} \to \mathbb{R}$ is $g(t) = (\cos(2\pi t), \sin(2\pi t))$. Then f is open because it is the product of two open maps (the openness of g was proved in Example 10.5). $\Diamond$

The next example for $n = 2$ also appeared at the beginning of this chapter.

Example 10.8 Define an equivalence relation on the unit closed disc $\mathbb{D}^n$ by

$$x \sim y \text{ if and only if } \begin{cases} x = y, & \text{or} \\ x, y \in \mathbb{S}^{n-1}. \end{cases}$$

We prove that $\mathbb{D}^n/\sim$ is homeomorphic to $\mathbb{S}^n$. If $n = 2$ we can think of $\mathbb{D}^2$ as made of tissue and we need to sew $\mathbb{S}^1$ into one point. Of course, the resulting space is the sphere $\mathbb{S}^2$. Instead of writing this gluing map, we prefer to expand the interior of $\mathbb{D}^n$ to $\mathbb{R}^n$ and then use that $\mathbb{R}^n$ is homeomorphic to the punctured sphere $\mathbb{S}^n \setminus \{N\}$. In such case, we will map $\mathbb{S}^{n-1}$ to the north pole N of $\mathbb{S}^n$. Define

$$f\colon \mathbb{D}^n \to \mathbb{S}^n, \quad f(x) = \begin{cases} N, & \text{if } |x| = 1 \\ \pi^{-1} \circ h(x), & \text{if } |x| < 1, \end{cases}$$

where $h\colon \mathbb{D}^n \setminus \mathbb{S}^{n-1} = B_1(0) \to \mathbb{R}^n$ is the homeomorphism

$$h(x) = \frac{x}{1 - |x|}$$

that appeared in Example 6.15 and $\pi\colon \mathbb{S}^n \setminus \{N\} \to \mathbb{R}^n$ is the stereographic projection. It is immediate that f is surjective.

On the other hand, f is closed because $\mathbb{D}^n$ is compact and $\mathbb{S}^n$ is Hausdorff. To establish that f is an identification, it remains to see that f is continuous. We do it point-by-point. Continuity of f is clear in the open set $\mathbb{D}^n \setminus \mathbb{S}^{n-1}$ and it remains to show that f is continuous at every point of $\mathbb{S}^{n-1}$. Let $x \in \mathbb{S}^{n-1}$ and let $(x_n) \subset \mathbb{D}^n$ such that $(x_n) \to x$. If $x_n \in \mathbb{S}^{n-1}$, its image is $f(x_n) = N$. If $|x_n| < 1$, then $(|x_n|) \to 1$, so $(h(x_n)) \to \infty$ and $(f(x_n)) \to N$ by the expression for π^{-1} in (6.6). In any case, $(f(x_n)) \to N$.

Finally, we see that $\sim_f$ coincides with $\sim$. Note that f is injective in $B_1(0)$. Assume $x \sim_f y$ so $f(x) = f(y)$. If $f(x) \neq N$, then $x, y \in B_1(0)$ where f is injective and thus $x = y$. If $f(x) = N$, then necessarily $x, y \in \mathbb{S}^{n-1}$. This proves that $x \sim y$. The converse is immediate. $\Diamond$

The next quotient space is the Möbius strip, a set that, like the torus, is famous in mathematics. The definition of the Möbius strip is the quotient space $Q = [0, 1] \times [0, 1]$ where the relation (10.2) is reversed in the sides of Q. See Fig. 10.7, left, where the arrows indicate the equivalence relation.

Example 10.9 The *Möbius strip* is the quotient space $Q/\sim$ where

$$(t, s) \sim (t', s') \text{ if and only if } \begin{cases} (t, s) = (t', s'), & \text{or} \\ t, t' \in \{0, 1\}, & s' = 1 - s. \end{cases} \tag{10.7}$$

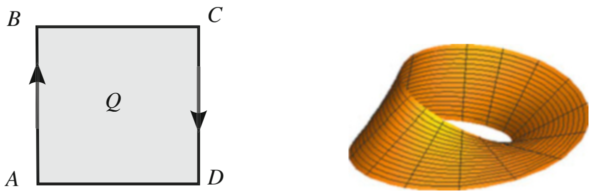


Fig. 10.7 The Möbius strip. Right, a subset of $\mathbb{R}^3$ homeomorphic to the Möbius strip

As in Example 10.6, we identify the Möbius strip with a set of $\mathbb{R}^3$. This set is a specific folding of Q. Let us label consecutively the vertices by A, B, C and D of Q (Fig. 10.7, left). To construct the Möbius strip, hold two opposite edges of Q, twist the AB side one half wind and bring to the CD side and finally glue along these edges with the rule that the vertex A coincides with C and B with D (Fig. 10.7, right). Folding the paper sheet like so, we embed $Q/\!\sim$ in $\mathbb{R}^3$.

We prove that the Möbius strip can be embedded in $\mathbb{R}^3$.

Although the analytic details to derive the embedding are somewhat involved, the idea of its construction is quite simple. For the sake of clarity, we replace the square Q by the rectangle $[0, 2\pi] \times [0, 1]$ where $(0, s) \sim (2\pi, 1 - s)$. Let C be a circle in the xy-plane and place a vertical segment L so the middle point p of L is supported at C. Label the endpoints of L by A and B. Then move L along C keeping the middle point at C so L is always contained in a vertical plane Π orthogonal to the velocity vector of C. Furthermore, L is rotated in the plane Π so when we come back to the initial position after one turn around C, the endpoints A and B of L are now situated in the reversed position.

The job is to describe analytically this process. Let $C = \{(x, y, 0) \in \mathbb{R}^3 : x^2 + y^2 = 4\}$ be the circle of radius 2 centered at the origin and positioned in the xy-plane. A parametrization of C is $\alpha(\theta) = 2(\cos\theta, \sin\theta, 0)$. The middle point of L is $\alpha(0) = (2, 0, 0)$ and the length of L is 2. This gives that when moving L along its movement, L never intersects the z-axis. The endpoints of L are $A = (2, 0, 1)$ and $B = (2, 0, -1)$ and a parametrization of L is

$$\beta\colon [0, 1] \to \mathbb{R}^3, \quad \beta(t) = A + t(B - A) = (2, 0, 1 - 2t).$$

We calculate the position of L at any point $\alpha(\theta)$. For convenience, we place L on the z-axis where we write easily the computations and next, we translate to the point $\alpha(\theta)$. On the z-axis, L is $\{(0, 0, 1 - 2t) : t \in [0, 1]\}$. We rotate L by an angle $\theta/2$ on the plane orthogonal to the velocity $\alpha'(0) = (0, 2, 0)$, obtaining

$$\begin{pmatrix} \cos\frac{\theta}{2} & 0 & -\sin\frac{\theta}{2} \\ 0 & 1 & 0 \\ \sin\frac{\theta}{2} & 0 & \cos\frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 - 2t \end{pmatrix} = \begin{pmatrix} -(1 - 2t)\sin\frac{\theta}{2} \\ 0 \\ (1 - 2t)\cos\frac{\theta}{2} \end{pmatrix}.$$

Then we translate to $\alpha(0)$ yielding $(2 - (1 - 2t)\sin\frac{\theta}{2}, 0, (1 - 2t)\cos\frac{\theta}{2})$. Finally we rotate about the z-axis by an angle equal to θ to situate L at $\alpha(\theta)$,

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 - (1 - 2t)\sin\frac{\theta}{2} \\ 0 \\ (1 - 2t)\cos\frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} 2\cos\theta - (1 - 2t)\sin\frac{\theta}{2}\cos\theta \\ 2\sin\theta - (1 - 2t)\sin\frac{\theta}{2}\cos\theta \\ (1 - 2t)\cos\frac{\theta}{2} \end{pmatrix}.$$

The set obtained is

$$\mathbb{M} = \left\{(2\cos\theta, 2\sin\theta, 0) + (2t-1)\left(\sin\tfrac{\theta}{2}\cos\theta, \sin\tfrac{\theta}{2}\sin\theta, -\cos\tfrac{\theta}{2}\right) : \right.$$
$$\left. \theta \in [0, 2\pi], t \in [0, 1]\right\}. \tag{10.8}$$

When L runs around C from $\theta = 0$ until $\theta = 2\pi$, the segment has reversed its endpoints because the rotation of L has been π. Define $f\colon [0, 2\pi] \times [0, 1] \to \mathbb{M}$ by

$$f(\theta, t) = (2\cos\theta, 2\sin\theta, 0) + (2t-1)(\sin\tfrac{\theta}{2}\cos\theta, \sin\tfrac{\theta}{2}\sin\theta, -\cos\tfrac{\theta}{2}).$$

See Fig. 10.7, right. This mapping is clearly continuous and surjective. Also f is closed because its domain is compact space and $\mathbb{M}$ is Hausdorff. This proves that f is an identification. We see that $\sim_f$ coincides with the relation $\sim$ given in (10.7).

1. Let $f(\theta, t) = f(\theta', t')$. Since f is 2π-periodic in the θ variable, there is not loss of generality in assuming $\theta = 0$. Thus $f(0, t) = f(t', \theta')$. By equating coordinate-by-coordinate,

$$2 = 2\cos\theta' - (1-2t')\sin\frac{\theta'}{2}\cos\theta'$$
$$0 = 2\sin\theta' - (1-2t')\sin\frac{\theta'}{2}\cos\theta'$$
$$1-2t = (1-2t')\cos\frac{\theta}{2}.$$

   From the first two equations, $1 - \cos\theta' = -\sin\theta'$. Then

$$1 = (\cos\theta')^2 + (\sin\theta')^2 = 1 + 2(\cos\theta')^2 - 2\cos\theta',$$

   so $\cos\theta'(\cos\theta' - 1) = 0$. Since $\cos\theta' \neq 0$ due to the first equation, we deduce $\theta' = 0$ or $\theta' = 2\pi$. In the first case, using the third equation, we conclude $t' = t$, so $(0, t) = (0, t')$. In the second case $\theta = 2\pi$, the third equation leads to $t' = 1 - t$, so $(0, t) \sim (2\pi, t')$.
2. The converse holds because the points $(0, t)$ and $(2\pi, 1-t)$ related under $\sim$ have identical images, $f(0, t) = f(2\pi, 1-t)$.

$\Diamond$

10.4 Maps Between Quotient Spaces

The aim of this section is to build a homeomorphism between two quotient spaces. As in Theorem 10.3.1, we revisit again the theory of equivalence relations to introduce the suitable context. Suppose that X and Y are two sets with two equivalence relations $\sim$ and $\approx$, respectively. A map $f\colon X \to Y$ is said to be *compatible* with $\sim$ and $\approx$ if satisfies the following property:

$$\text{if } x_1 \sim x_2 \text{ then } f(x_1) \approx f(x_2).$$

Under this situation, the mapping

$$\widetilde{f}\colon \frac{X}{\sim} \to \frac{Y}{\approx}, \quad \widetilde{f}([x]) = [f(x)]$$

satisfies $p' \circ \widetilde{f} \circ p = f$, where $p\colon X \to X/\sim$ and $p'\colon Y \to Y/\approx$ are the corresponding projections. In addition, if f satisfies the reversing property

$$\text{if } f(x_1) \approx f(x_2) \text{ then } x \sim x', \tag{10.9}$$

then $\widetilde{f}$ is injective. Notice that we have employed identical brackets to represent equivalence classes in different quotient spaces. The property that f is compatible with $\sim$ and $\approx$ implies that $\widetilde{f}$ is well defined. In other words, the expression $\widetilde{f}([x]) = [f(x)]$ does not depend on the choice of the representative x in the equivalence class $[x]$. We now equip each one of the sets with the structure of a topological space. The mapping $\widetilde{f}$ is the candidate of the desired homeomorphism between $X/\sim$ and $Y/\approx$.

Theorem 10.4.1 *Let X and Y be two topological spaces supporting two equivalence relations $\sim$ and $\approx$, respectively, and let $f\colon X \to Y$ be a compatible map with both relations. If f is continuous (resp. identification), then $\widetilde{f}$ is continuous (resp. identification).*

Suppose, in addition, that f satisfies (10.9). If f is an identification, then $X/\sim$ is homeomorphic to $Y/\approx$.

Example 10.10 Let $f\colon X \to Y$ be a homeomorphism and let $\sim$ be an equivalence relation on X. Consider the equivalence relation $\approx$ on Y by

$$y_1 \approx y_2 \text{ if and only if } f^{-1}(y_1) \sim f^{-1}(y_2). \tag{10.10}$$

Then $X/\sim$ is homeomorphic to $Y/\approx$. We give an example.

Consider the equivalence relation $\approx$ on $[a, b]$ that identifies the endpoints a and b. The homeomorphism

$$f\colon [0, 1] \to [a, b], \quad f(t) = (b - a)t + a$$

satisfies $f(\{0, 1\}) = \{a, b\}$. Thus the relation $\approx$ coincides with that of (10.10) concluding that $[a, b]/\approx$ is homeomorphic to $[0, 1]/\sim$. ♢

Example 10.11 Consider the unit square $Q = [0, 1] \times [0, 1]$ and the equivalence relation $\sim$ that identifies all points of its border. We see that $Q/\sim$ is homeomorphic to $\mathbb{S}^2$. We change Q to a homeomorphic space equipped with the equivalence relation (10.10) and prove that the new quotient space is homeomorphic to $\mathbb{S}^2$. Let $f: Q \to \mathbb{D}^2$ be the homeomorphism obtained by expanding $\mathbb{D}^2$ from the origin (Exercise 6.26). Let $\approx$ be the relation on $\mathbb{D}^2$ defined by $x \approx y$ if $f^{-1}(x) \sim f^{-1}(y)$. Then Example 10.10 implies that the quotient space $\mathbb{D}^2/\approx$ is homeomorphic to $Q/\sim$. The relation $\approx$ on $\mathbb{D}^2$ identifies the border of $\mathbb{D}^2$ at one point, that is, $f(\mathrm{Bd}(Q)) = \mathrm{Bd}(\mathbb{D}^2) = \mathbb{S}^1$. But it was proven that $\mathbb{D}^2/\approx$ is homeomorphic to $\mathbb{S}^2$ (Example 10.8). ◊

Example 10.12 Let $\sim$ be the equivalent relation (10.7) that defines the Möbius strip and let $\approx$ be the relation on $[0, 2\pi] \times [0, 1]$ that identifies $(0, s)$ with $(2\pi, 1 - s)$. Then $f: Q \to [0, 2\pi] \times [0, 1]$ given by $f(t, s) = (2\pi t, s)$ is a homeomorphism compatible with $\sim$ and $\approx$. So Theorem 10.4.1 implies that the Möbius strip is homeomorphic to the quotient space $[0, 2\pi]\times[0, 1]/\approx$. It was precisely this quotient space that was embedded in $\mathbb{R}^3$ in Example 10.9. ◊

Theorem 10.4.1 also permits us to relate the quotient topology and product topology. First we present the context from the point of view of set theory; here we change the notation of the equivalence relations by capital letters. Let X and Y be two sets and two equivalence relations R and R' on X and Y, respectively. Define an equivalence relation $R \times R'$ on the Cartesian product $X \times Y$ by

$$(x_1, y_1)\,(R \times R')\,(x_2, y_2) \text{ if and only if } x_1\, R\, x_2 \text{ and } y_1\, R'\, y_2.$$

Let us notice that if $\mathscr{R}$ and $\mathscr{R}'$ are the equivalence relations as subsets of $X \times X$ and $Y \times Y$, respectively, then $R \times R'$ is $\mathscr{R} \times \mathscr{R}' \subset X \times X \times Y \times Y$, up to reordering the factors. If p and p' denote the projections of the relations R and R', consider the product map $p \times p': X \times Y \to X/R \times Y/R'$. It is clear that the equivalence relation $\sim_{p\times p'}$ associated to $p \times p'$ coincides with $R \times R'$. We then know that there is a bijection

$$\overline{p \times p'}\colon \frac{X \times Y}{R \times R'} \to \frac{X}{R} \times \frac{Y}{R'}, \qquad \overline{p \times p'}([(x, y)]) = ([x], [y])$$

such that $(\overline{p \times p'}) \circ \pi = p \times p'$, where

$$\pi: X \times Y \to \frac{X \times Y}{R \times R'},$$

is the projection onto the quotient space. Once we have introduced the theoretical framework of set theory, we assume that X and Y are topological spaces and $X \times Y$ is the product space. Since p and p' are continuous, then $\overline{p \times p'}$ is also continuous. This proves that $\overline{p \times p'}$ is bijective and continuous between $(X \times Y)/(R \times R')$ and $X/R \times Y/R'$. Observe that $p \times p'$ may fail to be an identification even if p and p' are. Depending on the case, we can use previous theoretical results to conclude, if possible, that $\overline{p \times p'}$ is an identification.

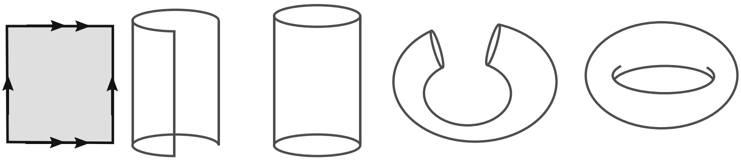


Fig. 10.8 The torus obtained as a quotient space of a square of $\mathbb{R}^2$

Example 10.13 Let $X = Y = [0, 1]$ be endowed with the relation R given in (10.6). Here the map

$$p \times p' \colon [0, 1] \times [0, 1] \to \frac{[0, 1]}{R} \times \frac{[0, 1]}{R} = \mathbb{S}^1 \times \mathbb{S}^1$$

is closed because the domain is compact and the codomain is Hausdorff. Thus

$$\frac{[0, 1] \times [0, 1]}{R \times R} \cong \frac{[0, 1]}{R} \times \frac{[0, 1]}{R} \cong \mathbb{S}^1 \times \mathbb{S}^1.$$

The relation $R \times R$ on the square $Q = [0, 1] \times [0, 1]$ identifies two-by-two the sides of Q such as it appears in Fig. 10.8. Therefore, the torus $\mathbb{T}^2$ is homeomorphic to a quotient space of the square Q. $\Diamond$

At this point, it deserves to be noted that the torus has been expressed in three ways in this book, all them topologically equivalent:

1. A topological product $\mathbb{S}^1 \times \mathbb{S}^1$. This was the definition given in (7.2), being also a subspace of the Euclidean space $\mathbb{R}^4$.
2. A surface of revolution in the Euclidean space $\mathbb{R}^3$ (Example 7.6). This is the way to think of the torus as a doughnut.
3. A quotient space of the square $[0, 1] \times [0, 1]$ (Example 10.13) and, as we will see in Exercise 10.5, as a quotient space of the Euclidean plane $\mathbb{R}^2$.

10.5 Worked Exercises

We offer a criterion to characterize if the projection on the quotient space is open or closed.

Exercise 10.1 Let $p \colon X \to X/\!\sim$ be the projection on the quotient space. Prove:

1. p is open if and only if $R[O]$ is open for all O open sets in X.
2. p is closed if and only if $R[F]$ is closed for all F closed sets in X.

In particular, (1) is true if we change open sets by elements of a basis for the topology.

Solution We will show the result for the case (1), since the other case is similar. The map p is open if and only if $p(O)$ is open for all open sets $O \subset X$. The set $p(O)$ is open in $\tau/\sim$ if and only if $p^{-1}(p(O)) \in \tau$, but this set is the saturation $R[O]$ of O.

Suppose now that β is a basis for the topology and that $R[B]$ is open for all $B \in \beta$. If O is an open set then $O = \bigcup_{i \in I} B_i$ for some set of indices I and $B_i \in \beta$. Then

$$R[O] = p^{-1}(p(\bigcup_{i \in I} B_i)) = \bigcup_{i \in I} p^{-1}(p(B_i)) = \bigcup_{i \in I} R[B_i]$$

is open because $R[B_i]$ is open hypothesis and $R[O]$ is union of open sets.

Let us observe that the saturation behaves well with respect to union of sets, $R[\cup_{i \in I} A_i] = \cup_{i \in I} R[A_i]$. □

Exercise 10.2 Using Theorem 10.4.1, prove that the quotient spaces $[0, 1]/\{0, 1\}$ and $\mathbb{R}/\sim$ are homeomorphic where $\sim$ is the relation

$$x \sim y \text{ if and only if } x - y \in \mathbb{Z}.$$

Solution We already know that $[0, 1]/\{0, 1\}$ and $\mathbb{R}/\sim$ are homeomorphic because they are each homeomorphic to the circle $\mathbb{S}^1$ (Examples 10.4 and 10.5). However, we now prove the result with the aid of Thm. 10.4.1. Recall that the equivalence classes for $\sim$ are $[x] = x + \mathbb{Z} = \{x + n : n \in \mathbb{Z}\}$. Define

$$f : [0, 1] \to \mathbb{R}, \quad f(x) = x,$$

and let $\sim$ be the relation on $[0, 1]$ that identifies $t = 0$ with $t = 1$. Then $t \sim s$ if and only if $f(t) \sim f(s)$. This induces an injective map

$$\widetilde{f} : \frac{[0, 1]}{\{0, 1\}} \to \frac{\mathbb{R}}{\sim}, \quad \widetilde{f}([x]) = [f(x)]$$

between the quotient spaces. Observe that f is not surjective so we cannot directly apply Theorem 10.4.1.

We prove that $\widetilde{f}$ is an identification. In order to distinguish the equivalence classes, let [ ] and [[ ]] denote the classes in $[0, 1]/\{0, 1\}$ and $\mathbb{R}/\sim$, respectively.

1. The map $\widetilde{f}$ is surjective. Let $[[t]] \in \mathbb{R}/\sim$. If $E(t)$ is the integer part of t, the number $t' = t - E(t)$ satisfies $t' \sim t$ and $t' \in [0, 1)$. By definition of $\widetilde{f}$,

$$\widetilde{f}([t']) = [[f(t')]] = [[t']] = [[t]].$$

2. The map $\widetilde{f}$ is continuous because $\widetilde{f} \circ p = f$ is continuous (Theorem 10.4.1).
3. The map $\widetilde{f}$ is closed. The domain of $\widetilde{f}$ is compact because it is the quotient space of a compact space. Thus $\widetilde{f}$ is closed if we prove that the codomain is

Hausdorff. Since we do not wish to use the homeomorphism $\mathbb{S}^1 \cong \mathbb{R}/\sim$, we go back to Example 10.2 to prove directly that $\mathbb{R}/\sim$ is Hausdorff. First, we see that the projection $p'\colon \mathbb{R} \to \mathbb{R}/\sim$ is open by using Exercise 10.1. Let $O \subset \mathbb{R}$ be an open set. Then p' is open if the saturation $R'[O]$ under $\sim$ is open in $\mathbb{R}$. But

$$R'[O] = p'^{-1}(p'(O)) = O + \mathbb{Z} = \bigcup\nolimits_{n\in\mathbb{Z}}(n + O).$$

Each one of the sets $n + O$ is open because $n + O = T_n(O)$, where $T_n\colon \mathbb{R} \to \mathbb{R}$ is the translation $T_n(x) = x + n$ and the translations are homeomorphisms. In terminology of topological groups (Exercise 7.26), and taking $(\mathbb{R}, +)$ as the group, we have $O + \mathbb{Z} = \bigcup_{n\in\mathbb{Z}} R_n(O)$, where R_n is the right translation from n. The second requirement in Example 10.2 is that the set

$$\mathscr{R}' = \{(t, s) \in \mathbb{R}^2 : t \sim s\} = \{(t, s) \in \mathbb{R}^2 : t - s \in \mathbb{Z}\}$$

is closed in $\mathbb{R}^2$. This is because $\mathscr{R}' = h^{-1}(\mathbb{Z})$, where h is the continuous function on $\mathbb{R}^2$ defined by $h(t, s) = t - s$ and $\mathbb{Z}$ is closed in $\mathbb{R}$. □

Exercise 10.3 Consider on $\mathbb{R}^2$ the equivalence relation $\sim$ that identifies all points of $\mathbb{R} \times \{0\}$. Prove that the projection $p\colon \mathbb{R}^2 \to \mathbb{R}^2/\sim$ is closed but not open. Prove $([(0, 1/n)]) \to [(0, 0)]$ but $([(n, 1/n)])$ does not converge to $[(0, 0)]$.

Solution

1. The saturation of a subset A of $\mathbb{R}^2$ is

$$R[A] = \begin{cases} A, & \text{if } A \cap (\mathbb{R} \times \{0\}) = \emptyset \\ A \cup (\mathbb{R} \times \{0\}), & \text{if } A \cap (\mathbb{R} \times \{0\}) \neq \emptyset. \end{cases}$$

   Thus if F is a closed set of $\mathbb{R}^2$, then $R[F]$ is F or $F \cup (\mathbb{R} \times \{0\})$. In either case, it is a closed set because in the second case, it is union of two closed sets. This proves that p is closed. However, p is not open using Exercise 10.1 again. For example, consider $O = B_1(0, 0)$ the ball centered at $(0, 0)$ and radius 1. Since $(0, 0) \in \mathbb{R} \times \{0\}$, then $R[O] = O \cup (\mathbb{R} \times \{0\})$. But this set is not open because there are no balls centered at $(2, 0)$ and contained in $R[O]$.
2. By the relation $\sim$, we have that $[(0, 0)] = \mathbb{R} \times \{0\}$ and $[(0, 1/n)] = \{(0, 1/n)\}$. Let $\widetilde{O}$ be an open neighborhood of $[(0, 0)]$. Then $p^{-1}(\widetilde{O})$ is open in $\mathbb{R}^2$ and contains $(0, 0)$ because $p(0, 0) = [(0, 0)]$. Then there is $r > 0$ such that $B_r(0, 0) \subset p^{-1}(\widetilde{O})$. Since $((0, 1/n)) \to (0, 0)$ in $\mathbb{R}^2$, there is $\nu \in \mathbb{N}$ such that $(0, 1/n) \in B_r(0, 0)$ for all $n \geq \nu$. Taking images under p,

$$[(0, 1/n)] \in p(B_r(0, 0)) \subset p(p^{-1}(\widetilde{O})) = \widetilde{O}$$

   for all $n \geq \nu$. This proves the desired convergence.
3. Consider $O = \{(x, y) \in \mathbb{R}^2 : 2xy < 1\}$, which is open in $\mathbb{R}^2$. Since $\mathbb{R} \times \{0\} \subset O$, then $R[O] = O \cup (\mathbb{R} \times \{0\}) = O$. Thus O is a saturated open set; in particular,

$\widetilde{O} = p(O)$ is open in the quotient space and contains $[(0, 0)]$. Then $\widetilde{O}$ is a neighborhood of $[(0, 0)]$. If $([(n, 1/n)]) \to [(0, 0)]$, there is $\nu \in \mathbb{N}$ such that $[(n, 1/n)] \in \widetilde{O}$ whenever $n \geq \nu$. Taking preimages under p, we have $(n, 1/n) \in p^{-1}(\widetilde{O}) = O$, which is not possible. □

Exercise 10.4 Let $A \subset X$ and consider $\sim$ the equivalence relation that identifies all points of A. Denote by X/A the quotient space. Prove that if A is open or closed, then $p\colon X \setminus A \to p(X \setminus A)$ is a homeomorphism. Give an example where p is not a homeomorphism for an arbitrary A. Show that if Y is a space and A is open or closed, then there is a bijective correspondence between the continuous maps $X \to Y$ that are constant on A and the continuous maps $X/A \to Y$.

Solution If $x \in X \setminus A$, then $[x] = \{x\}$, so $X \setminus A$ is bijective with $p(X \setminus A)$. This exercise gives conditions to ensure that both spaces are homeomorphic.

1. Suppose that A is open (an equal argument holds if A is closed). We know that $p\colon X \setminus A \to p(X \setminus A)$ is continuous and bijective. We see that p is closed. Let $C \subset X \setminus A$ be a closed set; in particular, because $X \setminus A$ is closed, we have that C is closed in X. Moreover C is saturated because $C \cap A = \emptyset$. By definition of quotient topology, $p(C)$ is closed in X/A. Because $p(C) \subset p(X \setminus A)$, it is closed in $p(X \setminus A)$.
2. Let $X = \mathbb{R}$ and $A = \mathbb{Q}$. Then $X \setminus A = \mathbb{R} - \mathbb{Q}$ is the set of irrationals. Let $O = (\mathbb{R} - \mathbb{Q}) \cap (0, 1)$, which is open in $\mathbb{R} - \mathbb{Q}$. If $p(O)$ is open in $p(\mathbb{R} - \mathbb{Q})$ there is an open set $G \subset \mathbb{R}/\mathbb{Q}$ such that $p(O) = G \cap p(\mathbb{R} - \mathbb{Q})$. Taking the preimage under p, $O = p^{-1}(G) \cap (\mathbb{R} - \mathbb{Q})$. Since $p^{-1}(G)$ is open in $\mathbb{R}$, $p^{-1}(G)$ must contain rationals, so $\mathbb{Q} \subset p^{-1}(G)$ because $p^{-1}(G)$ is saturated. If, for example, we take $x = 2$, there is $r > 0$ such that $(2 - r, 2 + r) \subset p^{-1}(G)$; in particular, $p^{-1}(G)$ contains irrationals outside $(0, 1)$. This is a contradiction with the identity $O = p^{-1}(G) \cap (\mathbb{R} - \mathbb{Q})$.
3. Let $f\colon X \to Y$ be a continuous map such that $f(a) = y_0$ for all $a \in A$. Then

$$g_f\colon X/A \to Y, \quad g_f([x]) = \begin{cases} f(x) & x \notin A \\ y_0 & x \in A \end{cases}$$

is well defined and it is continuous because $g_f \circ p = f$. Conversely, if $g\colon X/A \to Y$ is continuous, define $f\colon X \to Y$ to be $f(x) = g([x])$. Let us notice that f restricted to A is constant because A is one element in the quotient space. Moreover f is continuous because $f = g \circ p$. □

Exercise 10.5 Using Example 10.5 and letting R equal to $\sim$ for convenience in the notation, prove that

$$\frac{\mathbb{R}^2}{R \times R} \cong \mathbb{S}^1 \times \mathbb{S}^1.$$

Solution Consider the relation $R \times R$ on $\mathbb{R}^2$. If $(x, y)\,(R \times R)\,(x', y')$ then $(x, y) - (x', y') \in \mathbb{Z} \times \mathbb{Z}$, or equivalently, $x - x' \in \mathbb{Z}$ and $y - y' \in \mathbb{Z}$. On the other hand, if $p : \mathbb{R} \to \mathbb{R}/R \cong \mathbb{S}^1$ is the projection, then

$$p \times p : \mathbb{R} \times \mathbb{R} \to \frac{\mathbb{R}}{R} \times \frac{\mathbb{R}}{R} = \mathbb{S}^1 \times \mathbb{S}^1$$

is continuous and surjective. We see that $p \times p$ is open. To do this, let us first show that p is open. It was proved in Example 10.5 that

$$f : \mathbb{R} \to \mathbb{S}^1, \quad f(t) = (\cos(2\pi t), \sin(2\pi t))$$

is open. Let $\widetilde{f} : \mathbb{R}/R \to \mathbb{S}^1$ be the homeomorphism that induces f. Since $\widetilde{f} \circ p = f$, the composition $p = \widetilde{f}^{-1} \circ f$ of two open maps is open. Then $p \times p : \mathbb{R} \times \mathbb{R} \to \mathbb{R}/R \times \mathbb{R}/R$ is open (Exercise 7.5.5). Hence $p \times p$ is an identification and $\overline{p \times p}$ is a homeomorphism. This proves that

$$\frac{\mathbb{R}^2}{R \times R} \cong \frac{\mathbb{R}}{R} \times \frac{\mathbb{R}}{R} \cong \mathbb{S}^1 \times \mathbb{S}^1.$$

□

Exercise 10.6 Prove that the cone $C = \{(x, y, z) \in \mathbb{R}^3 : z^2 = x^2 + y^2\}$ is a quotient space of the cylinder $\mathbb{S}^1 \times \mathbb{R}$.

Solution If we bear in mind that a quotient space is a space resulting from a gluing process, it is conceivable that the cone is obtained by shrinking the horizontal circle at the height $z = 0$ reducing it at one point as it is shown in Fig. 10.9. Consider on $\mathbb{S}^1 \times \mathbb{R}$ the equivalence relation $\sim$ that identifies all points of the circle $\mathbb{S}^1 \times \{0\}$. We see that $(\mathbb{S}^1 \times \mathbb{R})/\!\sim$ is homeomorphic to the cone C. The gluing map will carry each circle of the cylinder in the circle of the cone at the same height by means of a dilation in the xy-plane, with the exception of the circle at the height $z = 0$, which maps to the vertex $(0, 0, 0)$ of the cone. This map writes as $(x, y, z) \mapsto (\lambda x, \lambda y, z)$. This yields $\lambda^2 x^2 + \lambda^2 y^2 = z^2$. Since the first two coordinates of (x, y, z)

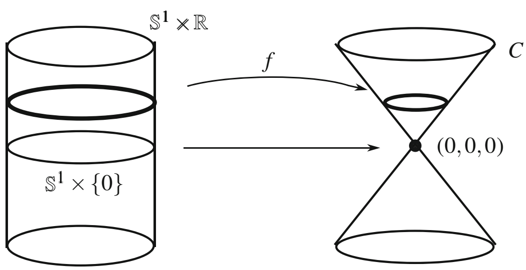


Fig. 10.9 The cone is a quotient space of the cylinder

lie contained in $\mathbb{S}^1$, we derive $\lambda^2 = z^2$, hence $\lambda = z$. Once the gluing map has been motivated, it remains to write it down in detail. Define

$$f : \mathbb{S}^1 \times \mathbb{R} \to C, \quad f(x, y, z) = (zx, zy, z).$$

The image f is included in C because

$$(zx)^2 + (zy)^2 = z^2(x^2 + y^2) = z^2.$$

We prove that $\sim_f$ coincides with $\sim$ and that f is an identification

1. We see that the relation $\sim_f$ coincides with $\sim$. Assume that (x, y, z) and (x', y', z') are two points of the cylinder with identical image. If this image is $(0, 0, 0)$, then both points belong to $\mathbb{S}^1 \times \{0\}$, so they are related by $\sim$. If the image is not the origin, from $(zx, zy, z) = (z'x', z'y', z')$ we deduce $z = z'$ and $zx = z'x'$ and $zy = z'y'$. Since $z, z' \neq 0$, we have $x = x'$, $y = y'$, hence both points coincide. Conversely, suppose that $(x, y, z) \sim (x', y', z')$. The only case to consider is that both points belong to $\mathbb{S}^1 \times \{0\}$. Then $z = z' = 0$, so $f(x, y, 0) = f(x', y', 0) = (0, 0, 0)$.
2. The continuity of f is immediate by composition with the projections. So, if p_i and q_i are the restrictions of the projections of $\mathbb{R}^3$ to $\mathbb{S}^1 \times \mathbb{R}$ and to C, respectively, then $q_1 \circ f = p_1 \cdot p_3$, $q_2 \circ f = p_2 \cdot p_3$ and $q_3 \circ f = p_3$.
3. We prove that f is onto. Let $(a, b, c) \in C$. If this point is $(0, 0, 0)$ then $f(1, 0, 0) = (0, 0, 0)$. Assume that $(a, b, c) \in C$ is not the origin. In particular, $c \neq 0$. Thus $a^2 + b^2 = c^2 \neq 0$ and

$$f\left(\frac{a}{\sqrt{a^2 + b^2}}, \frac{b}{\sqrt{a^2 + b^2}}, c\right) = \left(\frac{ac}{\sqrt{a^2 + b^2}}, \frac{bc}{\sqrt{a^2 + b^2}}, c\right) = (a, b, c).$$

4. The function f is closed. We prove that the image of a closed set F of the cylinder is closed in the cone C. Let $(a, b, c) \in \overline{f(F)}$. Then there is $((x_n, y_n, z_n)) \subset F$ such that $(f(x_n, y_n, z_n)) \to (a, b, c)$. Since $\mathbb{S}^1$ is compact, passing to a subsequence, we may suppose that $((x_n, y_n)) \to (a', b') \in \mathbb{S}^1$. We distinguish two cases:

   (a) Case $c \neq 0$. Thus $z_n \neq 0$ for all n sufficiently large. This implies $((x_n, y_n, z_n)) \cap (\mathbb{S}^1 \times \{0\}) = \emptyset$, so

   $$(f(x_n, y_n, z_n) = (z_n x_n, z_n y_n, z_n)) \to (a, b, c) = (a'c, b'c, c).$$

   Then $((x_n, y_n, z_n)) \to (a', b', c)$. Since F is closed, $(a', b', c) \in F$, so $f(a', b', c) \in f(F)$. But this images is $(a'c, b'c, c) = (a, b, c)$, as we desire to prove.

   (b) Case $c = 0$. From $f(x_n, y_n, z_n) \to (a, b, 0)$, and because $(a, b, 0) \in C$, then $a = b = 0$. By definition of f, $(z_n) \to 0$. Again, $((x_n, y_n, z_n)) \to (a', b', 0)$

for some $(a', b') \in \mathbb{S}^1$ and closedeness of F implies that $(a', b', 0) \in F$. Finally, $f(a', b', c) = (0, 0, 0) \in f(F)$. □

Exercise 10.7 Define on $[0, 1] \times \mathbb{R}$ the equivalence relation $(t, s) \sim (t', s')$ if are equal or $t, t' \in \{0, 1\}$ and $s = s'$. Prove that the quotient space is homeomorphic to the cylinder $\mathbb{S}^1 \times \mathbb{R}$.

Solution The gluing map f is as in Example 10.6 but the domain is now $[0, 1] \times \mathbb{R}$. Define

$$f : [0, 1] \times \mathbb{R} \to \mathbb{S}^1 \times \mathbb{R}, \quad f(t, s) = (\cos(2\pi t), \sin(2\pi t), s).$$

This map is continuous and surjective. It is also immediate that $\sim_f$ coincides with $\sim$. We see f is an identification proving that f is closed. The domain is not compact in comparison with Example 10.6. Let F be a closed set in $[0, 1] \times \mathbb{R}$ and we see that $f(F)$ is closed. Let $(x, y, z) \in \overline{f(F)}$. Then there is a sequence $((t_n, s_n)) \subset F$ such that $(f(t_n, s_n)) \to (x, y, z)$, or equivalently

$$((\cos(2\pi t_n), \sin(2\pi t_n), s_n)) \to (x, y, z).$$

Therefore $(s_n) \to z$. Since the sequence (t_n) is included in the compact set $[0, 1]$, there is $t \in [0, 1]$ and a subsequence $(t_{\sigma(n)})$ such that $(t_{\sigma(n)}) \to t$. Therefore $((t_{\sigma(n)}, s_{\sigma(n)})) \to (t, z)$. As a consequence, we derive that $(t, z) \in F$ because F is closed. By uniqueness of the limit, $(x, y, z) = f(t, z) \in f(F)$. □

In the following two exercises we define two quotient spaces based on the Cartesian product of a space with a closed interval.

Exercise 10.8 Let X be a space, $\mathbf{I} = [0, 1]$. On $X \times \mathbf{I}$ define the equivalence relation $\sim$ that identifies all points of $X \times \{1\}$. The quotient space $X \times \mathbf{I}/\sim$ is called the *cone over* X and denoted by $C(X)$. See Fig. 10.10.

1. Prove that the cone over a point is a closed bounded interval.
2. Prove that the cone over the circle $\mathbb{S}^1$ is homeomorphic to the cone Y given by $z = \sqrt{x^2 + y^2}$, $x^2 + y^2 \leq 1$.
3. Prove that the cone over a space is path connected (even if X is not).

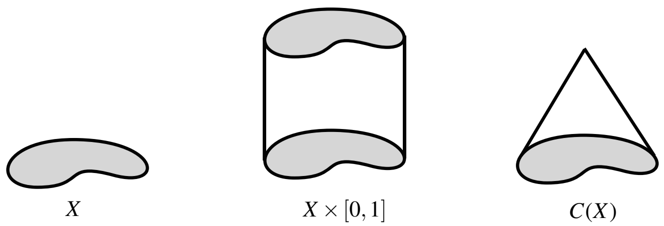


Fig. 10.10 A cone over a space X

Solution

1. If $X = \{p\}$ is a point then $X \times \mathbf{I} = \{p\} \times \mathbf{I}$. The equivalence relation that identifies all points of $\{p\} \times \{1\}$ has only one element as subset of $X \times \mathbf{I}$. Thus the relation is the identity and consequently, the quotient space is $X \times \mathbf{I} = \{p\} \times \mathbf{I}$, which is homeomorphic to $\mathbf{I}$.
2. Let $Y = \{(x, y, z) \in \mathbb{R}^3 : z = \sqrt{x^2 + y^2}, x^2 + y^2 \leq 1\}$. The gluing map carries each $\mathbb{S}^1 \times \{t\}$ in the circle of Y at height $z = 1 - t$. Each (x, y, t) is mapped into another point of the form $(\lambda x, \lambda y, 1 - t)$. In order for this point to belong to Y, we require $\lambda^2 x^2 + \lambda^2 y^2 = (1-t)^2$, or equivalently, $\lambda^2 = (1-t)^2$. Consequently, define

$$f : \mathbb{S}^1 \times \mathbf{I} \to Y, \quad f(x, y, z) = ((1-t)x, (1-t)y, 1-t).$$

We see that $\sim_f$ is the relation that defines the cone over $\mathbb{S}^1$. So, $f(x, y, t) = f(x', y', t')$ if and only if $(1-t)x = (1-t')x'$, $(1-t)y = (1-t')y'$ and $1-t = 1-t'$. From the last equation, it follows that $t = t'$. Thus $(1-t)x = (1-t)x'$ and $(1-t)y = (1-t')y'$. If $1 - t \neq 0$, that is, $t \neq 1$, then $x = x'$ and $y = y'$ and the two initial points coincide. The other case is $t = 1$ and the points are related with $\sim$.

We conclude the proof by observing that f is continuous and defined on a compact space whose codomain is Hausdorff.
3. We see that any point can be joined with $[(x, 1)]$. Let $[(x, t)] \in C(X)$ and define

$$\alpha : \mathbf{I} \to C(X), \quad \alpha(s) = [(x, (1-s)t + s)].$$

Note that $(1-s)t + s \in \mathbf{I}$ because $t, s \in \mathbf{I}$. It is clear that α joins $[(x, t)]$ with $[(x, 1)]$. It remains to prove that α is continuous. But this is clear because $\alpha = p \circ \widetilde{\alpha}$, where $p : X \to C(X)$ is the projection and $\widetilde{\alpha} : \mathbf{I} \to X \times \mathbf{I}$ is $\widetilde{\alpha}(s) = (x, (1-s)t + s)$ which is clearly continuous. □

Exercise 10.9 Let X be a space and on $X \times [-1, 1]$ define the equivalence relation $\sim$ that identifies $X \times \{1\}$ in one class and $X \times \{-1\}$ in another. The quotient space is called the *suspension of* X and denoted by $S(X)$. See Fig. 10.9.

1. Prove that the suspension of two points (endowed with the discrete topology) is a circle (observe that the suspension of a point is a closed interval).
2. Prove that the suspension of a circle is a sphere.

Solution

1. We may assume that $X = \{-1, 1\} \subset \mathbb{R}$. We think of $X \times [-1, 1]$ as two vertical segments where the top endpoints are glued and the two bottom points are glued

as well. We will assign the points $(-1, t)$ and $(1, t)$ to the points of $\mathbb{S}^1$ at height t. Define

$$f\colon X \times [-1, 1] \to \mathbb{S}^1, \quad f(x, t) = \begin{cases} (-\sqrt{1-t^2}, t), & x = -1 \\ (\sqrt{1-t^2}, t), & x = 1. \end{cases}$$

It is elementary that f is surjective and maps $\{-1\} \times I$ to the left halfcircle of $\mathbb{S}^1$ and $\{1\} \times I$, to the right one. We show that $\sim_f$ coincides with $\sim$. It is clear that two pairs of $X \times [-1, 1]$ related by $\sim$ have equal image. So, the image of all points $(x, 1)$ is $(0, 1)$ and the image of all points $(x, -1)$ is $(0, -1)$. Reciprocally, suppose $f(x, t) = f(x', t')$. Then $t = t'$. If $1-t^2 \neq 0$ then necessarily $x = x'$. If $1 - t^2 = 1$, then $t = 1$ or $t = -1$ and the image is $(0, 1)$ or $(0, -1)$, respectively. In the first case, we have $x, x' = 1$ and in the second, $x, x' = -1$.

Finally, f is an identification because f is continuous, surjective and closed (the domain is compact and the codomain is Hausdorff).

2. The equivalence relation of the suspension identifies the circles $\mathbb{S}^1 \times \{-1\}$ and $\mathbb{S}^1 \times \{1\}$ in two different classes. Thus the gluing map carries each horizontal circle of $\mathbb{S}^1 \times [-1, 1]$ to the corresponding circle of $\mathbb{S}^2$ of equal height. In this way, the image of the point (x, y, t) is $(\lambda x, \lambda y, t) \in \mathbb{S}^2$, thus $\lambda^2(x^2+y^2)+t^2 = 1$. This yields $\lambda = \sqrt{1-t^2}$. Define

$$f\colon \mathbb{S}^1 \times [-1, 1] \to \mathbb{S}^2, \quad f(x, y, t) = (\sqrt{1-t^2}x, \sqrt{1-t^2}y, t).$$

The reasoning is as in (1) and we leave it to the reader to check that the relation $\sim_f$ coincides with the relation of the suspension. Again the domain is compact and the codomain is Hausdorff, so f is closed. □

We prove a universal property of the quotient topology (Fig. 10.11). The following exercise links with Exercises 5.20 and 7.19.

Exercise 10.10 Let X be a topological space and let $\sim$ be an equivalence relation. Prove that the quotient topology is the unique topology on $X/\sim$ with the following property: for any topological space (Y, τ') and any map $f\colon X/\sim \to Y$, we have that f is continuous if and only if $f \circ p$ is continuous.

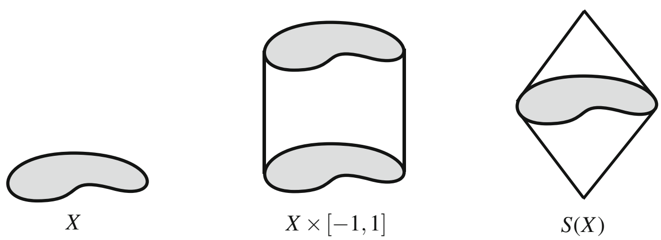


Fig. 10.11 A suspension on a space X

Solution First, let us observe immediately that the quotient topology satisfies the property thanks to Proposition 10.1.2. Suppose now that $\widetilde{\tau}$ is a topology on $X/\sim$ with that property. Consider two particular choices of (Y, τ') and f. First, let $(Y, \tau') = (X/\sim, \widetilde{\tau})$ and f the identity of the space $(X/\sim, \widetilde{\tau})$. Then f is continuous, so by hypothesis, $f \circ p = p$ is continuous. By Proposition 10.1.2, $\widetilde{\tau} \subset \tau/\sim$.

The second choice is $(Y, \tau') = (X/\sim, \tau/\sim)$ and $f = \mathbb{1}: (X/\sim, \widetilde{\tau}) \to (X/\sim, \tau/\sim)$. Then $f \circ p = p$ is the projection on $(X/\sim)$ is continuous. By hypothesis, f continuous, proving $\tau/\sim \subset \widetilde{\tau}$. □

Exercise 10.11 If X is a Hausdorff space and $A \subset X$ is finite, prove that the quotient space X/A is Hausdorff. Show an example where A is infinite but X/A is not Hausdorff.

Solution

1. Write $A = \{a_1, \ldots, a_n\}$. Since X is Hausdorff, all singletons are closed and thus A is closed and $X \setminus A$ is open in X. Let $[x], [y] \in X/A$, $[x] \neq [y]$, that is x and y are not related.

   (a) Case $x, y \notin A$. Then $[x] = \{x\}$ and $[y] = \{y\}$. Since X is Hausdorff, there are disjoint open sets O_1 and O_2 with $x \in O_1$ and $y \in O_2$. We do not know if O_1 and O_2 are saturated because they may intersect A. If so, we remove those points of A from O_i. To make this precise, let

   $$O_1' = O_1 \cap (X \setminus A), \quad O_2' = O_2 \cap (X \setminus A),$$

   which are open in X again. Moreover, $x \in O_1'$ and $y \in O_2'$ because x and y are not elements of A. Since $O_i' \cap A = \emptyset$, $R[O_i'] = O_i'$. Then $p(O_1')$ and $p(O_2')$ are disjoint open sets in the quotient space and separate $[x]$ from $[y]$.

   (b) Case $x \in A$ and $y \notin A$. For simplicity, suppose $x = a_1$. Because X is Hausdorff, there are disjoint open sets separating a_i from y, $1 \leq i \leq n$. So let $O_1, \ldots, O_n$ be open sets with $a_i \in O_i$, let $G_1, \ldots, G_n$ be open sets containing y and $O_i \cap G_i = \emptyset$ for $i = 1, \ldots, n$. Let $G = G_1 \cap \ldots \cap G_n$. Then $O_i \cap G = \emptyset$ but it is possible that G contains points of A. Let $G' = G \cap (X \setminus A)$, which is an open set containing y and saturated. On the other hand, let $O = O_1 \cup \ldots \cup O_n$. Then O is an open set that does not intersect G' and O is saturated since $A \subset O$. Under the projection, these two disjoint saturated open sets give open sets in the quotient space separating $[x]$ and $[y]$.

2. Let $X = \mathbb{R}$ and $A = \mathbb{Q}$. We prove that $\mathbb{R}/\mathbb{Q}$ is not Hausdorff. It suffices to verify that any two open sets of $\mathbb{R}/\mathbb{Q}$ must intersect. This is equivalent to seeing that every two saturated open sets O_1 and O_2 have nonempty intersection. Since O_1 and O_2 are open sets of $\mathbb{R}$ and that $\mathbb{Q}$ is dense in $\mathbb{R}$, then $O_i \cap \mathbb{Q} \neq \emptyset$ for $i = 1, 2$. Thus $R[O_i] = O_i \cup \mathbb{Q}$, $i = 1, 2$. As O_i is saturated, $O_i = O_i \cup \mathbb{Q}$, which implies that $\mathbb{Q} \subset O_i$, $i = 1, 2$. From this, $O_1 \cap O_2 \neq \emptyset$. □

Exercise 10.12 Apply the above exercise to study compactness of the quotient space of $\mathbb{R}$ by the equivalence relation that identifies all integers. Determine if the quotient space is Hausdorff.

Solution We denote by $\sim$ the relation that identifies in one point all integers. Recall that the notation of $\mathbb{R}/\mathbb{Z}$ employed in Example 10.5 does not correspond with the relation $\sim$.

1. We prove that $\mathbb{R}/\sim$ is not compact. For each $n \in \mathbb{Z}$, let $I_n = (n - \frac{1}{2}, n + \frac{1}{2})$ and $A = \bigcup_{n\in\mathbb{Z}} I_n$. Then A is open because it is union of open intervals. Further $\mathbb{Z} \subset A$ so $R[A] = A \cup \mathbb{Z} = A$ and thus, A is saturated. Also, for each $n \in \mathbb{Z}$, let $O_n = (n, n+1)$. This set is open and it is also saturated because $O_n \cap \mathbb{Z} = \emptyset$.
   Finally, consider the covering of $\mathbb{R}$ given by the saturated open sets $\{A\} \cup \{O_n : n \in \mathbb{Z}\}$. We see that it is not possible to extract a finite subcovering of $\mathbb{R}$. Certainly, in order to cover all integers, the only open set containing integers is A. Thus the finite subcovering would be $\mathbb{R} = A \cup O_{n_1} \cup \ldots \cup O_{n_m}$. This, however, is absurd since the point $k + 1/2$, where k is any integer bigger than n_i, $i = 1, \ldots, m$, belong neither A nor to $O_{n_1} \cup \ldots \cup O_{n_m}$.
2. We see that $\mathbb{R}/\sim$ is Hausdorff. We prove that for two given real numbers that are not related by $\sim$, it is possible to find disjoint saturated neighborhoods U and V of x and y, respectively. Let $E(x)$ denote the integer part of x. There are two alternatives regarding x and y.
   (a) Case $x, y \notin \mathbb{Z}$. If $E(x) \neq E(y)$, it suffices to take $U = (E(x), E(x+1))$ and $V = (E(y), E(y+1))$. Suppose $E(x) = E(y)$, which means that x and y belong to the interval $(m, m+1)$, where $m = E(x)$. Without loss of generality it can be assumed that $x < y$. Let c be a number with $x < c < y$. Then the saturated neighborhoods to consider are $U = (m, c)$ and $V = (c, m+1)$.
   (b) Case $x \notin \mathbb{Z}$ and $y \in \mathbb{Z}$. Since $x \notin \mathbb{Z}$, if $m = E(x)$ then $x \in (m, m+1)$. Let $r > 0$ be a number such that $m + r < x < m + 1 - r$. Then choose

$$U = (m + r, m + 1 - r), \quad V = \bigcup_{n\in\mathbb{Z}} (n - r, n + r).$$

   Then U and V are saturated and clearly $U \cap V = \emptyset$. □

Exercise 10.13 (Continuation) Prove that the above space $\mathbb{R}/\sim$ is not first countable, in particular it cannot be homeomorphic to any subset of a Euclidean space.

Solution The property of first countability fails at $[0]$. Suppose that $\beta_{[0]} = \{U_n : n \in J\}$ is a countable basis of neighborhoods of $[0]$ and without loss of generality assume that all U_n are open. Here J is a subset of $\mathbb{N}$ that can be finite or not. We distinguish both situations. First, suppose that J is infinite, $J = \mathbb{N}$. Let $V_n = p^{-1}(U_n)$. Observe that $\mathbb{Z} \subset V_n$ for all n. For each $n \in \mathbb{Z}$, let $\epsilon_n \in (0, 1)$ such that $(n - \epsilon_n, n + \epsilon_n) \subset V_n$. Then $V = \bigcup_{n\in\mathbb{Z}} (n - \frac{\epsilon_n}{2}, n + \frac{\epsilon_n}{2})$ is a saturated open set of $\mathbb{R}$, so $p(V)$ is open in $\mathbb{R}/\sim$ containing $[0]$. Then there is $m \in \mathbb{N}$ such that $[0] \in U_m \subset p(V)$. Thus $p^{-1}(U_m) = V_m \subset V$, so $(m - \epsilon_m, m + \epsilon_m) \subset \bigcup_{n\in\mathbb{Z}}(n - \frac{\epsilon_n}{2}, n + \frac{\epsilon_n}{2})$. This proves that $m + \frac{\epsilon_m}{2} \in (m - \frac{\epsilon_n}{2}, m + \frac{\epsilon_n}{2})$, a contradiction.

Assume that J is finite. The argument repeats again. Now $\mathbb{Z} \subset V_k$ for all $k \in J$. Since J is finite, for each $n \in \mathbb{Z}$, take $\epsilon_n \in (0, 1)$ such that $(n - \epsilon_n, n + \epsilon_n) \subset V_k$ for all $k \in J$. Then consider V as in the previous case and the reasoning is straightforward.

The quotient space $\mathbb{R}/\sim$ looks like the set formed by an infinite number of disjoint circles in $\mathbb{R}^2$ all passing through a point, this point being the class of integers in the quotient space. Compare with the Hawaiian earring of Exercise 9.4.13. However, the Hawaiian earring is first countable because it is a metric space (subspace of $\mathbb{R}^2$). □

Exercise 10.14 Define on $\mathbb{S}^n$ the equivalence relation $x \sim y$ if $x = y$ or $x = -y$. The quotient space $\mathbb{S}^n/\sim$ is called the *projective space* of dimension n and denoted by $\mathbb{RP}^n$. Prove that the projective space is connected and compact. Prove that the projection is open and deduce that the space is Hausdorff.

Solution

1. The space is connected and compact because it is the quotient space of $\mathbb{S}^n$, which it is connected and compact space.
2. We prove that the projection p is open. Consider A the antipodal map of $\mathbb{S}^n$. If O is a subset of $\mathbb{S}^n$, the saturation of O is $R[O] = O \cup A(O)$. Because A is a homeomorphism, if O is open in $\mathbb{S}^n$ then $A(O)$ is open, hence $R[O]$ is an open set in $\mathbb{S}^n$. This proves that p is an open map by Exercise 10.1.
3. Let $[x]$, $[y]$ two points of $\mathbb{RP}^n$, which means that $x \neq y$ and $x \neq -y$. Then $|x - y| > 0$ and $|x + y| > 0$. Let $\epsilon = \min\{|x - y|, |x + y|\}$ and
$$V_x = B_{\epsilon/2}(x) \cap \mathbb{S}^n, \quad V_y = B_{\epsilon/2}(y) \cap \mathbb{S}^n,$$
which are open neighborhoods in $\mathbb{S}^n$ of x and y respectively. Consider $W_{[x]} = p(V_x)$ and $W_{[y]} = p(V_y)$, that are neighborhoods of $[x]$ and $[y]$ respectively because p is open. See Fig. 10.12. We check that $W_{[x]} \cap W_{[y]} = \emptyset$. Otherwise, there would be $[z] \in \mathbb{RP}^n$ in both sets. Then $z \in V_x$ or $-z \in V_x$ and $z \in V_y$ or $-z \in V_y$. We study case-by-case.

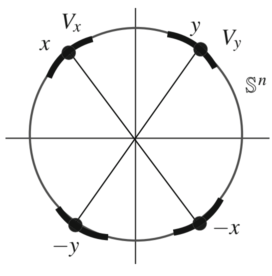


Fig. 10.12 The projective space is Hausdorff

(a) If $z \in V_x \cap V_y$ then $|x-y| \leq |x-z| + |z-y| < \epsilon \leq |x-y|$, that is impossible.
(b) If $-z \in V_x \cap V_y$, the same argument gives a contradiction.
(c) If $z \in V_x$ and $-z \in V_y$, then

$$|x+y| \leq |x-z| + |z+y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

a contradiction again. □

Exercise 10.15 Prove that $\mathbb{RP}^1$ is homeomorphic to $\mathbb{S}^1$.

Solution Consider the upper halfcircle of $\mathbb{S}^1$ given by $\mathbb{S}^1_+ = \{(x, y) \in \mathbb{S}^1 : y \geq 0\}$. We restrict $\sim$ on $\mathbb{S}^1_+$ and denote by $\sim^+$. The only case that two antipodal points belong to the halfcircle is that these points are $(-1, 0)$ and $(1, 0)$. Since $\mathbb{S}^1_+$ is compact (closed and bounded), the quotient space $\mathbb{S}^1_+/{\sim^+}$ is compact. Let $i : \mathbb{S}^1_+ \hookrightarrow \mathbb{S}^1$ be inclusion. If $x \sim^+ y$ then $i(x) \sim i(y)$, and consequently, there is a map

$$\widetilde{i} : \frac{\mathbb{S}^1_+}{\sim^+} \to \frac{\mathbb{S}^1}{\sim}.$$

Notice that i is not an identification because the inclusion i is not surjective. However, we prove directly that $\widetilde{i}$ is a homeomorphism. In order to avoid ambiguity with the notation of the equivalence classes, if $x \in \mathbb{S}^1_+ \subset \mathbb{S}^1$, we use $[[x]]$ and $[x]$ to represent the class of x with respect to $\sim^+$ and $\sim$, respectively.

1. The map $\widetilde{i}$ is injective. If $\widetilde{i}([[x]]) = \widetilde{i}([[y]])$ then $[x] = [y]$, so $x \sim y$. Since $\sim^+$ is the restriction of $\sim$, then $x \sim^+ y$, so $[[x]] = [[y]]$.
2. The map $\widetilde{i}$ is surjective. If $[x] \in \mathbb{S}^1/{\sim}$ then x or $-x$ belongs to $\mathbb{S}^1_+$. Thus a representative of $[x]$ lies in $\mathbb{S}^1_+$. If, for instance, is x, then $\widetilde{i}([[x]]) = [i(x)] = [x]$. For the case $-x$, we have $\widetilde{i}([[-x]]) = [i(-x)] = [-x] = [x]$.
3. The map $\widetilde{i}$ is continuous because the inclusion i is.
4. The map $\widetilde{i}$ is closed because $\widetilde{i}$ is continuous, $\mathbb{S}^1_+/{\sim^+}$ is compact and $\mathbb{S}^1$ is Hausdorff.

Once established that $\mathbb{RP}^1$ is homeomorphic to $\mathbb{S}^1_+/{\sim^+}$, the final steps are easy. We know that $\mathbb{S}^1_+ \cong [-1, 1]$ because $\mathbb{S}^1_+$ is the graph of the function $f(x) = \sqrt{1-x^2}$ defined on $[-1, 1]$. Let $\phi : \mathbb{S}^1_+ \to [-1, 1]$ be the homeomorphism $\phi(x, y) = x$. Let us use Example 10.10 to carry $\sim^+$ to the corresponding equivalence relation on $[-1, 1]$ via ϕ. Then it is clear that this relation on $[-1, 1]$ identifies the endpoints -1 and 1, so the quotient space is $\mathbb{S}^1$. □

We present various exercises involving the final topology.

Exercise 10.16 Let $f : (X, \tau) \to Y$ be a surjective map. Determine the final topology on Y when τ is the discrete topology, the trivial topology and the cofinite topology.

Solution

1. Case $\tau = \tau_D$. Let B be an arbitrary subset of Y. Then $f^{-1}(B)$ is open in X because any subset of X is. This proves that B is open in the final topology. Since B is arbitrary, the final topology $\tau_D(f)$ is discrete.
2. Case $\tau = \tau_T$. If $B \in \tau_T(f)$, $B \neq \emptyset$, then $f^{-1}(B)$ is open in X, so $f^{-1}(B)$ is $\emptyset$ or X. Since f is surjective, then $f^{-1}(B) = X$. Thus $f(X) = f(f^{-1}(B)) = B$ but $f(X) = Y$. This proves that $\tau_T(f)$ is the trivial topology on Y.
3. We show an example where the final topology on Y from the cofinite topology is not cofinite. Let $X = \mathbb{R}$, $Y = \{0, 1\}$ and the function

$$f\colon (\mathbb{R}, \tau_{CF}) \to (\{0, 1\}, \tau_{CF}(f)), \quad f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0. \end{cases}$$

If $B \subset Y$ is closed in $\tau_{CF}(f)$, then $f^{-1}(B)$ is finite or $\mathbb{R}$. But the only case that $f^{-1}(B)$ is finite is when $f^{-1}(B) = \emptyset$. This leads to $B = \emptyset$. Thus $\tau_{CF}(f) = \{\emptyset, Y\}$, proving that the final topology is the trivial topology. □

Exercise 10.17 Let $E\colon (\mathbb{R}, \tau_r) \to \mathbb{Z}$ be the integer part function and τ_r is the right order topology. Find the final topology on $\mathbb{Z}$.

Solution The elements of τ_r are formed by the trivial sets and the intervals of type $[x, \infty)$ and (x, ∞). For $n \in \mathbb{Z}$, it is clear that $E^{-1}(\{n\}) = [n, n+1)$. Thus if $B \subset \mathbb{Z}$,

$$E^{-1}(B) = \bigcup_{n \in B} [n, n+1).$$

If $E^{-1}(B)$ is a nontrivial open set in τ_r, then it must be an interval $[x, \infty)$, so the only case to consider is that B is formed by a set of consecutive integers converging to ∞. Thus B is $B_m = \{m, m+1, \ldots\}$ for some $m \in \mathbb{Z}$ so $E^{-1}(B) = [m, \infty)$. Definitively

$$\tau_r(f) = \{\emptyset, \mathbb{Z}\} \cup \{B_m : m \in \mathbb{Z}\}.$$

It is clear that this topology is the relative topology on $\mathbb{Z}$ of the right order topology, that is $\tau_r(f) = (\tau_r)_{|\mathbb{Z}}$. □

Exercise 10.18 Let $f\colon \mathbb{R} \to [-1, 1]$ be the function

$$f(x) = \begin{cases} \sin(\frac{1}{x}), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

Prove that the only neighborhood of $0 \in [-1, 1]$ in the final topology on $[-1, 1]$ is the interval $[-1, 1]$ itself.

Solution Let U be a neighborhood of $x = 0$ in the final topology. Since $f(0) = 0$, $f^{-1}(U)$ is a neighborhood of 0 in $\mathbb{R}$. Using that all balls form a basis of neighborhoods in $\mathbb{R}$, there is $r > 0$ such that $0 \in (-r, r) \subset f^{-1}(U)$. By definition of f, we have $f((-r, r)) = [-1, 1]$. Thus

$$[-1, 1] = f((-r, r)) \subset f(f^{-1}(U)) = U$$

and we conclude that $U = [-1, 1]$. $\square$

We show a final topology determined by the same map but from different topologies.

Exercise 10.19 Consider the function $f : \mathbb{R} \to [0, \infty)$ given by $f(x) = x^2$. Prove that the final topologies on $[0, \infty)$ given by $\tau_u(f)$ and $\tau_S(f)$ coincide.

Solution For both topologies, we prove that the final topology is the usual topology on $[0, \infty)$. It is clear that f is continuous if we consider on $[0, \infty)$ the usual topology. Since the Sorgenfrey topology is finer than the Euclidean one, the function $f : (\mathbb{R}, \tau_S) \to ([0, \infty), \tau_u)$ is also continuous. In view that the final topology is the finest topology that makes continuous f, we have $\tau_u \subset \tau_u(f)$ and $\tau_u \subset \tau_S(f)$. We show the other inclusions.

1. We prove that $\tau_u(f) \subset \tau_u$. Let $O \subset \tau_u(f)$ and $x \in O$. It suffices to verify that x is an interior point for the usual topology. We distinguish if $x = 0$ or $x > 0$. In case that $x = 0$, let $\epsilon > 0$ such that $0 \in [0, \epsilon) \subset f^{-1}(O)$. Then $f(0) = 0 \in f([0, \epsilon)) = [0, \epsilon^2) \subset O$. Here $[0, \epsilon^2)$ is a neighborhood of $x = 0$ in the usual topology on $[0, \infty)$, so we conclude that x is an interior point of O.

   Suppose $x > 0$. Since $f^{-1}(O) \in \tau_u$, there is an $\epsilon > 0$, which can be chosen such that $\epsilon < \sqrt{x}$, such that

   $$\sqrt{x} \in (\sqrt{x} - \epsilon, \sqrt{x} + \epsilon) \subset f^{-1}(O).$$

   In this chain of inclusions, take the image under f, obtaining

   $$x \in ((\sqrt{x} - \epsilon)^2, (\sqrt{x} + \epsilon)^2 \subset f(f^{-1}(O)) = O.$$

   This proves that x is an interior point of O with the usual topology.
2. Following a similar argument, if $x = 0$ there is $\epsilon > 0$ such that $0 \in [0, \epsilon) \subset f^{-1}(O)$. Then $0 \in [0, \epsilon^2) \subset f(f^{-1}(O)) = O$, proving that $x = 0$ is an interior point of O with the usual topology. Assume $x > 0$. Then there is $\epsilon > 0$ such that

   $$\sqrt{x} \in [\sqrt{x}, \sqrt{x} + \epsilon) \subset f^{-1}(O).$$

   Then

   $$x \in f([\sqrt{x}, \sqrt{x} + \epsilon)) = [x, (\sqrt{x} + \epsilon)^2) \subset O.$$

In the same fashion and taking $-\sqrt{x}$, there exists $\delta > 0$ such that

$$-\sqrt{x} \in [-\sqrt{x}, -\sqrt{x}+\delta) \subset f^{-1}(O).$$

We may select a δ sufficiently small so $-\sqrt{x}+\delta < 0$. Then

$$x \in f([-\sqrt{x}, -\sqrt{x}+\delta)) = ((-\sqrt{x}+\delta)^2, x] \subset O.$$

Combining this expression with the previous one, we deduce that

$$x \in ((-\sqrt{x}+\delta)^2, x] \cup [x, (\sqrt{x}+\epsilon)^2) \subset O,$$

that is,

$$x \in ((-\sqrt{x}+\delta)^2, \sqrt{x}+\epsilon)^2) \subset O.$$

Then x is an interior point of O with the usual topology. □

Exercise 10.20 Describe the final topology $\tau_r(f)$ on $[0,\infty)$ where $f : \mathbb{R} \to [0,\infty)$ is the function $f(x) = x^2$.

Solution A basis of neighborhoods of x for τ_r is $\{[x,\infty)\}$. Let $O \in \tau_r(f)$ and $x \in O$ be given.

1. Case $x > 0$. Then $-\sqrt{x} \in f^{-1}(O)$. Since $f^{-1}(O) \in \tau_r$, we have $[-\sqrt{x},\infty) \subset f^{-1}(O)$. Taking images under f, we have $f([-\sqrt{x},\infty)) = [0,\infty) \subset O$. This yields $O = [0,\infty)$.
2. Case $x = 0 \in O$. Because $f(0) = 0$, then $0 \in [0,\infty) \subset f^{-1}(O)$, hence $f([0,\infty)) = [0,\infty) \subset O$. This proves that $O = [0,\infty)$.

Thus the only nonempty open set of $\tau_r(f)$ is $[0,\infty)$, proving that the final topology is the trivial topology. □

In the following exercises, we show examples of homeomorphisms between a quotient space and a known space. The first exercise is intuitive because we construct a "football ball" by sewing two discs along the borders.

Exercise 10.21 Let $X = \mathbb{D}^2 \times \{0\}$, $Y = \mathbb{D}^2 \times \{1\}$ and $Z = X \cup Y$. Define the equivalence relation on Z by

$$(x,y,z) \sim (x',y',z') \text{ if } \begin{cases} (x,y,z) = (x',y',z'), \\ x = x', y = y', z, z,' \in \{0,1\}, \quad \text{if } (x,y), (x',y') \in \mathbb{S}^1. \end{cases}$$

Prove that $Z/\sim$ is homeomorphic to the unit sphere $\mathbb{S}^2$.

Solution The idea is geometric; however, to formalize the computations, we map each one of the discs X and Y to each one of the two hemispheres of $\mathbb{S}^2$. Next we

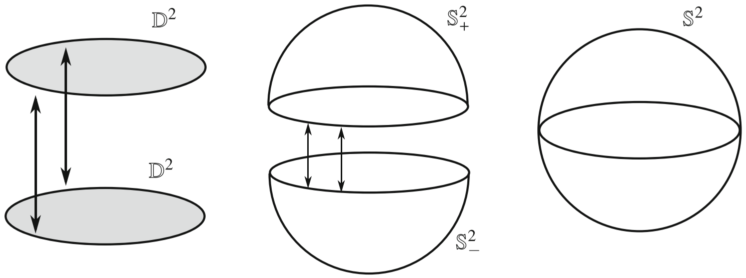


Fig. 10.13 Two copies of the disc along its boundary gets the sphere $\mathbb{S}^2$

will glue both hemispheres along the common equator that coincides with each one of the two borders of X and Y as indicated Fig. 10.13. Define

$$f\colon Z \to \mathbb{S}^2, \quad f(x, y, z) = \begin{cases} (x, y, -\sqrt{x^2+y^2}), & z = 0 \\ (x, y, \sqrt{x^2+y^2}), & z = 1. \end{cases}$$

Then f maps X to the lower hemisphere $\mathbb{S}^2_-$ and Y to $\mathbb{S}^2_+$. Moreover, if $x^2+y^2 = 1$, we have $f(x, y, z) = (x, y, 0)$ is a point of the equator of $\mathbb{S}^2$. The relation $\sim_f$ coincides with $\sim$ because the restriction of f in X and in Y is one-to-one, so the only possibility that $f(x, y, z) = f(x', y', z')$ for different points (x, y, z) and (x', y', z') is that $x^2 + y^2 = x'^2 + y'^2 = 1$, with $x = x'$, $y = y'$.

Finally, X and Y are closed in Z because they are closed in $\mathbb{R}^3$ (product of closed sets). By the Pasting Lemma, f is continuous. The domain of f is compact (union of two compact sets) and the codomain is Hausdorff. This yields the desired homeomorphism between $\mathbb{Z}/\sim$ and $\mathbb{S}^2$. □

Exercise 10.22 Define an equivalence relation on $\mathbb{S}^2$ by

$$(x, y, z) \sim (x', y', z') \text{ if and only if } z = z'.$$

Prove that $\mathbb{S}^2/\sim$ is homeomorphic to the closed interval $[-1, 1]$.

Solution The equivalence relation identifies all points on $\mathbb{S}^2$ with the same height, the z-coordinate. Thus each horizontal circle of $\mathbb{S}^2$ is a point in the quotient space and equivalence classes can be identified with the height of its points, which is a number of the interval $[-1, 1]$. The gluing map will take all points of each horizontal circle of $\mathbb{S}^2$ in the value of its height. Define

$$f\colon \mathbb{S}^2 \to [-1, 1], \quad f(x, y, z) = z.$$

It is clear that $\sim_f$ coincides with $\sim$. On the other hand, the function f is continuous and surjective. Further, f is closed because $\mathbb{S}^2$ is compact and the codomain $[-1, 1]$

is Hausdorff. Thus f is an identification and the corresponding map $\widetilde{f}$ between $\mathbb{S}^2/\sim$ and $[-1, 1]$ is a homeomorphism.

Also it can be proved that f has a continuous right-inverse: the map $g\colon [-1, 1] \to \mathbb{S}^2$ given by $g(t) = (0, \sqrt{1-t^2}, t)$ satisfies $(f \circ g)(t) = t$.

This exercise can be easily generalized to arbitrary dimensions. Consider on $\mathbb{S}^n$ the relation $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if $x_{n+1} = y_{n+1}$. Then $\mathbb{S}^n/\sim$ is homeomorphic to $[-1, 1]$ again. □

Exercise 10.23 Define an equivalence relation on $\mathbb{R}^n$ by

$$(x_1, \dots, x_n) \sim (y_1, \dots, y_n) \text{ if and only if } x_n = y_n.$$

Find the quotient space.

Solution The equivalence class of a point $(x_1, \dots, x_n) \in \mathbb{R}^n$ is the horizontal hyperplane containing the point $(x_1, \dots, x_n)$:

$$[(x_1, \dots, x_n)] = \{(y_1, \dots, y_n) \in \mathbb{R}^n : y_n = x_n\} = \mathbb{R}^{n-1} \times \{x_n\}.$$

Thus there is a one-by-one application that maps each equivalence class into the height of the hyperplane. See Fig. 10.14, left, for the case $n = 2$. Since the set of values of all these heights is $\mathbb{R}$, we prove that the quotient space is homeomorphic to the Euclidean real line. Define

$$f\colon \mathbb{R}^n \to \mathbb{R}, \quad f(x_1, \dots, x_n) = x_n.$$

This map is the nth projection of $\mathbb{R}^n$. Thus f is surjective, continuous and open; in particular, f is an identification. Clearly $\sim_f$ coincides with $\sim$ so $\mathbb{R}^n/\sim$ is homeomorphic to $\mathbb{R}$. Notice that in the preceding Exercise 10.22, the relation is the restriction of this relation $\sim$ to $\mathbb{S}^{n-1}$. □

Exercise 10.24 Define an equivalence relation on $\mathbb{R}^3$ by

$$(x, y, z) \sim (x', y, z') \text{ if and only if } (x, y) = (x', y').$$

Prove that $\mathbb{R}^3/\sim$ is homeomorphic to $\mathbb{R}^2$.

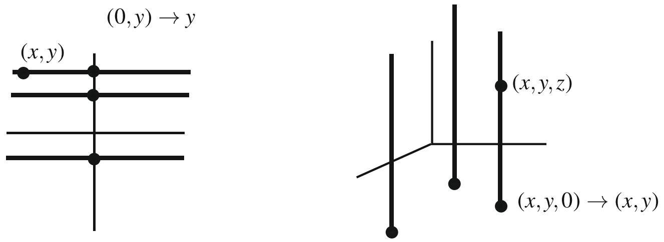


Fig. 10.14 Equivalence classes in Exercise 10.23 (left) and Exercise 10.24 (right)

Solution The equivalence class of a point is the vertical line through this point,

$$[(x, y, z)] = \{(x', y', z') \in \mathbb{R}^3 : x' = x, y' = y\} = \{(x, y)\} \times \mathbb{R}.$$

All these points of $\mathbb{R}^3$ are one element in the quotient space. Thus we glue all of them to the point (x, y) of the Euclidean plane $\mathbb{R}^2$, mapping each vertical line of equation $x' = x$ and $y' = y$ onto $(x, y, 0)$ of the plane of equation $z = 0$ (Fig. 10.14, right). Define

$$f : \mathbb{R}^3 \to \mathbb{R}^2, \quad f(x, y, z) = (x, y).$$

By definition of f, it is clear that the relation $\sim_f$ coincides with $\sim$.

Also it is immediate that f is surjective and continuous. Finally, f is open because f is the projection $\mathbb{R}^3 = \mathbb{R}^2 \times \mathbb{R} \to \mathbb{R}$. Thus f is an identification and $\widetilde{f} : \mathbb{R}^3/\sim \to \mathbb{R}^2$ is a homeomorphism. □

The last two exercises are special cases at more general settings. Let X and Y be two spaces and let $p : X \times Y \to X$ the first projection. Then p is an identification because p is open. Thus $X \times Y/\sim_p$ is homeomorphic to X. Here the relation $\sim_p$ writes as $(x, y) \sim_p (x', y')$ if $x = x'$.

Exercise 10.25 Define an equivalence relation on $\mathbb{R}$ by

$$x \sim y \text{ if } \begin{cases} x = y, & \text{or} \\ x, y \in [2n, 2n+1], & \text{for some } n \in \mathbb{Z}. \end{cases}$$

Prove that $\mathbb{R}/\sim$ is homeomorphic to $\mathbb{R}$.

Solution The equivalence class of a point x is $\{x\}$ if x does not belong to any interval $[2n, 2n+1]$. Otherwise, all points of an interval $[2n, 2n+1]$ are one element in the quotient space. Thus we can think that if we go walking through the real line, when we arrive to some interval $[2n, 2n+1]$, we shrink this interval to a point and we continue walking until the next one $[2n+2, 2n+3]$ shrinking this interval again. It is clear that the quotient space is homeomorphic to $\mathbb{R}$ because the only thing that we do is walk along $\mathbb{R}$ with different speed, depending on what interval we are situated. Define

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \begin{cases} n, & \text{if } x \in [2n, 2n+1] \text{ for some } n \in \mathbb{Z} \\ x - n - 1, & \text{if } x \in [2n+1, 2n+2] \text{ for some } n \in \mathbb{Z}. \end{cases}$$

See the graph of f in Fig. 10.15. Notice that the function f is well defined at the endpoints of $[n, n+1]$, $n \in \mathbb{Z}$. Also it is clear that f is surjective.

1. We see that $\sim_f$ coincides with $\sim$. Suppose $f(x) = f(y)$. We discuss according to x. If $x \in [2n, 2n+1]$ for some $n \in \mathbb{Z}$, then $f(x) = n$. Since $f(y) = f(x) = n$, necessarily $y \in [2n, 2n+1]$, proving that $x \sim y$. If $x \in [2n+1, 2n+1]$ for some $n \in \mathbb{Z}$, then $f(x) = x - n - 1$. Since $f(y) = f(x) = x - n - 1$, then necessarily

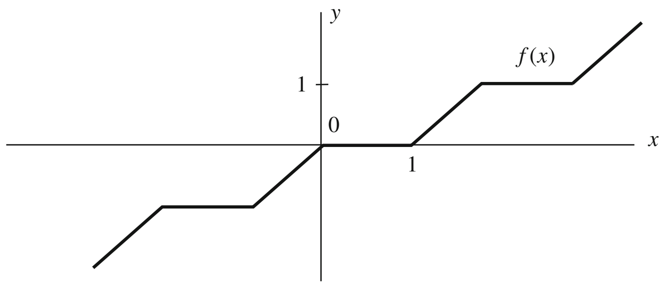


Fig. 10.15 The function f of Exercise 10.25

y belongs to the same interval, so $f(y) = y - n - 1$, hence $x = y$. Thus $x \sim y$ again. In the other direction, it is clear that if $x \sim y$, then $f(x) = f(y)$.

2. Notice that f is not open because $(2, 3)$ is open but not $f((2, 3)) = \{2\}$. We prove that f is closed. Let F be a closed set in $\mathbb{R}$ and $x \in \overline{f(F)}$. Then there is $(x_n) \subset F$ such that $(f(x_n)) \to x$. If x is not an integer, then $f(x_n) \notin \mathbb{Z}$ from a term of the sequence and each $f(x_n)$ belongs to a certain interval $(2m+1, 2m+2)$ with $m \in \mathbb{Z}$. Then $(f(x_n)) = (x_n - m - 1) \to x$, hence $(x_n) \to x + m + 1$. Because F is closed, then $x + m + 1 \in F$. Since $x + m + 1 \in [2m + 1, 2m + 2]$,

$$f(x + m + 1) = (x + m + 1) - m - 1 = x \in f(F).$$

Assume $x \in \mathbb{Z}$. From a certain term of the sequence (x_n), either the terms belong to $[2x, 2x + 1]$ or to $[2x - 1, 2x]$. If $x_n \in [2x, 2x + 1]$ then $f(x_n) = x$ and if $x_n \in [2x - 1, 2x]$ then $f(x_n) = x_n - x - 1$. For the terms of the first type, $x \in f(F)$. For the terms of the second type, $(x_n - x - 1) \to x$, hence $(x_n) \to 2x + 1$ and arguing as above, we conclude that $x \in f(F)$. □

Exercise 10.26 Define the equivalence relation $\sim$ on $X = [0, \infty)$ determined by the partition

$$\left\{\left\{x, \tfrac{1}{x}\right\} : x \in (0, \infty)\right\} \cup \{0\}.$$

Prove that the quotient space is compact and Hausdorff. Prove that the projection $p \colon X \to X/\sim$ is not closed. Prove that $X/\sim$ is homeomorphic to $[0, 1]$.

Solution By means of the function $f(t) = 1/t$, the saturation of any $A \subset X$ is

$$R[A] = \begin{cases} A \cup f(A), & \text{if } 0 \notin A \\ A \cup f(A \setminus \{0\}), & \text{if } 0 \in A. \end{cases}$$

1. Since any equivalence class has a representative in the interval $[0, 1]$, then $X/\sim = p([0, 1])$ and thus the quotient space $X/\sim$ is compact.
2. To prove that $X/\sim$ is Hausdorff, we first see that p is open. In $[0, \infty)$, consider the basis for open sets given by

$$\beta = \{(a, b) : 0 \leq a < b\} \cup \{[0, b) : b > 0\}.$$

Then it suffices to see that the saturation of any element of β is open in X (Exercise 10.1). There are three types of intervals for which we compute the saturation.

(a) If $a > 0$ then $R[(a, b)] = (a, b) \cup \left(\frac{1}{b}, \frac{1}{a}\right)$.
(b) If $a = 0$ then $R[(0, b)] = (0, b) \cup (1/b, \infty)$.
(c) Finally we have $R[[0, b)] = [0, b) \cup (\frac{1}{b}, \infty)$.

In any of the three cases, the saturation is an open set. Now we see that $X/\sim$ is Hausdorff.

(a) Let $[x] \neq [y]$ be two equivalence classes. As a first case, assume that $x, y \neq 0$. Notice that

$$f : (0, \infty) \to (0, \infty), \quad f(t) = 1/t,$$

is a homeomorphism whose inverse is f. Since $\mathbb{R}$ is Hausdorff and x is distinct from y and $x \neq 1/y$, there is $r > 0$ such that $1/y \notin (x - r, x + r)$ and $1/x \notin (y - r, y + r)$ and $(x - r, x + r) \cap (y - r, y + r) = \emptyset$. Let $U_r = (x - r, x + r)$ and $V_r = (y - r, y + r)$. Since f is bijective, $f(U_r)$ is disjoint from $f(V_r)$. Furthermore, it is possible to choose r sufficiently small so

$$(U_r \cup f(U_r)) \cap (V_r \cup f(V_r)) = \emptyset.$$

Thus $U_r \cup f(U_r)$ and $V_r \cup f(V_r)$ are two disjoint saturated neighborhoods of x and y respectively.
(b) Now consider the case $x = 0$ and $[x] \neq [y]$. For y, let $U = (y/2, 3y/2)$ which is a neighborhood of y. Let r be positive but sufficiently small so $[0, r) \cup (1/r, \infty)$ intersect neither U nor $f(U) = (\frac{2}{3y}, \frac{2}{y})$. This is possible because $\lim_{r\to 0} r = 0$ and $\lim_{r\to 0} 1/r = \infty$. Definitively, $[0, r) \cup (1/r, \infty)$ and $U \cup f(U)$ are two disjoint saturated neighborhoods of 0 and y.

3. The projection p is not closed. For instance, $F = [1, \infty)$ is closed but its saturation is $R[F] = (0, \infty)$, which is not closed in X.
4. Define

$$h : X \to [0, 1], \quad h(x) = \begin{cases} x, & x \in [0, 1] \\ \frac{1}{x}, & x \in [1, \infty). \end{cases}$$

By definition of h, the relations $\sim_h$ and $\sim$ coincide. The map h is an identification because f is continuous (Pasting Lemma) and $g\colon [0,1] \to X$ defined by $g(x) = x$, is an inverse from the right. This proves that $X/{\sim}$ is homeomorphic to $[0,1]$. □

Exercise 10.27 Consider on $\mathbb{R}^2$ the equivalence relation $x \sim y$ if $x = y$ or $|x|, |y| \geq 1$. Prove that $\mathbb{R}^2/{\sim}$ is homeomorphic to the unit sphere $\mathbb{S}^2$.

Solution The equivalence class of x is

$$[x] = \begin{cases} \{x\}, & \text{if } |x| < 1 \\ \mathbb{R}^2 \setminus B_1(0), & \text{if } |x| \geq 1. \end{cases}$$

Intuitively, points of the ball $B_1(0)$ remain fixed and points outside the ball are identified as a point, so we can close the outside thus forming, together with $B_1(0)$, a round sphere. For the gluing map, take any ray from 0 and keep the points that lie contained in the ball $B_1(0)$. The points of the ray outside the ball will be mapped onto the intersection point between the ray and $\mathbb{S}^1$. The ray determined by $x \neq 0$ is $\{\lambda x : \lambda \geq 0\}$ and the intersection point with $\mathbb{S}^1$ is $x/|x|$. Define

$$f\colon \mathbb{R}^2 \to \mathbb{D}^2, \quad f(x) = \begin{cases} x, & x \in \mathbb{D}^2 \\ \frac{x}{|x|}, & |x| \geq 1. \end{cases}$$

We prove that f is an identification. It is clear that f is surjective because the restriction of f to $\mathbb{D}^2$ is the identity. The function is continuous by the Pasting Lemma. Finally, the inclusion $i\colon \mathbb{D}^2 \to \mathbb{R}^2$ is continuous and $f \circ i$ is the identity of $\mathbb{D}^2$. This proves that f has a continuous right-inverse.

Consider the relation $\approx$ on $\mathbb{D}^2$ which identifies all points of $\mathbb{S}^1$. It is clear that $x \sim y$ if and only if $f(x) \approx f(y)$ because the points with $|x| \geq 1$ go to the unit circle $\mathbb{S}^1$, which are related by $\approx$. Thus $\widetilde{f}\colon \mathbb{R}^2/{\sim} \to \mathbb{D}^2/{\approx}$ is a homeomorphism. Finally, we have seen in Example 10.8 that $\mathbb{D}^2/\mathbb{S}^1$ is homeomorphic to the sphere $\mathbb{S}^2$.

Notice that this exercise extends easily to arbitrary dimension, proving that $\mathbb{R}^n/{\sim}$ is homeomorphic to $\mathbb{S}^n$. □

Exercise 10.28 On $X = \mathbb{R} \times \{0,1\}$ define $(x, y) \sim (x', y')$ if the points coincide or $x, x' \leq -1$ or $x, x' \geq 1$. Prove that $X/{\sim}$ is homeomorphic to $\mathbb{S}^1$ and investigate if the projection is open and closed.

Solution The relation $\sim$ identifies in one point the set $D^- = (-\infty, -1] \times \{0,1\}$ and in other point the set $D^+ = [1, \infty) \times \{0,1\}$. Thus it is easy to compute the saturation of any subset A of X,

$$R[A] = \begin{cases} A, & \text{if } A \cap D^- = A \cap D^+ = \emptyset \\ A \cup D^-, & \text{if } A \cap D^- \neq \emptyset, A \cap D^+ = \emptyset \\ A \cup D^+, & \text{if } A \cap D^- = \emptyset, A \cap D^+ \neq \emptyset \\ A \cup D^- \cup D^+, & \text{if } A \cap D^- \neq \emptyset, A \cap D^+ \neq \emptyset \end{cases}$$

The sets D^- and D^+ are closed in $\mathbb{R}^2$ since they are the product of two closed sets, thus closed in X. Hence $R[A]$ is closed if A is closed because $R[A]$ is union of a finite number of closed sets. This proves that the projection is closed by Exercise 10.1. However, the projection is not open because the saturation of $A = (2, 3) \times \{1\}$, which is open in X, is $R[A] = A \cup D^+ = D^+$ which is not open in X.

It is not difficult to see which is the gluing map. The points of D^- are all identified in the quotient space; in particular, the point $(-1, 0)$ with $(-1, 1)$ and similarly, $(1, 0)$ with $(1, 1)$. Thus we have the segments $[-1, 1] \times \{0\}$ and $[-1, 1]\times\{1\}$ where the endpoints in the left hand side are identified and analogously, the endpoints in the right hand side are identified. Instead of defining the gluing map with domain X, observe that $p(X) = p([-1, 1] \times \{0, 1\})$, so we prove that it is enough to consider the quotient space $[-1, 1] \times \{0, 1\}/\sim$. The problem that we now find is that we do not know if the quotient topology on $[-1, 1] \times \{0, 1\}/\sim$ coincides with the relative topology from $X/\sim$. To accomplish this, use Proposition 10.2.3. Since p is closed (and because $[-1, 1] \times \{0, 1\}$ is closed), we have assured the identity of both topologies: following the notation of Prop. 10.2.3, $A = [-1, 1] \times \{0, 1\}$ and $B = p(A)$.

Once we have done this, the final steps are immediate. Let $A = [-1, 1] \times \{0, 1\}$. We map the segment $[-1, 1] \times \{0\}$ to the lower halfcircle $\mathbb{S}^1_-$ and $[-1, 1] \times \{1\}$ to the upper halfcircle $\mathbb{S}^1_-$. Define

$$f\colon A \to \mathbb{S}^1, \quad f(x, y) = \begin{cases} (x, -\sqrt{1-x^2}), & y = 0 \\ (x, \sqrt{1-x^2}), & y = 1. \end{cases}$$

Then clearly $\sim_f$ and $\sim$ are equal, f is continuous by the Pasting Lemma, A is compact because it is the product of two compact sets and $\mathbb{S}^1$ is Hausdorff. This last part has appeared in (1) of Exercise 10.9 because $A/\sim$ is the suspension of $[-1, 1]$, and it was proved that the suspension is $\mathbb{S}^1$. It is also possible to define the gluing map from X. So $g\colon X \to \mathbb{S}^1$ is

$$g(x, y) = \begin{cases} (-1, 0), & (x, y) \in D^- \\ (x, -\sqrt{1-x^2}), & y = 0 \\ (x, \sqrt{1-x^2}), & y = 1 \\ (1, 0), & (x, y) \in D^+. \end{cases}$$

□

We modify a bit the above equivalence relation. We use the same notation.

Exercise 10.29 On $X = \mathbb{R} \times \{0, 1\}$ define $(x, y) \sim (x', y')$ if they coincide; if $x, x' < -1$; or if $x, x' > 1$. Prove that $X/\sim$ is not homeomorphic to $\mathbb{S}^1$ and investigate if the projection is open or closed.

Solution The saturation of a set is similar to that of the foregoing exercise, so p is open because the saturation is a union of open sets. However, p is not closed because $[2, 3] \times \{0, 1\}$ is closed and its saturation is $(1, \infty) \times \{0, 1\}$ which is not closed.

We see that the quotient space $X/\sim$ is not Hausdorff, so it cannot be homeomorphic to $\mathbb{S}^1$. Consider two particular points in the quotient space, namely, the equivalence classes $a = [(1, 0)]$ and $b = [(1, 1)]$. Two open sets O_0 and O_1 containing a and b, respectively, must intersect because $R[O_0] \cap R[O_1] \neq \emptyset$. Since $R[O_i]$ are open sets, there are $r_0, r_1 > 0$ such that

$$(1, 0) \in (1 - r_0, 1 + r_0) \times \{0\} \subset R[O_0], \quad (1, 1) \in (1 - r_1, 1 + r_1) \times \{1\} \subset R[O_1].$$

Thus

$$R[(1 - r_0, 1 + r_0) \times \{0\}] \subset R[O_0], \quad R[(1 - r_1, 1 + r_1) \times \{1\}] \subset R[O_1].$$

However, the intervals $(1-r_0, 1+r_0)$ and $(1-r_1, 1+r_1)$ intersect the interval $(1, \infty)$, so their saturations intersect D^+ and this implies that $D^+ \subset R[O_0] \cap R[O_1]$. $\square$

Exercise 10.30 Consider the equator of $\mathbb{S}^2$ given by $A = \{(x, y, z) \in \mathbb{S}^2 : z = 0\}$. Prove that the quotient space $\mathbb{S}^2/A$ is homeomorphic to two tangent spheres.

Solution As in Exercise 10.6, the gluing map shrinks the equator of $\mathbb{S}^2$ at one point, forming two tangent spheres. Consider two tangent spheres of identical radius 1/2, and centers $p = (0, 0, \frac{1}{2})$ and $q = (0, 0, -\frac{1}{2})$, respectively. Let $X_1 = \mathbb{S}^2_{1/2}(p)$ and $X_2 = \mathbb{S}^2_{1/2}(q)$. Then X_1 and X_2 are two spheres tangent at the origin of $\mathbb{R}^3$. Let $X = X_1 \cup X_2$ and we prove that $\mathbb{S}^2/A$ is homeomorphic to X. All the sets are compact (closed and bounded) and Hausdorff, so the only job is to find the gluing map $f : \mathbb{S}^2 \to X$ whose equivalence relation $\sim_f$ coincides with the relation $\sim$ that identifies all points of the equator A. We will map any horizontal circle of $\mathbb{S}^2$ to the corresponding circle of X at the same height. In particular, the equator of $\mathbb{S}^2$ will map to the origin of $\mathbb{R}^3$. See Fig. 10.16.

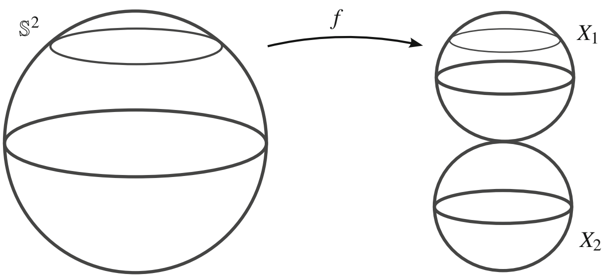


Fig. 10.16 The homeomorphism between $\mathbb{S}^2$ and the two tangent spheres $X_1 \cup X_2$ of Exercise 10.30

The map writes as $f(x, y, z) = (\lambda x, \lambda y, z)$. Here $(x, y, z) \in \mathbb{S}^2$ and we impose the requirement $f(x, y, z) \in X$. For the upper hemisphere of $\mathbb{S}^2$,

$$\lambda^2(x^2 + y^2) + (z - \tfrac{1}{2})^2 = \tfrac{1}{4},$$

where $x^2 + y^2 + z^2 = 1$. After some calculations,

$$\lambda = \frac{\sqrt{z - z^2}}{\sqrt{1 - z^2}} = \frac{\sqrt{z}}{\sqrt{1 + z}}.$$

Similarly, for the points of the lower hemisphere, we find $\lambda = \sqrt{-z}/\sqrt{1 - z}$. Define

$$f : \mathbb{S}^2 \to X, \quad f(x, y, z) = \begin{cases} \left(\frac{\sqrt{z}}{\sqrt{1+z}}x, \frac{\sqrt{z}}{\sqrt{1+z}}y, z\right), & z \geq 0 \\ \left(\frac{\sqrt{-z}}{\sqrt{1-z}}x, \frac{\sqrt{-z}}{\sqrt{1-z}}y, z\right), & z \leq 0. \end{cases}$$

Observe that at the points $(x, y, 0)$ of the equator A, f is well defined because its image is $(0, 0, 0)$. This map is continuous (Pasting Lemma) and closed. By the construction, f is surjective and injective on $\mathbb{S}^2 \setminus A$. The image of the points of A is $(0, 0, 0)$. Since f is injective in $\mathbb{S}^2 \setminus A$, we deduce that $\sim_f$ is the relation that identifies all points of A. Definitively, f is an identification and induces a homeomorphism between $\mathbb{S}^2/A$ and X. □

We revisit Example 10.8 and do a new proof for the case $n = 2$. In order to use Example 10.10, we assume that the radius of the disc is π.

Exercise 10.31 Consider the disc $\mathbb{D}^2_\pi = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq \pi^2\}$ and the relation $\sim$ that identifies the circle $\mathbb{S}^1_\pi$. Prove that $\mathbb{D}^2_\pi/\sim$ is homeomorphic to $\mathbb{S}^2$.

Solution We see $\mathbb{S}^2$ as a surface of revolution (Exercise 8.30) but with center at $p = (0, 0, 1)$,

$$\mathbb{S}^2(p) = \{(\cos t \cos\theta, \cos t \sin\theta, 1 + \sin t) : t \in [-\frac{\pi}{2}, \frac{\pi}{2}], \theta \in [0, 2\pi]\}.$$

Before stating the homeomorphism, we give the idea of the gluing map. See Fig. 10.17. Once we have situated the sphere supported on the xy-plane, we position $\mathbb{D}^2_\pi$ on the xy-plane so the center coincides with the origin. In the disc, consider any ray from the origin, making an angle θ with respect to the x-axis. Take the ray and lift it onto the sphere along a meridian for angle θ so that the endpoint coincides with the north pole of the sphere. In polar coordinates (r, θ) on the disc, first we change the interval $[0, \pi]$ that indicates the modulus of $\mathbb{D}^2_\pi$ into the interval $[-\pi/2, \pi/2]$ that measures the length along the meridians. For this, let the

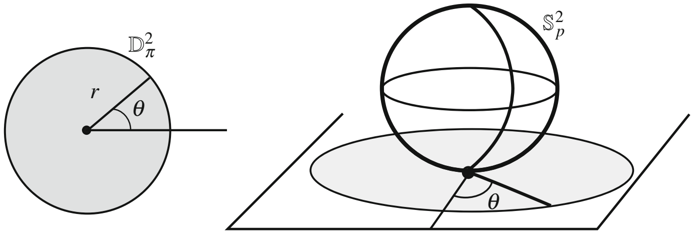


Fig. 10.17 The homeomorphism between the quotient space $\mathbb{D}^2_\pi/\sim$ and the sphere $\mathbb{S}^2$ of Exercise 10.31

translation $t \mapsto t - \pi/2$. We map a point $(r\cos\theta, r\sin\theta) \in \mathbb{D}^2_\pi$ to the point of the sphere

$$(\cos(r - \tfrac{\pi}{2})\cos\theta, \cos(r - \tfrac{\pi}{2})\sin\theta, 1 + \sin(r - \tfrac{\pi}{2})).$$

Since $(x, y) = (r\cos\theta, r\sin\theta)$, then $r = \sqrt{x^2 + y^2}$, $\cos\theta = x/r$ and $\sin\theta = y/r$. Set $\phi(x, y) = \sqrt{x^2 + y^2} - \frac{\pi}{2}$. Define $f : \mathbb{D}^2_\pi \to \mathbb{S}^2(p)$ by

$$f(x, y) = \begin{cases} \left(\dfrac{x\cos\phi(x, y)}{\sqrt{x^2 + y^2}}, \dfrac{y\cos\phi(x, y)}{\sqrt{x^2 + y^2}}, 1 + \sin\phi(x, y)\right), & (x, y) \neq (0, 0) \\ (0, 0, 0), & (x, y) = (0, 0). \end{cases}$$

From the construction, f is clearly surjective and $\sim_f$ coincides with $\sim$ because all points (x, y) of $\mathbb{S}^1_\pi$ satisfy $\sqrt{x^2 + y^2} = \pi$ and $\phi(x, y) = \pi/2$, so $f(x, y) = (0, 0, 2)$. The continuity of f is clear in $\mathbb{D}^2_\pi \setminus \{(0, 0)\}$, which is open in $\mathbb{D}^2_\pi$. Thus it remains to check the continuity at the origin. Let us observe that $f(0, 0) = (0, 0, 0)$. Then

$$|f(x, y) - f(0, 0)|^2 = |f(x, y)|^2 = 2 + 2\sin(\sqrt{x^2 + y^2} - \frac{\pi}{2}),$$

so if $((x_n, y_n)) \to (0, 0)$, then $|(f(x_n, y_n))| \to 0$. Finally f is closed because $\mathbb{D}^2_\pi$ is compact and $\mathbb{S}^2(p)$ is Hausdorff. □

Exercise 10.32 Let $X = [0, 3]$ and $A = (1, 2)$. Prove that X/A is not Hausdorff.

Solution The quotient space X/A is not Hausdorff because the equivalence classes [1] and [2] cannot be separated by disjoint open sets. Or equivalently, there are no disjoint saturated open sets U and V in X such that $1 \in U$ and $2 \in V$. Indeed, if U is an open set and $1 \in U$, then there is $r > 0$ such that $1 \in (1 - r, 1 + r) \cap X \subset U$. In particular, there is $x \in (1, 2) \cap U$. Analogously, there is $y \in (1, 2) \cap V$. This implies $U \cap A \neq \emptyset$ and $V \cap A \neq \emptyset$. Since U and V are saturated, $A \subset R[U] = U$ and $A \subset R[V] = V$. This implies $U \cap V \neq \emptyset$, a contradiction. □

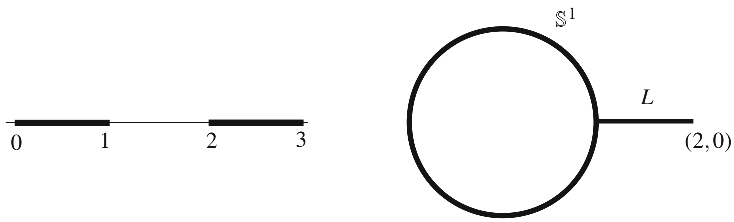


Fig. 10.18 The sets $[0, 1] \cup [2, 3]$ and $\mathbb{S}^1 \cup L$ of Exercise 10.33

Exercise 10.33 Let $X = [0, 1] \cup [2, 3]$ and $A = \{0, 1, 2\}$. Prove that X/A is homeomorphic to $\mathbb{S}^1 \cup \{(x, 0) \in \mathbb{R}^2 : 1 \leq x \leq 2\}$.

Solution We know that the quotient space $[0, 1]/\{0, 1\}$ is homeomorphic to the circle. In this case, the endpoint $x = 2$ of the interval $[2, 3]$ is also related to 0 and 1. Thus we can think of the interval $[2, 3]$ as being glued to the circle at the point $(1, 0)$. The gluing map will roll $[0, 1]$ in $\mathbb{S}^1$ as usually and we attach the interval $[2, 3]$ to $\mathbb{S}^1$ at the point $(1, 0)$ viewing $[2, 3]$ as the segment $L = [1, 2] \times \{0\}$. See Fig. 10.18. Define

$$f\colon [0, 1] \cup [2, 3] \to \mathbb{S}^1 \cup L, \qquad f(t) = \begin{cases} (\cos(2\pi t), \sin(2\pi t)) & t \in [0, 1] \\ (t - 1, 0) & t \in [2, 3]. \end{cases}$$

The function f is continuous by the Pasting Lemma because the intervals $[0, 1]$ and $[2, 3]$ are closed in $\mathbb{R}$, so closed in X. The domain of f is compact and the codomain is Hausdorff. Also, $f([0, 1]) = \mathbb{S}^1$ and $f([2, 3]) = L$. Thus f is an identification. The exercise concludes if we see that the equivalence relation $\sim_f$ is $\sim$. Assume $f(t) = f(s)$. We discuss three cases depending on the value of $f(t)$.

1. Case $f(t) \in \mathbb{S}^1 \setminus \{(1, 0)\}$. The identity $f(t) = f(s)$ implies $s \in (0, 1)$ and $t = s$.
2. Case $f(t) = (1, 0)$. Then $t \in A$. If $s \in [0, 1]$, then $s \in \{0, 1\}$ and if $s \in [2, 3]$ then $s = 2$. In any case, $s \in A$. This yields $t \sim s$.
3. Case $f(t) \in L \setminus \{(1, 0)\}$. Then $s \in (2, 3]$ and because f is bijective on $(2, 3]$, then $t = s$.

The converse is immediate because if $t, s \in A$, then $f(t) = f(s) = (1, 0)$. □

10.6 Suggested Exercises

10.6.1 Let A be a subset of a space X and let $p\colon X \to X/A$ be the projection. Prove that if A is open (resp. closed), then the projection on the quotient space is open (resp. closed). Investigate if the converse is true.

10.6.2 Define an equivalence relation on $\mathbb{R}^2$ by

$$(t, s) \sim (t', s') \text{ if and only if } t - t' \in \mathbb{Z} \text{ and } s = s'.$$

Show that the quotient space is homeomorphic to $\mathbb{S}^1 \times \mathbb{R}$.

10.6.3 Consider on $X = [-1, 1]$ the equivalence relation $\sim$ defined by the partition

$$\{\{-1\}, \{1\}, \{\{x, -x\} : x \in (-1, 1)\}\}.$$

Prove that $X/\sim$ is T_1 but not Hausdorff.

10.6.4 Let X be a space with an equivalence relation. If all equivalence classes are dense as subsets of X, show that the quotient topology is trivial.

10.6.5 Let (X, τ) where $X = \{1, \ldots, n\}$ and $\tau = \{\emptyset, \{1\}, \{1, 2\}, \ldots, \{1, 2, \ldots, n\}\}$. Consider the equivalence relation $\sim$ that identifies 1 and n. Show that $X/\sim$ is trivial.

10.6.6 Let X be a space and $Y = X \times \{0, 1\}$. Define an equivalence relation on Y by

$$(x, t) \sim (x', t') \text{ if and only if } x = x'.$$

Prove that $Y/\sim$ is homeomorphic to X.

10.6.7 Let $X = [-1, 2]$ and $A = [-1, 0] \cup [1, 2]$. Prove that $X/A \cong \mathbb{S}^1$.

10.6.8 Let $X = [-1, 2]$ and $A = [-1, 0]$. Prove that $X/A \cong [0, 1]$.

10.6.9 Let $X = [0, 2]$ and $A = \{0, 1, 2\}$. Prove that X/A is homeomorphic to two tangent circles of $\mathbb{R}^2$.

10.6.10 Define an equivalence relation on $\mathbb{R}^2$ by

$$(x_1, y_1) \sim (x_2, y_2) \text{ if and only if } x_1^2 + y_1^2 = x_2^2 + y_2^2.$$

Show that the quotient space is homeomorphic to $[0, \infty)$.

10.6.11 Define an equivalence relation on $\mathbb{R}^2$ by

$$(x_1, y_1) \sim (x_2, y_2) \text{ if and only if } x_1^2 + y_1 = x_2^2 + y_2.$$

Identify the quotient space with some known space.

10.6.12 A continuous map $\alpha\colon \mathbb{R} \to \mathbb{R}^n$ is called a *closed curve* if there is $T > 0$ such that $\alpha(t + T) = \alpha(t)$ for all $t \in \mathbb{R}$. If α is injective in $[0, T)$, we say that α is *simple*. If α is a closed curve, prove that $\alpha(\mathbb{R})$ is compact. If, in addition, α is simple, show then $\alpha(\mathbb{R})$ is homeomorphic to $\mathbb{S}^1$.

10.6.13 Let $A = \mathbb{S}^1 \times \{0\}$. Show that $\mathbb{S}^1 \times [0, \infty)/A$ is homeomorphic to the upper half cone $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2, z \geq 0\}$. Prove that $\mathbb{S}^1 \times [0, 1]/A$ is homeomorphic to the disc $\mathbb{D}^2$.

10.6.14 Define an equivalence relation on $\mathbb{R}$ by

$$x \sim y \text{ if and only if } \begin{cases} x = y, & \text{or} \\ x, y & \text{both are even natural numbers, or} \\ x, y & \text{both are odd natural numbers.} \end{cases}$$

Determine the connectedness and compactness of the quotient space.

10.6.15 Show that the cone over $\mathbb{S}^2$ (Exercise 10.8) is homeomorphic to $\mathbb{D}^3$.

10.6.16 Fix $\vec{v} \in \mathbb{R}^2 \setminus \{0\}$. Define an equivalence relation $\sim$ on $\mathbb{R}^2$ by

$$x \sim y \text{ if and only if there is } n \in \mathbb{Z} \text{ such that } y = x + n\vec{v}.$$

Prove that the quotient space is homeomorphic to $\mathbb{S}^1 \times \mathbb{R}$ and that the projection is open.

10.6.17 Fix $\vec{v}_1$ and $\vec{v}_2$ two linearly independent vectors of $\mathbb{R}^2$ and define the equivalence relation $\sim$ on $\mathbb{R}^2$ by

$$x \sim y \text{ if and only if there are } n, m \in \mathbb{Z} \text{ such that } y = x + n\vec{v}_1 + m\vec{v}_2.$$

Prove that the quotient space is homeomorphic to the torus $\mathbb{T}^2$.

10.6.18 Let $m \in \mathbb{Z}$ be a fix integer. Define an equivalence relation on $\mathbb{R}$ by

$$x \sim y \text{ if there is } k \in \mathbb{Z} \text{ such that } x - y = km.$$

Prove that the quotient space is homeomorphic to $\mathbb{S}^1$.

10.6.19 Let $X = \{x \in \mathbb{R}^2 : 1 \leq |x| \leq 2\}$. Define an equivalence relation on X by $x \sim y$ if they coincide, or if $|x - y| = 1$ and $x = \lambda y$ with $\lambda > 0$. Prove that the quotient space is homeomorphic to the torus $\mathbb{T}^2$.

Chapter 11
The Fundamental Group

Throughout this book, we have defined several topological invariants such as connectedness and compactness to classify topological spaces. However, these invariants do not allow us to distinguish spaces so simple as the sphere $\mathbb{S}^2$ from the torus $\mathbb{T}^2$. It is clear that they are not homeomorphic because the torus has a “hole” and the sphere does not. The hole is captured by considering loops with a fixed base point, in other words, paths that begin and end at the same point. On the sphere it is possible to stretch a given loop until it is reduced to a point by means of deformations on the sphere. The property of shrinking a loop to a point remains invariant by homeomorphisms. In contrast, those loops in the torus around the hole cannot be shrinked to one point.

In this chapter, we will endow the set of loops of a structure of a group, called the *fundamental group*, in such a way that if two spaces are homeomorphic, the corresponding groups are isomorphic. In the above case, the fundamental group of the sphere is trivial because every loop reduces into a point. The torus does not have this property, and consequently, it is not homeomorphic to the sphere.

11.1 Homotopy and Loops

The program to define the fundamental group comprises various steps. First we introduce the concept of loop and we establish an equivalence relation on the set of all loops on a space.

Definition 11.1.1 Let X be a space and $x_0, x_1 \in X$. Let Ω_{x_0,x_1} be the set of paths joining x_0 with x_1,

$$\Omega_{x_0,x_1} = \{\alpha\colon \mathbf{I} \to X : \alpha \text{ is continuous }, \alpha(0) = x_0, \alpha(1) = x_1\}.$$

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5_11

If $\alpha(0) = \alpha(1)$, we say that α is a *loop* with *base point* x_0. If $\alpha, \beta \in \Omega_{x_0,x_1}$, we say that α is *homotopic* to β, and we write $\alpha \sim \beta$, if there is a continuous map $F : \mathbf{I} \times \mathbf{I} \to X$ with the following properties:

(i) $F(t, 0) = \alpha(t)$ and $F(t, 1) = \beta(t)$ for all $t \in \mathbf{I}$.
(ii) $F(0, s) = x_0$ and $F(1, s) = x_1$ for all $s \in \mathbf{I}$.

The mapping F is called a *homotopy* between α and β.

A homotopy between two paths can be regarded as a continuous deformation of α into β by means of a family of paths, all of them with the same beginning point and the same endpoint. Indeed, fixing $s \in \mathbf{I}$, the maps

$$F_s : \mathbf{I} \to X, \quad F_s(t) = F(t, s)$$

are paths joining x_0 with x_1 (by (ii)) where the first curve F_0 is α and the last one F_1 is β by (i) (Fig. 11.1).

The binary relation $\sim$ defined on Ω_{x_0,x_1} is of equivalence. The elements of the quotient set $\Omega_{x_0,x_1}/\sim$ are called *homotopy classes*, where by $[\alpha]$ we mean the equivalence class of α. We are in position to find the homotopy classes in Euclidean spaces. The affine structure of $\mathbb{R}^n$ will be crucial in the next result.

Example 11.1 We prove that in $\mathbb{R}^n$ every two paths of $\Omega_{x_0x_1}$ are homotopic. In particular, every loop with base point x_0 is homotopic to the constant loop $c_{x_0} : \mathbf{I} \to \mathbb{R}^n$ defined by $c_{x_0}(t) = x_0$ for all $t \in \mathbf{I}$.

Let $\alpha, \beta \in \Omega_{x_0x_1}$. The idea of the homotopy is to deform the point $\alpha(t)$ in $\beta(t)$ along the segment $[\alpha(t), \beta(t)]$ as indicated in Fig. 11.2. The trick here is that this segment is included in the space $\mathbb{R}^n$. It suffices to define the homotopy $F(t, s) = (1 - s)\alpha(t) + s\beta(t)$. The continuity of F is as follows. The map $(t, s) \mapsto \alpha(t)$ is continuous (explained in the proof of the above proposition) and similarly the map $(t, s) \mapsto 1 - s$ is continuous. The product of both maps is $(t, s) \mapsto (1 - s)\alpha(t)$

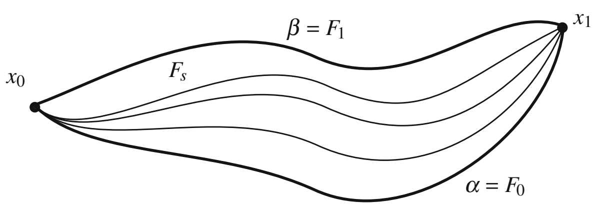


Fig. 11.1 A homotopy F between the paths α and β

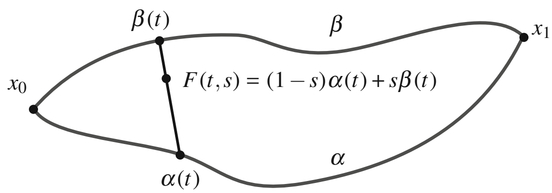


Fig. 11.2 A homotopy between two paths α and β of $\mathbb{R}^n$ along the segments $[\alpha(t), \beta(t)]$

which is continuous. In the same fashion, the map $(t, s) \mapsto s\beta(t)$ is continuous. Again, the sum of these maps, namely, the map F, is continuous.

Similarly, every loop in a convex set of $\mathbb{R}^n$ is homotopic to the constant loop. In a star-shaped set, a loop with base point the center of the set is homotopic to the constant loop. Notice that if we replace $\mathbb{R}^n$ for other subset, the arguments may fail. For example, if we take the circle $\mathbb{S}^1$ as subset of $\mathbb{R}^2$, then the above map $F = F(t, s)$ cannot be defined as a map $F\colon \mathbf{I} \times \mathbf{I} \to \mathbb{S}^1$ because the segment $[p, q]$ joining two points p, q of $\mathbb{S}^1$ is not included in $\mathbb{S}^1$. $\diamondsuit$

Having defined the equivalence classes, the next step in the construction of the fundamental group is to assign a group operation to these classes. If $\alpha \in \Omega_{x_0,x_1}$ and $\beta \in \Omega_{x_1,x_2}$, then we have defined $\alpha * \beta \in \Omega_{x_0,x_2}$ by

$$\alpha * \beta(t) = \begin{cases} \alpha(2t), & 0 \le t \le \frac{1}{2} \\ \beta(2t-1), & \frac{1}{2} \le t \le 1. \end{cases}$$

This operation $*$ is not associative in general, that is, $\alpha * (\beta * \gamma) \neq (\alpha * \beta) * \gamma$ because the mappings $\alpha * (\beta * \gamma)$ and $(\alpha * \beta) * \gamma$ are piecewise defined on three distinct subsets of $\mathbf{I} \times \mathbf{I}$. However, the operation $*$ is associative on the quotient space. To be more explicit, define

$$*\colon \frac{\Omega_{x_0,x_1}}{\sim} \times \frac{\Omega_{x_1,x_2}}{\sim} \to \frac{\Omega_{x_0,x_2}}{\sim}, \quad [\alpha] * [\beta] = [\alpha * \beta]. \tag{11.1}$$

In fact, our interest will turn to when the three points coincide. The definition of $*$ is independent of the chosen representative in each equivalence class.

Definition 11.1.2 The group $(\Omega_{x,x}/\sim)$ is called the *fundamental group of X with base point at x* and it will be denoted by $\pi_1(X, x)$.

The unit of $\pi_1(X, x)$ is the constant loop $[c_x]$, $c_x(t) = x$. The inverse of $[\alpha]$, denoted $[\alpha]^{-1}$, is $[\alpha^{-1}]$. Henceforth, we shall write simply $[\alpha][\beta]$ instead of $[\alpha] * [\beta]$. In a space X, we have associated a fundamental group for each element $x \in X$. For path connected spaces. however, all these groups are isomorphic and thus indistinguishable from the algebraic point of view.

Proposition 11.1.3 *If X is path connected then $\pi_1(X, x)$ and $\pi_1(X, y)$ are isomorphic for all $x, y \in X$.*

Proof Because X is path connected, there is a path $\gamma : \mathbf{I} \to X$ joining x with y. The idea of the isomorphism is that for each loop $\alpha \in \Omega_{xx}$, consider the loop on y that begins running towards x through γ^{-1}, then follow with α and return to y walking along γ, written $\gamma^{-1} * \alpha * \gamma$. See Fig. 11.3. Then

$$\phi\colon \pi_1(X, x) \to \pi_1(X, y), \quad \phi([\alpha]) = [\gamma^{-1}][\alpha][\gamma]$$

is a homomorphism of groups. $\square$

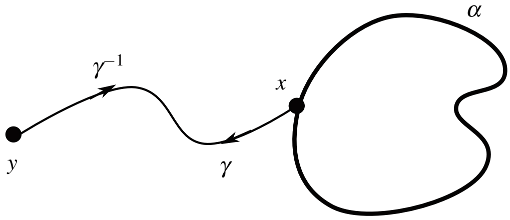


Fig. 11.3 The isomorphism between $\pi_1(X, x)$ and $\pi_1(X, y)$

Definition 11.1.4 Let X be a path connected space. The *fundamental group* of X, written $\pi_1(X)$, is defined as any group $\pi_1(X, x)$ with $x \in X$.

The other pillar in our program is that the fundamental group, as group, is preserved by homeomorphisms. First, let us examine the behavior of the fundamental group under continuous mappings.

Proposition 11.1.5 *If $f\colon X \to Y$ is continuous, then the map f_* defined by*

$$f_*\colon \pi_1(X, x) \to \pi_1(Y, f(x)), \quad f_*([\alpha]) = [f \circ \alpha]$$

is a homomorphism of groups.

Proof First we see that f_* is well-defined, in other words, f_* does not depend on the chosen representative in the equivalence class of α. If $\alpha \sim \beta$ and F is a homotopy between α and β, then $f \circ F$ is a homotopy between $f \circ \alpha$ and $f \circ \beta$, so $f \circ \alpha \sim f \circ \beta$. This proves $f_*([\alpha]) = f_*([\beta])$. We show that f_* is a homomorphism of groups. We have

$$f_*([\alpha][\beta]) = f_*([\alpha * \beta]) = [f \circ (\alpha * \beta)]$$

and

$$f_*([\alpha])f_*([\beta]) = [(f \circ \alpha) * (f \circ \beta)].$$

To conclude, a computation yields $f \circ (\alpha * \beta) = (f \circ \alpha) * (f \circ \beta)$. □

Theorem 11.1.6

1. *For the identity $\mathbb{1}_X\colon X \to X$, we have $(\mathbb{1}_X)_* = \mathbb{1}_{\pi_1(X)}$.*
2. *If $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous, then $(g \circ f)_* = g_* \circ f_*$.*

As a consequence, if X is homeomorphic to Y, then $\pi_1(X) \simeq \pi_1(Y)$.

Finally, we would like to compute the fundamental group for a given space. How can we calculate the fundamental group when the definition has been so complex? The purpose of this chapter is to calculate the fundamental group in those spaces where all this work is relatively easy.

Definition 11.1.7 A space is called *simply connected* if it is path connected and its fundamental group is trivial. Equivalently, a path connected space is simply connected if and only if any two paths joining two points are homotopic.

Example 11.2 From Example 11.1, star-shaped subsets of $\mathbb{R}^n$ are simply connected. In particular, $\mathbb{R}^n$ and any convex subset are simply connected.

As with any topological invariant, two spaces with the same fundamental group may not be homeomorphic. So, $\{1\}$, $[0, 1]$ and $\mathbb{R}$ are simply connected but they are not homeomorphic. $\diamondsuit$

We conclude this section by studying the behavior of the fundamental group with respect to the topological product. Recall from group theory that if G and G' are two groups, then the Cartesian product $G \times G'$ has a standard structure of group employing the operations of G and G' and defined by $(g, g') \cdot (h, h') = (gh, g'h')$.

Proposition 11.1.8 *For two spaces X and Y, we have $\pi_1(X \times Y) \simeq \pi_1(X) \times \pi_1(Y)$.*

Proof Denote by p and p' the projections of $X \times Y$. If $(x, y) \in X \times Y$, define

$$\phi\colon \pi_1(X \times Y, (x, y)) \to \pi_1(X, x) \times \pi_1(Y, y), \quad \phi([\gamma]) = \big(p_*([\gamma]), p'_*([\gamma])\big).$$

The inverse of ϕ is $\phi^{-1}([\alpha], [\beta]) = [\alpha \times \beta]$, where $\alpha \times \beta$ is the product map of α with β. We check that one is inverse of the other. Certainly,

$$\begin{aligned}\phi(\phi^{-1}([\alpha], [\beta]) &= \phi([\alpha \times \beta]) = \big(p_*([\alpha \times \beta]), p'_*([\alpha \times \beta])\big) \\ &= ([p \circ (\alpha \times \beta)], [p' \circ (\alpha \times \beta)]).\end{aligned}$$

But $p \circ (\alpha \times \beta)(t) = p(\alpha(t), \beta(t)) = \alpha(t)$ and $p' \circ (\alpha \times \beta)(t) = \beta(t)$. Reciprocally,

$$\phi^{-1}(\phi([\gamma]) = \phi^{-1}([p \circ \gamma], [p' \circ \gamma]) = [(p \circ \gamma) \times (p' \circ \gamma)].$$

Now $(p \circ \gamma) \times (p' \circ \gamma)(t) = (p(\gamma(t)), p'(\gamma(t))) = \gamma(t)$.

Finally we see that ϕ is a homomorphism of groups. By using the operation in $\pi_1(X, x) \times \pi_1(Y, y)$,

$$\begin{aligned}\phi_*([\gamma][\sigma]) &= (p_*([\gamma][\sigma]), p'_*([\gamma][\sigma])) = \big(p_*([\gamma])p_*([\sigma]), p'_*([\gamma])p'_*([\sigma])\big) \\ &= \big(p_*([\gamma]), p'_*([\gamma])\big) \cdot \big(p_*([\sigma]), p'_*([\sigma])\big) = \phi_*([\gamma])\phi_*([\sigma]).\end{aligned}$$

$\square$

Thanks to this result, $\pi_1(\mathbb{S}^1 \times \mathbb{R}) = \pi_1(\mathbb{S}^1)$ and $\pi_1(\mathbb{S}^1 \times \mathbb{S}^1) = \pi_1(\mathbb{S}^1) \times \pi_1(\mathbb{S}^1)$. However, we have not yet computed the fundamental group of $\mathbb{S}^1$. This is the objective of the subsequent section.

11.2 The Fundamental Group of $\mathbb{S}^1$

We calculate the fundamental group of the sphere of the smallest dimension, the circle $\mathbb{S}^1$. It is easy to have an intuitive idea of this group. A loop in $\mathbb{S}^1$ winds a number of times around the circle. If a loop winds one time, then it is not possible to deform it to a constant loop, or to a loop that winds twice around the circle. Likewise, it is also clear that if one loop winds, say three times, and we add another loop that winds four times, the resulting loop, after the $*$ operation, winds seven times. We also have a notion of "positive" and "negative" loops based on whether the loop is oriented counterclockwise or clockwise respectively. After these arguments, it seems reasonable that the number of times a loop "winds" around $\mathbb{S}^1$ identifies its equivalence class in $\pi_1(\mathbb{S}^1)$ so the fundamental group of $\mathbb{S}^1$ is $\mathbb{Z}$. However, we need an effort to translate all these ideas to rigorous definitions and proofs.

We begin by identifying $\mathbb{R}^2$ with the set of complex numbers $\mathbb{C}$ where $\mathbb{S}^1 \subset \mathbb{C}$ is a multiplicative subgroup. Let us employ the complex notation $e^{it} = (\cos t, \sin t)$ where $i^2 = -1$. Fix $x_0 = (1, 0) = 1$ as base point and we will compute $\pi_1(\mathbb{S}^1, x_0)$. Note that 1 is the unit element of the group $\mathbb{S}^1$. We collect various properties of the rolling map

$$p\colon \mathbb{R} \to \mathbb{S}^1, \quad p(t) = (\cos 2\pi t, \sin 2\pi t) = e^{2\pi i t}.$$

(i) The map p is surjective, continuous and 1-periodic in the sense that $p(t+1) = p(t)$ for all $t \in \mathbb{R}$.
(ii) The map $p\colon (\mathbb{R}, +) \to (\mathbb{S}^1, \cdot)$ is a homomorphism of groups, $p(s + t) = p(s) \cdot p(t)$. Furthermore its kernel, $\operatorname{Ker}(p)$, is $\mathbb{Z}$ because $p(t) = (1, 0)$ if and only if $t \in \mathbb{Z}$.
(iii) The map $p\colon (a, a + 1) \to \mathbb{S}^1 \setminus \{p(a)\}$ is a homeomorphism for every $a \in \mathbb{R}$ (Exercise 6.11).

The number of times that winds a loop α is obtained by "unrolling" α along $\mathbb{R}$ thanks to the map p and look the length of that path. For example, one time winding on $\mathbb{S}^1$ corresponds with an interval of length 1 in $\mathbb{R}$, and two times with an interval of length 2. When we unroll α we must fix the starting point at $\mathbb{R}$ and next, look the endpoint. The starting point will be $0 \in \mathbb{R}$, so the endpoint must be an integer. First, we need to formalize how to unroll a loop of $\mathbb{S}^1$ along $\mathbb{R}$.

Proposition 11.2.1 (Lifting Paths) *If $\alpha\colon \mathbf{I} \to \mathbb{S}^1$ is continuous and $\alpha(0) = 1$, then there is a unique path $\widetilde{\alpha}\colon \mathbf{I} \to \mathbb{R}$ such that $\widetilde{\alpha}(0) = 0$ and $p \circ \widetilde{\alpha} = \alpha$ (Fig. 11.4).*

$$\begin{tikzcd}
 & \mathbb{R} \arrow[dr, "p"] & \\
{[0,1]} \arrow[ur, dashed, "\widetilde{\alpha}"] \arrow[rr, "\alpha"] & & \mathbb{S}^1
\end{tikzcd}$$

Fig. 11.4 The scheme of lifting paths of $\mathbb{S}^1$ to $\mathbb{R}$

Proof The map $\alpha\colon \mathbf{I} \to \mathbb{S}^1$ is uniformly continuous because $\mathbf{I}$ is compact (Exercise 9.4.9). For $\epsilon = 2$, there is $\delta > 0$ such that if $|t-t'| < \delta$, $|\alpha(t)-\alpha(t')| < 2$. Since $\alpha(t)$ and $\alpha(t')$ are points of $\mathbb{S}^1$, the inequality $|\alpha(t) - \alpha(t')| < 2$ forces that they are not antipodal, or equivalently, viewed as complex numbers, $\alpha(t)/\alpha(t') \neq -1$.

Consider the homeomorphism $p\colon (-\frac{1}{2}, \frac{1}{2}) \to \mathbb{S}^1 \setminus \{-1\}$ and let $\psi = p^{-1}$. If $|t - t'| < \delta$, then $\alpha(t)/\alpha(t')$ is a complex number distinct from -1, so

$$\psi\left(\frac{\alpha(t)}{\alpha(t')}\right) \in \left(-\frac{1}{2}, \frac{1}{2}\right).$$

Let N be a natural number such that $N > 1/\delta$ and consider the division of $[0, 1]$ defined by $\{0, \frac{1}{N}, \frac{2}{N}, \ldots, \frac{N-1}{N}, 1\}$. Since $|\frac{j}{N}t - \frac{j-1}{N}t| \leq \frac{t}{N} \leq \frac{1}{N} < \delta$, then

$$\psi\left(\frac{\alpha(\frac{j}{N}t)}{\alpha(\frac{j-1}{N}t)}\right) \in \left(-\frac{1}{2}, \frac{1}{2}\right).$$

For each $1 \leq j \leq N$, define

$$\widetilde{\alpha}_j\colon \mathbf{I} \to \left(-\frac{1}{2}, \frac{1}{2}\right), \quad \widetilde{\alpha}_j(t) = \psi\left(\frac{\alpha(\frac{j}{N}t)}{\alpha(\frac{j-1}{N}t)}\right).$$

The maps $\widetilde{\alpha}_j$ are continuous because the quotient of complex number is continuous (Exercise 7.28). Moreover,

$$p \circ \widetilde{\alpha}_j(t) = \frac{\alpha(\frac{j}{N}t)}{\alpha(\frac{j-1}{N}t)}.$$

Define the continuous function

$$\widetilde{\alpha}\colon \mathbf{I} \to \mathbb{R}, \quad \widetilde{\alpha}(t) = \widetilde{\alpha}_1(t) + \ldots + \widetilde{\alpha}_N(t).$$

Then $\widetilde{\alpha}_j(0) = \psi(1) = 0$, hence $\widetilde{\alpha}(0) = 0$. Since p is a homomorphism of groups, and because $\alpha(0) = 1$,

$$p \circ \widetilde{\alpha}(t) = (p\widetilde{\alpha}_1(t)) \cdots (p\widetilde{\alpha}_N(t)) = \frac{\alpha(\frac{t}{N})}{\alpha(0)}\frac{\alpha(\frac{2t}{N})}{\alpha(\frac{t}{N})} \cdots \frac{\alpha(t)}{\alpha(\frac{N-1}{N}t)} = \frac{\alpha(t)}{\alpha(0)} = \alpha(t).$$

We prove the uniqueness of the lift $\widetilde{\alpha}$. Assume that $\widetilde{\beta}\colon \mathbf{I} \to \mathbb{R}$ is other path with $\widetilde{\beta}(0) = 0$ and $p \circ \widetilde{\beta} = \alpha$. If $h = \widetilde{\alpha} - \widetilde{\beta}\colon \mathbf{I} \to \mathbb{R}$, then

$$p \circ h(t) = \frac{p(\widetilde{\alpha}(t))}{p(\widetilde{\beta}(t))} = 1.$$

Thus $h(t) \in \operatorname{Ker}(p)$, that is, $h(\mathbf{I}) \subset \mathbb{Z}$. Since h is continuous, the domain $\mathbf{I}$ is connected and $\mathbb{Z}$ is discrete, the function h is constant by Proposition 8.1.2. As $h(0) = 0$, it follows that $h(t) = 0$ for all $t \in \mathbf{I}$, and this yields $\widetilde{\alpha} = \widetilde{\beta}$. □

The next step is the lifting process for homotopies.

Proposition 11.2.2 (Lifting Homotopies) *If $F\colon \mathbf{I} \times \mathbf{I} \to \mathbb{S}^1$ is a continuous map and $F(0, 0) = 1$, then there is a unique continuous map $\widetilde{F}\colon \mathbf{I} \times \mathbf{I} \to \mathbb{R}$ such that $\widetilde{F}(0, 0) = 0$ and $p \circ \widetilde{F} = F$.*

We define the times that a loop α winds around $\mathbb{S}^1$.

Definition 11.2.3 The *degree* of a loop α of $\mathbb{S}^1$ is defined as the integer number

$$\deg \alpha = \widetilde{\alpha}(1).$$

Example 11.3 For each integer $n \in \mathbb{Z}$, we find a loop that winds exactly n times. Define

$$\alpha\colon \mathbf{I} \to \mathbb{S}^1, \quad \alpha(t) = (\cos 2\pi nt, \sin 2\pi nt).$$

Then $\alpha(t) = p(nt)$. If $\gamma(t) = nt$ then $\gamma(0) = 0$. Therefore γ is actually the lift $\widetilde{\alpha}$ of α, hence the degree of α is $\gamma(1) = n$. If the direction of α is counterclockwise, then n is positive and if it is clockwise, then n is negative. ♢

Proposition 11.2.4 *Two loops α and β of $\mathbb{S}^1$ are homotopic if and only if* $\deg \alpha = \deg \beta$.

Proof Suppose that $\alpha \sim \beta$, and let F be a homotopy between α and β. Let $\widetilde{F}$ be the lift of F obtained by Proposition 11.2.2. Then $p \circ \widetilde{F}(t, 0) = \alpha(t)$ and $p \circ \widetilde{F}(t, 1) = \beta(t)$. Moreover, fixing $t = 0$, $p \circ \widetilde{F}(0, s) = F(0, s) = 1$, so $\widetilde{F}(0, s) \in \mathbb{Z}$. The mapping

$$\mathbf{I} \to \mathbb{Z}, \quad s \longmapsto \widetilde{F}(0, s)$$

is continuous, defined on the connected space $\mathbf{I}$ and the codomain $\mathbb{Z}$ is discrete. Thus the map must be constant. In view of $\widetilde{F}(0, 0) = 0$, then $\widetilde{F}(0, s) = 0$ for all $s \in \mathbf{I}$. By uniqueness of the lifting paths (Proposition 11.2.2), we deduce that $\widetilde{\alpha}(t) = \widetilde{F}(t, 0)$ and $\widetilde{\beta}(t) = \widetilde{F}(t, 1)$ are the corresponding lifts of α and β, respectively. On the other hand, the mapping

$$\mathbf{I} \to \mathbb{R}, \quad s \longmapsto \widetilde{F}(1, s)$$

is continuous with values on $\mathbb{Z}$ because $p(\widetilde{F}(1,s)) = F(1,s) = 1$, hence constant. We conclude

$$\deg\alpha = \widetilde{\alpha}(1) = \widetilde{F}(1,0) = \widetilde{F}(1,1) = \widetilde{\beta}(1) = \deg\beta.$$

Reciprocally, assume $\deg\alpha = \deg\beta$. Since $\widetilde{\alpha}(1) = \widetilde{\beta}(1)$, the paths $\widetilde{\alpha}$ and $\widetilde{\beta}$ defined on $\mathbb{R}$ have equal beginning point and equal endpoint. Since $\mathbb{R}$ is simply connected, $[\widetilde{\alpha}] = [\widetilde{\beta}]$, hence $[p\circ\widetilde{\alpha}] = [p\circ\widetilde{\beta}]$. This proves that $[\alpha] = [\beta]$. □

Theorem 11.2.5 *The fundamental group of* $\mathbb{S}^1$ *is* $\mathbb{Z}$.

Proof We show that

$$\phi\colon \pi_1(\mathbb{S}^1) \to \mathbb{Z}, \quad \phi([\alpha]) = \deg\alpha$$

is a isomorphism of groups. Proposition 11.2.4 asserts that ϕ is well-defined. Thus $\deg\alpha$ does not change under homotopies of α. To see that ϕ is a homomorphism,

$$\phi([\alpha][\beta]) = \phi([\alpha*\beta]) = \deg\alpha*\beta = \widetilde{(\alpha*\beta)}(1).$$

We prove that the lift of $\alpha*\beta$ is

$$\gamma(t) = \begin{cases} \widetilde{\alpha}(2t), & \text{if } 0 \le t \le \frac{1}{2} \\ \widetilde{\alpha}(1) + \widetilde{\beta}(2t-1), & \text{if } \frac{1}{2} \le t \le 1, \end{cases}$$

where $\widetilde{\alpha}$ and $\widetilde{\beta}$ are the corresponding liftings of α and β. It is immediate that γ is continuous and that $p(\widetilde{\alpha}(2t)) = \alpha(2t)$ when $0 \le t \le 1/2$. If $1/2 \le t \le 1$, and using that $p(\widetilde{\alpha}(1)) = \alpha(1) = 1$,

$$p(\widetilde{\alpha}(1) + \widetilde{\beta}(2t-1)) = p(\widetilde{\alpha}(1)) \cdot p(\widetilde{\beta}(2t-1)) = p(\widetilde{\beta}(2t-1)) = \beta(2t-1).$$

Then $p\circ\gamma = \alpha*\beta$. Moreover, $\gamma(0) = 0$. Thus γ is the unique lifted path of $\alpha*\beta$ determined by Proposition 11.2.1. Consequently,

$$\deg\alpha*\beta = \gamma(1) = \widetilde{\alpha}(1) + \widetilde{\beta}(1) = \deg\alpha + \deg\beta.$$

Finally, ϕ is injective by Proposition 11.2.4 and surjective by Example 11.3. □

Example 11.4 The fundamental group of the cylinder $\mathbb{S}^1\times\mathbb{R}$ is $\mathbb{Z}$. In the same way, the fundamental group of the n-dimensional torus $\mathbb{T}^n$ is $\mathbb{Z}^n$. ◊

This corollary allows us to give another example of two subsets of Euclidean spaces that are homeomorphic, but such that the homeomorphism cannot be expressed as the restriction of another one defined on the ambient space.

Example 11.5 Consider the spaces X and Y of $\mathbb{R}^2$ of Fig. 11.5, each one formed by the union of a circle $\mathbb{S}^1$ and a segment. These spaces are homeomorphic where

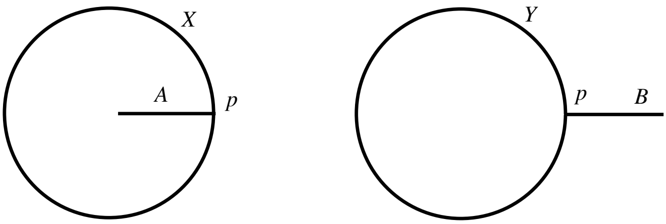


Fig. 11.5 The sets X and Y are homeomorphic but there is not a homeomorphism of $\mathbb{R}^2$ to itself carrying X into Y

the homeomorphism $\phi\colon X \to Y$ is defined piecewise and maps the circle into the circle and A into B, and under the condition that $\phi(p) = p$. However, there is no a homeomorphism $\Phi\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $\Phi_{|X} = \phi$. To show this, observe that $\mathbb{R}^2 \setminus X$ and $\mathbb{R}^2 \setminus Y$ have two connected components so we write $\mathbb{R}^2 \setminus X = C_1 \cup C_2$ and $\mathbb{R}^2 \setminus Y = D_1 \cup D_2$, where C_1 and D_1 are the bounded components. Since the boundary of C_1 contains A and $\phi(A) = B$, then necessarily, the boundary of $\Phi(C_1)$ must contain B. This proves that $\Phi(C_1) = D_2$, hence $\Phi(C_2) = D_1$. However, C_2 is not homeomorphic to D_1 because $C_2 \cong \mathbb{S}^1 \times \mathbb{R}$ and D_1 is a ball. $\Diamond$

11.3 The Fundamental Group of $\mathbb{S}^n$, $n \geq 2$

In this section we show that spheres $\mathbb{S}^n$ are simply connected for $n \geq 2$, which it is intuitive for $\mathbb{S}^2$. The proof requires the next lemma, which will be proved in Exercise 11.18.

Lemma 11.3.1 *Let $\alpha\colon \mathbf{I} \to X$ be a path and let $\bar{t} \in \mathbf{I}$ be a fixed number. Define*

$$\alpha_1\colon \mathbf{I} \to X, \quad \alpha_1(t) = \alpha(\bar{t}\,t)$$
$$\alpha_2\colon \mathbf{I} \to X, \quad \alpha_2(t) = \alpha(\bar{t} + t(1 - \bar{t})).$$

Then α is homotopic to $\alpha_1 * \alpha_2$.

It is immediate that this lemma can be generalized by taking a finite number of points $t_1, \ldots, t_n \in \mathbf{I}$ and construct the corresponding paths joining $\alpha(t_j)$ with $\alpha(t_{j+1})$. The next result is a particular case of a more general theorem, the so-called Seifert-Van Kampen theorem.

Theorem 11.3.2 (Seifert-van Kampen) *Let X be a space and let O_1 and O_2 be two open sets such that $X = O_1 \cup O_2$. If*

1. *O_1, O_2 and $O_1 \cap O_2$ are path connected, and*
2. *O_1 and O_2 are simply connected,*

then X is simply connected.

Proof Fix $x_0 \in O_1 \cap O_2$ and let $\alpha\colon \mathbf{I} \to X$ be a loop with base point x_0. As a preliminary step, we prove that there is a decomposition $a = t_0 < t_1 < \ldots < t_n = 1$ with the property that $\alpha([t_j, t_{j+1}])$ is entirely contained in O_1 or in O_2 for all $0 \leq j \leq n-1$. Indeed, let $t \in \mathbf{I}$, so $\alpha(t) \in O_1$ or $\alpha(t) \in O_2$. By continuity, there is $r_t > 0$ such that $\alpha((t - r_t, t + r_t))$ is included in O_1 or in O_2. The collection of open intervals $\{(t - \frac{r_t}{2}, t + \frac{r_t}{2}) : t \in \mathbf{I}\}$ covers $\mathbf{I}$ and by compactness, there is a finite subcovering. Then the partition we are looking for is formed by the endpoints of this subcovering, up to a reordering of them.

After this partition of $\mathbf{I}$, for each j define

$$\alpha_j\colon \mathbf{I} \to X, \quad \alpha_j(t) = \alpha((1-t)t_j + t\, t_{j+1}).$$

Let us notice that α_j is the restriction of α to $[t_j, t_{j+2}]$ but reparametrizating again so the domain is the interval $[0, 1]$. Then α_j is continuous, $\alpha_j(0) = \alpha(t_j)$ and $\alpha_j(1) = \alpha(t_{j+1})$, so $\alpha_j \in \Omega_{\alpha(t_j),\alpha(t_{j+1})}$. On the other hand, $\alpha(t_j) \in V_j \cap V_{j-1}$ and this intersection is O_1, O_2 or $O_1 \cap O_2$. Since the three sets are path connected, there is a path $\rho_j\colon \mathbf{I} \to V_j \cap V_{j-1}$ joining x_0 and $\alpha(t_j)$. By Lemma 11.3.1, we have $[\alpha] = [\alpha_0 * \alpha_1 * \ldots * \alpha_{n-1}]$, and this can be written as

$$\begin{aligned}[\alpha] &= [\alpha_0][\alpha_1]\ldots[\alpha_{n-1}] = [\alpha_0][\rho_1^{-1}][\rho_1][\alpha_1]\ldots[\rho_{n-1}^{-1}][\rho_{n-1}][\alpha_{n-1}]\\ &= [\alpha_0 * \rho_1^{-1}][\rho_1 * \alpha_1 * \rho_2^{-1}]\ldots[\rho_{n-1} * \alpha_{n-1}].\end{aligned}$$

Then $[\rho_j * \alpha_j * \rho_{j+1}^{-1}] = [c_{x_0}]$ because $\rho_j * \alpha_j * \rho_{j+1}^{-1}$ is a path in V_j that is simply connected. Also $[\alpha_0 * \rho_1^{-1}] = [\rho_{n-1} * \alpha_{n-1}] = [c_{x_0}]$. Therefore $\alpha \sim c_{x_0}$. □

Corollary 11.3.3 *The sphere $\mathbb{S}^n$ is simply connected for $n \geq 2$.*

Proof Let $O_1 = \mathbb{S}^n \setminus \{N\}$ and $O_2 = \mathbb{S}^n \setminus \{S\}$. They are homeomorphic to $\mathbb{R}^n$ which is simply connected. Since $O_1 \cap O_2 = \mathbb{S}^n \setminus \{N, S\} \cong \mathbb{R}^n \setminus \{0\}$ and $n \geq 2$, then $O_1 \cap O_2$ is path connected by Exercise 8.2. □

Example 11.6 The torus $\mathbb{S}^1 \times \mathbb{S}^1$ is not homeomorphic to the sphere $\mathbb{S}^2$. ◊

Example 11.7 For $n \geq 3$, the set $\mathbb{R}^n \setminus \{0\}$ is simply connected. This is because it was proved in Exercise 7.5.8 that the punctured space $\mathbb{R}^n \setminus \{0\}$ is homeomorphic to $\mathbb{S}^{n-1} \times \mathbb{R}$ for $n \geq 2$. Observe that $\mathbb{R}^2 \times \{0\}$ is an annulus so its fundamental group is of $\mathbb{S}^1 \times \mathbb{R}$, which is $\mathbb{Z}$. ◊

Example 11.8 If $n \geq 3$, the Euclidean space $\mathbb{R}^n$ is not homeomorphic to $\mathbb{R}^2$. Indeed, if $f\colon \mathbb{R}^2 \to \mathbb{R}^n$ is a homeomorphism, then $\mathbb{R}^2 \setminus \{p\} \cong \mathbb{R}^n \setminus \{f(p)\}$, but $\pi_1(\mathbb{R}^2 \setminus \{p\} = \mathbb{Z}$ and $\pi_1(\mathbb{R}^n \setminus \{f(p)\})$ is trivial.

As a consequence, the sphere $\mathbb{S}^n$ is not homeomorphic to $\mathbb{S}^2$ when $n \neq 2$. ◊

11.4 Retractions

In this section we give another method of computation of the fundamental group. The idea consists of deforming (not by homeomorphisms) a space X in a subset A in such a way that the fundamental group is preserved along the deformation. The trick is that in the final step of the deformation, the fundamental group of A can be easily computed.

Definition 11.4.1 Let A be a subset of a space X. A *retraction* from X onto A is a continuous map $r: X \to A$ such that $r(a) = a$ for all $a \in A$. We say that A is a *retract* of X. The set A is called a *deformation retract* if there is a retraction from X onto A and there is a continuous map $G: X \times \mathbf{I} \to X$, called a *deformation retraction*, such that

(i) $G(x, 0) = x$ for all $x \in X$,
(ii) $G(x, 1) = r(x)$ for all $x \in X$,
(iii) $G(a, s) = a$ for all $a \in A$.

If we fix the first variable in G, say $x \in X$, the map $G_x: \mathbf{I} \to X$ defined by $G_x(s) = G(x, s)$ is a path between x and the point $r(x) \in A$. See Fig. 11.6. This is the reason for saying that we are deforming X along G_x onto the set A. Furthermore, the points of A during the deformation are fixed. We show three illustrative examples.

Example 11.9 If A is a retract of X and X is path connected, then A is path connected: if $a, b \in A$ and $\alpha: \mathbf{I} \to X$ is a path joining a with b, then $r \circ \alpha: \mathbf{I} \to A$ is a path on A joining a with b. The converse is not true, for example, take $X = \{p, q\}$ a discrete space and let $A = \{p\}$, where $r: X \to A$ is the constant map. However, if A is a deformation retract of X, then X is connected if A is (see Exercise 11.4). ♢

Example 11.10 Consider the plane $X = \mathbb{R}^2$ and the x-axis $A = \mathbb{R} \times \{0\}$. Then A is a deformation retract of X. The idea is to map each point of $(x, y) \in \mathbb{R}^2$ to the point $(x, 0)$ by means of the map

$$r: X \to A, \quad r(x, y) = (x, 0).$$

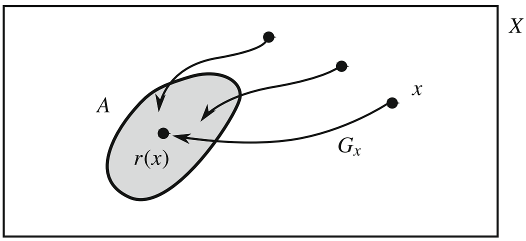


Fig. 11.6 The set A is a deformation retract of X

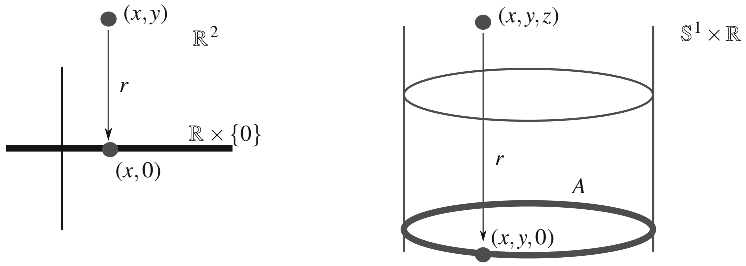


Fig. 11.7 The x-axis is a deformation retract of $\mathbb{R}^2$ (left) and the circle $A = \mathbb{S}^1 \times \{0\}$ is a deformation retract of the cylinder $X = \mathbb{S}^1 \times \mathbb{R}$ (right)

The deformation retraction is simply to move the point (x, y) of $\mathbb{R}^2$ into $(x, 0)$ along the segment $[(x, y), (x, 0)]$ that joins these points as indicated in Fig. 11.7, left. Thus $G\colon X \times \mathbf{I} \to X$ is defined by

$$G(x, y, s) = (1 - s)(x, y) + s\, r(x, y) = (x, (1 - s)y).$$

$\Diamond$

Example 11.11 Let $X \subset \mathbb{R}^n$ be a star-shaped set with center x_0. We prove that $\{x_0\}$ is a deformation retract of X. As a consequence, we have another proof that X is simply connected. The retract is clearly the constant map $r\colon X \to \{x_0\}$ and the deformation retraction shrinks the segment $[x, x_0]$ in the point x_0. Define

$$G\colon X \times \mathbf{I} \to X, \quad G(x, s) = (1 - s)x + sx_0.$$

Then $G(x, s) \in X$ because for each x, the set $\{G(x, s) : s \in \mathbf{I}\}$ is the segment $[x, x_0]$, which is included in X because X is star-shaped. $\Diamond$

Example 11.12 We see that the circle $A = \mathbb{S}^1 \times \{0\}$ is a deformation retract of the cylinder $X = \mathbb{S}^1 \times \mathbb{R}$. Define

$$r\colon X \to A, \quad r(x, y, z) = (x, y, 0).$$

See Fig. 11.7, right. Then r is continuous and $r(x, y, 0) = (x, y, 0)$ for all $(x, y, 0) \in A$. The deformation retraction parametrizes the segment joining (x, y, z) with $r(x, y, z) = (x, y, 0)$ because this segment in included in the cylinder. Thus the deformation retraction is

$$G((x, y, z), s) = (1 - s)(x, y, z) + sr(x, y, z) = (x, y, (1 - s)z).$$

$\Diamond$

Theorem 11.4.2 *Let A be a retract of X and $r\colon X \to A$ the corresponding retraction.*

1. *The map $r_*\colon \pi_1(X,a) \to \pi_1(A,a)$ is surjective and $i_*\colon \pi_1(A,a) \to \pi_1(X,a)$ is injective.*
2. *If A is a deformation retract, then $\pi_1(X) \simeq \pi_1(A)$.*

In particular, if X deforms to a point then it is simply connected.

Proof

1. Because r is a retraction, we have $r \circ i = \mathbb{1}_A$, where $i\colon A \hookrightarrow X$ is the inclusion. Then $(r \circ i)_* = \mathbb{1}_{\pi_1(A)}$, which also writes as $r_* \circ i_* = \mathbb{1}_{\pi_1(A)}$. Thus this composition is a bijective map and it is known from set theoretical arguments that r_* is surjective and i_* is injective.
2. It remains to prove that $i_*\colon \pi_1(A,a) \to \pi_1(X,a)$ is surjective, $a \in A$. We distinguish the notation for equivalence classes. Let $[[\alpha]]$ and $[\alpha]$ denote the class of α in $\pi_1(A,a)$ and $\pi_1(X,a)$, respectively. If $[\alpha] \in \pi_1(X,a)$, we prove that $i_*([[r \circ \alpha]]) = [\alpha]$. Note that

$$i_*([[r \circ \alpha]]) = [i \circ r \circ \alpha] = [r \circ \alpha].$$

Thus we must prove that $r \circ \alpha$ is homotopic to α viewed as loops in X. Define

$$F\colon \mathbf{I} \times \mathbf{I} \to X, \quad F(t,s) = G(\alpha(t), s),$$

where G is the deformation retraction from X to A. Then F is continuous and trivially satisfies the properties to be a homotopy between $r \circ \alpha$ and r.

□

11.5 Worked Exercises

Exercise 11.1 Let $\mathbb{D}^2$ be the unit disc and consider

$$A = \{x \in \mathbb{D}^2 : \mathbb{D}^2 \setminus \{x\} \text{ is simply connected}\}.$$

Prove that $A = \mathbb{S}^1$.

Solution The inclusion $\mathbb{S}^1 \subset A$ is clear because for any $x \in \mathbb{S}^1$, the set $\mathbb{D}^2 \setminus \{x\}$ is convex, so it is simply connected.

Now, let $a \in A$. By contradiction, suppose that $a \notin \mathbb{S}^1$. We make use of Exercise 6.27. There is a homeomorphism $f\colon \mathbb{D}^2 \to \mathbb{D}^2$ such that $f(a) = 0$. Then $\mathbb{D}^2 \setminus \{a\} \cong \mathbb{D}^2 \setminus \{0\}$, but this latter set is an open-closed annulus homeomorphic to $\mathbb{S}^1 \times (0,1]$ (Exercise 6.6.6). Thus $\pi_1(\mathbb{D}^2 \setminus \{a\}) = \mathbb{Z}$, a contradiction. □

We know that any homeomorphism $f: [a, b] \to [c, d]$ satisfies $f(\{a, b\}) = \{c, d\}$. We extend this result to dimension 2, replacing $[a, b]$ by $\mathbb{D}^2$ and $\{a, b\}$ by $\mathbb{S}^1$.

Exercise 11.2 If $f: \mathbb{D}^2 \to \mathbb{D}^2$ is a homeomorphism, prove that $f(\mathbb{S}^1) = \mathbb{S}^1$.

Solution By contradiction, assume that $f(\mathbb{S}^1) \not\subset \mathbb{S}^1$, so there is $a \in \mathbb{S}^1$ such that $f(a) \notin \mathbb{S}^1$. Since f is a homeomorphism, f induces an isomorphism $f_*: \pi_1(\mathbb{D}^2) \to \pi_1(\mathbb{D}^2)$. The restricted map

$$f_*: \pi_1(\mathbb{D}^2 \setminus \{a\}) \to \pi_1(\mathbb{D}^2 \setminus \{f(a)\})$$

is an isomorphism. However, $\pi_1(\mathbb{D}^2 \setminus \{a\})$ is trivial because $\mathbb{D}^2 \setminus \{a\}$ is convex but $\mathbb{D}^2 \setminus \{f(a)\}$ is not simply connected by Exercise 11.1. This yields a contradiction.

The same argument is valid for the inverse function, we obtain $f^{-1}(\mathbb{S}^1) \subset \mathbb{S}^1$, and consequently, $f(\mathbb{S}^1) = \mathbb{S}^1$. □

Exercise 11.3 Prove that the half disc $A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1, y \geq 0\}$ is not homeomorphic to the ball $B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$.

Solution Assume that $f: A \to B$ is a homeomorphism. Then $A \setminus \{(0, 0)\} \cong B \setminus \{f(0, 0)\}$. We see that the fundamental groups of these spaces do no coincide. The fundamental group of $A \setminus \{(0, 0)\}$ is trivial because the space is star-shaped from the point $(0, \frac{1}{2})$.

We see that the fundamental group of $B \setminus \{f(0, 0)\}$ is $\mathbb{Z}$. In Exercise 6.27 it was proved that, given two points of B, there is a homeomorphism from B into B mapping one point to the other. Taking the origin and $f(0, 0)$, we conclude that $B \setminus \{f(0, 0)\} \cong B \setminus \{(0, 0)\}$. The latter space is an annulus, so it is homeomorphic to the cylinder $\mathbb{S}^1 \times \mathbb{R}$ and its fundamental group is $\mathbb{Z}$. □

Exercise 11.4 If A is a deformation retract of X and A is path connected, prove that X is path connected.

Solution Denote by $G: X \times \mathbf{I} \to X$ the deformation retraction. Let $x, y \in X$. Since $r(x)$ and $r(y)$ belong to A, there is a path $\alpha: \mathbf{I} \to A$ joining $r(x)$ with $r(y)$. Consider the maps $\beta, \gamma: \mathbf{I} \to X$ defined by

$$\beta(t) = G(x, t), \quad \gamma(t) = G(y, t).$$

Then β is a map with $\beta(0) = G(x, 0) = x$ and $\beta(1) = G(x, 1) = r(x)$, and analogously, $\gamma(0) = y$ and $\gamma(1) = r(y)$.

We see that β is continuous. The map β is composition of two maps, $\beta = G \circ h$, where $h: \mathbf{I} \to X \times \mathbf{I}$ is defined by $h(t) = (x, t)$. Let us notice that $p_1 \circ h$ is constant and $p_2 \circ h$ is the identity. This proves that h is continuous and, consequently, β is continuous. A similar argument holds for γ. Then β and γ are paths and consequently, $\beta * \alpha * \gamma^{-1}$ is a path joining x with y. □

We see that the property of deformation retract is preserved by homeomorphisms.

Exercise 11.5 Let $f: X \to Y$ be a homeomorphism and let A be a deformation retract of X. Prove that $f(A)$ is a deformation retract of Y.

Solution The idea for defining the retraction as well as the deformation retraction on Y is to carry the problem to the domain X of f. Let $r: X \to A$ be a retract from X onto A, and let $F: X \times \mathbf{I} \to X$ be the deformation retraction. Let $g = f \circ r \circ f^{-1}: Y \to f(A)$, which is obviously continuous. For any $a \in A$, we have $g(f(a)) = f(r(a)) = f(a)$, hence $g_{|f(A)}$ is the identity in $f(A)$. Define

$$G: Y \times \mathbf{I} \to Y, \quad G(y,t) = f \circ F(f^{-1}(y), t).$$

Then G is continuous because it is composition of two maps, namely, $f \circ F$ and

$$H: Y \times \mathbf{I} \to X \times \mathbf{I}, \quad H(y,t) = (f^{-1}(y), t).$$

For the continuity of H, we compose with the projections p_1 and p_2 of $X \times \mathbf{I}$. Then $p_1 \circ H(y,t) = f^{-1}(y)$ and $p_2 \circ H(y,t) = t$. If q_1 and q_2 are the projections of $Y \times \mathbf{I}$, then $p_1 \circ H = f^{-1} \circ q_1$ and $p_2 \circ H = q_2$, which are continuous. Finally the mapping G obeys the properties to be a deformation retraction of Y in $f(A)$. □

We show that the deformation retract is transitive by inclusions.

Exercise 11.6 Let A and B be two subsets of a space X with $A \subset B$. If A is a retract (resp. deformation retract) of B and B is a retract (resp. deformation retract) of X, prove that A is a retract (resp. deformation retract) of X.

Solution

1. Denote by r and r' the corresponding retractions $r': B \to A$ and $r: X \to B$. Then $r' \circ r: X \to A$ is continuous and for all $a \in A$, $r'(a) = a$, so $r(r'(a)) = r(a)$ because $A \subset B$.
2. Let $G: B \times \mathbf{I} \to B$ and $F: X \times \mathbf{I} \to X$ be the corresponding deformation retractions of B in A and of X in B. Define

$$H: X \times \mathbf{I} \to X, \quad H(x,t) = \begin{cases} F(x, 2t), & 0 \le t \le \frac{1}{2} \\ G(r(x), 2t-1,) & \frac{1}{2} \le t \le 1. \end{cases}$$

The map is well-defined because at $t = 1/2$, $F(x,1) = r(x)$ and $G(r(x), 0) = r(x)$. Since H is defined piecewise, we apply the Pasting Lemma. The subdomains of H are $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$, which are closed sets. It is also obvious that H is continuous in each one of that closed sets. Finally, the properties for H of being a deformation retract are immediate. For example, if $a \in A$, we have

$$H(a,t) = \begin{cases} F(a, 2t) = a, & 0 \le t \le \frac{1}{2} \\ G(r(a), 2t-1) = r(a) = a, & \frac{1}{2} \le t \le 1. \end{cases}$$

□

Exercise 11.7 Let A and B be two subsets of a space X, with $X = A \cup B$. Assume that A and B are closed (or both open). If $A \cap B$ is a deformation retract of B, prove that A is a deformation retract of X.

Solution The result is intuitive because B deforms into $A \cap B$ which is also included in A, then $A \cup B$ will deform in $A \cap B$, keeping fixed the points of A. The points of B will be mapped to those points of $A \cap B$ by the retract of B.

We perform the proof assuming that A and B are closed; the arguments are similar when they are open. Let $r\colon B \to A \cap B$ be a deformation retract and let $F\colon B \times \mathbf{I} \to B$ be the corresponding deformation retraction. Define

$$r'\colon X \to A, \quad r'(x) = \begin{cases} x, & \text{if } x \in A \\ r(x), & \text{if } x \in B. \end{cases}$$

This map is well-defined because if $x \in A \cap B$, then $r(x) = x$. On the other hand, the domain of r' is formed by two closed sets. Then the Pasting Lemma asserts that r' is continuous if the restriction in each one them is continuous. But this is immediate because $r'_{|A} = \mathbb{1}_A$ and $r'_{|B} = r \circ i$, where $i\colon A \cap B \hookrightarrow A$ is the inclusion. Moreover, $r'(a) = a$ for all $a \in A$. For the deformation retract, define

$$G\colon X \times \mathbf{I} \to X, \quad G(x,t) = \begin{cases} x, & \text{if } x \in A \\ F(x,t), & \text{if } x \in B. \end{cases}$$

We see that G is a deformation retraction from X into A. Again, the mapping G is well-defined because if $x \in A \cap B$, $F(x,t) = x$. Continuity of G follows from the Pasting Lemma because the domain of G is decomposed in two closed sets, $A \times \mathbf{I}$ and $B \times \mathbf{I}$. Finally, we see that G obeys the properties of the deformation retraction. It is clear that $G(a,t) = a$ for all $a \in A$. Furthermore,

(i) $G(x,0) = \begin{cases} x, & \text{if } x \in A \\ F(x,0) = x, & \text{if } x \in B. \end{cases}$

(ii) $G(x,1) = \left.\begin{cases} x, & \text{if } x \in A \\ F(x,1) = r(x), & \text{if } x \in B \end{cases}\right\} = r'(x).$ □

Exercise 11.8 Consider the circles of Euclidean plane given by (Fig. 11.8)

$$C_1 = \{(x,y) \in \mathbb{R}^2 : (x-1)^2 + y^2 = 1\}, \quad C_2 = \{(x,y) \in \mathbb{R}^2 : (x+1)^2 + y^2 = 1\}.$$

Prove that $(0,0)$ is a deformation retract of $X = C_1 \cup C_2 \setminus \{(-2,0),(2,0)\}$.

Solution Using the notation of Exercise 11.7, let $A = C_1$ and $B = C_2$ and $X = C_1 \cup C_2$. The sets A and B are closed in X because $A = X \cap \{(x,y) \in \mathbb{R}^2 : x \geq 0\}$ and $B = X \cap \{(x,y) \in \mathbb{R}^2 : x \leq 0\}$. The intersection $A \cap B$ is $\{(0,0)\}$. This point is a deformation retract of B because $B \cong \mathbb{R}$ and any point is a deformation retract: here we are using Exercise 11.5. Thus by Exercise 11.7 we conclude that C_1 is a deformation retract of X. Using again Exercise 11.7, $\{(0,0)\}$ is a deformation

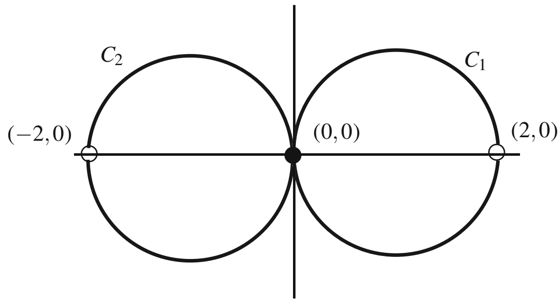


Fig. 11.8 Exercise 11.8. The set $(C_1 \cup C_2)\backslash\{(-2, 0), (2, 0)\}$ deforms onto the point $(0, 0)$

retract of C_1. Finally, we apply Exercise 11.6 to deduce that $\{(0, 0)\}$ is a deformation retract of $C_1 \cup C_2$. □

We show that retracts and deformation retracts are productive.

Exercise 11.9 Let A and B two retracts (resp. deformation retracts) of X and Y, respectively. Prove that $A \times B$ is a retract (resp. deformation retract) of $X \times Y$.

Solution If $r : X \to A$ and $r' : Y \to B$ are retracts, it is immediate that $r \times r' : X \times Y \to A \times B$ is a retract. Assume that $F : X \times \mathbf{I} \to X$ and $G : Y \times \mathbf{I} \to Y$ are the two deformation retractions from X in A and from Y in B, respectively. Define

$$H : (X \times Y) \times \mathbf{I} \to X \times Y, \quad H(x, y, t) = (F(x, t), G(y, t)).$$

For the continuity of H, we compose with the projections p and p' of $X \times Y$ and we see that are continuous maps (notice that $p \circ H \neq F$ and $p' \circ H \neq G$). Then $p(H(x, y, t)) = F(x, t)$. The map $p \circ H$ is composition of two maps. The first one is F and the second one is

$$X \times Y \times \mathbf{I}, \quad (x, y, t) \to (x, t).$$

This map is continuous because it is the first projection of the product space $(X \times \mathbf{I}) \times Y$. On the other hand, H satisfies the properties of a deformation retraction,

(i) $H(x, y, 0) = (F(x, 0), G(y, 0)) = (x, y)$, for all $(x, y) \in X \times Y$.
(ii) $H(x, y, 1) = (F(x, 1), G(y, 1)) = (r(x), r'(y)) = (r \times r')(x, y)$, for all $(x, y) \in X \times Y$.
(iii) If $(a, b) \in A \times B$ then $H(a, b, t) = (F(a, t), G(b, t)) = (a, b)$. □

Exercise 11.10 Let $x_0 \in \mathbb{S}^1$. Prove that $\mathbb{S}^1 \times \{x_0\}$ is a retract of $\mathbb{S}^1 \times \mathbb{S}^1$ but it is not a deformation retract.

Solution Define

$$r : \mathbb{S}^1 \times \mathbb{S}^1 \to \mathbb{S}^1 \times \{x_0\}, \quad r(x, y) = (x, x_0).$$

This map is continuous and the restriction to $\mathbb{S}^1 \times \{x_0\}$ is the identity $r(x, x_0) = (x, x_0)$. However, if $\mathbb{S}^1 \times \{x_0\}$ were a deformation retract of $\mathbb{S}^1 \times \mathbb{S}^1$, then these spaces would have the same the fundamental groups, which it is not true because $\mathbb{S}^1 \times \{x_0\} \cong \mathbb{S}^1$, so $\pi_1(\mathbb{S}^1 \times \{x_0\}) = \mathbb{Z}$ and $\pi_1(\mathbb{S}^1 \times \mathbb{S}^1) = \mathbb{Z} \times \mathbb{Z}$. □

We give a necessary condition for a subset to be a retract in a Hausdorff space.

Exercise 11.11 Prove that a retract of a Hausdorff space is a closed set.

Solution Let A be a retract of a Hausdorff space X and let $r : X \to A$ be the retraction. We prove that $X \setminus A$ is an open set. Let $x \in X \setminus A$ and we see that x is an interior point. Since $r(x) \in A$, then $r(x) \neq x$. We make use of the Hausdorff property of X to assert that there are two disjoint open sets O_1 and O_2 such that $r(x) \in O_1$ and $x \in O_2$.

The set $O_1 \cap A$ is open in the relative topology of A and $r : X \to A$ is continuous, so $r^{-1}(O_1 \cap A)$ is open in X. If we consider $O = O_2 \cap r^{-1}(O_1 \cap A)$, then O is open in X with $x \in O$. The proof is completed if we prove that $O \subset X \setminus A$. On the contrary, if $y \in O \cap A$ then $y \in O_2$ and

$$r(y) \in r(O) = r(r^{-1}(O_1 \cap A)) \subset O_1 \cap A.$$

Since $y \in A$, $r(y) = y$, hence $y \in O_1 \cap A \subset O_1$. This is contradictory because O_1 and O_2 are disjoint. □

Exercise 11.12 Consider the set X formed by $[0, 1] \times \{0\}$ together with all segments between $(0, 0)$ and the points $(1, \frac{1}{n})$, $n \in \mathbb{N}$. Let $p = (1, 0)$ and $A = \{p\}$. Prove that A is a retract of X but is not a deformation retract.

Solution Since A is formed by one point, the constant map $r : X \to A$ is a retraction. We prove that A is not a deformation retract of X. The idea is that during the possible deformation, all points $(1, \frac{1}{n})$ will deform towards p; in particular, the deformation must across the origin. In the limit, and by continuity, the point $(0, 0)$ remains fixed, which is impossible because the deformation retraction ends at p.

We argue by contradiction. Let $F : X \times \mathbf{I} \to X$ be a deformation retraction of X into A. Let $n \in \mathbb{N}$ be a fixed (but arbitrary) natural number. Set $p_n = (1, \frac{1}{n})$ and define $\alpha_n(t) = F(p_n, t)$. This map is continuous because it is composition of $t \mapsto (p_n, t)$ and F. Moreover,

$$\alpha_n(0) = F(p_n, 0) = p_n = (1, \tfrac{1}{n}), \quad \alpha_n(1) = F(p_n, 1) = p = (1, 0).$$

We claim that there is $t \in \mathbf{I}$ such that $\alpha_n(t) = (0, 0)$. Suppose on the contrary that $\alpha_n(\mathbf{I}) \subset X \setminus \{(0, 0)\}$. The set $X \setminus \{(0, 0)\}$ is not connected; its connected components are the segment between p with $(0, 0)$ (excluding the origin) and all the segments joining p_k with $(0, 0)$ (excluding the origin). In fact, we only need the existence of an open separation which is obtained by the line L with equation $y = 2x/3$.

This line leaves on either side of it the points $(1, 1)$ and $(1, \frac{1}{2})$. Then L separates $\mathbb{R}^2 \setminus \{(0, 0)\}$ in two open sets,

$$O_1 = \{(x, y) \in \mathbb{R}^2 : y > 2x/3\}, \quad O_2 = \{(x, y) \in \mathbb{R}^2 : y < 2x/3\}.$$

Both open sets induce an open separation in $X \setminus \{(0, 0)\}$ because L does not intersect $X \setminus \{(0, 0)\}$. Note that $p_n \in O_1$ and $p \in O_2$. The contradiction arrives because $\alpha_n(\mathbf{I})$ is connected where the endpoints of α_n are contained in distinct components.

As a result of the claim, we may select one solution, say t_n, of $\alpha_n(t) = (0, 0)$. With this process, we build a sequence $(t_n) \subset \mathbf{I}$ such that $\alpha_n(t_n) = (0, 0)$. Since $\mathbf{I}$ is compact, there is a convergent subsequence, that we still call (t_n). Let $(t_n) \to \bar{t}$ and the sequence $(\alpha_n(t_n)) = (F(p_n, t_n))$. By continuity,

$$\lim_{n\to\infty}(F(p_n, t_n)) = F(p, \bar{t}) = p$$

because $(p_n) \to p$. However, $\alpha_n(t_n) = (0, 0)$, a contradiction. □

Exercise 11.13 For each $q \in \mathbb{Q}$, let L_q be the segment of the Euclidean plane $\mathbb{R}^2$ given by $[(0, 0), (q, 1)]$. Let $X = \bigcup_{q\in\mathbb{Q}} L_q$. Prove that $(0, 0)$ is a deformation retract of X but $(0, 1)$ is not.

Solution

1. For the first part, it suffices to deform each segment L_q from $(q, 1)$ until the origin. So, define

$$F : X \times \mathbf{I} \to X, \quad F(x, y, t) = (1 - t)(x, y) + t(0, 0) = (1 - t)(x, y).$$

   The map F is well-defined for if $(x, y) \in X$ then there is $s \in [0, 1]$ and $q \in \mathbb{Q}$ such that $(x, y) = s(q, 1)$. Thus $F(x, y, t) = (1-t)s(q, 1)$ and $(1-t)s \in [0, 1]$, so $F(x, y, t) \in L_q$. The map F is continuous and it is immediate that F is a deformation retraction with $F(x, y, 1) = (0, 0)$.
2. For the second part, assume by contradiction that $(0, 1)$ is a deformation retract of X. Let G be such a deformation retraction. Now we cannot utilize a similar map as above because the segment $[(x, y), (0, 1)]$ is not included in X. On the other hand, X is path connected because it is star-shaped from the origin but $X \setminus \{(0, 0)\}$ is not connected, being the connected components each one of the pieces $L_q \setminus \{(0, 0)\}$ (Exercise 8.23). Let $q_n = 1/n$. For each $n \in \mathbb{N}$, let $t_n \in [0, 1]$ such that $G(q_n, 1, t_n)) = (0, 0)$. By passing to a subsequence if necessary, we may assume that $(t_n) \to \bar{t}$. By continuity

$$G(q_n, 1, t_n) \to G(0, 1, \bar{t}) = (0, 1)$$

   because $G(0, 1, t) = (0, 1)$ for all $t \in \mathbf{I}$. This is in contradiction with that $G(q_n, 1, t_n) = (0, 0)$ for all $n \in \mathbb{N}$. □

We extend the fixed-point theorem to dimension $n = 2$. The next result known as the Brouwer fixed point theorem is valid in arbitrary dimensions.

Exercise 11.14 (Brouwer Fixed Point Theorem) If $f : \mathbb{D}^2 \to \mathbb{D}^2$ is continuous, prove that there is $x \in \mathbb{D}^2$ such that $f(x) = x$.

Solution The proof is by contradiction. Assume $f(x) \neq x$ for all $x \in \mathbb{D}^2$. For each $x \in \mathbb{D}^2$, the points $f(x)$ and x determine a unique straight line. Consider the ray from $f(x)$ and passing through x. This ray meets $\mathbb{S}^1$ at a unique point that is denoted by $r(x)$. In this way, we establish a map $r : \mathbb{D}^2 \to \mathbb{S}^1$. Suppose at this point, that r is continuous and we conclude the proof. First, note that if $x \in \mathbb{S}^1$ then it is obvious by the definition of r that $r(x) = x$. Therefore, if $i : \mathbb{S}^1 \hookrightarrow \mathbb{D}^2$ is the inclusion, then $r \circ i = \mathbb{1}_{\mathbb{S}^1}$. In view of Theorem 11.4.2, $r_* : \pi_1(\mathbb{D}^2) \to \pi_1(\mathbb{S}^1)$ is surjective. This, however, is absurd since $\pi_1(\mathbb{D}^2)$ is trivial and $\pi_1(\mathbb{S}^1) = \mathbb{Z}$.

Now we prove the continuity of r. To accomplish this, it is enough to write down the map r rather explicitly. The point $r(x)$ is

$$r(x) = f(x) + \lambda(x - f(x)), \tag{11.2}$$

where the value of $\lambda = \lambda(x)$ is computed subject to the condition $|r(x)| = 1$ and $\lambda > 0$ because we are considering the ray from $f(x)$. Equation $|r(x)|^2 - 1 = 0$ is

$$|f(x)+\lambda(x-f(x)|^2-1 = |x-f(x)|^2\lambda^2+2\langle f(x), x-f(x)\rangle\lambda+|f(x)|^2-1 = 0.$$

We find two values of λ and we must choose the positive one,

$$\lambda(x) = \frac{-\langle f(x), x - f(x)\rangle + \sqrt{\langle f(x), x - f(x)\rangle^2 - |x - f(x)|^2(|f(x)|^2 - 1)}}{|x - f(x)|^2}.$$

If we replace this value of $\lambda(x)$ into (11.2), we obtain an algebraic expression of $r(x)$ in terms of continuous functions. This proves that r is continuous. As a final remark, notice that if $r(x)$ were defined as the intersection point between the ray from x through $f(x)$ with $\mathbb{S}^1$, we would not know if the restriction of r to $\mathbb{S}^1$ is the identity. □

Exercise 11.15 Find the fundamental group of the following subsets of the Euclidean space $\mathbb{R}^3$.

1. The solid cylinder $x^2 + y^2 \leq 1$.
2. The lower hemisphere $\mathbb{S}^2_-$.
3. The sphere $\mathbb{S}^2$ together with the straight line $\{(0, 1, z) \in \mathbb{R}^3 : z \in \mathbb{R}\}$.
4. The circle $\mathbb{S}^1$ together the segment $[1, 2] \times \{0\}$.
5. The circle $\mathbb{S}^1$ together the half-lines $[1, \infty) \times \{0\}$, $\{0\}[1, \infty)$, $[-\infty, -1] \times \{0\}$ and $\{0\} \times (-\infty, -1]$.
6. The solid cylinder $x^2 + y^2 \leq 1$ and the plane $z = 0$.

7. The lower hemisphere $\mathbb{S}^2_-$ and the circle $\{0\} \times \mathbb{S}^1$.
8. The union of the solid cylinders $x^2 + y^2 \leq 1$ and $x^2 + z^2 \leq 1$.

Solution

1. The solid cylinder is $\mathbb{D}^2 \times \mathbb{R}$, so its fundamental group is $\pi_1(\mathbb{D}^2) \times \pi_1(\mathbb{R})$. This group is trivial.
2. The lower hemisphere is homeomorphic to $\mathbb{D}^2$ because $\mathbb{S}^2_-$ is the graph of the function $f(x, y) = -\sqrt{1 - x^2 - y^2}$ defined on $\mathbb{D}^2$. Consequently, the fundamental group is trivial.
3. Apply Exercise 11.7 by letting $A = \mathbb{S}^2$ and let B the straight line. These sets are closed in $\mathbb{R}^3$, so closed in $A \cup B$. The set B is a straight line and intersects $\mathbb{S}^2$ precisely at $p = (0, 1, 0)$. Since $A \cap B = \{p\}$ is a deformation retract of B because B is convex, we deduce that $\mathbb{S}^2$ is a deformation retract of $\mathbb{S}^2 \cup B$. Using that $\pi_1(\mathbb{S}^2)$ is trivial, we conclude that $\mathbb{S}^2 \cup B$ is simply connected.
4. As in the above example, $\mathbb{S}^1$ is a deformation retract of the space, so the fundamental group is that of $\mathbb{S}^1$, that is, $\mathbb{Z}$.
5. As in the preceding item, $\mathbb{S}^1$ is a deformation retract of the space and, consequently, the fundamental group is $\mathbb{Z}$.
6. We employ Exercise 11.7 again where A is $x^2 + y^2 \leq 1$ and B is the plane $z = 0$. The intersection $A \cap B$ is the unit disc in the xy-plane. The sets A and B are closed in $X = A \cup B$ because they are closed in $\mathbb{R}^3$, specifically $A = \mathbb{D}^2 \times \mathbb{R}$ is the product of two closed sets and B is a plane of $\mathbb{R}^3$. On the other hand, $A \cap B = \mathbb{D}^2 \times \{0\}$ is homeomorphic to $\mathbb{D}^2$. Since B is convex, $A \cap B$ is a deformation retract of B. Definitively, A is a deformation retract of X. Finally, A is simply connected, as well as X.

   Another way is to observe that $A \cup B$ is star-shaped from the origin because A and B are convex sets and $(0, 0, 0) \in A \cap B$. Recall that the union of two not-disjoint convex sets is a star-shaped set from any point of the intersection.
7. Let $A = \mathbb{S}^2_-$ and $B = \{0\} \times \mathbb{S}^1$. Think of $A \cup B$ like a single-handle basket. Use Exercise 11.7. The sets A and B are closed sets in $\mathbb{R}^3$, so closed in $X = A \cup B$. The intersection is

$$A \cap B = \{(0, y, z) \in \mathbb{R}^3 : y^2 + z^2 = 1, z \leq 0\} = \mathbb{S}^1_- \times \{0\}.$$

   This set is homeomorphic to $\mathbb{S}^1_- \cong [-1, 1]$. To see that $A \cap B$ is a deformation retract of A, use Exercise 11.5. Recall that $A \cong \mathbb{D}^2$. The image under this homeomorphism of $A \cap B$ is the segment $[-1, 1] \times \{0\}$. We prove that $[-1, 1] \times \{0\}$ is a deformation retract of $\mathbb{D}^2$. It is enough to define

$$r \colon \mathbb{D}^2 \to [-1, 1] \times \{0\}, \quad r(x, y) = x,$$

$$F \colon \mathbb{D}^2 \times \mathbf{I} \to \mathbb{D}^2, \quad F((x, y), t) = (1 - t)(x, y) + t(x, 0).$$

Then Exercise 11.5 shows that $A \cap B$ is a deformation retract of A. Thus B is a retract of X, so $\pi_1(B) = \pi_1(X)$. Since $B \cong \mathbb{S}^1$, $\pi_1(X) = \mathbb{Z}$.
8. Each of the two solid cylinders are convex (product of two convex sets). Thus the set is star-shaped from any point of the intersection, therefore the space is simply connected. □

We establish an easy way to compute the degree of a loop of $\mathbb{S}^1$ in case that the loop is expressed by differentiable functions.

Exercise 11.16 Let $\alpha\colon [0, 1] \to \mathbb{S}^1$ be a loop in $\mathbb{S}^1$ with base point $1 \in \mathbb{S}^1$ and let $\alpha(t) = (x(t), y(t))$. If the functions $x = x(t)$ and $y(t)$ are differentiable, prove

$$\deg \alpha = \frac{1}{2\pi} \int_0^1 (x(t)y'(t) - x'(t)y(t))dt.$$

Solution Let $p\colon \mathbb{R} \to \mathbb{S}^1$ be the function $p(t) = (\cos 2\pi t, \sin 2\pi t)$. We know that p is a homeomorphism when restricting p to intervals of length less than 2π. Also it is clear that p is differentiable. We see that the inverse $\psi = p^{-1}$ of p is also differentiable. Because this is a local question, we check it in small intervals. If we set $x = \cos t$ and $y = \sin t$, then $y/x = \tan t$ and $x/y = \cot t$. Thus the inverse ψ is either $\tan^{-1}(y/x)$ or $\cot^{-1}(x/y)$, depending if $x \neq 0$ or $y \neq 0$, respectively. In either case, ψ is differentiable.

Once established that ψ is differentiable, we come back to look the details of the construction of the lifting $\widetilde{\alpha}$ in Proposition 11.2.1. The definition of $\widetilde{\alpha}$ is piecewise defined in subintervals and in terms of α and ψ. It is clear that $\widetilde{\alpha}$ is differentiable in all open subintervals $(jt/N, (j+1)t/N)$ and continuous at the points of $A = \{jt/N : j = 0, \ldots, N\}$. Writing α in coordinates,

$$\alpha(t) = (x(t), y(t)) = p(\widetilde{\alpha}(t)) = (\cos 2\pi\widetilde{\alpha}(t), \sin 2\pi\widetilde{\alpha}(t)),$$

then $x(t) = \cos 2\pi\widetilde{\alpha}(t)$ and $y(t) = \sin 2\pi\widetilde{\alpha}(t)$. Differentiating with respect to t,

$$x'(t) = -2\pi\widetilde{\alpha}'(t)\sin 2\pi\widetilde{\alpha}(t), \quad y'(t) = 2\pi\widetilde{\alpha}'(t)\cos 2\pi\widetilde{\alpha}(t).$$

These identities hold except perhaps at the points of A. Thus

$$x(t)y'(t) - x'(t)y(t) = 2\pi\widetilde{\alpha}'(t)$$

for all $t \in [0, 1] \setminus A$. By definition of the degree of a loop,

$$\deg \alpha = \widetilde{\alpha}(1) = \int_0^1 \frac{d}{dt}\widetilde{\alpha}(t)\, dt = \frac{1}{2\pi}\int_0^1 (x(t)y'(t) - x'(t)y(t))\, dt.$$

This integral is well-defined because the integrand is continuous and it is differentiable for all but finitely many points in $[0, 1]$. □

Exercise 11.17 Let $\alpha\colon \mathbf{I} \to X$ be a continuous map. A reparametrization of α is defined as a map $\beta = \alpha \circ \phi$, where $\phi\colon \mathbf{I} \to \mathbf{I}$ is a continuous map with $\phi(0) = 0$ and $\phi(1) = 1$. Prove that α is homotopic to β.

Solution Define

$$F\colon \mathbf{I} \times \mathbf{I} \to X, \quad F(t, s) = \alpha((1-s)t + s\phi(t)).$$

The map F is well-defined because, fixed $t \in \mathbf{I}$, $\{(1-s)t + s\phi(t) : s \in \mathbf{I}\}$ is the segment $[t, \phi(t)]$. The segment $[t, \phi(t)]$ is included in $\mathbf{I}$ because $t, \phi(t) \in \mathbf{I}$. On the other hand, the map F is continuous and obeys the properties of a homotopy between α and $\alpha \circ \phi$:

(i) $F(t, 0) = \alpha(t)$ and $F(t, 1) = \alpha(\phi(t))$, for all $t \in \mathbf{I}$,
(ii) $F(0, s) = \alpha(\phi(0)) = \alpha(0)$ and $F(1, s) = \alpha(\phi(1)) = \alpha(1)$ for all $s \in \mathbf{I}$. □

Exercise 11.18 Prove Lemma 11.3.1.

Solution Both maps are well-defined; that is, $\bar{t}t \in \mathbf{I}$ and $\bar{t} + t(1-\bar{t}) \in \mathbf{I}$ for all $t \in \mathbf{I}$. Moreover, $\alpha_1(1) = \alpha(\bar{t}) = \alpha_2(0)$, so it makes sense the operation $\alpha_1 * \alpha_2$. Let us also observe that $\alpha_1 * \alpha_2$ describes the same image that α, but with different velocities: in the domain of α_1, the parameter is $\bar{t}t$ and in of α_2 is $\bar{t} + t(1-\bar{t})$. Since $\alpha_1 * \alpha_2$ is simply traveling along α with different speed, we will use Exercise 11.17 to show that $\alpha_1 * \alpha_2$ is a reparametrization of α.

We only need to change the parameter so the interval $[0, \bar{t}]$ maps on $[0, \frac{1}{2}]$ and $[\bar{t}, 1]$ on $[\frac{1}{2}, 1]$. To do this, we use affinities $x \mapsto \lambda x + \mu$. In the first case the map is $t \mapsto 2\bar{t}t$ and in the second one, $t \mapsto 1 + 2\bar{t}(1-t)$. Define $\phi\colon [0, 1] \to [0, 1]$ by

$$\phi(t) = \begin{cases} 2\bar{t}\,t, & 0 \le t \le \frac{1}{2} \\ 1 + 2\bar{t}(1-t), & \frac{1}{2} \le t \le 1. \end{cases}$$

It is clear that ϕ is well-defined at $t = 1/2$ and that ϕ is continuous by the Pasting Lemma. Finally, $\alpha \circ \phi(t) = (\alpha_1 * \alpha_2)(t)$. □

Exercise 11.19 For $n \ge 2$, find the fundamental group of two n-spheres of $\mathbb{R}^{n+1}$ that are tangent at a point.

Solution Let A and B be two spheres such that $A \cap B = \{p\}$. Denote $X = A \cup B$ and apply the Seifert-van Kampen theorem. Let $a \in A$ and $b \in B$ be two distinct points of p. Let $O_1 = A \cup (B \setminus \{b\})$ and $O_2 = B \cup (A \setminus \{a\})$. We check that the hypotheses of the theorem hold.

1. The sets O_1 and O_2 are open in X. For O_1, $\{b\}$ is closed in X, so $X \setminus \{b\} = O_1$ is open, and likewise O_2 is open.
2. The sets O_1, O_2 and $O_1 \cap O_2$ are path connected. For O_1, the set A is a sphere and $B \setminus \{b\}$ is homeomorphic to $\mathbb{R}^2$. Since $A \cap (B \setminus \{b\}) = \{p\}$, the union $O_1 = A \cup (B \setminus \{b\})$ is path connected. For O_2 the argument is similar. We see

that $O_1 \cap O_2 = X \setminus \{a, b\}$ is path connected. But $X \setminus \{a, b\} = (A \setminus \{a\}) \cup (B \setminus \{b\})$ is union of two path connected sets with nonempty intersection.

3. We prove that O_1 and O_2 are simply connected. The proof will be for O_1 (and analogously for O_2). Let us utilize Exercise 11.7. Let $P = A$ and $Q = B \setminus \{b\}$.

   (a) The sets P and Q are closed in O_1. The sphere P is closed in $\mathbb{R}^{n+1}$. For Q (that it is not closed in $\mathbb{R}^{n+1}$), and using that B is closed in $\mathbb{R}^{n+1}$, we have

   $$\overline{B \setminus \{b\}}^{O_1} = \overline{B \setminus \{b\}} \cap O_1 = B \cap O_1 = B.$$

   (b) The point $\{p\} = P \cap Q$ is a deformation retract of Q. Certainly, Q is a punctured sphere, so $Q \cong \mathbb{R}^n$. Since every point of $\mathbb{R}^n$ is a deformation retract of $\mathbb{R}^n$, the same occurs for Q by Exercise 11.5.

   Thus P is a deformation retract of $P \cup Q = O_1$. Then $\pi_1(O_1) = \pi_1(P) = \pi_1(\mathbb{S}^n)$ is trivial because $n \geq 2$. □

Exercise 11.20 Let α, β and γ be three loops with the same base point such that

$$\alpha * (\beta * \gamma) = (\alpha * \beta) * \gamma.$$

Prove that α, β and γ are constant paths.

Solution We evaluate these paths at values $t \leq 1/2$. For $\alpha * (\beta * \gamma)$, the value at $t \in [0, \frac{1}{2}]$ is $\alpha(2t)$. For $(\alpha * \beta) * \gamma$ and at $t \leq 1/2$,

$$(\alpha * \beta)(2t) = \begin{cases} \alpha(4t), & \text{if } 0 \leq 2t \leq \frac{1}{2} \\ \beta(4t-1), & \text{if } \frac{1}{2} \leq 2t \leq 1. \end{cases}$$

Since $\alpha * (\beta * \gamma) = (\alpha * \beta) * \gamma$, we deduce that $\alpha(2t) = \alpha(4t)$ for all $0 \leq t \leq 1/4$.

Let $t \in \mathbf{I}$. Then $\alpha(t) = \alpha(t/2)$ and, similarly, $\alpha(t/2) = \alpha(t/4)$. Iterating this process, we deduce

$$\alpha(t) = \alpha\left(\left(\frac{t}{2^n}\right)\right)$$

for all $n \in \mathbb{N}$. Letting $n \to \infty$, and by continuity of α, we conclude $\alpha(t) = \alpha(0)$, that is α is a constant loop. The arguments for β and γ are similar. □

Exercise 11.21 Prove that the included point topology is simply connected.

Solution We choose the particular point p that defines the topology as base point to compute the fundamental group. Let $\alpha : \mathbf{I} \to X$ be a loop with $\alpha(0) = \alpha(1) = p$ and let c_p be the constant loop $c_p(t) = p$. Define

$$F : \mathbf{I} \times \mathbf{I} \to X, \quad F(t, s) = \begin{cases} p, & s < 1 \\ \alpha(t), & s = 1. \end{cases}$$

It is clear that $F(t, 0) = c_p(t)$ and $F(t, 1) = \alpha(t)$ for all $t \in \mathbf{I}$. Further, if $s \in \mathbf{I}$, $F(0, s) = F(1, s) = p$. It remains to prove the continuity of F. Let $O \subset X$ be a non-trivial open set. Then $p \in O$. In particular, $\mathbf{I} \times [0, 1) \subset F^{-1}(O)$. For the points with $s = 1$, $F^{-1}(O) \cap \{(t, 1) : t \in \mathbf{I}\} = \alpha^{-1}(O)$. Thus

$$F^{-1}(O) = ([0, 1] \times [0, 1)) \cup \left(\alpha^{-1}(O) \times \{1\}\right).$$

We prove that this set is open in $\mathbf{I} \times \mathbf{I}$. Although $[0, 1] \times [0, 1)$ is open in $\mathbf{I} \times \mathbf{I}$, the other set $\alpha^{-1}(O) \times \{1\}$ is not open in $\mathbf{I} \times \mathbf{I}$. However, it is obvious that

$$([0, 1] \times [0, 1)) \cup \left(\alpha^{-1}(O) \times \{1\}\right) = ([0, 1] \times [0, 1)) \cup \left(\alpha^{-1}(O) \times [0, 1]\right),$$

and $\alpha^{-1}(O) \times [0, 1]$ is open in $\mathbf{I} \times \mathbf{I}$ because it is the product of two open sets. This completes the proof that $F^{-1}(O)$ is open in $\mathbf{I} \times \mathbf{I}$. $\square$

11.6 Suggested Exercises

11.6.1 In a space X, let $\{A_n : n \in \mathbb{N}\}$ be a family of simply connected subsets such that $A_n \subset A_{n+1}$. Show that $\bigcup_{n \in \mathbb{N}} A_n$ is simply connected.

11.6.2 Give an injective (resp. surjective), continuous mapping $f : X \to Y$ such that $f_* : \pi_1(X) \to \pi_1(Y)$ is not injective (resp. surjective).

11.6.3 Consider the circle $\mathbb{S}^1$ as a multiplicative subgroup of the $\mathbb{C} \setminus \{0\}$. Let $n \in \mathbb{Z}$. For $f : \mathbb{S}^1 \to \mathbb{S}^1$ defined by $f(z) = z^n$, describe $f_* : \mathbb{Z} \to \mathbb{Z}$.

11.6.4 On $\mathbb{S}^2$, consider the meridians given by $\alpha(t) = (\cos(t), 0, \sin(t))$ and $\beta(t) = (0, \cos(t), \sin(t))$, $t \in [-\pi/2, \pi/2]$. Note that α and β join the south pole with the north pole. Construct an explicit homotopy between α and β.

11.6.5 Let $\bar{t} \in [0, 1]$. Consider $\phi : \mathbf{I} \to [0, \bar{t}]$, $\psi : \mathbf{I} \to [\bar{t}, 1]$ two continuous functions such that $\phi(0) = 0$, $\phi(1) = \bar{t} = \psi(0)$ and $\psi(1) = 1$. Let $\alpha : \mathbf{I} \to X$ be a path and define $\alpha_1, \alpha_2 : \mathbf{I} \to X$ by $\alpha_1(t) = \alpha(\phi(t))$ and $\alpha_2(t) = \alpha(\psi(t))$. Prove that α is homotopic to $\alpha_1 * \alpha_2$.

11.6.6 Show that the circle $\mathbb{S}^1_2$ is a deformation retract of the annulus $A(1, 3)$, proving again that the fundamental group of the annulus is $\mathbb{Z}$.

11.6.7 Find the fundamental group of the following spaces:

1. The paraboloid $z = x^2 + y^2$.
2. $(\mathbb{S}^1 \times \mathbb{R}) \cup (\mathbb{R}^2 \times \{0\})$.
3. $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq 1\} \cup \{(0, y, 0) : y \in \mathbb{R}\}$.
4. The complement of the unit disc, $\mathbb{R}^2 \setminus \mathbb{D}^2$.

5. The union of a sphere with a tangent plane: $\mathbb{S}^2 \cup \{(x, y, z) \in \mathbb{R}^3 : z = 1\}$.
6. The coordinate axes of $\mathbb{R}^2$: $(\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R})$.

11.6.8 Prove that $\{[(x, 1)]\}$ is a deformation retract of the cone $C(X)$ (Exercise 10.8).

11.6.9 Show that the loop $\alpha(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$, in the punctured plane $\mathbb{R}^2 \setminus \{(0, 0)\}$, and with base point $(1, 0)$, is not homotopic to the constant loop.

Bibliography

1. Adams, C.C., Franzosa, R.D.: Introduction to Topology. Pure and Applied. Pearson/Prentice Hall, London (2008)
2. Adamson, I.T.: A General Topology Workbook. Birkhäser, Boston (1996)
3. Armstrong, M.A.: Basic Topology. Undergraduate Texts in Mathematics. Springer, New York (1983)
4. Conway, J.: A Course on Point Set Topology. Springer, Berlin (2014)
5. Crossley, M.D.: Essential Topology. Springer Undergraduate Mathematics Series. Springer, London (2005)
6. Dixmier, J.: General Topology. Undergraduate Texts in Mathematics. Springer, New York (1984)
7. Dugundji, J.: Topology. Allyn and Bacon, Boston (1965)
8. Gaal, S.A.: Point Set Topology. Pure and Applied Mathematics, vol. XVI. Academic Press, New York (1964)
9. Gamelin, T.W., Greene, R.E.: Introduction to Topology. Dover, Mineola (1999)
10. Goodman, S.E.: Beginning Topology. Pure and Applied Undergraduate Texts. American Mathematical Society, Providence (2005)
11. Jänich, K.: Topology. Undergraduate Texts in Mathematics. Springer, New York (1984)
12. Kelley, J.L.: General Topology. Graduate Texts in Mathematics, vol. 27. Springer, New York (1975)
13. Krantz, S.G.: A Guide to Topology. Dolciani Mathematical Expositions. American Mathematical Society, Providence (2009)
14. Manetti, M.: Topology. Unitext, vol. 91. Springer, Cham (2015)
15. McCarty, G.: Topology: An Introduction with Application to Topological Groups. Dover, Mineola (2011)
16. Mendelson, B.: Introduction to Topology, 3rd edn. Dover, New York (1990)
17. Mortad, M.H.: Introductory Topology: Exercises and Solutions. World Scientific, Singapore (2016)
18. Munkres, J.R.: Topology. Prentice Hall, Upper Saddle River (2000)
19. Starbird, M., Su, F.: Topology Through Inquiry. AMS/MAA Textbooks. American Mathematical Society, Providence (2019)
20. Steen, L.A., Seebach, J.A.: Counterexamples in Topology. Springer, NewYork (1978)
21. Viro, O., Ivanov, O., Kharlamov, V., Netsvetaev, N.: Elementary Topology. Problem Textbook. American Mathematical Society, Providence (2008)
22. Waldmann, S.: Topology. An Introduction. Springer, Berlin (2014)
23. Willard, S.: General Topology. Addison-Wesley, Boston (1970)

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5

Index

R. López, *Point-Set Topology*, Springer Undergraduate Mathematics Series,
https://doi.org/10.1007/978-3-031-58513-5